LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXXI

THE CROSSING Full-Systems Playtest Integration

Generic Business Architecture, Universal Production, Contact-Graph Hiring, River Enterprise, Historical Event Control & Autonomous Settlement Growth

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Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XXXI preserves Canonical Project Bible Revision XXX in full except where this integration explicitly adds, deepens, clarifies, corrects or supersedes prior wording. The complete Revision XXX document follows this integration unchanged as the preserved historical layer. Where a conflict exists, Revision XXXI controls. Exact playtest prices, wages, rents, construction costs, land prices, vessel prices, traffic counts, hatch rates, flock thresholds, interest rates, business margins, completion dates and late-campaign accounting totals remain calibration unless this revision explicitly promotes the underlying rule to Canon.

PURPOSE

Revision XXXI integrates THE CROSSING, a 1870 northern-Dakota river-crossing scenario that began with one player, one cabin, $500 cash and no local service town, then developed through ferrying, retail, smithing, property development, construction, lumber, freight, warehousing, independent NPC enterprise, river transport and full portfolio delegation. The playtest completed all authored goals while the player ultimately managed zero ordinary businesses.

The central architectural result is a stronger symmetry between Persons and Businesses: occupation describes a Person but does not mechanically define every task they may attempt; a Business label describes an enterprise but does not mechanically define every activity it may operate. Actual people, physical resources, legal rights, agreements, information and time remain authoritative.

REVISION XXXI MASTER DOCTRINE

- ROLE DESCRIBES; STATE PERMITS. A Person occupation is descriptive rather than a class gate, and a Business label is descriptive rather than a business-class gate. Physical, economic, legal and relational state controls what can actually occur.
- Create Business creates a generic legal/economic Business authority. BusinessOperation, CommercialCapability, BusinessCapabilityRegistry or successor records may organize or describe current activity, but may never become replacement hard classes or permission flags.
- For ordinary production, any Person may attempt a task when its physical prerequisites are available. Competence changes time, quality, waste, failure, rework and safety rather than granting the button. Worker agency, law and contract remain separate reasons an assignment may be refused or prohibited.
- Production is object/method-centric. ProductionMethod plus inputs, equipment, workspace, labor/time and conditions determine feasible execution. Workshops and business labels do not own arbitrary recipe whitelists.
- Recruitment is contact- and evidence-driven. Add Employee exposes actual known/contactable Persons and legitimate recruitment channels, not a world-wide omniscient applicant pool.
- Full delegation can free the player completely from ordinary operations. Managers act through the same Tasks, Agreements, cash, inventory, equipment, information and authority as any other actor; autonomy creates no hidden efficiency.
- Vertical integration removes outside transactions only when real internal capacity replaces them. It never creates free goods, duplicate capacity or consolidated profit.
- Historical Events are configurable. The default is ON; disabling a named authored event prevents that authored event from injecting its causal inputs but does not disable endogenous recessions, weather, disease, fire or other ordinary simulation outcomes.
- Scenario objectives observe authoritative state and ownership. They never spawn people, enterprises, cargo, passengers or capital and must distinguish player-controlled, jointly owned and genuinely independent NPC businesses when the wording requires it.

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PART I - AUTHORITY, SUPERSESSION & CALIBRATION

1.1 Playtest findings versus generated numbers

CANON LOCK - THE CROSSING promotes repeatable causal rules, not its generated dollar amounts or success path. A playtest value may remain useful calibration evidence without becoming a universal wage, rent, land price, construction quote, vessel price, hatch rate, capacity, profitability or timing constant.

1.2 Role describes; state permits

CANON LOCK - Person occupation and Business label are both descriptive layers over deeper state. Occupation does not grant a universal action list. Business name/type does not grant a universal commercial action list. Actual Persons, equipment, workspace, inventory, property rights, agreements, legal authority, information, route access, capital and time determine what can be attempted or operated.

1.3 Descriptive records may not become hidden classes

BusinessOperation, CommercialCapability, BusinessCapabilityRegistry, occupation profiles, tags, templates and similar records may support UI organization, manager reasoning, analytics or authoring. They do not become independent permission authorities. A renamed or newly composed descriptive record must not recreate the rigid class system this revision removes.

1.4 Worker agency remains independent

"Any Person may attempt" means the simulation does not mechanically forbid an ordinary production task merely because the Person lacks the expected occupation or skill. It does not mean an employee must obey. A Person can refuse, negotiate, abandon, escalate or report work because of employment terms, danger, relationships, fatigue, legality, conscience, authority or other ordinary causes.

1.5 Reconciled state still controls exact claims

Revision XXX state-discipline doctrine remains controlling. THE CROSSING also exposed small ledger-presentation discrepancies, reinforcing that narration cannot make unreconciled balances authoritative. Corrections require explicit provenance or replay; repeated downstream use is not reconciliation.

PART II - GENERIC BUSINESS, RETAIL & OPERATING IDENTITY

2.1 Create Business creates a generic economic entity

CANON LOCK - The ordinary Create Business action creates Business identity, ownership/governance and the legal/economic ledger relationship. It does not require choosing Blacksmith, General Store, Sawmill, Saddlery, Construction Company, Freight Company, Pharmacy or another hard business class. Formation costs arise only from real legal, property, equipment, inventory, staffing, finance and working-capital requirements.

2.2 Business form follows actual function - completed

The existing Canon rule that business form follows actual function is extended to its logical conclusion. A Business behaves as a retailer because it offers inventory for sale; as a smithing operation because it possesses and uses forge resources; as a carrier because it controls transport assets and sells transport; as a landlord because it owns/controls property and leases space; and as a mixed enterprise when several such activities coexist.

2.3 Labels, templates and inferred identity

Player-facing names such as General Store, Smith & Repair, Sawmill, Saddlery or Transportation & River Company remain useful. Templates may suggest premises, equipment, supplier lines, stock policies or workflows. Inferred labels may summarize what an enterprise mostly does. None of these may be authoritative for what actions are available.

2.4 Retail and storefront

Retail means that a Business legitimately possesses inventory and offers some of it for sale through an accessible selling arrangement. The inventory may be purchased finished goods, internally manufactured output, farm output, processed food, salvage, used equipment, materials or other lawful goods. A broad assortment may be described as a General Store; a narrow assortment may be described as grocery, hardware, feed, produce or dry goods.

Storefront describes physical customer-facing commercial space, not a business class. Retail can also occur from a workshop, yard, warehouse, landing, home, wagon, stall or other physically valid place.

2.5 Opening, operation and profitability remain separate facts

CANON LOCK - Creating a Business record does not make it operating. A Business opens only when the actual people, time, space where needed, stock/equipment, legal/contract authority, access and transaction path needed for its real activity exist. Building completion, Business opening and profitability remain distinct states. Do not replace this with a hidden Capability=true opening gate.

2.6 One Business may combine activities

One Business may buy/resell goods, manufacture, repair, provide services, rent property, warehouse, haul freight, operate a ferry and provide river transport when its actual resources and agreements support those activities. Activities remain distinct for capacity and accounting even when they share one Business ledger.

2.7 Enterprise grouping versus reorganization

Administrative enterprise/portfolio grouping does not merge the member Businesses. A true merger/reorganization changes legal/economic ownership, cash authority, assets, liabilities, employees and contracts according to actual agreements and creditor requirements. Predecessor history remains.

PART III - UNIVERSAL PRODUCTION, TASK EXECUTION & INTERNAL SUPPLY

3.1 Universal production permission

CANON LOCK - For ordinary production and repair, the simulation may not use occupation or specialist capability as a universal hard permission gate. If the physical method can be attempted with the available materials, equipment, workspace and time, any Person may be assigned to attempt it. This supersedes preserved wording that could be read as requiring specialist capability merely to attempt ordinary production.

3.2 ProductionMethod is the causal center

Each producible object may expose one or more ProductionMethods. A method can define inputs, tools/equipment, workspace, task sequence, labor/time, environmental tolerances, output quantity/quality, waste paths, rework and safety risks. The available catalog is resolved from methods and state, not from a workshop class or occupation recipe list.

3.3 Competence changes execution

Capability, experience, instruction, supervision, task difficulty, equipment condition, material quality, fatigue, environment and time pressure may alter speed, quality, consistency, waste, failure, injury and rework. Novice work is therefore possible without being equally effective.

3.4 Production policy

A manager/player may configure item-level standing policy such as Enabled, Target Stock, Maximum Stock, Priority, Minimum Input Reserve, Make to Stock, Make to Order, External Sale, Internal Orders and customer-order priority. These controls organize work; they do not manufacture capacity.

3.5 Shared physical capacity remains exclusive

A forge, saw, wagon, workbench, worker, stall, bay or workspace cannot satisfy two simultaneous incompatible tasks because multiple operations reference it. Stocked production competes with customer work and internal projects for the same labor and equipment time.

3.6 Internal supply is real production

Smith-made nails, mill-made lumber, internal construction and controlled freight are valid vertical integration only when they consume real inputs, labor, equipment capacity, transport and time. Internal transfer pricing may be zero, at cost or another supported amount, but consolidated reporting eliminates manufactured internal revenue/profit.

3.7 Free owner labor reduces cash cost, not creates revenue

Owner-performed labor can reduce the cash expenditure needed to complete a task. It does not create a fictional wage, sale or mark-up that increases consolidated wealth merely because the owner worked.

PART IV - CONTACT-GRAPH HIRING, EVIDENCE & EMPLOYMENT

4.1 Add Employee uses actual contactability

CANON LOCK - The ordinary employee-selection surface should expose actual Persons the recruiting actor can legitimately contact or reach through a real channel. It must not expose an omniscient global labor list.

- Directly met or previously corresponded Person.
- Employee, associate, supplier, customer, family, friend or former coworker referral.
- Person who answered a handbill, newspaper advertisement or other recruitment campaign.
- Person whose address or contact path was legitimately supplied by another actor.
- Person reached through local businesses, travel networks, mail, messenger or telegraph where available.

4.2 Clicking a stranger does not grant omniscience

World interaction may allow Observe, Approach, Introduce Yourself, Ask Name, Ask About Work or similar actions. Merely clicking an unknown Person does not reveal hidden identity, address, occupation or capabilities. Contact and evidence must be acquired.

4.3 Recruitment channels create awareness and response

Mail, word of mouth, in-person visits, sent messengers/employees, handbills, newspaper advertisements and telegraph are information channels with cost, reach and latency. They can produce a lead or no lead; recipients may reply, ignore, negotiate, visit or decline.

4.4 Known skills are evidence, not full stats

Candidate UI should lead with capabilities relevant to the current role, followed by other known general evidence. Unknown capabilities remain unknown. References, work history, observation and practical demonstrations change evidence/confidence rather than directly exposing hidden truth.

4.5 Capability confidence rendering remains exact

CANON LOCK - Human capability bands remain Ordinary -> Great -> Excellent -> Mastered. High confidence displays exactly one band; Moderate exactly two adjacent bands; Low exactly three adjacent bands; insufficient evidence displays Not enough information. A single-band reading paired with Moderate or Low confidence is invalid.

4.6 Employment offers are agreements

Employment offers may specify wage/pay, housing or board, trial/probation, guaranteed versus conditional employment, start date, expected duties, authority and other real terms. Housing, employment, tenancy and ownership remain distinct relationships unless the agreement explicitly links them.

4.7 Relationship provenance should persist

The world should retain who introduced whom and by what channel. The same relationship graph can support hiring, tenant leads, suppliers, land sellers, lenders, business partners, specialists and migration chains.

PART V - DELEGATION, MANAGEMENT & OWNER ATTENTION

5.1 Zero personally managed businesses is fully supported

Revision XXX full-autonomy doctrine is validated and deepened. The player may act only as proprietor/investor while managers continue ordinary staffing, procurement, production, scheduling, pricing within policy, maintenance, collections, customer service, contracting, property development and opportunity response.

5.2 Standing instruction modes

Player intent should distinguish at least three useful semantics: Do Once for a one-time action; Manager Judgment for an ongoing delegated objective where the manager chooses ordinary means; and Owner Locked for a persistent strategic prohibition or boundary that managers cannot override without explicit change.

5.3 Manager Activity remains visible

Autonomy is not invisible passive income. Manager reporting should expose material decisions, reasons, authority/policy used, resulting commitments and exceptions. Routine activity can stay compact; unusual borrowing, asset sales, owner-lock conflicts or strategic deviations may escalate according to policy.

5.4 Portfolio/general manager

A large portfolio may employ a Person to coordinate managers, screen opportunities, investigate property, arrange suppliers/finance, reconcile cross-business priorities and route capital requests. This is real labor with finite attention, information and authority, not a global automation bonus.

5.5 Employees may perform multiple compatible task families

A carpenter can construct when projects exist and perform repair/joinery when internal construction is idle. A smith can alternate stocked fabrication, customer repair and farriery. Occupation profiles inform fit and evidence but do not lock workers to one action family.

PART VI - PROPERTY, CONSTRUCTION, LUMBER & SETTLEMENT DEVELOPMENT

6.1 Mixed-use commercial/residential development

THE CROSSING validated the two-story storefront/residence as a useful but non-mandatory frontier form. Commercial ground-floor space and upstairs housing can solve both premises and household accommodation for an entrepreneur, but the building never determines which business may occupy it.

6.2 Autonomous property development

A property manager may investigate land, negotiate purchases/options, arrange ordinary financing, commission repeat building types, recruit tenants, lease units, collect rent and service property obligations within delegated risk bounds. Vacancy, weak inquiry, debt service, reserves and construction capacity are real brakes.

6.3 Land acquisition supports more than Buy

Acquisition workflows may include cash purchase, seller financing, mortgage/security, option, lease, parcel severance, easement and other historically supported arrangements. Seller willingness remains causal; strategic land is not automatically available.

6.4 Public-land interest remains distinct from title

Filing, lawful possession/claim, compliance, proof and patent remain separate states. Visually empty land is never equivalent to claimable land, and an unperfected public-land claim cannot be treated as unrestricted fee-simple inventory.

6.5 Generic construction business

A player may create a generic Business, hire carpenters/builders and then operate construction/repair activities from actual tools, workers, materials and contracts. "Construction Company" is a useful description, not an action permission class. Internal projects may receive priority while spare capacity is sold externally.

6.6 Lumber chain

A generic Business with log supply, mill machinery, fuel, workers, storage and hauling can function as a sawmill/lumber supplier. Logs -> hauling -> milling -> lumber inventory -> hauling -> construction is a physical chain. Specialty dimensions, finish stock, glass, roofing and other materials may still require outside procurement.

6.7 Vertical-integration savings must be causal

Owning carpentry, milling, smithing and freight can reduce outside markup and transaction cost only to the extent real internal resources replace those purchases. No VerticalIntegrationCostReduction modifier is authorized.

6.8 Housing enables migration but does not create it

Rental houses and employer lodging create accommodation capacity. Population increases only after actual Persons/Households decide, travel, arrive and establish residence. Employment and tenancy remain separate even when the same Person participates in both.

PART VII - FREIGHT, STORAGE, FERRY & RIVER ENTERPRISE

7.1 Micro-freight entry remains valid

One teamster, two horses, one wagon, harness, feed/stabling access and route rights can support a legitimate local carrier. A yard or warehouse is not required merely to haul freight.

7.2 Warehouse capacity is physical

CANON LOCK - A freight or forwarding business cannot store cargo merely because it exists. Warehousing requires actual suitable space, capacity, access and handling labor/equipment. Building a warehouse can add consolidation, staging, storage and handling services that were previously unavailable.

7.3 Livery, freight, ferry and river service are distinct activities

These activities may coexist in one Business or in dedicated Businesses because they perform different physical work. Livery boards/feeds animals; freight sells cargo transport; ferry provides a crossing; warehousing stores/handles cargo; river service provides trunk transport. Shared horses, wagons, workers, docks and spaces cannot be double-booked.

7.4 Internal transport policy

A controlled portfolio may set a $0 internal freight or ferry tariff. This removes internal cash charges but not the trip, labor, feed, vessel time, wear, queueing or opportunity cost. External customers continue to transact according to ordinary agreements.

7.5 Combined transportation enterprise

Ferry, wagon freight, drayage, warehousing, forwarding and river transport may become operations of one reorganized Business when assets, contracts, employees, debts and cash authority are actually combined. An administrative grouping alone does not perform that merger.

7.6 Staged river development is an example, not a tech tree

THE CROSSING progressed from landing -> forwarding relationships -> warehouse -> opposite-bank frontage -> improved terminal -> chartered steamer capacity -> owned powered vessel. This is a validated systemic path, not a required sequence. Players/NPCs may skip or reorder steps when real resources and opportunities permit.

7.7 Charter before ownership can create evidence

Leasing/chartering outside capacity can reveal actual volume, passenger demand, seasonality, rates, route imbalance, fuel and scheduling needs before major capital purchase. The same evidence-first principle can apply to other expensive equipment and facilities.

7.8 Seasonal navigation changes route state

Ice, flood, wind and navigation season affect actual crossing/river capacity and safety. Seasonal closure is not a generic WinterRevenue penalty; route/service state changes first and the economy reacts afterward.

PART VIII - POULTRY, BIOLOGICAL PRODUCTION & FARM POLICY

8.1 Poultry lineage is operational state

Poultry must preserve parentage/source, fertility, incubation, hatch, sex when known, juvenile stage, mortality and disposition. Breeding policy can deliberately rotate unrelated roosters before daughters enter the breeding group, making lineage information economically useful rather than decorative.

8.2 Breeding flock and commercial laying flock

A poultry operation may prioritize reproduction while small, then shift toward a dedicated breeding group plus a larger commercial laying group as productive scale grows. The exact transition threshold is scenario calibration, not a universal flock-size rule.

8.3 Male birds remain physical

Cockerels are not deleted because they do not lay eggs. Surplus males can remain juveniles, be retained for breeding, sold, slaughtered at appropriate size or otherwise disposed through ordinary systems.

8.4 Biological growth does not manufacture owner equity

Own-produced chicks, eggs, butter, crops, timber/firewood and similar outputs should not automatically create cash or arbitrary mark-to-market profit. Sale, consumption, transfer and the chosen inventory/accounting method determine recognized economic results.

PART IX - HISTORICAL EVENT CONFIGURATION & CAUSAL REGIMES

9.1 Historical Events setting

CANON LOCK - Scenarios that use researched named historical events expose a Historical Events setting. Default = ON. When ON, applicable authored events/regimes may inject their dated, geographic causal inputs. When OFF, those named authored events do not trigger merely because the calendar reaches the historical date.

9.2 Advanced individual overrides

An advanced configuration may permit individual named events/regimes to be enabled or disabled separately when scenario design supports it. The setting belongs to scenario configuration/save state and must not retroactively rewrite events that already occurred unless the user explicitly restarts/rebuilds the scenario.

9.3 Event effects remain causal

The Panic of 1873 remains the controlling example: credit appetite, lender failure, railway finance, projects, migration and demand can change; the event may not directly post a universal percentage penalty. The Great Dakota Boom similarly affects awareness, migration, land, finance, freight and business formation through ordinary systems.

9.4 Historical Events OFF is not No Adversity

Disabling authored historical events does not disable endogenous recessions, bank stress, disease, drought, blizzard, fire, flood, crop failure or other outcomes generated by ordinary simulation. The toggle controls authored historical injection, not causal simulation generally.

PART X - SCENARIO OBJECTIVES, ACCOUNTING & STATE DISCIPLINE

10.1 Objectives observe; they do not produce

Population, independent business, freight, passenger, ownership and equity objectives query authoritative state. They do not create settlers, entrepreneurs, cargo, passengers, land, financing or revenue needed to satisfy themselves.

10.2 Historical/event objectives normally persist

Once a milestone such as first independent business, first household, cumulative freight or cumulative passenger crossings is completed, it normally remains complete unless the objective explicitly states a maintained/current condition.

10.3 Ownership predicates must be explicit

CANON LOCK - Objectives that require NPC-owned or independently owned businesses must exclude player-controlled subsidiaries and other disallowed ownership structures according to the authored predicate. Joint ownership, wholly player-owned, wholly NPC-owned and independent operation are not interchangeable.

10.4 Owner equity definition is scenario authority

Scenarios must explicitly define which starting assets, personal property, business interests, land interests, cash and liabilities count toward owner equity. A supplied horse/tack or similar starting asset may not be inconsistently included/excluded mid-campaign.

10.5 Debt funds assets but does not create equity

Borrowing can accelerate growth by financing real assets and working capital. Loan proceeds alone do not increase owner equity because the corresponding liability remains. Objective equity uses the authoritative asset/liability definition.

10.6 Land appreciation versus realized value

Estimated market value may be exposed analytically, but scenario objectives must state whether they use book/cost basis, supported appraised transferable value, realized sale proceeds or another authoritative valuation rule. Town growth may not silently mark land upward merely to satisfy an objective.

10.7 Explicit correction transactions

Detected ledger mistakes such as price transcription or omitted asset classification require explicit adjustment/reconciliation provenance. They may not be silently rewritten in historical reports.

PART XI - NPC ENTERPRISE, PARTNERSHIP & SERVICE COMPOSITION

11.1 NPC entrepreneurs use the same generic Business architecture

A Person who sees unmet demand may investigate premises, capital, equipment, suppliers, staffing and customers and then form a generic Business. The resulting observed identity - saddlery, bakery, dry goods, tinwork or other trade - follows the same state-based doctrine as player Businesses.

11.2 Viability comes from reachable market, not town headcount

Resident population is only one source of demand. Rural farms, transient travelers, ferry traffic, freight teams, surrounding settlements and route position can support a business that local headcount alone could not. No universal Population >= X business formula is authorized.

11.3 50/50 silent partnership

Ownership percentage, capital contribution, profit/loss allocation and operating authority remain separate. A player may hold 50 percent while another owner manages day-to-day operations. Material governance rights and approval thresholds follow the actual agreement rather than ownership implying constant owner clicks.

11.4 Physician and pharmacy may be separate or combined

THE CROSSING used a physician tenant and separate player-owned druggist/pharmacy, which is valid but not universal. Under generic-business doctrine, medical consultation and pharmacy/dispensing may be separate or combined when actual ownership, people, premises, equipment, inventory, agreements and dated law support the arrangement.

11.5 Contact graph extends beyond hiring

Known/contactable relationships can also generate tenant prospects, land opportunities, supplier introductions, financing, partnership offers, specialists and migration leads. A request such as "Do you know anyone who..." should query real relationships and may return no suitable lead.

PART XII - THE CROSSING PLAYTEST FINDING REGISTER

CROSS-CAN-001 Role describes; state permits. Person occupation and Business label are symmetrical descriptive layers, not class permissions.

CROSS-CAN-002 Create Business is generic. Actual operations emerge from people, assets, space, inventory, rights, agreements, information and time.

CROSS-CAN-003 Ordinary production is universally attemptable when physical prerequisites exist; specialist capability no longer acts as a universal permission gate.

CROSS-CAN-004 Worker agency is independent of mechanical feasibility; assignment may still be refused or abandoned.

CROSS-DES-001 Retail assortment and storefront are activity/space state, not GeneralStore class authority.

CROSS-DES-002 BusinessOperation/CommercialCapability records may describe or organize work but cannot become replacement hard classes.

CROSS-DES-003 Production policy supports enabled recipes, targets, maxima, input reserves, priorities and make-to-order/stock behavior over real capacity.

CROSS-DES-004 Hiring uses a known/contactable Person graph and real channels rather than a global applicant pool.

CROSS-DES-005 Historical Events setting defaults ON and can disable authored named-event injection while preserving endogenous simulation.

CROSS-VALID-001 Smithing hardware production consumed iron, fuel, worker time and forge capacity while reducing outside purchases.

CROSS-VALID-002 Construction and sawmilling worked as generic Businesses whose observed identities followed owned equipment and tasks.

CROSS-VALID-003 A mixed-use storefront/residence supported independent entrepreneur migration without dictating business identity.

CROSS-VALID-004 Keller saddlery viability depended on rural/ferry/freight catchment, not seven local residents.

CROSS-VALID-005 Contact/referral provenance generated hires, tenants and specialists without spawning Persons.

CROSS-VALID-006 Full portfolio autonomy operated for extended periods with zero ordinary player management.

CROSS-VALID-007 Property development correctly paused when vacancies, leverage or reserves made further speculation unattractive.

CROSS-VALID-008 Freight, ferry, warehouse, forwarding and river service coexisted as distinct physical activities inside one reorganized Business.

CROSS-VALID-009 Chartered river capacity produced demand evidence before owned-vessel investment.

CROSS-VALID-010 Poultry breeding required lineage, sire rotation, maturation and disposition; commercial laying policy changed as flock scale grew.

CROSS-VALID-011 Independent-business objectives required genuinely independent ownership and could not be satisfied by player subsidiaries.

CROSS-VALID-012 Debt financed growth without itself increasing equity; property and enterprise value had to remain net of associated liabilities.

CROSS-QA-001 Capability confidence formatting must be structurally validated: High one band, Moderate two adjacent, Low three adjacent.

CROSS-QA-002 Ledger discrepancies must be corrected explicitly; narration/repetition is not reconciliation.

CROSS-QA-003 Scenario equity must define treatment of supplied starting assets before play begins.

CROSS-CAL-001 Exact rents, wages, land/building/vessel prices, interest, hatch rates, flock thresholds and profit levels remain calibration only.

PART XIII - RESEARCH & CALIBRATION HOLDS

- Exact production-time, quality, waste, injury and failure distributions by task, skill, equipment and supervision.
- Exact period productivity and cost for hand-forged nails/hardware, small sawmills, frontier lumber quality and log-hauling chains.

- Exact recruitment conversion, referral reach, mail/telegraph timing and evidence reliability by geography/date.
- Exact employment terms, trial/probation practice, employer housing and guaranteed-work conventions by trade/place/date.
- Exact land, rent, construction, warehouse, dock, ferry, freight and shallow-draft river-vessel economics by corridor/year.
- Exact poultry fertility, incubation, mortality, maturity, laying curves and commercially rational breeding-group sizes by breed/conditions.
- Exact historical event severity/probability and local exposure. The configuration toggle changes authored-event injection, not these causal datasets.
- Exact valuation basis for scenario owner-equity objectives remains authored scenario policy; no universal mark-to-market rule is promoted.
- Exact legal/licensing limits on medical, pharmacy, craft, transport and other activities remain LegalRegime data and can still make an otherwise physically possible action unlawful.

PART XIV - REVISION XXXI LOCK SUMMARY & DEEPENING MAP

- Revision XXX Business Form Follows Actual Function doctrine is completed into a generic Business architecture. Labels, archetypes, BusinessOperation and capability registries may organize state but do not grant permissions.
- Preserved occupation-is-descriptive and role-is-not-capability doctrine is mirrored at Business level: Person occupation does not determine what the Person may attempt; Business label does not determine what the Business may operate.
- Preserved task/capability doctrine is superseded where it could be read as requiring specialist capability merely to attempt ordinary production. Capability instead modifies execution; worker agency and legality remain separate.
- Existing Physicality, Task, Equipment, FunctionalSpace, Inventory, Agreement, Property, Logistics, Finance and LegalRegime authorities remain controlling and are reused rather than replaced.
- Recruitment/referral doctrine is deepened into contact-graph employee selection, evidence-limited skill display and relationship provenance across hiring, tenants, suppliers, partners and migration.
- Portfolio autonomy is deepened with explicit one-time versus Manager Judgment versus Owner Locked semantics and visible Manager Activity.
- Property/construction doctrine is deepened with generic construction and lumber examples, causal vertical-integration savings, options/land assembly and vacancy-aware repeat development.
- Freight/river doctrine is deepened with physical warehouse capacity, true business reorganization, combined transport activities, charter-before-ownership evidence and seasonal navigation.
- Poultry lineage doctrine is deepened with sire rotation, breeding-versus-laying policy and explicit surplus-male disposition.
- Historical regime doctrine is deepened with player/scenario configuration: Historical Events default ON, optional OFF/individual override, while all effects remain causal and endogenous incidents remain possible.
- Scenario objective doctrine is deepened with explicit ownership predicates, persistent event milestones, scenario-defined equity treatment and no objective-generated state.

END OF REVISION XXXI INTEGRATION

The preserved Canonical Project Bible Revision XXX follows unchanged and remains authoritative except where Revision XXXI explicitly adds, deepens, clarifies, corrects or supersedes a rule.

LAND & LEDGERS
CANONICAL PROJECT BIBLE - REVISION XXX

The Market Town Full-Systems Playtest Integration

Autonomous Portfolios, Rental Development, Private Banking, Demographic Continuity, Health, NPC Enterprise & State Discipline

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PURPOSE

Revision XXX integrates the complete MARKET TOWN playtest. The campaign tested a small service settlement growing into a more mature town while the player delegated nearly all ordinary operations, built a rental-development pipeline, formed a construction company, evolved a loan-and-exchange office into a private banking house, added pharmacy/barber/butchery services, relied on NPC entrepreneurs for some missing businesses and allowed population to mature through migration, aging, relationships, marriage, birth, illness and death.

The main result is not a fixed “best build order.” It is a set of stronger causal rules for portfolio autonomy, cash discipline, rental development, bank governance, NPC enterprise formation, mixed-use professional premises, health-service demand and generational continuity.

REVISION XXX MASTER DOCTRINE

- Owner strategy can be expressed as persistent policies. Full Autonomy means managers act through ordinary authority, information, cash, labor and contracts rather than receiving hidden efficiency.
- Cash, protected cash, debt, deposits, owner equity and business equity are different facts. Earmarking cash never creates a second cash balance.
- Rental development is a causal loop: acquire lawful site, finance, build through real labor/materials, complete accommodation, recruit/lease, receive rent, service obligations and only then reinvest.
- Housing can make migration feasible but never directly creates population. External-household preference is a player policy, not a population generator.
- A player-owned construction company may prioritize internal capital projects and sell spare capacity externally. Internal work can capitalize real cost but cannot manufacture consolidated profit.
- Finance can begin below full banking scale through own-account lending, note discounting/placement, collections, exchange and correspondent relationships. Public deposits introduce a new liability and liquidity problem.
- A player-owned bank should expose deep policy control. Lawful limits should be visible; historically possible unlawful actions should generally remain performable when the player deliberately orders them, with evidence and consequences handled by LegalRegime/Enforcement rather than by an invisible prohibition.
- NPC entrepreneurs use the same economic world. They may capture opportunities the player ignores, then become suppliers, customers, borrowers, tenants, employers, competitors or acquisition targets.
- Growing settlement demand must propagate into real workload, queues, staffing, inventory and supplier relationships. Existing businesses do not process unlimited new demand with frozen staffing.
- Health, injury and treatment are causal Person systems. Illness can remove labor availability, create physician/pharmacy demand and sometimes end in impairment or death.

- Children and teenagers already in the world age into new work, relationship and household states. Marriage, births, deaths and household splits are persistent demographic events, not flavor text.
- Scenario objectives measure accomplishments. They should not secretly spawn the customers, households, businesses or infrastructure needed to complete themselves.

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PART I - PLAYTEST AUTHORITY, STATE DISCIPLINE & CALIBRATION

1.1 Design findings versus generated numbers

CANON LOCK - Playtest-generated economic values are evidence for calibration, not automatic Canon. A repeatable causal behavior may be promoted while the exact dollar amount, wage, rent, construction quote, interest balance, completion date or profitability remains scenario data until separately justified.

The MARKET TOWN campaign exposed several valuable systems while also exposing a serious text-simulation continuity hazard. The campaign's late narrative state was useful for design discovery, but an earlier unresolved accounting checkpoint meant that exact downstream totals could not be treated as audited historical truth. Revision XXX therefore preserves the systems findings while explicitly refusing to promote the exact soft-continuity late-campaign balances into canonical calibration.

1.2 Reconciled checkpoints control exact state

CANON LOCK - An autonomous/text playtest may not silently continue exact financial or world-state claims from a checkpoint known to be unreconciled, voided or contradictory. Later exact balances derived from that checkpoint remain non-authoritative until the branch is replayed/reconciled or explicitly accepted as soft continuity. Design observations may survive when they do not depend on the disputed number.

Required discipline:

- Never invent a balancing transaction merely to make a ledger close.
- Never silently delete or recreate Persons, inventory, debt, leases, deposits, loans or ownership to repair continuity.
- Objective completion dates that depend on equity, population, ownership or debt require authoritative supporting state.
- A soft-continuity campaign can continue for design exploration, but reports must not later relabel those exact figures as audited truth.

1.3 Clear owner reporting

Player-facing long-run reports should present new playtest notes first, then date/population/objectives and a clear ledger. At minimum distinguish:

- owner-personal cash;
- each business's cash;
- protected/committed cash inside that same balance;
- principal debt and material liabilities;
- bank customer deposits separately from ordinary debt;
- consolidated owner equity separately from liquidity.

Equity is never presented as spendable cash.

PART II - PORTFOLIO AUTONOMY, OWNER ATTENTION & CAPITAL ALLOCATION

2.1 Full portfolio autonomy

A mature player may personally manage zero ordinary businesses. Managers can control routine staffing, procurement, schedules, pricing within policy, ordinary maintenance, collections, customer service and small capital decisions when authority permits. Owner attention is reserved for material exceptions, strategic changes, unusual borrowing, major asset sales or explicit owner locks.

2.2 Persistent owner policies

Standing instructions can include reserve floors, reinvestment mandates, preferred lead routing, rental-first disposition, internal-project priority, hiring boundaries, bank risk policy and escalation thresholds. Policies constrain delegated decisions but do not create outcomes directly.

2.3 Operating reserves and owner sweeps

CANON LOCK - A wholly owned business may distribute excess cash after genuine operating, payroll, inventory, maintenance and debt needs are protected. Reinvesting that distribution into another controlled business is recorded as Business Distribution -> Owner Cash -> Owner Capital Contribution unless a real intercompany loan is explicitly intended. The movement creates no consolidated revenue or profit.

Reserve floors are targets/read models, not magical lockboxes. A manager may temporarily retain more cash when known obligations justify it.

2.4 Settlement growth must alter workload

If population, transient traffic or rural catchment activity grows, existing businesses should experience the consequences through actual customers, orders, inventory demand, queues, cleaning, bookkeeping, route load and employee time. Managers may add workers, extend shifts or change supplier relationships only when telemetry supports it. The MARKET TOWN saloon was a controlling example: more people increased patronage, but also increased bartender/server, housekeeping, food, liquor and cleaning costs.

PART III - RENTAL HOLDINGS, HOUSING DEVELOPMENT & MIGRATION

3.1 Rental holdings as an autonomous property operation

A property business may own multiple residential and commercial assets while preserving per-property rent, debt, condition, vacancy, maintenance and tax histories. A dedicated property manager may also perform third-party agency work under real Agency/Management agreements; third-party property does not become player inventory.

3.2 Rental cash waterfall

The MARKET TOWN standing policy is promoted as a supported player-configurable pattern:

1. pay property-level operating obligations;
2. reserve enough actual cash for the next known property debt-service obligation according to policy;
3. retain maintenance/tax/contingency liquidity;
4. use genuinely uncommitted surplus toward the next acquisition, down payment or construction project.

CANON LOCK - A debt-service reserve is an earmark within authoritative cash, not a duplicate treasury balance. “Available cash” may subtract protected commitments for decision support, but only one cash authority exists.

3.3 Repeat-development authority

A property manager can be granted standing authority to investigate and acquire ordinary suitable residential sites and repeat an approved development type within financing, price, reserve and risk bounds. Major departures - unusually large projects, sale of core rental assets, extraordinary leverage or nonstandard ownership structures - may still escalate.

3.4 Housing and external migration

CANON LOCK - Housing does not increment population. A household becomes a permanent resident only through an actual migration/occupancy path: awareness or referral -> opportunity evaluation -> acceptable work/income path where needed -> available accommodation -> travel -> physical arrival -> residence stabilization.

The owner may prioritize external households when the strategic aim is population growth. Local households remain eligible. External preference must not force indefinite vacancy when suitable local demand exists.

3.5 Build first, recruit near credible delivery

For speculative rental housing, aggressive recruitment should normally intensify when completion is close enough to be a credible offer, rather than months before a highly uncertain project. Earlier awareness/referral can exist, but no tenant or resident is guaranteed.

3.6 Employment, tenancy, construction labor and ownership remain separate

An employee may help build a dwelling that they later rent without acquiring equity. A tenant may work for another employer. A buyer may purchase a completed house if the owner chooses to sell. These are separate Agreement/Employment/Property relationships unless an explicit contract links them.

3.7 Rental-first does not mean sale-prohibited

The player may adopt a rental-first disposition policy while still authorizing sale when a buyer strongly prefers ownership and the terms create an acceptable return or strategic benefit. A completed rental milestone remains historical unless the scenario explicitly requires continued ownership.

3.8 Vacancy is a real brake

A successful first duplex does not prove infinite demand. Persistent vacancy, weak inquiry quality, inadequate reserves, financing stress or construction bottlenecks can delay the next acquisition. Managers should respond to absorption evidence rather than a perpetual-build flag.

3.9 Housing form can vary

Two independent dwellings on one property are a valid efficiency strategy when site, sanitation, household fit and economics support it. The same policy can result in a duplex, two detached cottages or another historically appropriate configuration; no single prefab is the rule.

PART IV - CONSTRUCTION AS AN INTERNAL CAPITAL PIPELINE

4.1 Internal-first construction company

A player-owned construction enterprise may operate under the policy “internal owner projects first, outside work with spare capacity.” A foreman can schedule crews, specialists, materials and outside contracts autonomously within authority.

4.2 No manufactured internal profit

CANON LOCK - When a controlled construction company builds an asset for another controlled business, consolidated reporting preserves actual labor, material, transport, specialist and overhead cost while eliminating internal revenue/profit. The receiving property’s basis may capitalize legitimate project cost.

4.3 Physical time remains real

Repeated designs can reduce planning/rework modestly through familiarity, but they do not erase labor, material, weather, specialist or logistics requirements. Construction timing should continue to use the Historical Work, Construction & Infrastructure Calibration Reference as a calibration source, distinguishing active work from elapsed, lead/queue and interruption time.

PART V - FROM LOAN & EXCHANGE OFFICE TO PRIVATE BANKING HOUSE

5.1 Capital-light finance is a legitimate early form

A small finance office may begin with own-account lending plus note/loan placement, collections, exchange/remittance, correspondence and outside lender relationships. It need not accept public deposits on day one.

5.2 Outside-funded placements remain outside assets

CANON LOCK - When the office originates or services a loan funded and owned by an outside lender, the loan is not an asset of the player’s finance business and the lender’s funding is not the player’s cash. The office records its actual fees, servicing obligations and any retained exposure.

5.3 Conversion to deposit banking is a qualitative step

Transition from lender/exchange office to private banking house should depend on dated legality, genuine capital, capable staff, secure records/premises, correspondent relationships, liquidity and market demand. No universal playtest dollar threshold is Canon.

5.4 Deposits are liabilities

CANON LOCK - Public deposits create liabilities owed to depositors. Bank cash and correspondent balances protect withdrawals and settlement; loans and other earning assets can generate income while reducing immediate liquidity. Deposit growth is not owner wealth.

A bank panic or withdrawal burst should be resolved through actual liquid assets, correspondent access, loan maturities, asset sales, new capital, suspension/failure rules and depositor behavior - never through a single reputation check.

5.5 Deep bank policy control

A player-owned bank should expose unusually deep control over:

- liquidity/correspondent targets and withdrawal preparedness;
- deposit products/terms where lawful;
- rate/pricing policy within dated law;
- collateral standards and appraisal evidence;
- loan maturity preferences;
- maximum borrower exposure and concentration by sector/location/collateral family;
- connected-party lending policy;
- approval thresholds and delegated lender authority;
- delinquency, restructuring, enforcement and write-off policy;
- dividends/retained earnings and capitalization;
- staffing, opening hours, reconciliation and control procedures.

A single “Risk” or “Reserve” slider is insufficient.

5.6 Connected lending remains real

A bank loan to the owner or another controlled business is still a real loan on each entity’s books, with genuine underwriting, collateral, liquidity and concentration implications. For consolidated owner reporting, the internal receivable/liability pair is eliminated, while the bank-level exposure remains visible because depositors and bank solvency still care about it.

5.7 Intentional illegality

CANON LOCK - Where an action is physically and institutionally possible, the game should not generally protect the player from historically meaningful illegality by silently disabling the action. A player may deliberately direct unlawful deposit-taking, disclosure, lending, recordkeeping or other conduct when the dated LegalRegime defines it. The order/action creates evidence and can trigger employee refusal, concealment, discovery, civil/criminal exposure, closure, loss or reputation consequences according to the world state.

This is not permission for laws to become optional. It is a simulation rule: law operates through authority, evidence and enforcement rather than invisible design rails.

PART VI - NPC ENTREPRENEURS, PRESSURE & COMMERCIAL NETWORKS

6.1 Opportunities are not reserved for the player

CANON LOCK - When unmet demand, supplier failure, congestion, customer travel or other evidence creates a business opportunity, an NPC entrepreneur may investigate and capture it through the same property, finance, staffing, inventory and information systems available to the player.

The independent bakery in MARKET TOWN is the controlling playtest example. The player did not open it, but it later became a wholesale supplier to Pike Hall and a banking customer. An NPC business can therefore be competitor, supplier, customer, borrower, tenant, employer or later acquisition target simultaneously.

6.2 Opportunity pressure should resolve causally

When a new NPC business successfully serves a missing need, the underlying pressure should change because real transactions, supply and capacity changed - not because an Opportunity flag was manually cleared.

PART VII - MIXED-USE PROFESSIONAL SPACE, PHARMACY, BARBER & BUTCHERY

7.1 Mixed-use commercial buildings

One building may contain separate FunctionalSpaces used by distinct businesses. The MARKET TOWN health/trade building combined a player-owned pharmacy, an independent physician’s consulting room and a player-owned barber room. Shared structure does not merge cash, ownership, employees or service authority.

7.2 Pharmacy is both professional service and retail

A Druggist/Pharmacist requires actual capability, compounding/dispensing equipment, inventory, records and supplier access. Pharmacy demand can arise from physician treatment, household self-purchase, injury care and ordinary retail health demand.

7.3 Patent/proprietary medicines

CANON LOCK - Historically sold patent/proprietary medicines may be stocked and marketed when dated availability, supply and law support them. Product advertising claims, pharmacist belief and actual treatment effect remain separate facts. A cure-all does not gain medical efficacy because its label says so.

Exact inventory, formulation, toxicity, controlled-substance status and legal sale requirements remain dated research data.

7.4 Health-treatment chain

Illness or injury can create absence, reduced duty, physician/attendant tasks, pharmacy purchases, treatment, recovery, lasting impairment or death. Treatment affects outcome only where the actual intervention plausibly matters. Owning a doctor/pharmacy does not automatically prevent mortality.

7.5 Barber does not imply dentist

A Barber is primarily a grooming/service occupation. Dentistry remains a separate capability/profession. A particular person may perform limited dental work only when evidence, capability, tools and the dated legal/social context support it; the Barber title alone never grants dental treatment ability.

7.6 Whole-animal butchery remains controlling

Butcher economics continue to require allocation across desirable cuts, lower-value product, preservation, hides/tallow and true waste/spoilage. Growing town demand can support a meat market, but no carcass becomes 100 percent premium retail output.

PART VIII - DEMOGRAPHIC CONTINUITY & GENERATIONAL TOWNS

8.1 Existing children age into economic actors

CANON LOCK - Long-run simulation must periodically reconsider existing Persons as they age. A teenager can move from dependent/no formal work into family work, irregular paid work, apprenticeship or regular employment according to age, household need, law, opportunity, capability and local norms. There is no universal Age18CreatesWorker event.

8.2 Relationship formation and marriage

Older teenagers and young adults can form relationships, court, marry and create new households. Ages 18-19 are possible but should not become the default center of the distribution merely because the simulation exposes them. Dated law and calibrated demographic distributions govern eligibility and frequency.

8.3 Household split is not population growth

Two existing residents who marry and rent a new unit change household/occupancy structure but do not increase town population. A spouse migrating from outside may increase population on arrival. The distinction must remain explicit.

8.4 Births and deaths are persistent Person events

Pregnancy schedules a possible later birth; birth creates a new Person. Death removes a living Person while preserving historical identity, estate/kinship and business consequences. Population summaries are derived from actual living resident Persons.

PART IX - SCENARIO DESIGN: THE MARKET TOWN & A TOWN FROM NOTHING

9.1 THE MARKET TOWN as a systems fixture

The campaign validated a scenario in which objectives primarily measure ownership, equity and settlement-population accomplishments while the world creates difficulty through housing, labor, customers, lenders, competitors and physical projects. Exact completion dates and generated late-campaign balances are not locked.

9.2 A TOWN FROM NOTHING / THE PROMOTER

CANON LOCK - The next scenario concept is preserved and deepened: begin at an effectively undeveloped remote site rather than inside an already functioning settlement. The scenario's goals should focus on creating a durable town through lawful site control, access, structures, housing, employment, businesses/services, permanent households and institutions.

The starting world may contain distant farms, routes, legal land interests, external nodes and potential counterparties, but it should not quietly provide the player with a complete service town. Scenario objectives must observe development; they must not spawn guaranteed settlers or enterprises to satisfy themselves.

Objective families may include first lawful site/control, first permanent non-player household, first operating service business, accommodation capacity, population milestones, independent/NPC enterprise formation, infrastructure access, durable finance or other researched outcomes. Exact objective list and deadlines remain authored scenario data.

PART X - PLAYER-FACING UX DOCTRINE

10.1 Long-run report order

For autonomous simulation, surface new design/playtest notes first, then date/population/objectives, then the opening ledger and major causal events. Routine manager activity can stay compact unless it creates an exception.

10.2 Ledger presentation

The default portfolio summary should show at least:

- personal owner cash;
- business cash by entity;
- protected/committed portion where meaningful;
- bank deposits and other material liabilities;
- principal debt separately from accrued interest when material;
- non-cash assets where needed to explain equity;
- consolidated equity with a warning that it is not liquidity.

10.3 Bank UI

Bank management should provide both a compact health summary and drill-down views for liquidity, deposit liabilities, maturity ladder, concentration, connected exposure, delinquency/workouts, correspondent balances, capital/equity and legality warnings. The UI may warn that a proposed policy is unlawful without necessarily preventing deliberate execution.

PART XI - RESEARCH & CALIBRATION HOLDS

The following remain calibration/research rather than universal rules:

- exact 1878-1884 wages, rents, residential land prices, construction costs, freight rates and business margins;
- private banking capitalization needed for credibility/profitability in a specific settlement;
- bank charter, examination, disclosure, reserve, insolvency and enforcement rules by exact jurisdiction/date/institution type;
- exact probability, detection and consequences for unlawful banking behavior;
- marriage-age distributions by region, sex, class/community and period, including early marriage tails;
- illness/injury incidence, treatment efficacy, medicine availability, drug law and patent-medicine product catalogs by date/place;
- service-entry thresholds for bakery, butcher, barber, pharmacy, dentist and similar businesses; use catchment evidence rather than population formulas;
- rental vacancy, maintenance, taxes, insurance, financing, rent levels and development yields;
- exact staffing thresholds for growing businesses; workload telemetry controls implementation;
- exact migration response to available housing, jobs, kin/referrals and incentives;
- exact construction timing; use task/scope-based historical calibration rather than one building timer.

PART XII - MARKET TOWN PLAYTEST FINDING REGISTER

Provenance	Integrated finding
MARKET-UX-011	Ledger must separate owner cash, business cash, committed/reserved cash, debt and equity; equity is not spendable cash.
MARKET-VALID-037	Small rental portfolios may justify scalable/part-time agency management before a full-time manager.
MARKET-POL-006	External-household priority is a configurable housing/recruitment policy, not a migration bonus.
MARKET-VALID-038	Referral/job/housing/travel/arrival/stabilization is the qualifying migration loop.
MARKET-POL-007	Managers may receive repeat-development authority inside explicit acquisition/financing bounds.
MARKET-RES-007/008/009	Banking legality and institution form are dated/jurisdictional; national-bank-scale rules must not be applied to every private banking house.
MARKET-CAL-009	A very small owned loan book cannot support expensive banking staff on interest spread alone; fee/intermediation business matters during early scale.
MARKET-VALID-043	Outside-funded loan placement can scale financial activity without misclassifying outside funds/notes as player assets.
MARKET-POL-008	A finance office may deliberately operate without public deposits until the owner enables a lawful deposit-taking institution.
MARKET-POL-009	Connected lending is explicit and reviewable; common ownership does not erase credit/liquidity risk.
MARKET-POL-010	Excess-cash sweeps use real distributions and capital contributions; rental cash can remain under a separate reinvestment mandate.
MARKET-VALID-044/045	Finance scales through both transaction volume and staffing capacity; administrative workload can become the bottleneck.
MARKET-VALID-046	Rental growth can pause when vacancy/reserves/financing do not support another build.
MARKET-VALID-047/048	Portfolio sweeps and financial intermediation can become a coherent capital-allocation system without creating fake revenue.
MARKET-VALID-049/050	Vacancy and competing capital priorities create genuine delays; not every desired project starts immediately.
MARKET-VALID-051	Missing-business pressure must use actual service catchment and transaction evidence, not a simple population formula.
MARKET-VALID-052	Settlement growth must feed back into real workload and staffing at existing businesses.
MARKET-VALID-053	Mixed-use professional buildings can host independent businesses without merging their economic identities.
MARKET-CAN-009 / MARKET-DES-010	Bank policy deserves deep player control, including deliberate unlawful choices where physically possible and dated law can respond.
MARKET-VALID-054	NPC-created businesses can become player suppliers, customers, borrowers, tenants or acquisition targets.
MARKET-CAL-010	Young-adult work/relationship/marriage behavior requires a distribution and lifecycle model, not an age-18 switch.
MARKET-VALID-055	Health events create real labor loss, treatment demand and pharmacy transactions.
MARKET-RES-011	Patent/proprietary medicines belong in period pharmacy commerce, while marketing claims remain separate from actual effect.
MARKET-VALID-056–060	Long-run towns should mature through NPC counterparties, generational demography, health continuity, vacancy brakes and autonomous portfolio management.
MARKET-DEF-023	The original low double-house quote demonstrated the need to calibrate construction from real labor/material scope before commitment.
State-discipline defects	The repeated continuation from an unreconciled/void branch is promoted into a hard governance rule: invalid exact state cannot silently become authoritative through repetition.

PART XIII - REVISION XXX LOCK SUMMARY & DEEPENING MAP

- Revision XXIX finance doctrine is deepened with staged finance-office-to-bank development, external loan placement, deposit-liability UX, bank policy control and intentional illegality through LegalRegime/Enforcement.
- Earlier Manager Activity/Full Autonomy doctrine is deepened with portfolio-wide reserve/sweep policy, property development authority and owner-attention reporting.
- Earlier housing/migration doctrine is deepened with debt-service cash waterfalls, external-household priority, recruitment-near-delivery and vacancy-based construction brakes.
- Construction doctrine is deepened with internal-project priority and explicit consolidated elimination of controlled construction revenue/profit.
- NPC parity is deepened so unmet opportunity can form new NPC businesses that become multi-role counterparties to the player.

- Person/Household continuity is deepened with aging into work, relationship formation, marriage/household split and health-service chains visible in long simulations.
- Commercial space doctrine is deepened with mixed-use professional buildings and separate physician/pharmacy/barber economic identities.
- Autonomous/text QA doctrine is hardened: unreconciled checkpoints cannot become authoritative exact state by narration alone.
- A TOWN FROM NOTHING / THE PROMOTER is preserved as the next scenario concept and explicitly begins without a convenient pre-existing service town.

END OF REVISION XXX INTEGRATION

The preserved Canonical Project Bible Revision XXIX follows unchanged and remains authoritative except where Revision XXX explicitly adds, deepens, clarifies, corrects or supersedes a rule.

LAND & LEDGERS
CANONICAL PROJECT BIBLE - REVISION XXIX

Frontier Historical Systems Integration: Land Entry, Transport Shocks, Information Networks, Rural Markets, Credit, Cooperatives, Distress & Regional Chronology

Revision date: October 5, 2026
Source baseline: Canonical Project Bible Revision XXVIII - October 5, 2026
New source integrated: Historical Research Passes I-IV and conversation synthesis - October 2026
Status: Active integration

PRESERVATION GUARANTEE

REVISION XXIX preserves Canonical Project Bible Revision XXVIII in full except where this integration explicitly adds, deepens, clarifies, corrects or supersedes prior wording. The complete Revision XXVIII document follows unchanged as the preserved historical layer. Where a conflict exists, Revision XXIX controls. Exact local prices, traffic counts, loss percentages, adoption rates and procedural details remain dated evidence or calibration unless explicitly promoted to a Canon lock.

PURPOSE

Revision XXIX consolidates the frontier-history research developed after the 1870 historical-envelope work. It does not add a disconnected event-card layer. Instead it maps public-land law, route change, environmental shocks, communications, wholesale commerce, agricultural institutions, credit, insurance, cooperatives, distress and regional legal chronology onto the existing Property, Agreement, Logistics, Information, Evidence, Finance, Institution, Project, Inventory, Person and LegalRegime authorities.

REVISION XXIX MASTER DOCTRINE

- Historical change normally alters physical resources, legal rights, routes, information, counterparties, credit or institutions first; prices, migration, profit and business survival react afterward.
- Southeastern Dakota remains the strongest default Open Frontier anchor, but nearby Minnesota, Nebraska, northern Dakota, West River/Black Hills and selected Wyoming corridor scenarios may use the same systems with different historical priors.
- Longitude alone never determines modernity. Corridor position, rail construction history, river crossings, outside markets and institutional access can make a farther-west location better connected than a nominally eastern one.
- Public land is a family of legal processes. Filing, possession, compliance, proof, patent, railroad ownership, private title, lease, reservation/federal restriction and tax/creditor interests remain distinct states.
- Rail, stage, ferry, river, wagon, mail, express and communications networks can create temporary boom corridors and can later strand or bypass investments without scripted town-death events.
- Information latency is historical gameplay. The world may know a fact while the player, manager, lender or supplier learns it later through mail, newspaper, telegraph, telephone, commercial report, fair, traveling salesman or observation.
- Mail order expands supplier discovery, not teleportation. Large goods still require payment, acceptance, freight/express movement, depot handling and local delivery.
- Commercial credit, security interests, warehouse receipts, insurance, bankruptcy, tax sale and foreclosure are relationship/legal processes rather than generic debt or distress flags.
- Cooperative and mutual capital formation is a reusable ownership/governance pattern that can support creameries, elevators, insurance, building-and-loan associations and communications networks.

- Environmental shocks act through crops, animals, water, routes, buildings, inventory and worker availability. Historical events never authorize universal price/yield multipliers.
- Sears kit houses are outside the 1905 Historical Horizon: Sears Modern Homes begin in 1908. Earlier movement of lumber, components and dismantled structures remains historically valid when actually sourced and transported.

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PART I - REGIONAL GEOGRAPHY, CORRIDORS & HISTORICAL REGIMES

1.1 Default regional anchor and adjacent-world latitude

CANON LOCK - The default Open Frontier remains a Dakota-style southeastern frontier centered on Missouri/Sioux City-oriented trade and a sparse 1870 local economy. Scenario authors may move the focal settlement roughly one regional step east, south, north or west only after recalibrating actual rail chronology, water, settlement density, institutions, land law, market access and outside nodes. The simulation may not carry a southeastern-Dakota starting template unchanged into another state.

Regional option	Expected historical difference	Gameplay consequence
Southeastern Dakota	Sparse 1870 opening; river/wagon orientation; rail and settlement accelerate later.	Best default for watching the full 1870-1905 transformation.
Southwestern Minnesota	Earlier and denser rail/market/institution network.	More competition and supplier depth at start; less frontier isolation.
Nebraska	Union Pacific corridor already mature while western/Sandhills districts remain sparse.	Earlier rail and wholesale access; Black Hills supply traffic; late Kinkaid ranch-land variation.
Northern Dakota	Northern Pacific reaches Bismarck early; later Great Northern competition and wheat/elevator intensity.	Stronger rail/wheat/bonanza/elevator play; long-distance rural settlement still matters.
West River / Black Hills	Mining, ranching and supply-route booms; different treaty/land/water context.	More transient freight, livestock and mining; less representative ordinary farm-service town.
Southern Wyoming corridor	Union Pacific already present before 1870 in key nodes.	Rail/coal/ranching economy can be advanced despite western longitude; corridor history overrides simple east-west assumptions.

1.2 Historical regimes remain causal contexts

Revision XXV regime doctrine remains controlling. Revision XXIX deepens it with named event/institution cases. A regime may change crop potential, credit appetite, route reliability, land demand, migration awareness, information speed or legal options. It may not directly post revenue, customers, deaths, business failures or price percentages.

Period	Expanded authoring character
1870-72	Sparse territorial opening; wagon/river logistics; telegraph/weather networks are emerging; horse-power dependence is extreme; the 1872 horse epizootic can stress nearly every physical transport chain.
1873-77	Panic of 1873 and railway-finance interruption; Homestead plus Timber Culture era; grasshopper risk; Black Hills legal/geopolitical transformation; barbed wire begins diffusing.
1878-87	Great Dakota Boom; rapid rail/town building; bonanza agriculture; telephone/water/electric experiments; grain/elevator institutions deepen; 1880-81 winter and 1881 flood are major stress anchors.
1887-95	Statehood; drought/debt pressure; cattle-winter aftermath; ICC rules; experiment stations; cooperative dairying; Babcock testing; Panic of 1893; foreclosure and migration reversal.
1896-1905	RFD, deeper telephone networks, stronger institutions and late-campaign bankruptcy law; Nebraska Kinkaid Act; rare early internal-combustion equipment remains peripheral. Sears kit houses remain outside the normal horizon.

PART II - PUBLIC LAND, TENURE, TOWNSITES & SETTLEMENT RIGHTS

2.1 Public-land entry is a legal relationship, not instant ownership

CANON LOCK - Homestead, Timber Culture, Desert Land and other dated public-land programs create different entry rights, obligations, evidence and eventual title outcomes. Filing may grant a lawful claim/possession basis without patented title. Actual residence, cultivation, improvement, tree survival or reclamation requirements must be evaluated from world history and LegalRegime data.

2.2 Land-status mosaic

The world can contain unperfected claims, patented farms, railroad/company land, private speculative holdings, leases, tax-sale interests, mortgages, estates, federal land, treaty/reservation land and disputed or unknown title simultaneously. Property UI may expose only what the actor legitimately knows. Promotional maps and visually vacant terrain never override the right-holder state.

2.3 Railroad/townsite land promotion and paper towns

Railroads and townsite companies may market land because settlement generates future traffic. They may plat a depot town, advertise exaggerated prospects, grant favorable terms or organize land-seeker excursions. A promoted town may fail. Lots and brochures create opportunities and expectations, not automatic households, businesses or institutions.

2.4 Farm tenure beyond owner-operation

By the late campaign, owner-operator, cash tenant, share tenant and part-owner/operator are all valid rural arrangements. Share tenancy is a farming/operating relationship, not wage employment. Landlord, operator, equipment owner and crop owner/share claimant may be different parties according to the lease.

2.5 Indigenous land chronology requires exact geography

The Black Hills and other Indigenous/federal geographies cannot be treated as ordinary vacant land. The 1868 treaty context, 1874 gold expedition/rush, federal conflict and later taking/allotment policies are date- and place-specific. Dawes-era allotment requires a separate research-backed authoring record for the affected reservation/community; no generic OpenReservationLand event is permitted.

PART III - TRANSPORT CORRIDORS, RAILHEAD ECONOMIES & SERVICE CHANGE

3.1 Route value can be temporary

A stage, freight, mail or construction-front corridor can generate large but temporary traffic. The Bismarck-Deadwood route and end-of-track towns are controlling examples: equipment, horses, employees, hotels, liveries and merchant demand can rationally expand, then collapse or relocate when the rail network changes.

3.2 Crossings and service stages are distinct

CANON LOCK - A rail line reaching a river is not the same as an all-weather bridge connection. Ferry transfer, seasonal ice crossing, temporary transfer track and permanent bridge must remain distinct route/service states where the geography justifies them. Each changes capacity, delay, cost and reliability independently.

3.3 Rail bypass and supply-network opportunity

A bypass can redirect freight, hotel traffic, institutions and property demand without explicitly deleting the bypassed town. Conversely, a settlement can profit from a distant boom as an outfitting/supply node even when the resource itself lies elsewhere. Network position, not local resource labels, creates the opportunity.

3.4 Rail regulation and claims

The Interstate Commerce Act of 1887 changes published tariff/rate/classification evidence and the legality of specified rebates/preferences while creating formal complaint/damages avenues. Carrier disputes over lost/damaged freight, livestock, right-of-way or contract performance use ordinary claims and legal processes, not a generic service-quality penalty.

PART IV - ENVIRONMENTAL, BIOLOGICAL & INFRASTRUCTURE SHOCKS

4.1 Horse-power dependency and the 1872 epizootic

CANON LOCK - Draft capacity belongs to healthy usable animals, not wagons alone. A widespread equine outbreak may leave freight assets, stages, farm machinery and fire equipment physically present but unable to operate. Replacement horses, rest, veterinary care and prioritization become real scarcity decisions.

4.2 Weather shocks act through physical state

The Long Winter, 1881 Missouri flooding/ice jams, the 1888 blizzard, drought and other severe weather events remain authoring anchors for route closure, livestock/feed stress, building damage, inventory shortage and worker exposure. The event is not an outcome card; affected objects and networks determine consequence.

4.3 Fire, locust and compound failure

Locust/grasshopper outbreaks damage crops/forage geographically. Prairie fire uses fuel, wind, ignition and suppression/backfire capacity. Town fire uses building material, spacing, ignition and response. Multiple weak seasons/events can compound until a household/business/settlement chooses to leave without any forced ghost-town transition.

4.4 Water projects can succeed or fail

Wells, pumping windmills, artesian drilling, irrigation ditches and municipal/private waterworks are Project/Property/Utility investments. Drilling can consume capital without useful yield. Successful access can expand viable settlement/ranching

geography and fire/sanitation capacity.

PART V - INFORMATION, COMMUNICATIONS, MAIL & MARKET KNOWLEDGE

5.1 Information latency is historical gameplay

CANON LOCK - True world state and actor knowledge remain separate. Mail, messenger, newspaper, railway mail, telegraph, telephone, commercial reports, fairs and traveling salesmen differ in speed, cost, reach and reliability. Better communications narrow delay and uncertainty; they do not grant omniscience or a universal management bonus.

5.2 Postal and express financial/logistics services

Postal money orders, telegraph money transfer where available, railway mail, private express and COD can support distant trade before local banking and parcel-post systems mature. The payment instrument and the goods shipment are separate obligations and can arrive at different times.

5.3 Mail order and the Sears anachronism guard

Montgomery Ward-era mail order is valid inside the campaign as a supplier-discovery/order channel. Heavy goods still move by express/rail/wagon. Rural Free Delivery arrives late and changes mail access more than heavy-goods logistics. Sears kit houses begin in 1908 and therefore must not appear during the normal 1870-1905 horizon; earlier freight of lumber/components remains valid.

5.4 Weather and market information

Federal weather observations/forecasts and telegraphed commodity quotations create dated evidence. A person can receive yesterday's newspaper price, a faster telegraph price or no outside quote at all. Local price formation still depends on actual local supply, buyers, transport and contracts.

5.5 Commercial credit information

R.G. Dun-style commercial reports and later formal bank reporting/examination allow distant counterparties to obtain evidence about business standing. Such reports are dated, partial and potentially stale. They can support credit decisions without giving the creditor direct access to the simulation truth.

PART VI - RURAL PRODUCTION, AGRICULTURAL SERVICES & MARKET INSTITUTIONS

6.1 Mechanization creates service and consumable economies

Bonanza farms demonstrate that annualized staffing and peak harvest concurrency can differ enormously. Custom threshing/reaping outfits, implement dealers, repair shops and seasonal crews can serve many farms. Machinery creates demand for draft animals, fuel where applicable, parts, twine, skilled operation, freight and maintenance rather than becoming a one-time productivity unlock.

6.2 Fencing, water and winter resilience

Barbed wire, wells and windmills change actual stock control and watered acreage. Ranch survival through the 1886-87 winter depends on stocking, condition, shelter, hay/forage and weather. Investments reduce specific vulnerabilities rather than raising a generic RanchResilience stat.

6.3 Grain grading, elevators, receipts and futures

CANON LOCK - Grain custody, ownership, grade, weight, storage receipt and sale remain separable. A warehouse/elevator may own purchased grain while merely holding other grain for depositors. Warehouse receipts can represent claims and may be transferable/pledgeable where law permits. Terminal-market/futures information can influence elevator bids and hedging without replacing physical local commerce.

6.4 Dairy cooperation and testing

Cooperative creameries can aggregate milk and capital to justify separators and other processing equipment. Babcock-era butterfat testing changes payment evidence and farmer incentives. The organization, equipment, member farms and

transactions remain separate economic entities/relationships.

6.5 Agricultural information and temporary markets

Experiment-station bulletins, agricultural fairs, stock shows, seed catalogs and machinery demonstrations are information/commercial channels. Awards can become evidence about animals/products. Seed firms may serve broad territories through catalog/mail/rail and should preserve variety/source suitability rather than a generic SeedQuality score.

6.6 Finite rural side-economies

Buffalo-bone collection is a controlling example of a temporary exhaustible rural trade. Physical resource availability supports collectors, buyers and freight until the stock is depleted, after which the business must move, pivot or close.

PART VII - WHOLESALE COMMERCE, TRAVELING SALES & MERCHANT NETWORKS

7.1 Regional wholesaling is a real middle layer

CANON LOCK - Manufacturer -> regional jobber/wholesaler -> local merchant is an ordinary supply chain. Regional hubs such as Omaha, Sioux City, Minneapolis/St. Paul and Chicago may differ by product, route, price and lead time. A merchant is not assumed to buy every line directly from the original manufacturer.

7.2 Traveling salesmen and sample rooms

A wholesaler representative may physically travel a rail/stage circuit, stay at a hotel, store/display samples, visit merchants, negotiate terms and take orders. The visit creates information and commitments. The actual goods arrive later through a shipment. Hotels can therefore participate in business-to-business commerce rather than serving only overnight travelers.

7.3 Implement dealers and supplier lines

Implement/hardware dealers can carry multiple product lines and commercial capabilities. Manufacturer relationships influence stock, credit, warranties/repairs, parts and replacement lead time. One business may combine implements, hardware, lumber, grain buying or livestock trade when actual people/assets/premises support it.

7.4 Collection and trade-credit continuity

A remote supplier can use correspondence, local attorneys/agents and secured remedies to collect frontier debt. Default can lead to workout, revised terms, demand for security, lawsuit or foreclosure rather than instant write-off or repossession. Liberal credit can increase sales and bad-debt exposure simultaneously.

PART VIII - FINANCE, INSURANCE, INSOLVENCY & DISTRESSED ASSETS

8.1 Multiple credit institutions

Commercial banks, private lenders, merchant credit, building-and-loan associations, manufacturer notes and cooperative/mutual arrangements can coexist. Financial instruments keep their own rights, collateral, payment schedules and creditor identities rather than merging into one Debt object.

8.2 Secured personal-property finance

CANON LOCK - Chattel mortgages/security interests attach to identified movable property only when the dated legal regime supports the instrument and requirements. Default does not transfer unrelated assets. Future-crop collateral must obey the law in force; the 1889 Dakota example is a warning against assuming universal enforceability.

8.3 Insurance is balance-sheet risk transfer

Fire, hail and mutual insurance policies create premiums/assessments, reserves, coverage limits and claims. Physical loss occurs first. Concentrated disasters can threaten the insurer itself. Insurance payment does not restore destroyed property or crops.

8.4 Distress and insolvency have multiple pathways

Workout, private liquidation, mortgage/chattel foreclosure, judgment execution/sheriff sale, tax sale, assignment, estate process and bankruptcy are distinct. The 1878-1898 federal bankruptcy gap is a material campaign condition; the 1898 Act adds a later federal route but does not create modern Chapter 7/11 behavior.

8.5 Distressed assets remain ordinary market objects

Foreclosed farms, machinery, livestock, stores and industrial plants remain physical assets with condition, tenants, encumbrances, redemption rights and carrying costs. Distress can create opportunity but does not guarantee a discount or clean title.

PART IX - COOPERATIVES, INSTITUTIONS, UTILITIES & SETTLEMENT ORGANIZATION

9.1 Cooperative/mutual capital formation

CANON LOCK - A cooperative or mutual is an ownership/governance pattern, not a bonus type. Members may pool capital because they need a shared service such as cream processing, grain storage/marketing, insurance, housing finance or telephone connectivity. The organization owns its assets/cash and contracts separately from the members.

9.2 County seats, capitals and institutional relocation

Courthouses, county seats and state-capital contests create real legal traffic, records access, construction, lodging, land speculation and public debt. Institutions can relocate; supporting economic activity should move or decay through actual journeys/businesses rather than a permanent AdministrativeCenter modifier.

9.3 Utilities may be public, private or mixed

Water, irrigation, electric light/power and telephone systems can be privately capitalized, cooperative, municipal or mixed according to local law and finance. Incorporation/franchise only creates authority/organization; physical plant, connections and capacity still have to be built and maintained.

9.4 Corporate and communal diversity

Territorial incorporation records should calibrate the variety of real organizations, including banks, elevators, utilities, insurers and agricultural associations. Communal organizations such as Hutterite colonies must fit the same general ownership/governance framework without erasing Person/family identity.

9.5 Migration promotion and chain networks

Railroad, territorial, county and private immigration campaigns create advertising, correspondence and leads. Kinship, religious/community and language networks can lower information, travel and initial-housing barriers. Ethnicity is never converted into an economic-stat bonus.

PART X - HISTORICAL RESEARCH FINDING REGISTER

The register below is intentionally comprehensive. It preserves the discrete findings from the October historical-research conversation and states how each enters the game. A row is not permission to universalize a local observation; exact geography/date/source still controls.

Date / period	Finding	Canon treatment
1862 onward	Homestead Act land entry	Public-land entry is a process, not instant ownership: eligibility, surveyed/status-qualified land, residence, improvement/cultivation, proof and patent remain distinct. Witness/evidence requirements belong to dated LegalRegime data.
1873 onward	Timber Culture Act	Support a distinct tree-culture public-land path where applicable. Planting/survival obligations use real acreage, labor, climate and evidence; do not treat it as a free bonus parcel.
1877 onward	Desert Land Act / irrigation	In qualifying western/semi-arid contexts, land viability can depend on water rights and reclamation capital. Do not apply the regime universally to southeastern Dakota.
1870-1905	Public-land mosaic	Homestead entries, patented farms, railroad land, private/speculative property, federal/reservation land and disputed/unknown rights coexist. Visual vacancy never implies claimability.
1870s-90s	Railroad land sales and immigration promotion	Railroads/townsite companies may sell land on terms partly to generate future traffic. Promotion creates awareness and biased evidence, not guaranteed population.

Date / period	Finding	Canon treatment
1870s-90s	Paper towns and townsite speculation	A platted/promoted settlement may exist before durable demand. Lots, advertising, depot expectations and county-seat hopes can precede actual business survival.
1904	Kinkaid Act in Nebraska	Qualifying Nebraska Sandhills scenarios can allow much larger homestead entries than the ordinary quarter-section pattern. This is region-specific late-horizon law, not a Dakota-wide rule.
1887 onward	Dawes allotment policy	Federal allotment materially changes some Indigenous land relationships, but implementation must be reservation-, treaty-, date- and jurisdiction-specific. No generic 'land opens' mechanic.
1874-77	Black Hills gold / treaty conflict	Pre-rush Black Hills cannot be ordinary purchasable mining land. Gold discovery, unauthorized entry, federal action, war and later land-status change must respect treaty/reservation geography and historical chronology.
1873	Panic of 1873	Credit contraction and railway-finance failure propagate through lenders, projects, migration and demand. Northern Pacific construction interruption is a key regional example; no direct economy-wide percentage modifier.
1878-87	Great Dakota Boom	Migration, land filings, rail construction, town development, freight, banking and business formation can reinforce one another through ordinary systems. The boom is a regime, not a guaranteed local outcome.
1886-95	Dakota drought / agricultural depression	Yield and water stress reduce farm income, debt service and trade, then propagate into merchant receivables, lending, migration, land prices and freight.
1893 onward	Panic of 1893	Outside financial stress can tighten correspondent liquidity, credit appetite and railway/industrial finance while local agricultural weakness may amplify losses.
1886-87	Cattle winter catastrophe	Open-range exposure, animal condition, forage/hay, water, shelter and stocking determine losses. Historical severe losses are calibration evidence, not a scripted mortality percentage.
1872	Great Horse Epizootic	Equine disease can temporarily remove draft/transport capacity and thereby disrupt freight, farm work, stage service, fuel/food delivery and fire response. Healthy teams and veterinary capacity become scarce assets.
1880-81	Long Winter / rail isolation	Repeated storms can close rail and road links for long periods, causing shortages of fuel, flour, feed and other inventory. Local reserves and production matter more than abstract cash while routes are blocked.
1881	Missouri River flood and ice jams	Flood/ice events can destroy low-lying property, livestock, ferries, bridges and inventories and may motivate relocation or infrastructure rebuilding.
1888	Children's Blizzard	Severe weather can simultaneously block rail/roads and damage communications. Network components fail separately; surviving local telephone/other links can have temporary strategic value.
1870s	Rocky Mountain locust / grasshopper outbreaks	Crop and forage state is physically consumed/damaged by geographically scoped outbreaks. Relief, seed/feed scarcity, migration and credit consequences arise afterward.
1870-1905	Prairie fire	Fire spread uses ignition, dry fuel, wind, terrain, spacing and response. Backfires/firebreaks are real mitigation; weak settlements can be destabilized by compound crop/fire losses.
1870-1905	Town fire / rebuilding	Urban fire damages actual structures/inventory, can create insurance claims and reconstruction booms, and can raise demand for brick, waterworks, hydrants and organized fire protection.
1878	Washburn A Mill explosion	Large outside industrial accidents can temporarily alter processing capacity, contract destinations, congestion and regional commodity demand without occurring on the player's local map.
1874 onward	Barbed wire diffusion	Wire changes fencing cost and therefore crop protection, stock control, herding burden, property conflict and hardware/freight demand. Adoption depends on supply, capital and labor.
1870s-1900s	Windmills / pumping	Mechanical windmills and wells expand actual water access and can change viable farm/ranch geography. They require capital, installation, maintenance and replacement parts.
1882 onward	Dakota artesian wells	Deep-well projects may create abundant water or fail after expensive drilling. Yield/pressure/depth are uncertain evidence; successful water infrastructure can change settlement and utility viability.
1870s-1900s	Natural ice economy	Winter ice cutting, storage loss, icehouses and delivery support hospitality, butchery/dairy and later refrigerated transport. Ice remains physical inventory with seasonal production and storage burden.
1870s-90s	Refrigerated meat / rail	Local slaughter plus ice/refrigerated shipment competes with live-animal shipment. The Marquis de Mores model is a dated high-risk example, not proof of automatic profitability.
1884 onward	Bureau of Animal Industry	Livestock disease control and interstate movement increasingly interact with inspection/quarantine. Exact disease rules remain dated authoring data.
1870s-1900s	Railhead / end-of-track booms	Construction-front traffic can create temporary hotel, livery, saloon, freight and supply booms that fade when the rail front moves.
1877-79	Bismarck-Deadwood route	A major stage/freight/mail corridor can support large temporary employment and horse-team demand, then collapse when a better rail gateway changes the network.
1873-82	Bismarck Missouri crossing states	Rail-to-river transfer, ferry, seasonal ice crossing and permanent bridge are distinct route states with different capacity, cost and reliability.
1870s-1900s	Rail bypass and town relocation	A new depot/route can redirect freight, institutions and Main Street activity away from an older node. Settlement decline emerges from changed accessibility rather than a GhostTown flag.
1870s-90s	Supply towns outside the boom zone	Mining/ranch booms create opportunity along their supply corridors. A town such as Sidney can profit from Black Hills traffic without itself being a mining settlement.
1883	Standard Railway Time	Time standardization changes schedule coordination and record interpretation. It is a dated information/operations event, not a productivity bonus.
1887	Interstate Commerce Act	Rail rate posting, classifications, anti-rebate rules and complaint/damages procedures change permissible/evident carrier behavior. Early enforcement limitations remain calibration context.
1860s onward	Railway Mail Service	Mail can move and be sorted on trains; mail service, passenger service and freight service are related but distinct capabilities. Some locations can exchange mail without a normal passenger stop.
1864 onward	Postal money orders	The post office can provide a limited long-distance remittance instrument before every town has a bank. Limits, office availability and procedure are dated data.
1871 onward	Telegraph money transfer	Telegraph firms can transmit both information and, where supported, money transfers. Service requires actual network/office access.
1872 onward	Mail-order procurement	Catalogs expand supplier awareness. Orders still require correspondence/payment, supplier acceptance and physical freight/express fulfillment.
pre-1913	Express companies / COD	Heavy parcels and valuable shipments often use private express rather than ordinary mail. COD, baggage/package custody and express agency are separate transport/commercial capabilities.
1896 onward	Rural Free Delivery	RFD changes information/mail accessibility and household trip patterns but does not teleport merchandise. Local rollout is route-specific and late-period.

Date / period	Finding	Canon treatment
1908 outside horizon	Sears kit houses	Sears Modern Homes begin after the normal 1905 horizon. Do not backdate the product. Earlier shipment of lumber/components or dismantled structures remains valid when sourced.
1876 onward	Telephone private networks	Large farms/businesses can justify private telephone lines before public exchanges; access is connection-specific.
1881 onward	Telephone exchanges	Local exchanges have finite subscribers/operators/capacity. A town having an exchange does not mean every household is connected.
c.1900-05	Farmer telephone lines	Cooperative/party-line networks can extend rural service cheaply with capacity/privacy tradeoffs.
1870 onward	Federal weather service	Weather knowledge improves through telegraphed observations/forecasts. Forecast evidence has date, source, coverage and confidence; underlying weather remains physical simulation.
1870s-90s	Telegraphed market quotations	Newspapers and telegraph channels can expose outside cash/futures prices with different latency. Local actors are not omniscient and may see stale quotations.
1871 onward	Indigenous-language information networks	Indigenous communities have their own publications, institutions and relationship networks. Iapi Oaye is a dated regional example; do not make settler media the only information system.
1840s onward	R.G. Dun commercial credit reporting	Distant suppliers/lenders can obtain third-party evidence about business standing, capital, debt and reputation. Reports are source/date/confidence records, not omniscient ratings.
1870s-1900s	Newspaper / formal bank reporting	Dated territorial rules can require published bank statements or later official examination. Regulatory information availability changes over time.
1887 onward	Dakota public bank examiner	Formal examination becomes a source of institutional evidence; it does not guarantee solvency.
1880s onward	Building-and-loan associations	Mutual thrift/home-finance institutions create a historically distinct housing-credit channel using member shares/dues and loan rights.
1870s-90s	Trade credit and manufacturer finance	Wholesalers/manufacturers can extend terms, notes or secured credit to merchants/farmers; payment history and commercial references matter.
1880s	Chattel mortgages	Movable assets such as horses, machinery and livestock may secure debt where law permits. Foreclosure notices and sale procedures are real legal processes.
1889 Dakota example	Future-crop collateral limits	What property can secure a loan depends on dated law. Do not assume future crops are universally valid collateral.
1889 Dakota	Warehouse receipt bailment	Stored grain under receipts can remain owned by the depositor even when physically commingled at an elevator. Purchased grain and stored grain must not share one ownership truth.
1880s onward	Warehouse-receipt finance	Receipts may be transferred/pledged as security. Inventory custody, title and secured-credit rights stay separate.
1881 onward	Minneapolis grain exchange / futures	Major grain merchants/elevators can use terminal cash and futures information. Do not require ordinary farmers to day-trade; futures primarily support pricing/hedging context.
1880s onward	Grain grading / inspection	Weight and grade affect settlement proceeds and can generate disputes. Public inspection becomes more formal in mature markets.
1880s onward	Elevator buyer concentration	One rail-line buyer can gain local power; alternative elevators/cooperatives can emerge from real producer dissatisfaction and capital.
1880s-1900s	Cooperative creameries	Farmers can pool milk and capital to support equipment they could not individually justify. Cooperative ownership is real governance, not a production bonus.
1880s	Cream separator diffusion	Mechanical separation changes feasible dairy processing and centralization; adoption depends on capital, supplier access and throughput.
1890 onward	Babcock butterfat test	Milk purchasing can increasingly use measured butterfat quality. Testing changes transaction evidence and producer incentives.
1887 onward	Hatch Act / experiment stations	Agricultural research enters the information system through bulletins, demonstrations, newspapers and correspondence. Advice can be ignored, misunderstood or unaffordable.
1870s-90s	Immigration promotion	Territorial/county/railroad agents can advertise land and opportunities. Campaigns create awareness, correspondence and migration leads rather than direct population increments.
1870s-1900s	Chain migration	Kinship, church, language and community ties change information, lodging and relocation feasibility. Do not convert ethnicity into productivity modifiers.
1874 onward	Hutterite communal colonies	Ownership/governance must be flexible enough for communal organizations to own land/assets while Persons/families remain identifiable. No one-off special economy is required.
1870s-90s	Portable/prefabricated building components	Settlers may transport pre-cut lumber/components or even dismantled structures before Sears kit houses. Logistics and assembly remain real work.
1870s-1900s	Rail access and building materials	Frame construction becomes easier where finished lumber and components can arrive cheaply. Architecture should respond to actual material access rather than a building-era unlock.
1870s-80s	Buffalo-bone commerce	A finite, exhaustible resource can support collection -> local buyer -> wagon -> rail -> distant industrial market. The trade should disappear as local physical stock is depleted.
1880s	County-seat competition	Courthouse/record location changes legal traffic, hotels, lawyers, land values and construction. Administrative-center status is an institution with economic flows, not a cosmetic role.
1880s	Courthouse relocation	Institutions can physically relocate when politics/transport changes; the world object and local demand move with them.
1889	Statehood	North/South Dakota statehood changes legal/administrative authority, offices and regulatory datasets. It does not reset businesses or property relationships.
1889-96	South Dakota prohibition	Alcohol legality changes during the campaign. Saloons must query dated jurisdictional law rather than a timeless license rule.
1880s-1900s	Capital/county-seat promotion debt	Civic competition can create construction, hotel and land booms while leaving local public debt. Institutional location can be economically valuable and financially costly.
1880s	Private water / irrigation companies	Utilities can be privately capitalized, cooperative or municipal. Incorporation/franchise does not mean a physical network exists.
1880s	Combined steam/power operations	A boiler/engine may support time-separated industrial functions such as milling by day and electric lighting at night. Shared capacity remains physical.
1870s-1900s	Fire insurance / Sanborn risk	Insurance risk should respond to materials, spacing, ignition, water/hydrant coverage and neighboring exposure. Sanborn maps are a strong calibration dataset.
1880s-90s	Mutual and hail insurance	Farmers can organize mutual insurers; premium/reserve/claim concentration matters. Hail insurance is a real risk-transfer business, not guaranteed restitution.
1870s-1900s	Bank runs / correspondent stress	Depositor withdrawals are claims on actual bank liabilities. Outside correspondent strain can propagate inward; conservative banks can survive by tightening credit.
late 1880s-90s	Foreclosure and absentee creditors	Farm/business collateral can transfer to distant creditors after default. Recovered property remains an asset that must be managed, rented, sold or restructured.
1878-1898	Federal bankruptcy gap	There is no continuous modern bankruptcy button. During the gap, workouts, assignments, foreclosure, attachment and state/territorial remedies dominate.

Date / period	Finding	Canon treatment
1898 onward	Bankruptcy Act of 1898	A new federal bankruptcy route appears late in the campaign. It is not modern Chapter 7/11 and must use dated procedure.
1870s-1900s	Sheriff sales / execution	Judgment enforcement can physically liquidate property and return assets to the market. Sale does not erase preexisting rights that survive under law.
1870s-1900s	Tax sales and redemption	Tax delinquency creates a distinct property-transfer path. Purchasers may hold an interest subject to redemption rather than instant undisputed ownership.
1880s-1900	Farm tenancy / share rent	Owner-operated, cash-rented, share-rented and part-owned farms are distinct relationships. A share tenant is still a farm operator, not a wage laborer.
1880s-1900s	Distressed-asset market	Private liquidation, estate sale, foreclosure, sheriff sale, tax sale and bankruptcy can all feed ordinary acquisition opportunities. Distress constrains the seller; it does not guarantee a bargain.
1880s-1900s	Patent medicines	Druggists, newspapers, mail order and traveling vendors can sell proprietary medicines under weak pre-1906 federal regulation. Claims/reputation are evidence, not magical health buffs.
1870s-1900s	Medicine shows	Traveling entertainment can generate crowds, lodging/livery demand, advertising and medicine sales through ordinary transient and retail systems.
1870s-90s	Traveling salesmen / drummers	Wholesaler representatives physically travel circuits, stay in hotels, show samples and take orders. They create supplier information and purchase opportunities, not instant inventory.
1870s-90s	Hotel sample rooms	Hotels can provide temporary commercial display/storage space for traveling salesmen. This creates a commercial-traveler customer class distinct from ordinary tourism.
1870s-1900s	Regional jobbers / wholesalers	Manufacturer -> jobber/wholesaler -> local merchant is an ordinary supply chain. Omaha, Sioux City, Minneapolis/St. Paul and Chicago have different network advantages.
1870s-1900s	Farm implement dealers	Local/regional dealers can carry multiple equipment lines and related hardware/lumber/grain capabilities. Dealer stock, supplier relationships and parts support matter.
1870s-90s	Collection attorneys	Remote manufacturers/wholesalers can hire local lawyers/agents to locate debtors, negotiate, sue or foreclose. Default can become a long workflow rather than instant repossession.
1870s-1900s	Agricultural fairs and stock shows	Fairs are temporary market/information events: machinery demonstrations, livestock judging/sales, awards, advertising and relationship formation.
1885 onward	Regional seed houses / catalogs	Seed firms such as Oscar H. Will can serve broad markets by catalog/mail/rail. Variety suitability and source evidence matter.
1870s-1900s	Indigenous horticultural knowledge	Northern Plains crop knowledge and varieties can enter settler commercial networks through real exchange. Research must retain community/source provenance rather than credit all adaptation to later firms.
1880s	Territorial incorporation records	Dakota incorporation archives show a wide range of elevators, banks, insurers, utilities, agricultural organizations and other firms. Use them as generator calibration, not proof every town had such entities.
1880s	Livestock brands	Brands/marks are ownership evidence and can support disputes, sales and recovered animals. Registration and geographic rules are dated legal data.
1870s-80s	Estray law	Wandering livestock can trigger notice, brand identification, appraisal and recovery workflows. Western-county exceptions show that rules may vary within the same territory.
1880s onward	Stockyards / commission merchants	Terminal yards create pens, weighing, rail, banking, commission agency and packing demand. Producers may consign livestock to agents rather than sell directly.
1880s	South Omaha / Sioux City livestock nodes	Closer terminal markets can compete with Chicago through freight, time and shrinkage economics; industrial clusters can generate new settlement growth.
1890s	Binder twine supply	Self-binders create real consumable-input demand. Government/penitentiary twine production near 1900 is a region-specific supplier development, not a technology unlock.
1870s-1900s	Commercial claims against carriers	Lost/damaged freight, livestock injury, contract breach and right-of-way disputes can create ordinary claims/litigation against railroads or carriers.
1870-1905	Information latency as modernization	Management scale improves partly because mail, newspapers, telegraph, telephone, standardized time, credit reports and market quotations reduce information delay. Better information changes decisions, not hidden efficiency stats.

PART XI - ACCEPTANCE DOCTRINE, RESEARCH HOLDS & SUPERSESSION

11.1 Acceptance rules

- A drought, panic, blizzard, epizootic, fire, flood or locust outbreak must first change physical/economic drivers; market outcomes follow from ordinary systems.
- A farther-west scenario can receive rail earlier than an eastern one when actual corridor chronology supports it. No West = Later Rail rule is allowed.
- A public-land filer can possess/use land without yet holding patent; sale, mortgage and inheritance rights depend on the actual interest and law.
- A railhead business can be rational and profitable before becoming stranded by route change. The game must not protect the player from historically plausible network risk.
- A mail-order purchase cannot enter inventory until payment/acceptance and physical fulfillment occur.
- A town can have telegraph/telephone/weather-service access while many residents/businesses remain unconnected or receive information late.
- A distant wholesaler can deny or extend credit based on incomplete third-party evidence without seeing true hidden player finances.
- Stored grain can be physically inside an elevator while owned by a depositor; warehouse custody cannot silently become title.

- A cooperative can own expensive equipment while members remain independent farms/households/businesses.
- Foreclosure, tax sale, sheriff sale and bankruptcy cannot be collapsed into one DistressedSale state when surviving rights differ.
- Sears kit houses cannot appear before 1908 in the normal 1870-1905 historical envelope.

11.2 Research/calibration holds

- Exact homestead/Timber Culture/Desert Land/Dawes procedure by county, land district and date beyond the general legal structure.
- Exact railroad-land sale terms, immigration-agent budgets, townsite lot prices and promotional conversion rates.
- Quantitative probabilities/severity for the 1872 epizootic, grasshoppers, drought, prairie fire, flood, blizzard and cattle-winter events.
- Exact Bismarck-Deadwood traffic by date, individual stage-company rates, schedules and asset rosters beyond the cited anchor evidence.
- Artisan/utility/telephone/water adoption rates by settlement size and year; artesian drilling costs and failure distributions.
- Express/COD tariffs, postal-money-order office coverage, RFD route effects and telephone subscription prices by date/place.
- Warehouse charges, grain-grade discounts, inspection rules and actual hedging/futures use by Dakota elevators.
- Cooperative creamery capital structures, patronage/distribution practice, Babcock adoption and separator prices by locality/year.
- Building-and-loan dues/loan conventions, chattel-mortgage notice/perfection rules, tax-redemption periods and execution practice by jurisdiction.
- Insurance premiums/reserves/assessment structures for fire, hail and mutual companies and how catastrophic concentration affected failures.
- Wholesale jobber/dealer catalogs, discounts, credit terms, manufacturer lines, sample-room rents and drummer circuits by hub/year.
- Farm lease/share-contract templates, obligations for seed/equipment/fencing/repairs and exact tenancy prevalence below state level.
- Indigenous land, allotment and agricultural-knowledge treatment requires community-, treaty- and reservation-specific research before detailed mechanics are locked.
- Complete extraction of Dakota territorial incorporation records and Sanborn maps into quantitative generator datasets.

11.3 Supersession/deepening map

Revision XXIX deepens Revision XXV historical-envelope doctrine, Revision XXVI property/market/evidence doctrine, Revision XXVII catchment/workforce doctrine and Revision XXVIII construction/project doctrine. Existing HistoricalRegime, DatedHistoricalEvent, LegalRegime, PropertyRight, Agreement, Finance, Insurance, Institution, Logistics, Information/Evidence, Person/Household, Business/CommercialCapability and Project authorities remain active. This revision does not create a parallel historical-events economy.

RESEARCH BASIS - REVISION XXIX

The following source families support the October research findings. They are research provenance, not universal constants. Local observations must retain date, geography, source type and confidence when converted into authoring data.

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   https://www.archives.gov/milestone-documents/homestead-act
2. National Archives - Dawes Act - Federal allotment chronology; detailed local implementation remains a research hold.
   https://www.archives.gov/milestone-documents/dawes-act
3. National Archives - Interstate Commerce Act - 1887 rail-rate, posting and complaint framework.
   https://www.archives.gov/milestone-documents/interstate-commerce-act

4. National Archives - Bankruptcy history - 1878 repeal and 1898 federal bankruptcy chronology.
   https://www.archives.gov/publications/prologue/2014/fall/bankruptcy
5. National Park Service - 1868 Fort Laramie Treaty / Black Hills - Treaty geography and Black Hills chronology.
   https://www.nps.gov/places/1868-treaty-field-area.htm
6. National Park Service - Windmills - Water-pumping and railroad windmill context.
   https://www.nps.gov/articles/windmills.htm
7. National Park Service - Water in the West / Desert Land Act - Western water/reclamation context.
   https://www.nps.gov/articles/2-water-in-the-west.htm
8. National Park Service - Marquis de Mores - Dakota refrigerated-meat experiment.
   https://home.nps.gov/thro/learn/historyculture/marquis-de-mores.htm
9. National Park Service - Kinkaid settlement context - 1904 Nebraska Sandhills land regime.
   https://www.nps.gov/niob/learn/historyculture/people.htm
10. National Park Service - Natural ice / icebox - Ice harvesting/storage economy.
    https://www.nps.gov/articles/000/the-icebox-the-predecessor-of-modern-refrigeration.htm
11. National Park Service - Minneapolis milling / Washburn explosion - 1878 milling disaster and regional processing context.
    https://www.nps.gov/miss/learn/historyculture/river-of-hisory-chapter-6.htm
12. North Dakota Studies - Great Dakota Boom - Settlement boom chronology.
    https://www.ndstudies.gov/gr4/early-settlement-north-dakota/part-2-great-dakota-boom/section-5-great-dakota-boom
13. North Dakota Studies - Drought - Late-1880s/1890s agricultural stress.
    https://www.ndstudies.gov/gr8/content/unit-iii-waves-development-1861-1920/lesson-1-changing-landscapes/topic-5-blizzards-floods-and-drought/section-2-drought
14. North Dakota Studies - Northern Pacific - Rail chronology, Bismarck and Panic-era interruption.
    https://www.ndstudies.gov/gr8/content/unit-iii-waves-development-1861-1920/lesson-1-changing-landscapes/topic-6-railroads/section-2-northern-pacific-railway-north-dakota
15. North Dakota Studies - Bonanza farms - Seasonal scale and mechanization.
    https://www.ndstudies.gov/gr8/content/unit-iii-waves-development-1861-1920/lesson-2-making-living/topic-3-farming/section-4-bonanza-farms
16. North Dakota Studies - Rural telephone - Private farm lines and later farmer networks.
    https://www.ndstudies.gov/gr8/node/382
17. South Dakota State Historical Society - Agricultural context - Drought, farm systems, grading and rural development.
    https://history.sd.gov/preservation/docs/SDAgriculturalContext2013.pdf
18. South Dakota State Historical Society - Water towers/waterworks - Artesian and municipal water-project evidence.
    https://history.sd.gov/preservation/docs/SDWaterTowers.pdf
19. South Dakota State Historical Society - Hutterite colonies - Communal settlement/ownership context.
    https://history.sd.gov/preservation/docs/HutteriteColonies.pdf
20. South Dakota State Archives - Territorial Articles of Incorporation - Corporate-form and business-type calibration.
    https://history.sd.gov/archives/AI.aspx
21. Smithsonian National Postal Museum - Mail order - Catalog/express constraints.
    https://postalmuseum.si.edu/mail-order
22. Smithsonian National Postal Museum - Railway Mail - Railway-mail service mechanics.
    https://postalmuseum.si.edu/exhibition/about-postal-operations-transportation/railway
23. Smithsonian National Postal Museum - Money orders / transactions - Postal financial services.
    https://postalmuseum.si.edu/exhibition/america%E2%80%99s-mailing-industry-impacting-lives/conducting-transactions
24. Smithsonian National Postal Museum - Catalogs and RFD - Catalog chronology and rural delivery.
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    https://www.loc.gov/item/today-in-history/november-18/
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https://www.usgs.gov/publications/artesian-water-supply-dakota-sandstone-north-dakota-special-reference-edgeley

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https://www.cftc.gov/About/HistoryoftheCFTC/history_precftc.html

34. Federal Reserve History - Gilded Age banking panics - Panic propagation and liquidity context.
https://www.federalreservehistory.org/essays/banking-panics-of-the-gilded-age

35. Harvard Library - R.G. Dun credit reports - Commercial credit-information network.
https://library.harvard.edu/collections/rg-dun-company-credit-reports

36. Federal Reserve Bank of Richmond - Building and loan associations - Mutual home-finance structure.
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37. Dakota Territory / North Dakota session laws - Chattel mortgage, estray, tax, warehouse and related legal chronology.
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38. Sioux City Public Museum - Stockyards - Livestock terminal, commission and industrial-cluster development.
https://www.siouxcitymuseum.org/history-website/stock-yards

39. History Nebraska - South Omaha stockyards - Livestock-market geography and packing development.
https://history.nebraska.gov/the-birth-of-the-south-omaha-stockyards/

40. History Nebraska - Jobbers Canyon - Regional wholesaling/jobbing system.
https://history.nebraska.gov/big-ugly-red-brick-buildings-the-fight-to-save-jobbers-canyon-daniel-d-spegel/

41. History Nebraska - Samuel M. Chapman papers - Collections for distant manufacturers and secured debt.
https://history.nebraska.gov/collection_section/samuel-m-chapman-1839-1907-guide-to-microfilm-rg2974-am/

42. History Nebraska - Francis E. Newton papers - Implement-dealer supplier network and terms.
https://history.nebraska.gov/collection_section/francis-e-newton-rg4269-am/

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https://www.census.gov/programs-surveys/decennial-census/technical-documentation/questionnaires/1900/1900-instructions.html

44. USDA Census of Agriculture historical tables - South Dakota farm-tenure context.
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45. NOAA / National Weather Service history - 1870 weather-service chronology and forecast networks.
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PRESERVED HISTORICAL REVISION LAYERS

The complete prior Canon follows unchanged. The master table of contents links each preserved revision layer so the document can be navigated without manually paging through every historical integration.

LAND & LEDGERS
CANONICAL PROJECT BIBLE - REVISION XXVIII

Construction, Contracting & Project Delivery Integration: Delivery Structures, Estimating, Resource Execution, Payment, Change, Default & Claims

Revision date: October 5, 2026

Source baseline: Canonical Project Bible Revision XXVII - September 26, 2026

New source integrated: Construction, Contracting & Project Delivery Research Passes I-III - October 2026

Status: Active integration

PRESERVATION GUARANTEE

REVISION XXVIII preserves Canonical Project Bible Revision XXVII in full except where this integration explicitly adds, deepens, clarifies, corrects or supersedes prior wording. The complete Revision XXVII document follows unchanged as the preserved historical layer. Where a conflict exists, Revision XXVIII controls. Historical prices, wages, crew outputs, contract sums and legal procedures below remain dated evidence or calibration unless explicitly promoted to a Canon lock.

PURPOSE

Revision XXVIII consolidates the three construction research passes into durable gameplay doctrine. The research strongly validates the existing Project, Agreement, WorkPackage, Task, Property, Finance, Inventory, Evidence and LegalRegime foundations while deepening delivery structure, bottom-up estimating, crew/resource execution, working-capital risk, public procurement, defects, claims, liens, surety and recovery after default.

REVISION XXVIII MASTER DOCTRINE

- A construction contract promises obligations; it does not purchase guaranteed progress.
- Construction delivery is compositional: owners may self-perform, hire separate trades, hire a prime contractor or use a mixed arrangement according to actual capability, agreement and period practice.
- Project cost and duration arise from physical scope, materials, labor, methods, logistics, equipment, site conditions, information and dependencies rather than universal construction points or cost-per-area formulas.
- Estimate, bid/proposal, Agreement, approved Baseline, Current Forecast and Actual outcome remain distinct historical/economic states.
- Contract value, owner cash outlay, resource cost consumed and completed-asset value are not automatically the same number.
- Crew productivity arises from task/method, worker capability, crew composition, tools, available work fronts and environment. Extra labor has task-specific diminishing returns.
- Backlog is committed future capacity, not revenue. Profitable growth can create a liquidity crisis when payroll, materials and subcontractors must be financed before customer receipts arrive.
- Payment to a prime contractor does not prove workers, subcontractors or suppliers beneath that contractor have been paid.

- Change, delay, defect, acceptance, payment and responsibility remain separately auditable. Rework consumes new labor and materials.
- Default, abandonment, fire or cancellation do not reset the site. Completed work, defects, materials, claims, debts, salvage and unfinished improvements persist.
- Mechanic/material claims, liens, performance security and surety exist only through historically/date/jurisdiction-appropriate Agreement and LegalRegime rules; they are real contingent/encumbrance relationships, not abstract penalties.
- Public/institutional procurement uses the same ordinary economic systems but may add plans/specifications, advertised bidding, responsible-bidder screening, security, progress certification, withholding/retention and formal acceptance.

PART I - CONSTRUCTION DELIVERY, ESTIMATING & CONTRACT FORMATION

1.1 Construction delivery is a relationship structure

CANON LOCK - The same physical building may be delivered by owner/self-performed labor, directly contracted trades, a coordinating prime/general contractor or a mixed arrangement. The delivery structure changes responsibility, procurement, working-capital burden, coordination and counterparty risk; it is not a technology tier.

A carpenter, mason, painter, plumber, builder or other specialist may operate independently or under a prime. A prime contractor coordinates obligations and capacity; the role does not erase the underlying trade businesses or Persons.

1.2 Contractor is a commercial role, not a universal occupation

Contractor/builder status is ordinarily a business/relationship/authority role layered over real trade, estimating, supervision, commercial and management capabilities. Do not require every contractor to have a unique Contractor occupation profile, and do not infer construction capability from the title alone.

1.3 Estimate, proposal and agreement stay separate

CANON LOCK - An internal Estimate is an actor's forecast. A Bid/Proposal is the commercial offer presented to a counterparty. An Agreement is the accepted obligation structure. Baseline preserves the authorized scope/budget/schedule; Current Forecast changes as information changes; Actual records what occurred. None should silently overwrite the previous state.

1.4 Bottom-up estimating and uncertainty

Construction estimates should derive from meaningful measurable scope: excavation/site work, masonry/foundation quantities, lumber/board measure, roofing, openings, plaster/finish area, fixtures/systems and comparable work packages. Floor area may help infer quantities but is not the economic truth.

Estimate uncertainty should be explainable through categories such as quantity uncertainty, unit-cost/quote uncertainty, productivity uncertainty, schedule/dependency uncertainty and collection/payment uncertainty. Better expertise, inspection and records narrow justified uncertainty; they do not guarantee profit.

1.5 Procurement responsibility is contractual

The Agreement may assign labor, materials, tools, equipment, hauling, temporary works, specialist installations and other inputs to the owner, contractor, subcontractor or a mixture. Owner-supplied and contractor-supplied material are therefore different commercial arrangements even when the finished structure is identical.

PART II - PHYSICAL EXECUTION, CREWS, METHODS & ENVIRONMENT

2.1 Quantity/resource construction

CANON LOCK - Authoritative construction progress comes from executed physical WorkPackages/Tasks and changed Building/site state, not from spending construction points. A summary completion percentage is allowed only as a read model.

2.2 Active-work time versus elapsed project time

Calendar duration may include waiting for finance, material delivery, a specialist, curing, weather, inspection, customer decisions or prerequisite work. Actual active-work intervals remain separate and continue to control objectives that measure overlapping work or crew continuity.

2.3 Crew composition and diminishing returns

A skilled trade lead can coordinate appropriate helpers on work that permits it. General labor can stage, carry, excavate and support skilled work, but specialist requirements remain real. Productivity depends on available work fronts, supervision, method, congestion and equipment; adding Persons does not scale output linearly.

2.4 Construction methods change input grammar

New or diffusing methods should alter required materials, tools, skills, task sequence and labor burden rather than grant abstract speed bonuses. Dimensional/plank framing, manufactured doors/windows, wire nails, Portland cement, tar paper and other period changes are availability/supportability inputs, not global unlock multipliers.

2.5 Logistics is construction

Lumber, stone, brick, hardware, windows, fixtures and equipment must come from actual suppliers/producers and reach the project through physical logistics. Rail access can improve construction because it changes supplier choice, delivered cost, lead time and available components; it does not apply a generic ConstructionSpeed bonus.

2.6 Weather acts on tasks and supply

Do not use a universal winter construction shutdown. Frozen ground, low temperature, precipitation, storms, daylight and route condition affect the particular tasks/material methods they physically constrain. Interior work may continue while excavation or masonry is blocked. Severe weather can also interrupt delivery even when on-site work would otherwise be possible.

2.7 Completed work becomes persistent property history

Installed work changes the Building/improvement and remains after contractor departure, default or project suspension. Hidden or known construction quality, defects, repair burden and previous corrective work remain part of future diligence and condition history.

PART III - PROJECT FINANCE, PAYMENT, WORKING CAPITAL & CAPACITY

3.1 Four distinct economic views

Measure	Meaning
Contract / Agreement Value	Amount or pricing structure the counterparty agreed to pay, before or after authorized amendments.
Project Cash Outlay	Cash actually paid by the owner/project entity for purchases, payroll, contractor bills, freight, fees and other project expenses.
Resource / Cost Attribution	Labor, owned materials, equipment use, donated/family work and other real resources consumed even when no cash transaction occurs.
Asset Basis / Value	Historical/carrying/market valuation of the resulting improvement under the applicable property/valuation system; not automatically equal to any one figure above.

3.2 Progress billing follows the Agreement

Physical progress does not directly create cash. A contract may use deposits/advances, payment for delivered materials, milestone/progress certificates, completion balances, withholding/retention or another historically supported structure. Billable/certified amount, invoice/claim, receivable, cash received, retained/withheld amount and disputed amount may all differ.

3.3 Contractor cash remains ordinary business cash

CANON LOCK - Do not create a magical project wallet merely to prevent poor treasury decisions. Contractor cash is ordinary business cash unless a real agreement, escrow, trust, security or restricted-fund relationship says otherwise. Project accounting may allocate costs and commitments without creating a second money authority.

3.4 Profitability is not liquidity

A contract can be profitable while the contractor cannot finance payroll, materials, freight or subcontractors until the next receipt. Supplier credit, customer advances, lines of credit, owner capital and payment terms can bridge the gap. Expansion and simultaneous projects can increase working-capital stress faster than reported profit.

3.5 Backlog means committed capacity

Backlog should be understood through future commitments of specific Persons/capabilities, supervisor attention, equipment, project windows, materials and liquidity rather than a flat BacklogScore. A contractor can be capable but unavailable for the required schedule.

3.6 Downstream payment remains independent

CANON LOCK - Owner payment to a prime contractor does not automatically settle the prime's suppliers, subcontractors or workers. Each obligation settles only through its own authoritative transaction/claim path. This preserves historically plausible lien and nonpayment exposure.

PART IV - CHANGE, DELAY, DEFECT, ACCEPTANCE & RESPONSIBILITY

4.1 Changes append history

Owner changes, discovered conditions, design changes, substitutions and corrective work can create Change Orders/Amendments that record changed scope, price, schedule and risk. The original baseline remains visible. Authorized extras are not silently absorbed into the original estimate.

4.2 Delay is causal

Late completion does not by itself prove breach. Contractor scheduling failure, supplier failure, owner-caused change, owner failure to furnish required access/input, weather, route disruption, labor scarcity and other causes can have different contractual consequences. Time extensions, daily damages or other remedies apply only where the Agreement/LegalRegime supports them.

4.3 Physical completion, usability, occupancy, acceptance and settlement are separate

CANON LOCK - A building may be physically usable or occupied while defects remain, acceptance is disputed or final payment is outstanding. Conversely, contractual acceptance does not prove that every downstream supplier has been paid.

4.4 Defect truth and responsibility are separate

A physical defect can arise from workmanship, wrong/defective material, design/specification error, unauthorized substitution, site condition, environmental damage or an unresolved cause. Evidence and later investigation determine what actors know and who is responsible; the game should not instantly expose omniscient fault attribution.

4.5 Rework consumes new resources

Corrective work creates real Tasks/WorkPackages, consumes additional labor/material/equipment and changes schedule. Original inputs are not refunded. Financial responsibility for rework follows causation, contract terms, negotiation and dated law.

4.6 Negotiated resolution before full legal escalation

Disagreements may resolve through negotiation, architect/superintendent interpretation, agreed referee/expert, settlement, arbitration or court/legal process when supported. The game should preserve time, cost, uncertainty and enforceable outcome without requiring a courtroom simulator.

PART V - DEFAULT, LIENS, SURETY, ABANDONMENT & RECOVERY

5.1 Default changes relationships; it does not erase physical history

A Project can be suspended, abandoned, underfunded, damaged, disputed or continued under a replacement contractor. Agreement termination does not require Project deletion. Completed work, site materials, defects, temporary works, salvage and outstanding obligations remain.

5.2 Replacement execution uses the remaining real problem

A replacement contractor should inspect and price the actual remaining work, acceptable completed work, defective work, on-site materials and stabilization needs. Do not convert the unfinished project into an abstract remaining percentage that ignores physical state.

5.3 Construction liens/claims are real encumbrances where law permits

CANON LOCK - Dated Dakota/South Dakota law supports construction claims by labor/material suppliers. The LegalRegime determines who may claim, what property/improvement interest can be reached, notice/timing requirements, withholding rights and enforcement. A lien/claim is not a reputation debuff and must interact with title, sale, refinancing and diligence.

5.4 Surety/security transfers specified contingent risk

Performance bonds and other security do not make failure impossible. The contractor can still default and work can still stop. Security creates a third-party or pledged obligation that may respond after specified default/notice conditions, subject to coverage limits and defenses. It should be modeled as contingent exposure, not instant project completion.

5.5 Default does not confiscate unrelated property

Owner rights to project-site materials, equipment or other assets after default depend on the Agreement, security arrangement and law. Do not automatically transfer a contractor's unrelated wagons, shop assets or cash because one project failed.

5.6 Fire and risk of loss follow real state and agreement

Fire or other casualty destroys/damages actual work, materials and equipment first. Contract terms, ownership, legal regime and insurance determine who bears the economic loss and whether rebuilding is required. Insurance creates a claim/possible later payment; it does not restore the destroyed work.

5.7 Distress can become ordinary market opportunity

Incomplete or encumbered projects can feed normal property, lending, acquisition and restructuring systems. A distressed developer, creditor or replacement contractor may sell rights/assets or negotiate a recovery rather than triggering a special construction-failure screen.

PART VI - PUBLIC / INSTITUTIONAL PROCUREMENT

6.1 Same systems, greater formality

Public and institutional construction remains ordinary Agreement/Project execution but can impose historically valid additional procedure: prepared plans/specifications, public notice/advertising, sealed or formal proposals, responsible-bidder selection, cost ceilings/appropriations, performance security, superintendent/architect certification, progress payment limits, withholding/retention and final acceptance.

6.2 Public work creates a real risk/reward tradeoff

Large public contracts can provide scale, visibility and credible counterparties while imposing tighter pricing, qualification, security, inspection and slower cash conversion. Do not model government work as a generic revenue bonus.

6.3 Qualification is evidence and financeability

A customer may reasonably consider relevant accepted work, references, specialist Persons, equipment/assets, geography, present commitments, liquidity/credit and ability to furnish required security. This deepens existing contractor qualification doctrine without creating a ContractorLevel gate.

PART VII - HISTORICAL / CALIBRATION EVIDENCE RETAINED

The following observations are retained as dated evidence and QA anchors. They are not universal prices, wages, crew recipes or legal constants.

Anchor	Observed evidence	Use
1870 Dakota wage guide	Ordinary day labor about $2/day; carpenter about $3.50/day in the cited guide.	Early-period wage-order anchor; local market still controls.
1882 Sioux Falls dwelling contract	$4,770 base dwelling; $1,021.76 claimed extras; separate $630 barn/outbuilding; April 3 to Sept. 1 target; milestone/change/security language.	Private-contract scale, changes and milestone structure.
1887 Dakota Reform School	$30,000 statutory ceiling; 30-day advertised proposals; security; no payment until at least one-third complete; cumulative payment capped at 80% of certified value before final acceptance.	Public procurement and working-capital stress anchor.
South Dakota barn study	Period guidance: journeyman carpenter with about four men could raise a 40x60 plank-framed barn in about one week, with about another week for finishing.	Order-of-magnitude productivity fixture; not a universal recipe.
1884 Pierre store / Albright	$1,290 prime contract; owner paid prime in full while material supplier remained unpaid and pursued a lien.	Downstream-payment independence and lien fixture.
1904 Milbank sewer	$11,276.35 bid plus $1,000 adjustment; about 500 ft completed before abandonment; city completion roughly $1,700 above contract price; $7,000 performance bond.	Underbid/default/replacement/surety fixture.

PART VIII - ACCEPTANCE DOCTRINE, RESEARCH HOLDS & SUPERSESSION

8.1 Acceptance rules

- Two visually identical buildings may have different costs, schedules and risk because delivery structure, source responsibility and actual supply chains differ.
- A low bid can win a project and later produce a loss, liquidity crisis, breach or abandonment without automatic repricing.
- A project can be physically advanced while billable amount or cash received remains low or zero under the Agreement.
- A building can be occupied while final acceptance, defects and financial settlement remain unresolved.
- Payment to the prime cannot automatically clear unpaid downstream obligations.
- Default preserves the partially completed site and allows replacement execution through the remaining physical work.
- A performance bond changes contingent financial exposure but cannot prevent delay, physical failure or contractor insolvency.
- Construction claims/liens, risk of loss and public-procurement procedure must query dated law/Agreement data rather than use one timeless global rule.
- Scenario-specific crew minimums from THE CONTRACTOR remain scenario definitions and do not become universal construction crew requirements.

8.2 Remaining research/calibration holds

Keep open as data/calibration rather than Canon constants: detailed 1870-1905 local wage series by trade; project-type material quantity libraries; Dakota building-cost series by locality/year; exact productivity distributions by method/crew/weather; local supplier inventories and freight spreads; detailed historical lien notice/enforcement procedure by date; bond premium/underwriting practice; builders-risk coverage terms; public procurement thresholds by jurisdiction; and exact allocation of disputed design-versus-workmanship responsibility in specific legal regimes.

8.3 Supersession/deepening map

- Revision XVI physical staged construction and Construction & Contracting archetype are deepened by delivery-structure, bottom-up estimating, crew/capacity, working-capital and downstream-payment doctrine.
- Revision XVII Project financial reporting, change-order and project-failure persistence are deepened by contract-value/cash/resource distinctions, replacement execution, defect responsibility and claims.
- Revision XXIII Agreement/Obligation/Project lifecycle is deepened by construction-specific billing, acceptance, default, lien, surety and risk-of-loss behavior.
- Revision XXII contractor qualification is deepened by explicit security/bonding feasibility and committed-capacity interpretation.
- All preserved Property, Finance, LegalRegime, Insurance, Inventory, Logistics and Evidence authorities remain active and are reused rather than duplicated.

END OF REVISION XXVIII INTEGRATION

The complete Canonical Project Bible Revision XXVII follows unchanged as the preserved baseline and remains authoritative except where Revision XXVIII explicitly adds, deepens, corrects or supersedes it.

LAND & LEDGERS
CANONICAL PROJECT BIBLE - REVISION XXVII

Town, Catchment & Workforce Integration: Economic Population, Market Territory, Family Labor, Business Density & Demand-Driven Generation

Revision date: September 26, 2026

Source baseline: Canonical Project Bible Revision XXVI - September 26, 2026

New source integrated: Historical Research Findings - September 26, 2026

Status: Active integration

PRESERVATION GUARANTEE

REVISION XXVII preserves Canonical Project Bible Revision XXVI in full except where this integration explicitly adds, deepens, clarifies, corrects or supersedes prior wording. The complete Revision XXVI document follows this integration unchanged as the preserved historical layer. Where a conflict exists, Revision XXVII controls. Numeric catchment ranges, workforce percentages, staffing bands and business-prevalence observations from the new research remain calibration/authoring evidence unless this integration explicitly identifies a Canon lock.

PURPOSE

Revision XXVII integrates the September 26 Historical Research Findings on Dakota Territory town economics and workforce calibration. Its central correction is structural: the number of people sleeping inside a town is not the same thing as the population supporting that town economically. The revision therefore separates built-up residents, rural service population, effective market population, economic participation and wage employment; makes market territory and settlement role causal; strengthens family/seasonal labor doctrine; and rejects hidden population-to-business formulas.

REVISION XXVII MASTER DOCTRINE

- Town population is a residence count, not a direct proxy for customers, workers or viable businesses.
- Built-up population, rural service population and effective market population are separate read models over actual Persons, Households, Farms, routes and market relationships.
- Residents, economically active Persons, primary occupation and wage employees are separate measures. Employment must not be treated as synonymous with employees.
- Market territory emerges from physical rural settlement, route access, competing centers, institutions, transient traffic and actual trade. No RuralPopulationMultiplier or BusinessCount = Population / X rule is allowed.
- Settlement role can outweigh size. Agricultural service town, railhead, county/administrative center, mining node, river center and industrial node generate different demand even at similar resident population.
- Business formation and survival follow actual demand and capability. One enterprise may expose several commercial capabilities and productive work may occur in houses, barns, yards, shared premises or mobile/route operations.
- Most local enterprises should be able to remain owner-operated or very small. Staffing grows from task workload, seasonality and simultaneous demand rather than a fixed employee recipe.
- Farm labor distinguishes operator, family enterprise labor, permanent hired labor, seasonal/casual labor and contract/service labor. Unpaid family work is economically real without becoming wage employment.
- Labor demand is worker-time by task and season, with peak simultaneous headcount distinct from annualized FTE.

- Farms are mixed enterprises; raising animals, processing products and retailing finished goods remain distinct economic stages.
- Historical percentages, catchment brackets and staffing bands in the source are evidence-anchored authoring envelopes, not universal Canon constants.

PART I - SETTLEMENT POPULATION, ECONOMIC POPULATION & MARKET TERRITORY

1.1 Town population is not economic population

CANON LOCK - The resident population attached to a settlement must not directly determine the number of viable businesses, the number of customers those businesses can reach or the size of the local workforce.

A small town can be the commercial node for a much larger dispersed farming district. Rural households may travel into town for grain shipment, blacksmithing, lumber, implements, groceries, mail, banking, lodging, saloons, medical care and other services. Conversely, a badly placed or bypassed settlement can underperform despite a larger resident count.

1.2 Three population views

Settlement reporting should distinguish at least: **Built-up population** - Persons whose primary residence lies inside the settlement; **Rural service population** - rural Persons/Households economically connected to the settlement through reachable trade/service relationships; and **Effective market population** - the combined reachable market relevant to local commerce. These are read models over world state, not independent counters that can mutate the world.

1.3 Person-side economic measures

CANON LOCK - Residents, Economically Active Persons, Primary Occupation and Wage Employees are separate measures.

A farmer or proprietor can be economically active without being anyone's employee. A spouse or older child can contribute economically meaningful family labor without conventional wages. A Person can have one primary livelihood while still performing secondary household, seasonal, civic or business tasks.

1.4 Market territory is physical and competitive

A settlement catchment is shaped by farms and households, roads and river crossings, travel time, wagon economics, railway access, nearby competing towns, county or institutional boundaries, terrain, ferries/bridges, production flows and existing commercial relationships. Do not implement a fixed radius or multiplier merely because two settlements have the same resident population.

PART II - SETTLEMENT ROLE, BUSINESS ECOLOGY & DEMAND

2.1 Demand-driven generation

CANON LOCK - Do not generate businesses directly from town population. Generate viable opportunities from resident demand + surrounding rural demand + access + settlement role + competition + transient traffic + local production + date + actual unmet demand.

Population is a useful prior and constraint, but not a cap or a business recipe. The causal chain is: physical rural settlement -> farms/households/output -> roads and travel reach -> settlement catchment -> town residents + transients + institutions -> actual demand -> businesses form, survive, expand or fail -> staffing emerges from workload.

2.2 Institution-generated demand

Courthouses, depots, mills, mines, military/federal facilities and other institutions create demand in their own right. A courthouse can support legal, clerical, lodging, restaurant, printing, transport and finance activity; a depot can support freight, drayage, hotel, livery, grain and warehousing; a mill can create transport, fuel, maintenance and labor demand. Institutional demand is causal rather than an abstract settlement bonus.

2.3 Settlement role outranks a universal Town Level

Agricultural service towns, rail junctions/railheads, administrative centers, mining towns, river centers and industrial nodes must not be cosmetic skins over one population template. Role and history alter who travels there, what is produced, which institutions exist and which capabilities can be supported.

2.4 Business prevalence is a distribution, not a ratio

Historical town rosters show high variance at small populations. Therefore rules such as one general store per 100 residents, GeneralStores = Population / 150 or similar hidden formulas are expressly rejected. Use empirical distributions and actual demand drivers instead.

2.5 Historical observations used as authoring anchors

The new research documents very small settlements with substantial business rosters and also shows larger towns with fewer establishments of particular types. These records establish plausibility and variance, not mandatory templates.

Observation	Resident population / date	Research use
De Smet post-office application	50 town residents; 500 population served (Feb. 1880)	High-end documented service-population example; never a universal 10x multiplier
Menoken	45 (1886)	3 stores, grain elevator, blacksmith and school demonstrate rural-market support
Sterling	~100	Multiple stores, drugstore, lumber, grain warehouses and newspaper in farming country
Alexandria	~99 (1880/1881 evidence)	Extensive directory roster in a county ~13x the town population; strong catchment/role example
Elk Point	~710 (1880)	Fewer general-merchandise firms than Alexandria despite much larger resident population; rejects monotonic ratios
Bismarck	~1,200 (Feb. 1874)	Railhead/frontier supply node with unusually high hospitality and saloon demand from transients

PART III - BUSINESS ENTITY, COMMERCIAL CAPABILITIES, PREMISES & STAFFING

3.1 One business may provide several services

A Business is an ownership/accounting/governance entity. Its commercial capabilities can include general goods, groceries, hardware, implements, mail/post functions, credit, produce purchasing and other supported services according to actual Persons, stock, tools, premises, agreements and authority. Do not satisfy every demand category by spawning a separate firm.

3.2 Business is not building

A dedicated shop is not required when the work can genuinely occur in a house, barn, yard, mixed-use room, shared premises or mobile/route operation. The 1870 opening should therefore contain more productive enterprises than storefronts where the physical capability supports it. This deepens Revision XXVI premises doctrine.

3.3 Small-enterprise structure

Most ordinary local businesses can remain tiny. Owner-operated microenterprises and 2-6 Person establishments should be common; larger 6-20+ local establishments appear where throughput and industrial/service scale justify supervision, shifts, clerical work and specialization. These size descriptions are mental models, not hard tiers or unlocks.

3.4 Staffing is workload-driven

Staffing follows labor-hours/days, task collision, customer/service load and seasonal peaks. Historical manufacturing evidence in the source - roughly 3.4 workers per Dakota manufacturing establishment in 1880, 2.9 per flour/grist mill and about 7.4 per sawmill - is retained as dated calibration evidence, not a universal staffing formula.

PART IV - FARM LABOR, HOUSEHOLD PRODUCTION & MIXED ENTERPRISE

4.1 Five farm labor relationships

CANON LOCK - Farm work must distinguish: (1) farm operator; (2) family enterprise labor; (3) permanent hired hand; (4) seasonal/casual hired labor; and (5) contract/service labor.

Family enterprise labor is real economic work even when no conventional wage changes hands. Contract/service labor includes hired threshing crews, freight carriers, veterinarians, machine providers and comparable external services. Permanent employment is only one way labor enters a farm.

4.2 Seasonal labor is worker-time, not employee slots

A farm can average modest annual FTE while requiring many Persons at once during harvest, haying, threshing, fencing or other peaks. Labor demand should therefore be expressed by task, date/season, duration and simultaneous capacity, with FTE only as a reporting abstraction.

4.3 Primary occupation does not capture all work

Historical census occupation tables omit or understate much household and family labor. A Person without a recorded wage occupation may still perform poultry, dairy, garden, preserving, bookkeeping, childcare, repairs or other productive work. Demographic/workforce calibration must not erase that labor merely because it lacks a formal occupation label.

4.4 Farms are mixed enterprises

Do not create rigid Chicken Farmer, Wheat Farmer or Beef Farmer classes for ordinary farms. The farm's actual land, crops, livestock, equipment and task schedule determine current work. Raising livestock, processing meat/dairy/grain and retailing finished goods are separate economic stages that can be performed by different businesses or combined only where real capability exists.

PART V - 1870 OPENING & LONGITUDINAL SETTLEMENT CHANGE

5.1 1870 opening character

The normal 55-95 Person 1870 settlement should remain sparse, dispersed and less rail-industrial than the 1880s directory examples. Authoring should lean toward wagon freight, stage/ferry/river connections, smithing, one or a few merchants where the route/catchment supports them, combined post functions, school/church/shared community space, home/yard carpentry and repair and small-scale teamster activity. Bank, grain elevator, lumber yard, newspaper and hotel are not automatic population-based fixtures.

5.2 Generate the countryside first

A stronger opening rule is to generate surrounding farms/households and their routes before inferring the settlement's market. Two 70-Person towns can have radically different economies if one serves dozens of farm households while the other is an isolated ferry/stage node with little nearby settlement.

5.3 Transformation is physical

As rail, population, finance, institutions and production grow, denser Main Street form should emerge through actual businesses, buildings, mixed-use conversion, traffic and investment rather than through a scripted Town Level change.

PART VI - HISTORICAL ANCHORS, CALIBRATION ENVELOPES & NON-LOCKED NUMBERS

6.1 De Smet high-end service-population anchor

The source records a February 24, 1880 post-office application stating De Smet town population 50 and post-office population served 500, with rapid growth reported soon after. This is a documented proof that service population can greatly exceed resident population. It is not permission to apply a 10x multiplier to every town.

6.2 Occupational context anchors

The source records 1880 Dakota Territory occupational shares of 49.3% agriculture, 24.2% professional/personal services, 10.8% trade/transport and 15.7% manufacturing/mechanical/mining, while warning that territory-wide figures include mining, larger towns, transport and government and may understate agriculture in rural districts. It also records 1890 South Dakota farmers plus agricultural laborers at about 64.0% of gainfully occupied males and 1900 South Dakota agricultural pursuits at 60.4% of gainful occupations. These are context anchors, not direct small-town generator percentages.

6.3 Latest workforce-sector table remains calibration

The source's latest settlement-plus-catchment sector percentages for 10/50/100/200/500/1,000/2,000 resident towns are explicitly BRIDGE ESTIMATES. They remain authoring/calibration envelopes with uncertainty, not Canon locks. Separate town-proper and rural-district workforce distributions are preferred to one blended population formula.

6.4 Latest catchment table remains calibration

The source's latest rural-service ranges - for example a 100-resident agricultural market village serving roughly 300-600 surrounding rural Persons and a 200-resident farm-service town serving roughly 500-1,000 - are retained as provisional authoring envelopes for normal 1878-1890 agricultural settlements. They do not apply automatically to mining camps, division points, military posts or major river centers and must never be stored as multipliers or rigid radii.

6.5 Research provenance boundaries

Promotional directories and one-town records prove that a configuration could exist; they do not prove every same-size settlement had it. Derived arithmetic, bridge estimates and historical observations must retain source, geography, date, denominator and confidence under the existing Canon provenance doctrine.

PART VII - SOURCE OPEN QUESTIONS & FUTURE RESEARCH

The new source explicitly leaves several items unresolved. Revision XXVII preserves them as research/calibration work rather than silently closing them.

Open item from source	Revision XXVII treatment
Business prevalence distributions across 15-30 Dakota settlements and multiple population brackets	Research priority; use distributions, not point ratios
Population -> Rural Catchment -> Workforce -> Business Demand matrix by year and rail/non-rail state	Authoring/calibration target; not yet Canon numbers
Workers-per-business by type; business prevalence per rural population; family vs hired farm labor over time	Evidence gap retained
Reconciliation of catchment and industry tables; unlabeled computed fragments and denominators	Do not canon-lock until provenance is confirmed
Exact retail percentage; exact agricultural-generation probabilities and settlement/business distributions	Remain research/calibration
100-job ranking noted as pending in this research thread	No Canon effect in this integration; does not supersede separate occupation documentation

PART VIII - ACCEPTANCE DOCTRINE & SOURCE TRACEABILITY

8.1 Acceptance rules

- A 50-Person or 100-Person town can support more commerce than resident headcount suggests when a real rural catchment, transit flow or institution supplies the demand.
- Increasing resident population alone cannot automatically add businesses, workers or customers.
- A larger town may have fewer firms of a given type than a smaller town when market territory, role, competition or timing differ.
- Family enterprise labor contributes real task time without creating wage payroll entries.
- A farm can require several simultaneous seasonal workers even when its annualized FTE is much lower.

- One business can satisfy several commercial capabilities when its people, stock, tools, rights and premises support them.
- A Person can be a resident and economically active without being a wage employee.
- No catchment multiplier, business-per-population ratio, Town Level or fixed staffing recipe may bypass the causal systems above.

8.2 Complete source-section disposition register

Source section	Disposition in Revision XXVII
1. Town population is not economic population	General Canon doctrine; three population views and four Person-side economic measures
2. Documented historical anchors	Retained as dated authoring evidence; not universalized
3. Documented town business profiles	Plausibility/variance evidence; not templates
4. Business density and market-territory principle	Canonized: market territory/role/competition replace population ratios
5. Business size and staffing	Canonized as workload-driven small-enterprise doctrine; exact bands remain calibration
6. Farm labor	Canonized five relationships + worker-time/seasonality + mixed-enterprise doctrine
7. Industry percentage calibration	Calibration only; latest table retained as authoring envelope
8. Catchment model	Calibration only; physical generation rule canonized, no multiplier/radius
9. 1870 opening calibration	Integrated as default authoring guidance, not automatic business list
10. Generator philosophy and demand drivers	General Canon lock: demand-driven generation
11. Named concepts glossary	Terminology synchronized into this revision
12. Open questions and pending work	Preserved as explicit research/calibration backlog

END OF REVISION XXVII INTEGRATION

The complete Canonical Project Bible Revision XXVI follows unchanged as the preserved baseline and remains authoritative except where Revision XXVII explicitly adds, deepens, corrects or supersedes it.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXVI

Ghost Town Playtest Integration: Open Entrepreneurship, Property Rights & Agency, Physical Commerce, Migration, Delegation, Valuation & Evidence

Revision date: September 26, 2026

Source baseline: Canonical Project Bible Revision XXV - September 22, 2026

Status: Active integration

PRESERVATION GUARANTEE

REVISION XXVI preserves the supplied baseline in full except where this integration explicitly adds, deepens, clarifies, corrects or supersedes prior wording. The complete baseline document follows this integration unchanged as the preserved historical layer. Where a conflict exists, this integration controls. Scenario-specific values remain scenario data or calibration unless this integration explicitly promotes them to a general rule.

PURPOSE

Revision XXVI consolidates every recorded finding from the September 22-26 Ghost Town playtest into durable design doctrine, scenario-specific corrections, calibration boundaries, research holds and explicit supersessions. The playtest exposed particularly strong needs around open starting choice, rural catchment economics, property rights, real business formation, perishable inventory, recruitment, property agency, delegated portfolio management and enterprise valuation. Intermediate in-session references to possible Revision XXVI/XXVII labels are superseded by this single consolidated Revision XXVI.

REVISION XXVI MASTER DOCTRINE

- Open-ended scenarios should let the player investigate credible opportunities before irreversible starting investment unless forced commitment is an intentional premise.
- Vacancy, abandonment, deterioration and non-use never erase property rights. Property right, physical possession, legal basis and enforcement are separate states.
- A business becomes operating through actual commerce supported by real people, premises, assets, inventory and counterparties - not through a setup button or a business-level flag.
- One business may use several sites and one property may host several businesses or operations. Functional space and shared physical capacity remain real.
- Perishable inventory, prepared food and long processes retain lot provenance, condition and calendar time. Quality and food-safety risk are separate.
- Recruitment and migration arise through actual people, relationships, information channels, jobs, ownership opportunities and housing feasibility; applicant and migrant buttons do not conjure people.
- Human capability presentation is Ordinary -> Great -> Excellent -> Mastered. Underlying capability remains continuous and confidence controls the width of what the player sees.
- Procurement begins with real suppliers and real quoted goods. Landed cost, package terms, deposits, freight and payment terms precede retail pricing.
- Delegated managers act through explicit authority and evidence. Competent rivals use the same systems as the player and receive no hidden economic bonuses or omniscience.
- Enterprise evaluation distinguishes gross going-concern value from the estimated equity/sale proceeds attributable to the owner after business-specific liabilities. Business cash is displayed separately.
- Maintainable operating profit, normalized for replacement labor and seasonality, is the primary ordinary valuation evidence; productive assets matter without being double-counted.
- Property agency, commercial tenancy, chair/room/station leasing, post-sale supply agreements and related-party contracts are ordinary contractual relationships rather than special-case ownership shortcuts.

- Railroad promotion, freight adaptation, advertising and market discovery operate through real counterparties, capacity, information and incentives rather than success rolls or percentage bonuses.
- Historical law and specific prices remain dated authoring data with provenance. Scenario-generated quotes and playthrough prices are calibration unless explicitly locked.
- World and portfolio reporting is evidence, not omniscience. Important changes can surface through inspection, records or delegated monitoring without turning every state change into a blocking pop-up.

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- Part 2: Property Rights, Use, Enforcement & Residence
- Part 3: Business Formation, Premises & Ownership
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- Part 5: Markets, Procurement, Pricing & Service Capacity
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- Part 13: Playtest Corrections, Acceptance Doctrine & Traceability

PART I - SCENARIO FRAMING, STARTING CHOICE & OBJECTIVES

1.1 Scenario narration

CANON LOCK - Every authored scenario should have brief opening narration that establishes premise and stakes, plus short route-agnostic completion narration that recognizes what changed without prescribing the strategy used.

Playtest provenance: GHOST-CAN-001

1.2 Player-selected starting opportunity

When a scenario is intended to support open economic strategy, it should not force the player into a specific industry merely because the scenario preselects a building or asset. The author may provide a constrained pool of credible starting opportunities, but the player chooses among them. Once selected, the choice becomes persistent world history; unchosen opportunities remain in the world and may be taken, altered or lost through NPC action.

Where the scenario is explicitly structured around a pre-start choice, the campaign clock may remain paused until that choice is made. This is an authoring tool, not a universal rule that every campaign must pause at start.

Playtest provenance: GHOST-CAN-008

1.3 Starting investment follows ordinary investigation

CANON LOCK - Open entrepreneurship should permit ordinary available market research before scarce capital is committed unless forced commitment is deliberately part of the scenario premise.

The player may investigate traffic, rural demand, existing suppliers, customer sourcing, property condition, title status, nearby production and competing businesses using whatever information channels are actually available. Investigation is allowed to kill a weak idea before money is spent; a failed investigation is a successful use of the system, not a content failure.

Playtest provenance: GHOST-CAN-009, GHOST-DES-019, GHOST-DEF-001

1.4 Historical starting year is gameplay

Scenario dates should be chosen for the economic regime they create. The Panic of 1873, railway capital conditions, settlement waves and infrastructure availability change actual opportunities, credit and counterparties. Starting in 1874 is therefore not a cosmetic reskin of 1880. Scenario authors should select dates intentionally and describe the resulting context through the existing HistoricalRegime and evidence doctrine.

Playtest provenance: GHOST-DES-086

1.5 Ghost Town objective correction

SCENARIO LOCK - GHOST TOWN SCENARIO LOCK - By December 31, 1885, own 100% of three operating businesses with a cash-free, debt-adjusted estimated sale value of the owner interest of at least $1,000 each.

The earlier $500-per-business revision is superseded. The separate December 31, 1876 objective requiring one $1,000 business remains unchanged. A business qualifies only if it is genuinely operating through commerce and the valuation uses transferable owner equity rather than gross financed assets or cash parked in the account.

Playtest provenance: GHOST-DES-038, GHOST-DES-043, GHOST-DES-049, GHOST-DEF-002

PART II - PROPERTY RIGHTS, USE, ENFORCEMENT & RESIDENCE

2.1 Vacant is not ownerless

CANON LOCK - Vacancy, abandonment, deterioration and non-use never by themselves establish that a parcel or building is unowned, free to claim or available for ordinary purchase.

Property UI must show the ownership/control status the player legitimately knows, the player's current legal basis for possession or use and the feasible avenues known to exist. Unknown or disputed is an acceptable state. Visually empty does not become legally unowned.

Playtest provenance: GHOST-CAN-002, GHOST-CAN-003

2.2 Unauthorized use is physically possible

Legal right and physical possibility are distinct. If access and physical requirements permit, a Person may occupy, store goods, improve property or conduct business without adequate premises rights. That action can create civil, financial, reputational and, where applicable, criminal exposure. Unauthorized occupation never silently creates title or a valid lease.

Playtest provenance: GHOST-CAN-004

2.3 Legal rule is not instant enforcement

Discovery, notice, awareness, travel, service, negotiation, litigation and enforcement consume ordinary world time. A legal rule can exist while an infringement continues because nobody knows, nobody has acted yet or the enforcement chain has not completed. The game must preserve the distinction among legality, detection, proceeding and consequence.

Playtest provenance: GHOST-CAN-005

2.4 Good-faith and adverse possession states remain distinguishable

Color of title, defective documentation, honest boundary error, disputed rights and knowing trespass are not one Unauthorized flag. They can carry different knowledge, bargaining, legal and reputational consequences. Property troubleshooting should therefore support record searches, owner/agent/executor contact, lease or sale negotiation, mortgage resolution, tax-sale paths, public-land entry where lawful, claim challenges, settlement or withdrawal.

Playtest provenance: GHOST-CAN-006, GHOST-CAN-007

2.5 Property opportunity provenance

CANON LOCK - Every market property opportunity must have a physical property, current right-holder or justified unknown status, a reason/channel by which it became known and a traceable path into the market.

Do not fabricate generic cottages, storefronts or farms as ungrounded purchase cards. A broker, newspaper, owner, creditor, estate, tax proceeding, personal relationship or direct observation can introduce an opportunity. The current owner and sale/lease motivation may be partly unknown, but the world object and information source must exist.

Playtest provenance: GHOST-DEF-011, GHOST-DES-096

2.6 Primary residence is distinct from tonight's sleep location

The player, like other Persons, can have a continuing Primary Residence/Home while temporarily sleeping elsewhere for travel, work or lodging. One hotel night does not rewrite residency. Where a legal regime such as homestead entry depends on actual residence, compliance uses real residence, presence, cultivation and abandonment history rather than a checkbox or merely leaving a bed on site.

Playtest provenance: GHOST-DES-084, GHOST-DES-085

2.7 Property change is information, not a mandatory pop-up

NPC property purchases, transfers and encumbrance changes should appear when the player inspects the relevant records/property or when a delegated watcher reports them. They should not automatically interrupt play. This does not make the strategic camera omniscient: direct inspection, records and delegated observation can provide different levels of detail and confidence.

Playtest provenance: GHOST-DES-096, GHOST-DES-097

PART III - BUSINESS FORMATION, PREMISES & OWNERSHIP

3.1 Universal Create Business workflow

CANON LOCK - Creating a Business creates an economic/legal entity and its ownership/governance relationships; it does not make the business operational by itself.

The ordinary workflow is Businesses -> Create Business -> choose business type -> assign owner/ownership structure -> establish premises if required -> assign assets/working capital/authority as needed. Premises choices can include no dedicated premises required, owned space, leased space or an acquired property. An already-used parcel is not automatically unavailable; the player assigns real functional space where compatible.

Playtest provenance: GHOST-DES-029, GHOST-DES-031, GHOST-TECH-004

3.2 Operating status comes from actual commerce

A business crosses into operating status when it can actually perform its economic activity and real commerce occurs: inventory or services are produced/provided, customer/supplier counterparties exist and transactions settle through ordinary systems. There is no universal Open Business button that bypasses capability or commerce.

Playtest provenance: GHOST-DES-009

3.3 Production and retail space may separate

One business may span several sites when operations have materially different sanitation, nuisance, storage, access or customer-facing requirements. Slaughter, hide preservation, tanning, meat retail and food service are a controlling example. Conversely, one building can host several businesses or operations when actual rooms, floors, yards, storefronts and shared facilities support them.

Playtest provenance: GHOST-DES-003, GHOST-DES-004, GHOST-DES-031

3.4 Add operation versus add separate business

Inside an existing business, Add Operation/Capability expands what the same ledger and ownership entity does. Add Separate Business Here creates a distinct business with its own cash, ownership, staff and agreements while preselecting the same property and requiring an actual space assignment. Enterprise may exist as an optional management umbrella/read model, but it is not another source of cash or ownership truth unless the businesses are formally reorganized.

Playtest provenance: GHOST-DES-033

3.5 Partnerships and partial ownership

Formation and acquisition workflows may create sole ownership, partnerships/partial ownership and later historically appropriate corporate structures. Ownership percentage, capital contribution, profit/loss participation, creditor exposure and managing authority are separate facts. Internal economic sharing of a liability does not rewrite what the external creditor can legally claim.

Playtest provenance: GHOST-DES-035

3.6 Shared assets and multi-unit enterprises

One business may contain multiple operating units under one ledger, such as freight and wagon repair. Shared workers, bays, animals, wagons, yards and buildings remain one physical resource pool. A wagon in repair cannot haul freight at the same time, and a worker reserved for one task is not simultaneously available elsewhere. Internal work consumes time and materials but does not create fake revenue to the same enterprise.

Playtest provenance: GHOST-DES-071, GHOST-DES-072, GHOST-DES-073

3.7 Parcel expansion and common building costs

Unused yard area is not generic spare capacity. Planning must preserve real space for circulation, deliveries, future structures, utilities/privies/water, fire separation, storage and additions. In mixed commercial buildings, common heating/water, stairs, roof, exterior maintenance and common cleaning remain landlord/property costs unless a real agreement reallocates them.

Playtest provenance: GHOST-DES-064, GHOST-DES-080

PART IV - PHYSICAL PRODUCTION, PERISHABLES & PROCESSING

4.1 Rural production can support town-facing enterprise

Opportunity evaluation for a small settlement must include its wider rural catchment. Livestock buying, slaughter, aggregation, processing and trade can be viable because surrounding farms/ranches and travelers use the town even when permanent town population alone is tiny. The market is the actual network of reachable people and producers, not the municipal boundary.

Playtest provenance: GHOST-DES-002, GHOST-DEF-001

4.2 Poultry is biological state, not a growth percentage

Poultry growth preserves egg production, fertility, incubation/brooding, hatch events, sex, juvenile stages, mortality, retained females, surplus males and slaughter outcomes. Individual/source identity and known parentage remain available where economically material. Same-source birds can be treated conservatively as potentially related until evidence says otherwise; breeding decisions can therefore respond to actual lineage rather than a calendar timer.

Playtest provenance: GHOST-DES-005, GHOST-DES-022, GHOST-DEF-003, GHOST-DEF-004

4.3 Butchery and carcass allocation

A butcher requires a real minimum workstation/tool set rather than ButcherTools=true. Slaughter produces a physical carcass and byproducts that can be allocated among fresh retail, wholesale, preservation, household use, tallow/fat use, hide processing and waste. Cheap livestock can still be a poor purchase when throughput is lower than carcass size and preservation/customer reach are insufficient.

Playtest provenance: GHOST-DES-008, GHOST-DES-010, GHOST-DES-011, GHOST-DES-012

4.4 Tanning and leatherworking are separate process/capability chains

Tanning is a multi-stage, time-consuming process using actual tools, inputs and work-in-progress state. Producing usable leather does not automatically confer the separate leatherworking/saddlery capabilities required to cut, punch, stitch and finish goods.

Playtest provenance: GHOST-DES-006, GHOST-DES-007, GHOST-CAN-013

4.5 Perishable condition and food safety

CANON LOCK - Perishable lot degradation and food-safety risk are separate causal states. Neither should be represented only by an expiry timer.

Taste, texture, appearance and saleability can deteriorate before complete spoilage, but contamination or poor handling can also produce safety risk before obvious sensory failure. Mixing new and old stock cannot rejuvenate old product. Losses, discards and downgraded sales must remain visible in inventory and ledgers.

Playtest provenance: GHOST-CAN-010, GHOST-CAN-011, GHOST-TECH-002

4.6 Preservation transforms; it does not reset

Curing, smoking, salting, tanning and comparable transformations create persistent work-in-progress batches carrying source condition, handling history, process quality, ingredients, storage and elapsed calendar time. Successful preservation changes future deterioration; it does not erase earlier damage. Prepared foods such as boiled eggs are likewise physical batches with quantity, age and condition.

Playtest provenance: GHOST-CAN-012, GHOST-CAN-013, GHOST-DES-063

4.7 Work-in-process can create real bridge-income pressure

Long production chains do not create revenue before sale. Businesses such as butcher/tannery operations may be profitable in the long run while remaining cash-starved between livestock purchase, curing, leather production and final sale. That gap is an intended working-capital problem.

Playtest provenance: GHOST-DES-015

PART V - MARKETS, PROCUREMENT, PRICING & SERVICE CAPACITY

5.1 Procurement is supplier-first

CANON LOCK - The game must not generate purchasable stock as an anonymous product-price pair when a real supplier/channel should exist.

A procurement option resolves supplier, exact product/grade, package and quantity, minimum order, bulk tier, supplier price, refundable container/deposit terms, freight/handling/other charges, landed consumable cost and payment terms. Refundable deposits consume cash but are not COGS unless lost.

Playtest provenance: GHOST-DES-056, GHOST-DES-065

5.2 Supplier qualification and delegated opening purchases

Where a supplier will provide one, a sample can be requested, transported and inspected. Quality becomes evidence from the actual sample/use rather than a hidden permanent supplier score. Delegated buyers may choose among acceptable suppliers, grades, package sizes and opening quantities within owner constraints using landed cost, quality evidence, expected demand, availability and working-capital burden.

Playtest provenance: GHOST-DES-062, GHOST-DES-066, GHOST-DES-067

5.3 Retail pricing follows landed cost

Menu and retail pricing should be evaluated only after the actual procurement terms, serving yield and landed cost are known. UI may then show margin, cash burden and demand consequences of candidate prices; it should not reverse-engineer a fictional supplier price from the desired retail price.

Playtest provenance: GHOST-DES-060

5.4 Internal sourcing remains physical and contractual

A wholly owned supplier does not create free goods. Internal requisitions draw from physical inventory after other obligations and production needs. Existing third-party agreements remain binding until legitimately renegotiated or ended. Inter-business transfers can be recorded for business-level economics, but internal transfer pricing cannot manufacture consolidated profit.

Playtest provenance: GHOST-DES-061, GHOST-DEF-008

5.5 Staffing follows observed workload

Hospitality and service businesses should not use fixed recipes such as Saloon=2 bartenders. Start with the staff actually hired, then observe customer volume, service delay, cleaning/restocking, bookkeeping, manager overtime and concurrent tasks. Add labor when workload supports it. One Person may cover compatible duties when real time permits.

Playtest provenance: GHOST-DES-050

5.6 Physical capacity, patron density and congestion

Premises capacity derives from actual tables, chairs, bar-standing positions, usable open floor and service points. Crowd effects primarily arise through occupied positions, queues, blocked circulation, table scarcity and wait time. Individual customers can prefer quiet, moderate or lively establishments while separately tolerating crowding and waiting. Preference response should be gradual, and expected crowd based on prior knowledge may differ from experienced crowd after arrival.

Playtest provenance: GHOST-DES-054, GHOST-DES-058, GHOST-DES-059

5.7 Business reporting is evidence

The simulation can know true occupancy, workload and inventory internally, but the player learns through direct observation, records and employee/manager reports. Multiple competent observations can improve confidence without simply adding skill scores together. This evidence doctrine applies equally to management dashboards and property information.

Playtest provenance: GHOST-DES-055

5.8 Advertising is a real media market

Newspapers and other paid media should have publication location, circulation/reach, issue frequency, readership geography, placement/size inventory, rates and repeat terms. Advertising creates awareness according to actual distribution and placement; it does not create better applicants or tenants. Effectiveness is observed through a chain such as insertion -> inquiry -> qualified inquiry -> visit -> hire/lease/sale rather than a hidden +20% bonus.

Playtest provenance: GHOST-DES-076, GHOST-DES-079

PART VI - LABOR, RECRUITMENT, CAPABILITY & CAREER

6.1 Casual specialist/helper relationships

Short-term helpers and specialists can be hired for bounded tasks without creating permanent employment. Standing delegation may authorize routine casual hires when a task genuinely benefits from extra labor or experience. Major or long-term employment can remain an escalation point.

Playtest provenance: GHOST-DES-013, GHOST-DES-014

6.2 Hiring is a recruitment workflow

Hire Worker should expose actual channels: contact a known Person, ask employees or associates for referrals, post handbills/notices, use local businesses/travel networks, send correspondence or buy newspaper advertising where available. These actions create awareness and possible inquiries from actual Persons over time. They do not spawn a candidate list on demand.

Playtest provenance: GHOST-DES-051

6.3 Referral and correspondence recruitment

NPCs can be asked whether they genuinely know family, former coworkers, friends or business contacts who may be interested in a job, property, business or relocation. A relationship can produce a plausible lead or no lead. Letters take real travel time; recipients may reply, ignore, negotiate or travel to inspect. The player action remains manual unless delegated; separate passive word-of-mouth can spread awareness without a click.

Playtest provenance: GHOST-DES-021, GHOST-DES-034, GHOST-DES-039, GHOST-DES-041

6.4 Capability bands and candidate evidence

CANON LOCK - Human player-facing capability bands are Ordinary -> Great -> Excellent -> Mastered. "Good" is not a human capability band, though it may remain a valid condition adjective for assets/buildings. All preserved seven-band human capability wording is superseded.

Underlying capability remains continuous or sufficiently granular. Weak, novice or unproven workers are described through experience, mistakes, task fit and evidence rather than Bad/Poor labels. Candidate presentation is ordered Capability | Current Reading | Confidence | Evidence. High confidence displays one band, Moderate exactly two adjacent bands, Low exactly three adjacent bands and insufficient evidence displays Not enough information.

Playtest provenance: GHOST-CAN-014, GHOST-CAN-015, GHOST-DES-052, GHOST-TECH-005

6.5 Capability advancement uses one system for player and NPCs

Improvement arises from relevant task exposure, aptitude, frequency, variety, instruction/supervision, task difficulty and accumulated experience, with diminishing gains as capability becomes exceptional. The player is not given a separate RPG XP progression. Promotions or titles do not manufacture competence.

Playtest provenance: GHOST-DES-091

6.6 Player employment is a valid starting state

A scenario may begin with the player as a wage employee such as a clerk, carpenter, teamster, farmhand, bartender or assistant. The NPC employer owns and controls the business; the player earns real wages, consumes real time, gains experience through ordinary work and can later accumulate capital for entrepreneurship. Starting self-knowledge of the player's own relevant capabilities may be high confidence where scenario design intends it.

Playtest provenance: GHOST-DES-090

6.7 Employees can carry personal relationships

A brokerage or other relationship-heavy employee may leave, join a rival or become independent. Business-held exclusive agreements remain with the business unless legitimately transferred, while personal relationships and future opportunities can follow the Person. Non-compete protection is not assumed without a real agreement and historical/legal support.

Playtest provenance: GHOST-DES-046

PART VII - MIGRATION, SETTLEMENT & RURAL CATCHMENT

7.1 Housing is feasibility, not sufficient cause

Vacant housing alone does not create migration. People move for durable work, wages, business/land opportunity, kinship or another plausible pull, then housing determines whether and how the move can occur. Settlement strategy should therefore target viable households and their economic reasons for moving rather than only a raw headcount threshold.

Playtest provenance: GHOST-DES-016, GHOST-DES-018, GHOST-DES-025

7.2 Business ownership can drive migration

A skilled outsider may relocate because they can acquire/control a viable business, not only because the player offers wages. Genuine business sales, partial ownership and employee/manager financing can therefore become migration causes when the household can feasibly relocate.

Playtest provenance: GHOST-DES-020

7.3 Recruitment and housing can occur at distance

A prospect can learn about a property through correspondence, receive descriptions and known terms, rent sight-unseen, request more information or travel to inspect before moving the household. Information travel and physical travel remain distinct.

Playtest provenance: GHOST-DES-032

7.4 A settlement does not require one dominant anchor

Durable settlement can emerge from several ordinary businesses and opportunities rather than one mine, mill or railroad. A useful causal loop is work/business opportunity -> feasible housing -> household migration -> relationship/word-of-mouth spread -> more workers/customers/businesses -> more opportunity. One business serving only its owner is not yet a settlement anchor.

Playtest provenance: GHOST-DES-017, GHOST-DES-037

7.5 Abandoned extraction is evidence, not a binary resource truth

A vanished placer or mining boom should lower confidence in an easy profitable restart but does not prove that no recoverable resource remains. Investigation must resolve actual geology/tailings, rights, process, cost and market conditions rather than reading AbandonedMine=false/true as a resource oracle.

Playtest provenance: GHOST-DES-026

PART VIII - PROPERTY AGENCY, RENTALS & COMMERCIAL TENANCY

8.1 Property agency is a first-class contractual business

A real-estate/land/property agent may represent privately owned property for sale or letting, collect rent, care for absentee property, coordinate taxes/records and arrange related transactions when authorized. An Agency Agreement grants authority and compensation. Exact commission rates remain market/scenario calibration unless specifically sourced and locked.

Playtest provenance: GHOST-DES-027, GHOST-DES-028

8.2 Listings are represented property, not owned inventory

Agency UI should distinguish Active Listings/Managed Properties from Leads. A represented property remains owned by its actual right-holder. The agency gains only the authority granted in the agreement; it does not acquire inventory ownership by listing the parcel.

Playtest provenance: GHOST-DES-030

8.3 Related-party brokerage conflicts require disclosure

When an agent represents a seller and the agent/player becomes the buyer or otherwise acts on both sides, the conflict must be exposed. A commission is not automatically earned on the agent's own acquisition; compensation requires the actual agreement and whatever disclosure/consent the historical/legal setting requires.

Playtest provenance: GHOST-DES-036

8.4 Rental holdings can be an operation inside a property business

An agency business may also own rental property without creating a new business entity. Each property retains separate rent, debt, condition and maintenance history. Lead-routing policies - such as offering suitable company vacancies before third-party listings - are configurable management policy, not a universal hidden preference. Any playtest commission percentage or employee split remains scenario/company calibration.

Playtest provenance: GHOST-DES-047, GHOST-DES-048

8.5 Commercial space leasing supports independent businesses

A landlord may lease a furnished room, station/chair or other functional commercial space to an independent operator who keeps customer revenue and pays rent. This is premises use, not disguised employment. A building can have occupied and vacant stations simultaneously. The same pattern can support barber, professional office or other historically plausible small-space uses.

Playtest provenance: GHOST-DES-074, GHOST-DES-075

8.6 Professional tenant recruitment is not hiring

Commercial tenant recruitment follows advertise space -> awareness -> inquiry -> inspection/information -> lease negotiation -> relocation/decline. The tenant remains an independent business. Exact claims about how common individual barber-chair rental was in 1870s Dakota remain a research hold; the general space-use agreement is retained without universalizing that specific custom.

Playtest provenance: GHOST-DES-077

8.7 Commercial tenancy has operating evidence

Landlord reporting should separate rent collection, tenant retention, tenant business health, customer traffic and space utilization. Full occupancy does not guarantee that tenants are economically healthy, and a strong tenant is not the same fact as a high landlord occupancy rate.

Playtest provenance: GHOST-DES-078

PART IX - DELEGATION, PORTFOLIO AUTONOMY & RIVAL ENTREPRENEURS

9.1 Delegated authority can include credit

Authority assignments may explicitly cover ordinary purchasing, hiring, contracts, business borrowing and, where the owner and lender permit, personal co-borrowing/guaranty. A manager title alone does not grant those powers. Authority to borrow for one business does not silently permit sale of the company or mortgage of unrelated personal/property assets.

Playtest provenance: GHOST-DES-044

9.2 Management optimization operates on evidence

A freight or other manager can monitor real operating evidence such as loaded/empty mileage, asset-days, worker load, refused work, delays, route/customer margin, maintenance, queues and stockouts. Better management changes actual scheduling, pricing, purchasing, staffing and investment decisions; it does not apply an Efficiency +X% bonus.

Playtest provenance: GHOST-DES-045

9.3 Portfolio autonomy and owner attention

The player may place businesses under standing delegated operation so routine purchasing, scheduling, maintenance, staffing and cash handling continue without owner clicks. Reports surface periodic state and material exceptions, with detail weighted toward current owner attention. Major ownership transfers, extraordinary debt, unrelated collateral and other owner-locked actions still escalate.

Playtest provenance: GHOST-DES-057, GHOST-DES-068

9.4 Standing portfolio policies are configurable

Owners may issue persistent policies such as reinvest all rental cash after obligations, maintain a minimum reserve or prioritize particular lead types. These policies constrain delegated choices but are not global rules. The Ghost Town 100% rental reinvestment mandate is therefore an example of player policy, not the default behavior of every rental business.

Playtest provenance: GHOST-DES-069

9.5 Venture Manager / Business Development Agent

A mature portfolio can employ a Person to investigate opportunities, inspect property, research markets, negotiate ordinary deals within authority, recruit specialists, arrange suppliers, prepare startup budgets, establish operations and hand them to a manager. The Person still requires time, information, capability, capital and granted authority. They cannot create enterprises from an abstract Opportunity score.

Playtest provenance: GHOST-DES-089

9.6 Strong rivals use the same rules

CANON LOCK - When competition is too weak, prefer stronger actual rival behavior over player nerfs or hidden NPC bonuses.

A strong rival can investigate more opportunities, preserve liquidity, negotiate finance, hire capable managers, acquire distressed companies, bid for strategic property, vertically integrate and react to evidence. Their edge comes from capability, capital, information, relationships and policy. Entrepreneur capital-structure preferences - debt-averse, conservative, liquidity-focused, growth-oriented or reckless - are real behavioral inputs rather than one generic aggression stat.

Playtest provenance: GHOST-DES-053, GHOST-DES-089, GHOST-DES-093

PART X - FREIGHT, RAIL STRATEGY & RELATED-PARTY COMMERCE

10.1 Farm-route freight is first-class commerce

A carrier can canvass farms, establish standing pickup relationships and combine compatible cargo into routes rather than waiting only for one-off jobs. Volume remains real, cargo remains physical and route economics must support the service.

Playtest provenance: GHOST-DES-040

10.2 Related-party exclusive service agreements

Player-controlled or partly owned businesses may enter exclusive or preferred service contracts when ordinary governance approves them. Minority owners must be protected from value-stripping terms, and the agreement can include performance windows or fallback rights. Any specific playtest discount is a negotiated scenario term, not a universal related-party rate.

Playtest provenance: GHOST-DES-042

10.3 Internal repair and shared physical capacity

Freight and wagon-repair operations can coexist in one enterprise, but shared wagons, workers, stables, yards and bays cannot be double-booked. Repairs to the enterprise's own wagons consume labor/materials and change asset condition without creating internal sales revenue. External repair customers create revenue.

Playtest provenance: GHOST-DES-071, GHOST-DES-072, GHOST-DES-073

10.4 Railroad promotion / civic lobbying

A settlement may organize a railroad promotion committee and pursue a route, station, siding or freight stop through real persuasion, rights, traffic evidence, land, finance and public authority. Success depends on railway economics, engineering and strategy. Public bonds/subsidies and depot/shop commitments execute through real civic finance rather than a Lobby Railroad roll.

Playtest provenance: GHOST-DES-082

10.5 Rail-resilient freight strategy

Wagon carriers can reason about territory as likely rail-displaced, rail-feeder or rail-insulated and deliberately develop non-rail collection/distribution routes to future depots. Rail arrival can destroy some long-haul corridors while increasing drayage, farm collection and secondary-node demand. It remains a strategic reconfiguration, not a generic penalty.

Playtest provenance: GHOST-DES-083

PART XI - FINANCE, ACQUISITIONS & ENTERPRISE VALUATION

11.1 Financing structure is negotiable

Property and business financing can negotiate price, cash down, maturity, amortization, payment timing, collateral scope, guaranty/co-borrowing and prepayment terms. Preserving startup working capital can rationally justify more total interest, but counterparties accept or reject terms based on actual risk, security, liquidity and alternatives.

Playtest provenance: GHOST-FIN-001

11.2 Debt assumption is distinct from purchase price

A buyer may assume secured debt, subject to the real lender/legal requirements, while a seller note or cash covers the remaining consideration. Ownership transfer, underlying lender claim, assumed liability and seller credit remain separate relationships. Debt continuity must survive sale, save/load and reporting.

Playtest provenance: GHOST-FIN-002, GHOST-DEF-010

11.3 Enterprise evaluation shows two values

CANON LOCK - Enterprise Evaluation must distinguish estimated gross going-concern/business sale value from estimated owner equity/sale proceeds after business-specific liabilities.

A heavily financed business and a debt-free business with the same operating assets are not economically identical to the owner. Objective checks that intend to measure what the player has actually built should use the transferable value attributable to the required ownership interest after directly associated liabilities.

Playtest provenance: GHOST-DES-043, GHOST-DES-081

11.4 Business cash is separate from business value

Cash held in the business account is economically owned but contributes zero to the Ghost Town business-value objective and should be displayed separately. Parking cash inside a weak business cannot satisfy a valuation milestone. Receivables/payables can be handled separately at closing according to the deal.

Playtest provenance: GHOST-DES-087

11.5 Maintainable operating profit is primary valuation evidence

Ordinary operating valuation starts with maintainable operating profit, then adjusts for seasonality, consistency, margins, trend, customer concentration, repeat contracts, supplier stability, competition, operating history, records, reputation, management continuity, owner dependence, capital replacement needs, legal/licensing risk and premises security. Do not blindly annualize the best short period.

Owner labor is normalized at replacement cost. If the owner works extensive unpaid hours, the transferable business profit is what remains after paying a competent replacement for the actual work. Employee count itself adds no value; capable staff can increase transferability while unnecessary payroll can reduce it.

Playtest provenance: GHOST-DES-088

11.6 Productive assets matter without double counting

Machinery, tools, vehicles, fixtures and owned real estate affect sale value, replacement-capital needs and asset floors. Do not simply add original asset purchase price on top of an earnings value that already assumes those productive assets are included. Asset-heavy enterprises can lean more strongly on asset/equity evidence; service/agency businesses may lean more strongly on earnings and transferable clientele. Treatment of inventory at business sale remains an open design point: durable inventory may transfer at agreed current value while perishables need realizable value, age, condition and spoilage risk.

Playtest provenance: GHOST-DES-092

11.7 Seller motivation and distress are evidence

Acquisition negotiation should model known evidence about seller age/plans, liquidity need, attachment, alternatives, creditor pressure and willingness to remain as employee/manager. Do not invent a retirement motive after the fact to justify a deal. Distressed businesses can be surfaced by owners, creditors, employees or brokers, but the player receives evidence about problems rather than an answer key; some businesses cannot be fixed economically.

Playtest provenance: GHOST-DES-094, GHOST-DES-095

11.8 Plain-language finance UI

Underlying structures such as seller note, mortgage, lien, guaranty and assumed debt remain distinct. Default player-facing presentation should additionally state in ordinary language who lent the money, amount owed, rate, next payment, maturity, collateral and any guaranty/assumption exposure.

Playtest provenance: GHOST-UX-001

PART XII - HISTORICAL LAW, CALIBRATION, RESEARCH HOLDS & OPEN QUESTIONS

12.1 Historical legal gates can precede operation

The Ghost Town research established a scenario-relevant Dakota liquor-law gate: the January 1873 regime requires a license and a $3,000 bond with at least two sufficient sureties before on-premises liquor sales. The bond is security/liability, not automatically a $3,000 cash deposit. Historical law should be represented through dated, sourced authoring records and actual compliance workflows rather than a generic SaloonUnlocked flag.

Playtest provenance: GHOST-LEGAL-001

12.2 Exact playtest prices remain calibration

Opening purchase prices, repair estimates, land values, poultry/livestock prices, exact tool/material prices, first-slaughter yields, local meat prices and horse/wagon prices used in Ghost Town remain scenario calibration unless later research explicitly promotes a particular value to a dated authoring anchor. They should persist consistently within the playthrough but not become universal Canon constants.

Playtest provenance: GHOST-CAL-001, GHOST-CAL-002, GHOST-CAL-003, GHOST-CAL-004, GHOST-CAL-005

12.3 Generated local quotes are not archival facts

Where historical supplier/commodity/freight/tax evidence constrains a fictional town but no exact local quote survives, the game may generate an authoritative scenario quote calibrated against that evidence. It must remain labeled generated scenario data rather than presented as a surviving historical quotation.

Playtest provenance: GHOST-RES-003

12.4 Exact 1872 rail construction milepost remains a research hold

The playtest accepted the historical Sioux City-Vermillion construction front but did not establish an exact physical milepost on the scenario date. Do not fabricate that precision. If the exact front becomes economically decisive, perform targeted research or use an explicitly uncertain authoring range.

Playtest provenance: GHOST-RES-001

12.5 Camera versus direct observation remains open

No final Canon lock is made on how much the strategic/sky camera itself counts as observation. The current design direction is that records, employees and obvious visible development can surface strategic facts while direct inspection yields stronger condition/evidence detail. This remains an explicit open question rather than an accidental omniscience rule.

Playtest provenance: GHOST-DES-097

12.6 Ghost Town scenario historical calibration

For the current Ghost Town playtest history, Morrow Creek is calibrated at roughly 85 permanent residents at its boom-era peak, roughly 120 persons physically present during the busiest transient/mining periods and 14 non-player permanent residents at the 1872 opening. The player household is separately real rather than omitted from domestic simulation. These values belong to this scenario unless separately promoted.

Playtest provenance: GHOST-DES-097, GHOST-TECH-001, GHOST-DEF-006

PART XIII - PLAYTEST CORRECTIONS, ACCEPTANCE DOCTRINE & TRACEABILITY

13.1 Corrections that become future acceptance rules

- Opportunity evaluation must include the rural catchment before concluding that a tiny town cannot support a business.
- Poultry ledgers must preserve male/female juvenile pathways and must not invent retrospective revenue when a missing classification is corrected.
- Egg and other sales require real counterparties or a legitimate remote market channel; anonymous infinite demand is invalid.
- The player household/residence and ordinary personal consumption must exist when scenario design says the player is a resident; population objectives must state whether the player counts.

- Wholly owned internal sourcing should use flexible requisitions over physical stock unless a real fixed contract exists; internal transfer prices cannot create consolidated profit.
- Property-management labor must be charged to the business/operation receiving the work rather than disappearing into another business payroll.
- Debt balances and obligations must carry forward exactly; correcting a missing debt entry must not invent repayments or rewrite transaction history.
- Every property opportunity must preserve physical identity, owner/right-holder provenance and the information channel that exposed it.

Playtest provenance: GHOST-DEF-001, GHOST-DEF-003, GHOST-DEF-004, GHOST-DEF-005, GHOST-DEF-006, GHOST-DEF-008, GHOST-DEF-009, GHOST-DEF-010, GHOST-DEF-011

13.2 Validated architecture carried forward

The playtest positively validated population represented by identifiable Persons and household/accommodation arrangements rather than an independent population counter, stable Person IDs with kinship separate from household membership, perishable lot provenance and premises requirements that depend on the actual business capability. These confirmations deepen rather than replace Revision XXV doctrine.

Playtest provenance: GHOST-TECH-001, GHOST-TECH-002, GHOST-TECH-003, GHOST-TECH-004

13.3 Complete Ghost Town finding disposition register

Every recorded Ghost Town finding is listed below. The table distinguishes general Canon integration from scenario-only calibration, research holds and technical implementation work. The register is traceability, not a second source of truth: the controlling rules are the sections above and the preserved baseline where not superseded.

Finding	Title	Disposition	Integrated at
GHOST-DES-002	Rural Production Must Generate Town-Facing Opportunities	Integrated design rule	Production, perishables and processing
GHOST-DES-003	Production Space and Retail Space Must Be Allowed to Separate	Integrated design rule	Production, perishables and processing
GHOST-DES-004	Multi-Site Microenterprise Works Well	Integrated design rule	Production, perishables and processing
GHOST-DES-005	Poultry Expansion Needs Biological Accounting	Integrated design rule	Production, perishables and processing
GHOST-DES-006	Tanning must be a process, not a crafting recipe.	Integrated design rule	Production, perishables and processing
GHOST-DES-007	Leather production and leatherworking are distinct capabilities.	Integrated design rule	Production, perishables and processing
GHOST-DES-008	A butcher startup has a meaningful minimum tool floor.	Integrated design rule	Production, perishables and processing
GHOST-DES-009	A business becomes operating through actual commerce, not setup.	Integrated design rule	Production, perishables and processing
GHOST-DES-010	First-animal economics expose carcass-balance risk immediately.	Integrated design rule	Production, perishables and processing
GHOST-DES-011	Slaughter should create a carcass-allocation decision.	Integrated design rule	Production, perishables and processing
GHOST-DES-012	Byproducts can stabilize a butcher's economics.	Integrated design rule	Production, perishables and processing
GHOST-DES-013	Casual specialist/helper relationships are useful.	Integrated design rule	Labor and working-capital continuity
GHOST-DES-014	Casual labor policy:	Integrated design rule	Labor and working-capital continuity
GHOST-DES-015	Startup businesses need bridge income.	Integrated design rule	Labor and working-capital continuity
GHOST-DES-016	Housing alone should not create migration.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-017	Our butcher can become part of an anchor, but it is not one yet.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-018	The ghost-town bottleneck is now clearly employment, not buildings.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-019	Anchor investigation can fail productively.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-020	Business sale can itself drive migration.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-021	Referral & Correspondence Recruitment.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-022	Poultry lineage management.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-023	Business-sale supply agreements.	Integrated design rule	Migration, settlement and opportunity evidence

Finding	Title	Disposition	Integrated at
GHOST-DES-024	Post-sale supplier relationships.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-025	Population strategy should target households, not raw headcount.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-026	Abandoned mining is evidence, not a binary state.	Integrated design rule	Migration, settlement and opportunity evidence
GHOST-DES-027	Property brokerage/relocation service.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-028	Property Agency Agreements.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-029	Create Business UI.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-030	Agency “inventory” is represented property, not owned inventory.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-031	Universal Create Business workflow.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-032	Recruitment + housing workflow validated.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-033	Two ways to expand from the Business UI.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-034	Referral chains become standing migration infrastructure.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-035	Partnership / partial-ownership formation.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-036	Related-party brokerage conflict.	Integrated design rule	Property agency, business formation and ownership
GHOST-DES-037	A settlement does not require one dominant “anchor.”	Integrated design rule	Settlement growth, recruitment and freight
GHOST-DES-038	Three-business objective now has a value floor.	Ghost Town scenario objective	Settlement growth, recruitment and freight
GHOST-DES-039	Passive migration awareness is separate from manual recruiting.	Integrated design rule	Settlement growth, recruitment and freight
GHOST-DES-040	Farm-route freight should be a first-class commercial pattern.	Integrated design rule	Settlement growth, recruitment and freight
GHOST-DES-041	Referral clarification.	Integrated design rule	Settlement growth, recruitment and freight
GHOST-DES-042	Related-party exclusive freight agreements.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-043	Business-value objectives should use transferable equity value, not gross financed assets.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-044	Delegated borrowing authority.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-045	Management optimization should operate on evidence.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-046	Brokerage employees can build portable relationships.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-047	Property Agency can add a Rental Holdings operation.	Integrated design rule	Related parties, delegation, rentals and objectives
GHOST-DES-048	Rental-first lead routing.	Mechanic integrated; playthrough values remain scenario data	Related parties, delegation, rentals and objectives
GHOST-DES-049	Three-business objective strengthened again.	Ghost Town scenario objective	Related parties, delegation, rentals and objectives
GHOST-DES-050	Saloon staffing should scale from observed workload, not a fixed employee recipe.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-051	Hiring is a recruitment workflow, not an applicant-generation button.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-052	Candidate capability display corrected.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-053	Strong-rival balancing doctrine.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-054	Premises utilization needs physical capacity + imperfect knowledge.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-055	Business reporting is evidence, not omniscience.	Integrated design rule	Hospitality, recruitment, reporting and procurement

Finding	Title	Disposition	Integrated at
GHOST-DES-056	Procurement must be supplier-first.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-057	Portfolio autonomy / attention filtering.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-058	Patron density preference.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-059	Service congestion replaces abstract crowd penalties.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-060	Retail pricing follows procurement, not the reverse.	Integrated design rule	Hospitality, recruitment, reporting and procurement
GHOST-DES-061	Internal food sourcing competes with existing contracts.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-062	Opening equipment purchasing delegated to Samuel.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-063	Prepared-food batches are physical inventory.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-064	Parcel expansion planning needs a reserved buildable envelope.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-065	Procurement needs full landed-cost truth.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-066	Supplier sampling/qualification.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-067	Opening stock can be delegated after constraints are set.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-068	Full portfolio autonomy.	Integrated design rule	Internal sourcing, inventory, premises and autonomy
GHOST-DES-069	100% rental reinvestment mandate.	Mechanic integrated; playthrough values remain scenario data	Internal sourcing, inventory, premises and autonomy
GHOST-DES-070	Miners' Rest becomes mixed-use.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-071	Freight + Wagon Works can be one enterprise.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-072	Shared assets do not duplicate capacity.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-073	Internal repairs are not fake revenue.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-074	Chair/room leasing model.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-075	Station leasing.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-076	Newspaper advertising needs a real media market.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-077	Professional tenant recruitment.	Rule integrated with explicit open/research hold	Mixed-use enterprise, leasing and media market
GHOST-DES-078	Commercial tenancy needs operating evidence.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-079	Advertising effectiveness is observable, not deterministic.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-080	Shared commercial-building costs belong to the landlord.	Integrated design rule	Mixed-use enterprise, leasing and media market
GHOST-DES-081	Enterprise value needs two numbers.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-082	Railroad promotion / civic lobbying.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-083	Rail-resilient freight strategy.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-084	Player Primary Residence versus nightly sleep.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-085	Homestead compliance must use actual residence, not a checkbox.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-086	Historical starting year is gameplay.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-087	Cash does not count toward Business Value.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-088	Operating profit is the primary valuation evidence.	Integrated design rule	Valuation, rail strategy and residence
GHOST-DES-089	Venture Manager / Business Development Agent.	Integrated design rule	Entrepreneurship, rivals, acquisition and information

Finding	Title	Disposition	Integrated at
GHOST-DES-090	Player may begin as an employee.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-091	Player and NPC capability advancement use the same system.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-092	Business sale value must include productive assets, but avoid double counting.	Rule integrated with explicit open/research hold	Entrepreneurship, rivals, acquisition and information
GHOST-DES-093	Entrepreneur debt policy.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-094	Distressed-business opportunities.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-095	Seller motivation belongs in acquisition negotiations.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-096	Property changes are information, not pop-ups.	Integrated design rule	Entrepreneurship, rivals, acquisition and information
GHOST-DES-097	Camera versus direct observation remains open.	Mechanic integrated; playthrough values remain scenario data	Entrepreneurship, rivals, acquisition and information
GHOST-CAN-001	Scenario Narration	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-002	Vacant Is Not Ownerless:	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-003	Property Action Clarity.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-004	Unauthorized Use Is Physically Possible.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-005	Legal Rule Is Not Instant Enforcement.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-006	Property Troubleshooting Is a Workflow.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-007	Good-Faith Claim Is Not Knowing Trespass.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-008	Player-Selected Starting Opportunity	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-009	Starting Investment Follows Investigation	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-010	Perishable Condition & Food Safety.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-011	Quality and safety must be separate.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-012	Preservation transforms inventory; it does not reset it.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-013	Preservation work must consume calendar time.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-014	Capability bands changed.	Canon lock / supersession	Relevant Part II-VI
GHOST-CAN-015	Capability bands revised.	Canon lock / supersession	Relevant Part II-VI
GHOST-DEF-001	Missing Rural Catchment	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-002	Three-business objective is much easier than its 1885 deadline suggests.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-003	Male poultry pipeline was underreported.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-004	Male poultry accounting omission.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-005	Egg sales lacked proper counterparties.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-006	Player household was missing from the simulation.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-008	South Garden egg arrangement corrected.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-009	Property management labor wasn't being fully charged.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-010	Debt continuity correction.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-DEF-011	Property source/provenance must be explicit.	Playtest correction + future acceptance rule	Part XIII - Corrections and acceptance
GHOST-CAL-001	CALIBRATION	Calibration only; not universal canon	Part XII - Calibration and holds
GHOST-CAL-002	Current poultry, land and livestock prices remain calibration values	Calibration only; not universal canon	Part XII - Calibration and holds
GHOST-CAL-003	All exact tool, livestock and material prices below remain scenario calibration.	Calibration only; not universal canon	Part XII - Calibration and holds
GHOST-CAL-004	First-slaughter yields and local meat prices remain calibration values.	Calibration only; not universal canon	Part XII - Calibration and holds

Finding	Title	Disposition	Integrated at
GHOST-CAL-005	Horse/wagon pricing remains calibration.	Calibration only; not universal canon	Part XII - Calibration and holds
GHOST-TECH-001	VALIDATED SYSTEM	Canonical implication acknowledged; implementation delegated	Technical IX / relevant canon section
GHOST-TECH-002	Perishable Lot State.	Canonical implication acknowledged; implementation delegated	Technical IX / relevant canon section
GHOST-TECH-003	Persistent Person IDs + family trees.	Canonical implication acknowledged; implementation delegated	Technical IX / relevant canon section
GHOST-TECH-004	Premises are capability-dependent.	Canonical implication acknowledged; implementation delegated	Technical IX / relevant canon section
GHOST-TECH-005	Technical terminology now needs synchronization.	Canonical implication acknowledged; implementation delegated	Technical IX / relevant canon section
GHOST-FIN-001	Startup Financing Should Support Structure Negotiation	Finance canon rule	Part XI - Finance and valuation
GHOST-FIN-002	Debt-assumption sales.	Finance canon rule	Part XI - Finance and valuation
GHOST-RES-001	RESEARCH HOLD	Research hold / provenance rule	Part XII - Calibration and holds
GHOST-RES-003	Exact 1873 local quotes remain calibration.	Research hold / provenance rule	Part XII - Calibration and holds
GHOST-LEGAL-001	Saloon opening has a major legal gate we had not yet surfaced.	Historical legal authoring anchor	Part XII - Legal gates
GHOST-UX-001	Plain-Language Finance.	Presentation rule	Part XI - Plain-language finance

END OF REVISION XXVI INTEGRATION

The complete Canonical Project Bible Revision XXV follows unchanged as the preserved baseline and remains authoritative except where Revision XXVI explicitly adds, deepens, corrects or supersedes it.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXV

1870 Frontier Transformation Integration: Historical Envelope, Railway & Transport Networks, Technology Diffusion, Environmental Regimes, Settlement Evolution & Research Provenance

Revision date: September 22, 2026

Source baseline: Canonical Project Bible Revision XXIV - September 21, 2026

Status: Active integration

PRESERVATION GUARANTEE

Revision XXV preserves Revision XXIV in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XXIV document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XXV controls. Exact prices, wages, freight tariffs, farm distributions, equipment ownership shares, technology adoption rates, weather coefficients, route timings, clerical error frequencies and other generated values remain calibration unless this integration explicitly identifies a design or historical lock.

PURPOSE

Revision XXV integrates the dedicated 1870-start historical research program and the architecture cleanup performed after Revision XXIV. The central change is that the normal Open Frontier historical envelope now begins in spring 1870 rather than spring 1880. This is not a cosmetic date shift. It establishes an earlier wagon-, river- and emerging-network frontier and lets the player experience thirty-five years of transport, information, finance, settlement and technology transformation through 1905.

The research also closes several architectural ambiguities: physical railway construction is distinct from service opening; railway infrastructure, legal corporation, owner, operator and controller are distinct relationships; technology progresses through availability, access and supportability rather than universal unlocks; non-rail transport remains economically meaningful after rail arrives; environmental and financial eras alter causal inputs rather than applying flat modifiers; and historically constrained regional geography can coexist with procedural local economies.

REVISION XXV MASTER DOCTRINE

* **The normal Open Frontier historical envelope is spring 1870 through 1905.** Scenarios may begin later or use shorter windows.
* **1870 is an early-network frontier, not a primitive world.** Hand, animal, water, wind and steam power are mature; rail, banking, utilities and rapid communications are sparse or uneven locally.
* **Historical geography is constrained where it matters.** Major rivers, treaty/reservation/federal geography, outside market nodes and dated regional infrastructure approaches form a historical skeleton; local settlements, businesses, Persons and many roads/projects can remain procedural.
* **Track laid is not service open.** Proposed, incorporated, surveyed, financed, graded, track-laid and operating railway states remain distinct.
* **Railway property, corporate identity, operation, lease/control and later system identity are distinct.** Corporate succession does not rewrite the history of a physical segment, contract or Person.
* **Rail complements and reconfigures wagon, stage, ferry and river transport.** It does not apply a generic Freight Obsolescence penalty.
* **Technology is not a global unlock tree.** Existence, commercial production, regional/local availability, acquisition/connection, supportability and operation are separate states.
* **Historical regimes are causal contexts.** Panic, drought, grasshoppers, favorable rainfall or boom conditions alter actual credit, projects, crops, migration and markets rather than multiplying all outcomes by a scripted percentage.
* **Business records have physical methods.** Handwriting, forms, typewriters, copying devices, registers and calculating equipment change real clerical tasks and evidence quality rather than granting abstract administration bonuses.
* **Historical evidence itself needs provenance.** A first-known date, one-town observation or unusual capital operation must not silently become a universal regional rule.

PART I - 1870-1905 HISTORICAL ENVELOPE & OPENING WORLD

1.1 Default Opening

CANON LOCK - Open Frontier normally begins in spring 1870 and retains 1905 as the authored Historical Horizon. The exact opening day may remain world/scenario data. Reaching 1905 still produces the Historical Record/retrospective; continued simulation may proceed under the existing continuation doctrine.

All preserved references that define spring 1880 as the ordinary Open Frontier start or define 1880-1905 as the full normal historical envelope are superseded by this section. Scenario-specific dates remain unchanged unless separately revised. References describing an 1880 condition remain historically usable as mid-campaign state rather than opening state.

1.2 1870 Opening Character

The opening regional economy should be sparse, physically dispersed and highly dependent on overland and river movement. Southeastern Dakota and Missouri-oriented settlement supply the strongest ordinary opening analog. The wider United States already contains railroads, mechanized farming, industrial steam, telegraphy, sophisticated printing and mature manufactured goods; the frontier constraint is local access, capital, transport, information and support, not ignorance of those technologies.

A small settlement around the existing 55-95 Person target remains plausible. The early economy can contain more productive activity than dedicated commercial buildings: artisans may work from a house, barn, yard or mixed-use premises when the actual tools, space and rights support the work.

1.3 Initial Connectivity

Ordinary 1870 local transport is built around walking, riding, horse/ox teams, wagons, ferries, stage/freight roads and river landings. Sioux City is a major outside rail interface. The Sioux City-Yankton telegraph connection arrives in 1870, but communication access remains geographic. A settlement being in a territory with telegraph service does not mean every business has a telegraph office or direct wire.

1.4 Indigenous Land, Public Land & Restricted Geography

The 1870 opening precedes the Black Hills gold rush and the 1877 federal taking. The Black Hills cannot appear as an ordinary purchasable mining district at start. Treaty, reservation, agency, military, federal/public-domain and ordinary private property remain distinct legal/geographic states. A visually empty parcel is not automatically claimable. Public-land entry requires the applicable survey/status and entry process; restricted land never enters the ordinary Make Offer workflow solely because it is visible.

1.5 Historical Transformation Phases

Period	Working historical character
1870-72	Sparse territorial frontier; wagon/river logistics; thin finance; first telegraph and railway promotion/construction.
1873-77	First local rail service plus Panic/depression; uneven settlement; drought/grasshopper risk; Black Hills transformation.
1878-87	Great Dakota Boom; rapid rail/town expansion; rising machinery access; early telephone, waterworks and electric-light systems in advanced nodes.
1887-95	Agricultural drought/debt pressure; mature regional rail networks; statehood; Panic of 1893; uneven town fortunes.
1896-1905	Recovery and deeper institutions; RFD and mature communications in selected areas; West River investment; first rare internal-combustion machines; temporary Pierre rail crossing in 1905.

These phases are contextual priors, not scripted outcome bands. A local town may boom or decline against the broader regime if its actual resources, routes, projects and actors justify it.

PART II - HISTORICALLY CONSTRAINED PROCEDURAL WORLD

2.1 Three-Layer World Generation

CANON LOCK - Open Frontier combines a historical regional skeleton with historically constrained procedural geography and ordinary emergent simulation.

Layer	Examples	Rule
Historical skeleton	Missouri River; major outside cities/market nodes; treaty/reservation/federal geography; major dated rail approaches; statehood and major national/regional events.	Must remain consistent with the historical envelope unless a scenario explicitly uses an alternate premise.
Constrained procedural geography	Fictional/local settlements; crossroads; secondary roads; farms; small ferries/landings; plausible station sites.	May vary by seed inside geographic, legal and historical constraints.
Emergent simulation	Persons; businesses; property control; exact projects; station competition; migration; firm births/deaths; local winners/losers.	Resolved by ordinary systems and may diverge from actual history.

2.2 Historical Baseline Does Not Predetermine Future Local Outcomes

Revision XXIV historical-divergence doctrine remains active. A historical railway intention, townsite or business opportunity enters the world through real actors, rights, finance and projects. If player/NPC action changes the prerequisites, later local history can diverge. Historical dates may introduce technology, outside actors, laws or external projects; they do not force an impossible local result.

2.3 Settlement Role Instead of Universal Town Level

Settlement physical form and business mix should depend on economic role and history. Useful archetypes include river/administrative center, rail-platted agricultural service town, mining supply center, rail junction, rural market town, industrial/company node and bypassed/declining settlement. These are authoring/generation descriptors, not progression tiers or bonuses.

2.4 Railway-Created Settlements

A railway may establish a depot/townsite away from an existing settlement. Land platting, depot siting, elevator/warehouse development and new businesses can create a competing node. Existing settlements may connect, shift toward the station, decline, lobby for service or retain a separate niche. Rail does not automatically attach itself to the nearest existing town.

PART III - RAILWAY DEVELOPMENT, SERVICE & CORPORATE IDENTITY

3.1 Railway Development Lifecycle

CANON LOCK - Proposed/Incorporated, Surveyed, Financed/Authorized, Right-of-Way Secured, Grading, Track Laid, Facilities Ready and Service Open are different states. Track that physically exists may remain commercially useless when no service operates.

The 1872-73 Winona & St. Peter extension provides the controlling example: track could physically extend well into Dakota while normal trains operated only to an earlier point for years. Scenarios, markets and objectives must query the state actually required.

3.2 Physical Segment & Corporate Relationships

A physical railway segment persists independently from the corporate relationships around it. Historical builder, infrastructure owner, legal railroad corporation, lessee/operator, controller/parent and later system identity may differ. Merger, lease or acquisition changes relationships; it does not rewrite who built a segment, which corporation signed an earlier contract or which Persons performed the work.

3.3 Working Regional Chronology

Date	Regional railway anchor
1870	No ordinary in-state common-carrier network in the normal South Dakota opening region; Sioux City functions as an external railhead/interface.
1872	Dakota Southern construction reaches Vermillion; additional eastern penetration projects appear.
1873	Dakota Southern reaches Yankton; national financial panic weakens railway finance.
1878	Rail construction resumes strongly; Sioux Falls receives service; new townsite/railroad settlement effects accelerate.
1879-80	Multiple Iowa/Minnesota penetration fronts expand; Dakota Central reaches Pierre in 1880.
1885	Common-carrier rail reaches Buffalo Gap on the southern Black Hills approach.
1886	Rail reaches Rapid City.
1887	Rapid City-Whitewood connection extends Black Hills access.
1890	Deadwood/Belle Fourche gain major common-carrier connections while mature East River networks already exist.
1905	Temporary railway bridge/crossing at Pierre allows the first official trains across the Missouri shortly before Christmas; permanent bridge lies beyond the 1905 horizon.

3.4 Railway Town Competition & Civic Participation

Railway route and station decisions may involve local inducements, right-of-way, depot land, bonds/subsidies, traffic expectations and political/legal authority. Towns can compete, refuse, overpay or lose. A civic financing decision must execute through real public authority/finance rather than a Town Wants Railroad flag.

3.5 Railway Technology Evolves Through Assets

Railroad modernization should arise through actual track, bridges, locomotives, rolling stock, brakes, couplers, yards, water/fuel facilities, dispatch/timekeeping and signaling. Older equipment may remain usable on lighter service while becoming inefficient, incompatible or legally restricted. The 1893/1903 federal safety-appliance regime supplies a later-campaign legal pressure toward power brakes and automatic couplers; it is not an instant whole-network conversion.

PART IV - ROAD, STAGE, FERRY, RIVER & MULTIMODAL TRANSPORT

4.1 Non-Rail Transport Is First-Class Infrastructure

Roads/trails, ferries, stage lines, freight routes, way stations, river landings and steamboat service are real transport infrastructure. A stage or freight service requires vehicles, animals, crews, feed, maintenance, route access, schedules and appropriate facilities. River service requires vessels, landings, navigation season and cargo/passenger handling.

4.2 Multimodal Logistics

One economic trip may combine wagon -> river landing -> steamboat -> railhead -> distant market, or farm wagon -> depot -> rail -> destination drayage. Each leg preserves time, custody, cost, cargo and risk. Transshipment is an actual handling/transfer event rather than a free route change.

4.3 Rail Reconfigures Rather Than Deletes Wagon Work

CANON LOCK - Rail arrival changes route economics causally. Long wagon hauls can contract while farm-to-depot collection, station drayage, local delivery, hotel/livery demand and feeder freight increase. Businesses survive, change, move or fail according to actual demand and capacity; there is no generic pre-rail-industry obsolescence modifier.

4.4 Route Conditions & Seasonality

Road surface, mud, snow, river ice/high/low water, ferry availability and weather can alter travel time/capacity or close a route. Exact historical speeds and rates remain calibration data. The architecture must allow route availability and capacity to change without inventing or deleting cargo.

PART V - TECHNOLOGY DIFFUSION, ACCESS & SUPPORTABILITY

5.1 No Universal Technology Unlock

CANON LOCK - Technology progression uses at least the following distinctions: Known/Technically Exists -> Commercially Produced -> Regionally Obtainable -> Locally Obtainable or Networked -> Acquired/Connected -> Supported/Maintainable -> Operational.

A patent or national commercial date proves possibility, not local adoption. A technology already used by a well-capitalized rail-connected operation does not imply ordinary small-town or rural availability.

5.2 Supportability

Equipment may be owned yet temporarily unusable because fuel, parts, consumables, compatible infrastructure, trained operators or repair capability are unavailable. Supportability is therefore distinct from ownership and condition. Frontier technology should fail or sit idle for explainable logistical reasons rather than through arbitrary Tech Level rules.

5.3 Confirmed Historical Anchors

Technology/system	Historical rule for authoring
Reapers/mowers	Already established before 1870. Adoption depends on capital, acreage, draft power, dealer/repair access and economics.
Hand/animal/water/wind/steam power	All valid productive power sources in 1870. Steam growth does not erase older methods.
Barbed wire	Commercially transformative 1874-era development; regional adoption should spread rather than appear universally at once.
Typewriter	Commercial 1874; improved upper/lowercase model 1878; office adoption uneven. Treat mainly as document-production equipment.
Telephone	Commercial exchange era begins 1878; Deadwood has first South Dakota exchange in 1878; requires actual local network/subscription.
Waterworks	Deadwood 1879; Yankton and Sioux Falls 1884 anchors; other settlements remain wells/cisterns/hauling until real projects exist.
Electric light/power	Deadwood electric lighting by 1883; adoption begins in advanced nodes and industrial sites, not as universal household infrastructure.
Cash register	Ritty-era 1878 invention/commercialization; stronger business adoption later, especially after NCR development.
Bicycles	Useful personal transport grows in the late 1880s/1890s; not a freight replacement.
Mechanical refrigeration	Later/capital-heavy industrial option; ordinary ice/icehouse systems remain important throughout much of the campaign.
Automobile/gasoline tractor	Rare/cutting-edge at the 1900-05 horizon; horses remain the normal transport/agricultural core.

5.4 Technology Changes Tasks, Capacity & Information

Technology should primarily alter feasible task methods, throughput, labor burden, reliability, information speed, quality/control or network reach. It should not grant free output. A machine still needs a Person/operator, location, inputs, time, authority and often maintenance.

PART VI - AGRICULTURAL MECHANIZATION & CONSTRUCTION MATERIAL ECONOMY

6.1 Farm Development Is Multidimensional

Farm scale must not collapse into one Farm Level. Land/right, improved acreage, buildings, fencing, draft power, productive livestock, equipment, Household labor, hired labor, working capital, debt and output remain separate. 1870 agricultural records show wide variation between property value and livestock value, supporting genuinely different economic structures at similar apparent scale.

6.2 Machinery Growth Is Economic, Not a Tech Tier

For the area that became South Dakota, farm implement/machinery investment rises sharply from about $1.55 million in 1880 to $8.37 million in 1890 and $12.22 million in 1900. The 1880s are therefore a major mechanization/diffusion decade, but advanced machinery can appear earlier in exceptionally capitalized and connected operations. The game should allow uneven adoption rather than a fixed regional unlock date.

6.3 Equipment Access Methods

Revision XXIII access doctrine is reaffirmed and expanded for the 1870 start: ownership, borrowing, rental, dealer/service access, hired machine-and-crew operations and contractor-supplied equipment are all legitimate where history and agreements support them. Traveling threshing and similar services are especially important because major machinery can be commercially useful without every farm owning it.

6.4 Construction Materials Follow Logistics

Early settlements can rely on local timber/logs, earth/sod, stone where suitable, rough-sawn lumber and expensive wagon-hauled manufactured material. Rail connection can sharply improve access to dimension lumber, shingles, doors, windows, hardware and standardized components. A sophisticated frame building therefore emerges from material supply, skilled labor, equipment, money and transport, not SettlementLevel >= N.

6.5 Building and Business Are Still Separate

The early frontier strengthens the existing doctrine that a Business is not synonymous with a dedicated storefront. Productive work can occur in a house, barn, yard, temporary shop or mixed-use premises when the required physical capability and legal/use rights exist.

PART VII - INFORMATION, MAIL, TELEGRAPH, TELEPHONE & MAIL-ORDER COMMERCE

7.1 One Information World, Multiple Delivery Channels

Letters/mail, messenger, newspaper/print, telegraph, telephone, in-person conversation and later reports all use the shared Information/Evidence doctrine. Delivery channel changes speed, cost, reach, documentary strength and access; it does not create separate knowledge truth.

7.2 Postal Access

Postal service is a real information network. A rural household may need to travel to collect/send mail, combining that errand with shopping, banking and social activity. Rural Free Delivery begins late in the period; South Dakota has an early listed RFD route at Ellis on May 1, 1899. RFD can reduce the Person-time cost of mail access without teleporting goods.

7.3 Mail-Order Procurement

Mail-order commerce is a legitimate procurement channel beginning during the campaign. Montgomery Ward begins mail-order business in 1872; Sears expands the form later. Information/order travel may occur by post, while heavier goods continue to require rail/express or other physical shipment because Parcel Post lies beyond the normal horizon. Catalog competition therefore expands choice but retains delivery time, freight, station collection and payment/credit constraints.

7.4 Standard Time Transition

Before November 18, 1883, local/company railway time standards can differ. The simulation keeps one authoritative absolute time; historical schedules/documents may carry a TimeStandard reference for conversion/presentation. Standard railway time begins November 18, 1883 and gradually simplifies schedule presentation. The player is not required to manually maintain multiple clocks.

PART VIII - BANKING, CREDIT & MACROECONOMIC REGIMES

8.1 Thin 1870 Institutional Finance

The 1870 frontier should not assume ubiquitous formal banks. Dakota's first bank begins as a private Yankton operation in 1869 and the First National Bank of Yankton is chartered in late 1872. Early financing can therefore rely more heavily on merchant/supplier credit, promissory notes, mortgages, seller finance, wealthy private lenders, private bankers and distant institutions. Institutional depth grows with settlement.

8.2 Panic of 1873

CANON LOCK - The Panic of 1873 is a major historical macro event whose effects propagate through actual lenders, railway finance, projects, businesses, labor and asset markets. It is not a global revenue penalty.

Railroad financing can fail, projects can pause, lenders can protect liquidity, borrowers can lose renewal/refinancing options, distressed assets can appear and financially strong actors can exploit opportunities. Each outcome must still be attributable to actual balance sheets, agreements and counterparties.

8.3 Panic of 1893 & Later-Campaign Financial Stress

The Panic of 1893 provides a second large national credit/railroad shock at a stage when the player may control a much larger organization. The same financial authorities apply at greater scale: liquidity, debt maturity, collateral, lender behavior, securities/capital needs and business cash remain real.

8.4 No Single Credit Availability Meter

A historical credit regime can influence lender priors and regional capital conditions, but individual lenders still decide using their own liquidity, knowledge, collateral, relationship, legal rights and risk. One lender tightening does not automatically remove every possible source of finance.

PART IX - ENVIRONMENTAL REGIMES & CAUSAL SHOCKS

9.1 Environment Is Economic State

Rainfall/soil moisture, temperature, snow, wind, grasshopper pressure, prairie-fire conditions, river conditions and comparable environmental variables may be represented at the resolution needed to create real agricultural/logistics consequences. Do not model historical shocks as flat revenue or yield multipliers.

9.2 Working Historical Context

The opening years sit near the end of an early Dakota settlement/agricultural boom. The 1873 financial crisis, weak farm prices, drought and grasshopper disasters contribute to hardship and abandonment in the mid-1870s. Broadly favorable rainfall from roughly the mid-1870s through mid-1880s supports the Great Dakota Boom. Drought and agricultural depression from the late 1880s into the 1890s increase debt stress, then overlap with the Panic of 1893.

9.3 Causal Chains

Examples: drought -> lower crop/hay yield -> feed shortage -> livestock condition/culling -> commodity prices/cash pressure -> debt/migration choices. Grasshoppers consume actual crop/forage state. Prairie fire uses fuel/weather/ignition/mitigation and can be reduced by real fireguards. Flood/river conditions alter routes/landings/ferries rather than directly changing abstract transport scores.

9.4 Historical Regime Is Context, Not Fate

HistoricalRegime-style authoring may supply regional/date priors for climate, migration, credit, rail investment and commodity context. It cannot override actual local simulation. Good water, diversified business, low leverage or strong logistics can make one actor resilient inside a bad regional period.

PART X - SETTLEMENT INFRASTRUCTURE, UTILITIES & FIRE PROTECTION

10.1 Infrastructure Is Project History

Telegraph, telephone, waterworks, electric light/power, roads, bridges, rail facilities and organized fire protection emerge through actual projects, networks, institutions and service relationships. Earliest-known local dates are authoring anchors, not universal unlocks.

10.2 Water & Fire

Deadwood provides an 1879 waterworks anchor; Yankton and Sioux Falls provide 1884 anchors. Earlier settlements may rely on wells, cisterns, hauled water or natural sources. Fire response can progress from household/ad hoc bucket response to volunteer organizations, hose carts/engines and later more developed apparatus as water supply, population and institutional capacity justify it.

10.3 Electricity

Deadwood electric lighting by 1883 confirms rapid but localized adoption. Electricity can first appear in mines, mills, central stations or dense commercial nodes while nearby rural households remain non-electric. Generating capacity, motive power, wiring/distribution, service agreements and maintenance remain physical constraints.

10.4 Longitudinal Settlement Evidence

Sanborn sequences and other dated maps should be used to author/validate settlement form over time. Sheet counts or map coverage are research-index signals rather than direct simulation formulas. The intended use is comparative extraction of business mix, building material, density, stables/outbuildings, industrial sites, utilities and rail interfaces.

PART XI - RECORDS, PENMANSHIP & OFFICE EQUIPMENT

11.1 Records Have Method & Quality

Important written records can distinguish content accuracy, arithmetic accuracy, legibility, copy fidelity and completeness/organization where those differences matter. A skilled clerk can produce excellent handwritten records; poor penmanship can create interpretation errors. Equipment changes specific work rather than applying one OfficeEfficiency statistic.

11.2 Typewriters & Copying

Typewriters become commercially possible from 1874 but diffuse unevenly. Their primary ordinary effects are faster/clearer document production, greater legibility and improved copy workflows when used with appropriate copying methods. A typewriter does not make the underlying number or bookkeeping classification correct. A typed wrong number remains wrong.

11.3 Other Office Equipment

Preprinted forms can improve completeness/standardization; cash registers can strengthen transaction capture and cash control; adding/calculating machines can reduce arithmetic workload; safes/files improve custody. Equipment should matter most when workload justifies it. A tiny owner-operated shop may reasonably use handwritten books for the entire campaign.

PART XII - HISTORICAL EVIDENCE, AUTHORING & CONFIDENCE

12.1 Historical Claims Need Provenance

Research observations used to configure the game should preserve source, place, date/range, relationship type where relevant, source type and confidence. An observation from one unusual business or town should remain an observation until broader evidence justifies generalization.

12.2 Preferred Research Objects

Research object	Purpose
Evidence Record	One sourced historical observation with exact date/geography and confidence.
Historical State Record	What was true for a place/network/entity at a particular time.
Game Authoring Record	Usable availability/range/prerequisite data derived from evidence.
Canon Decision	Only the durable rule that should govern simulation behavior.

12.3 Observation vs General Rule

Examples: an 1886 Yankton advertisement proving telephone use establishes local evidence, not universal Dakota telephone coverage. An exceptionally mechanized 1877 bonanza farm proves early feasibility under exceptional capital/connectivity, not typical ownership. A railway map labeled by a later system name does not prove that later corporation was the legal builder/operator in the earlier year.

12.4 Research Data Should Remain Replaceable

Historical prices, capacities, adoption curves, tariffs and rates should be data-driven/configurable where practical. Better evidence can replace a record without architecture rewrites or retroactively changing the meaning of fundamental authorities.

PART XIII - REPOSITORY/ARCHITECTURE CLEANUP CARRIED FORWARD

13.1 Repository Audit Still Precedes Implementation Claims

Revision XXIII/Technical VI architecture remains authoritative as target behavior, but implementation must still map current code before migration. The earlier audit identified several classes of duplicate/synthetic truth that remain priority checks if still present: Person/Household membership stored from both directions, employment split between Person and business worker slots, household demand directly posting retail sales, synthetic periodic household income, synthetic startup business cash, service resolution without real Person travel/time and aggregate/town-pulse sales that can create revenue without a legitimate customer/channel.

13.2 Economic Inflow Must Have Provenance

CANON LOCK - Ordinary household or business cash may not replenish solely because a week/day/startup threshold passed. Wage, owner contribution, loan, sale, remittance, outside employer payment, investment or other inflow must have a legitimate source/economic event. Off-map counterparties may remain coarse, but the source class/reason persists.

13.3 Relationship Authorities

Household membership, employment and boarding/residential service are different relationships. A boarder can receive accommodation/meals without becoming part of the host household's ordinary clothing/medicine/general-goods demand. Employment should eventually have one authoritative relationship from which Person workplace/wage and business roster views derive.

13.4 Customer Execution

Existing household-need logic may remain a planning source, but local embodied purchase execution must continue through Revision XXIII's ProcurementNeed -> acting Person/channel -> Activity/Journey -> supplier availability -> Transaction -> possession/use chain. Opening a store or increasing aggregate demand never creates a customer by itself.

PART XIV - RESEARCH & CALIBRATION HOLDS

Revision XXV closes architecture and historical-envelope questions while deliberately leaving the following as data/calibration rather than universal constants:

* exact rail freight/passenger tariff tables, timetables and route speeds by year/company;

* complete legal/corporate succession for every railway branch, locomotive roster, rail weight and bridge rating;

* exact farm-value percentile bands, machinery ownership percentages and agricultural generation probabilities;

* historical wages, equipment prices, freight costs and dealer inventories by place/date;
* weather/yield response coefficients, grasshopper frequency/severity and prairie-fire probabilities;
* technology adoption percentages and exact first local use for typewriters, cash registers and other office equipment;
* complete Sanborn extraction and settlement-building distributions;
* mail-order penetration, express tariffs and exact RFD effects on household trip behavior;
* all uncommon 1870-1905 legal edge cases and every mining/industrial machine configuration;
* exact macro-regime parameter strength; historical regimes never authorize direct universal outcome multipliers.

PART XV - REVISION XXV LOCK SUMMARY & SUPERSESSION MAP

15.1 Hard Locks

* Open Frontier normally begins in spring 1870 and runs through the 1905 Historical Horizon.
* Historical regional skeleton + constrained procedural geography + emergent local simulation is the world-generation model.
* Railway development states are distinct; track laid does not equal service open.
* Railway physical ownership, legal corporation, operator/lease/controller and historical system identity remain distinct.
* Road/stage/ferry/river transport are first-class networks and rail reconfigures rather than numerically obsoletes them.
* Technology progresses through availability, access, supportability and operation, not global unlocks.
* Environmental and macroeconomic eras alter causal world inputs rather than scripted revenues/yields.
* Agricultural mechanization remains uneven and task/equipment/capital/connectivity driven.
* Mail, telegraph, telephone and in-person information share one evidence world; mail-order still requires physical fulfillment.
* Settlement infrastructure/utility development remains project/network state, not Town Level.
* Record equipment changes real clerical tasks/evidence quality rather than abstract administration bonuses.
* Synthetic local sales and unprovenanced recurring/startup cash are rejected where they remain in implementation.

15.2 Explicit Supersessions

All prior general Canon wording that makes spring 1880 the ordinary Open Frontier start, defines 1880-1905 as the complete normal historical envelope, or describes 1880 equipment reference as the opening-year baseline is superseded to 1870-1905 by this integration. Historical statements specifically about conditions in 1880 remain valid as dated mid-campaign evidence. Scenario-specific dates are not automatically shifted.

15.3 Deepening Map

Revision XXIV historical divergence/1:1 geography is deepened with historically constrained world generation and a 1870 baseline. Revision XXIII equipment/capability doctrine is deepened with technology diffusion/supportability. Existing Project/Agreement/Transaction doctrines are reused for railway development, civic inducements, mail-order procurement, utility connection and transport services. Revision XXIV environmental/livestock causality is generalized into broader environmental regimes. Revision XXIII Information/Evidence is deepened with postal/telegraph/telephone channels and historical-source provenance.

END OF REVISION XXV INTEGRATION

The preserved Canonical Project Bible Revision XXIV follows unchanged and remains authoritative except where Revision XXV explicitly adds, deepens, clarifies or supersedes a rule.

RESEARCH BASIS - REVISION XXV

The integration above is based on the closing research program conducted after Revision XXIV. These sources support historical authoring and chronology; exact numeric tuning remains subject to the calibration holds above.

[R25-1] U.S. Census Bureau, 1870 Census instructions and agriculture/manufacturing reports - power categories, small establishments, Dakota agriculture/population context.

[R25-2] South Dakota State Historical Society, South Dakota Railroads: An Historic Context - 1872-1905 rail construction chronology, Great Dakota Boom/Bust and network maps.

[R25-3] Library of Congress, Rice Map of Dakota Territory (1872); 1882 Dakota railroad map; Dakota Railroad Commissioners map (1886); 1890 Census railroad atlas.

[R25-4] South Dakota State Archives, 1870 Dakota Territory Agricultural Census index - individual farm/property/livestock observations.

[R25-5] South Dakota State Historical Society marker/historic-context collections - Dakota Southern, Sioux Falls/Brandon railway arrival, first Yankton banking, stage routes and settlement histories.

[R25-6] National Archives, Homestead Act materials; National Park Service, 1868 Fort Laramie Treaty / Black Hills historical material.

[R25-7] Federal Reserve History, Banking Panics of the Gilded Age - Panic of 1873 and Panic of 1893 context.

[R25-8] Smithsonian National Museum of American History - reapers/agricultural machinery, barbed wire, typewriters, cash registers, bicycles and early mechanization.

[R25-9] South Dakota historical contexts for waterworks, electricity, telephone, agriculture, schools/buildings, mining and fire protection.

[R25-10] Smithsonian National Postal Museum / USPS historical material - Montgomery Ward/mail-order development, pre-Parcel-Post constraints and Rural Free Delivery.

[R25-11] Library of Congress, November 18 standard railway time historical note.

[R25-12] South Dakota Sanborn Fire Insurance Map collections / Library of Congress - longitudinal settlement morphology and infrastructure evidence.

LAND & LEDGERS

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXIV

ASHES & NOTES Rural-Systems Integration: Capability Calibration, Livestock Biology & Markets, Land Expansion, Delegated Authority, Calendar UX, World Scale & Historical Divergence

Revision date: September 21, 2026

Source baseline: Canonical Project Bible Revision XXIII - September 19, 2026

Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XXIV preserves Revision XXIII in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XXIII document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XXIV controls. Exact playtest prices, wages, farm values, cattle prices, land prices, mortality rates, skill-distribution percentages, yields, freight rates, carrying capacities, task durations and other generated values remain calibration unless this integration explicitly identifies a design, historical or scenario lock.

PURPOSE

Revision XXIV integrates the complete ASHES & NOTES Crisis & Recovery / Agriculture & Property playtest and the subsequent full playtest-integration review. The campaign progressed from a burned and indebted single farm through herd rebuilding, dairy-market development, rail access, delegated operation, stocker cattle, multiple farms, a small freight company, land assembly and more than 100,000 pounds of qualifying rail-shipped cattle. The principal result is not a fixed cattle progression or a new table of historical prices. It is a deeper causal connection among people, capability, animal biology, land, finance, markets, transport, information and delegation.

REVISION XXIV MASTER DOCTRINE

- **Average means average.** Capability bands must retain population meaning; ordinary competent workers must not drift into Great merely through routine tenure.
- **Confidence is presentation of knowledge, not capability.** High shows one band, Moderate two adjacent bands, Low three adjacent bands, and insufficient evidence remains Not enough information.
- **Delegation is the authority mechanic.** Manager, foreman, superintendent and general manager are optional descriptive titles rather than progression tiers or sources of power.
- **Livestock are persistent biological capital.** Breeding, growth, disease, mortality, production, culling and sale arise from actual animal state and environment.
- **Mortality is causal.** There is no universal calf-death roll. Reproductive and calf loss emerges from biological, environmental, management, disease and weather conditions, including correlated events.
- **There is no universal cattle market.** Breeding stock, stockers/feeders, finished cattle and culls can face different buyers, evidence, timing and transport economics.
- **Stocker cattle are a legitimate business model.** Profit comes from buying biological weight, adding value through feed/pasture/time and selling through real markets after all costs and risks.
- **Farm production is not automatically sold.** Fresh milk, butter, cheese, eggs and other outputs require real outlets, finite buyer capacity, processing, storage and logistics.
- **Land control is layered.** Owned land, leases, grazing agreements and nonexclusive/open-range use are distinct rights with different security.
- **Represented geography is 1:1.** Off-map resolution exists because the continental world cannot all be rendered at full detail, not because distance is compressed.
- **Historical chronology is a causal baseline, not predestination.** Legitimate player and NPC action may alter future commercial and infrastructure history.

- **Information should have provenance.** Where an actual worker, buyer, agent, lender or institution is the source, the player should normally learn through that source rather than an omniscient generic popup.
- **Scenario objectives measure accomplishments.** ASHES & NOTES validates the existing doctrine that ordinary systems create difficulty and objectives should not forbid legitimate strategy merely because it is unexpected.

PART I - CAPABILITY DISTRIBUTION, EXPERIENCE & CONFIDENCE

1.1 Capability Bands Retain Population Meaning

CANON LOCK - Average must represent ordinary capability within the relevant reference population. Great, Excellent and Mastered are not ordinary endpoints of employee tenure.

Player-facing bands remain Abysmal / Bad / Poor / Average / Great / Excellent / Mastered. Exact population percentages remain calibration, but the underlying distribution should be centrally concentrated around ordinary capability, with Abysmal and Mastered occupying relatively rare tails.

Candidate pools may legitimately differ. An experienced cattle-hand applicant pool should be stronger in cattle work than a random adult population. A young or inexperienced applicant should generally begin farther left in relevant occupational capabilities. Selection effects change who appears in the pool; they do not redefine what the bands mean.

1.2 Experience Does Not Equal Excellence

Capability growth requires relevant learning, challenge, feedback and actual participation. Calendar time or repetition of already-routine work must not automatically increase capability. An employee can work a trade for decades and remain Average. Another can become Great or Excellent through aptitude, broad difficult experience and consistently strong outcomes.

Learning should generally diminish as a task becomes routine and as a Person approaches their practical ceiling in that domain. Domain familiarity remains separate enough that expertise in one family of work does not create universal competence.

1.3 Mastered

Mastered represents exceptional career-level capability. It is not the final rung of a generic XP progression. Long relevant experience is normally necessary but never sufficient. As a calibration direction, Mastered will usually be associated with multiple decades of meaningful relevant work, often in the approximate 20+ year minimum and 25-35 year career territory discussed during the playtest, plus unusual breadth and repeated exceptional results. Exact thresholds remain calibration.

1.4 Abysmal

Abysmal represents exceptional inability in one capability. It is not the normal rating for inexperience and is not a global intelligence or moral label. Extreme lack of aptitude, poor learning, injury/disability where individually modeled, catastrophic judgment or other specific causes can contribute. A Person can be Abysmal in one capability and ordinary or strong elsewhere.

1.5 Confidence Presentation

CANON LOCK - High confidence displays one band; Moderate displays two adjacent bands; Low displays three adjacent bands; insufficient evidence displays Not enough information.

Confidence describes the width of the observer's plausible estimate, not the actual skill. Confidence can improve without capability improving, and capability can change while the player remains uncertain. Evidence may come from references, demonstrations, observed work, outcomes and reputation, with source quality and recency mattering.

1.6 Playtest Defect - Skill Inflation

ASHES & NOTES exposed a calibration defect in which too many employees drifted into Great across many unrelated capabilities. This is explicitly rejected. Future simulation and QA should treat population-wide upper-band inflation as a regression failure rather than desirable progression.

PART II - DELEGATION, AUTHORITY & OWNER ABSENCE

2.1 Delegation Is Primary

CANON LOCK - Delegated authority is the canonical mechanic. Manager, general manager, foreman, superintendent and similar titles are optional descriptions of real assignments; titles do not create authority or capability.

An authority arrangement should answer: who may act, for which business/site/task or decision family, within what financial or procedural limits, under what standing policy, and what must be escalated. A trusted farm hand can hold broad operating authority

without a formal managerial title. A titled manager can have narrow authority.

2.2 No Mandatory Management Ladder

There is no required Manager -> General Manager progression. A Person may oversee several sites when workload, geography, information, capability, communication and travel permit it. The title used by UI or narrative remains a presentation choice under active design.

2.3 Layered Delegation

A delegated Person may coordinate subordinate work when their authority permits it. Scheduling, purchasing, routine livestock decisions, staffing, freight dispatch and similar work can be passed downward without hidden owner clicks, while owner-reserved actions continue to escalate according to policy.

2.4 Cross-Business Authority Does Not Merge Businesses

One Person may hold authority across several controlled businesses. Each business retains separate cash, debt, property, inventory, employees, agreements, transactions and operating history unless a real merger or consolidation changes those facts.

2.5 Delegated Activity

The existing Manager Activity concept should be broadened in design toward Delegated Activity unless later UX testing establishes that the older label is clearer. The player-facing purpose is to answer: What did delegated people decide, perform or resolve so the owner did not have to? Events should identify actor, scope, authority source, trigger, decision/action, escalation where any and measurable consequence where defensible.

2.6 Owner Withdrawal Is a Valid Mature State

A mature agricultural or commercial portfolio may operate with zero routine owner labor when real delegated people, schedules, authority, records and controls exist. This is not passive income. The people continue to work and make decisions, obligations continue to mature and the owner retains only the authority that has not been delegated.

PART III - LIVESTOCK IDENTITY, INFORMATION & RECORDS

3.1 Persistent Animal Identity

Where individual identity matters, livestock retain stable IDs across birth, ownership changes, reproduction, sale and death. Individual identity is particularly important for breeding animals, valuable sires, animals under reproductive or health observation, purchased stockers with meaningful provenance, sale candidates and objective-tracked animals.

Economically useful state can include species/breed context, sex, birth date or estimate, birthplace, breeder/source, ownership history, parentage, reproductive history, health history, production history, acquisition basis, commercial class and final disposition.

3.2 Historical Records Survive Disposition

Sold or dead animals can collapse into lightweight historical records where pedigree, transaction history, genetic relationships or later audit still depends on them. Animal IDs are never reused.

3.3 Actual State vs Known State

Underlying animal truth and player knowledge are separate. Examples include actual weight vs estimated weight, actual pregnancy vs suspected pregnancy, actual pedigree vs claimed pedigree and actual illness vs signs/diagnosis. Unknown pedigree does not mean unrelated. Sellers, breeders, employees and written records may be incomplete, mistaken or deceptive.

3.4 Terminology

Male cattle presentation should use Breeding Bull, Breeding Bull Prospect, Bull Calf and Steer. Female presentation should use Type = Cow or Heifer, with reproductive state shown separately. A heifer remains a heifer until first calving; pregnancy and lactation are states rather than alternate biological types.

PART IV - GENETICS, BREEDING & HERD DEVELOPMENT

4.1 Genetics Are Inherited State

Breeding quality must not reduce to one generic bonus. Hidden inherited traits can influence supported characteristics such as growth, mature size, structure, fertility, calving characteristics, milk production, temperament and survivability. Exact genetic depth is an implementation/calibration question, but outcomes should emerge through inherited state and environment.

4.2 Pedigree Inbreeding

Pedigree relationship should affect risk through increased homozygosity and inherited trait expression rather than one direct Inbreeding Penalty. Genetic problems can become visible during gestation, birth, growth, breeding age or adult production depending on the underlying trait.

4.3 Premium Sire Economics

A superior sire can improve expected offspring because he contributes inherited material to every calf he sires, but purchase price alone does not establish superiority. Evaluation can use credible pedigree, breeder reputation, conformation, feet/legs, temperament, growth, fertility and progeny evidence.

Replacing the herd sire with an unrelated animal makes current matings easier but does not permanently reset inbreeding. Once daughters are retained, future sire selection must again account for relationship.

4.4 Sire Return Is Realized

Return on an expensive breeding animal emerges through real offspring outcomes, retained daughters, breeding-stock premiums, growth, fertility and later herd value. There is no hidden Premium Bull percentage bonus. Retained daughters can be a larger part of long-term sire value than extra pounds on steers.

4.5 Breeding-Stock Sales

A bull or heifer that is genetically redundant inside one controlled herd may still have substantial value to an unrelated outside buyer. Breeding-stock sale and meat-market sale remain different commercial decisions.

PART V - MALE CATTLE PATHWAYS & CASTRATION

5.1 Bull Calves Do Not Share One Decision Date

Not every male calf should remain intact until one universal evaluation age. Where pedigree, parental performance, structure, growth and other evidence provide little reason to preserve breeding optionality, early castration and entry into the steer pathway can be rational.

5.2 Preserve Genuine Breeding Optionality

Promising males may remain intact while evidence accumulates. The value of waiting is the possibility of breeding-stock value; the cost includes management difficulty, unwanted breeding risk, market-channel differences and the possibility that later castration is less convenient.

5.3 Bull vs Steer Markets

Intact males and steers can grow differently and can face different buyer preferences. A breeding bull commands value only where a buyer believes the reproductive evidence. A commercial intact male does not automatically receive a premium over a steer merely because it remained uncastrated.

PART VI - LIVESTOCK MORTALITY, CALVING & HEALTH

6.1 Causal Mortality

CANON LOCK - Do not implement one flat calf mortality percentage. Reproductive and calf loss must emerge from causal biological, environmental, management, disease and weather state.

Relevant stages include late pregnancy, calving, the first hours, neonatal life, pre-weaning growth and weaning. Relevant inputs can include dam nutrition/condition, age/parity, fetal size, presentation, inherited defects, labor duration, observation delay, assistance quality, temperature, wind and wetness, shelter, calf vigor, nursing/colostrum success, maternal behavior, bedding, sanitation, crowding, infection, respiratory disease, injuries, parasites where supported, predators where relevant and weaning stress.

6.2 Correlated Risk

Livestock deaths are not always independent events. Contaminated crowded maternity space can expose several calves. One cold wet storm can strike several births. Purchased cattle can raise pathogen pressure. Shared feed or water problems can affect a cohort. The simulation must allow clustered outcomes that share one underlying cause.

6.3 Shelter Capacity Is Not Health Capacity

Nominal cattle capacity does not fully describe safe maternity or disease-control capacity. Actual usefulness depends on animal size, group composition, bedding, drainage, ventilation/drafts, cleanliness, separation ability, traffic through the space and weather. A farm can have enough square footage and still have poor calf-health conditions.

6.4 Calibration Direction

The playtest research did not establish one defensible exact Dakota-Territory calf-mortality constant. A provisional historical tuning envelope around roughly 8-12 percent loss from late pregnancy/full-term birth through weaning was identified only as a calibration target for ordinary operations, with well-run operations capable of doing materially better and bad weather, disease or management producing materially worse outcomes. This is a calibration hold, not a Canon probability.

6.5 ASHES & NOTES Mortality Defect

The original campaign produced far too few calf deaths for the scale and period. That result is explicitly rejected as a simulation-quality baseline. The later corrected season demonstrated the intended kind of causal mix: late-pregnancy loss, stillbirth, exposure, enteric disease and respiratory loss, including clustered outcomes.

PART VII - LIVESTOCK COMMERCE & STOCKER/FEEDER CATTLE

7.1 Stocker/Feeder Operations

Buying young or unfinished cattle, adding economically valuable biological weight and reselling them is a first-class rural business strategy distinct from maintaining a breeding cow-calf herd.

Profit arises from purchase cost, starting weight and condition, biological gain, pasture, stored feed, water, labor, health losses, land/access costs, financing, transport, market conditions and sale timing. There is no Stocker Profit bonus.

7.2 Scaling Stocker Operations

A successful small lot can justify a larger later commitment, but cash alone does not create carrying capacity. Pasture, winter feed, water, shelter, labor, disease separation, transport, working capital and buyer depth remain binding. The player may knowingly overstock or overleverage and accept the consequences.

7.3 Livestock Working-Capital Credit

Purchased cattle may be financed through historically appropriate seller notes, livestock-secured credit, merchant arrangements or other supported structures. Financing cost reduces actual contribution. The financed animals remain assets and may be collateral.

7.4 Creditor Workouts

A maturing livestock note need not force an automatic sale if the creditor has reason to extend, restructure or refinance. Any extension remains a real agreement with cost and risk rather than a scripted rescue.

PART VIII - LIVESTOCK MARKETS, SALES & SEASONALITY

8.1 No Universal Sell-Cattle Market

CANON LOCK - Breeding stock, stocker/feeder animals, finished commercial cattle and cull cattle may face different buyers, evidence, costs and prices. The game must not collapse them into one universal cattle price or Sell button.

Valid channels can include private farm sale, local dealer/aggregator, feeder/stocker buyer, slaughter/cull buyer, railhead buyer, distant commission market, breeding-stock private sale, herd dispersal/breeder sale and historically appropriate fair or exposition channels.

8.2 Dated Market Opportunities

Some livestock opportunities can exist as dated events with location, entry deadline, sale date, eligible classes, documentation/evidence, fees, buyer pool, transport requirement and settlement terms. Special sales gather different buyers; they do not guarantee better prices.

8.3 Commission Merchants

A producer may use a commission merchant or agent to reach a distant livestock or produce market. The relationship can include market information, consignment, commission, shipment coordination, inspection, handling/yard expenses and settlement. The agent improves access; the agent does not guarantee profit.

8.4 Seasonal Markets Are Causal

Do not apply month-based price modifiers simply because a season historically tended to be stronger or weaker. Seasonal outcomes should arise through actual supply, feed pressure, pasture conditions, animal finishing, buyer demand, credit and transport availability.

8.5 Holding vs Selling

A sale-timing policy should compare expected marginal value from further gain against feed, land, labor, health risk, market risk, capital tie-up and opportunity cost. The player or manager never knows the future price top with certainty.

8.6 Rail Qualification

Rail-shipment objectives qualify only from actual qualifying physical shipment and recorded liveweight/origin evidence. Booking a shipment, owning a railway connection or selling to a rail buyer is not sufficient if the cattle have not physically moved according to the authored predicate.

PART IX - DAIRY, EGGS & FARM PRODUCT PROCESSING

9.1 Production Is Not Sale

CANON LOCK - Milk produced is not automatically milk sold, and eggs laid are not automatically eggs sold. Farm products require actual customers, agreements, internal use, storage, processing or loss/spoilage.

9.2 Finite Buyer Capacity

A recurring buyer can have practical quantity limits that vary with season, their own customers, working capital, spoilage risk, storage and resale performance. There is no invisible infinite buyer behind a standing account.

9.3 Dairy Allocation

Milk can flow among calf nursing, household/employee consumption, fresh sale, butter, cheese, other supported processing, animal-feed/byproduct use and spoilage. Processing allocation should use actual available milk rather than one abstract Dairy Output value.

9.4 Fresh Milk

Fresh milk has short practical market reach unless real transport speed, handling/cooling and receiving arrangements extend it. Nearby high-net-price customers can rationally receive priority before milk is converted into more durable products. A wagon route can widen the fresh-milk market when actual time and conditions permit.

9.5 Butter

Butter converts highly perishable milk into a denser and more durable product but requires labor, equipment, packing, storage and quality control. Rail or distant buyer access creates market reach, not free processing capacity.

9.6 Cheese

Cheese is a legitimate processing path where history, capability and demand support it. Cheesemaking may require specialized knowledge, workspace, vats or vessels, rennet/ingredients, curd handling, pressing, curing/storage and delayed sale. It must not be a one-click Milk -> More Valuable Product conversion. Larger cheese/creamery operations may aggregate milk from surrounding farms where historically and economically justified.

9.7 Dairy Byproducts

Whey, buttermilk, skim milk and other residues should be represented only where they create meaningful feed, disposal or customer decisions. Their existence can support integrated livestock choices such as feeding pigs, but should not force unnecessary micromanagement.

9.8 Poultry

Egg output remains biological and seasonal. Eggs set for incubation are unavailable for sale. Chicks do not immediately become adult laying inventory. Poultry breeding, household use, mortality and surplus males remain physical outcomes rather than a fixed Eggs Per Day business stat.

PART X - LAND, RANGE & AGRICULTURAL EXPANSION

10.1 Rural Control Types

Agricultural scale can expand through fee ownership, purchase contracts, tenancy/lease, seasonal grazing agreements, other lawful use rights and nonexclusive/open-range use where historically applicable. These relationships must not be collapsed into one Owned Acres number.

10.2 Parcel-Level Acquisition

The player should be able to inspect nearby parcels and pursue control through the appropriate owner or legal process. A private owner can be approached for sale or lease. Corporate or railroad land may require an agent. Public-domain land uses the applicable entry process rather than a private Make Offer action. Restricted land does not become purchasable merely because it is visible.

10.3 Open Range Is Insecure

CANON LOCK - Nonexclusive open-range use is not ownership, not a lease and not permanent carrying capacity.

A ranch can benefit from land it does not securely control while accepting that settlement, legal claims, fencing, competing grazing or policy change can reduce or eliminate that access. Losing such use is not confiscation of a player asset if no secure right existed.

10.4 Land Assembly vs Whole-Farm Acquisition

A growing operation may acquire another farm or assemble additional land around an existing operation. A whole farm may add buildings, wells, fences, labor and livestock but creates another operating site. Adjacent land strengthens one site but may require new fencing, water and shelter. Neither route is universally superior.

10.5 Adjacency Has Operational Value

Nearby land and farms can reduce travel, herding, fencing, dispatch and supervision burden. Adjacency therefore affects real economic value rather than merely map aesthetics.

10.6 Land Market Agency

A neighboring owner can refuse to sell, ask more, offer a lease or have legal/title constraints. Land does not become available simply because the player clicks it.

10.7 Calibration Hold - Farm and Acre Prices

ASHES & NOTES exposed likely underpricing in several farm-acquisition values. Those playtest prices are calibration placeholders and must not be treated as historical anchors. Rural valuation and financing terms require dated regional calibration.

PART XI - FREIGHT, HORSES & PORTFOLIO LOGISTICS

11.1 Micro-Freight Company

A legitimate local freight business can begin with one team, one wagon and one teamster and may have no yard or warehouse. Capability derives from actual animals, vehicle, driver, stable/feed access, cargo handling and route rights. Lack of a yard means lack of magical storage.

11.2 Portfolio Freight

An owned freight company may haul for controlled farms and take outside work when capacity is available. Internal service still consumes animals, labor, feed, wear and time. Separate businesses may record real intercompany charges, while consolidated reporting must not treat internal transfers as external profit.

11.3 Rail Complements Wagon Freight

Rail can create local collection, drayage and depot-interface demand rather than eliminating wagons. Freight can also extend fresh-product selling radius and reduce dependence on ad hoc trips by farm labor.

11.4 Expansion Follows Persistent Utilization

One seasonal spike does not automatically justify another freight team. Repeated profitable refusals during ordinary weeks are stronger evidence that permanent capacity is constrained.

11.5 Horse Capital

Horse ownership should solve real work or travel constraints. A dedicated riding/management horse can be valuable because it leaves a work team available. Hiring or borrowing draft power can outperform ownership for seasonal peaks. Permanent animals carry year-round feed and care cost.

PART XII - CALENDAR, HUD & CHARACTER-SOURCED INFORMATION

12.1 Universal Calendar

Land & Ledgers should provide one player-facing calendar with year/month overview and drill-down to month, week and day. The interface may use familiar modern calendar conventions for usability even though the world itself remains historical.

Known events can include payroll, debt payments, supplier credit, rents/leases, agreement deadlines, planned shipment departures/arrivals, projects, expected calving windows, weaning/evaluation dates, market events and player-created notes.

12.2 Delegated Events Remain Visible

Delegating work does not remove important dates from the owner's calendar. A delegated calving or shipment can remain visible with the responsible Person/business identified.

12.3 Exact Dates vs Forecast Windows

CANON LOCK - The calendar must preserve uncertainty. Known due dates display as dates; biological, shipment or market forecasts may display as windows/ranges.

12.4 Calendar Filters

Players may hide categories without changing simulation. Candidate families include payroll, finance/debt, property/rent, contracts, freight, livestock, markets, projects and personal notes. Exact defaults remain UX calibration.

12.5 Main Financial HUD

The universal top-level financial presentation should prioritize Date, Owner Cash, current Business Cash and Debt/Liabilities. Scenario-specific state such as cattle, hay, shelter or stocker lots belongs in contextual panels rather than the universal strip.

12.6 Information Should Have a Source

Where a legitimate Person, business or institution is the source of commercially useful information, player-facing presentation should preserve that provenance. A buyer can say they cannot take more butter. A farm hand can report bedding problems. A freight agent can report a delayed car. A lender can warn of an obligation. Generic system notices remain appropriate for UI-owned information such as objective completion and calendar organization.

12.7 Guidance Without Solving the Game

Because period business practice is unfamiliar to modern players, NPCs and contextual actions may explain plausible mechanisms such as contacting a station agent, hiring a commission merchant or processing surplus milk. Guidance should expose options and mechanics rather than identify the mathematically optimal strategy.

PART XIII - WORLD SCALE, HISTORICAL BASELINE & SCENARIO BASIS

13.1 One-to-One Spatial Scale

CANON LOCK - Represented world distance uses 1:1 physical scale. Off-map/coarse simulation exists because a continental world cannot all be rendered at full detail, not because local geography is compressed.

When a represented road is one mile in the simulation, it should correspond to approximately one real mile of economic/travel distance. Off-map transitions preserve travel time, route, cargo, identity and cost.

13.2 Historical Baseline Is Causal, Not Predestined

Historical chronology establishes starting geography, institutions, outside actors, technologies, known projects and plausible intentions. It does not reserve future outcomes against legitimate player or NPC action.

If the player buys land another actor historically acquired, establishes a railway, changes a route, founds a competing business or otherwise changes the prerequisites that produced historical development, later commercial and infrastructure history may diverge. The game must not reject a valid action merely because history later used that land or opportunity differently.

13.3 Historical Actors Persist in the Changed World

Historical actors continue pursuing plausible goals according to their resources, knowledge, rights and incentives after divergence. Historical dates may introduce actors, technologies, political conditions or outside projects, but must not force a local result whose prerequisites no longer exist.

13.4 Historical, Fictional & Emergent Settlements

Real towns, fictional scenario settlements and settlements that emerge from simulation may coexist. Historical settlements preserve their historical starting identity and geography where relevant. Fictional settlements allow authored scenarios within the same historical economy. Sufficient player or NPC development may create a durable new settlement that can receive an appropriate world-generated or player-selected name.

13.5 Scenario Basis Labels

Scenario selection should distinguish historically based and fictional authored scenarios. Current preferred badges are HISTORICAL and FICTIONAL. Historical means the opening premise/geography/actors are materially based on documented history; it does not mean outcomes are scripted. Fictional means the authored premise or local setting is invented but still follows the historical world's technology, law and economy. Historical scenarios may expose a short Historical Basis section.

PART XIV - ASHES & NOTES CANONICAL SCENARIO UPDATE

14.1 Scenario Identity

ASHES & NOTES remains a Crisis & Recovery / Agriculture & Property scenario. The player begins with an operating farm and meaningful secured/seller debt after a major physical fire destroys much of the productive/residential base. Recovery uses ordinary systems: rebuild, sell, haul, contract, refinance, negotiate, lease, restructure, partner, change commercial direction or expand into new businesses.

14.2 Objective Philosophy

Objectives continue to measure accomplishments rather than prescribe strategy. The player is not required to employ a specific number of workers, retain every asset, use stocker cattle, acquire multiple farms or become debt-free unless such a restriction is explicitly authored as an objective.

14.3 Tested Objective Structure

Deadline	Objective	Playtest status
Dec. 31, 1870	Earn $10 livestock revenue	Completed in test
Dec. 31, 1872	Earn $100 cumulative livestock revenue	Completed in test
Dec. 31, 1874	Own 20 cattle	Completed in test
Dec. 31, 1877	Earn $500 cumulative livestock revenue	Completed in test
Dec. 31, 1879	Own 30 cattle	Completed in test
Dec. 31, 1883	Rail-ship/sell 15,000 lb of North Fork-raised cattle to an off-map market	Completed June 14, 1875 in test
Dec. 31, 1886	Earn $2,000 livestock revenue in one calendar year across player-controlled farms	Completed Dec. 22, 1875 in test
Dec. 31, 1890	Rail-ship/sell 100,000 lb of cattle from controlled farms to off-map markets	Completed May 20, 1878 in test

14.4 Removed / Rejected Objectives

The former requirement to sell ten cattle in one sale is removed. A proposed employee-count objective is rejected because staffing should emerge from workload. A debt-payoff-without-missed-payment objective is not currently part of the locked structure because ordinary debt service already exists as a causal business constraint.

14.5 Objective Presentation

Incomplete objectives should show exact numeric progress rather than generic In Progress text. Once an event objective is completed, the scenario UI should retain its completion date and evidence instead of replacing it with later current totals. Separate lifetime metrics may continue updating.

14.6 Calibration Holds

The playtest completed the 100,000-pound objective by May 1878, well before the 1890 deadline. This may indicate generous pacing, successful leveraged strategy, underpriced land/cattle or a combination. Deadline and economic rebalance require comparative runs before any further objective change. Exact opening farm price, later acquisitions and most generated economic values remain calibration.

PART XV - RESEARCH & CALIBRATION HOLDS

Revision XXIV locks system behavior while explicitly leaving the following as researched/calibrated data rather than universal constants:

- capability-band population shares and the final rarity of Abysmal/Mastered;
- exact experience envelope associated with Mastered;
- historical calf mortality by region, year, management type and stage;
- dystocia, stillbirth, scours, respiratory disease and treatment frequencies;
- regional cattle growth rates, prices, breeding premiums and stocker spreads;
- milk, butter, cheese and egg yields, prices, spoilage and buyer capacities;
- rural wages, farm prices, acre values, railroad-land terms and seller-finance customs;
- the private/public/railroad-land mix for specific regions and dates;
- open-range legal/customary access by place and year;
- forage carrying capacity, hay yields and winter-feed requirements;
- shelter costs, capacities and maternity-space performance;
- horse/wagon purchase and operating costs and local freight rates;
- ASHES & NOTES objective pacing and the economic effects of aggressive leverage.

PART XVI - REVISION XXIV LOCK SUMMARY & DEEPENING MAP

16.1 Revision XXIV Hard Locks

- Average retains ordinary population meaning; capability bands cannot inflate until Great becomes the default.
- High/Moderate/Low confidence must render one/two/three adjacent bands respectively.
- Delegation is primary; management titles are descriptive and do not create capability or authority.
- Livestock identity, pedigree, biological state and player knowledge remain distinct authoritative/observed facts.
- Mortality emerges causally and may be correlated across animals.
- Stocker/feeder cattle are a legitimate livestock-commerce lane.
- Livestock markets are differentiated by commercial use, buyer/channel and physical delivery.
- Farm production is not automatically sold; repeat buyers have finite capacity.
- Owned, leased, contractual grazing and open-range use are distinct land-control relationships.
- Represented geography uses 1:1 physical scale; off-map is resolution, not compressed geography.
- Historical chronology is a causal baseline and may diverge through legitimate world action.
- A universal calendar projects known and forecast events without becoming a second authority.
- Important commercial information should preserve a legitimate in-world source where one exists.
- ASHES & NOTES uses the revised 15,000-pound and 100,000-pound rail objectives; the ten-head single-sale and employee-count objectives remain removed.

16.2 Deepening Map

- Revision XXIII continuous-Person and capability doctrine is deepened by population-calibrated capability meaning and anti-inflation rules.
- Revision XXI/XXII delegation and Full Autonomy are deepened by a delegation-first authority model and optional management titles.
- Existing agriculture biological-capital doctrine is deepened by persistent animal identity, genetics, male pathways and causal mortality.
- Existing livestock-commerce doctrine is deepened by explicit stocker/feeder strategy and differentiated livestock markets.
- Existing physical logistics and freight doctrine is deepened by micro-carriers, portfolio hauling and rail/wagon complementarity.
- Existing property/provenance doctrine is deepened by parcel-level agricultural acquisition, railroad land, grazing rights and insecure open-range use.
- Existing historical-skeleton doctrine is clarified so player action can causally alter future commercial/infrastructure history.
- Existing large-world/on-map/off-map doctrine is deepened by an explicit 1:1 distance requirement.
- Existing UI/read-model doctrine is deepened by a universal calendar, exact financial HUD and character-sourced information.
- ASHES & NOTES is deepened from a general crisis-recovery concept into the tested agriculture/rail objective structure above.

END OF REVISION XXIV INTEGRATION

The preserved Canonical Project Bible Revision XXIII follows unchanged and remains authoritative except where Revision XXIV explicitly adds, deepens, clarifies or supersedes a rule.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXIII

Commercial & Human Continuity Integration: Continuous Personhood, Physical Capability & Equipment, Agreement/Transaction/Project Depth, Institutional Authority & Long-Run Continuity

Revision date: September 19, 2026

Source baseline: Canonical Project Bible Revision XXII - September 18, 2026

Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XXIII preserves Revision XXII in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XXII document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XXIII controls. Exact wages, prices, equipment prices, equipment counts, task durations, tool wear rates, household-consumption quantities, insurance premiums, legal time limits, utility tariffs, project allowances and other generated values remain calibration unless this integration explicitly identifies a design or historical lock.

PURPOSE

Revision XXIII integrates the post-Contractor depth review, including the dedicated continuous-Person pass, the occupation/equipment research pass and the subsequent shared-authority cleanup. It does not add a new progression ladder. It deepens the connective tissue of the existing economy so that people, work, tools, contracts, transactions, organizations and institutions remain physically and economically continuous.

The principal result is a stricter doctrine: local economic activity must be attributable to real Persons, households, businesses and institutions; work capability must arise from real people, tools, equipment, space, materials and authority; and broad systems such as Agreements, Projects, governance, succession, internal controls, law, insurance, distress and utilities must use reusable shared lifecycles instead of shallow bespoke flags.

REVISION XXIII MASTER DOCTRINE

- Every local Person is a persistent human actor through time. Work, unemployment, shopping, meals, drinking, sleep, family life, travel, illness, migration and business interaction are states and activities of that same Person.
- No anonymous local customer. Ordinary local sales and services require a real economic principal and, when embodied action is required, a specific acting Person. Aggregate demand may summarize history; it may not fabricate transactions.
- Time is exclusive across life, not only across paid work. A Person cannot work, shop, travel, sleep or socialize in physically incompatible places at the same time.
- Households remain ordinary domestic-economic consumption units, but individuals perform physical actions. The payer, acting Person and beneficiary may be different legitimate entities.
- Business capability emerges from actual people, tools/equipment, suitable space, materials/inputs, legal or organizational authority, information and available time. A business label does not grant a capability by itself.
- Equipment requirements attach primarily to tasks/capabilities rather than rigid occupations. Ownership is not universally required; equipment can be owned, leased, borrowed, rented, contractor-supplied or obtained through hired service where historically plausible.
- Customer traffic, market demand, Opportunity & Pressure and business-attractiveness summaries are derived observations over real needs, awareness, choices and transactions. They are not independent sale generators.
- Agreement existence, obligation performance, physical fulfillment, acceptance, economic recognition, cash settlement and bookkeeping are different states. Partial performance remains partial rather than collapsing to success/failure.
- Projects preserve baseline estimate, current forecast and actual outcome as distinct histories. Changes, delays and cost growth should not overwrite the original promise.

- Governance, succession, internal controls, insurance, dispute resolution, distress, civic projects, utilities and labor markets use real counterparties, rights, resources, information and procedures rather than generic scores.
- Simulation resolution can reduce detail but cannot change ontology. Coarse/off-map activity must reconcile to the same Persons, assets, cash, inventory, obligations, relationships and histories that exist on-map.

PART I - SHARED-AUTHORITY CLEANUP & ONE-TRUTH DOCTRINE

1.1 One Authority Per Consequential Fact

When several systems need the same consequential truth, they should query or derive it from one authority rather than store competing copies. Read models may summarize or forecast authoritative state but cannot become a second causal source.

1.2 Economic Event, Bookkeeping & Lifetime Ledger

Economic events own what actually happened: goods moved, a service was accepted, cash changed hands, a receivable arose, an asset was acquired or a loss was recognized. Bookkeeping records those events. The Lifetime Ledger aggregates qualifying history. Neither bookkeeping nor scenario summaries may manufacture an economic event that did not occur.

1.3 Agreement & Obligation Authority

Supply agreements, construction contracts, freight contracts, employment terms, leases, concessions, finance agreements, utility service arrangements and similar commitments share a common conceptual grammar: parties, obligations, conditions, performance windows, pricing/consideration, allocation of cost/risk, amendment, breach/cure, termination and evidence of performance. Domain-specific terms remain specialized data rather than separate contract truth.

1.4 Person Time & Activity Authority

Employment scheduling, management work, travel, shopping, sleep, domestic work, school, medical care and leisure must consume one authoritative Person time budget. Domain systems may request or create activities, but none may independently decide that a Person is free when another authoritative activity already occupies incompatible time.

1.5 Information & Evidence Authority

Awareness, market research, references, competitor intelligence, lender diligence and reputation evidence should use one information/evidence grammar where practical: subject, source, observer, time learned, evidence type, confidence, staleness and access. A rumor, signed agreement, ledger entry, witnessed event and manager estimate are not equally strong evidence.

1.6 Shared Resource Reservation

Workers, wagons, horses, machines, workstations, inventory lots, project stockpiles, freight capacity and committed funds all require defensible reservation/commitment semantics. A resource cannot be promised twice merely because two domain systems use different labels.

1.7 Rights & Relationship Query Layer

Ownership, possession, occupancy, leasehold, employment, management authority, corporate office, security interest, easement and service rights remain legally/economically distinct. Systems should use a common way to ask what right or authority a Person/entity has over a thing without collapsing those distinct relationships into one Owner field.

1.8 Derived Demand, Traffic & Opportunity

MarketDemand, CustomerTraffic, LaborPressure, HousingPressure and Opportunity/Pressure may exist as read models or synthesized evidence, but they are not independent causal authorities. A traffic chart summarizes Persons who actually visited. Lost-sales evidence comes from failed or abandoned real purchase attempts. Opportunity analysis interprets existing conditions rather than spawning the conditions.

PART II - CONTINUOUS PERSONHOOD & DAILY LIFE

2.1 Continuous Personhood

Every Person who participates in the simulated regional economy retains persistent identity and history across employment, household membership, housing, travel, consumption, relationships, health and aging. Employment loss, retirement, injury or temporary travel changes state; it does not suspend personhood.

CANON LOCK - No temporary anonymous worker, shopper, drinker, hotel guest, resident or local customer may be created solely to satisfy a demand, traffic or labor calculation when the activity is meant to occur inside the represented regional economy.

2.2 Economic Principal, Acting Person & Beneficiary

A household, business or institution may be the economic principal while a specific Person performs the physical action. A father can buy boots for a child; a manager can order lumber for a business; a boarder can receive a meal paid through a boarding arrangement. Transactions and activities should preserve these roles when materially relevant.

2.3 The Day Belongs to the Person

A Person has one continuous day. Paid work, management, travel, household duties, errands, job search, school, meals, social/leisure activity, medical care and sleep compete for time. The player need not see minute-by-minute life simulation, but the underlying schedule cannot permit physical contradictions.

2.4 Sleep & Nightly Shelter

Each Person needs a legitimate nightly state such as household accommodation, rented room, boarding bed, hotel room, employer bunk, camp, temporary hosting, travel lodging, precarious shelter or explicit no-stable-shelter state. Residential Occupancy answers the continuing shelter arrangement; actual Sleep answers where and when the Person rested that night. They are related but not identical facts.

2.5 Food, Meals & Nutrition

Food remains primarily a household/institution inventory and budget problem rather than a bite-by-bite minigame. Meals must nevertheless have a legitimate source. Household food, board-included meals, employment meals, purchased prepared food, restaurant/saloon meals or another real source can satisfy Persons. Nutrition/food adequacy is a consequence of actual access, reserves, affordability and regularity rather than an independent FoodPoints inventory.

2.6 Water & Alcohol Are Distinct

Ordinary hydration can be represented through practical access at home, work, institutions and travel without simulating every cup. Alcohol is discretionary Person-specific consumption. Saloon drink revenue requires a real patron, real inventory, service and payment/credit. Alcohol can affect spending, time, sleep, fatigue and behavior where relevant without requiring a perfect biological BAC simulation or moral score.

2.7 Family & Domestic Labor

Cooking, cleaning, laundry, water/fuel handling, childcare, elder care, household animal care, gardens, coops, boarding work and similar domestic production consume real Person time. Allocation depends on actual household composition, capability, availability, relationships and historically calibrated behavior rather than fixed gender penalties. Family labor remains labor even when no ordinary wage is paid.

2.8 Children, Elderly, Sick & Disabled Persons

A child is not merely a dependency number. Age-appropriate schooling, chores, care, consumption, skill formation and family work can occur where supported. Elderly, sick and disabled Persons remain consumers, owners, relatives, knowledge holders and potential contributors even when wage labor capacity falls. Reduced labor participation never deletes a Person from society.

2.9 Unemployment Does Not Suspend Life

An unemployed Person can still have savings, household support, prior wages, casual earnings, remittances, credit, property income or assets. They may have more daytime availability for job search, errands, shopping, social contact and services while simultaneously facing greater future-income uncertainty. No Job does not mean No Consumption or Idle Until Hired.

2.10 Travelers, Visitors & Transients

A traveler who physically enters or participates in the represented regional economy should be a persistent or promoted Person with origin, destination/purpose, means, transport, lodging/food needs and timing. Hotel, livery, meal, saloon and store transactions should attach to that Person. Returning visitors should retain identity where practical.

2.11 Coarse Simulation Preserves Ontology

A Person may be represented through activity blocks, scheduled events or coarse travel outside the active region, but the same identity, location continuity, money/household state and commitments persist. Coarse simulation may compress

footsteps; it cannot fabricate time, purchases or duplicate a Person.

PART III - REAL CUSTOMER DEMAND, SHOPPING & SERVICE USE

3.1 Household/Person Procurement Loop

Ordinary local consumption follows a causal loop: need/reserve change -> awareness of possible sources -> decision whether to act now -> selection of acting Person or delivery mechanism -> travel/contact -> supplier availability -> service/choice -> transaction or abandonment/substitution -> possession/use -> future need state. Needs can be postponed or go unsatisfied.

3.2 Opening Hours Create Availability, Not Customers

A business being open does not spawn traffic. Opening hours only make service possible to Persons whose needs, schedules, location, knowledge and means overlap with those hours. A late-closing store may capture workers who cannot shop earlier; a closed store may lose the sale, delay it or push the customer to a rival.

3.3 Feasible Choice Sets

A supplier is not in a customer's feasible choice set if the customer does not know about it, cannot reach it, cannot afford/finance the purchase, the store is closed, stock/service is unavailable or timing is unacceptable. Price, quality, distance, wait, trust, credit, prior experience, reliability and suitability can then shape choice among feasible options.

3.4 Substitution, Reduction & Abandonment

When the preferred purchase fails, the Person/household may accept a substitute, use another supplier, reduce quantity, delay, borrow/use credit, produce internally or go without depending on urgency, alternatives and circumstances. Failed purchase attempts should create evidence for merchants and Opportunity/Pressure without automatically generating a sale elsewhere.

3.5 Repeat Business & Relationship Memory

Repeat trade should emerge from actual experiences: reliable stock, fair dealing, product quality, useful credit, service speed, convenience and social/commercial relationships. Marketing primarily changes awareness and consideration; operations and experience create retention.

PART IV - PHYSICAL CAPABILITY, EQUIPMENT & WORKSTATIONS

4.1 Capability Formula

A task is executable only when the relevant combination exists: capable Person(s) + required tool/equipment method + suitable workspace/location + materials/inputs + motive power where needed + authority/right + available time. Information/diagnosis may also be required for tasks whose problem is not yet known.

> **CANON LOCK - A job title or business label never grants physical capability by itself. Capability emerges from the actual resources and people that can perform the task.**

4.2 Requirement Types

- Hard requirement - the task is physically impossible without a suitable item/setup.
- Alternative equipment set - several historically plausible methods can perform the same task.
- Capacity equipment - manual work remains possible but equipment materially changes throughput, acreage, crew need or time.
- Workstation - a fixed or prepared operating place such as forge, oven, print press, compounding bench, saw line or bar.
- Shared capital asset - a wagon, thresher, steam engine, locomotive, hoist, dynamo or other scarce durable productive asset.
- Personal/small tool kit - ordinary hand tools grouped for usability while preserving their real contents.
- Consumable - fuel, nails, shoes, twine, ink, chemicals, medicine ingredients and similar items consumed through work.
- Motive power - human, horse, mule, ox, water, steam or electric power where historically and physically appropriate.
- Support equipment - ladders, scaffolds, lighting, storage, jacks, pumps, animal restraint or other enabling gear.

4.3 Task Requirements, Not Occupation Gates

A Farmer does not require every farm implement at all times. PlowField may require a plow, draft power and harness; HandHarvestGrain may use a cradle/sickle method; MechanicalHarvest may require a reaper/binder and team. A Person may perform compatible tasks across occupations when capability and circumstances allow.

4.4 Access Instead of Mandatory Ownership

The performer may use employer-owned, household-owned, rented, leased, borrowed, customer-supplied, contractor-supplied or hired-service equipment. A small farm can hire a traveling threshing outfit rather than own one. A contractor can rent a scraper. A repairperson can travel with a field kit while large machine work remains shop-bound.

4.5 Location, Reservation & Condition

Tools and equipment have location and availability. A wagon at another worksite is unavailable. One exclusive machine cannot perform two jobs simultaneously. Condition, maintenance and defects can reduce capacity or make a method unusable, feeding the existing repair/failure systems.

4.6 Tool-Kit Abstraction

Ordinary small tools may be grouped into historically defined kits to avoid inventory micromanagement. A Carpenter Hand Tool Kit still canonically contains saws, planes, chisels, brace/bits, square and related hand tools; the player need not maintain twenty-seven individual chisels unless an individual object becomes materially important.

4.7 Mechanization Is Usually Scale, Not a Binary Tech Gate

Hand methods may remain physically possible at small scale while animal-powered or powered machinery changes throughput and labor economics. Large acreage, industrial volume or contract deadlines can make the slower method commercially infeasible without making it physically impossible.

PART V - OCCUPATION & EQUIPMENT REFERENCE

The following profiles cover current Canon business roles, named future-pool occupations and recurring supporting assignments. They are researched starting profiles for 1880-1905 gameplay. Exact brands, counts, prices, regional adoption and specialist sub-tools remain data/calibration unless separately locked. Future occupations must follow the same task-level rule rather than receiving a generic Tools=true flag.

Occupation / capability	Core tools / workstation / physical requirements	Capacity, scale or optional equipment
Crop farmer / farmhand	Hoe; spade/shovel; fork; rake; hand seed sacks; knives; sharpening file/stone; access to field and seed. Soil-working tasks require suitable implement and draft/human power.	Moldboard/steel plow; harrow; cultivator; grain drill/broadcast seeder; farm wagon; larger draft team.
Grain planting crew	Seed sacks; scoops/measures; field markers where used; harness/handling gear when animal-powered.	Grain drill or broadcast seeder; team; wagon for seed supply.
Grain harvest hand	Sickle or grain cradle; whetstone; forks; binding material.	Horse-drawn reaper or self-binder; team; twine; field wagons.
Threshing crew	Forks; shovels; grain sacks; scales; adjustment/repair hand tools.	Threshing separator; horse power or steam engine/traction power where available; grain/straw wagons; hired traveling outfit.
Hay worker	Scythe; whetstone; hay fork; rake.	Horse-drawn mower; horse rake/tedder; hay wagon; larger stacking/lifting gear.
Rancher / stock hand / cowboy	Saddle; bridle/reins; rope/lariat; halter; grooming/hoof tools; appropriate riding horse or on-foot access.	Corrals; gates; chutes/handling pens; troughs; hay/feed storage; stock scales; stock wagon.
Poultry keeper	Feed scoop; water vessels; egg basket/crates; cleaning tools.	Coop; nesting boxes; run/fencing; feed bins; brooding/expanded housing where supported.
Dairy / milking hand	Milk pails; stool; strainer; containers; washing/cleaning vessels.	Churn or separator where appropriate; cooling/storage area; wagon/crates for milk movement.
Farm owner / foreman	Field/financial records; writing materials; measuring/inspection tools; access to farm equipment and workers.	No unique production machine; value comes from authority, scheduling, inspection and coordination.
Seasonal farm laborer	Uses the task-appropriate employer/crew tool set.	Need not personally own major machinery.
Teamster / freight driver	Suitable wagon/cart; draft team; harness/collars; reins; hitching hardware; ropes/chains/tarps; lantern; feed/water provisions; basic repair kit.	Heavier wagon; additional animals; spare harness/parts; specialized cargo equipment.

Occupation / capability	Core tools / workstation / physical requirements	Capacity, scale or optional equipment
Drayman / local delivery	Dray/cart; team; harness; ropes/chains; loading aids.	Hand truck/dolly; larger platform wagon; warehouse/loading dock access.
Livery keeper	Stalls; tack racks; saddles/harnesses; rental rigs; feed bins; troughs; ledger/keys.	Multiple rental vehicles/teams; wash/grooming areas; repair relationships.
Stable hand / hostler	Hoof pick; curry comb; brush; mane comb/whisp; grooming cloth; fork; rake; broom; shovel; buckets/tubs; wheelbarrow.	Dedicated wash/grooming bay; larger manure/feed handling equipment.
Blacksmith	Forge/hearth; bellows or blower; anvil; hand/cross-peen hammers and sledges; tongs; vise; punches; chisels; hardy tools; swages; files/rasps; slack tub; fuel and iron/steel stock.	Multiple anvils/forges; drill/lathe or powered machine tools later; lifting gear; larger stock racks.
Farrier	Forge/anvil access; shoeing hammer; hoof knife; rasp; nippers; clinching/punch tools; tongs; shoes/nails; safe animal tie/restraint area.	Mobile shoeing kit; dedicated shoeing stall; larger stock of shoe sizes/material.
Wheelwright / wagon maker	Rip/crosscut/backsaws; planes; chisels; brace/bits/augers; drawknife; spokeshave; adze; hammers/mallets; clamps; measuring tools; workbench; wheel jig.	Tire bender; forge access; wheel/axle lifting gear; oiling trough; powered woodworking tools later.
Wagon repairer	Jacks/supports; wrenches; saws; brace/bits; hammers; chisels; wheel/axle tools; grease/oil; spare hardware/components.	Shop lift/gantry; forge/wheelwright access; stocked standardized replacement parts.
Harness maker / saddler	Leather knives/shears; awls; punches; needles; stitching clamp/horse; edge tools; mallet; rivet/set tools; workbench; leather; thread; buckles/rings.	Specialized saddle forms; larger cutting bench; sewing machine where supported later.
Carpenter / joiner	Rip/crosscut/backsaws; hammer/mallet; planes; chisels/gouges; brace/bits/augers; square; bevel; rule; level; marking gauge; drawknife/spokeshave; hatchet/adze; clamps; sharpening stone.	Portable benches; larger saws; block/tackle; powered woodworking equipment in mature shops.
Framing crew	Carpenter set plus axes/adzes; larger augers; ropes; lifting blocks; ladders/scaffolds.	Horse/steam lifting assistance on large work; dedicated material staging.
Mason / bricklayer	Trowels; mortar hoe; mixing box; buckets; brick hammer; chisels; line/pins; level; plumb bob; straightedge; wheelbarrow; scaffold.	Mortar mixer/powered hoist where later/large; stone lifting gear.
Painter	Brushes; scrapers; paint pots; mixing containers; cloths; ladders/scaffolds; surface-prep tools.	Larger staging/scaffold system; spray methods are outside ordinary early-period profile.
Glazier	Glass cutter; straightedge; glazing knife; putty; points; measuring rule; safe glass handling; ladders/scaffold.	Glass rack/cart; shop cutting table.
Roofer / shingler	Shingling hatchet/hammer; knife; saw; rule; marking/chalk tools; ladder/scaffold.	Material-specific metal tools; hoisting gear; larger scaffold.
Construction laborer	Shovel; pick; wheelbarrow/cart; pry/crow bar; sledge; buckets; ropes; assignment-specific hand tools.	Horse-drawn scraper/grader; hoist; larger earthwork equipment.
Working foreman	Plans/drawings; measuring rule; level; notebook/ledger; ordinary trade tools when personally working.	Communication/clerical assistance on larger jobs.
General contractor / estimator	Plans/specifications; measuring tools; estimate/record books; desk/writing/correspondence; site inspection access.	Office staff; survey/engineering subcontract access; rented/owned project equipment.
Surveyor / civil engineer	Surveyor transit; tripod; survey chain/tape; level; leveling rod; compass; ranging rods/stakes; field notebook; drawing instruments.	Additional crews/instruments; drafting office; specialized railroad/bridge instruments.
Logger / feller	Felling axe; crosscut saw; wedges; sledge; cant hook/peavey; files/saw-set; chains/ropes.	Draft skidding team; logging sled/wagon; larger loading gear.
Skid / log-haul worker	Draft animals; sled/log wagon or skidding gear; chains; hooks/tongs; handling bars.	Winch/steam donkey only where later/regional and researched; larger landings.
Sawyer / sawmill operator	Sash/circular saw line appropriate to mill; log carriage; drive system; log handling tools; lubrication; guards/work area.	Edger/trimmer; multiple saw lines; larger steam/water power; mechanical log handling.
Mill engineer / power operator	Steam engine/boiler or water-power equipment; belts/shafting/gearing; oiling tools; gauges/valves; fuel/water handling.	Backup power; larger boiler plant; dedicated maintenance shop.
Saw filer / mill maintenance	Files; saw set/swage tools; vise/bench; gauges; sharpening station; general mechanical kit.	Dedicated filer room; replacement saws; machine shop access.
Lumber-yard worker	Rule/tally book; hand saw; cant hook; carts/wagon access; stacking supports; yard/storage space.	Loading platform; derrick/hoist; rail siding; larger covered sheds.
Prospector / placer miner	Pick; shovel; gold pan; hammer; sample bags; basic camp/travel gear.	Rocker/sluice; pump; pack animals; larger prospecting outfit.
Hard-rock miner	Pick; shovel; sledge; hand drill/drill steel; wedges; tamping tools where blasting is authorized; candles/lamps; ore bucket/cart.	Compressed/steam drilling later where researched; mine rail/ore cars; hoist; ventilation/pumping.

Occupation / capability	Core tools / workstation / physical requirements	Capacity, scale or optional equipment
Mine laborer / mucker	Shovel; pick; ore car/bucket; wheelbarrow; loading/handling tools.	Mine rail, larger ore cars, mechanical haulage later where supported.
Hoist operator	Windlass/horse/steam hoist according to scale; rope/cable; bucket/cage; brake/control apparatus; inspection/oiling tools.	Larger engine/hoist house; duplicate ropes/cages; signaling system.
Pump operator	Pump; motive power; hoses/pipes; valves; repair/lubrication tools.	Multiple pumps; steam/electric drive later; sump/reservoir system.
Ore / stamp-mill worker	Crusher/stamp machinery; feed bins; water handling; power source; ore carts; maintenance tools.	Multiple batteries/lines; concentrating equipment; larger power plant.
Assayer	Precision assay balance/weights; sample tools; mortar/pestle; crucibles/cupels; assay furnace; tongs; sample containers; records.	Dedicated laboratory room; chemical/glassware set; ventilation/storage.
Miller / grain-mill operator	Millstones or other period mill system; hopper/feed; sifters/bolters/screens; scales; sacks; scoops; power source; lubrication/maintenance tools.	Elevators/conveyors; roller-milling equipment where date/place supports; larger storage bins.
General merchant / proprietor	Counter; shelves; bins/drawers; barrels/boxes; scales/weights; dry/liquid measures; scoops; ledger; pen/ink; cash drawer/lockbox; wrapping paper/string; lamps.	Safe; platform scale; coffee grinder; specialized fixtures; cash register/security recorder later where adopted.
Store clerk	Counter access; scales/measures; wrapping materials; account slips/ledger access; cash drawer; product-retrieval tools.	Telephone/postal/message equipment where the store offers those services.
Stock / receiving clerk	Receiving table; knives/opening tools; hand cart; storage bins/shelves; counting/record equipment.	Platform scale; hand truck/dolly; loading dock/yard equipment.
Bookkeeper / credit clerk	Desk; ledger/journal; invoices/account books; pen/ink; lockable records.	Safe/file systems; later adding/calculating devices only where adopted.
Feed merchant	Bulk/platform scales; scoops; sacks; bins; hay/grain storage; wagon-loading space.	Warehouse; rail siding; hay handling gear.
Grain buyer / dealer	Scales; sampling tools/containers; sacks; records; storage access; wagon/rail loading access.	Warehouse/elevator capacity; grading/testing equipment where implemented.
Grain warehouse / elevator worker	Scales; scoops/shovels; sacks; bins; lifting/conveying system appropriate to facility; loading platform.	Powered elevators/conveyors; rail siding; dust/fire-control improvements.
Baker	Oven/firebox; bread pans; bowls/crocks; dough trough; kneading table; scraper/cutter; wooden peel; rolling pin; brush; proofing rack; scales; water vessel; flour storage.	Larger mixer/oven systems where researched; delivery wagon; expanded racks.
Butcher	Butcher knives; cleaver; meat saw; sharpening steel/stone; block/table; hooks/rail; scales; tubs; wrapping/storage.	Smokehouse; salting barrels; ice/cold-storage capability; slaughter/handling facilities where lawful and supported.
Cook	Stove/range/hearth; pots/pans; knives; cutting table/board; spoons/ladles; roasting/baking vessels; storage; water vessels.	Larger range; pantry; icebox/cold storage; specialized hotel/saloon kitchen gear.
Bartender	Bar/counter; bottles; barrels/kegs; taps/spigots; glasses/cups; measuring vessels; cash/ledger access; wash basin/cleaning supplies.	Ice/cooling; larger backbar/storage; draft system; extra service stations.
Server / waiter	Trays; serving dishes; glassware/tableware; order/cash handling access.	Additional service stations; bussing carts where applicable.
Drink solicitor / entertainer	Premises access; ordinary seating/social space; service relationship with bar; no special magical sales equipment.	Stage/music equipment only when the establishment genuinely offers entertainment.
Faro dealer / banker	Faro table/layout; cards; dealing apparatus; casekeeping system; counters/chips; separate bankroll/cash box.	Dedicated concession space; security/cash controls.
Poker host / dealer	Cards; table; counters/chips where used; lighting/seating.	Dedicated room/table; security/cash controls when stakes are handled by house.
Brewer	Mash tub; brewing kettle/copper; fermentation vessels; stirring/transfer equipment; water source; cooling capability; casks/barrels; cleaning tools.	Pumps; malt/grain handling; larger storage/cellar; steam power where supported.
Cellar / keg handler	Barrel handling tools; taps; storage racks; hand truck/dolly; loading access.	Hoist/ramps; larger cellar cooling/ice capability.
Boarding-house / hotel keeper	Beds/mattresses/bedding; tables/chairs; linens; tableware; cookware; stove; pantry/storage; lamps; account ledger; keys.	Reception/front desk; larger kitchen; bath facilities; bell system; laundry room.
Housekeeper / chamber worker	Broom; brushes; pails; cloths; linens; wash basin; chamber/privy cleaning equipment.	Larger linen storage; cleaning cart; plumbing access later.
Laundress	Wash tubs/sinks; laundry stove/boiler/kettle; washboard; soap/lye; clothesline/drying area; irons.	Wringer; dedicated laundry room; larger hot-water system.

Occupation / capability	Core tools / workstation / physical requirements	Capacity, scale or optional equipment
Porter / hotel general hand	Hand cart/luggage equipment; cleaning kit; lamps; keys; basic maintenance hand tools.	Bell/cart systems; larger baggage room.
Independent room-based sex worker (future scope)	Legitimate room/premises access; bedding/linens; wash/hygiene facilities; personal effects; ordinary transaction/record arrangements where modeled. No invented special vice-equipment meter.	Premises security, separate entry or manager/service arrangements only when physically present and historically supported.
Doctor / town physician	Medical bag/pocket case; examination surface/chair; bandages/dressings; medicine cabinet; forceps/scissors/knives/needles and procedure tools appropriate to care.	Larger surgical set; office treatment room; horse/buggy for calls; assistant.
Veterinarian / animal doctor	Veterinary bag; examination/handling tools; knives/forceps; bandages/medicines; safe animal restraint; travel capability.	Large-animal stocks/chute; mobile call-out rig; specialized surgical/obstetric tools where researched.
Medical assistant / nurse	Bandages/dressings; basins; linens; medicine handling; patient-care tools appropriate to date.	Clinic storage; sterilizing/cleaning setup as period practice evolves.
Pharmacist / apothecary	Mortar/pestles; balance/weights; measures; jars/bottles; funnels; compounding slab/bench; labels; glassware; heating equipment where needed.	Expanded laboratory/storage; pill/tablet tools; distillation/extraction gear where supported.
Barber	Barber chair; straight razors; strops; clippers; shears; combs; brushes; shaving mugs; towels; wash basin; mirror.	Bath tubs/hot-water capability; multiple chairs; later powered/electric tools outside early baseline.
Tailor / seamstress	Shears; needles; thread; measuring tape/rule; chalk; thimble; pins; irons; cutting table; patterns.	Sewing machine; larger pressing table; stock of cloth/notions.
Typesetter / compositor	Movable type; type cases; composing stick; galley/tray; rules/leads/spaces; work stand.	Multiple cases/fonts; proof press; improved composing equipment later.
Press operator	Hand/platen/cylinder press appropriate to scale; chases/forms; ink rollers/inking equipment; ink; paper; drying/stacking space.	Power press; paper cutter; larger bindery/finishing equipment.
Editor / reporter	Desk; pen/ink; paper/notebook; reference files; correspondence access.	Telegraph/telephone access; printing office; travel horse/rail access.
Telegraph operator	Key; sounder/receiver; desk; paper/forms; battery/cell; connected wire/insulator network.	Multiple-line switch equipment; relay/repeater equipment; office clerk.
Telegraph line worker	Wire; insulators; poles/supports; pliers/cutters; splicing tools; climbing/access equipment; repair wagon.	Larger line crew; test instruments; spare pole/wire stock.
Telephone operator	Exchange switchboard; operator station; cords/plugs/connecting apparatus; subscriber directory/records.	Larger switchboard sections; multiple operators; message desk.
Telephone line worker	Poles; wire; insulators; subscriber sets; batteries/magnetos where required; climbing and hand-repair tools.	Line wagon; test equipment; exchange construction crew.
Rail surveyor	Transit; tripod; chain/tape; level/rod; notebook; stakes.	Larger survey party; drafting office; bridge/grade instruments.
Rail grader / earthwork laborer	Pick; shovel; sledge; wheelbarrow/cart.	Horse-drawn plow/scraper/grader; dump wagons; steam equipment where later supported.
Rock / tunnel crew	Hand drills; sledges; picks/shovels; lifting/hauling tools; lighting.	Authorized blasting setup; powered drill/compressor later where supported.
Track layer / section hand	Spike maul; track gauge; lining/tamping/crow bars; rail tongs; track jack; picks; shovels; wrenches; hand drills; level; saws.	Hand/pump car or work car; larger rail-lifting gear; section tool car.
Section foreman	Section tool set; gauge/inspection tools; timetable/records; warning/signal equipment.	Handcar/work car; crew tool inventory; telegraph access.
Locomotive engineer	Locomotive controls/gauges; timetable/train orders; required signals/communications.	No separate handheld motive equipment; capability depends on locomotive, track rights, fuel/water and crew.
Locomotive fireman	Fuel shovel/firing tools; locomotive fuel/water systems; basic firebox/boiler observation tools.	Mechanical stoker only where much later/appropriate; larger fueling facilities.
Brakeman / switchman	Lantern/flags as appropriate; switch key; hand brakes/switches; coupling/inspection tools.	Yard communication/signaling infrastructure.
Conductor	Train orders/timetable; paperwork/tickets/manifests; signaling/communication equipment.	Clerk support on larger operations.
Depot / station agent	Desk/ledgers; telegraph where connected; scales; freight platform; manifests; lamps; secure office/storage.	Warehouse; livestock pens; loading crane/derrick; ticket/baggage equipment.
Freight handler	Hand truck/cart; scale; loading platform/ramps; ropes; cargo-handling tools.	Hoist/derrick; larger warehouse/siding facilities.
Electric-plant operator	Dynamo/generator; steam engine/boiler or water turbine as plant requires; belts/shafting; switchboard/switches; gauges; oiling/maintenance tools.	Additional generators; storage/batteries where system uses them; larger distribution switchgear.
Boiler / stationary-engine operator	Firing/fuel tools; gauges/valves; oiling tools; feedwater equipment; maintenance hand tools.	Coal handling; feedwater pump; reserve engine/boiler.

Occupation / capability	Core tools / workstation / physical requirements	Capacity, scale or optional equipment
Electrical line worker	Poles; wire; insulators; ladders/climbing gear; pliers/cutters; splicing/repair tools.	Line wagon; test instruments; transformer-related equipment only as period network evolves.
Waterworks operator	Pump/engine; well/intake; tank/reservoir; mains/valves; gauges where fitted; pipe/valve tools.	Multiple pumps; larger reservoir; steam/electric drive; treatment equipment only when historically appropriate.
Pipe / main worker	Pick/shovel; pipe wrenches/tongs; cutting/joining tools; lifting gear; carts/wagons.	Trench/hoisting equipment; larger pipe stock yard.
Sewer / drainage laborer	Pick/shovel; grading/level tools; carts; pipe/masonry tools.	Horse-drawn excavation/haul equipment; larger pumps.
Firefighter - early/ad hoc	Buckets; ladders; hooks; axes; access to wells/cistern/water source.	Organized hose cart or hand engine when institution exists.
Firefighter - organized company	Hose; hand/steam engine depending date/place; hose cart; ladders; axes/hooks; horses/harness where horse-drawn.	Dedicated station; alarm/telegraph; reserve apparatus.
Owner / proprietor	Records/ledger access; writing/correspondence; keys/authority documents; access to site. No universal Owner tool.	Office, telephone/telegraph, travel, clerical staff as scale grows.
Manager / general manager	Desk/records; correspondence/communication access; inspection access; authority/policy records.	Clerical staff; telephone/telegraph; travel support.
Auditor / inspector	Ledger/contracts/receipts; field notebook; counting/measuring tools appropriate to the assets being verified.	Specialist inspection instruments for boilers, structures, utilities or other technical domains.
Bank teller / cashier	Teller counter; cash drawer; ledgers/account books; secure coin/currency storage; safe/vault access.	Cash/security register or improved controls later; multiple teller stations.
Bank manager / lender	Desk; loan files/contracts; ledgers; correspondence; collateral/security records; safe/vault access.	Appraisal/inspection specialists; telegraph/telephone; board/committee meeting space.
Lawyer / notary	Desk; law/reference books; pen/ink/paper; document files/forms; seal where required; secure records.	Clerk; courthouse/records access; travel/correspondence.
Insurance agent / adjuster	Policy/claim ledgers; forms; desk; correspondence; property inspection notebook/measuring tools.	Travel capability; specialist inspection reports; office safe/files.
Land agent / promoter	Maps/plats; deeds/options/contracts; desk; correspondence; measuring/inspection access.	Surveyor relationship; advertising/print material; travel.
Teacher	Teacher desk; blackboard/chalk; books/readers; pupil slates/slate pencils; pen/ink/paper; school stove/lamps as building equipment.	Additional books/maps; larger classroom furniture; later institutional supplies.
Postmaster / mail clerk	Sorting shelves/pigeonholes; mail bags; desk/ledger; scales; stamps/canceling supplies; secure mail storage.	Postal counter; safe; telegraph/telephone where connected; cart/wagon handoff.
Constable / law officer	Badge/authority documents; notebook; lantern; horse/travel where needed; restraint equipment; legal notices.	Issued sidearm where role/local practice supports; office/jail access; telegraph/telephone later.
Private detective / reward agent	Notebook; maps; correspondence; travel funds; horse/rail access; case papers/evidence containers.	Camera/technical aids only where dated; lawful restraint/weapon access depends on actual authority and setting.
Civic clerk / official	Desk; ledgers/registers; forms; pen/ink; filing/secure record storage.	Meeting room; printing/notice access; telephone/telegraph later.
Soldier / federal field staff where represented	Issued uniform and duty-specific government equipment; field gear or office/records depending assignment.	Weapons, horses, wagons, telegraph or institutional equipment only according to actual unit/role authority.
General laborer	No universal kit. Uses assignment-specific tools, materials and protective/support equipment.	Capacity depends entirely on the task/site equipment provided.
Child / household helper	Buckets/yoke; broom; feed scoop; garden tools; school slate when attending; other age-appropriate task tools.	Uses household/farm equipment only when physically and historically appropriate.

5.17 Household Domestic Equipment Baseline

People can exist without formal employment, but an ordinary household needs physical domestic capability for the way it lives. The simulation should distinguish access to sleeping arrangements, heating, cooking, water, washing, laundry, sanitation, food storage and light where those functions are materially simulated.

Domestic function	Physical capability
Sleep	Bed/bunk or legitimate sleeping arrangement; bedding.
Heating	Stove/hearth or other supported heat source; fuel; safe vent/chimney where required.

Domestic function	Physical capability
Lighting	Daylight plus candles/lamps, or later electric connection when actually available.
Cooking	Stove/hearth; cookware/utensils; food and water access; preparation surface.
Water	Well/pump/cistern/network/other legitimate source plus vessels or direct service.
Washing	Basin/buckets, soap/cleaning supplies and water.
Laundry	Tub/washboard/soap; drying area; irons; larger households may use boiler/stove/wringer.
Sanitation	Privy/waste arrangement or later legitimate sewer/plumbing service.
Food storage	Pantry/shelves/barrels/bins; root cellar/ice/cold storage where present.
Fuel work	Axe/saw and fuel storage where the household cuts/handles its own wood or coal.

PART VI - AGREEMENTS, ECONOMIC EVENTS & PROJECT EXECUTION DEPTH

6.1 Commercial Commitment Lifecycle

Use a reusable lifecycle: inquiry/proposal -> negotiation/counterproposal -> conditional or final acceptance -> effective agreement -> individual obligations mature -> attempted performance -> inspection/acceptance -> payment/deficiency -> cure/waiver/dispute -> completion/renewal/termination. Conditions precedent can delay effectiveness or performance without pretending no agreement exists.

6.2 Obligations Are Smaller Than Agreements

An agreement can contain multiple obligations with different obligors, due dates, performance standards and consequences. A lumber contract can obligate the seller to deliver, the buyer to provide unloading, the buyer to pay after accepted delivery and the seller to replace rejected material. Performance should be measured obligation by obligation.

6.3 Partial Performance, Defect & Breach

Delayed, incomplete, defective, unauthorized-substitution, non-payment and repudiation outcomes remain distinct. Accepted quantity, rejected quantity, outstanding quantity, invoice amount and paid amount can all differ. Cure, waiver, renegotiation, withholding or termination must result from actual rights/choices.

6.4 Economic Event Lifecycle

Use the sequence Intent/Order -> Commitment/Reservation -> Fulfillment -> Acceptance -> Economic Recognition -> Settlement -> Adjustment/Reversal -> Historical Record. A deposit is cash but may not yet be fully earned revenue. Credit sale can create revenue/receivable without cash. Receivable collection is cash but not a second sale. Return, bad debt, spoilage or later-discovered error can append correcting events rather than erasing history.

6.5 Capital Assets Versus Operating Inputs

Buying a wagon, boiler, building or machine changes cash and asset state but is not economically identical to consuming flour or paying wages. Asset cost basis, continued ownership, condition, productive use and later disposal should remain traceable enough to measure real gain/loss without turning the game into a modern accounting textbook.

6.6 Project Baseline, Forecast & Actual

Every consequential Project should preserve the original approved estimate/schedule as Baseline, the best current expectation as Current Forecast and completed outcome as Actual. Hidden site conditions, supplier delay, customer change, rework or weather can change the forecast without rewriting the original bid.

6.7 Work Packages, Change Orders & Acceptance

Projects decompose into meaningful Work Packages and Tasks with dependencies, resources, materials, location and acceptance conditions. Customer changes, design changes, unknown conditions, substitutions and defects can create explicit change proposals/orders. Completion can pass through substantial physical completion, inspection, punch-list/corrective work, acceptance, final account/payment and any supported warranty/defect responsibility.

PART VII - NPC PLANNING, CUSTOMER CHOICE & INFORMATION DEPTH

7.1 Persistent NPC Plans

NPC owners/managers should form and revise plans rather than resolve every decision through one instantaneous utility score. A plan may progress through awareness, investigation, inspection, estimates, financing, negotiation, commitment, development, opening and later reassessment. Plans can die or change because land is lost, finance fails, prices change, a rival enters, household circumstances change or a better opportunity appears.

7.2 Capital Allocation Competition

One dollar cannot simultaneously fund maintenance, inventory, debt repayment, wages, reserves, equipment and expansion. NPC enterprises should face the same competing uses of scarce cash/capacity as the player and act according to policy, risk tolerance, time horizon, obligations and information.

7.3 Customer Decision Processes Are Contextual

Staple groceries, emergency equipment repair, hotel lodging, freight service and a long-term construction contract should not share one hidden attractiveness formula. Each purchase type can weight price, distance, urgency, reliability, wait, quality, trust, credit, suitability and prior experience differently while using the same general feasible-choice doctrine.

7.4 Information Can Conflict

Observed fact, rumor, advertisement, signed agreement, newspaper report, customer statement, manager estimate and official record can coexist and conflict. Actors can investigate. Communication technology changes access and latency; it does not make information automatically true or omniscient.

7.5 Reputation Is Evidence-Specific

Supplier-payment reputation, lender standing, employer reputation, customer-service reputation and project-completion reliability can differ. A single universal Reputation score should not overwrite these dimensions, although player-facing summaries may exist when they are traceable to evidence.

PART VIII - GOVERNANCE, SUCCESSION & INTERNAL CONTROLS

8.1 Governance Decision Lifecycle

Corporate/partnership governance should support Proposal -> authority classification -> notice/information -> required approvals -> negotiation/amendment -> approval/refusal -> execution -> record. Economic ownership, voting rights, office, board authority, reserved decisions and creditor restrictions remain separate.

8.2 Capital Calls & New Financing

A company needing capital can propose contributions, borrowing, new equity or other financing. Investors can participate, refuse or negotiate according to their rights. New money is not automatically filled pro rata. Dilution, covenants, collateral and control consequences must be real where applicable.

8.3 Succession & Continuity Case

Death, incapacity, retirement, relocation or partner exit redistributes ownership and authority; it does not automatically delete a business. When a consequential Person disappears from an ownership/authority role, create a continuity problem that identifies surviving operating coverage, ownership claims, obligations, signing authority, heirs/partners, manager coverage and viable outcomes such as family continuation, hired management, buyout, sale, dormancy or liquidation.

8.4 Internal Controls Are Operational Structure

Cash custody, purchase authority, record entry, receiving and verification can belong to different Persons. Very small firms may legitimately combine them; larger firms can reduce error/theft/concealment risk through separation, approvals, physical counts and reconciliation. A discrepancy can remain ambiguous until evidence distinguishes mistake, shrinkage, short delivery, spoilage or theft.

8.5 Acquisition Diligence Consumes Controls & Records

A buyer evaluating a well-controlled business should have stronger evidence than one evaluating informal, unreconciled books. Internal controls do not guarantee honesty; they change what can be checked and how reliably discrepancies can be explained.

PART IX - LAW, INSURANCE, DISTRESS & INSTITUTIONAL DEPTH

9.1 Commercial Dispute Lifecycle

A legal right does not automatically produce an outcome. Commercial disputes should support claim/demand -> response -> negotiation/settlement -> formal proceeding where needed -> decision/judgment -> enforcement. Notice, proof, professional expertise, cost, delay and enforcement capacity matter. Exact court procedure, lien priority, redemption and filing law remain dated research.

9.2 Insurance & Risk Transfer

Insurance should begin with historically supported property/fire and cargo/commercial-asset coverage, then expand where research supports it. Policy terms identify insured interest, covered risk/property, term, premium, limit, retained loss/deductible concept where period-appropriate, conditions/exclusions and insurer. Insurance never prevents physical loss. Claims follow loss -> notice -> investigation/assessment -> coverage decision -> valuation -> payment/partial payment/denial -> dispute where applicable.

9.3 Distress, Workout & Insolvency

Distinguish illiquidity, insolvency and operational failure. Creditors have agency and may demand cure, extend, waive, restructure, require collateral, restrict activity or enforce based on expected recovery and legal rights. Recovery can use asset sales, owner capital, refinancing, new partners, inventory reduction, layoffs, lease/supplier workout, partial liquidation or going-concern sale. Commercial Ruin remains an endpoint of no viable recovery, not a single cash threshold.

9.4 Civic Project Institutions

Civic infrastructure follows need -> promoter/proposal -> authority/franchise -> financing -> engineering/source -> procurement/construction -> initial service -> maintenance/expansion. Residents, merchants, utilities, railroads, lenders or civic bodies can promote or oppose a project. Public procurement should reuse ordinary Agreement/Project systems rather than a civic minigame.

9.5 Utility Service Economics

Physical generation/source, network and connection remain authoritative. Add the commercial layer: request/offer for connection, connection cost, service agreement/tariff, billing, arrears, disconnection/reconnection, capacity/load commitments and network expansion economics. Large customers may justify line extension through real expected load/revenue rather than a Boolean UtilityAvailable flag.

9.6 Labor-Market Lifecycle

Vacancy -> information/search -> candidate awareness -> mutual evaluation -> offer -> negotiation/counteroffer -> acceptance/refusal -> start/probation -> retention/promotion -> resignation/dismissal/layoff. Poaching changes both employers because the same Person leaves one organization and joins another. Housing, commute, family obligations, safety, wages, meals/board, reputation and alternatives all remain relevant.

PART X - PERFORMANCE, RESEARCH & REVISION XXIII LOCK SUMMARY

10.1 Life Simulation Must Remain Scalable

Continuous Personhood is a causality requirement, not permission to frame-poll every Person. Use event-driven schedules, activity blocks, household-level batching and coarse-resolution execution so long as identity, time, money, inventory, relationships and outcomes remain reconcilable.

10.2 Equipment Research Is a Starting Data Set

The occupation table establishes historically grounded categories and required physical methods. Exact equipment ownership prevalence, prices, model adoption, counts, wear, maintenance intervals and productivity effects remain regional/date calibration unless separately researched. The game should prefer plausible alternative methods over false universal minimum loadouts.

10.3 Systems Already Strong

Revision XXIII does not replace the existing strong household/housing ontology, asset condition/maintenance/failure system, core physical logistics, provenance, basic demography or banking/liquidity architecture. These are deepened through shared Person, Agreement, Transaction, Equipment and institutional rules rather than rebuilt.

10.4 Research Basis for Revision XXIII Equipment Additions

- National Park Service - Wheat Farms, Flour Mills, and Railroads: A Web of Interdependence - https://home.nps.gov/articles/wheat-farms-flour-mills-and-railroads-a-web-of-interdependence-teaching-with-historic-places.htm

- National Museum of American History - McCormick Reaper Model - https://americanhistory.si.edu/collections/object/nmah_861230
- Library of Congress - Aultman-Taylor threshing machine photograph, c.1881 - https://www.loc.gov/item/93510114/
- National Park Service - Fort Scott Stables: Tools of the Trade - https://home.nps.gov/fosc/learn/education/stabletools.htm
- National Park Service - Fort Scott Quartermaster: Supplies and Transportation - https://www.nps.gov/fosc/learn/education/qmsuppfort.htm
- National Park Service - Herbert Hoover Blacksmith Shop - https://www.nps.gov/places/jesse-hoover-blacksmith-shop.htm
- National Park Service - Fort Scott Post Bakehouse: Tools of the Trade - https://www.nps.gov/fosc/learn/education/bakerytools.htm
- National Museum of American History - Surveyor's Transit - https://americanhistory.si.edu/collections/object/nmah_745958
- National Museum of American History - Surgical Set, 1886-1900 - https://americanhistory.si.edu/collections/object/nmah_1062410
- National Museum of American History - Telegraph Sounder, ca.1880 - https://americanhistory.si.edu/collections/object/nmah_891412
- National Museum of American History - Composing Stick, ca.1880 - https://americanhistory.si.edu/collections/nmah_1282334
- National Park Service - Site of the First Telephone Exchange - https://home.nps.gov/subjects/nationalhistoriclandmarks/site-of-the-first-telephone-exchange.htm
- The Henry Ford - Shopping at an 1880s General Store - https://www.thehenryford.org/collections/explore/articles/shopping-at-an-1880s-general-store
- National Museum of American History - West Virginia General Store Ledger, 1879-1880 - https://americanhistory.si.edu/collections/archival-collection/sova-nmah-ac-0142
- National Park Service - Lowell Boarding Houses - https://www.nps.gov/lowe/learn/historyculture/the-boarding-houses.htm
- National Park Service - Longfellow House Service Ell / Laundry - https://home.nps.gov/long/learn/historyculture/service-ell.htm
- National Park Service - West Branch Schoolhouse - https://www.nps.gov/places/west-branch-schoolhouse.htm
- National Museum of American History - Cash and Credit Registers - https://americanhistory.si.edu/collections/object-groups/cash-and-credit-registers

10.5 Revision XXIII Hard Locks

- Continuous Personhood is authoritative; local customers and workers are not anonymous demand/labor tokens.
- Local in-person transactions require a legitimate buyer principal and acting Person.
- Household consumption and individual action remain separate but causally linked.
- Person time/location are exclusive across work and non-work life.
- Customer traffic and demand summaries are derived, not transaction generators.
- Business capability derives from people, equipment/tools, space, materials, authority, information and time.
- Equipment requirements attach to tasks/capabilities and can use alternative historically plausible methods.
- Agreement/Obligation, Economic Event, Activity/Time, Information/Evidence and Resource Reservation are preferred shared grammars.
- Projects preserve baseline, current forecast and actual outcome separately.
- Governance, succession, internal controls, insurance, disputes, distress, civic projects, utilities and labor markets require real institutional processes.
- Coarse simulation preserves the same entities and economic truth.

END OF REVISION XXIII INTEGRATION

The preserved Canonical Project Bible Revision XXII follows unchanged and remains authoritative except where Revision XXIII explicitly adds, deepens, clarifies or supersedes a rule.

LAND & LEDGERS
CANONICAL PROJECT BIBLE - REVISION XXII

The Contractor Full-Systems Playtest Integration: Scenario Accomplishment Doctrine, Contract Access, Settlement Integrity, Delegated Capital & Opportunity Provenance

Revision date: September 18, 2026

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Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XXII preserves Revision XXI in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XXI document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XXII controls. Exact Contractor playtest prices, wages, rents, populations, project costs, contract values, interest rates, margins, work durations, crew sizes, timber quantities, equipment counts and other generated values remain calibration unless this integration explicitly identifies a design or scenario lock.

PURPOSE

Revision XXII integrates THE CONTRACTOR full-systems playtest and the subsequent four-pass Master Playtest Integration. The campaign moved from broke handyman work through hired labor, multi-crew construction, railway contracting, delegated operations and the causal creation of Cinder Creek. The principal result is not a fixed contractor progression or a new table of historical prices. It is a tighter set of rules connecting scenario accomplishments, opportunity provenance, organizational qualification, external infrastructure progress, physical settlement state, delegated capital policy and persistent obligations.

The review also validates substantial existing Canon from Revisions XIX-XXI: Construction & Contracting, Full Autonomy, Manager Activity, NPC parity, exact-known-information doctrine, physical logistics, maintenance/failure causality, migration, housing, sawmilling, Lifetime Ledger authority, separate owner/business cash and scenario failure without world deletion. Those systems remain authoritative and are not duplicated here.

REVISION XXII MASTER DOCTRINE

- **Scenario objectives normally measure accomplishments, not designer-preferred routes.** A difficult but legitimate strategy remains valid unless the restriction itself is an authored objective or premise.
- **Opportunities require causal world provenance.** A scenario may author circumstances and schedules, but it should not spawn a qualifying contract merely because the player needs one for a checklist.
- **External infrastructure projects progress through persistent world state.** Railways, roads and other projects may advance, delay or change independently of player objective completion.
- **Contract access may depend on evidence.** Prior work, references, relationships, specialist staff, equipment, capital, geography and current commitments can affect whether a counterparty considers the player capable.
- **Settlement population remains physical.** Reported residents must correspond to actual Persons and legitimate accommodation/residence relationships.
- **Building completion, business opening and business profitability are separate facts.** Scenario predicates must ask the authority that actually owns the required truth.
- **Delegated policy can include recurring capital mandates.** A manager may be told to reinvest, expand or protect liquidity while still executing real projects, transactions and financing decisions.
- **Financing authority is delegable when explicitly granted.** Major borrowing is commonly owner-reserved, but it is not universally owner-only.
- **Settlement durability is multidimensional.** Do not collapse housing, employment, services, transport, imports and boom exposure into one authoritative Self-Sufficiency score.
- **Migration incentives alter ordinary economics and information.** Free rent, relocation aid or discounted lots can improve feasibility and attractiveness; they do not directly create population.
- **Destroyed collateral does not automatically erase surviving debt.** Asset loss, insurance, security value, lender response and the underlying obligation remain separate facts.

PART I - SCENARIO ACCOMPLISHMENTS, OPPORTUNITY & EXTERNAL PROJECTS

1.1 Objectives Measure Accomplishments

Primary objectives should describe the economic or world accomplishments required for completion. They should not prohibit a strategy merely because it is faster, more leveraged, more vertically integrated, more decentralized or otherwise different from the route the designer expected, provided the ordinary systems, costs, authority, labor, time, law, provenance and consequences actually execute.

CANON LOCK - Gamey player optimization remains valid when the simulation is not cheating. Scenario restrictions are legitimate when the restriction itself is part of the authored challenge, historical premise or qualifying accomplishment.

1.2 Opportunity Provenance

A material contract, acquisition, supplier proposal, property opportunity or customer project should exist for a world reason. Useful provenance can include an actual counterparty need, project stage, shortage, seller motive, route change, maintenance burden, customer request, public program, financing condition or other causal source. Scenario authors may shape initial circumstances and broad timing, but should not create otherwise unsupported qualifying transactions solely because an objective requires them.

1.3 Contract Access Through Demonstrated Capability

Larger or more specialized work can require credible evidence that a contractor can perform it. Relevant evidence may include completed work, references, known personnel, specialist employees, equipment, working capital, insurance/security or bonding where historically appropriate, current workload, geography, commercial standing and relationships with engineers, customers or other counterparties. There is no abstract Contractor Level that replaces those facts.

1.4 Increasingly Demanding Opportunities Are Not Unlock Tiers

A scenario may deliberately arrange a plausible sequence of increasingly demanding opportunities. THE CONTRACTOR reference arc used ordinary building work, railway structures, grading, bridges and multi-site execution. Completing an earlier objective does not magically unlock the next contract. The next opportunity must still exist through world schedule, geography, counterparty need, information and qualification.

1.5 External Infrastructure Has Its Own Timeline

A scenario-authored railway, road, canal, utility or similar external project can maintain its own route, work fronts, construction stages, delays and service openings. The player may follow, ignore, service, invest around or relocate relative to that project. Objective completion must not teleport the infrastructure front to the player or freeze it until a checklist condition is met.

1.6 Active Work Is Distinct From Agreement Existence

Where an objective requires overlapping work, simultaneous crews or maintained execution, qualification depends on actual active work across the authored period. Merely having two signed contracts, two named crews or two planned projects is insufficient unless the authored predicate explicitly says otherwise.

PART II - SETTLEMENT, POPULATION & PROPERTY-STATE INTEGRITY

2.1 Resident Population Must Reconcile to Persons

A settlement resident count is a read model over actual persistent Persons and legitimate residence/accommodation relationships. The simulation must not maintain an independent population number that can exceed or contradict the people physically represented by the economy.

A practical audit question is: **Where does this Person sleep tonight?** A valid answer may be a house, rented room, boarding house, hotel, bunkhouse, employer lodging, temporary camp or another historically and physically legitimate accommodation. Homeless/displaced Persons may exist, but they should be represented explicitly rather than silently counted as housed residents.

2.2 Permanent, Transient and Daytime Presence Remain Distinct

Permanent residents, transient overnight residents, temporary camp populations and daytime workers/visitors are different economic facts. Employment in a settlement does not automatically make the worker a resident. Settlement reports should preserve the distinction when it materially affects housing, consumption, labor supply or durability.

2.3 Building State and Business State Are Separate

Preserve the existing Parcel -> Building/Improvement -> Functional Space -> occupancy/business-use separation and deepen it with explicit state semantics. A construction project may be complete, the resulting building may be physically usable, a business may later occupy it, the business may open, and it may later become profitable or fail. These events can occur at different times.

CANON LOCK - A permanent-building objective should qualify from authoritative physical/project state, not from whether an occupying business has opened, unless the objective explicitly requires operation.

2.4 Settlement Durability Is Multidimensional

A settlement can be strong in some dimensions and fragile in others. Cinder Creek demonstrated that a place may have stable housing and repair capacity while remaining dependent on imported food, imported finish lumber, railway access or transient construction demand. Do not create one authoritative Self-Sufficiency meter that hides those dependencies.

Useful explanation dimensions include employment concentration, permanent versus transient demand, housing fit, essential services, transport access, import dependence, local production diversity, vacancy, infrastructure continuity and exposure to temporary booms. These are read models over real state, not substitute economic truth.

PART III - DELEGATED CAPITAL POLICY & FINANCING AUTHORITY

3.1 Standing Capital Mandates

Delegation can include recurring strategic policy such as reinvesting distributable rental earnings into qualifying housing, protecting a liquidity band, replacing failed equipment, expanding within a defined geography or staging projects when evidence supports them. A standing mandate changes who is authorized to decide; it does not spawn buildings, loans, workers, inventory or cash.

The responsible manager must still identify real opportunities, evaluate information, arrange land, labor, materials and financing, create actual projects/agreements, and accept the consequences of their judgment.

3.2 Financial Authority Need Not Use a Universal Dollar Cap

Delegated financial authority may use dollar limits, transaction classes, risk constraints, approved collateral, geography, business scope, strategic mandate, escalation rules or broad discretionary authority. A fixed numerical ceiling is optional rather than mandatory.

3.3 Major Borrowing Can Be Explicitly Delegated

Full Autonomy does not automatically grant authority over every mortgage, major loan, ownership change or sale. Those acts remain subject to the owner/governance authority actually granted. However, the player may explicitly authorize a trusted manager or executive to arrange borrowing, refinancing, mortgages, seller finance or other capital transactions within defined scope. Lenders, collateral, legal powers, liquidity and governance still apply normally.

3.4 Management Remains Decision-Making Labor

The Contractor validates that recurring reinvestment and financing work should appear through actual manager decisions, Manager Activity, negotiations, Projects and Transactions. Do not replace this with a passive Property Growth or Management Efficiency modifier.

PART IV - MIGRATION INCENTIVES & SETTLEMENT PROMOTION

4.1 Incentives Enter the Ordinary Migration Chain

Free or discounted introductory rent, relocation assistance, employer-paid travel, discounted lots, cheap commercial leases and similar incentives can affect destination awareness, expected housing cost, relocation feasibility, household budget and perceived opportunity. They do not directly increment population.

Households still evaluate available work, household needs, housing fit, travel, family constraints, information quality, alternatives and expected durability, then physically move and either remain or leave according to subsequent conditions.

4.2 Growth Is Not Durable Merely Because a Threshold Was Touched

Settlement-development scenarios may require a population threshold to be sustained, paired with economic viability or maintained after extraordinary subsidies expire. The exact threshold and duration remain scenario calibration. The design purpose is to distinguish durable settlement from a one-day or heavily subsidized temporary boom.

PART V - PHYSICAL LOSS, COLLATERAL & SURVIVING OBLIGATIONS

5.1 Asset Destruction and Debt Resolution Are Different Events

Fire, accident, theft, disaster or deterioration can destroy or materially reduce the value of collateral without extinguishing the debt it secured. The obligation continues according to the Agreement, insurance, lender response, collateral/security rights, repayment, settlement, discharge and dated law.

CANON LOCK - Destroyed collateral is not debt forgiveness. Loss of security may increase lender exposure, trigger contractual remedies, reduce refinancing options or produce a deficiency after collateral proceeds, but the liability changes only through an authoritative financial/legal event.

5.2 Crisis Recovery Uses Shared Systems

A crisis scenario should create severe circumstances rather than a parallel recovery minigame. Rebuilding, asset sale, hauling, contracting, refinancing, workouts, leasing, partnership capital, occupation change and other responses should use the same systems available elsewhere. Investigating sabotage or arson, when present, is a separate legal/information thread and does not erase the balance sheet by fiat.

PART VI - CANONICAL SCENARIO ADDITIONS

6.1 THE CONTRACTOR

THE CONTRACTOR is a canonical Construction & Infrastructure / Settlement Development scenario concept built around the arc **broke handyman -> contractor -> employer -> railway contractor -> multi-crew organization -> town builder**. An external railway project already exists and advances physically through the region. The player succeeds by building organizational capability alongside the changing opportunity rather than by unlocking contractor tiers.

The tested reference objective structure is preserved below. The dates are scenario-authored requirements for this scenario structure and do not replace the ordinary Open Frontier start-year doctrine:

1. By June 30, 1875 - complete one paid contract with all construction labor personally performed by the player.
2. By December 31, 1875 - complete eight paid contracts.

3 By December 31, 1876 - complete five paid contracts using hired labor.

4 By December 31, 1877 - complete two paid permanent new-building contracts.

5 By December 31, 1878 - complete one paid contract with zero manual construction labor by the player.

6 By December 31, 1879 - have active work on two paid contracts overlap for at least seven consecutive days.

7 By December 31, 1880 - complete one paid contract at least twenty road miles from the starting settlement.

8 By December 31, 1881 - complete one railroad-issued construction contract.

9 By December 31, 1882 - maintain two separate construction crews at two active worksites for at least thirty consecutive days.

10 By December 31, 1883 - complete five railroad-issued contracts in total.

11 By December 31, 1883 - complete five cumulative route miles of railroad grading/earthwork.

12 By June 30, 1884 - complete one permanent railroad bridge.

13 By September 30, 1884 - complete twelve permanent buildings in one developing railroad town/railhead.

14 By December 31, 1884 - complete twenty-five permanent new-building contracts in total.

15 By December 31, 1884 - maintain four separate construction crews at four active worksites for at least thirty consecutive days.

For the maintained crew objectives, a qualifying crew in the tested structure contains at least three employees at one worksite with a designated working lead/foreman capable of routine progress without player manual labor. That definition is scenario-specific and does not create a universal minimum construction-crew rule.

6.2 ASHES & NOTES

ASHES & NOTES is a Crisis & Recovery / Agriculture & Property scenario concept. The player begins with an operating farm and meaningful seller debt or other secured obligation. A major physical fire destroys much of the productive/residential base while leaving some surviving assets and the financial obligation intact. Recovery remains open-ended through ordinary systems: rebuild, sell, haul, contract, refinance, negotiate, lease, restructure, seek partners or change commercial direction. Exact opening debt, surviving assets, fire cause and deadline remain authored scenario calibration.

6.3 A TOWN FROM NOTHING / THE PROMOTER

A TOWN FROM NOTHING / THE PROMOTER is a Settlement Development scenario concept. The player begins with land and limited capital but no durable town under their control. The objective is to create enough economic gravity that households and independent businesses choose to arrive, remain, invest and expand. Possible tools include industrial anchors, freight access, housing, free/discounted introductory rent, relocation assistance, cheap lots/leases and commercial promotion. The player may vertically own much of the settlement or act mainly as promoter/landowner; neither route is mandatory.

A useful completion pattern is a permanent-population threshold plus a sustained retention or viability requirement after extraordinary recruitment support ends. Exact population, duration and subsidy rules remain scenario calibration.

PART VII - CONTRACTOR PLAYTEST VALIDATION & CALIBRATION

7.1 Existing Canon Strongly Validated

- Construction & Contracting scales through real workload, people, supervision, equipment, transport, cash and backlog rather than abstract crew slots.
- Full Autonomy and Manager Activity can support mature operations with zero routinely owner-managed businesses.
- Owner cash, business cash, consolidated reporting, profit, cash flow, book equity and market value remain distinct.
- Employee capability is evidence-limited; role or promotion does not manufacture competence.
- Task assignment does not guarantee execution; people, animals, vehicles and equipment remain exclusive physical resources.
- Maintenance, defects and downtime produce causal work, lost capacity and repair decisions rather than generic debuffs.
- Sawmill/logging/lumber and internal-supply chains remain physical and cost-bearing.
- Housing pressure is household-type-specific; transient labor and durable population remain different.
- NPC entry and competition use the same economic systems and can succeed, fail or misjudge opportunities.
- Lifetime Ledger evidence is necessary for long scenarios; narrative reconstruction is not authoritative.

7.2 Contractor Calibration Holds

The following Contractor playtest values remain calibration/research rather than universal historical constants:

- exact wages, room rents, business margins, debt terms and supplier credit;
- construction contract prices, bridge costs, grading prices and project contribution margins;
- crew sizes, horse/team counts, equipment ratios and exact output rates;

- timber acreage, species mix, yields, local-versus-imported lumber economics and sawmill throughput;
- Cinder Creek population-growth rate, service-entry timing and the number of residents needed to support any particular business;
- railway-contract frequency and the exact pace from small contractor to regional/railway contractor;
- exact migration-incentive response and post-subsidy retention thresholds.

PART VIII - REVISION XXII LOCK SUMMARY & DEEPENING MAP

8.1 Revision XXII Locks

- Scenario objectives normally measure accomplishments rather than prescribe one expected strategy.
- Material opportunities require causal world provenance.
- External infrastructure projects progress through persistent world state independent of objective checkmarks.
- Contract access can depend on demonstrated capability, capacity, relationships and prior performance.
- Settlement resident counts reconcile to actual Persons and legitimate accommodation.
- Physical building completion and business opening/performance remain separate state.
- Recurring manager reinvestment/capital mandates are valid delegated policy.
- Major financing authority may be explicitly delegated within real scope and constraints.
- Settlement durability is multidimensional rather than one Self-Sufficiency score.
- Recruitment incentives affect ordinary migration economics rather than directly creating population.
- Collateral destruction does not automatically extinguish the secured obligation.
- THE CONTRACTOR, ASHES & NOTES and A TOWN FROM NOTHING / THE PROMOTER are preserved scenario concepts.

8.2 Deepening Map

- Revision XXI Part I / objective semantics is deepened by accomplishment-versus-route doctrine and active-work qualification.
- Revision XIX Construction & Contracting and Railway Project doctrine is deepened by contract qualification, opportunity provenance and independently progressing external project fronts.
- Revision XXI Part V and earlier Housing/Settlement doctrine is deepened by hard population/accommodation reconciliation and explicit physical-building versus operating-business state.
- Revision XXI Part IV and Revision XIX Manager Activity/Autonomy are deepened by recurring capital mandates and explicitly delegated financing scope.
- Existing migration and settlement-development doctrine is deepened by explicit treatment of recruitment incentives and multidimensional durability.
- Existing finance, collateral, fire and crisis doctrine is clarified so physical collateral loss cannot be interpreted as automatic debt deletion.

END OF REVISION XXII INTEGRATION

The preserved Canonical Project Bible Revision XXI follows unchanged and remains authoritative except where Revision XXII explicitly adds, deepens, clarifies or supersedes a rule.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XXI

Saloon Circuit Full-Systems Playtest Integration: Objective Semantics, Person/Delegation, Provenance, Finance, Property & Saloon Operations

Revision date: September 15, 2026

Source baseline: Canonical Project Bible Revision XX - September 8, 2026

Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XXI preserves Revision XX in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XX document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XXI controls. Exact Saloon Circuit playtest prices, wages, traffic counts, project costs, interest rates, holding periods, town populations, construction durations and other generated values remain calibration unless this integration explicitly identifies a design or scenario lock.

PURPOSE

Revision XXI integrates the complete SALOON CIRCUIT - THREE TOWNS systems playtest and the subsequent Master Playtest Integration. The campaign stress-tested scenario objectives, physical player time, labor autonomy, capability evidence, delegation, mixed-use property, food service, gambling, marketing, vertical integration, physical logistics, historical finance, employee succession and final liquidation. The principal result is not a new set of universal prices or a fixed saloon progression. It is a stronger causal connection among existing authorities: Persons perform real work, goods preserve provenance, agreements allocate rights, finance preserves obligations and scenario objectives consume authoritative state rather than narrative intent.

REVISION XXI MASTER DOCTRINE

- Scenario objectives should be explicit economic/world predicates. Event, current-state and maintained-state objectives have different completion semantics.
- A legitimately completed historical event milestone remains completed unless the authored objective explicitly requires continuing ownership or maintenance.
- Unusual or deliberately gamey player strategies are valid when the real systems, costs, labor, time, provenance and consequences actually execute.
- The player and NPCs should use the same ordinary causal economic actions. Difference comes from information, resources, skill, authority, goals and judgment rather than exclusive mechanics.
- Employees are autonomous Persons. Assignment creates work and authority; it does not guarantee compliance, competence or completion.
- Delegation transfers real work and decision authority. Full Autonomy is a convenience configuration over real tasks, limits, records and escalation rather than magic automation.
- Vertical integration is proven by physical/economic execution - production, inventory, transport, receipt, transformation/use and customer sale - not by ownership labels alone.
- Debt, seller finance, guaranties, collateral and contingent exposure remain distinct. A debt-free objective must define what it counts.
- Marketing changes information and customer choice through causal channels; it does not create guaranteed demand.
- Known transaction facts remain exact. Forecast uncertainty, hidden capability and competitor knowledge remain information-limited.

PART I - SCENARIO OBJECTIVE SEMANTICS & COMPLETION

1.1 Event, Current-State & Maintained Predicates

Scenario authors should define objectives from explicit authoritative facts. Three common predicate families are:

- Event predicate - a qualifying event occurred, such as a customer sale, acquisition, disposition or completed project.
- Current-state predicate - a condition is true now, such as 100 percent ownership, zero qualifying principal debt or current market rank.
- Maintained-state predicate - a condition remained true for an authored duration, such as profitable autonomous operation for a specified period.

CANON LOCK - Do not treat these predicate families as interchangeable. Historical events persist; current-state conditions re-evaluate; maintained-state objectives require the authored duration/evidence.

1.2 Historical Milestones Persist

If the player legitimately completes a historical milestone using real underlying systems, later sale, destruction or loss of the enabling asset does not retroactively erase the event unless continued ownership is an explicit part of the objective. This allows build -> integrate -> prove -> sell strategies without rewriting history.

1.3 Headline Objectives & Attached Criteria

A scenario may present a readable headline objective while evaluating several attached predicates. Every mandatory criterion must be discoverable from the beginning through objective details, information affordances or equivalent UI. The player should not encounter surprise victory requirements at the deadline.

1.4 Countdown Presentation

Objective UI may show current date, deadline, status, remaining time and early-completion margin. Presentation granularity may become more precise as the deadline approaches, for example months -> days -> hours. Exact thresholds remain UX calibration; the authored deadline remains authoritative.

1.5 Creative Completion

A player may deliberately satisfy an objective through a minimal, odd or inefficient sequence if every underlying action is real. The Saloon Circuit one-chicken test is the reference example: one owned chicken can be processed, physically moved, prepared and sold to one customer to satisfy a distinct-product event if the authored predicate says one qualifying sale. The simulation should not add an unstated minimum batch merely because the designer expected a larger operation.

CANON LOCK - Gamey player optimization is allowed. Gamey simulation shortcuts are not.

1.6 Scenario Failure Remains Separate From Commercial Ruin

Missing an authored objective or deadline can fail the scenario without destroying the persistent world, enterprise or save. Scenario failure and Commercial Ruin remain separate concepts.

PART II - PHYSICAL PLAYER, TIME & SYSTEMIC PARITY

2.1 Player Is a Physical Person

The general Land & Ledgers player remains one physical commercial Person. The player may personally perform ordinary work where capability, tools, location and law permit. A 15-minute task consumes approximately 15 minutes of authoritative simulation time; travel also consumes time. Do not create a generic Owner Hours resource pool.

Scenario-specific role restrictions may still prohibit or discourage personal labor when deliberately authored. Such restrictions do not become a universal rule merely because an earlier scenario used them.

2.2 Opportunity Cost Creates the Delegation Curve

The natural progression from hands-on work to hiring, supervision and capital allocation should emerge because owner time becomes scarce and economically valuable. Delegation is useful because another Person actually performs or decides the work, not because an owner-action-point penalty disappears.

2.3 Systemic Player/NPC Parity

Ordinary systemic actions available to NPC owners, managers, competitors and workers should generally be available to the player through the same underlying authorities. This includes hiring, bargaining, property transactions, supplier competition, credit, market research and - where the legal/crime systems eventually support them - deception, interference or sabotage. The consequences must also be shared.

CANON LOCK - Difficulty may tune decision quality, information search, reaction speed or risk tolerance. It should not give NPCs hidden economic physics unavailable to the player.

2.4 Incarceration Does Not Pause the Economy

If the player is arrested or incarcerated, time continues. Businesses with adequate delegation may continue operating; poorly delegated businesses may miss decisions, consume inventory, fail obligations or deteriorate. Incarceration is not a pause screen.

PART III - LABOR AUTONOMY, CAPABILITY & EMPLOYMENT RELATIONSHIPS

3.1 Employees Are Autonomous Persons

An assignment is not a guarantee. Employees may complete, delay, make mistakes, neglect, partially perform, refuse, abandon, complain, escalate or quit according to real circumstances. Relevant causes can include capability, workload, fatigue, task fit, safety, legality, pay, reliability, fairness, treatment, relationships and available alternatives.

3.2 No Universal Refusal Threshold

Extreme scheduling should create rising fatigue and human response rather than one global cutoff. One Person may tolerate a long day that another refuses. Prior treatment, current need, relationship, danger and alternatives can matter.

3.3 Jobs Are Assignments, Not Rigid Classes

A worker may be assigned several compatible duties and may also be assigned work outside their strongest capability when physically and legally possible. The result should emerge from actual capability, conditions, tools and supervision rather than a Wrong Class prohibition.

3.4 Employment Rapport

Employment and supervisory relationships retain causal history. Fair treatment, pay, scheduling and prior interactions may influence cooperation, disclosure, retention and tolerance for unusual requests. Rapport is not a universal Productivity multiplier.

3.5 Capability & Confidence Presentation

Player-facing capability bands are locked as: Abysmal / Bad / Poor / Average / Great / Excellent / Mastered.

Confidence is Low / Moderate / High. High confidence should normally display one band, Moderate two adjacent bands and Low three adjacent bands. When evidence is insufficient, display Not enough information rather than false precision.

3.6 Skill Changes Outcome Distribution

Skill should shift the probability distribution of plausible task outcomes. Task difficulty, tools, information, fatigue, environment and other relevant conditions still matter. Avoid a universal +X percent Skill bonus.

3.7 Evidence Creates Knowledge

Actual capability remains hidden. Interviews, references, reputation, observed work, practical demonstrations, records and outcomes create evidence from which the player forms an estimate. A practical demonstration is an ordinary task/observation sequence, not a separate magical interview minigame.

3.8 Family Work Is Explicit

Household membership does not silently create free business labor. Spouses, children and relatives may work under explicit employment/family-work relationships where historically and economically appropriate; their time and competing household obligations remain real.

3.9 Advances Do Not Create Labor Ownership

An employee loan, wage advance, lodging benefit or other financial relationship does not automatically create forced labor authority. Debt and employment remain distinct relationships.

PART IV - DELEGATION, FULL AUTONOMY & OWNER DEPENDENCY

4.1 Granular Delegation

Ordinary decisions and tasks may be delegated to qualified people unless ownership, law, specialist authority or the player's explicit reservation prevents it. Delegation can cover individual tasks, categories or standing policy such as inventory counts, reorder limits, receiving, cash reconciliation, scheduling, routine hiring recommendations, minor maintenance, freight coordination, routine bills, inspections and price changes inside limits.

4.2 Management Is Still Labor

A delegated inventory check still consumes the manager's time and information. Travel, conversation, paperwork, calculation and inspection remain real. Small actions may take minutes; accumulated workload can delay decisions.

4.3 Full Autonomy Is a Preset

Full Autonomy is a convenience preset that assigns broad routine authority and standing policy. It does not create a separate abstract manager simulation. Material debt, sale, mortgage, ownership changes or other owner-reserved acts still escalate as appropriate.

4.4 Zero Personally Managed Businesses

A mature portfolio can validly reach zero routinely owner-managed businesses. Crown House operating for an extended period under Tom Harrow without routine owner decisions is the reference playtest validation. The exact duration is calibration; the principle is continuity through real people and policies.

PART V - BUSINESS CAPABILITY, PROPERTY & PHYSICAL SPACE

5.1 Business Form Follows Actual Function

A business's capabilities should emerge from its spaces, equipment, labor, expertise, legal authority, inventory/supply, agreements and market access rather than rigid labels alone. A saloon can therefore also support food service, lodging or gambling when the actual conditions exist.

5.2 Parcel, Building, Functional Space & Business Use Remain Separate

One enterprise may span multiple structures, and one building may contain several functional/economic uses. Preserve the existing Parcel -> Building/Improvement -> Functional Space -> occupancy/business-use separation.

5.3 Building Condition Bands

Use the visible condition bands Critical / Poor / Fair / Good / Excellent. Recommended shared domains are Structural, Exterior, Interior and Functional. More detailed hidden component state may exist where materially useful.

5.4 Access Remodeling & Concealment

Real improvements may create or remove exterior stairs, partitions, sealed access, controlled routes, concealed doors or concealed spaces when geometry and construction permit. Concealment affects discoverability and security; it does not create magical invisibility.

5.5 Cosmetic Improvements Are Causal

Painting or visible repair can change presentation, customer expectations and reputation through observation. Do not encode a direct CustomerDemand +X percent upgrade independent of awareness and experience.

PART VI - MARKETING, MARKET RESEARCH & COMPETITION

6.1 Marketing Is an Information Chain

The core causal chain is: channel -> message -> audience -> exposure -> awareness -> consideration -> choice -> transaction. Newspapers, handbills, signage, workplace solicitation, relationships, promotions and word of mouth can create awareness without guaranteeing purchase.

6.2 Customer Awareness Has Sources

Where useful, customer awareness may derive from prior visits, signage, print, coworkers, employer/foreman relationships, personal recommendation or other real information paths. The player's attribution remains imperfect.

6.3 Competition Is Segment & Daypart Specific

A business can be weak at midday and strong at night, or appeal to workers but not travelers. Competitive analysis should support segment/daypart observation rather than reducing the market to one total customer number.

6.4 Market Research Is a Real Action

The player may ask customers and staff what they observe, request or hear. Answers are Person-based evidence, not omniscient market statistics. Employees may also suggest candidates, suppliers, products, promotions or repairs when they possess a plausible information basis, and those suggestions may be wrong.

6.5 Attractiveness & Persuasion Are Contextual

Physical attractiveness or presentation may influence persuasion where a particular customer is receptive and the social context makes it relevant. It is not a universal sex-based sales bonus. The Saloon Circuit +15 concept remains calibration only.

PART VII - VERTICAL INTEGRATION, INVENTORY & FREIGHT

7.1 Integration Requires Execution

Vertical integration should be evidenced by actual production, owned inventory, physical transfer, destination receipt, transformation/use and a qualifying customer transaction. Ownership of compatible enterprises alone does not prove integration.

7.2 Provenance Matters

When a scenario or analysis depends on origin, the underlying goods/inputs must preserve enough provenance to determine where they came from and how they reached the finished customer-facing product. Distinct-product objectives should count qualifying product definitions, not repeated lots of the same item.

7.3 Internal Sourcing Is Optional

Owning a supplier and customer enterprise does not force internal sourcing. Outside supply may still be cheaper, better, faster or more reliable. Internal transfers preserve real production, transport, spoilage, labor and other cost without manufacturing portfolio revenue.

7.4 Freight Remains Physical

Important freight preserves cargo, origin, destination, vehicle/team, driver, loading, route, time, feed, delay, unloading and cost. Useful operating analysis can distinguish loaded and empty mileage, labor, feed, maintenance, delay, rejected work and utilization.

PART VIII - FINANCE, CREDIT, BUYOUTS & CONTINGENT EXPOSURE

8.1 Historical Credit Is Information-Limited

Avoid a universal modern consumer-credit formula for nineteenth-century business lending. Creditworthiness can depend on character/reputation, repayment history, cash flow, obligations, collateral/security, liquidity, purpose, maturity, lender exposure, lender capital/liquidity, relationships and local conditions.

8.2 Financial Institution Type Matters

Where legal powers differ, the institution's legal/charter type must matter. Do not assume every bank in every jurisdiction/year can make the same loans or hold the same security. Exact banking powers remain a dated research/data question.

8.3 Lenders Are Not Omniscient

A lender learns private obligations through legitimate information sources, diligence and representations. The player or NPC may potentially omit or misrepresent information where the interaction permits; discovery can create refusal, acceleration/default disputes, civil/fraud exposure, reputation damage or other consequences according to law and contract.

8.4 Employee/Manager Buyout Financing

A buyer may finance an acquisition through a third-party lender. The seller may receive cash at closing while separately providing a guaranty/surety or collateral support where historically and legally appropriate. Buyer debt, seller note, retained equity, guaranty and pledged collateral remain distinct structures.

8.5 Guaranties Are Contingent Exposure

A guaranty can remain after the seller owns zero percent of the business. It is real exposure that can affect future credit and become payable when triggered, but it is distinct from current principal debt until the agreement and events make it an actual liability.

8.6 Seller Objectives Affect Deal Choice

A seller seeking liquidity, certainty and speed may rationally accept a lower all-cash bid over a higher nominal seller-financed offer. Employee buyers are not entitled to artificial preference over better outside offers.

8.7 Working Cash Remains Where Obligations Exist

Aggressive owner debt repayment does not justify stripping operating businesses below required payroll, supplies or known obligations. There is no universal $150 or other fixed owner/business reserve; protected liquidity depends on real commitments and strategy.

PART IX - SALOON, GAMBLING & INDEPENDENT PREMISES RELATIONSHIPS

9.1 Saloon Capabilities Reuse Shared Systems

Food service, lodging, poker, faro and other historically supported activities should emerge from shared Person, Premises/Space, Task, Inventory/Asset, Agreement, Transaction, LegalRegime and Information systems rather than isolated minigames.

9.2 Free Poker

A saloon may offer ordinary poker without rake or table fee. Such a game creates no direct gaming revenue; any commercial value arises through real customer attraction, dwell and food/drink purchases.

9.3 Faro Banking & Concessions

Faro is a banking game and requires real bankroll exposure. A saloon may contract with an independent dealer/banker who supplies bankroll and keeps game wins/losses while the premises owner supplies space/services and receives concession rent and ordinary food/drink revenue. Exclusivity, licensing responsibility and duration are real agreement terms.

9.4 Occupancy Does Not Automatically Create Employment

An independent worker using rented business-controlled space is not automatically an employee. Premises/occupancy, employment and business ownership remain separate relationships. This is especially important for historically supported vice-related room arrangements.

PART X - RECORDS, MONEY & INFORMATION PRESENTATION

10.1 Known Facts Stay Exact

When the player legitimately knows a transaction amount, cash balance or recorded payment, present the exact known fact. Uncertainty belongs to estimates, forecasts, hidden capability and unknown competitor information rather than recorded facts.

10.2 Exact Cents Remain Authoritative

Compact UI may suppress cents or use abbreviated displays, but affordability and all economic authority use exact underlying values. Display rounding can never make an unaffordable purchase affordable.

10.3 Competitor Recon Uses Observable Information

The player may observe customer counts, prices, signage, hours, public records, reputation and other legitimate evidence. Hidden competitor ledgers remain hidden unless a real information path reveals them.

PART XI - LAW, ENFORCEMENT & ECONOMIC CONSEQUENCES

11.1 Legal Rule and Enforcement Remain Separate

An activity can be illegal yet weakly enforced or locally tolerated, or lawful/licensed yet heavily scrutinized. Preserve the chain legal rule -> enforcement posture/capability -> detection -> proceeding -> consequence rather than Illegal = instant arrest.

11.2 Crime Can Propagate Into Economic State

Where crime, sabotage or property damage systems exist, their real consequences can affect rival capacity, inventory, opening status, reputation, finances and customer choice. Scenario evaluators should consume the resulting authoritative economic state rather than special-case whether the designer intended the route. Crime still carries evidence, investigation, legal, reputation, restitution/insurance and retaliation consequences where applicable.

PART XII - SALOON CIRCUIT SCENARIO LOCK

12.1 Scenario Identity

SALOON CIRCUIT - THREE TOWNS is a canonical scenario concept built around progression through increasingly complex saloon markets. This revision locks the tested completion structure but not every generated economic number, settlement population or exact historical start year.

12.2 Final Completion Structure

- Historical milestone: 100 percent Ash Creek saloon ownership plus qualifying customer sale by the authored deadline.
- Historical milestone: 100 percent Prairie Crossing saloon ownership plus qualifying customer sale by the authored deadline.
- Historical milestone: qualifying sale using product/input supplied by another owned enterprise.
- Historical milestone: 100 percent Crown House ownership plus first qualifying customer sale.
- Historical milestone: three distinct internally supplied Crown customer products sold.
- Historical milestone: five distinct internally supplied Crown customer products sold.
- Final-state predicate: Crown House reaches authored #1 standardized Granite Junction saloon traffic rank.
- Final-state predicate: player owns 100 percent of Ash Creek, Prairie Crossing and Crown House.
- Final-state predicate: qualifying scenario principal debt equals $0.00.

The final rank, ownership and debt predicates re-evaluate current state. The historical customer-sale and integration milestones remain completed after legitimate execution. Supporting farms, breweries, freight enterprises or other suppliers therefore need not remain owned after their event milestones unless a variant explicitly requires retention.

12.3 Strategic Freedom

The scenario should not prescribe which supporting enterprises must be owned, which managers must be hired, which customer segment must dominate or whether Crown wins through meals, lodging, gambling, service, marketing or another valid combination. Objectives constrain outcomes; the shared simulation determines routes.

12.4 Scenario Calibration Holds

- The 1875 playtest start does not yet replace the broader project opening-date doctrine. Exact supported earlier start remains a design/research hold.
- Settlement population ranges, road distances, traffic thresholds and the observation window required to establish #1 rank remain scenario calibration.
- All exact wages, prices, sale values, loan rates, project costs, profit margins, concession terms and room rents from the playtest remain calibration unless separately researched and locked.

PART XIII - AUTONOMOUS/TEXT PLAYTEST STATE DISCIPLINE

- Never invent missing cash or silently reconcile incompatible balances. Label unresolved state.
- Never silently create inventory. Physical products require legitimate source and provenance.

- Never silently delete debt, guaranties, leases, seller notes or other obligations.
- Do not confuse profit with cash, business cash with owner cash, ownership with management authority, effective control with 100 percent ownership, household membership with employment or occupancy with ownership.
- After a player decision, execute the decision and advance to the next consequential date/event unless a material exception or choice requires intervention.

PART XIV - RESEARCH & CALIBRATION HOLDS

- Supported opening year earlier than the current ordinary 1880 recommendation.
- Saloon/server/cook/freight/farm/brewery compensation bundles by place, date and occupation.
- Commercial meal ingredient/labor economics and small-brewery capacity/cost/wholesale economics.
- Bank charter/institution powers, guaranty/surety practice and enforcement by jurisdiction/date.
- Faro legality/licensing/enforcement and prostitution/room-use law by jurisdiction/date.
- Construction pacing, marketing effectiveness, traffic-rank evidence windows and simulation-speed presentation.
- Any numeric attractiveness/persuasion magnitude; no universal +15 modifier is authorized.

PART XV - REVISION XXI LOCK SUMMARY

- Objective semantics explicitly distinguish historical event, current-state and maintained-state predicates.
- Historical event milestones persist after enabling assets are sold unless retention is explicit.
- Employees may refuse, fail or quit; jobs are flexible assignments; capability knowledge is evidence-based.
- Full Autonomy is granular delegation, not magic automation; zero personally managed businesses is a valid mature state.
- Building condition uses Critical/Poor/Fair/Good/Excellent across Structural/Exterior/Interior/Functional domains.
- Marketing creates awareness rather than guaranteed demand; competition can be segment/daypart specific.
- Vertical integration requires a real provenance chain through physical logistics and customer sale.
- Debt, guaranty, collateral, seller note and retained equity remain distinct financial concepts.
- Saloon gambling/vice-related arrangements reuse shared Person, Agreement, Premises, Transaction and Legal systems.
- Known transaction facts and exact cents remain authoritative despite compact presentation.
- The Saloon Circuit final objective structure is locked while its exact generated economic values remain calibration.

END OF REVISION XXI INTEGRATION

The preserved Revision XX document follows unchanged and remains authoritative except where Revision XXI explicitly adds, deepens, clarifies or supersedes a rule.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XX

Autonomous Enterprise Ladder Playtest Integration: Scenario Geography, Transferability, Corporate Control & Completion Presentation

Revision date: September 8, 2026

Source baseline: Canonical Project Bible Revision XIX - September 7, 2026

Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XX preserves Revision XIX in full except where this integration explicitly adds, deepens, clarifies or supersedes prior wording. The complete Revision XIX document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XX controls. Exact autonomous-playtest prices, wages, traffic, project costs, interest rates, holding periods, map shares and narration timing remain calibration unless this revision explicitly identifies a design lock.

PURPOSE

Revision XX integrates the autonomous Enterprise Ladder systems playtest and the subsequent scenario/world-scale design notes. The autonomous campaign intentionally used a different business path from the prior full playtests and therefore tested whether the scenario doctrine survives alternate player strategy. The result reinforces the existing economic foundation while adding stronger rules for geographically bound objectives, very large multi-settlement maps, pre-operating major enterprises, relationship transferability, internal segment reporting, effective control, lifetime attribution and scenario-completion presentation.

REVISION XX MASTER DOCTRINE

- Scenario objectives may be bound to a designated on-map settlement, district, corridor or other geographic scope. Satisfying the same economic condition elsewhere does not qualify when geography is part of the authored objective.
- The playable map is intended to support multiple meaningful settlements and a very large on-map economy. Off-map remains meaningful, but it is not a substitute economy that bypasses physical truth.
- The Enterprise Ladder is defined by scale, transferability and ownership problems rather than by one required sequence of industries.
- Effective control, economic ownership, voting rights, board authority, management office and creditor claims remain distinct.
- A major enterprise can become the player's controlled enterprise during development before ordinary revenue begins.
- A relationship may fail to transfer when a business changes hands even when physical assets and legal ownership transfer successfully.
- A consolidated enterprise may use internal segment reporting without manufacturing internal revenue or creating fake separate businesses.
- Scenario completion should acknowledge the player's economic arc and lasting world impact without reciting a checklist or assuming one specific business route.

PART I - SCENARIO GEOGRAPHY & VERY LARGE WORLD DOCTRINE

1.1 Designated Scenario Settlement

A scenario may define one or more designated settlements, districts, corridors or regions as part of its objective authority. The Enterprise Ladder may therefore require a qualifying enterprise, property, development or disposition to exist inside the named scenario settlement rather than anywhere on the map.

Geography is an authored constraint, not a universal requirement. Open Frontier remains open-ended, and individual scenarios decide whether location is part of qualification.

CANON LOCK - A scenario objective is not satisfied merely because its economic numbers are met somewhere else when the objective explicitly names a geographic scope.

1.2 Geographic Qualification Can Combine With Economic Qualification

Scenario conditions may combine location with any other legitimate requirement: ownership of the property, debt state, operating history, sale value, control transfer, population, infrastructure connection or other persistent fact. The geographic requirement should be evaluated against the same authoritative world state as the economic requirement.

- Example structure: qualifying enterprise is located in the designated town + player owns the required property + debt condition is satisfied + disposition condition is satisfied.
- Do not expose geography as a cosmetic objective label if the player can actually satisfy the rule elsewhere.

1.3 Multi-Settlement Playable Map

The intended world may contain several substantial on-map towns, smaller settlements, rural clusters, industrial sites and corridors in the same campaign. A scenario can focus attention on one primary settlement while other on-map settlements continue to trade, migrate, compete, invest and develop under normal simulation rules.

1.4 On-Map Primacy With Off-Map Continuity

The design direction is that the overwhelming majority of ordinary business activity should occur on the playable map when practical. Off-map remains essential for distant suppliers, regional markets, migration, correspondence, finance, travel beyond the represented region and wider network connections.

This is a qualitative world-design priority, not a locked 95 percent or 99 percent numerical requirement. The exact balance depends on final map scale, performance and business scope.

CANON LOCK - Off-map simulation remains part of the same causal economy. It may reduce representation detail but must not create a second set of economic physics.

PART II - ENTERPRISE LADDER AUTONOMOUS-PLAY CLARIFICATIONS

2.1 Business Path Is Not Predetermined

The three Enterprise Ladder rungs describe increasing organizational and capital complexity, not a required bakery, freight, bank, railway or other fixed industry sequence. A valid completion can use materially different businesses so long as the authored rung requirements are met.

2.2 Pre-Operating Major Development Can Occupy the Enterprise Slot

A controlled major-enterprise development company can count as the player's one primary operating enterprise before ordinary external revenue begins when the player has formed the entity, committed meaningful capital, obtained control and begun real development toward operation.

This prevents the one-enterprise rule from being evaded by calling a railway, mine, utility or industrial project merely a pre-operating project while simultaneously controlling an unrelated operating company.

CANON LOCK - Enterprise Ladder control restrictions follow substantive control and committed enterprise development, not only whether the company has booked its first sale.

2.3 Control Transfer Governs Disposition

A seller note, retained receivable or other creditor claim does not by itself preserve operating control. Conversely, selling some shares or resigning from an office does not qualify if the player retains effective governance or operating control.

Qualifying disposition occurs when the authored scenario's required control has genuinely transferred through the applicable ownership and governance structure.

2.4 Control-Block Transactions Are Valid

A major-company disposition may be a whole-company acquisition, merger or control-block transaction rather than a sale of every outstanding share. Minority investors may remain after the player's control ends. The player receives only the economic consideration attached to the interest actually owned or otherwise lawfully conveyed.

2.5 Sale Readiness Is Evidence, Not Certification

Enterprise Evaluation may communicate operating history, customers, staff, management coverage, records, working capital, relationship continuity and owner dependency. It should not become an opaque percentage that magically certifies a business as sellable.

An actual buyer, financing structure, diligence result and control transfer remain necessary where the scenario calls for a disposition.

2.6 Pre-Opening Commercial Discovery

A founder may approach prospective customers, suppliers, carriers, lenders or other counterparties before committing all startup capital. Expressions of interest, quotes, letters, conditional commitments and negotiated preliminary agreements can provide legitimate evidence about demand and feasibility.

Interest is not guaranteed demand. A non-binding conversation cannot become future revenue merely because the player asked first.

CANON LOCK - Founding a business should not require either blind capital expenditure or accidental omniscience. Legitimate pre-opening discovery is allowed, but commitment remains distinct from fulfillment.

PART III - TRANSFERABILITY, RELATIONSHIPS & OWNER DEPENDENCY

3.1 Relationship Transferability

A change of business ownership does not automatically transfer every relationship surrounding the enterprise. Supplier terms, customer arrangements, leases, licenses, service agreements, personal trust and informal credit can have different continuation rules.

- Automatically continuing under valid terms.
- Assignable by the current holder.
- Requires counterparty consent.
- Terminable or renegotiable on change of control.
- Likely to continue but not contractually guaranteed.
- Strongly personal to a departing owner, manager or worker.
- Unknown until diligence or negotiation produces evidence.

CANON LOCK - Buying the enterprise does not teleport another person's trust or consent to the buyer.

3.2 Owner Dependency Is a Capability Map

Acquisition and sale diligence should ask what disappears if the owner leaves. Owner dependency can include physical labor, technical expertise, purchasing judgment, management, customer relationships, supplier relationships, reputation, local knowledge, records knowledge and informal coordination.

A replacement wage alone can understate the problem when the departing owner supplied several different capabilities or relationships.

3.3 Technical Leadership and General Management Remain Distinct

A skilled operating lead can control a technical domain without automatically becoming the enterprise's commercial or financial manager. Mature enterprises may legitimately divide authority among technical superintendents, commercial managers, clerks and general management according to workload and competence.

3.4 Employee and Manager Succession

The existing management/employee buyout doctrine is reinforced. A worker can progress from employee to delegated authority to prospective buyer to independent entrepreneur when financing, capability and personal choice support the transition. The seller is not required to prefer an employee buyer over a stronger external offer.

3.5 Sold Enterprises Persist as Economic History

A sold enterprise remains available to become a future supplier, customer, competitor, borrower, employer or acquisition target. Ownership change should enrich world history rather than erase prior player work.

PART IV - CONSOLIDATED ENTERPRISE REPORTING

4.1 One Enterprise Can Have Internal Segments

A genuinely consolidated enterprise may maintain diagnostic reporting by site, department, route, product family, operating function or other meaningful segment while retaining one authoritative enterprise treasury and external ledger where the legal/economic structure requires it.

4.2 Segment Reporting Does Not Create Internal Revenue

Internal segment reports may attribute cost, workload, inventory usage, margin contribution and capital consumption for diagnosis. They must not manufacture external revenue, duplicate cash or create profit merely because value moved from one part of the same consolidated enterprise to another.

CANON LOCK - Internal reporting is analytical attribution inside one enterprise, not permission to create a second economic truth.

4.3 Segment Reports Should Explain Bottlenecks

The purpose of internal reporting is to answer practical ownership questions: which site consumes working capital, which function is capacity-constrained, which product or route earns contribution, where labor is concentrated and whether integration is actually helping. It is not a requirement for formal modern management-accounting detail.

PART V - MAJOR ENTERPRISE DEVELOPMENT, CAPITAL & STAGING

5.1 Generic Development Stage

The railway development chain from Revision XIX establishes a broader pattern for capital-intensive enterprises. Mines, utilities, industrial plants, large property concerns and other major ventures may have meaningful investigation, promotion/formation, development, capital assembly, construction/setup, commissioning and operation stages before mature operation.

5.2 Physical Commissioning Is Not Standalone Economic Viability

A project stage can be physically complete, legally usable and capable of serving customers while still failing to cover its operating, maintenance and financing burden as an independent business. Revenue from an early stage may help the next stage, but the game must not assume that it will.

CANON LOCK - Physical staging and economic staging are separate facts.

5.3 Outside Capital Must Bring Outside Rights

The player may legitimately control a major enterprise while personally funding only a minority of its total capital. That scale transition is allowed only when outside investors and creditors have real claims, governance, covenants, repayment priority, information rights or other period-appropriate protections. Outside capital is not free owner leverage.

5.4 Redundancy Has Contextual Economic Value

Spare or reserve equipment can look inefficient during normal operation yet become valuable during failure, peak demand or maintenance. The value should emerge from avoided lost capacity and continuity, not from an abstract Backup Equipment bonus.

PART VI - LIFETIME LEDGER & CORPORATE ATTRIBUTION

6.1 Enterprise Activity Versus Owner Activity

As outside shareholders become important, the Lifetime Ledger must distinguish the activity of enterprises the player controls from the economic income actually attributable to the player. Controlling a large corporation does not make its entire revenue the player's personal income or wealth.

- Controlled-enterprise external revenue.
- Controlled-enterprise operating profit.
- Owner distributions/dividends actually received.
- Interest personally received.
- Realized disposition proceeds and gains/losses.
- Value of the player's actual equity or creditor claim.
- Enterprise Value and debt remain separate from owner Net Worth.

6.2 Lifetime Ledger Is Persistent Scenario Authority

The autonomous text playtest again showed that cumulative taxes, payroll and other lifetime statistics become unreliable when reconstructed from narrative summaries after the fact. The canonical Lifetime Ledger should be maintained continuously from authoritative transactions and events when a scenario depends on historical accumulation.

6.3 Scenario Pacing Is Calibration

Outside equity and debt can allow a skilled player to reach major-enterprise scale far earlier than a personal-cash-only progression. That is not inherently an exploit. The intended pace of a specific scenario should be calibrated deliberately through opportunity availability, investor standards, financing, governance, risk and objective design rather than by forbidding outside capital.

PART VII - SCENARIO COMPLETION PRESENTATION

7.1 Completion Should Summarize the Arc

A completed scenario may end with a short cinematic, narration and result presentation that acknowledges what the player changed in the world. For The Enterprise Ladder, the emotional arc is small ownership -> larger organization -> major enterprise -> lasting economic legacy.

7.2 Narration Must Remain Route-Agnostic

Because players can satisfy the scenario through different industries and financing structures, standard completion narration should describe the scale and meaning of the accomplishment rather than naming a bakery, freight company, railway or other path that may not have occurred.

7.3 Do Not Read the Objective Checklist Back to the Player

Completion presentation should not sound like a report that lists every fulfilled condition. The objective UI already serves that function. The ending should provide closure, acknowledge the persistent world and hand control back to the player or campaign flow.

CANON LOCK - Scenario completion presentation is recognition of the player's economic story, not a narrated checklist.

PART VIII - CALIBRATION & RESEARCH HOLDS

8.1 Map Activity Share

The planning estimate that perhaps 95-99 percent of ordinary business activity should be on-map is not a locked numerical rule. Preserve only the qualitative priority of on-map primacy until final world scale and performance are proven.

8.2 Completion Cinematic Timing and Voice

Exact narration length, camera duration, voice actor/accent, music, subtitle timing and final cinematic structure are production calibration, not Canon. A shorter roughly half-minute target is currently favored for prototyping but remains subject to player-attention testing.

8.3 Corporate and Railway Law

Revision XIX research holds concerning railway corporate law, control transactions, right-of-way, interchange, shareholder procedure and related-party conduct remain active. Revision XX does not invent missing period law.

PART IX - REVISION XX LOCK SUMMARY

9.1 New Locks

- Scenario objectives may be geographically bound to designated on-map places.
- A very large map may contain multiple meaningful settlements; off-map remains the same causal economy at reduced representation.
- Enterprise Ladder phases are structural and do not require one fixed industry path.
- Substantive control and committed development can occupy the one-enterprise slot before revenue begins.
- Control transfer, not note repayment or job-title change, determines disposition.
- Relationship continuation after sale is not automatic.
- Owner dependency includes capability, knowledge and relationships, not just labor hours.
- Consolidated-enterprise segment reporting must not create fake internal revenue or cash.
- Physical project staging and standalone economic viability remain distinct.
- Controlled-enterprise activity and owner-attributable income remain distinct in the Lifetime Ledger.
- Scenario-ending narration should recognize the arc and remain compatible with alternate valid routes.

END OF REVISION XX INTEGRATION

The preserved Revision XIX document follows unchanged and remains authoritative except where Revision XX explicitly adds, deepens, clarifies or supersedes a rule.

LAND & LEDGERS

CANONICAL PROJECT BIBLE - REVISION XIX

Third Full Playtest Integration: Enterprise Ladder, Rail Capital, Retail Labor, Historical Events & New Business Research

Revision date: September 7, 2026

Source baseline: Canonical Project Bible Revision XVIII - August 30, 2026

Status: Active Canon integration

PRESERVATION GUARANTEE

Revision XIX preserves Revision XVIII in full except where this integration explicitly adds, deepens, corrects, parks or supersedes prior wording. The complete Revision XVIII document follows this integration unchanged as the preserved baseline. Where a conflict exists, Revision XIX controls. Exact playtest dollar amounts, wages, traffic thresholds, construction costs, interest rates and durations remain calibration unless this revision explicitly identifies a historical or design lock.

PURPOSE

Revision XIX integrates the third and longest full text systems playtest, The Enterprise Ladder. The playtest ran from Spring 1880 through June 1890 and moved through three materially different enterprises: Merritt Bakery, Dakota Union Grain Company and Dakota Gateway Railway Company. It validated existing Canon authorities around workload, delegation, Physical Truth, working capital, projects, logistics, records, information and business disposition while exposing major additions needed for action discoverability, task granularity, dynamic liquidity, corporate finance, staged railway development, historical event delivery, General Store labor, barter, early telephone use, saloon/vice systems and reward/private-detective business design.

The common rule remains causal simulation over abstract game bonuses. Complexity is welcome when the systems are connected, discoverable and materially consequential. Complexity is not an excuse for hidden mechanics, repetitive owner clicks or isolated novelty features.

REVISION XIX MASTER DOCTRINE

- The player should virtually always be able to find something meaningful to do, but should not be forced to personally decide everything that could be done.
- Feature breadth is desirable when features reuse shared economic, physical, legal, labor, information and management systems.
- Mechanics should be discoverable through ordinary play, UI explanation, visible prerequisites, causal consequences and manager/contractor recommendations rather than outside guides.
- Routine execution belongs to workers, managers, contractors and standing policy. Owner attention belongs to strategy, exceptions, capital allocation, negotiation and decisions deliberately reserved by the player.
- Projects, agreements, tasks, shipments, legal rights, corporate claims and historical events remain persistent world state. A summary screen is never the source of truth.
- Historical occurrence, not modern moral comfort or legality alone, determines whether an economic activity can exist in the simulated world. Illegal activity remains illegal and exposed to the applicable enforcement system.
- Exact numbers from the playtest remain calibration unless independently established as a dated historical rule.

PART I - ENTERPRISE LADDER & SCENARIO DOCTRINE

1.1 Scenario identity

The Enterprise Ladder remains a canonical scenario family. Its core loop is:

Build value -> prove the enterprise works -> transfer control -> reinvest into a materially larger enterprise.

The intended learning progression is not merely bigger revenue. Each rung should introduce qualitatively different ownership problems.

1.2 One primary operating enterprise

The standard scenario uses one primary active operating enterprise at a time. Passive investments, seller notes, property holdings, securities and other non-operating interests may persist where the scenario allows them.

A Consolidated Enterprise may contain several divisions, locations or legally distinct controlled ledgers when they are genuinely integrated into one operating strategy. Merely placing unrelated businesses under one administrative screen does not make them one enterprise for scenario purposes.

1.3 Value creation before disposition

A qualifying Enterprise Ladder disposition requires meaningful value creation or organizational improvement rather than passive appreciation alone. Valid value creation can include profitable expansion, stronger customer relationships, added productive capacity, improved records, improved property, better logistics, lower risk, improved management continuity, new market access or other causal improvements.

1.4 Qualifying going-concern disposition

A qualifying disposition should normally demonstrate:

- real operations and an operating history;
- genuine outside customers or revenue where the business model requires them;
- required employees, contractors, suppliers and service relationships;
- adequate records for a serious buyer to understand the business;
- no hidden owner subsidy disguising an economically broken enterprise;
- a legitimate buyer, merger partner or other counterparty;
- actual transfer of control;
- a realized economic outcome for the seller after relevant costs.

A formal manager is not universally required. Transferability is the real test. A scenario may add a stricter autonomy objective, but that objective is scenario-specific rather than a universal law of business sale.

1.5 Enterprise Evaluation presentation

A simple player-facing Enterprise Evaluation may summarize:

- PROFIT
- CUSTOMERS
- OPERATIONS
- TRANSFERABILITY
- ESTIMATED SALE VALUE

These are read models built from real evidence. Do not replace the underlying systems with a single Business Quality score.

1.6 Scenario completion at scale

The final enterprise does not have to exhaust every future growth path. A railway does not need to reach every intended destination merely because additional profitable expansion remains possible. The scenario can complete once the enterprise has been materially built, proven, made transferable and legitimately transferred.

1.7 Corporate completion

If the player owns only a minority economic interest, the player cannot personally sell the entire corporation. A qualifying final transfer can occur through a merger, whole-company acquisition, stock transaction or other valid corporate disposition requiring the appropriate governance and creditor treatment.

PART II - PLAYER ROLE, COMPLEXITY & DISCOVERABILITY

2.1 High Action Visibility, Low Mandatory Interruption

New hard design doctrine:

The player should almost always be able to discover meaningful actions, but available actions should not become mandatory interruption spam.

At a business, project, property or institution, the player should be able to understand:

- what can be done;
- why the action might matter;
- what it requires;
- what it may accomplish;
- what uncertainty remains;
- what happens if it is ignored;
- whether it can be self-managed, delegated or subcontracted.

2.2 Feature breadth doctrine

Revision XIX supersedes any interpretation that feature count itself is a design problem. Large feature breadth is acceptable and desirable when features are:

- causal;
- interconnected;
- economically meaningful;
- reusable across systems;
- visible enough to learn through play;
- compatible with the scale-transformation and delegation doctrines.

Reject isolated novelty mechanics that do not affect the wider simulation. Do not reject a deep mechanic merely because the project already contains many systems.

2.3 Discoverability requirement

The game should not require an external wiki to reveal basic possibilities such as subcontracting a railway survey, using public-land right-of-way, arranging a freight contract, checking an operating bottleneck, hiring a manager or examining a business for sale. UI, reports, world evidence, NPC advice, manager proposals and workflow surfaces should reveal legitimate options when the player has enough information to know they exist.

2.4 Text-playtest option count is not Canon

The repeated use of exactly nine strategic options was a text-playtest format. It is not a final-game rule.

PART III - TASKS, WORK & FAILURE

3.1 Meaningful task granularity

Business execution should not collapse a whole production chain into one abstract task when intermediate work has real labor, equipment, timing, quality or failure consequences.

A bakery may generate meaningful tasks such as retrieve ingredients, measure, mix, knead, proof, shape, prepare oven, bake, unload, cool, package, stage, load and deliver. Other industries should follow the same principle.

The target is meaningful granularity, not animation granularity. Do not turn every hand motion into a simulation object.

3.2 Continuous time and parallel work

Tasks use the authoritative continuous clock. A 37-minute task finishes 37 minutes later, not at the next arbitrary hourly or daily tick. Independent workers, equipment and work areas can proceed in parallel when prerequisites and resources permit.

3.3 Task failure inputs

Task outcome can depend on relevant:

- capability and experience;
- difficulty;
- fatigue;
- supervision;
- equipment and material condition;
- environment and weather;
- time pressure;
- information quality;
- other domain-specific causes.

3.4 Failure creates consequences, not generic penalties

Task failures should produce domain-specific results such as rework, delay, material loss, batch loss, weak quality, injury, property damage or fire where causally justified. Do not resolve every failure through a generic percentage debuff.

A failure can create actual follow-up work. A repairable mistake should create the corrective task rather than merely reducing Efficiency.

PART IV - MANAGEMENT, DELEGATION & AUTONOMY

4.1 Manager titles do not impose hierarchy

Manager and General Manager are descriptive authority roles, not RPG tiers.

A reasonable naming convention is:

- Manager: routine authority over one business, site or operating unit.
- General Manager: routine authority across several businesses or substantial operating units.

This does not require a General Manager to have subordinate managers. One capable manager can cover several units when workload, geography, expertise and decision volume allow it.

4.2 Management remains real labor

Management consumes finite attention. More businesses, locations, workers, contracts, exceptions and travel create real administrative burden. The player can personally manage when workload permits and can avoid management payroll until the economics justify delegation.

4.3 Manager optimization

Capable managers may improve schedules and operating practice by changing actual decisions such as start times, recurring task timing, delivery windows, assignment order, equipment sequencing, stock timing and customer service windows. The benefit comes from better execution, not a Management Efficiency multiplier.

4.4 Transferability audit

When preparing a business for disposition, ask:

If the current owner disappeared tomorrow, what consequential decisions would stop being made?

The answer identifies genuine authority gaps. Fix them through policy, staffing, governance or other real continuity measures rather than forcing a manager checkbox.

4.5 Autonomy testing

A business may prove transferability through a period in which the owner takes no routine operating decisions. The exact duration is scenario calibration. The test succeeds only if routine decisions continue, obligations are met and the enterprise remains stable without hidden owner intervention.

4.6 Manager Activity

Add a player-visible Manager Activity concept separate from the financial ledger. It answers:

What did delegated people decide or handle so the owner did not have to?

Useful information can include date/time, manager, business/site, decision, trigger, standing policy or authority, outcome, whether an escalation occurred and directly measurable financial impact where that impact is defensible.

Do not fabricate counterfactual dollar savings for every decision. When value cannot be measured honestly, report the activity without an invented number.

PART V - BUILDING DEFECTS, INSPECTION & KNOWLEDGE

5.1 Ordinary defects can be discovered through use

Workers using a building can notice observable problems such as leaks, damaged floors, sticking doors, obvious chimney trouble, visible deterioration or other symptoms. Ordinary property should not require a generic monthly Building Inspection simply to reveal what normal occupants would notice.

5.2 Specialist inspection remains valid

Hidden or high-risk systems can require specialists, including boilers, electrical systems, water systems, mines, structural questions, rail infrastructure and other technically difficult domains.

Inspection changes knowledge. Repair changes physical state.

5.3 Poor records reduce information, not reality

Bad bookkeeping should produce missing, stale, incomplete or unreconciled information. It should not randomly fuzz facts that were actually known. Better records improve Recorded State and decision support; they do not change Actual State by magic.

PART VI - ANIMALS & BIOLOGICAL WORK ASSETS

6.1 Preventive animal care

Working animals require real feed, water, rest and ordinary observation. Depending on species and use, recurring work can include grooming, hoof observation, harness checks, condition observation and other practical care.

6.2 Specialist care

Farriers and veterinarians may operate on-map or off-map. Travel, call-out burden and group visits can matter. A standing service agreement does not magically service every animal; actual animal-level or cohort-level work still occurs.

6.3 Persistent animal state and imperfect information

Animals retain biological health, fatigue, injury, feed/nutrition, disease exposure, age and relevant temperament/behavior. Inspection improves knowledge rather than revealing perfect hidden truth. Seller descriptions can be truthful, mistaken or dishonest.

Player-facing descriptions can map internal continuous values into understandable bands, but exact band boundaries remain calibration.

6.4 Working cats and dogs

Where historically and practically supported, cats can reduce actual rodent pressure and dogs can contribute to guarding, herding or ranch work. Their value must arise from real tasks and conditions, not passive magical bonuses.

PART VII - B2B, SUPPLIERS & PRODUCT ECONOMICS

7.1 B2B relationship workflow

Preserve a full business-to-business commercial loop:

identify -> approach -> learn need -> propose -> negotiate quantity/price/cadence/credit/delivery -> accept/refuse/counter -> perform -> renew/end.

The player-facing UI should explain B2B rather than assume the acronym is universally understood.

7.2 Item-level sourcing

Different merchandise can legitimately use different suppliers. Supplier evaluation can consider invoice price, landed cost, quality, minimum order, credit terms, lead time, freight responsibility, reliability, seasonal capacity and origin.

7.3 Consolidated freight economics

A supplier with a higher invoice price can still be economically superior when consolidated freight lowers landed cost. Freight burden must be allocated using real transport/storage/handling burden rather than arbitrary equal division.

7.4 Capital turnover

Margin percentage alone is not enough. Faster payment, shorter inventory holding periods and better capital turnover can make a lower-spread relationship more profitable than a slower high-margin buyer.

Product and buying reports may therefore distinguish invoice cost, landed cost, current selling price, target/minimum margin, financing cost, inventory age and sales velocity.

7.5 Pricing changes customer behavior

Higher prices can cause customers to buy from local rivals, travel, source off-map, substitute, postpone or stop buying. Sustainable profit is the objective; maximizing margin percentage is not.

PART VIII - LIQUIDITY, LINE OF CREDIT & TREASURY

8.1 Dynamic liquidity requirement

Revision XIX deepens the Operating Reserve doctrine. Do not treat reserve policy as a universal fixed cash floor.

A business should protect the cash actually needed for foreseeable near-term obligations and risk, including payroll, supplier bills, debt service, committed purchases, transport/animal needs, project draws, seasonal requirements and a justified contingency.

A reserve can rise before harvest or a major payment and fall afterward. Protected liquidity is a policy/read model derived from real commitments, not a separate money pool.

8.2 Line of Credit

Line of Credit becomes a first-class credit facility where historically appropriate.

A Line of Credit should conceptually track:

- lender;
- authorized limit;
- outstanding amount;
- undrawn availability;
- rate or discount terms;
- purpose;
- collateral/security;
- draw and repayment history;
- maturity or renewal;
- covenants or lender discretion.

Unused availability is neither cash nor debt. Financing cost should reflect actual time outstanding rather than treating an annual rate as a cost on every inventory turn.

8.3 Owner-capital backstop

A standing policy may allow owner capital to support an owned business if projected liquidity becomes unsafe. Material capital calls should still escalate rather than silently emptying the owner's personal funds.

8.4 Debt repayment policy

Highest-effective-rate debt repayment was successful in the playtest and is a valid player policy. It is not a universal Canon mandate. Other strategies can be rational depending on liquidity, covenants, maturity, collateral, opportunity cost and risk.

PART IX - BUSINESS SALES & CONTINUITY

9.1 Cash inclusion is an explicit transaction term

Sale of a business does not silently transfer all business cash. The agreement must state what cash, working capital, receivables, payables, inventory and other assets or obligations are included.

A buyer must have sufficient working capital to operate after closing, whether that capital comes from included cash, the buyer's own funds, financing or another explicit arrangement.

9.2 Continuing contractual rights after sale

A seller may retain a limited contractual right or service arrangement after disposition where legally and commercially valid. Such a right must be explicit, bounded and distinct from retained operating control. The bakery employee-meal arrangement from the playtest is an example of the structure, not a universal clause.

9.3 Employees after sale

Employees remain independent Persons. The buyer may offer continued employment; workers may accept, negotiate, refuse, leave or retire. Business ownership does not transfer ownership of people.

PART X - RAILWAY CAPITAL LADDER

10.1 Low-entry railway path

Railway ownership should not require the player to begin with a giant trunk-line construction project. Historically plausible entry can include:

lease/access -> operate -> retain earnings -> selectively own rolling stock -> buy track/facilities -> improve terminals -> build short extensions -> build larger regional corridor.

This is a possibility set, not a mandatory linear technology tree.

10.2 Leasing is strategic

Track access, terminals, locomotives, cars and backup equipment can be leased or contracted where historically supported. Ownership is justified by economics and strategic control, not by an abstract progression reward.

Emergency equipment access can be superior to owning an underutilized backup asset.

PART XI - RAILWAY PROJECT DEVELOPMENT

11.1 Railway development chain

A serious railway corridor can involve:

corridor selection -> reconnaissance/traffic study -> route alternatives -> engineering/survey -> alignment/gauge decisions -> anchor customers/traffic commitments -> connection/interchange arrangements -> finance -> right-of-way/control -> construction -> facilities/equipment -> opening service -> operating history -> later extension.

The exact sequence can overlap where reality allows it.

11.2 Specialist work

Small railways can hire outside engineering, traffic, land, legal and financial specialists rather than permanently employing every function. Prime contractors may coordinate several specialist tasks when the contract scope genuinely covers them.

11.3 Financeability and Economic Viability

New hard Canon lock:

Financeability asks whether enough capital can be assembled to build the project.

Economic Viability asks whether the completed railway can generate enough value to support operation, maintenance, financing and continued reinvestment.

A project can pass one gate and fail the other. Both must remain visible.

11.4 Commitments are not track

Surveys, options, rights, traffic commitments and financing intentions are valuable development state but do not become operational railway until the physical and legal work is completed.

PART XII - STAGED RAILWAY CONSTRUCTION & ECONOMIC NODES

12.1 Build in economic stages

A railway corridor can be constructed and opened in sequential segments. The company does not have to complete the final intended destination before earning revenue from an already useful section.

12.2 Intermediate Service Nodes

A stage can terminate at a real economic node such as:

- town or settlement;
- major farm cluster;
- stock-loading point;
- grain concentration;
- mill or mine;
- timber camp;
- wagon-road junction;
- industrial site;
- other traffic-generating location.

12.3 No artificial profitable checkpoint rule

The game must not manufacture a profitable station every few miles merely to support staged gameplay. Some intermediate nodes will fail the economic test. The correct stage length comes from geography, traffic and cost.

12.4 Segment economics

Each stage should be evaluated on its own capital requirement, traffic, revenue, operating burden, equipment need, financing burden and strategic value. A completed stage can generate cash and information that help finance or justify the next stage.

PART XIII - RAILWAY RIGHTS, PARALLEL TRACK & INTERCHANGE

13.1 Same-company additional track

A railway may add passing sidings, yard tracks, partial double track or full double track as traffic and geography warrant.

13.2 Competing parallel construction

A different railway may build a parallel line where it can lawfully obtain its own right-of-way and construction authority. The existence of another railway does not give a competitor free use of that railway's land.

13.3 Physical proximity does not grant legal authority

Track touching another track, crossing it or running beside it does not automatically grant operating rights.

Keep distinct:

- crossing right;
- connection;
- interchange;
- terminal access;
- joint use;
- trackage rights;
- lease;
- ownership.

Dakota-specific 1880s crossing/connection law remains a dated legal implementation/research matter where exact procedural detail is needed.

PART XIV - RAILWAY CONSTRUCTION TIME & PHYSICAL EXECUTION

14.1 Historical-speed baseline

For projects rather than biology, planning duration should begin from the fastest reasonably achievable historical baseline under competent execution and the actual site/technology conditions. Variance then comes from weather, material delivery, labor, engineering, ground conditions, financing, defects and contractor performance.

Do not make every major project slow simply because it is large. Do not make every project fast merely because exceptional historical construction speeds existed.

14.2 Winter is task-specific

Winter can make some work impractical while allowing other tasks to continue, such as material preparation, shop fabrication, structures, administrative work or selected construction. No universal winter shutdown rule.

PART XV - RAILWAY OPERATING CAPACITY

15.1 Diagnose the real bottleneck

High locomotive utilization does not automatically mean the company needs another locomotive. Scheduling, staging, yard geometry, interchange timing, car availability, crew coverage or customer windows may be the true constraint.

15.2 Peak versus average utilization

Reports should distinguish peak-period strain from full-day or full-period utilization.

15.3 Capital evidence

Expansion decisions can consider:

- queue duration;
- late movements;
- profitable traffic lost or deferred;
- maintenance downtime;
- utilization;
- service reliability;
- expected future committed traffic.

The company should not buy an asset merely because it is the most obvious object associated with the bottleneck.

15.4 Yard geometry as infrastructure

Leads, crossovers, sidings, staging tracks and customer layout can materially affect throughput. Small physical improvements can outperform expensive equipment purchases.

15.5 Freight-car availability is its own capacity problem

Locomotive availability and freight-car availability are separate constraints. Connecting-road equipment, leases and owned cars can coexist. A railway does not need to own every car that touches its network.

15.6 Minimal station design

A new freight node can open with the minimum facilities needed to perform the traffic safely and reliably. Do not require an elaborate passenger depot when the business case only supports siding, freight shelter, loading platform or livestock facilities.

PART XVI - GOVERNMENT, LAND, MAIL & RAILWAY TAXATION

16.1 Government is not a generic railway subsidy button

Public support must follow actual dated law, program, contract or land authority. The major congressional land-grant/bond programs for specific railroads do not create a universal subsidy available to every small railway.

16.2 Federal public-land right-of-way

The General Railroad Right-of-Way Act of 1875 is a canonical candidate for route economics where the route crosses qualifying federal public domain and the company satisfies the dated legal requirements. The route must distinguish land status rather than treating every mile as ordinary private purchase.

Useful land-status concepts include:

- private property;
- federal public domain;
- railroad grant land;
- reservation/trust/restricted land;
- existing transportation right-of-way;
- other legally distinct land.

16.3 July 30, 1886 corporate-aid restriction

A dated legal event should reflect the federal restriction on territorial/local-government stock subscriptions, credit lending or borrowing for corporate benefit after July 30, 1886. Do not casually offer Dakota Gateway a county stock subscription in 1888.

16.4 Postal contracts

The Post Office can be a paying transportation customer. Mail service uses real schedules, handoff, reliability and contract terms. It is not free subsidy income.

16.5 Railway-specific taxation

Dakota railway taxation should use dated railway-specific authority where validated. The 1883 territorial gross-earnings tax framework identified in playtest research should be represented after reconciling the applicable line/company start date. Exact applicability and rates remain a legal-data issue rather than a generic system constant.

PART XVII - CORPORATE FINANCE, EQUITY & CONTROL

17.1 Corporate treasury

Corporate cash belongs to the corporation, not to the player's personal wallet. A minority shareholder cannot use company cash for a personal purchase or unrelated enterprise without a lawful corporate transaction.

17.2 Cap table and dilution

New share issuance changes every holder's economic percentage. Before issuance, the player should be shown current ownership, post-issue ownership if they do not participate and the amount required to preserve the current percentage where a pro-rata opportunity exists.

17.3 Distinct corporate concepts

Keep separate:

- contributed capital;
- economic ownership;
- voting rights;
- board control;
- officer/manager authority;
- creditor claims.

Do not infer 51% control merely from being founder or President. Do not invent modern super-voting founder shares without historical/legal support.

17.4 Bond finance

Bond face amount, coupon, maturity, security and outstanding debt are distinct from net cash proceeds. Issuance discount and costs can make net proceeds lower than face debt.

PART XVIII - RAILWAY DIVERSIFICATION & TOWNSITE DEVELOPMENT

18.1 Corporate-purpose constraint

Railway cash should not casually buy unrelated businesses. An investment must be permitted by the company's legal/charter authority and governance and should have a defensible corporate purpose.

18.2 Rail-adjacent ventures

Historically and economically plausible ventures can include:

- freight warehouses;
- team tracks/public loading facilities;
- stockyards;
- station/freight property;
- townsite/land development;
- selected hotels/eating houses where demand supports them;
- communications or utility relationships;
- other traffic-supporting infrastructure.

18.3 Own versus enable

The railway should compare direct ownership with letting an independent entrepreneur invest. Providing a site, siding and service to an independent elevator can outperform tying up railway capital in the elevator itself.

18.4 Double benefit must be causal

A townsite, warehouse or other adjacent investment may earn direct profit and create railway traffic, but neither benefit is automatic. The actual town or business must succeed.

18.5 Strategic land options

The railway can use legitimate options or conditional control to preserve strategically important land without purchasing every future parcel immediately.

18.6 Unrelated businesses

A saloon, bakery, General Store or other unrelated enterprise should ordinarily use the player's personal capital or a separate company rather than silently consuming railway treasury.

PART XIX - CORPORATE SALE, MERGER & FINAL TRANSFER

19.1 Minority owner cannot sell the company alone

A player owning a small percentage of a railway may support a whole-company sale but cannot personally convey every share or asset without the appropriate corporate authority.

19.2 Whole-company transaction

A serious corporate sale or merger can require:

- buyer diligence;
- board approval;
- shareholder approval where applicable;
- creditor/bond treatment;
- contract assignment or consent;
- treatment of subsidiaries/property;
- employee transition;
- transfer of officer/management authority.

19.3 Enterprise value versus equity value

Headline enterprise value and shareholder consideration are not the same when debt is assumed or retained. Player proceeds come from the economic claim actually owned.

19.4 Pre-close distributions

A pre-closing shareholder distribution is a separate transaction from the purchase consideration.

19.5 Transfer ends control

A qualifying sale or merger must actually transfer control. No hidden continued owner authority is allowed after closing unless a separate new employment, board or contractual relationship explicitly grants it.

PART XX - GENERAL STORE RETAIL LABOR

20.1 Clerk-mediated retail

The 1880s General Store should generally be modeled as clerk-mediated rather than as a modern self-service shop. Customers may browse visible goods, but ordinary purchases can require the proprietor or clerk to retrieve merchandise, weigh or measure bulk goods, cut or grind where applicable, wrap/package purchases, record credit/barter and settle payment.

20.2 Staffing emerges from workload

Do not hard-code two employees. A low-traffic store can operate with one person. Customer service consumes real time and competes with receiving, stocking, ordering, bookkeeping, cleaning, deliveries, post-office/telephone work and other tasks. As traffic rises, queues and deferred back-room work make additional clerks economically valuable.

20.3 Family-run stores

General Stores were often family businesses, but the whole family is not universally required. Valid configurations include one proprietor, proprietor plus spouse, wider family operation, proprietor plus outside clerks or a much larger mercantile staff.

Family members are real Persons. Their work consumes time, creates fatigue, competes with household duties and may substitute for outside paid labor without becoming invisible free labor.

PART XXI - BARTER, STORE CREDIT & IN-KIND VALUE

21.1 Barter is a valid transaction form

Rural commerce may use cash, store credit and barter simultaneously. A customer can exchange eggs, butter, honey, produce, game, labor or other accepted goods for merchandise where historically and economically appropriate.

21.2 Bartered goods remain real goods

Goods received in trade become real owned inventory or another real asset. They can be resold, shipped, consumed, spoiled, rejected for quality or otherwise handled physically. Do not convert barter directly into invisible cash.

21.3 Mixed payment

A transaction may combine goods-in-kind, cash and customer account credit.

21.4 Valuation

Barter value is market/relationship/quality-dependent and can be negotiated. The transaction record should preserve both the physical goods and the agreed economic valuation.

21.5 Historical change

Barter may decline as cash and financial systems deepen, but it does not disappear through one global technology unlock.

21.6 In-kind compensation

Meals, lodging, goods or other in-kind consideration can be components of employment or service compensation where historically appropriate. They remain distinct from cash wages.

PART XXII - TELEPHONE, TELEGRAPH & GENERAL STORE COMMUNICATIONS

22.1 Telephone versus telegraph

Telegraph is already a mature long-distance written communications system at the 1880 start. Telephone is a newer spoken communications network beginning to spread unevenly through business subscribers and exchanges.

They remain parallel systems rather than one replacing the other.

22.2 General Store as early telephone hub

A General Store can plausibly be the first or one of the first local telephone subscribers when telephone service reaches a settlement. Other early candidates can include hotels, post offices, railway offices, banks and major merchants.

22.3 Public-access/message role

Where most households lack telephones, a connected business can become a place for public calls or message receipt. This can increase foot traffic and change ordering/market-information speed while also consuming clerk labor.

22.4 Network-dependent adoption

No global TELEPHONE UNLOCKED event gives every building service. Use:

invention/existence -> regional network/exchange -> local availability -> business subscription/connection -> household or rural-network diffusion.

Rural/farmer mutual telephone systems become increasingly important later in the campaign as historically supported; exact adoption timing remains regional data.

PART XXIII - DATED HISTORICAL WORLD EVENTS

23.1 Historical events should change systems

Major historical, technological, institutional or legal developments should appear as dated world events when they materially alter gameplay. They are not trivia cards.

A useful event notice tells the player:

- what changed;
- when it changed;
- where it applies;
- which mechanics are affected.

23.2 Railroad Standard Time

North American railroad standard time adoption on November 18, 1883 is a canonical candidate major event.

Before adoption, railway/local time conventions can differ where gameplay requires the distinction. After adoption, relevant railway timetables, connections and time references use the standardized railway-time regime. Civil/local adoption may still vary where historically appropriate.

The event matters even if the player does not own a railway. A merchant depending on rail schedules can experience the change.

PART XXIV - SALOONS, GAMBLING, DRINK SOLICITATION & PROSTITUTION

24.1 Historical occurrence governs eligibility

Revision XIX supersedes the Revision XVII production-scope interpretation to the extent it could be read as excluding historically documented player-facing vice businesses solely because they were illegal.

New hard principle:

If an economic activity is historically documented in the relevant place/time, illegality alone does not exclude it from the simulated economy or eventual player ownership. Legality and enforcement instead determine risk, consequences and available arrangements.

This does not make every historically occurring activity a near-term implementation priority.

24.2 Legal Status versus Enforcement Posture remains authoritative

An activity can be illegal and widely known or locally tolerated at the same time. Tolerance does not legalize the activity or create permanent immunity. Enforcement can change with police/prosecutor practice, politics, complaints, reform pressure, serious incidents and other grounded causes.

24.3 Saloon gambling

Where supported by local/date-specific history, a saloon or related establishment may operate gambling through real tables/games, equipment, dealers, bankroll/stakes, customers, wins/losses, theft/fraud exposure and legal/enforcement risk.

The exact game catalogue, house edge, legality and enforcement remain research/data rather than universal constants.

24.4 Female drink solicitation / entertainment labor

Historically supported establishments may employ women whose primary job includes socializing with patrons and encouraging drink purchases rather than ordinary bartending. Compensation can include wages, commission or a combination.

Drink solicitation and prostitution are not automatically the same occupation. They can coexist or overlap in some historical establishments, but one must not imply the other by default.

Workers remain Persons with agency. They can negotiate compensation, refuse terms, leave, change jobs or work part-time where circumstances allow.

24.5 Prostitution and brothel ownership

Historically supported prostitution/house-of-ill-fame activity becomes in-scope for future player business design rather than permanently excluded. The earlier legal research remains authoritative: such operation is not an ordinary licensed lawful business in the normal campaign, no generic Brothel License is authorized and fines do not transform illegality into lawful operation.

Coercive recruitment is not authorized as a normal player-facing economic mechanic. Adult participants remain people with agency.

Implementation priority remains separate from Canon eligibility. A dedicated brothel management system may remain later-phase work until it receives a full design/research pass.

PART XXV - SALOON SCENARIO FAMILY

25.1 Saloon Circuit

Add a scenario concept built around progressively larger saloon markets. The player begins in a very small settlement and expands or opens additional saloons in larger towns, with each location exposing different limiting problems rather than repeating the same business at higher numbers.

Possible progression themes include:

- very small town: inventory, supply, getting enough customers;
- small town: competition, staffing, evening traffic;
- growing town: entertainment, higher-volume purchasing, customer segmentation;
- large town: management, internal controls, theft, multiple shifts and reputation;
- regional center: large operation, managers, complex suppliers, ancillary businesses and institutional pressures.

The exact number of towns/saloons remains scenario calibration.

25.2 Saloon Enterprise Ladder

A separate variant can use build/improve/sell progression through increasingly large saloon enterprises.

25.3 Vertical integration

Supporting businesses such as bakery, butcher, freight, lodging or other historically plausible services can supply a saloon network when the logistics and economics actually make sense.

Do not canonize the playtest's placeholder $1,000-per-quarter target or any placeholder drink/commission figures.

PART XXVI - REWARD / PRIVATE DETECTIVE AGENCY

26.1 Candidate business archetype

Add Reward / Private Detective Agency as a researched candidate business family for the 1880-1905 setting.

A possible commercial loop is:

reward or private case becomes known -> agency accepts/solicits case -> choose agent(s) -> fund travel/expenses -> investigation/pursuit -> capture/recovery/failure -> reward or client payment -> agent compensation -> agency margin.

26.2 Agent differentiation

Agents can differ through actual capabilities such as tracking, investigation, riding, firearms, negotiation, local knowledge, languages or other case-relevant skills.

26.3 On-map and off-map cases

The same persistent agent/case identity can support locally manifested pursuit and longer off-map work.

26.4 Historical/legal distinctions

Do not collapse:

- posted criminal rewards;
- private detective work;
- bail/surety fugitive recovery;
- law-enforcement rewards;
- company-sponsored pursuit.

Exact arrest authority, jurisdiction, reward entitlement and ordinary compensation structures remain research holds before full business lock.

PART XXVII - CALIBRATION HOLDS

The following playtest numbers remain calibration, not historical constants:

- all exact bakery acquisition, wage, rent, expansion and sale figures;
- all Dakota Union purchase, farm, loan, LOC, wage and inventory numbers;
- all exact reserve targets;
- all exact interest rates and lender limits unless attached to a specific authored scenario;
- all exact railway stock authorizations, share prices, bond sizes, coupon rates and underwriting discounts;
- all exact railway acquisition prices;
- all exact track mileage, extension cost, traffic tonnage and return thresholds from the playtest;
- all exact locomotive, freight-car and lease prices;
- all exact construction-duration values;
- all exact management wages;
- all exact autonomy-test durations;
- all exact final railway valuation and player proceeds;
- all saloon drink cost, price, commission and profit examples;
- exact number of saloons in a scenario;
- exact manager-value attribution totals;
- exact animal descriptor thresholds.

PART XXVIII - RESEARCH HOLDS

Preserve the following questions for later research or data calibration:

1. Exact saloon hours by town/business type/date.
2. Exact prevalence and compensation of female drink solicitors.
3. Distinction between ordinary saloons and dance/concert halls in staffing and vice activity.
4. Gambling game types, legality and enforcement by jurisdiction/date.
5. Detailed prostitution/brothel economics and organization by place/date.
6. Exact drink prices, COGS and margins.
7. Private detective/reward-agency compensation and arrest authority.
8. Dakota railroad crossing/interchange rights and procedure where deeper legal simulation is needed.
9. Railway gross-earnings-tax applicability by company/line start date.
10. Public-land ROW filing/completion procedures to the implementation depth actually required.
11. Railway superintendent/manager wage calibration by scale.
12. Construction-speed distributions by project type and season.
13. General Store labor minutes by customer/order type.
14. Barter prevalence and valuation by region/year.
15. Early telephone pricing, message/public-call economics and network diffusion timing.

PART XXIX - EXPLICITLY REJECTED OR SUPERSEDED INTERPRETATIONS

The following are not Canon requirements:

- Manager required for every business sale.
- General Manager automatically requires subordinate managers.
- Fixed universal $200 or other static business cash reserve.
- Railway must reach Dry Willow to complete Enterprise Ladder.
- Entire 43-mile or 17-mile corridor must be built before any railway revenue.
- Every few miles of railway must contain a profitable stopping point.
- Ownership is always better than leasing.
- A second locomotive should be purchased solely because peak utilization is high.
- The railway should own every station business.
- Railway treasury can casually buy unrelated businesses.
- Business cash automatically transfers in a sale.
- Corporate treasury is personal owner cash.
- Board control is the same as economic ownership.
- Every General Store requires two employees.
- The 1880s General Store is modern self-service retail.
- The whole family is required to run every General Store.
- Universal monthly building inspection.
- Inspection automatically repairs condition.
- One giant Make Bread task is sufficient for all bakery labor.
- Failures should resolve as generic percentage penalties.
- One-click subcontracting may magically complete an underlying project.
- Feature count should be minimized merely to avoid feature creep.
- Exactly nine strategic options is a game rule.
- Historical events should be decorative trivia popups.
- Telephone is a global technology unlock.
- Telephone is a replacement for telegraph.
- Drink solicitation automatically implies prostitution.
- Illegality alone excludes a mechanic from the game.
- Every saloon should have gambling or prostitution.
- The bakery-oven household-rental idea remains parked unless stronger evidence/design need emerges.

PART XXX - REVISION XIX RESEARCH BASIS

The following sources were used during the playtest research and should remain audit references for the new historical additions. The Canon conclusions above control gameplay; exact source interpretation should be rechecked if a deeper implementation requires additional legal detail.

- National Park Service, period/general-store interpretation: https://www.nps.gov/places/000/withers-and-spauldings-general-store.htm
- National Park Service, rural store barter/credit interpretation: https://www.nps.gov/grsm/learn/historyculture/stores.htm
- Smithsonian/NMAH material on general stores, barter and retail development: https://www.si.edu/spotlight/cash-and-credit-registers/introduction
- Henry Ford, J.R. Jones 1880s Storekeeper, including early local telephone example: https://www.thehenryford.org/collections/explore/articles/detail/articles/2019/10/28/jr-jones-1880s-storekeeper
- North Dakota Studies, telephone history and early Fargo-Moorhead exchange: https://www.ndstudies.gov/gr8/content/unit-iv-modern-north-dakota-1921-present/lesson-3-building-communities/topic-4-communications/section-1-telephone
- Library of Congress, November 18 standard railway time historical note: https://www.loc.gov/item/today-in-history/november-18/
- National Park Service, parallel railroad grading example: https://www.nps.gov/places/parallel-grading.htm
- 43 U.S.C. historical right-of-way framework / 1875 act reference: https://www.law.cornell.edu/uscode/text/43/934
- Dakota Territory / North Dakota legislative archives used during railway-law research: https://ndlegis.gov/
- U.S. historical postal/rail mail material used in playtest research: https://www.govinfo.gov/ and https://history.state.gov/
- South Dakota historical material on railroad/townsite development: https://www.sdhspress.com/
- Historical Houston detective/reward-agency archive example: https://historical.ha.com/
- Project Gutenberg period reform/vice source used to establish drink-solicitation/commission evidence: https://www.gutenberg.org/
- Revision XVII prostitution legal basis and Deadwood tolerance sources remain authoritative and are preserved in Revision XVIII.

PART XXXI - REVISION XIX LOCK SUMMARY

Revision XIX hard locks:

- Complexity should be interconnected and discoverable, not artificially minimized.
- Action visibility and interruption are separate design problems.
- Meaningful task granularity and causal rework are authoritative.
- Dynamic liquidity supersedes fixed universal reserve assumptions.
- Line of Credit is a distinct facility; undrawn capacity is not cash or debt.
- Manager titles do not impose mandatory organizational hierarchy.
- Manager Activity is distinct from the Ledger.
- Railway Financeability and Economic Viability are separate gates.
- Railway corridors may open in economically valid stages.
- Intermediate nodes are not guaranteed to be profitable.
- Track proximity does not grant land or operating rights.
- Corporate cash, economic ownership, voting control, board control and managerial authority remain distinct.
- Railway diversification must have lawful corporate purpose and causal economics.
- General Store retail is predominantly clerk-mediated; staffing follows workload.
- Barter is a real transaction that transfers real goods.
- Telephone availability is local/network-dependent and can make a General Store an early communications hub.
- Historical events should change simulation state; Railroad Standard Time is a major 1883 candidate event.
- Historically documented illegal activity can exist in simulation; illegality creates legal/enforcement risk rather than automatic exclusion.
- Drink solicitation and prostitution are distinct activities.
- Saloon scenario variants and a reward/private-detective agency are preserved as new scenario/business directions.

END OF REVISION XIX INTEGRATION

LAND & LEDGERS

Canonical Project Bible - Revision XVIII

System Audit Integration - August 30, 2026

Authority and preservation rule

Revision XVIII preserves **Revision XVII in full** except where this new integration section explicitly adds, deepens or changes a prior rule. Nothing from Revision XVII is removed, condensed, simplified or silently replaced. The seven completed system audits below are now Canon. Where an audit says KEEP, the prior rule remains authoritative. Where an audit says DEEPEN, the new material adds authority without deleting the prior rule. Where an audit says SIMPLIFY, the simplification applies to player-facing presentation or implementation burden only unless the text explicitly changes simulation behavior. Where an audit states a new lock, doctrine or distinction, that rule supersedes any older wording that would otherwise conflict.

These audits are intended to consolidate existing Canon rules into shared cross-system grammars so that future business, AI, UI and implementation work does not invent incompatible local interpretations.

SYSTEM AUDIT 01 - ECONOMY & TRANSACTION CORE

Final verdict: KEEP + DEEPEN.

The economic foundation is already unusually strong. The Canon distinguishes owner cash from business cash, cash from Net Worth, physical stock from recorded stock, credit sales from cash receipts, internal transfers from genuine consolidated portfolio profit, landed cost from invoice price, and capital contribution from intercompany advance and bank borrowing. The weakness is not lack of economics; it is the absence of one authoritative transaction grammar tying those rules together across every business.

1. System purpose

The Economy & Transaction Core answers:

When something economically meaningful happens, what changed, who owns it, who owes whom, when was value earned or consumed, when did cash actually move, and what does each party legitimately know about it?

The same economic truth must support a one-store owner and a mature portfolio with dozens of businesses, properties, obligations, contracts, internal supply chains, banks and investments.

2. Universal economic loop

Need / Demand -> Offer / Price / Terms -> Order / Agreement -> Commitment / Reservation -> Physical Fulfillment -> Economic Transaction -> Cash / Receivable / Payable / Inventory / Asset changes -> Bookkeeping -> Settlement -> Reconciliation -> Historical Memory.

The steps are not interchangeable.

An order is not revenue.

A shipment leaving the supplier is not automatically inventory at the destination.

An invoice is not cash.

Collection of an old account is not a second sale.

Borrowing is not revenue.

Buying a wagon is not the same economic event as paying wages.

Moving value between controlled businesses does not create new consolidated wealth.

3. Current economic strengths and gaps

Area	Canon status after audit
Owner vs business cash	KEEP - excellent
Net Worth vs liquidity	KEEP - excellent
Physical inventory	KEEP - excellent
Working-capital pressure	KEEP + elevate
Accounts receivable/customer credit	KEEP - strong
Accounts payable/supplier credit	DEEPEN
Pricing causality	KEEP - strong
Landed cost	KEEP - strong
Cost basis through time	

DEEPEN

Bookkeeping/reconciliation

KEEP - excellent

Internal transfers

KEEP + DEEPEN accounting

Revenue/profit definition

DEEPEN

Capital expenditure treatment

DEEPEN

Portfolio consolidation

NEW CANON LOCK

Transaction reversals/losses/write-offs

DEEPEN

Manager automation

KEEP

NPC parity

KEEP - excellent

4. Cash architecture

The following economic concepts remain distinct:

Liquid Owner Cash: immediately spendable owner-side money.

Business Cash: cash belonging to a particular business ledger.

Bank Cash: cash belonging to the banking institution, not to the owner personally.

Restricted/Committed Funds: cash constrained by a real obligation or arrangement where applicable.

Operating Reserve: a standing management policy protecting desired liquidity.

Capital Authorization: permission for a particular capital expenditure or project.

Net Worth: grounded assets less liabilities.

Book/Ledger Equity: accounting carrying-value concept, not synonymous with Net Worth.

Derived Discretionary Cash report

Add **Discretionary Cash** as a derived management report, never as a new money pool.

Example:

Business Cash: $340

Known obligations due soon: $120

Operating Reserve target: $100

Approximate discretionary cash: $120

This report may be uncertain when future collections, expenses or records are uncertain.

5. Profit, cash flow and wealth are separate

New explicit Canon lock:

Profit, cash flow and change in Net Worth are three different measurements.

A business can be profitable and short of cash. It can receive cash while losing money. A player can exchange cash for property without immediately reducing wealth by the full purchase price. Borrowing increases liquidity without creating equivalent wealth. Credit sales can create earnings before collection and therefore produce apparent success alongside cash strain.

Every serious business report should conceptually support three perspectives:

Performance: did operations create or destroy economic earnings during the period?

Cash Flow: where did actual money come from and where did it go?

Position: what does the entity own and owe now?

6. Canonical Transaction Classification Rule

Every economically meaningful event must have a classification determining its effect on cash, assets, inventory, obligations, revenue, expenses and consolidated reporting.

Event

Cash effect

Asset / inventory effect

Claim / liability effect

Performance effect

Cash retail sale

Cash +

Inventory -

-

Revenue earned; merchandise cost recognized

Credit retail sale

-

Inventory -

Receivable +

Revenue earned; merchandise cost recognized

Customer later pays account

Cash +

-

Receivable -

No new revenue
Buy inventory for cash
Cash -
Inventory +
-
Not a second loss merely because cash moved
Buy inventory on account
-
Inventory +
Payable +
Acquisition recorded without immediate cash payment
Pay supplier bill
Cash -
-
Payable -
Not a second purchase
Owner contributes capital
Business Cash + / Owner Cash -
-
Equity relationship changes
Not revenue
Owner distribution/draw
Business Cash - / Owner Cash +
-
-
Not ordinary operating expense
Bank loan received
Cash +
-
Loan liability +
Not revenue
Loan principal repaid
Cash -
-
Liability -
Not ordinary operating expense
Interest paid
Cash -
-
Accrued obligation reduced where applicable
Financing cost
Wagon/property/equipment purchased
Cash -
Asset +
Maybe financing obligation +
Capital acquisition, not ordinary consumption
Inventory spoiled/stolen
-
Inventory -

-

Loss

Bad receivable written off

-

-

Receivable -

Credit loss

Internal inventory transfer

Depends on terms

Source - / destination +

Maybe intercompany balance

No consolidated portfolio profit

This is underlying economic authority. It does not require exposing formal double-entry accounting to the player.

7. Revenue recognition

New common principle:

Revenue is recognized when the business has actually earned the consideration under the relevant transaction or contract, not merely because an order exists or cash happened to move.

Examples:

Ordinary retail: goods physically transferred and sale completed.

Services: qualifying service performed.

Construction: contractually earned work or milestone according to the agreed structure.

Freight: qualifying transport service completed or agreed billable milestone reached.

Customer deposits/prepayments can create cash before all revenue is earned. Credit sales can create revenue before cash is collected. Backlog is not revenue.

8. Accounts Payable / supplier trade credit

Accounts Payable becomes a first-class working-capital system equal in importance to customer credit.

Possible supplier terms include:

cash before shipment

cash on delivery

partial deposit

supplier account

scheduled settlement

overdue account with relationship consequences

A business can be well stocked while owing substantial money. Supplier credit therefore changes working capital without creating free inventory.

9. Working Capital as a named foundation

Opening commercial gameplay should explicitly teach and report the cycle:

Cash -> Inventory -> Sale -> Receivable -> Cash

while supplier credit can create:

Supplier Delivery -> Inventory -> Payable -> Later Cash Payment.

Working Capital Health should be explained through components rather than one magic score:

cash

inventory

receivables

receivable aging

payables

upcoming payroll

debt payments

committed contracts

operating reserve

stock replacement needs

Managers may produce explanations such as:

sales are strong but too much money is tied up in customer accounts;

cash looks healthy because replacement inventory has been delayed;

business is profitable but bills due exceed expected collections.

10. Inventory ownership and availability

New general doctrine:

Physical possession is not necessarily ownership, and physical presence does not mean inventory is freely available.

A location may contain:

owned available inventory

owned inventory reserved for customers or agreements

owned project stock

owned stock unavailable because of condition/spoilage

customer-owned property in storage or service

goods in transit

The economic question is always: **what is here, who owns it, and what is it available for?**

11. Historical cost vs replacement cost

Reports may distinguish:

Acquisition/carrying cost: what existing stock actually cost the business.

Current replacement cost: what comparable stock would cost now.

Selling price: current asking or realized price.

Realized margin: what was actually earned on sold inventory.

Replacement margin: whether current selling economics can replace stock at current sourcing cost.

A business can therefore report a positive realized margin while replacement economics are poor after a supply shock.

12. Pricing

KEEP the existing causal pricing system. Price information may include, when legitimately known:

current selling price

recent realized prices

acquisition cost of current stock

current replacement quote

observed competitor/local range

recent sales velocity

stockout frequency

estimated demand response and confidence

Never expose a magical **Optimal Price** answer.

13. Demand is not a sale

New lock:

There is no free-floating quantity called Demand that automatically becomes sales.

A need becomes a sale only when buyer need exists, buyer can act, seller is known/reachable, terms are acceptable, stock/service capacity exists and physical execution occurs.

14. Internal transfers and carrying value

A free or low-cash internal transfer does not make the input economically free.

If a sawmill transfers lumber that cost $60 to produce to an owned construction business for $0 cash, the construction business does not receive economically costless lumber. The physical inventory carries relevant cost/history and the portfolio has created no new wealth.

Cash transfer price, accounting treatment and economic opportunity value may differ, but all must remain truthful.

15. Portfolio consolidation

New hard Canon lock:

Controlled-business transactions must not be double-counted as consolidated external revenue, profit, receivables or liabilities.

Reports distinguish:

External Revenue: value earned from outside the controlled economic group.

Internal Transfer Activity: transactions among controlled ledgers.

Consolidated Portfolio Revenue: external revenue after elimination of internal duplication.

Intercompany receivables and liabilities are likewise eliminated in consolidated portfolio position while remaining real at the individual-ledger level.

16. Capital purchases

Explicit distinction:

Operating expenditure consumes ordinary operating resources.

Capital acquisition exchanges cash/obligation for a durable asset.

Financing cash flow changes debt/capital structure.

Owner/portfolio capital movement transfers value among owners and controlled ledgers.

Buying an asset is not automatically an operating loss on the purchase date. The asset can later wear, lose value, require repair, be damaged, become obsolete, be sold or be salvaged.

Detailed historical depreciation schedules remain unneeded unless later research/design requires them.

17. Losses

Common loss treatment:

spoilage reduces inventory and creates loss;

theft reduces actual cash/inventory and may first appear as a discrepancy;

bad debt reduces receivables and creates credit loss;

destroyed asset reduces asset value;

fire can destroy several asset/inventory states at once.

Actual State and Recorded State may temporarily disagree until discovery and reconciliation.

18. Bookkeeping

KEEP existing doctrine. Known transaction facts remain exact. Poor bookkeeping can cause missing, stale or unreconciled information; it does not make a known twelve-cent transaction become an approximate number.

Bookkeeping improves knowledge and control, not real sales or production by magic.

19. NPC parity

KEEP. NPCs can overstock, understock, price badly, misjudge supplier terms, extend excessive credit, hold too little cash, delay capital spending or allocate capital poorly under the same economic rules as the player.

20. Scale transformation

Early game: Can I afford this reorder? Can I extend this tab? Can I make payroll?

Growing business: What reserve should management protect? Which supplier terms are best? How seasonal should inventory be?

Portfolio: Which business receives scarce capital? Equity, advance or bank borrowing? Which business is profitable but illiquid?

Capital command: Which divisions retain cash, distribute cash, borrow, reinvest or restructure?

21. Historical transformation

Underlying economic truth remains stable from 1880-1905. The environment changes through rail, telegraph, telephone, banking, larger firms, improved market reach, deeper specialization and higher organizational complexity.

22. Canon locks from Audit 01

Commitment is not fulfillment.

Revenue is not cash.

Cash flow is not profit.

Profit is not Net Worth.

Borrowing is not revenue.

Principal repayment is not ordinary expense.

Owner capital contribution is not revenue.

Owner distribution is not ordinary expense.

Asset acquisition is not ordinary operating consumption.

Internal transfer is not external portfolio revenue.

Possession is not necessarily ownership.

Current replacement cost does not rewrite historical acquisition cost.

Known transaction facts remain exact even when books are incomplete.

Audit 01 Canon status: LOCKED.

SYSTEM AUDIT 02 - PEOPLE / LABOR / SKILLS

Final verdict: KEEP + DEEPEN, with targeted SIMPLIFICATION of presentation and repetitive control.

The Canon correctly abandoned occupation-as-character-class and now separates Aptitude, Skill, Experience and Knowledge / Domain Familiarity. Businesses generate workload rather than fixed employee slots, employees provide finite compatible labor, schedules interact with travel and rest, compensation and poaching are real market forces, and skills grow from actual work rather than completion-day XP bonuses.

1. System purpose

The People / Labor / Skills system answers:

Who can realistically do the work, who is available to do it, how well will they probably do it, what does employing them cost, what happens to them through work, and why do they stay, negotiate or leave?

Employees are persistent economic people, not interchangeable production modifiers.

2. Universal labor loop

Work Need Exists -> Required capabilities/location/timing identified -> Potential workers considered -> Worker assigned -> Travel/arrival/availability -> Task performed -> Time/quality/material/safety consequences -> Work outcome -> Experience/knowledge/reputation generated -> Fatigue/injury/satisfaction/relationship consequences -> Compensation -> Retention/advancement/future availability.

This same grammar applies across store service, team driving, bookkeeping, construction, repair, mining, agriculture, management and other work.

3. Aptitude, Skill, Experience and Knowledge

KEEP the following distinctions:

Aptitude: tendency to learn or perform related capabilities well.

Skill: present capability.

Experience: what the person has actually encountered and done.

Knowledge / Domain Familiarity: understanding of a particular trade, industry, market, place, organization, procedure or customer base.

A highly skilled generic negotiator does not automatically possess deep lumber-market knowledge. A banking veteran and an ordinary bookkeeper may have similar numerical skill but very different domain familiarity.

4. Capability architecture and catalogue control

To prevent skill-stat explosion, use three conceptual layers:

Capability Family

Broad transferable ability, such as:

Commercial Interaction

Numerical/Clerical

Organization/Management

Construction

Mechanical/Repair

Animal Handling

Transport

Food Processing

Extraction/Industrial

Agricultural

Task Capability

What materially affects execution, such as framing, finish carpentry, loading, dispatching, diagnosis, painting or masonry.

Domain Familiarity

Contextual knowledge such as residential construction, mine timbering, railroad work, banking, livestock freight or local customer relationships.

Player-facing presentation should usually be descriptive rather than a wall of numeric RPG attributes.

Example:

Experienced carpenter. Strong framing and layout. Adequate finish work. Limited masonry experience. Familiar with residential construction.

Detailed numbers may exist internally where useful.

5. Occupation is descriptive, not class

Occupation summarizes current work or history. It does not restrict the person to one hard class.

A teamster may also possess animal handling, loading, repair familiarity, route knowledge and customer experience. Those capabilities remain relevant if their job changes.

6. Attempt Rule and task types

KEEP the rule that low competence normally changes time, quality, waste, rework, supervision need, safety and customer response rather than creating arbitrary locks.

Distinguish:

Ordinary attemptable

A novice can reasonably try.

Attemptable but dangerous/specialist-sensitive

Possible but poor competence can have serious consequences.

Actually restricted

Action is impossible without real authority, equipment, access, information or specialist capability.

Hard locks must come from reality, not progression gates.

7. Workload-based staffing

New elevated doctrine:

Businesses create work, not abstract employee requirements.

A business's staffing need emerges from concurrent work, timing and required capabilities. A store may need one person Tuesday morning and three Saturday afternoon without an artificial employee-slot change.

8. Work Demand grammar

Each meaningful work demand may identify:

what needs doing

where

when

expected time burden

required/relevant capabilities

urgency

whether it can wait

whether several people can collaborate

whether one worker can cover compatible work in the same period

consequence if missed

Examples include customer service, restocking, unloading, month-end reconciliation and weather-sensitive construction work.

9. Role is not capability

New lock:

Role grants responsibility and authority. Capability determines how well that responsibility is exercised.

A Manager or Foreman title does not manufacture judgment, scheduling skill, leadership or technical skill. An untitled experienced worker can coordinate effectively.

10. Management is real labor

Managers perform finite administrative and decision work. Management overload should appear through causal symptoms:

delayed decisions

unfilled vacancies

weak scheduling

late supplier response

stale records

unresolved exceptions

weak supervision

poor hiring

excessive travel/attention burden

Avoid abstract Management Capacity bonuses where actual workload can explain the outcome.

11. Mentorship consumes time

New explicit rule:

Teaching consumes the mentor's time and attention.

A veteran who demonstrates, checks, explains and corrects work may produce less personally while accelerating another worker's development. Training is therefore an investment with opportunity cost.

12. Learning

KEEP continuous learning from relevant exposure, novelty/difficulty, aptitude, instructor competence, teaching ability, time together and feedback.

Use diminishing value:

new exposure: strong learning

developing familiarity: normal learning

routine mastery: small refinement

novel difficult situation: renewed learning opportunity

Do not add routine generic skill decay. Long absence, injury, age or specific circumstances may matter where physically justified.

13. Experience memory

Experience should preserve meaningful career history rather than every microscopic task event.

Useful memory may include:

years/seasons in domain

major task exposure

difficult problems encountered

supervisory responsibility

businesses worked for

important projects

unusual failures/successes

market/local familiarity

14. Recruitment needs full loop

Recruitment becomes:

Vacancy / Work Pressure -> Employer defines need -> Potential candidates become discoverable -> Information gathered -> Candidate evaluates employer -> Employer evaluates candidate -> Offer -> Negotiation / acceptance / refusal -> Employment begins -> Actual performance reveals more information.

Candidate sources can include:

local unemployed workers

employed workers

referrals

employee networks

kin/household contacts

migrants

newspaper notices

professional contacts

other settlements

There is no omniscient global candidate list.

15. Labor supply is not population

New lock:

Effective labor supply depends on availability, compatibility, willingness, wages/conditions, housing, commute, household responsibilities, health, season, competing employers and migration feasibility.

Town population is not the same as available labor.

16. Wage markets are capability-specific

Avoid one universal town wage multiplier. Demand for teamsters, clerks, carpenters, bookkeepers or miners can move differently depending on overlapping alternatives and industries.

Booms should propagate through actual competition for compatible workers.

17. Employment relationship

Treat employment as a persistent simulation relationship. It can include:

employer

worker

primary responsibilities

compensation basis

board/meals/lodging where applicable

schedule expectations

bonus/commission/profit-sharing where supported

authority

start date/status

material special conditions

This does not imply every worker has a literal written contract.

18. No universal Morale bar

Do not collapse pay reliability, wage level, hours, fatigue, safety, commute, housing, manager relationship, broken promises, promotion opportunity, reputation and outside offers into one player-facing Morale score.

Use explanations such as:

Generally satisfied. Concerned about long hours. Commute is difficult. Believes pay is below alternatives.

19. Retention and poaching

KEEP. Valuable workers can request better compensation, report competing offers or leave. Responses can include wage, bonus, promotion, duties, hours, accommodation, stability, status and other grounded terms. Money does not have to solve every retention problem.

20. Promotion

New lock:

Promotion changes authority and responsibility; it does not automatically increase capability.

A brilliant craft worker may be a poor manager. A moderately skilled craft worker may be an excellent foreman because of organization and communication.

21. Cross-business labor and cost attribution

A person can temporarily work across controlled businesses without fictional firing/rehiring where authority allows. However:

legal employer/payroll payer remains real;

economic consumer of the worker's time can differ;

internal allocation/reimbursement may be tracked;

consolidated profit must remain truthful.

22. No double-booking

New hard lock:

A person's time and physical location are authoritative. One person cannot perform simultaneous incompatible work in different places.

This applies to employees, managers and mentors.

Some decision/information tasks can occur remotely only when communication and authority make them physically plausible.

23. Fatigue

KEEP cumulative causal fatigue. Long shifts create slower work, errors, injury exposure, absence, irritability and turnover rather than a generic production penalty.

Repeated overwork matters more than one unusual day.

24. Sleep

KEEP as a lightweight causal connection among work hours, commute, housing and labor performance. Do not require player-visible minute-level sleep simulation.

25. Nutrition

KEEP the economic connection but simplify presentation heavily.

Track aggregate food adequacy/regularity such as:

well supplied

adequate

poor

severely strained

Causes remain physical: household reserves, affordability, meals, boarding and shortages. Do not build a nutrition minigame.

26. Commute

KEEP. Geography changes practical labor supply through travel time, rest, punctuality, availability and job attractiveness. Better transport can expand the usable labor market because people can actually reach work.

27. Household responsibilities

A person's labor availability can be affected by real household responsibilities such as dependent care, illness, domestic work, pregnancy/recovery, elder care, childcare, seasonal household work and household enterprise.

These effects must come from actual household/person circumstances rather than gender stereotypes or fixed demographic penalties.

28. Family labor

New lock:

Family labor is still labor.

A spouse or adult child can perform real business work without a conventional wage transaction, but their time, fatigue, household contribution and outside earning opportunity remain economically meaningful.

29. Illness/injury

KEEP person-specific continuity effects. Reduced availability matters according to actual role coverage. The loss of the only qualified operator is much more serious than the same injury in a redundant organization.

30. Task-specific mistakes

KEEP. Clerk errors, driving errors, construction defects, banking mistakes and other failures create domain-specific consequences, not one generic Quality penalty.

31. Worker information doctrine

Separate:

Known facts

Current employment, wage, attendance, performed jobs, observed failures, recorded training.

Evidence

References, prior-employment claims, reputation, interview impression.

Inference

Likely aptitude, fit, retention, management potential.

Unknown

Unobserved or unsupported information.

The player does not automatically know true hidden skill values.

32. Labor pressure explanation

Labor shortages should be explainable through actual vacancies, wage offers, housing, competing employers, migration, worker exits and capability scarcity rather than a raw Labor Pressure score.

33. Scale transformation

Owner-operator: can one worker cover the counter while I unload?

Small business: hire another clerk or work longer myself?

Growing business: who should supervise?

Multi-business owner: can a worker cover another business temporarily?

Mature portfolio: why is retention poor in this division?

Capital command: retain senior manager, promote internal candidate or recruit externally?

34. Canon locks from Audit 02

Workload is not headcount.

Role is not capability.

Occupation is not class.

Aptitude is not skill.

Training is work.

Promotion is not training.

Employment is a relationship.

Labor supply is not population.

Time is exclusive.

Family labor is still labor.

Human strain is causal.

Worker agency persists.

Capability information is imperfect.

Audit 02 Canon status: LOCKED.

SYSTEM AUDIT 03 - HOUSEHOLDS / CONSUMPTION / HOUSING / MIGRATION

Final verdict: KEEP + DEEPEN, with simplification guardrails.

The household-housing architecture is one of the strongest systems in the project. Household is not House. Kinship is distinct from economic household membership. Residential Occupancy is distinct from Accommodation Space, Building and Parcel. The biggest need is a common household-allocation grammar and a complete information-aware migration loop.

1. System purpose

The combined system answers:

Who lives together economically, what resources and obligations they share, what they need, how they respond when they cannot afford everything, where they can realistically live, why they enter or leave a settlement and how those choices affect businesses and labor markets?

Households can be consumers, labor suppliers, tenants, landlords, debtors, family-business labor sources, property owners, care networks and migration units.

2. Universal household loop

Household composition/resources -> Income/production/reserves arrive -> Needs and obligations mature -> Household decides what can be satisfied -> Use reserves / produce internally / purchase / use credit / substitute / postpone / go without -> Individual member performs physical action -> Goods/services/housing obtained -> Household state changes -> Shortfall/surplus consequences -> Work/housing/credit/migration decisions adjust -> History persists.

3. Household identity

KEEP. A household is a persistent domestic-economic unit. It can consist of one adult or many people. Household size and composition emerge through birth, marriage, death, adult children leaving/staying, migration, kin hosting, splitting and merger rather than fixed templates.

4. Household / family / occupancy separation

KEEP foundational hierarchy:

Person

Household

Residential Occupancy

Accommodation Space

Building

Parcel

Household membership, kinship, property ownership and where somebody sleeps can change independently.

5. Household finance and allocation under scarcity

Needs do not create money. Households can spend actual cash, use available credit, sell labor/goods, consume reserves, substitute, delay, reduce quantity or go without.

New **Household Allocation grammar**: when a household cannot satisfy everything, decisions consider:

urgency

severity

amount

substitutability

deferrability

existing reserves

credit/relationship consequences

household preference/circumstances

There is no rigid universal priority such as Food > Rent > Fuel > Debt. Different households can make different grounded choices.

6. No household budgeting minigame

NPC households decide their ordinary consumption autonomously. The player influences them only through legitimate authority and economic relationships: wages, prices, rents, credit, employment, housing supply, boarding, services, property and business availability.

Player-facing household information should be diagnostic and limited by legitimate knowledge.

7. Consumption reserves

KEEP target/reserve/critical concepts. Purchasing urgency rises as household reserves fall rather than every household shopping on a fixed weekly script.

8. Consumption category restraint

Track a category only when it creates meaningful commercial decisions. Potential categories include staple food, meat/protein, preserved food, fuel, clothing/textiles, household consumables, tools/hardware where relevant, animal feed, medicines/services and occasional durable purchases.

Do not simulate every fork, button, sock or bar of soap as an independent household need unless the product is economically meaningful elsewhere.

9. Household demand remains physical and competitive

Need becomes a purchase only when:

household has reason to act;

it has cash/credit or another legitimate means;

an acceptable supplier is known/reachable;

stock/service exists;

somebody can physically shop or receive delivery;

the transaction actually occurs.

Households do not automatically buy from the player.

10. Household production

KEEP economically meaningful small production such as coops, gardens, root-cellar storage and boarder rooms where appropriate.

Such activities use real labor, inputs, seasonality and space. Do not turn households into manually controlled homestead minigames.

11. Domestic workload

New lock:

Domestic work consumes real household labor time even when no wage payment occurs.

Domestic workload can include cooking, cleaning, childcare, fuel handling, water hauling, laundry, animal care, garden work and boarding labor. Who performs it depends on actual household circumstances and calibrated historical behavior, not arbitrary stereotypes.

12. Boarding

KEEP. Boarding is not simply bed capacity. Where board includes meals/services, capacity also depends on food, fuel, water, cooking, cleaning, laundry and labor. Boarding can convert regional labor into locally usable labor without a fake recruitment bonus.

13. Housing ontology

KEEP exactly:

Person -> Household -> Residential Occupancy -> Accommodation Space -> Building -> Parcel.

Double Houses remain one Building with two independent accommodation units unless actual legal/property facts say otherwise.

14. Accommodation types are arrangements, not tiers

KEEP owner-occupied, rental, room rental, boarding/lodging, employer lodging, kin hosting, doubled-up/shared, transient lodging, temporary camp, precarious accommodation and no stable shelter as non-linear arrangements.

15. No universal Housing Quality score

Housing suitability is household-specific and may depend on:

shelter capacity

weather protection

heating

water

sanitation

cooking

storage

privacy/shared status

condition

fire exposure

yard/outbuilding access

proximity

seasonal accessibility

Summary suitability can exist but cannot replace the real factors.

16. Crowding

KEEP. Capacity creates rising strain rather than a hard occupancy wall. People can double up, host relatives, use camps or accept cramped rooms with real consequences such as privacy loss, domestic burden, sanitation strain, wear, fire exposure and move pressure.

17. Rental market

KEEP landlord-set asking rents and actual household choice. New construction responds with delay because it requires land, capital, labor, materials and time. Several investors can respond simultaneously and oversupply the market.

Vacancy as information

Vacancy is evidence about rent/location/suitability/demand, not an automatic instruction to change rent.

18. Rent agreement memory

New lock:

Rent is an occupancy/tenancy agreement, not a daily spot-price reset.

Current tenant rent and current market asking rent can differ. Changes occur when the actual arrangement and applicable law permit.

19. Housing pressure is match-specific

KEEP. A town can have empty bachelor lodging while families face severe shortage. Housing pressure compares households with acceptable reachable accommodation, not Population - Beds.

20. Migration requires full loop

New universal migration loop:

Push / Opportunity exists -> Potential migrant becomes aware of destination -> Destination evaluated -> Job/business/kin/property opportunity assessed -> Housing feasibility assessed -> Travel cost/difficulty assessed -> Move/wait/reject decision -> Physical travel -> Temporary arrival state -> Employment/housing/consumption stabilization -> Stay / onward move / return / leave.

21. Migration obeys Information Distance

People cannot respond to opportunities they have no plausible way to know about. Awareness may come through kin, former coworkers, newspapers, recruiters, travelers, merchants, rail/transport networks, telegraph where appropriate, labor agents where supported, project reputation or direct offers.

22. Individual vs household migration

A single worker and a household face different moving burdens. A worker may move first, board near work and bring family later. Household migration may depend on family housing, spouse opportunities, travel, reserves, kin and job stability.

23. Relocation costs

Moving has friction:

travel

time

lost work

moving possessions

advance rent/deposits where applicable

uncertainty

separation from support networks

Small wage differences therefore do not cause constant town-to-town ping-pong.

24. Migration inertia

People accumulate real reasons to stay:

property

stable rent

employer relationship

kin

spouse employment

children/dependents

business ownership

local reputation

debt

customers

community familiarity

This is not a Loyalty stat; it is actual embeddedness.

25. Population growth decomposition

Settlement population analysis should distinguish:

natural change: births/deaths

individual labor migration

whole-household migration

household formation

household dissolution/merger

temporary/transient presence

A town gaining temporary workers is not automatically experiencing durable population growth.

26. Retention matters as much as attraction

New lock:

Arrival is not successful settlement growth.

A town can attract workers through high wages but fail to retain them because of poor housing, food costs, employment instability, winter strain, weak family accommodation, debt or stronger alternatives elsewhere.

27. Temporary boom camps and durable conversion

Boom activity can create:

job pulse -> transient workers -> temporary lodging -> supporting merchants/services -> permanent housing investment -> family migration -> durable settlement

or reverse:

job pulse ends -> workers leave -> boarding vacancy -> speculative housing vacancy -> merchants weaken -> town contracts.

28. Employer lodging

KEEP. Employer housing works by creating actual acceptable accommodation and therefore changes recruitment feasibility. It also creates real construction, maintenance, property, fuel/service and vacancy burdens.

29. Housing investment is speculative

Developers act on expectations, not future truth. High rents, boarding saturation or expected project expansion can motivate construction. If expectations fail, vacancy and distress can follow.

30. Household hardship is causal

Do not create a Poverty meter. Household hardship should propagate through actual events:

job loss -> reduced purchases -> more credit -> rent pressure -> altered work search -> possible relocation.

31. Credit connection

Merchant credit bridges temporary income timing but does not make households wealthier. Unpaid household debt becomes merchant receivable risk and can become merchant loss.

32. Winter/fuel pressure

KEEP. Housing economics include rent, fuel, heat retention, stove capability, reserve levels, substitution and winter strain. A larger/draftier dwelling can be more expensive in total even if nominal rent is attractive.

33. Household information limits

The player does not automatically know exact household cash, food inventory, rent arrears, disputes or migration intention. Knowledge depends on customer, tenant, employment, credit, landlord, reputation or other legitimate channels.

34. Household UI

Keep household panels lightweight. Show composition, broad reserves/pressure, housing and known relationships where appropriate, not dozens of tabs.

35. Demographic lifecycle

KEEP births, deaths, marriages, household formation/split/merge, migration, displacement and guardianship as relationship changes preserving person identity.

36. World generation

KEEP person-first generation:

generate people

establish kinship

generate prior family history

form households

assign housing

validate aggregate output

Do not generate fixed stereotypes such as Family of Five and then manufacture people to fit the template.

37. Canon locks from Audit 03

Household need is not purchasing power.

Consumption is reserve- and urgency-driven.

Household action remains physical.

Domestic work is labor.

Accommodation is a relationship.

Housing availability is match-specific.

Crowding is consequence, not prohibition.

Rent is an agreement.

Migration requires awareness.

Migration requires feasibility.

Moving has cost and inertia.

Arrival is not retention.

Transient population is economically real but analytically separate.

Housing supply responds with delay and uncertainty.

Household hardship propagates causally.

Audit 03 Canon status: LOCKED.

SYSTEM AUDIT 04 - PROPERTY / LAND / BUILDINGS / CONSTRUCTION / DEVELOPMENT

Final verdict: KEEP + DEEPEN.

Property already expresses the ownership fantasy exceptionally well: persistent parcels, real sellers, private approaches, negotiation, diligence, flexible use, physical project execution and persistent property history. The main need is one common Property Lifecycle authority connecting legal geography, control, acquisition, project authorization, construction, use, redevelopment and disposition.

1. System purpose

The system answers:

What exactly do I control, what can I realistically do with it, what will it cost to change, what risks am I inheriting and why is this land more or less valuable to me than to someone else?

Property value can arise from frontage, traffic, road/depot access, parcel depth, yard capacity, drainage, neighbors, current improvements, title/control, occupants, taxes/debt, conversion potential and expansion potential.

2. Unified Property Lifecycle

Land/property exists -> Ownership and legal rights exist -> Opportunity becomes known -> Site/property evaluated -> Owner approached -> Terms negotiated -> Diligence -> Financing/conditions -> Closing -> Actual ownership/control transfers -> Carry property as-is OR authorize project -> Materials/labor/contractors committed -> Physical work -> Building/site changes -> Productive use -> Condition/occupancy/value evolve -> Expand / convert / refinance / sell / abandon / lose.

3. Survey/legal geography

KEEP parcel identity, parcel type, acreage, jurisdiction, ownership form, transfer status and historical provenance. Current ownership/transferability remains separate from historical provenance.

Restricted federal, reservation, trust-allotment or other land must not become a normal Buy button merely because it is visible.

4. Ownership vs control

New lock:

Owning land and possessing a legitimate right to use or control land are not always the same thing.

Control can arise through ownership, tenancy/lease, occupancy, contract, right-of-way, corridor rights, management authority or another historically valid relationship.

5. Parcel / Building / Functional Space separation

KEEP:

Parcel = legal/property geography.

Building/Improvement = physical structure.

Functional Space/Residential Unit/Business Space = usable internal physical/economic space.

Occupancy/Business/Lease = relationship using that space.

Double Houses remain one Building with independent residential units unless facts establish otherwise.

6. Suitability is use-specific

New explicit lock:

Use + Site -> Fit. There is no universal parcel Commercial Potential score.

Examples:

General Store

frontage, traffic, receiving, storage, nearby households.

Livery

stable/yard depth, road access, water, feed handling, nuisance/space.

Freight Yard

loading space, major route/depot proximity, maneuvering room, stable/storage.

Boarding House

accommodation space, kitchen/service capability, employment access.

Sawmill

large site, log access, freight, fire/nuisance separation, water where relevant.

7. Flexible conversion

KEEP the rule: anything sensible into anything sensible. Buildings may be reused when their physical arrangement can support the new use. No conversion simply because a player clicked a button.

Conversion Feasibility report

Show strengths, weaknesses and required work rather than only Valid/Invalid.

Example residence -> General Store:

strong frontage

adequate customer space

weak rear receiving

limited storage

storefront/shelving/counter modifications required

8. Deal discovery

KEEP public listings, private approaches, introductions, rumors/leads, estate situations, debt pressure, retirement, relocation, distress and bank-held property.

Property markets are not perfect universal listing databases.

9. Negotiation

KEEP exchange-based negotiation. A seller can trade price against closing certainty, cash, financing, contingencies, earnest money, transition, equipment inclusion and other terms. No generic Negotiation Skill discount.

10. Deal structure

KEEP headline price separate from full transaction terms:

cash at closing

financed portion

seller-carried balance

earnest money

closing date

exclusivity

contingencies

included/excluded property

inventory/equipment

tenant/account treatment

transition assistance

11. Diligence

KEEP issue-specific diligence instead of one percentage meter. Diligence can answer title, books, physical condition, employee retention, supplier continuity, tax status and market sustainability questions. Closing with unresolved issues remains possible.

12. Property encumbrance burden

Deepen diligence to include, where applicable:

title/ownership

mortgages/security interests

taxes due/delinquent

special assessments

occupants/tenants

leases/use arrangements

access/corridor rights

other claims

survey/boundary uncertainty

included/excluded improvements

New principle:

Closing transfers whatever was actually conveyed, not whatever the buyer assumed existed.

13. Valuation categories

New explicit separation:

Asking Price: seller's requested terms/price.

Expected Market Value: informed estimate of likely transaction value under current market conditions.

Buyer-Specific Strategic Value: value to a particular buyer because of adjacency, integration or strategy.

Carrying/Book Value: ledger carrying concept.

Distressed/Liquidation Value: expected realizable value under time pressure.

These can diverge substantially.

14. Buyer-specific strategic value

A parcel ordinary buyers value near $300 might be worth $600 to an owner who needs it to combine two adjacent sites. The seller may infer that value from real circumstances. This is not arbitrary player-targeted price inflation.

15. Land assembly

Elevate land assembly to a first-class strategy. Larger projects may require several parcels or rights. A player can face holdouts, high prices, rival acquisition, title issues, occupants or access problems.

Owning most of a site does not authorize construction across land not controlled.

16. Land banking and carrying cost

Land speculation remains valid, but holding land carries real opportunity cost:

capital tied up

taxes

financing cost

maintenance/security where improved

assessments where applicable

forgone investment

There is no automatic fixed annual appreciation.

17. Geography-driven revaluation

Infrastructure changes land value through actual spatial effects. Rail/depot development changes traffic, freight, clustering, housing and road patterns rather than applying a universal percentage increase to all land.

18. Capacity vs authorization

Hard lock:

Physical Development Capacity: what the site could plausibly support.

Authorized Projects: what an owner has actually approved.

Work in Progress: what is physically being built.

Never collapse these.

19. Project Authorization

A major project authorization may define:

scope

site

budget authority

funding route

sourcing policy

contractor/internal work policy

quality/specification target

schedule/priority

owner-reserved decisions

acceptable contingency/overrun range

After authorization, routine continuation should proceed automatically until genuine exceptions need escalation.

20. Project execution

KEEP staged physical project logic. Stages/dependencies vary by project rather than using one immutable universal stage list. Possible work includes site work, foundation/masonry, framing, enclosure, roofing, openings, interior work, finish and final preparation.

Overall completion percentage can exist only as a summary. Work-package state remains authoritative.

21. Project Stockpiles

KEEP. Materials allocated to a project cannot simultaneously be ordinary sale inventory. Prebuying protects against shortages but ties capital and creates storage, theft, fire, weather and deterioration exposure.

22. Direct-to-project production

KEEP. Owned producers can manufacture directly against project requirements without fictional intermediate retail steps. Real logs/materials, labor, machinery, freight, loading/unloading, ledger attribution and opportunity cost remain.

23. Construction project financial reporting

New four-number framework:

Estimated Cost: original/updated estimate.

Committed Cost: orders/contracts already entered into.

Spent to Date: money/resources already paid or consumed.

Forecast Cost to Complete / Forecast Final Cost: best current estimate based on known remaining burden.

Forecasts use ranges where records or estimates do not support exact precision.

24. Estimating remains uncertain

Estimating skill improves information about material quantities, freight, labor duration, difficult site work, specialists, weather and rework. It does not directly make work cheaper.

25. Scope changes / Change Orders

Major new work or discovered conditions should create revised scope, additional authorization or changed forecast. Projects do not silently absorb unlimited new work under the original approval.

Possible responses include:

maintain scope and accept higher cost

reduce scope

change design

pause

cancel

26. Project failure persists

A project can become paused, underfunded, abandoned, partially complete, damaged, disputed or foreclosed with the property.

New lock:

Cancellation/failure does not reset the site or generate a magical refund.

Unfinished work, materials, salvage and deterioration remain in the world.

27. Construction quality becomes property history

Defects or weak construction can create hidden or known roof, structural, drainage, heating or other issues later. Project history remains part of Building/Asset history and can affect future diligence.

28. Condition vs Presentation

KEEP separate:

Condition: structural/mechanical/weatherproofing/reliability state.

Presentation: cleanliness, paint, frontage, visible care and appearance.

A building can be attractive but unsafe or ugly but mechanically sound.

29. Maintenance vs improvement

Distinguish:

maintenance/repair: preserve or restore existing function

refurbishment: improve condition/presentation

expansion: add capacity

conversion: change use

new construction: create new physical capability

30. Construction & Contracting business

KEEP full business archetype based on people, tools, working capital, suppliers, subcontractors and reputation. Scale emerges organically; no Contractor Level gates.

31. Contractor backlog

KEEP: backlog is not revenue. Awarded work still needs labor, material, time, equipment, working capital and supervision. Overbooking can create delays, losses and reputation damage.

32. Development speculation

KEEP. Residential/commercial development responds to expected demand with construction delay and uncertainty. Completion does not guarantee occupancy.

33. Development Feasibility report

A serious project can evaluate:

Site

owned/control status, suitability, grading/site work, access, capacity.

Project

materials, labor, contractor availability, duration.

Finance

cash requirement, funding plan, carrying burden.

Market

expected rent/revenue, current vacancy, known competing projects, household/business pressure.

Risks

estimate confidence, unresolved site/supply/labor/timing issues.

Use Unknown/ranges rather than fabricated ROI precision.

34. Vacant land carries economic state

Unimproved land still ties capital, incurs tax/financing/assessment burdens and may provide yard, storage, access, expansion or strategic control.

35. Property disposition

Add a real selling lifecycle mirroring acquisition:

Owner decides to sell -> property prepared/information assembled -> potential buyers become aware -> offers -> negotiation -> buyer diligence -> financing/title conditions -> closing.

Do not use a formulaic instant Sell button for serious property.

36. NPC property parity

NPCs can hold, refuse to sell, speculate, develop, rent, convert, sell, become distressed, assemble sites and compete for property under the same rules.

37. Canon locks from Audit 04

Ownership is not universal use permission.

Parcel is not Building.

Building is not Occupancy.

Suitability is use-specific.

Asking Price is not Value.

Market Value is not Buyer-Specific Strategic Value.

Development Capacity is not Project Authorization.

Project Authorization is not Completion.

Project Estimate is not Final Cost.

Materials are consumed when work occurs.

Internal supply is not free.

Partial construction persists.

Construction quality persists into property history.

Property markets move through real geography.

Disposition requires a counterparty.

Audit 04 Canon status: LOCKED.

SYSTEM AUDIT 05 - SUPPLY / AGREEMENTS / LOGISTICS / TRANSPORT

Final verdict: KEEP + DEEPEN.

The Canon already separates standing agreements from physical shipments, prevents global inventory teleportation, uses composite transport capacity, supports multi-leg and multi-stop routes, and now correctly distinguishes Livery from Freight/Hauling while allowing combined businesses. The key need is one universal Logistics Execution grammar and explicit ownership-versus-custody authority.

1. System purpose

The system answers:

What needs to move, why is it moving, who owns it, who promised to move it, what capacity is actually available, what route will it use, what happens while it travels and when is the commercial obligation truly fulfilled?

2. Universal logistics loop

Need / Demand -> Supplier / carrier identified -> Terms agreed -> Goods/capacity committed -> Source inventory available -> Shipment Order -> Transport capacity assembled -> Loading -> Departure -> Travel Leg -> Transfer/Intermediate Stop -> Additional Leg(s) -> Arrival -> Unloading / Receiving -> Inventory ownership/location update -> Commercial settlement -> Performance/relationship history.

3. Agreement vs Shipment

KEEP hard separation.

A standing agreement describes an ongoing commercial relationship. Each actual delivery is a separate physical Shipment. An agreement can remain active while individual shipments succeed, partially fill, miss, delay or fail.

4. Agreement hierarchy

Expand commercial commitment types conceptually:

Spot Order

Recurring Supply Agreement

Requirements Agreement

Allocation/Percentage Agreement

Reserved Supply

Freight Contract

Preferred Carrier / Preferred Supplier

Project Supply Commitment

These structures differ because commitments differ, not because the game needs separate incompatible systems.

5. Inventory reserve and availability

For transport-relevant goods:

On Hand - Protected Reserve - Already Committed - Unavailable/Condition Restricted = Available to Allocate.

Priority and reserve rules remain real management policy.

6. Fulfillment Priority policy

When promises collide, managers may prioritize based on:

contract priority

internal strategic priority

due date

customer importance

margin

breach consequence

perishability

owner lock

There is no hidden universal maximize-profit rule.

7. Internal ownership does not automatically win

Owned businesses are not automatically entitled to every internal input before outside contracts. External contract economics, breach costs, opportunity value and strategic policy matter.

8. Shipment as persistent authority

A Shipment can conceptually know:

source

destination

cargo

quantity

owner

originating agreement/order

carrier

current custodian

assigned vehicle/team

assigned labor

route/legs

current location/progress

loading state

departure

ETA/expected arrival

actual arrival

delay/failure state

freight charges

cargo condition where relevant

Shipments persist even when no related scene is loaded.

9. Ownership vs custody

New hard doctrine:

Possession, custody and ownership are separate economic facts.

A carrier may have custody of customer-owned goods. A warehouse/elevator may physically hold goods owned by someone else. Repair providers can hold customer property. Physical control does not automatically transfer ownership.

10. Ownership transfer timing

Physical location change does not automatically imply legal ownership change. Ownership may transfer at purchase, loading, delivery, acceptance or another agreed point according to historically appropriate terms.

Architecture must permit this variation even where exact legal calibration remains future research.

11. Loading/unloading are work

Transport does not begin or end instantaneously. Loading and unloading consume compatible labor, time, space, equipment and access. Receiving businesses can become bottlenecks.

12. Receiving capacity

A destination may be economically willing to buy but physically unable to receive because storage is full, loading space is blocked, workers are unavailable, arrival is outside usable hours or special handling is missing.

13. Transport capacity is composite

Usable trip capacity emerges from compatible:

vehicle

draft power

harness/equipment

driver/teamster

cargo compatibility

route feasibility

time

condition

Do not store a magical wagon-count logistics capacity.

14. Available Transport Capacity report

Player-facing report can explain owned vs ready vs in-use vs under-repair assets rather than one abstract number.

15. Operational assignment vs asset ownership

KEEP. A wagon/team can be legally owned by one ledger and operationally assigned elsewhere under authorized portfolio policy. Asset condition, financing and ownership history remain attached to the owning ledger.

16. Transport cost attribution

When one controlled business's transport asset serves another business, distinguish legal owner/payroll/maintenance burden from economic consumer of the service. Internal reimbursement/transfer pricing may vary but consolidated economics remain truthful.

17. Dispatch is policy-driven

Routine wagon dispatch should normally be handled by managers under standing policy. Player attention belongs to:

commitment conflicts

major delay

rush job

capacity shortage

repeated empty mileage

expansion request

route failure

breach risk

No requirement to manually route every ordinary shipment.

18. Route vs Shipment

New lock:

A Route is a reusable operating plan. A Shipment is actual cargo movement.

The same Monday route can carry different cargo each week.

19. Multi-stop routes / through-cargo / backhauls

KEEP. Routes can carry through-cargo, reserve capacity for later pickups, wait conditionally and use spare capacity for eligible cargo.

20. Empty mileage

Elevate empty mileage as a central freight economic. Full outbound utilization can still produce weak economics if return travel is unpaid. Lower-margin return cargo can be valuable because it converts otherwise empty mileage into paid work.

21. Utilization is diagnostic, not objective

KEEP. 100% utilization can still lose money; 60% utilization can be highly profitable. Reports should show loaded/empty mileage, revenue, feed, labor, maintenance, delays, rejected work and margin.

22. Rush work

KEEP rush freight as real priority displacement. Rush premiums compensate for scarce capacity, schedule disruption and urgency; they are not arbitrary modifiers.

23. Freight pricing

Ground quotes in:

distance

road/terrain

load/cargo

team requirements

return opportunity

urgency

loading/unloading burden

expected delays

current capacity

relationship

risk

Do not reduce all freight to distance x constant.

24. Minimum-charge/service-zone policies

KEEP legitimate carrier policies such as minimum service charge, minimum load, service zone, target margin and rush premium. These are business policies, not designer restrictions.

25. Livery vs Freight/Hauling

New Revision XVIII lock preserving Research Item 6:

Livery core

stabling

feeding

horse/team/rig hire

possible drivers

possible sales

possible passenger/hack/bus work

Freight/Hauling core

customer cargo movement

route/contracts/loading/utilization/backhauls/teams/drivers/feed/wagon condition/scheduling

Drayage/Cartage

legitimate short/local hauling specialization, especially depot/warehouse/local transfer

Combined operations

Livery & Freight

Livery with Dray Line

Livery, Feed & Sale Stable

other evidence-supported combinations

Shared assets create coordination, not an abstract synergy bonus.

26. Feed links transport to agriculture

Transport animals consume real hay/oats/bran/feed/water. Feed shortages can raise transport costs, tighten margins, raise freight rates, increase landed cost and affect local prices through actual resource chains.

27. Draft-animal work/recovery

Animals are not infinitely reusable motive power. Work creates fatigue, feed/water needs, injury exposure and recovery requirements. This shares causal philosophy with human fatigue but remains a biological system.

28. Roads are infrastructure

KEEP route effects from grade, surface, drainage, bridges, crossings, weather, feed/water/stopping infrastructure. Physical improvement changes travel economics rather than granting abstract freight bonuses.

29. Complete closure can occur

Some weather/route states should make travel practically impossible rather than merely applying speed penalties.

30. Seasonal stockpiling

Deepen working-capital/logistics interaction. Businesses may stockpile before unreliable seasons, trading resilience against tied capital, storage burden, spoilage, fire and theft risk.

31. Logistics information and ETA

New lock:

Departure/arrival facts are facts; ETA is a forecast.

ETA confidence depends on route, weather, communication, carrier and last known information. Telegraph/telephone/rail systems can improve information speed but do not create omniscience.

32. Exception-based logistics alerts

Surface missed departure, partial fill, blocked route, receiving constraint, team shortage, repeated empty returns, deteriorating carrier reliability and repeatedly rejected profitable work. Do not notify every routine movement.

33. Supplier evaluation

KEEP price, freight, lead time, reliability, quality and volume limit. Add supplier flexibility dimensions such as:

minimum order

credit willingness

rush capacity

seasonal capacity

reserve-stock willingness

recurring terms

transport responsibility

34. Supplier relationship memory

Repeated successful/failed business can affect trust, reserved stock, credit, shortage priority and knowledge of requirements through real relationship history rather than a generic Supplier Level.

35. Partial fulfillment

Agreements can result in partial shipment, accepted short delivery, backup sourcing, delay or priority allocation. Do not reduce every shortage to Agreement Failed.

36. Backup suppliers

Maintaining a secondary supplier can trade slightly worse ordinary economics for resilience. This is a real relationship and may have weaker terms or low use, not a free resilience stat.

37. Contracted capacity is not completed capacity

Freight companies can overbook. Contracts do not create wagons, teams or drivers. Excess commitments can lead to failures, lost reputation, expansion requests or contract loss.

38. Overbooking remains allowed

Do not hard-block a player from accepting work simply because current capacity looks insufficient if they could add assets, hire, subcontract or reorganize before performance is due. Consequences determine whether the choice was smart.

39. Subcontract freight

A primary carrier can subcontract excess work where appropriate, retaining customer obligation and some margin while giving up control/profit.

40. Rail integration

KEEP depot as transfer machine, not teleport portal. Rail still requires receipt, unloading, documentation, storage and local drayage/final delivery. Local terminal/platform/storage capacity can bottleneck traffic.

Rail transforms wagon routes rather than deleting the wagon economy:

distant railhead long haul

nearer railhead

local depot

final-mile/drayage/feeder specialization

41. Intermodal journeys

Support multi-leg chains such as supplier -> rail -> railhead -> wagon freight -> warehouse -> local dray -> customer. Each leg can have different carrier, cost, delay and custody.

42. Storage is part of logistics

Warehousing buffers timing mismatch between arrival and consumption/onward shipment. Storage creates cost, capacity, damage/fire/theft exposure and handling burden.

43. Pipeline inventory

Add explicit **Inventory in Transit** state. Distinguish:

on hand

reserved

inbound

outbound/in transit

unavailable

customer-owned/custody

Inbound stock cannot satisfy immediate local demand until physically received.

44. Opportunity & Pressure integration

Repeated logistics failures can create freight opportunity, carrier expansion pressure, depot drayage gaps, remote-camp supply pressure or excess-carrier distress through actual evidence.

45. Canon locks from Audit 05

Agreement is not Shipment.

Shipment is not Route.

Possession is not Ownership.

Custody is not Ownership.

Inbound inventory is not on-hand inventory.

Contracted capacity is not available capacity.

Transport capacity is composite.

Loading and unloading are work.

Time and capacity are exclusive.

Vertical integration does not make logistics free.

Rail changes the network rather than deleting older transport.

ETA is forecast.

Routine dispatch belongs to management policy.

Audit 05 Canon status: LOCKED.

SYSTEM AUDIT 06 - CONDITION / MAINTENANCE / REPAIR / FAILURE

Final verdict: KEEP + DEEPEN, with strong anti-micromanagement protection.

This is one of the strongest completed systems in the Canon. It already separates actual condition from known condition, makes inspection an information action, allows deferred maintenance, requires physical maintenance work and uses causal failure chains. The main addition is a separate Operating State and one universal Work Order lifecycle.

1. System purpose

The system answers:

What physical state is this asset in, what do we know, what work does it need, can we continue using it, what are we risking if we do, and when does repair stop making sense compared with overhaul or replacement?

2. Universal asset lifecycle

Asset exists -> Use/exposure -> Component wear/damage -> Underlying condition changes -> Symptoms may appear -> Worker/operator notices -> Inspection/diagnosis -> Defect becomes known -> Operating response -> Continue/restrict/shut down -> Service/repair authorized -> Parts/labor/facility assembled -> Work -> Repair quality verified -> Return to service -> History persists -> Maintain again / overhaul / replace / sell / salvage.

3. Condition domains

KEEP economically meaningful component groups, not one Health number and not microscopic part simulation.

Examples:

Wagon

wheels/running gear

axle/bearings

frame/body

harness

Steam sawmill

boiler/pressure system

engine/mechanical state

drive system

saw/cutting system

building/support condition

4. Condition vs Defect

New explicit lock:

Condition describes broader accumulated physical state. Defect describes a specific problem.

An asset can be worn without a particular acute defect, or otherwise good while suffering one serious sudden defect.

5. Operating State

Add a separate authoritative **Operating State** such as:

Available

Available with restriction

Monitoring

Scheduled maintenance

Out of service - planned

Out of service - failed

Temporary service

Retired/stored

Salvage/disposal

Condition, defect and operating state are separate facts.

6. Restricted operation

A damaged asset can remain usable under grounded limits such as reduced load, reduced pressure, reduced throughput or restricted space occupancy. This creates real Continue / Restrict / Stop decisions.

7. Wear causality

KEEP wear from operating time, workload, load, terrain, heat, vibration, lubrication, material characteristics, storage, weather, moisture, corrosion and other asset-specific exposure.

Idle assets avoid use wear but can still deteriorate environmentally.

8. Protection remains physical

Shelter, drainage, lubrication, road quality and other protective conditions affect physical exposure rather than giving abstract lifespan bonuses.

9. Construction quality connection

Newly constructed assets are not automatically physically perfect. Weak construction can create known or latent defects that enter asset history and may be discovered later.

10. Actual vs Known condition

KEEP as an A+ information system. The simulation knows reality; businesses know only what operation, inspection, repair, records and observation reveal. Inspection knowledge becomes stale as additional use/exposure accumulates.

11. Inspection is not a health-bar reveal

Inspection should produce evidence and diagnosis such as wear, looseness, cracks, leaks, vibration, unsafe conditions or recommended operating limits. Avoid magical exact failure percentages unless genuinely measurable.

12. Inspector capability

Different people notice different things. Ordinary operators may observe symptoms; specialists may identify causes and repair scope. Better capability improves information, not physical condition.

13. Symptom vs diagnosis

New lock:

Observed symptom is not automatically the diagnosed defect.

Example: vibration may result from mounting, bearing, alignment or drive damage. Diagnosis is a separate information step.

14. Work Order authority

Add universal **Work Order** for repair/service. A Work Order can identify:

target asset/component

reported symptom

known/diagnosed defect

requested work

urgency

required capability

parts/material

location

assigned provider

estimated labor/time

quote/agreed price

operational restriction

status

completion

verification

payment

Use the same shared lifecycle for blacksmith work, wagon repair, building maintenance, machinery service and later rail/utility maintenance.

15. Maintenance Backlog

Backlog is a set of real outstanding work, not a score. Reports should show actual service/repair needs, next actions, skills/parts, downtime and status.

16. Backlog Age / Criticality

Recognized maintenance needs should consider:

severity

urgency

contextual criticality

consequence of delay

downtime opportunity cost

parts/provider availability

17. Asset Criticality

New lock:

Criticality is contextual, not inherent. It depends on what relies on the asset and what alternatives/redundancy exist.

The only mine pump is more critical than one of four redundant pumps even if physical condition is identical.

18. Dependency chains

Failure changes physical capability; dependent systems then react. A sawmill drive failure can affect lumber throughput, customer orders, log accumulation, freight and construction without one universal failure penalty.

19. Maintenance policy

KEEP standing policies for spending limits, reserve protection, approval thresholds, shutdown rules, inspection cadence and spare-parts targets. Player may deliberately defer maintenance.

20. Maintenance Stance

A convenient player-facing stance such as Conservative / Balanced / Run Hard may bundle real policies underneath but must never create magic bonuses. Players may inspect/override the underlying policies.

21. Deferred maintenance is a real gamble

Cash-poor businesses can rationally postpone work. The consequence is unresolved physical state and increased exposure where causally justified, not an arbitrary Neglect penalty.

22. Boom maintenance debt

Boom periods can create physical maintenance debt because equipment is run hard and downtime is scarce. Recessions can provide time but little cash. This tension should emerge naturally.

23. Routine service consumables

Track economically meaningful lubricants, belts, timber, roofing material, packing, protective material or other consumables where shortages/sourcing matter. Do not simulate trivial maintenance clutter.

24. Spare parts

KEEP resilience trade-off. Stocking parts ties cash and storage but can prevent long downtime. No universal Spare Parts inventory that repairs any machine.

25. Cannibalization

Allow compatible failed assets to donate usable parts where physically plausible. Donor asset loses corresponding rebuild/salvage potential.

This may be a later implementation priority but is Canon-valid.

26. Redundancy

KEEP separate from spare parts. Backup installed/usable assets can preserve operation but require capital, storage, upkeep and may sit idle.

Backup capacity may be partial rather than identical.

27. Downtime

Treat downtime as real economic cost alongside parts/labor:

lost production

missed freight

delayed contract

displaced worker time

lost rent/occupancy

hired substitute equipment

emergency costs

28. Repair Decision report

Serious repair decisions may show:

known issue

operating state/restriction

repair estimate

expected downtime

parts/provider availability

consequence of delay

business commitments currently exposed

29. Seasonal maintenance

Schedule maintenance around actual economic/seasonal windows such as harvest, winter hauling, roofing, mine pumping, lodging occupancy or traffic cycles. Timing matters because downtime opportunity cost changes.

30. Maintenance labor is real

Maintenance consumes finite labor time. It does not happen for free at night unless people actually perform it and scheduling permits.

31. Proper vs Temporary/Improvised repair

Keep distinct:

Proper repair: addressed defect restored to expected service.

Temporary repair: limited/restricted service restored pending proper work.

Improvised repair: available materials/capability restore some function with limitations/follow-up need.

Temporary repair cannot become a permanent cheap exploit.

32. Repair quality

Repair work can be excellent, adequate, weak or incomplete according to actual capability/materials/supervision. Customer may not know immediately. Later use can reveal problems.

33. Verification

Work Completed and Defect Confirmed Resolved do not always have to be identical information events. Critical work can require test/inspection/operational verification.

34. Component replacement

Replacing one component does not reset the whole asset. The component receives new history/state while other systems retain their age/condition.

35. Overhaul

Overhaul is a bundle of substantial real work on specified systems, not Set Condition = 90%. Different overhaul scopes are possible and recorded.

36. Repair vs Overhaul vs Replace vs Sell vs Salvage

Late-game capital decisions should compare:

repair

overhaul

replacement

sale as-is

salvage

based on cost, downtime, remaining service, throughput needs, financing, input efficiency, parts, market outlook and resale/salvage.

37. Replacement is not a tech-tier mandate

Newer equipment is not automatically economically superior. Old equipment may remain rational to repair; newer equipment may be rational to buy early depending on actual economics.

38. Remaining service life is a forecast

Do not expose deterministic timers. Experts may estimate likely service horizon based on condition, history, workload and inspection, but future failure remains uncertain.

39. Failure causality

KEEP. Failure exposure comes from design, condition, unresolved defects, load, environment and operating practice. Probability primarily resolves timing among already plausible stressed outcomes rather than generating random healthy-asset explosions.

40. Failure categories

Distinguish:

wear-driven failure

defect-driven failure

accident damage

catastrophic destruction

A new asset can still be destroyed by accident without violating the Condition system.

41. Secondary damage

New explicit rule:

Deferred or continued operation with a defect can cause additional physical damage where causally appropriate.

Examples: loose wheel -> hub/axle damage; leak -> moisture/structural damage; overheating bearing -> shaft damage; chimney defect -> fire.

No universal monthly maintenance-cost escalator is required.

42. Defects do not all worsen identically

Some remain stable; some worsen only under heavy load, moisture, heat, freeze/thaw or continued use. Avoid deterministic countdown behavior.

43. Catastrophic failure remains rare but meaningful

Routine wear should more often create repair burden/downtime than catastrophe. Serious failure requires credible physical conditions. Maintenance is risk management, not tribute to RNG.

44. Safety-critical governance

Businesses can set safety shutdown policies and manager discretion rules where physically/legal appropriate. Player override cannot make impossible operation physically possible.

45. No universal Safety score

Safety exposure emerges from actual defects, affected systems, people exposed and operating conditions.

46. Fire integration

Maintenance defects can become fire-system precursors such as cracked stovepipe, damaged chimney, overheating bearing, boiler problem or later electrical fault. No generic Poor Maintenance fire modifier.

47. Buildings

Building condition can include structure, roof/weatherproofing, chimney/heating, loading platforms and specialized systems. Buildings simply use different time scales from machinery.

48. Condition vs Presentation

KEEP separate. Cosmetic improvement does not repair structural problems. Structural maintenance does not automatically improve appearance.

49. Animals are not machine assets

Hard boundary lock:

Living animals use biological health/fatigue/nutrition/injury/age/recovery authority, not machinery durability.

The systems can interact but must not collapse.

50. Used equipment markets

KEEP uncertain used-asset condition, inspection, refurbishment, operation, resale and salvage. Cheap used equipment can be a bargain or deferred-maintenance trap.

51. Maintenance records and diligence

Service/overhaul records are evidence, not guarantees. Poor or misleading records can coexist with actual condition. Diligence combines records with inspection.

52. Condition and value

Condition affects acquisition/collateral/resale through actual repair cost, productive capability, downtime, replacement scarcity and salvage, not a universal Poor Condition percentage.

53. Failed equipment persists

Failed assets remain physical objects that can be repaired, stored, sold, parted out, scrapped, abandoned or used as limited backup.

54. Maintenance-service market

Growth of wagon, machinery, rail or utility assets creates real repair/service demand and can support blacksmiths, mechanics, machinists or internal maintenance departments.

55. Specialist bottlenecks

Parts and money may exist while no qualified local provider is available. Options include wait, import specialist, send component away, improvise temporary repair, train staff or replace equipment.

56. Internal maintenance scaling

A portfolio may rationally establish internal repair staff/shop only when actual workload supports it. Centralized maintenance does not merge asset ownership.

57. Manager capital requests

Maintenance systems can generate grounded requests for backup assets, overhauls, spare parts, staffing or facility investment based on actual downtime and rejected work.

58. Information doctrine

Reports distinguish known facts from forecasts:

known defect

repair quote

current unavailable state

versus

expected completion

likely progression if delayed

projected reliability benefit of replacement

59. Preventive maintenance is not perfect

Good maintenance reduces avoidable exposure but cannot eliminate age, accidents, latent defects or unusual stress. Neglected assets can sometimes continue longer than expected.

60. Maintenance is not a designer money sink

Hard philosophy lock:

Every maintenance cost must correspond to real work, material, downtime, risk reduction or physical restoration.

61. Canon locks from Audit 06

Condition is not Defect.

Condition is not Operating State.

Symptom is not Diagnosis.

Inspection is information.

Maintenance is work.

Routine service should normally be automated.

Maintenance preserves capability; capital improvement changes capability.

Repair scope persists.

Temporary repair is not full restoration.

Failure requires causal exposure.

Secondary damage is possible.

Criticality is contextual.

Downtime is an economic cost.

Spare parts and redundant assets are different resilience strategies.

Failed assets persist.

Animals are not machine assets.

Audit 06 Canon status: LOCKED.

SYSTEM AUDIT 07 - MARKETS / OPPORTUNITY & PRESSURE / RIVAL AI

Final verdict: KEEP + DEEPEN.

This system determines whether the world is a living economy or merely a player-facing simulation. Existing Canon already uses bounded choice, localized markets, real alternatives, equal-rule NPC competition and discoverable rival intent. The critical new rule is that Opportunity & Pressure is an analytical/synthesis layer; it observes and interprets economic reality but does not create it.

1. System purpose

The combined system answers:

Where is economic value being left on the table, where is strain accumulating, who can perceive it, who can realistically act on it, what competing responses are underway, and how does the opportunity change as actors commit resources?

Three distinct jobs remain:

Markets: counterparties actually choose and transact.

Opportunity & Pressure: synthesize persistent evidence.

Rival AI: economic actors perceive some evidence and act using real resources.

2. Fundamental market loop

Need / Supply / Opportunity exists -> Actor has reason to reconsider -> Actor becomes aware of alternatives -> Feasible alternatives identified -> Shortlist evaluated -> Choice/negotiation -> Physical/economic execution -> Transaction succeeds/partially succeeds/fails -> Evidence accumulates -> Prices/availability/relationships/profitability change -> Businesses and entrepreneurs reconsider.

3. Opportunity & Pressure is non-causal synthesis

New foundational lock:

Opportunity & Pressure never manufactures demand, supply, labor, customers, money, property, capacity or rival resources. It analyzes evidence produced by authoritative systems.

A shortage exists because customers cannot obtain what they need. The analytical layer can identify and report that shortage; it does not create it.

4. Market specificity

KEEP competition specific to:

good/service + customer group + reachable geography.

There is no one empire-wide Market Share score.

A store can simultaneously lead workwear, trail grocery staples and barely participate in remote-camp supply.

5. Market share is derived

Market share describes completed transactions and estimated relevant market activity. It does not directly buff pricing power or determine the next customer choice.

6. Outside option

KEEP. Customers can substitute, postpone, repair, produce at home, travel, buy from another settlement or let demand leak off-map. Improved rail/transport can expose local firms to outside competition as well as create sourcing opportunities.

7. Market power = alternatives

Elevate to foundational doctrine:

Bargaining power arises primarily from the quality of credible alternatives, not a Monopoly/Influence stat.

This applies to sellers, buyers, lenders, borrowers, employers, workers, carriers and property owners.

8. Opportunity / Pressure lifecycle

Formalize persistent states:

Background Condition

Emerging Signal

Confirmed Opportunity / Pressure

Intensifying

Contested / Responding

Easing

Resolved

Reversed

Former shortages can become oversupply.

9. Opportunities can disappear without completion

An opportunity can fade because demand disappears, a route changes, an existing firm expands, outside supply arrives or another structural condition changes. It does not require a quest-completion flag.

10. Opportunity evidence dimensions

Internal synthesis may consider:

unmet demand

urgency

prices

observed margins

lost sales

refused jobs

leakage/distance

missed recurring commitments

capacity utilization

expected persistence

entry cost

labor availability

site availability

supply access

financing access

Player-facing reports should explain causes rather than show a raw Opportunity Score.

11. Pressure evidence

Housing, labor, freight and other pressures should be explained through actual symptoms such as crowding, vacancies, wages, overtime, missed loads, premium prices, leakage, failed hiring and delay.

12. Confidence and evidence quality

The simulation can know more than the player. Opportunity reports should be limited by legitimate information and can use confidence language/ranges when evidence is incomplete.

13. Player and rival information differ

Different actors know their own books, customers, suppliers, property discussions and relationships. Two entrepreneurs can see the same settlement and reach different conclusions without either cheating.

14. Rival economic decision loop

Formalize:

Trigger -> Actor notices possible problem/opportunity -> Gathers/recalls information -> Feasible consideration set -> Estimates economics -> Compares alternatives -> Forms Intent -> Seeks resources -> Makes commitments -> Execution -> Observes results -> Adjusts/continues/abandons.

15. Rival Intent is persistent

Rival plans can progress through states such as:

Considering

Investigating land

Seeking finance

Negotiating parcel

Project authorized

Construction underway

Opening preparation

Operating

No hidden countdown is required.

16. Rival plans can fail

Intent does not guarantee entry. Plans can fail because of seller refusal, finance denial, labor shortage, cost, supplier failure, better alternatives, liquidity or market change.

17. Rivals can be wrong

NPCs can overestimate demand, underestimate construction cost, overexpand, price badly, overstock, choose poor property, take too much debt, misread temporary demand, lose employees or overpay for acquisitions. These are decisions under shared rules, not hidden inefficiency penalties.

18. Rivals can also be highly competent

Some NPCs can identify opportunities early, have strong supplier knowledge, deep capital, trade expertise, good hiring, strong acquisitions or strategic patience. Advantages arise from actual resources, skill, knowledge, relationships and policy, not difficulty modifiers.

19. Owner preference shapes strategy

NPC decisions can respond to reserves, obligations, expected return, uncertainty, risk tolerance, commitments, relationships, expertise, time horizon and feasible alternatives.

Avoid giant personality-bonus systems. Preferences should bias real choices.

20. Opportunity cost applies to rivals

NPCs have finite capital, management time and property. Choosing one opportunity can prevent another. Rival strategy therefore changes when the player acquires critical resources first.

21. Entry requires assembled capabilities

KEEP. New firms require real combinations of skill, knowledge, capital/finance, property, suppliers, labor and confidence. Strong demand alone does not spawn a business.

22. Local opportunity does not require local startup

Unmet demand can resolve through local expansion, outside supply, regional entrant, substitution, demand destruction or customer migration.

23. Existing firms can respond before new entrants

New doctrine:

Opportunity response must consider expansion or adjacent capability growth by existing businesses as well as startup formation.

Examples: General Store expands assortment; carrier buys another team; livery adds drayage; mill adds flour/feed sales.

24. Adjacent expansion

Capability-based business architecture should let historically/economically adjacent services grow organically where existing assets, labor and knowledge support them.

25. Market saturation / oversupply

Add explicit pressure detection for:

falling utilization

persistent vacancy

growing unsold inventory

discounting

declining margins

idle equipment

closures

Oversupply is not simply Shortage Resolved.

26. Saturation does not automatically close firms

Firm survival depends on cash reserves, debt, property ownership, owner labor, other profitable lines, expectations and creditor tolerance. Do not despawn the weakest firm mechanically.

27. Price wars

KEEP as real competition using cash, inventory, margins and reserves. Deep-reserve rivals can survive longer. Customers can benefit. No generic Price War modifier is necessary.

28. Competition is multidimensional

Businesses compete through:

price

availability

assortment

reliability

credit

location

hours

delivery

service quality

specialization

relationships

speed

capacity

reputation

29. Labor competition

Rivals can recruit, poach, raise wages and seek managers or specialists. Hiring activity can itself become evidence of rival intent.

30. Supplier competition

Rivals can reserve supplier output, sign agreements, buy scarce stock or secure transport capacity. Competitive harm emerges through actual allocation rather than sabotage events.

31. Property competition

Rivals can pursue frontage, depot sites, industrial land, rental development and strategic adjacent parcels under the same acquisition rules as the player.

32. Distressed acquisitions and recombination

KEEP. Business closure/failure releases property, machinery, wagons, inventory, employees, managers, customer relationships and supplier capacity. Rivals, creditors, employees or the player can acquire/recombine them.

33. Rival exit decision

NPC owners can choose to continue, shrink, change service mix, sell assets, seek partners, sell the business, refinance, close or liquidate based on actual recovery capacity and alternatives rather than Profit < 0 for N months.

34. Closure changes the market

When a business closes, customers seek alternatives, workers move, property/equipment becomes available, competitor backlogs change and new opportunities can emerge automatically.

35. Employees/managers can become entrepreneurs

New deepening:

Experienced employees/managers with sufficient savings/finance, skill, knowledge, property access and willingness can become business owners. Entrepreneurship is not limited to a hidden investor pool.

36. Off-map entrepreneurs

Regional/outside investors remain valid but must obey Information Distance, migration/travel, property, finance and setup requirements. They do not materialize because a town has a need slot.

37. Entry lead time

Even after deciding to enter, actors must obtain property, finance, construction, staffing, inventory/equipment and other required capabilities. Competition therefore develops through real lead time.

38. Contested opportunities

An opportunity can remain active while rival response is already underway. Player may learn that another actor is seeking land, finance, employees or suppliers and decide whether to act, differentiate, wait or abandon.

39. Rival intent confidence

Player-visible rival information can be:

confirmed fact

strong evidence

rumor/weak evidence

Do not present all intent equally.

40. Rivals can conceal plans

Sophisticated actors do not need to publicly declare strategy. Intent becomes visible through credible traces such as property inquiries, lender conversations, recruitment, supplier activity, construction and public records where appropriate.

No espionage minigame is required.

41. Opportunity & Pressure does not reveal hidden intent automatically

Hard lock:

The simulation’s knowledge of a rival plan does not make that plan player-known.

42. Forecast uncertainty

Market/opportunity reports should show assumptions and risks.

Example:

Current evidence: 95% boarding occupancy; recent families unable to secure rooms.

Forecast assumption: mine expansion proceeds.

Known risk: two housing projects already under construction.

43. Pipeline response awareness

New core rule:

Opportunity analysis should account for known committed response already in the pipeline.

Housing under construction, freight fleets ordered, recruiting underway and supplier expansions can change the future outlook even before current shortage fully resolves.

This prevents analysis from pretending nobody is reacting, while still allowing overbuilding because information and forecasts are imperfect.

44. Speculative boom/bust

Opportunity, delayed projects and forecast error can naturally create cycles such as housing shortage -> simultaneous projects -> demand weakens -> vacancy -> distress -> acquisitions.

No scripted crash event is required.

45. Available capacity vs nominal capacity

Known business assets do not equal market availability. Existing wagons may be committed; contractor crews may have full backlog; lodging may be unsuitable. Opportunity analysis should use actual available capacity where information permits.

46. Service gaps

KEEP service opportunities such as finance, printing/news, medicine/pharmacy, barber/bath, fuel and repair, not only commodity shortages.

47. Institutional pressure

Some pressures involve civic/public institutions that private owners cannot directly build/own. Player opportunities may instead involve contracts, land, financing or indirect support.

48. No Town Needs checklist

Avoid UI such as Doctor 0/1 or Blacksmith 0/1. Report actual economic symptoms:

Repair backlog has doubled. Several jobs were sent to another node.

49. Opportunity categories

Analytical categories may include:

trade/inventory

labor

housing

services

land/frontage

freight/infrastructure

credit/capital

production inputs

regional-node viability

These are synthesis categories, not separate economic universes.

50. Persistent trend evidence

Use rolling evidence and periodic synthesis rather than full rescan/notification spam. Stockouts, failed purchases, missed orders, failed hires, vacancy changes, boarding saturation, route delays and closures update trend state.

One ordinary stockout does not automatically create a business opportunity.

51. Structural events can matter immediately

Extreme discrete changes such as bridge collapse, bank failure or major mine closure may instantly create large pressure without needing weeks of rolling evidence.

52. Rival response changes the signal

When competitors order capacity, hire, build or secure property, the opportunity forecast should change. Once new capacity operates, underlying evidence changes and pressure can ease or reverse.

53. Competitive Response / Market Movement memory

Add an analytical view of material responses already underway, such as:

rival acquiring land

housing under construction

carrier expanding fleet

bank tightening lending

merchant exiting

This helps explain where opportunities/pressures are heading.

54. Rivals do not react perfectly to every signal

Even strong opportunities may persist because actors lack savings, property, credit, willingness, family flexibility, confidence or compatible skill.

55. Entrepreneur confidence

Interpret confidence as willingness to commit under evidence, uncertainty, reserves, experience, risk tolerance and circumstances. Do not create an arbitrary 0-100 entrepreneur meter as the primary authority.

56. Entry friction is real requirements

Some businesses are easier to enter than others because they need different combinations of site, capital, equipment, labor and institutional form. Do not represent this only as a generic Entry Difficulty stat.

57. High profit attracts attention only if observable

Exact private profits are not automatically public knowledge. Rivals may infer success from traffic, hiring, deliveries, expansion, supplier volume, reputation or other observable consequences.

58. Commercial Standing and opportunity access

A well-known/trusted owner may hear about deals, partnerships, supplier arrangements, financing or private property earlier because counterparties know them. Do not convert this into artificial opportunity tiers.

59. Rival reputation

NPC businesses build localized reputation through actual quality, reliability, promises, credit behavior and service. Reputation affects future choices through memory and evidence.

60. No mandatory named nemesis

Important rivals may emerge naturally and can change over the campaign. Do not require a permanent designated Rival character.

61. Competition and cooperation can coexist

New lock:

Economic actors can compete in one relationship and cooperate in another.

Rivals can subcontract, share suppliers, syndicate, lend, buy/sell assets, refer overflow work, partner or merge while competing elsewhere.

62. Consolidation after market failure

Overcapacity can lead to closure, asset sales, contract transfers, worker movement and acquisition rather than simple deletion.

63. Regional competition

Local firms compete with nearby nodes and later rail-connected outside suppliers. Improved infrastructure is a two-way door: it can reduce sourcing costs while increasing outside competition.

64. Regional specialization

Different nodes can become stronger in livestock, lumber, mining, finance, distribution, lodging, agricultural supply or repair because of actual geography, investment and history, not fixed Town Type labels.

65. Market reach is product/service-specific

No universal market radius. Reach depends on product/service, urgency, transport, information, delivery capability and customer willingness.

66. No scripted anti-player rival spawn

Player dominance does not trigger a rival merely because market share crosses a threshold. Rival entry requires genuine attractive opportunity plus a feasible actor.

67. Monopoly is a strategic condition, not a policy

Sole suppliers can choose high price, moderate price, good service, rationing or other strategies. Entry pressure responds to actual conditions and alternatives.

68. Predatory/very-low pricing carries real cost

If permitted by applicable law, a wealthy business can price aggressively, but cash burns, inventory still costs money and the rival may outlast the player. No rival hit points.

69. Distress opportunities

Rivals can notice weak businesses, debt, valuable property or strong customer relationships and attempt acquisition under the same diligence/finance rules as the player.

70. Opportunities can be lost causally

The player can lose an opportunity because another actor buys the lot, secures the contract, hires the worker, acquires the business or signs the supplier. Do not use magical expiration timers.

71. Player-facing Opportunity / Pressure / Movements

A useful high-level presentation can separate:

Opportunities: attractive openings.

Pressures: current strain.

Movements: material known responses underway.

Do not turn every minor NPC action into an alert.

72. Persistent history

Opportunity state, pressure state and rival intent persist across saves. Major market events can feed the Lifetime Ledger selectively, while granular transactions remain in underlying ledgers/history.

73. Canon locks from Audit 07

Opportunity is evidence, not causation.

Pressure is persistent strain, not a random debuff.

Market share is derived.

Markets are specific by good/service, segment and geography.

Alternatives create market power.

Choice is bounded.

NPCs obey equal economic physics.

Rival information is bounded.

Rival intent is not execution.

Player knowledge of rivals requires evidence.

Existing firms can respond through expansion.

Opportunities can reverse.

Pipeline response matters.

Forecasts remain uncertain.

Economic actors can compete and cooperate simultaneously.

Failure recombines economic assets.

Audit 07 Canon status: LOCKED.

REVISION XVIII - CROSS-SYSTEM CONSOLIDATION LOCKS

The seven audits above create several shared authorities that now apply across all future business and system design unless a later Canon revision explicitly supersedes them.

A. Shared execution-object pattern

Several systems now have authoritative persistent execution objects:

Project for construction/development work.

Shipment for physical cargo movement.

Work Order for maintenance/repair/service work.

Employment relationship + Work Demand/Assignment for human labor.

Commercial Agreement/Commitment for standing economic promises.

These objects are not interchangeable. Agreements create commitments; execution objects carry out actual physical/economic work.

B. Shared separation of reality, relationship and information

Across all systems:

ownership is separate from possession/custody/control;

physical location is separate from legal ownership;

actual state is separate from recorded/known state;

forecast is separate from fact;

authority is separate from capability;

commitment is separate from fulfillment.

C. Shared anti-teleportation / exclusivity doctrine

People, assets, goods, animals, projects, property rights and transport capacity cannot be used simultaneously in physically incompatible ways merely because separate UI screens request them.

Time, location, ownership, custody, assignment and capacity remain authoritative.

D. Shared portfolio-accounting doctrine

Controlled businesses remain separate economic ledgers even when labor, goods, transport, maintenance or capital move among them. Internal activity may be reported locally but is eliminated or appropriately consolidated when calculating portfolio-wide external revenue, receivables, liabilities and profit.

Vertical integration transforms dependencies into internal responsibilities. It does not make labor, logistics, materials, maintenance, property or capital free.

E. Shared information doctrine

The simulation may know exact underlying reality. Player/NPC knowledge depends on legitimate evidence.

Reports may distinguish:

Known Fact

Recorded State

Evidence

Inference

Forecast

Unknown

Unknown is preferable to invented precision.

F. Shared automation doctrine

Routine execution should usually follow:

Player Decision -> Standing Policy -> Manager/Routine Execution -> Exception Escalation.

The player should make strategically meaningful decisions, not repeatedly click routine continuation, dispatch, maintenance, household purchases or ordinary staffing actions.

G. Shared causal-failure doctrine

Adversity should normally arise from real shortages, defects, financial structures, market changes, competing commitments, information gaps, bad decisions or physical disruption.

Avoid global random debuffs when the simulation can express the underlying cause directly.

H. Shared scale-transformation doctrine

Every mature system should change the nature of player decisions from owner-operator detail toward policy, capital allocation, management selection and exception handling as the portfolio grows.

The game should not simply multiply early-game clicks by fifty.

I. Shared historical-transformation doctrine

Core economic physics remain coherent from 1880-1905. Historical change should alter infrastructure, information speed, available institutions, technology, asset complexity, market reach, regional connectivity and organizational scale rather than arbitrarily unlocking stronger numeric tiers.

J. Revision XVIII preservation rule restated

No prior Revision XVII rule is deleted by this integration unless an Audit explicitly says a prior interpretation is changed. Where the audits add a distinction that prior sections lacked, both bodies remain authoritative, with Revision XVIII controlling the new distinction. Where an audit says KEEP, the existing Revision XVII text remains part of the active Canon unchanged.

Revision XVIII System Audit Integration status: COMPLETE THROUGH SYSTEM AUDIT 07.

LAND & LEDGERS

CANONICAL PROJECT BIBLE

ACTIVE CANON - REVISION XVII

Commerce Classification, Local Enforcement & Final Playtest Research Reconciliation Pass

Revision date: August 30, 2026
Source baseline: Canonical Project Bible Revision XVI - August 30, 2026
Status: Active Canon

PRESERVATION GUARANTEE

Revision XVII preserves Revision XVI in full except where this integration pass explicitly deepens, corrects or supersedes its wording. The preserved Revision XVI document follows this integration section unchanged as the authoritative baseline. Where a conflict exists, Revision XVII integration text controls. The Technical Implementation Bible remains subordinate for implementation architecture.

REVISION XVII PURPOSE

Revision XVII closes the final two historical research holds created by the first full playtest reconciliation. It resolves the business-form boundaries around feed, grain, livery, freight, drayage and related services, and it establishes the historically supportable legal/institutional treatment of prostitution and houses of ill fame without promoting that subject into active production scope. All unrelated Revision XVI mechanics remain authoritative, including human capability, construction, private lending, banking, taxation, acquisition, portfolio management, Double Houses and the Lifetime Ledger.

This revision follows the same evidence hierarchy used in Revision XVI: historical/legal locks govern behavior where the sources are sufficiently strong; design locks define the intended mechanic; calibration remains tunable; and parked material is preserved without authorizing implementation.

CANON LOCK - RESEARCH CLOSURE DOES NOT AUTHORIZE FEATURE BLOAT

- Feed, grain, livery and freight are resolved by capability and economic function rather than by multiplying shallow business classes.
- Brothel/prostitution research establishes legal and world-simulation truth, but player ownership and dedicated management remain parked.
- The 1880-1905 campaign remains governed by causal physical and accounting truth. Historical labels never substitute for real assets, services, custody, contracts, labor and legal exposure.
- No tuning number discovered in playtest or research becomes a historical constant merely because the underlying business form is now confirmed.

1. RESEARCH ITEM 6 - FEED, GRAIN & RELATED MERCHANT FORMS

1.1 No universal Feed & Grain archetype

Revision XVI Section 15.1 is superseded. Period Dakota evidence distinguishes flour-and-feed merchants, grain dealers/buyers, general merchants, agricultural-implement dealers, mills and grain-storage/elevator operations while also showing that individual proprietors could combine several of those lines. The game should therefore treat feed retail, grain buying, storage, milling and adjacent agricultural merchandise as related but separable commercial capabilities rather than one atomic Feed & Grain business class. [R6-1] [R6-2]

CANON LOCK - BUSINESS FORM FOLLOWS ACTUAL FUNCTION

- A standalone Feed or Flour-and-Feed Merchant is valid.
- A General Merchant may carry feed, flour, grain or adjacent farm-facing merchandise if its real stock policy, storage and suppliers support those lines.
- A mill may sell flour/feed through an attached or remote commercial outlet.
- A Grain Dealer/Buyer is distinct from a feed retailer because the core relationship is producer-to-market aggregation and resale/shipment.
- A Grain Warehouse/Elevator is a storage and handling capability that may coexist with dealing but is not synonymous with it.
- Agricultural implements and broader farm merchandise may overlap with feed/grain commerce, but the game must not force a modern one-stop farm-supply-store model backward into the period.

1.2 Feed / Flour-and-Feed Merchant

Feed commerce is a valid independent business role in Dakota and Black Hills settlements. It can serve households, farms, ranches, livery operators, freight carriers, construction operations using draft animals and other commercial customers. The exact stock mix is location- and period-dependent; common supported lines can include hay, oats, bran, mill feed, flour and other historically justified merchandise. A dedicated feed merchant should compete with broader merchants and direct producer relationships rather than receiving artificial exclusivity.

1.3 Grain Dealer / Grain Buyer

Grain dealing is producer-facing commercial aggregation. A dealer buys or otherwise arranges acquisition of grain, bears working-capital and market exposure, and then sells or ships into a larger market. The activity depends on actual farm surplus, quality/grade knowledge where represented, storage/handling access, prices, transport and outside relationships. The same proprietor may also sell feed, but the two revenue loops remain distinguishable. Rapid City evidence in the 1890s explicitly described grain business and hay/feed business separately. [R6-3]

1.4 Grain Warehouse / Elevator and custody truth

Grain storage requires an ownership distinction that ordinary retail inventory does not always require. Dakota warehouse law recognized grain delivered to a warehouse/elevator under receipt arrangements as a bailment rather than automatically a sale to the operator. Land & Ledgers must therefore be able to represent grain physically present at a facility while owned by another party. [R6-4]

CANON LOCK - PHYSICAL POSSESSION IS NOT AUTOMATIC OWNERSHIP

- Operator-owned commodity inventory and customer-owned/stored commodity must remain separate property states.
- Warehouse/elevator receipts or equivalent claims represent a delivery/custody obligation rather than magically converting customer grain into merchant stock.

- A warehouse/elevator may also conduct dealing where its actual business configuration supports it.

1.5 Settlement scale, rail and elevator development

No date alone unlocks a grain elevator. The appropriate economic preconditions are substantial local or feeder-area production, storage/aggregation demand, working capital, market access and a transport connection capable of moving enough bulk grain. A wagon-era gateway can have hay/feed commerce and open-lot storage without possessing a mature rail elevator. Rail can make elevator investment much more attractive, but local service, traffic and crop structure still matter. This is a world-economy condition, not a technology-tree step.

1.6 Feed as a transport-industry input

Feed is an important business-to-business input for the horse economy. Period livery evidence records substantial stocks of hay, oats and bran. Feed shortages and price movements should therefore propagate into livery and freight costs through actual consumption: weather or crop trouble can reduce feed supply; operating costs rise; carriers respond through margins, rates, route choices, reduced capacity or failure. Conversely, freight and livery expansion create real demand for feed. [R6-2]

1.7 Commission and terminal relationships

Commission merchants and larger outside grain-market intermediaries are historically valid, but this research does not require them to become an immediate standalone player storefront. Their strongest current role is as counterparties and market-access relationships through which local dealers, elevators and producers reach larger markets. Direct ownership may be promoted later only if it passes the project depth-before-width test.

2. RESEARCH ITEM 6 - LIVERY, FREIGHT, DRAYAGE & EXPRESS

2.1 Livery and professional hauling are distinct functions

Revision XVI Section 15.2 and the classification wording inherited from Revision XV Section 9.5 are superseded to the extent they imply that Livery / Freight is one inherent business form. Federal Census instructions explicitly distinguished livery stable keepers from hostlers and separately distinguished expressmen, teamsters, draymen and carriage/hack drivers. Period South Dakota advertising shows both separate livery and dray businesses and combined livery establishments with a dray line in connection. [R6-5] [R6-6] [R6-7]

2.2 Livery Stable

A Livery Stable is primarily an animal-care and mobility-service business. Historically grounded services may include transient/feed stabling, longer boarding, hire of horses or teams, hire of rigs, supplied drivers where offered, horse/stock sales and passenger/hack/bus service where supported. A livery needs actual stable capacity, feed, water, animal care, equipment and labor. It does not need to haul customer freight merely because it owns teams and wagons.

2.3 Freight / Hauling Carrier

A Freight / Hauling Carrier sells transportation capacity for customer cargo. The existing Canon mechanics governing cargo class, distance, route difficulty, road condition, wagon fit, team requirements, driver labor, loading/unloading, feed, water/stopping access, upkeep and lead time remain authoritative. The correction is classification: those mechanics belong to professional hauling rather than automatically to every livery. A carrier may operate its own private stable without selling public livery services.

2.4 Drayage / Cartage / Transfer

Drayage/cartage is a valid short-haul specialization within the professional hauling family. Typical work includes depot-to-merchant, depot-to-warehouse, warehouse transfer, local commercial delivery, baggage and other short-distance movements. It becomes particularly important after rail arrives because rail can reduce some long-haul wagon traffic while increasing feeder and final-mile movement around the depot.

2.5 Express and baggage

Express/baggage service is a distinct service profile where historically appropriate. The Census distinguished expressmen from teamsters and draymen, while period businesses could combine express, baggage and dray service. The current Canon should support this as a specialization or contract/service type without automatically promoting Express Company into a separate shallow business archetype.

2.6 Combined enterprises are fully valid

CANON LOCK - COMBINATION IS ALLOWED, CAPACITY IS NOT DUPLICATED

- Dedicated Livery, dedicated Freight/Hauling, dedicated Dray/Transfer and combined Livery + Hauling configurations are all historically valid.
- A combined establishment can share stable space, teams, wagons, feed stocks, hostlers, drivers and repair relationships.
- Shared ownership does not create a synergy modifier. The same horse, team, wagon or worker cannot perform two jobs at once.

- A livery hire can displace a freight job and a freight commitment can make rigs unavailable to livery customers. Opportunity cost is real.

2.7 Rail transforms route demand rather than deleting hauling

The existing rail-transition doctrine is reaffirmed. A new rail connection can destroy or weaken specific long-haul wagon routes while creating depot drayage, feeder routes, local delivery, farm/ranch collection and secondary-node connection work. Do not apply a generic Railroad -50% Freight Revenue modifier. The contracts, routes, cost structure and competitive alternatives change first; profitability follows.

3. CROSS-SYSTEM BUSINESS CAPABILITY DOCTRINE

Research Item 6 reinforces a broader design rule already implicit in Revision XVI: an economically meaningful business should be defined by the capabilities and relationships it actually possesses, not by an immutable RPG-style business class. A single legal/business ledger can legitimately combine several compatible lines, while a shared physical facility can serve several ledgers or activities without erasing ownership.

CANON LOCK - LABELS DESCRIBE, SYSTEMS EXPLAIN

- Business names and historical categories are player-facing descriptions of real capabilities.
- Adding a service requires the assets, labor, storage, market access, authority and working capital that service actually needs.
- A General Store carrying feed does not become a different legal entity merely because its merchandise mix changed.
- A freight carrier adding a public livery line must actually add public stabling/hire capacity; the new label alone grants nothing.
- NPC businesses use the same capability logic and may broaden, narrow, combine or exit lines when their resources and market conditions support it.

4. RESEARCH ITEM 7 - PROSTITUTION / HOUSES OF ILL FAME

4.1 Historical/legal status through the normal campaign

Revision XVI Section 20 is superseded: the legal/institutional research hold is now complete. Dakota Territorial law prohibited keeping bawdy houses, houses of ill fame, houses of assignation or prostitution, and the prohibition was enforceable in territorial courts. South Dakota municipalities received express power in 1890 to suppress bawdy/disorderly houses and houses of ill fame, and South Dakota continued prosecution through the end of the normal campaign horizon, including State v. Cambron in 1905. [R7-1] [R7-2] [R7-3]

HISTORICAL/LEGAL LOCK - NOT AN ORDINARY LICENSED BUSINESS

- During the normal 1880-1905 campaign, a house of prostitution / house of ill fame is not treated as an ordinary lawful licensed business.
- No generic Brothel License may be created from Western-game convention.
- A locally tolerated establishment remains legally exposed unless a specific dated authority actually changes its legal status.
- The 1913 South Dakota nuisance-abatement/closure regime is outside the normal campaign and must not be imported backward.

4.2 Deadwood: tolerance is an enforcement condition, not legalization

Deadwood demonstrates that unlawful premises could nevertheless be widely known, durable and openly tolerated for long periods. The correct simulation distinction is therefore Legal Status versus Enforcement Posture. An establishment can be unlawful, commonly known and currently tolerated at the same time. Tolerance may change with police/prosecutor practice, political leadership, complaints, reform pressure, serious incidents or other local causes. Prior tolerance does not create a property right or permanent immunity. [R7-4]

4.3 Property owner and premises-use exposure

Territorial law could reach an owner or other person who knowingly let or permitted premises to be used for prohibited purposes. Property ownership, occupancy/lease, operating control, actual premises use and owner knowledge must therefore remain separate facts. A mixed-use building can contain a prohibited upstairs use without the entire parcel becoming a special Brothel Property class. [R7-1]

4.4 Fines are not licenses

This research did not establish a sufficiently authoritative, universal 1880-1905 Deadwood system in which payment of a recurring prescribed fee legally authorized prostitution. Period or local fines may be represented when town-specific evidence supports them, but payment of a fine does not itself change the underlying legal status. Do not implement Pay Fine -> Legal for 30 Days.

4.5 Selective tolerance is not general immunity

A town can tolerate the existence of a prohibited establishment while still enforcing assault, theft, disorder, coercion or other crimes, and it can later prosecute the establishment itself. No Vice District blanket immunity is permitted. The legal and enforcement systems should remain capable of selective or changing priorities.

4.6 Worker agency and prohibited recruitment

The period law also addressed forms of coercive or deceptive recruitment into prostitution. Historical recognition of that exploitation does not create a desirable player management lever. If prostitution is ever represented beyond background/world context, adult participants must remain persons with agency rather than purchasable production inputs. Coercive recruitment is not an authorized player-facing economic mechanic in this Canon pass.

4.7 Production status remains parked

PARKED - WORLD TRUTH WITHOUT PLAYER-OWNED BROTHEL MANAGEMENT

- Historically appropriate settlements may represent prostitution premises as part of the local world/economy if future scope justifies it.
- The game does not currently require a Brothel construction menu, prostitution staffing screen, vice skill tree or dedicated player business archetype.
- Research closure authorizes accurate treatment if encountered; it does not create production priority.
- Any future promotion into active ownership must receive a separate design/scope review against the project depth-before-width standard.

5. REMAINING RESEARCH STATUS

The playtest historical/legal research sequence identified in Revision XVI is now complete. Research Items 1-5 remain incorporated through Revision XVI; Research Items 6-7 are incorporated here. There are no remaining historical research holds from this particular playtest reconciliation docket. Future questions may still require research as new systems are proposed, but they are not unresolved carryovers from Revision XVI.

6. CALIBRATION HOLDS - DO NOT CONVERT INTO HISTORICAL LOCKS

Revision XVI Section 21 remains authoritative and is extended by the following Research Item 6/7 calibration holds:

- Exact prevalence of standalone feed merchants, grain dealers, elevators and combined merchant forms by settlement/year.
- Exact grain-elevator capacity, storage charges, dealer margins, feed margins, grade discounts and shrink/spoilage assumptions.
- Exact hay/oat/bran consumption by animal class, work intensity and season.
- Exact livery boarding rates, rig-hire rates, freight/dray charges and utilization targets.
- Exact number of teams, stalls, wagons and staff required for profitable livery or freight operations.
- Exact prevalence of dedicated versus combined livery/dray/freight enterprises.
- Any town-specific prostitution fine amount, cadence, enforcement probability or tolerated-establishment frequency.
- Any prostitution revenue, pricing, rent or customer-volume assumptions if the parked content is ever simulated.

7. SUPERSESSION & INTEGRATION MAP

Revision XVII is intentionally narrow. It supersedes only the unresolved business-form and prostitution/legal-treatment holds while preserving Revision XVI in full elsewhere.

- Revision XVI 15.1 Feed & Grain research hold -> superseded by Revision XVII Sections 1 and 3.
- Revision XVI 15.2 Livery versus Freight research hold -> superseded by Revision XVII Sections 2 and 3.
- Revision XV / preserved Revision XIV Section 9.5 label Livery / Freight -> classification corrected; its grounded professional-hauling mechanics remain authoritative under Freight / Hauling / Cartage.
- Revision XVI 20 remaining historical research holds -> closed for Research Items 6 and 7.
- Revision XVI statements parking brothel/prostitution content -> deepened with a researched illegal/tolerated/enforcement doctrine; production status remains parked.
- All Revision XVI playtest systems and Research Items 1-5 remain unchanged unless an explicit Revision XVII sentence above conflicts.

APPENDIX XVII-A - RESEARCH BASIS

The following sources support the new historical/legal locks. They are listed for auditability; the Canon conclusions above remain the player-facing design authority.

ID	Source / authority	Use in Revision XVII
R6-1	1881 Dakota business-directory material for Elk Point and Parker, separating Flour and Feed, Grain Dealers, General Merchandise, Agricultural Implements and Mills. Genealogy Trails South Dakota. https://www.genealogytrails.com/sdak/union/elkpoint.html ; https://genealogytrails.com/sdak/turner/hx.html	Separate but overlapping merchant categories.
R6-2	1881 Sioux Falls business history, including flour/feed store, elevators and livery feed stocks. https://genealogytrails.com/sdak/minnehaha/siouxfallsbiz.html	Standalone feed commerce, elevators and B2B horse-feed demand.
R6-3	The American Elevator and Grain Trade, Dec. 15, 1897, reporting Rapid City grain business and hay/feed business separately. https://upload.wikimedia.org/wikipedia/commons/3/38/The_American_elevator_and_grain_trade_%28IA_CAT31053470078%29.pdf	Grain dealing distinct from feed dealing.
R6-4	Dakota Territory Session Laws, 1889, elevator/warehouse receipt provisions treating specified stored grain relationships as bailment. https://ndlegis.gov/assembly/sessionlaws/1889t/pdf/elections.pdf	Custody/ownership distinction for warehouse grain.
R6-5	U.S. Census Bureau, 1900 Instructions to Enumerators, paragraphs 199-200. https://www.census.gov/programs-surveys/decennial-census/technical-documentation/questionnaires/1900/1900-instructions.html	Livery keeper/hostler and expressman/teamster/drayman/hack-driver distinctions.
R6-6	Dakota Farmers Leader, Canton, Feb. 17, 1905: Palace Livery Stable advertisement, "Dray Line in Connection." Library of Congress NDNP. https://tile.loc.gov/storage-services/service/ndnp/mnhi/batch_mnhi_volvo_ver01/data/sn00065127/00295876554/1905021701/0393.pdf	Direct evidence for combined livery + dray operation.
R6-7	Period South Dakota newspaper evidence for separate City Livery and Dray Line in Hurley, 1898. Library of Congress NDNP. https://tile.loc.gov/storage-services/service/ndnp/sdhi/batch_sdhi_eagle_ver01/data/sn2001063133/00415623276/1898010601/0579.pdf	Direct evidence for separate livery and dray businesses.
R7-1	Compiled Laws of Dakota Territory (1887), Penal Code provisions on bawdy houses/houses of ill fame/prostitution and knowingly permitting premises. https://en.wikisource.org/wiki/Page:1887_Compiled_Laws_of_Dakota_Territory.pdf/1113	Territorial prohibition and property-owner exposure.
R7-2	South Dakota Codified Law 9-29-4 source history: SL 1890, ch. 37, municipal power to suppress bawdy/disorderly houses and houses of ill fame. https://sdlegislature.gov/api/Statutes/9.html?all=true	Statehood-era municipal suppression authority.
R7-3	State v. Cambron, 20 S.D. 282, 105 N.W. 241 (1905). https://case-law.vlex.com/vid/state-v-cambron-929887433	Confirmed prosecution of keeping a house of ill fame at the 1905 campaign horizon.

ID	Source / authority	Use in Revision XVII
R7-4	City of Deadwood / National Park Service historical interpretation on Deadwood brothel history and vice district. https://www.cityofdeadwood.com/sites/default/files/fileattachments/historic_preservation_office/page/35748/nps_exceptiona_places_2022_brothel_deadwood.pdf	Deadwood-specific evidence for persistent illegal but tolerated operation and mixed commercial setting.

APPENDIX XVII-B - REVISION XVII CANONICAL LOCKS

- Feed retail, grain dealing, grain storage/elevator operation, milling and agricultural merchandise are related but separable capabilities, not one universal Feed & Grain class.
- Customer-owned grain in storage must remain separate from operator-owned inventory.
- Livery, freight/hauling and drayage are distinct functions; dedicated and combined enterprises are all valid.
- Existing professional-hauling mechanics remain, but belong to Freight / Hauling / Cartage rather than every livery stable.
- Rail changes route demand and business opportunities causally rather than applying a generic freight-obsolescence modifier.
- Houses of prostitution are unlawful/legal-exposure uses during the 1880-1905 normal campaign; Deadwood-style tolerance is an enforcement posture, not legalization.
- No generic Brothel License or pay-fine-to-become-legal mechanic is authorized.
- Brothel/prostitution content remains parked as a player-owned business system despite the completed historical research.
- All playtest historical research holds from Revision XVI are now closed.

LAND & LEDGERS

CANONICAL PROJECT BIBLE

ACTIVE CANON - REVISION XVI

Human Capability, Construction, Capital Architecture & Lifetime Ledger Pass

Revision date: August 30, 2026
Source baseline: Canonical Project Bible Revision XV - August 29, 2026
Status: Active Canon

PRESERVATION GUARANTEE

Revision XVI preserves Revision XV in full except where this integration pass explicitly deepens, corrects or supersedes its wording. The preserved Revision XV body follows this integration section as the authoritative baseline. Where a conflict exists, Revision XVI integration text controls. The Technical Implementation Bible remains subordinate for implementation architecture.

REVISION XVI PURPOSE

Revision XVI formalizes the strongest design discoveries from the first full text-based playtest and the historical/legal research conducted immediately afterward. It does not replace the systemic spine established by Revision XV. It adds connective tissue that the playtest proved was necessary: people as transferable human capability rather than occupation bundles; continuous learning and mentorship; project work as physical staged execution; Construction & Contracting as a real business role; fallible manager-originated capital proposals; a cleaner separation of owner equity, intercompany finance and bank debt; deeper acquisition economics; historically grounded private banking progression; taxation; and a persistent Lifetime Ledger capable of remembering the player's actual commercial career.

The common rule remains causal simulation over abstract game bonuses. A skilled worker matters because the person can perform or understand real work. A contractor succeeds because the estimate, crew, materials, schedule and execution were good. A manager becomes valuable because good judgment reduces unresolved exceptions and improves decisions. A bank becomes powerful because it controls real capital under real balance-sheet constraints. A large portfolio becomes manageable because the player has built an organization, not because an empire level grants free efficiency.

Research interpretation rule

Revision XVI distinguishes four levels of authority. Historical/legal locks are supported strongly enough to govern simulation behavior. Design locks describe the intended game mechanic even where numerical values remain tunable. Calibration holds remain deliberately data-driven and should not be mistaken for historical claims. Parked concepts are preserved without authorizing implementation. When a playtest assumption conflicted with stronger period evidence, the researched interpretation controls.

CANON LOCK - PHYSICAL AND ACCOUNTING TRUTH REMAIN THE COMMON TEST

- People, money, materials, vehicles, authority, information and obligations must remain attributable to actual entities and relationships.
- Internal ownership may improve control and coordination, but it never makes labor, materials, capital, transport or risk free.
- A forecast or professional opinion remains an estimate even when it comes from a highly competent manager.
- Ordinary convenience automation may remove repetitive clicks, but it must preserve the underlying ledger transaction and consequences.
- The same meaningful economic rules apply to NPCs and the player unless a legal or informational difference genuinely explains an exception.

1. OPENING CALIBRATION & EARLY COMMERCIAL PRESSURE

1.1 Day-One starter doctrine

The normal introductory owner-operator experience should feel like the beginning of a career, not a small established empire. The strongest default remains one meaningful starter business, normally the General Store, paired with limited Liquid Owner Cash and enough business working capital to operate without making the first year trivial. The opening should not routinely include the cash, inventory, transport fleet, debt structure and managerial depth of an already mature merchant.

Exact dollar amounts remain balance values. The Canon locks the relationship rather than a universal number: the player should have enough to make decisions immediately, but not enough to neutralize inventory pressure, supplier reliability, labor, credit exposure and early expansion tradeoffs.

CANON LOCK - EARLY PRESSURE COMES FROM OPERATING REALITY, NOT ARTIFICIAL POVERTY

- Low liquidity should interact with real commitments rather than merely forcing the player to wait for money.
- Early opportunity should arrive alongside scarcity: a recurring account can improve demand certainty while reserving stock and exposing the business to fulfillment risk.
- The introductory economy should teach working capital, supplier reliability, stock discipline, customer commitments and opportunity cost before it teaches empire administration.
- Exceptional starts or later scenarios may begin wealthier; this is a default introductory calibration doctrine, not a universal scenario restriction.

2. UNIVERSAL HUMAN CAPABILITY ARCHITECTURE

2.1 Occupation is context, not character class

Every meaningful person should be capable of carrying a broad profile of human abilities. Occupation describes what that person currently does and where much of their experience was accumulated. It must not function as a hard class that determines which skills the person is allowed to possess.

The period record itself supports this granularity: census instructions distinguished actual professions, trades and kinds of work rather than accepting vague labels, and differentiated bank clerks from officers, salesmen from clerks, hostlers from livery keepers, foremen from helpers and different transport occupations. Land & Ledgers should therefore model people from task capability upward rather than from a small set of game-job archetypes downward.

2.2 Aptitude, Skill, Experience and Knowledge

Revision XVI separates four authorities that may correlate but must not collapse into one rating.

- Aptitude - the person's tendency to learn or perform related capabilities well. Aptitude can accelerate development or make difficult learning easier; it does not create fictional prior training.
- Skill - the person's present capability at a real class of task.
- Experience - the relevant work, problems, seasons, responsibilities and unusual situations the person has actually encountered.
- Knowledge / Domain Familiarity - what the person understands about a particular industry, place, customer base, market, procedure, organization or technical domain.

A naturally numerate employee can be promising for bank work without already knowing banking. A strong negotiator can still be a poor lumber buyer if they do not understand grades, suppliers, seasonality or freight. An experienced carpenter can have useful adjacent competence without becoming a mason simply because both work on buildings.

2.3 Transferable skills and task-derived catalogue

Skills should be derived from economically meaningful tasks and allowed to transfer wherever the same capability truly matters. The eventual master catalogue should be broad enough to express real differences without duplicating the same human ability under every industry.

- Commercial/social examples: selling, customer service, negotiation, interviewing, correspondence, relationship maintenance.
- Clerical/financial examples: literacy, numeracy, bookkeeping, reconciliation, cash handling, documentation, estimating, record organization.
- Management examples: supervision, scheduling, delegation, conflict handling, quality judgment, planning and task coordination.
- Physical/trade examples: measuring/layout, framing, finish carpentry, painting, roofing, masonry, blacksmithing, mechanical diagnosis, repair, loading and safe material handling.
- Transport/animal examples: driving, route judgment, animal handling, stable work, loading, dispatch and freight documentation.
- Industry familiarity remains separate so a person can have transferable competence without magically understanding every industry that uses it.

2.4 Attempt rule - competence shapes consequences, not arbitrary permission

Low skill should not normally make a physically and legally possible task unavailable. A weak painter can paint a shed. A novice can assist a skilled mason. The economic consequence emerges through time, finish quality, material waste, rework, supervision burden, customer response, durability and safety. Hard prohibitions should come from genuine legal authority, physical impossibility, equipment requirements or specialist knowledge where the task truly cannot be performed competently without them.

The simulation should also avoid bizarre adjacent zeroes. A person with years of general construction work is likely to have at least some measuring, material handling, basic finishing and adjacent task familiarity even when one trade remains their strongest capability. Generation and learning should reflect plausible exposure rather than isolated RPG skill silos.

2.5 Mastery scarcity and UI restraint

Extremely high mastery should be rare. Internal systems may use detailed numerical values for tuning and simulation, but ordinary competent workers should occupy the broad middle rather than constantly appearing as ninety-plus-point masters. Exact scale bands remain calibration. The player-facing experience should favor qualifications, work history, observed performance, references, traits and informed estimates over a wall of exposed numbers.

3. CONTINUOUS LEARNING, MENTORSHIP & PROFESSIONAL DEVELOPMENT

3.1 Learning by doing

Employees improve from actual relevant work. Calendar time alone does not create competence. The learning authority should consider task relevance, difficulty, novelty, repetition, feedback and whether the person was genuinely involved. The first months of meaningful exposure can teach more than the thousandth repetition of a simple mastered task.

Failure can teach without undoing the loss. A worker who makes and corrects an estimating error may improve future judgment. A clerk who reconciles a difficult discrepancy gains more useful experience than one who simply copies routine figures. Adjacent work can build limited adjacent capability when the employee actually participates.

3.2 Continuous rather than binary training

Training is not a completion switch. If an employee spends three weeks of a planned four-week instruction period working with a competent mentor, the learning that occurred during those three weeks remains. Completion can matter for a credential, a recognized status, confidence, authority or social standing where historically appropriate, but it does not suddenly manufacture the underlying competence on the final day.

3.3 Instructor quality and teaching ability

Learning can depend on instructor competence, the instructor's ability to teach, learner aptitude, learner prior knowledge, task variety, real time together, opportunities to perform the work, quality of feedback and difficulty of the material. A brilliant practitioner who communicates poorly may be a weaker teacher than a somewhat less exceptional practitioner who can explain, supervise and correct effectively.

3.4 Mentorship and outside professional services

Employers may use internal mentorship or purchase outside instruction, assessment or temporary specialist support when historically plausible. Formal apprenticeship is a valid period institution but must not become the only path to learning. Informal shop training, supervised work, temporary specialists and professional consultation can all improve capability through real time and exposure.

CANON LOCK - NO TRAINING BUFF

- Training changes what the person knows or can do because instruction and practice occurred.
- It does not create an unexplained percentage productivity modifier.
- A competent assessment can improve the organization's knowledge of the worker without revealing perfect hidden truth.
- Promotion may follow training and demonstrated capability, but training never forces promotion and promotion never creates competence.

4. TASK QUALITY, MISTAKES & CAPABILITY-SPECIFIC REPUTATION

4.1 Quality is broader than mistake severity

Revision XV already permits hidden minor, moderate and major mistakes. Revision XVI adds ordinary quality variation. A job can be completed without a discrete mistake and still be excellent, acceptable, visibly mediocre, defective or prone to early failure. Relevant causes include task difficulty, skill, experience, supervision, equipment, material quality, fatigue, weather and other task-specific conditions.

Severity must be domain-specific. Weak ordinary painting may cause uneven finish, slow work, excess paint use and mild customer dissatisfaction. Weak structural masonry can create cracking, costly reconstruction or safety risk. The simulation must never resolve every low-skill outcome through the same generic penalty table.

4.2 Reputation remembers what actually happened

Service and contractor reputation may preserve evidence by capability rather than collapsing every outcome into one universal trust number. A construction company can be known for strong carpentry, weak painting, excellent reliability and fair dealing. These dimensions should emerge from job history, customer experience and observed failures rather than from purchased reputation upgrades.

Reputation remains localized. A failed roof should damage the relevant contractor/site/capability reputation. It does not automatically cause customers to distrust an unrelated bakery owned by the same principal. Broader Owner or Commercial Standing transfers only where counterparties have a legitimate reason to treat the owner's wider history as evidence.

5. CREW LEADERSHIP, FOREMEN & ORGANIZATIONAL COVERAGE

5.1 No mandatory foreman slot

A crew does not require a formal Foreman position solely because it reaches an arbitrary employee count. Small capable crews can coordinate informally. Practical leadership can shift to the person with the strongest relevant skill, experience, planning judgment or communication ability for the work currently underway.

The strongest framer may direct framing while a mason directs masonry and an experienced smith handles specialized hardware. This is not a free efficiency system. Informal or committee leadership can produce unclear accountability, conflicting instructions, waiting, duplicated work and sequencing errors.

5.2 Formal foremen become valuable through real complexity

A dedicated foreman or project leader becomes increasingly valuable as crews grow, experience becomes uneven, work is geographically distributed, several tasks must be sequenced or specialist coordination becomes difficult. A strong foreman combines enough trade understanding with planning, communication, scheduling, delegation, conflict handling and quality judgment to reduce actual organizational failure.

CANON LOCK - TITLE DOES NOT CREATE LEADERSHIP

- Appointing a weak leader can make the operation worse.
- An untitled experienced worker can still provide real coordination.
- Formal supervision changes authority and accountability; any performance benefit must arise from better decisions and execution.

6. PHYSICAL STAGED CONSTRUCTION & PROJECT EXECUTION

6.1 Projects have physical stage state

The existing project system is deepened from generic sequencing into a physical workflow. A project should know what currently exists and what work remains. Depending on building type and period, stages can include site preparation, foundation or masonry, framing, enclosure, roofing, windows and doors, applicable period systems, interior carpentry, finishing, fixtures and final preparation. Not every project uses every stage, and exact sequencing remains data-driven by project type.

The player should be able to understand why a building is delayed without reading a generic progress percentage. The project may be waiting for stonework, framing lumber, a roofer, a delivered fixture, better weather, financing or completion of a prerequisite stage.

6.2 Stage consumption and stage funding

Labor and materials are consumed when the corresponding work occurs. Approving a $300 project does not necessarily remove $300 immediately. The owner approves scope, budget and funding authority; actual cash leaves through purchases, payroll, contractor invoices, delivered materials and progress payments as the work proceeds.

This makes construction financing structurally different from many lump-sum acquisitions. A project can begin with less than its ultimate cost sitting idle in cash when future funding is credible. The risk is real: a project can later pause because money, materials or labor fail to arrive.

6.3 Automatic continuation after authorization

Once a project is approved, routine continuation should not require repeated strategic clicks. Work proceeds while authorization, labor, materials, prerequisite stages and site conditions permit it. Escalation is reserved for genuine exceptions: cost overruns beyond policy, unexpected site condition, missing specialist, funding shortfall, major delay, scope change or other owner-reserved decision.

6.4 Project Stockpile versus Operating Reserve

A Project Stockpile is physical inventory earmarked for a named project. It is distinct from the producing business's ordinary operating reserve, target stock or sale inventory. Allocated project material is not simultaneously available for normal sale unless the allocation is deliberately released.

Pre-buying materials can protect schedule but ties up capital and storage and can create theft, fire, deterioration, weather, warping, handling and other material-specific risks. Project stockpiling is therefore a strategic hedge, not a dominant no-cost tactic.

6.5 Direct-to-project sourcing and sourcing policy

The player should be able to instruct an owned producer to fulfill a project requirement directly. Lumber can leave the mill for the building site without an unnecessary intermediate inventory-management click. Physical production, loading, travel, unloading and ledger attribution still occur.

A project can source requirements from project stockpile, owned producers, direct-to-project production, external spot purchase, standing supplier agreements, contractor-supplied materials, subcontractor-supplied components or customer-supplied materials. Sourcing may differ by component.

6.6 Internal supply is not free

A vertically integrated project must preserve real cost and opportunity cost. Sawmill output consumed internally still required logs, labor, equipment and yard capacity. Freight still consumed teams, drivers, feed, time and vehicle condition. Forge hardware still consumed iron, fuel and smithing time. Internal transfer can reduce markup, delay or bargaining risk, but it converts an external dependency into internal responsibility rather than eliminating scarcity.

7. CONSTRUCTION & CONTRACTING BUSINESS ARCHETYPE

7.1 Business identity

Construction & Contracting is promoted to a supported business archetype because it creates a distinct commercial relationship: project capability can be bought and sold. A small contractor can exist with modest fixed property because much of its productive value resides in people, tools, reputation, supplier relationships, subcontractors, scheduling and working capital.

Supported work can include new buildings, additions, repairs, structural work, painting, roofing, renovations, outbuildings, property maintenance and project coordination where the firm has the capability. Individual trades can also remain independent businesses or workers rather than every job passing through one general contractor.

7.2 Bidding and estimating

A contractor quote should be built from expected labor, materials, transport, subcontracting, tools/equipment burden, duration, contingency and desired return. The estimate is a forecast informed by records and expertise. Underestimating can win work and destroy margin. Overestimating can lose a profitable contract. Estimating expertise improves understanding of the work; it does not directly add profit.

7.3 Backlog, work in progress and payment timing

Contract backlog is not revenue. Awarded work consumes calendar time, crews, materials, working capital, equipment and supervision before it becomes completed work. Contracts may use deposits, delivered-material payments, milestone/progress payments, completion balances or other historically appropriate terms. Exact percentages are contract/project data rather than universal Canon values.

7.4 Subcontractors and temporary specialists

A firm does not need to keep every specialist permanently on payroll. Period practice supports prime contractors, subcontractors, material suppliers and specialist installation work. Temporary specialists are therefore a normal response to capability gaps. The commercial tradeoff is availability, price, coordination, quality control and dependence on outside capacity.

7.5 Organic scaling

A second crew, larger project or wider territory becomes possible only when real people, supervision, tools, cash, transport and workload support it. There is no Contractor Level that grants another crew. Taking more work than the organization can execute creates delays, cash strain, quality failures, breached commitments and reputation damage.

8. UTILIZATION, JOB ECONOMICS & BOTTLENECK DIAGNOSIS

8.1 Utilization is diagnostic, not victory

A highly utilized asset can be busy doing low-value work. A less utilized asset can become more profitable after weak jobs are refused. Business reports should therefore distinguish utilization from contribution and expose the causal components where records support them: revenue, direct cost, labor, maintenance, feed/team burden, empty positioning, idle time, margin, rejected demand and opportunity cost.

8.2 Minimum service policies

Businesses may use economically grounded policies such as minimum job charges, call-out charges, minimum loads, service zones, target margins and rush premiums. These are market-facing policies whose success depends on customers and competitors, not designer-enforced restrictions.

8.3 Expansion follows persistent evidence

Managers should not recommend capital expansion merely because one month was busy. Evidence can include sustained profitable utilization, repeated profitable work refused, recurring commitments, seasonality, route economics, driver/team availability, empty mileage and credible future demand. Expansion should target the actual bottleneck rather than defaulting to buying the next obvious asset.

9. DELEGATION, OWNER LOCKS, CAPITAL REQUESTS & DIVISIONS

9.1 Persistent owner policy modes

Delegated administration needs a practical distinction among one-time orders, ordinary manager discretion and persistent owner locks.

- Do Once - execute a specific instruction once, then return to the standing policy.
- Manager Judgment - the manager retains ordinary discretion within authority.
- OWNER LOCKED - the strategic policy remains fixed until the owner changes it. A manager does not silently override it; conflict must surface as an actual authority/personnel problem.

9.2 Multidimensional authority

Delegation should not reduce every decision to one dollar threshold. Contracting, lending and treasury authority can consider customer or borrower history, cargo or business type, route familiarity, price/margin policy, credit exposure, collateral, connected-party status, whether new equipment is required and whether the choice displaces a materially better commitment. Opportunity cost itself can create an escalation even when the nominal transaction is small.

9.3 Manager-originated Capital Requests

Managers can identify bottlenecks and propose investment without waiting for the player to inspect every business. A Capital Request should state the amount, purpose, why now, evidence, expected result, alternative if rejected, preferred funding route and urgency where relevant.

The manager's forecast remains a professional opinion. Competence, records, optimism, incentives and bias can all affect it. Over time the owner can learn which managers are conservative, aggressive, accurate, cost-blind or unusually good at seeing future demand.

9.4 Division / Operating Group layer

Large portfolios may introduce a Division or Operating Group between the owner and individual businesses. A division is an administrative grouping under a senior manager; it is not a new legal super-ledger that erases the books of its member businesses. Each business retains its cash, debt, assets, employees, contracts and operating history.

Division-level reporting can aggregate results and coordinate staff, purchasing, logistics or treasury only where actual authority and resources exist. Senior managers can have distinct thresholds for intercompany transfers, borrowing, capital spending and strategic commitments.

9.5 Zero personally managed businesses is a valid mature state

The ownership arc explicitly permits the player to reach a state in which no ordinary business is personally managed. The player may act primarily as investor, capital allocator, governance authority, acquisition decision-maker and strategic principal.

The businesses continue to hire, buy, sell, maintain, contract, err, request capital and react to markets through real managers and staff. Mature delegation is active simulation, not passive-income autoplay.

10. CAPITAL ARCHITECTURE, RESERVES & INTERNAL TREASURY

10.1 Three distinct internal funding relationships

Revision XVI separates three transactions that must not be collapsed into one generic fund transfer.

- Owner Capital Contribution - personal owner cash is permanently invested into the business. Business cash/equity rises; no artificial repayment date or interest obligation is created.
- Intercompany Advance - one controlled ledger temporarily finances another. The advancing ledger records a receivable/claim and the receiving ledger records an obligation. Repayment and terms remain visible.
- Bank Loan - a bank advances institutional funds to a borrower, creating a real bank asset and borrower liability subject to underwriting, liquidity, concentration and legal rules.

10.2 Funding route follows economic purpose

Permanent equipment or expansion may sensibly use retained business cash or owner capital. A temporary receivable or seasonal mismatch may fit an intercompany advance or bank credit better. Debt may be attractive when preserving owner liquidity or organizational independence matters. The player is not forced into one hierarchy, but managers should recognize the structural differences among these funding routes.

10.3 Operating Reserve is not a Capital Budget

A standing operating reserve expresses ordinary resilience policy. A separately approved capital project or machine purchase is a special authorization. The player does not need to permanently raise a $50 reserve to $190 merely because a $140 one-time purchase was approved. After the purchase, ordinary reserve policy remains what it was unless deliberately changed.

10.4 Reserve floors, excess cash and owner sweeps

Treasury policy may distinguish a minimum/target operating reserve, protected obligations, an excess-cash threshold, owner distribution rule and temporary capital authorization. Sweeps and transfers must remain real ledger entries. Managers do not silently move major sums across controlled businesses without authority.

10.5 Automatic owner draws as a convenience layer

Owner cash and business cash remain separate. When the player makes a legitimate personal purchase and Liquid Owner Cash is insufficient, the game may automatically execute the necessary owner distribution from an authorized owned business if available distributable funds exist and the transfer would not materially endanger protected payroll, debt, supplier or other obligations. The books still record Business Distribution -> Liquid Owner Cash -> Personal Purchase.

10.6 Routine debt-payment automation

Scheduled required payments may occur automatically by default when the paying ledger has authorized funds. The player can disable this convenience policy. Extra principal repayment remains a strategic capital-allocation choice. If funds are insufficient, the obligation becomes missed/delinquent according to its contract and dated legal regime; unrelated ledgers are not silently raided without an explicit treasury policy.

11. PRIVATE LENDING, UNDERWRITING & CONTRACT-SPECIFIC DEFAULT

11.1 Personal/private lending

The owner may lend personal money directly where the dated legal regime permits, without owning a bank. This is distinct from merchant account credit and from institutional bank lending. A private loan records borrower, principal, interest terms, payment schedule, security, guarantors/sureties where used, payments, remaining balance, delinquency and enforcement state.

11.2 Flexible promissory-note structures

Period evidence supports multiple note structures rather than one universal loan template. Notes may be payable on or before a maturity date, use installment or maturity structures, be renewed, carry security, include guarantors/sureties and contain contractual acceleration after specified defaults. Historical law also permitted some agreed interest to be deducted in advance.

CANON LOCK - EARLY REPAYMENT IS CONTRACT-SPECIFIC

- Do not create a generic rule that all future promised interest remains due after early repayment.
- Some notes permit payment on or before maturity with interest until paid; other lawful arrangements can price interest differently.
- The contract and dated interest/usury law determine the payoff economics.

11.3 Missed payment does not have one universal consequence

Some contracts can accelerate the entire balance after a missed installment; others can be renewed, tolerated, renegotiated or worked out. The simulation should therefore resolve missed obligation -> contractual lender rights -> lender response/workout -> enforcement where pursued. The existing Commercial Ruin doctrine remains unchanged.

Acceleration is contract-dependent rather than universal. Installment debt can sometimes be cured as to amounts then due; a lender may have an optional acceleration right, an automatic clause or no acceleration mechanism for that default. Debt UI should therefore distinguish delinquent installment, accelerated balance and enforcement rather than collapsing all three into DEFAULTED.

Real-estate mortgages and chattel mortgages are security instruments rather than automatic title transfers. Foreclosure or collateral sale follows the dated legal process; sale proceeds can first satisfy lawful costs and secured debt, with surplus returning to the debtor where applicable, while a shortfall can leave a deficiency. Guarantors/sureties, renewals, consolidation, refinancing and workouts are valid credit structures.

11.4 Underwriting is a negotiation over structure

A lender can approve less than requested, shorten maturity, request more collateral, require a guarantor, stage advances, alter pricing or refuse. Borrower quality, collateral quality, title/information quality and lender liquidity are separate dimensions. Borrower-reported information can be false, optimistic, incomplete or honestly mistaken. The player should see evidence and professional judgment, not an omniscient default percentage.

12. ACQUISITION DILIGENCE, SELLER KNOWLEDGE & DEAL STRUCTURE

12.1 Normalize owner labor

Acquisition diligence must distinguish accounting profit from the economic burden of replacing a departing owner's labor. A store earning $35 per month while the seller performs substantial skilled full-time work is not an absentee $35-profit investment. Reports should identify owner labor contribution, replacement coverage and normalized earnings where evidence allows.

12.2 Seller misrepresentation and imperfect books

Sellers can exaggerate sales, omit costs, misstate inventory, conceal deferred maintenance, overvalue receivables, omit obligations or present unpaid owner labor as profit. Deliberate fraud is possible but should be rarer than ordinary incomplete or optimistic records. Diligence changes what the buyer knows; it does not rewrite the underlying business.

12.3 Transition assistance and person-specific knowledge

The seller or key employee may remain for a defined transition period to transfer customer relationships, supplier knowledge, procedures, introductions, technical practice and unwritten routines. Knowledge can genuinely reside in individuals. Ownership transfer does not automatically copy that knowledge into the new organization.

12.4 Partial acquisitions and retained ownership

Acquisition workflows may support purchases of inventory, equipment, wagons, land, particular assets or other transferable rights without purchasing the whole business. Where appropriate, a seller may remain manager, retain a minority interest, enter a partnership or agree to a staged buyout. Existing succession rules continue to support manager/employee buyouts.

12.5 Earn-outs and contingent seller obligations

An earn-out, profit share or other contingent seller payment is not ordinary bank debt. The agreement should define the measured base closely enough to prevent meaningless manipulation: permitted operating expenses, manager compensation, capital expenditure treatment, owner compensation, related-party transactions, reserve policy, access/audit rights and termination conditions may matter. The obligation persists until satisfied, released or otherwise resolved by its contract.

12.6 Fraud suspicion can remain unresolved

Internal-control systems may generate genuine uncertainty. A discrepancy can result from error, shrinkage, poor records or deliberate theft. A worker may become suspected without the simulation revealing a visible THIEF tag. Wrongful suspicion and

false positives are valid outcomes. External delinquency can sometimes be the first evidence that a manager made an error, diverted money or allowed cash to run short.

13. PROPERTY INVESTMENT & SMALL MULTI-HOUSEHOLD DEVELOPMENT

13.1 Investment-analysis reporting

Where records and forecasts support it, property/project reports can show invested capital, rental income, avoided rent, maintenance, vacancy, taxes, financing burden, simple payback, cash-on-cash style return and estimated market value. Operational benefit and direct rental income remain separate. If evidence is insufficient, the report should say Unknown or present a justified range rather than inventing a precise payoff period.

13.2 Double House / Two-Family House

Revision XVI supports a purpose-built two-household rental form under historically appropriate terminology such as Double House or Two-Family House. A modern player-facing synonym such as duplex may be used if helpful, but the Canon should not imply that the modern term or form was universal.

The mechanic supports two independent households in one structure with more privacy than boarding and some shared structural/land economy. Direct South Dakota evidence is strongest by 1900, so this form should not be assumed as a common spring-1880 opening-town archetype. Frequency can rise later or where authored evidence supports it.

The physical hierarchy should remain explicit: one Parcel may contain one shared Building with two Residential Units, each with its own Occupancy relationship and potentially a different Household. The building form does not determine whether the occupants are kin, owner plus tenant or two unrelated tenant households.

Two independent residential units do not automatically create two separately saleable deeds. The safe ordinary model is one property/building containing two economic occupancy units unless a different title arrangement actually exists. The archetype also does not imply poverty or luxury: quality comes from size, materials, finish, utilities, condition, location and maintenance.

Any land or construction economy arises from the actual shared structure and parcel geometry rather than a fixed Duplex ROI or Construction Cost bonus. Exact rents, construction cost, vacancy and yield remain market/calibration data.

13.3 Staged tract development

A player may buy land capable of several future dwellings and develop it incrementally. Land acquisition and project authorization are separate decisions. Ownership of a tract does not automatically begin every building the land might eventually support.

14. OPPORTUNITY & PRESSURE, NPC ENTRY & PROCEDURAL NAMING

14.1 Unmet demand is not reserved for the player

When a business or service gap exists, an NPC may enter if they can plausibly assemble enough skill, knowledge, capital/finance, property, suppliers, labor and confidence. Rival entry should not spawn solely because a hidden shortage meter crossed a threshold. The same feasibility tests that constrain the player should constrain the NPC.

14.2 Rival intent can become information

Investigation can reveal evidence that another actor is already considering entry: a lot inquiry, lender conversation, supplier approach, equipment purchase or recruiting effort. That creates urgency through the world rather than through an arbitrary countdown timer.

14.3 Service and institutional gaps

Opportunity & Pressure should evaluate service and institutional gaps as well as commodity production: financial services, printing/news, medicine/pharmacy, barber/bath, fuel supply and other historically supported services can become opportunities when actual households and businesses create demand.

14.4 Procedural settlement names

Open Frontier settlements should normally receive generated names from curated region/period-appropriate structures rather than gibberish syllables. Multiple grammatical forms may be used, but component pools must be structurally compatible. The

generator should filter nearby duplicates, absurd combinations and unintended inappropriate outputs. The player may keep or override the generated name where setup permits; authored scenarios may retain fixed names. Exact one/two/three-word weighting remains tuning.

15. BUSINESS-FORM RESEARCH HOLDS - NOT YET SUPERSEDED

15.1 Feed & Grain classification remains under research

The playtest demonstrated that feed-and-grain commerce can matter as a distinct commercial opportunity, but Revision XVI does not yet impose a new business classification. Research Item 6 will determine how often feed dealers, grain dealers, warehouses, agricultural-supply merchants, mills and broader General Stores overlapped in Dakota/Black Hills settlements. Until that pass is complete, existing Revision XV agricultural-supply and merchant authority remains in force.

15.2 Livery versus Freight classification remains under research

The playtest demonstrated that livery/stabling and freight/cartage can plausibly be operated separately or together, but the historical prevalence and exact business-form boundary has not yet been researched in this checkpoint. Research Item 6 must resolve that question before Revision XVI creates a new classification rule. Existing Revision XV Livery / Freight authority remains unchanged in the meantime.

15.3 Whole-animal butcher economics is reaffirmed

The existing butcher system is reaffirmed: profitability depends on monetizing the animal across desirable cuts, lower-value cuts, preservable product and supported byproducts such as hides or tallow while managing spoilage and true waste. Do not evaluate livestock purchases as though every pound becomes the highest-value retail cut.

16. PRIVATE BANKING, STATE-BANK GOVERNANCE & FINANCIAL SERVICES

16.1 Private Banking House as an earlier/smaller institutional form

Historical research strongly supports private banking houses in Dakota and South Dakota. The game should not require national-bank-scale capitalization merely to begin legitimate local deposit/lending services. Institution type is a strategic/legal choice, not a cosmetic upgrade ladder.

A private banking house can have a much lower formal statutory entry barrier than an incorporated state bank, but that does not make it economically easy. The owner still needs actual capital/liquidity, records, staff, premises, depositor confidence, lending opportunities and outside relationships. Private ownership can also expose the proprietor more directly to failure and can be subject to a different supervision regime.

Paid-in bank capital must represent real qualifying contributed value. An owner cannot capitalize the institution by writing a personal IOU to the bank and then treating that IOU as cash or paid capital. Exact practical capitalization for a private house remains balance-sensitive even when no national-bank-scale statutory minimum applies.

16.2 Banking regulation changes during the campaign

The legal environment is not static. Dakota Territory recognized and regulated banking houses during the territorial era, South Dakota created a formal state-bank framework in 1891, and the 1892 Scougal decision confirmed that private bankers were not simply forced out of existence by the new corporate form. National banks continued under a separate federal regime. The bank system should therefore query dated institutional authority rather than apply one timeless set of requirements to 1880 through 1905.

16.3 Incorporated state-bank requirements after 1891

For the South Dakota state-bank form, the 1891 regime supplies concrete later-campaign anchors including minimum paid capital varying with town scale, reserve requirements, examination/reporting, a single-borrower concentration limit and a prohibition on insolvent banks continuing to receive deposits. Exact thresholds belong to the applicable dated institution type and should not be generalized backward to every private banking house.

Private banking did not simply disappear when the state-charter regime arrived; private banking houses and chartered banks can coexist as different institutional forms. State-bank shareholders and national-bank shareholders can also face period-specific liability beyond the amount initially invested, making bank ownership potentially more dangerous than a modern limited-liability assumption.

National banking is a separate federal form rather than Bank Level 3. In small towns the period national-bank framework required capitalization on the order of $50,000, far above a small private banking house, along with federal supervision and United States-bond/note-circulation requirements. National-bank status should therefore be a major strategic institutional choice rather than a cosmetic upgrade.

16.4 Bank Ledger and reporting

A banking institution should be capable of producing a period-style balance-sheet view that separates assets such as cash, correspondent balances, loans, supported bonds/securities, banking premises and other real estate from liabilities/equity such as deposits, paid capital, surplus/profits and other obligations. Cash reconciliation remains especially important because recorded books and actual cash can diverge through error or fraud.

Chartered-bank rules can require minimum reserves, qualifying correspondent balances, concentration limits and transfers of earnings into surplus. Profit is therefore not automatically distributable owner cash. Solvency and liquidity remain separate: an institution can have valuable assets but insufficient immediately available money, or have cash on hand while losses have impaired its capital.

Knowingly insolvent banking institutions must not be allowed to continue accepting deposits indefinitely merely to postpone failure. Dated law and institutional form determine when deposit-taking authority, examination or closure consequences change.

16.5 Related-party lending

A bank loan to the owner or another controlled business remains a real loan. The connected relationship is identified. Underwriting, liquidity, concentration and solvency constraints still apply. Favorable terms may be possible when lawful, but administrative cost, credit risk, opportunity cost and liquidity cost do not disappear because both businesses share an owner. A delegated bank manager can refuse a connected loan that breaches policy or law.

16.6 Supported fee services and correspondents

Period evidence supports banks acting in general banking, collection and loan business; buying/selling exchange; handling drafts/remittances and collections; providing safe-keeping services; and using outside banking relationships. Correspondent access therefore supports ordinary settlement, remittance and collection work in addition to panic liquidity. Fee income need not come only from loan interest.

16.7 Proprietary debt-securities portfolio

Where the institution type permits it, the bank may hold supported debt securities such as government, municipal or other lawful bonds/evidences of debt. These are a distinct asset class from cash and loans. They can earn income while carrying issuer, maturity, market-value and liquidation risk. Forced sale under stress can realize losses.

CANON LOCK - NO GENERIC BANK STOCK-TRADING BUSINESS

- National banks did not possess a general power to speculate in corporate stocks.
- The ordinary bank portfolio should therefore be data-driven by institution type and emphasize historically lawful debt instruments rather than a generic stocks menu.
- Specific instrument availability remains tied to year, legal form and market access.

16.8 Customer securities intermediation is conditional, not universal

The research does not support giving every ordinary bank a modern retail brokerage desk. Historically appropriate private bankers, banking-and-investment houses or other financial intermediaries may arrange customer investment transactions where their legal form, market relationships and dated authority support it. This is an optional institutional capability, not a default Bank tab.

Whenever a financial intermediary acts for a customer, ownership remains explicit. Owner personal investments, Bank proprietary investments and Customer-owned/agency securities are three separate property domains and must never be merged. Specific customer-investment products remain data-driven by institution, year and market access.

16.9 Physical bank security and premises

A bank can legitimately require secure cash/record storage, controlled employee access and period safes or vaults; later and larger institutions can use more substantial dedicated premises. Brick/stone/fire-resistant construction and prominent premises may causally affect physical security, fire exposure and public perceptions because customers observe real evidence of permanence. Do not implement a direct Stone Bank +10 Confidence modifier.

17. TAXATION - DAKOTA TERRITORY TO SOUTH DAKOTA, 1880-1905

17.1 Taxation follows property, jurisdiction and dated law

Land & Ledgers should not import a modern general business-income-tax model backward. The ordinary 1880-1905 tax system relevant to the player is primarily an annual assessment of legally taxable real and personal property, combined with local levies, specific licenses or occupation charges where applicable, road obligations, special assessments and dated enforcement rules. The system should query the owning ledger, asset type, jurisdiction and year rather than applying one empire-wide Tax Rate.

17.2 Real property and personal/business property are distinct

Real property covers land and improvements/buildings. Period personal-property taxation could also reach economically important movable or financial property such as livestock, wagons, machinery, merchandise/commercial capital, money, credits and other supported categories. The game may simplify assessment categories for usability, but it must not reduce the historical tax base to land alone. A mortgage does not simply reduce the taxable real-property value to the owner's equity.

17.3 Annual assessment and imperfect records

Taxation should have a real annual assessment cycle rather than accruing continuously as a modern percentage skim. Dated law determines the assessment/listing date and collection schedule. Because assessors could rely on required property statements and estimates, poor or incomplete records can affect the information and dispute process without turning taxation into a filing minigame. Actual State, Recorded State and Assessed State can therefore diverge for causal reasons.

17.4 Exemptions and household-scale relief

The broad taxable base coexisted with exemptions. Governmental, educational, religious, charitable and other protected property can remain outside ordinary taxation where the dated law provides. South Dakota's original constitutional framework also protected a limited amount of personal property for individuals. The simulation should therefore avoid absurdly taxing every household possession while ignoring major commercial inventories and equipment.

17.5 Layered public levies without paperwork simulation

Territorial/state, county, municipal, school or other lawful local layers can contribute to a property-tax obligation. There is no single timeless Dakota Tax Rate. The player-facing Ledger may consolidate the bill while preserving payer, taxable property, jurisdiction and component for audit/history. Exact millages remain dated data.

17.6 Specific licenses, road obligations and special assessments

Selected businesses and activities can face period-specific municipal licenses or occupation charges, but there is no universal annual Business License applied to every enterprise. Territorial road obligations can also involve a small money-or-labor civic burden for eligible persons under the applicable early regime. Public improvements can create separate special assessments against benefited or abutting property; those assessments are project-linked property obligations rather than another generic annual tax percentage.

17.7 Delinquency, liens and tax-sale exposure

Tax obligations enter a staged collection process rather than causing instant confiscation. Unpaid taxes can become delinquent, accrue penalties/interest, create liens, expose personal property to lawful collection and real property to tax sale, and create redemption/title-history consequences under the dated regime. Tax status therefore belongs in acquisition/property diligence.

17.8 Statehood changes authority, not the genre of taxation

The 1889 transition from Dakota Territory to South Dakota changes the governing legal authority but does not turn the economy into a modern income-tax system. Property-based taxation and local levies continue under the new constitutional/statutory framework. Bank capital and banking property remain within the taxable system subject to the institution-specific rules researched elsewhere in Revision XVI.

17.9 No general retail sales tax or general South Dakota income tax during the normal campaign

Ordinary General Store purchases should not receive a modern retail sales-tax line during 1880-1905; South Dakota's general retail sales tax arrives decades later. Likewise the normal campaign should not impose a generic Business Profit Tax or Owner Income Tax. Specific period excises, licenses or other lawful charges remain separate dated systems.

17.10 1905 inheritance-tax introduction

South Dakota introduces inheritance taxation in 1905. Because Land & Ledgers already simulates death, estates, succession and ownership transfer, this can become a late-horizon dated legal event when a taxable succession actually occurs. Exact rates, exemptions and administration belong to dated legal data rather than a generic wealth tax.

17.11 Tax information in the Lifetime Ledger

The Lifetime Ledger should preserve Taxes Paid by payer and meaningful category, including property tax, supported licenses/occupation charges, special assessments and inheritance tax where applicable. Exact liability should come from the dated tax authority rather than from retrospective estimation at the end of the campaign.

18. LIFETIME LEDGER, PORTFOLIO REPORTING & HISTORICAL RECORD

18.1 Persistent Lifetime Ledger

The game should accumulate career statistics from the beginning of the save. The 1905 Historical Record and scenario completion summaries must not attempt to reconstruct twenty-five years of commerce from whatever balances happen to remain at the end. The Lifetime Ledger is the persistent historical authority that feeds both in-play reporting and final retrospectives.

18.2 Headline career statistics

The first layer should prioritize career-defining information rather than trivia. Supported headline categories can include current Net Worth, peak Net Worth, Liquid Owner Cash, lifetime revenue, lifetime profit, owner capital invested, businesses founded/acquired/sold/failed, properties acquired/built/sold, employees hired/promoted/lost, payroll paid, principal borrowed, interest paid, principal lent/originated, interest earned, defaults/losses and taxes paid.

Exact availability depends on what the simulation legitimately tracks and how a metric is defined. A category should not be printed merely because it sounds interesting if the underlying accounting cannot support it.

18.3 Business-specific lifetime statistics

Each industry can extend the shared historical record with meaningful statistics. A construction company can track bids, wins, estimate variance, jobs completed and rework. A bank can track deposits, withdrawals, loan originations, interest, credit losses and lawful securities activity. Freight can track loads, tonnage, distance and rejected work. Hospitality can track occupancy/lodging nights. A sawmill can track production. Irrelevant categories remain hidden.

18.4 Debt, lending and business chronology

History should preserve flows, not only ending balances. A player who ends debt-free after financing twenty expansions has a different career from one who never borrowed. Useful history includes peak debt, total principal borrowed, loans repaid, restructurings/defaults, collateral lost, private loans made, principal recovered and lending losses.

Each business should preserve founding/acquisition date, purchase/founding cost where known, managers, major expansions, important finance events and sale/closure/disposition. Long-serving people and major projects can become part of the Historical Record because the underlying career data already exists.

18.5 Filters and hierarchy

The Ledger should allow the same history to be examined at useful scopes such as All Time or a selected year/period, and Portfolio -> Division -> Business -> Property/Asset/Account where relevant. The exact UI can change, but the data should support these questions without creating separate incompatible reporting systems.

18.6 Liquid Owner Cash, Book Equity and Net Worth

Player-facing labels must remain unambiguous as the portfolio grows. Liquid Owner Cash means immediately spendable owner-side cash. Business cash remains on its own ledger. Bank cash belongs to the bank. Book/Ledger Equity is an

accounting carrying-value concept; Net Worth remains the Canon's grounded portfolio economic-value metric. Illiquid valuation can be estimated and should not pretend to false precision.

18.7 Information hierarchy and zero suppression

The main portfolio view should show the most decision-relevant information first and suppress irrelevant zero-value categories. A one-store owner should not see a permanent wall of zero banking, securities, railway and contingent-liability rows. Detail and historical screens can still preserve those categories when they become relevant.

18.8 Forecast discipline

Manager capital proposals, construction estimates, property payback, acquisition earnings and lending forecasts all remain forecasts. Known facts remain exact where legitimately known. Insufficient data can produce Unknown, wide ranges or low confidence. A sophisticated organization should become better at explaining uncertainty rather than converting estimates into omniscient truth.

19. RESEARCH-BASED CORRECTIONS TO PLAYTEST PROPOSALS

The following playtest ideas are deliberately modified or parked because the historical research produced a stronger interpretation.

1. Customer securities intermediation is not a universal ordinary-bank power. Permit it only for historically appropriate banking/investment intermediaries whose legal form, market relationships and dated authority actually support the service.
2. Bank proprietary investment is narrowed to institution-appropriate debt securities/bonds and other verified instruments. Do not create a generic bank stock-speculation menu.
3. Private-loan early repayment is contract-specific. Do not create a generic loan toggle that automatically preserves all unearned future interest after prepayment.
4. The small two-household rental archetype is canonized as Double House / Two-Family House, with duplex only an optional modern synonym. It should not be assumed common at the 1880 opening.
5. Feed & Grain and Livery/Freight classification remain unresolved at this checkpoint. Research Item 6 must determine the historically plausible business forms, overlaps and separations before Revision XVI supersedes Revision XV on those boundaries.
6. Private Banking House is positively confirmed as a valid smaller/earlier banking form; low formal statutory barrier does not eliminate the need for economic bankability.
7. Brothel/prostitution content remains parked. No legal, licensing or tolerance model is Canonized in this checkpoint; if the project brings it into active scope, Research Item 7 must establish the territorial/state/local legal and institutional treatment first.

20. REMAINING HISTORICAL RESEARCH HOLDS AT THIS CHECKPOINT

Revision XVI is intentionally a checkpoint, not a claim that every playtest question has been researched. The following subjects remain open and should be researched one at a time before a later superseding revision changes their detailed Canon treatment.

- Research Item 6 - Feed & Grain and Livery/Freight business classification: determine the historically plausible business forms, overlaps and separations in Dakota/Black Hills settlements from 1880-1905.
- Research Item 7 - Brothel/prostitution legal and institutional treatment, only if the project intentionally chooses to bring this business/service family into active scope. Research territorial/state law, local ordinance, tolerated versus illegal operation, policing, property and economic treatment rather than relying on generic Western-game assumptions.

All other completed research blocks in this checkpoint are incorporated into the sections above. Numerical tuning and implementation architecture remain separate from historical research.

21. CALIBRATION HOLDS - DO NOT CONVERT THESE INTO HISTORICAL LOCKS

Research is sufficient for the Revision XVI design architecture. The remaining open questions are primarily tuning/data rather than conceptual historical uncertainty. These should remain configurable and tested against long-run simulation rather than hard-coded into prose.

- Exact opening Liquid Owner Cash, General Store cash, inventory and debt.
- Expected time to second/third business, first manager, division structure, institutional ownership and player-founded bank.
- Net-worth compounding and late-game pacing across the full 1880-1905 horizon.

- Internal numerical skill scale, mastery thresholds, learning rates and mentorship effectiveness.
- Trade wages, workday/shift length by year/occupation, project labor times and crew-size norms.
- Task-quality probabilities, rework frequency and manager forecast error.
- Project durations, material carrying-loss rates and stockpile/storage risk.
- Reserve floors, sweep thresholds and automatic-draw safety rules.
- Private-loan rates/amounts and market pricing within the already validated legal ceiling chronology.
- Double-house construction cost, rent, vacancy and world-generation frequency.
- Exact settlement-name generation weights.
- Exact tax millages, due dates and penalties by jurisdiction/year.
- Specific debt securities available to each supported bank type by year and market access.

PLAYTEST PACING WARNING: the test run reached roughly eleven businesses, ten properties and a private bank by 1885. Because the playtest used provisional values and accelerated decision-making, this is not proof of an economy defect. It is nevertheless a required calibration target. A twenty-five-year game should not routinely deliver nearly the complete capital-command fantasy in the first five years unless an exceptional run genuinely earns it through unusually favorable conditions and strong play.

22. TECHNICAL BIBLE BOUNDARY

Revision XVI does not move scene-loading architecture into the design Canon. The Technical Implementation Bible already establishes that scene objects are manifestations of persistent simulation state and that scene unload is not an economic event. A later Technical revision may add the playtest-specific example that an interior activation boundary can extend slightly outside a building so the visual interior is ready before entry. Employees, inventory, condition, active work and financial state remain persistent regardless of visual interior activation.

APPENDIX XVI-A - REVISION XVI CANONICAL LOCKS

- Occupation is descriptive context, not a hard character class; human capability is task-derived and transferable.
- Aptitude, Skill, Experience and Knowledge/Domain Familiarity are distinct authorities.
- People learn continuously through relevant work and instruction; there is no magical training-completion competence bonus.
- Low competence normally changes time, quality, waste, supervision, safety and rework rather than arbitrarily forbidding a task.
- Formal foremen are optional; leadership can emerge by task, while coordination complexity makes formal supervision increasingly valuable at scale.
- Construction is staged physical execution with real labor/material consumption, project stockpiles, sourcing policy and causal pauses.
- Approved construction continues automatically within policy; exceptions rather than repetitive continuation clicks reach the owner.
- Construction & Contracting is a supported business archetype with bidding, estimating, backlog, subcontracting and organic multi-crew scale.
- Utilization is diagnostic rather than a success target; businesses should distinguish activity from economic contribution.
- Managers may originate Capital Requests and their forecasts remain fallible professional judgments.
- Large portfolios may use Divisions/Operating Groups without merging underlying business ledgers.
- It is valid for a mature player to personally manage zero ordinary businesses.
- Owner Capital Contribution, Intercompany Advance and Bank Loan are distinct financial relationships.
- Operating Reserve is not Capital Budget; one-time capital authorization does not permanently change the standing reserve target.
- Automatic owner draws and routine debt payments may remove repetitive clicks only while preserving the real ledger transaction and protected obligations.
- Personal/private lending is distinct from merchant credit and bank lending; note/payoff/default behavior follows the contract and dated law.
- Acquisition diligence normalizes departing owner labor and can encounter incomplete or deliberately misleading information.
- Knowledge transfer, partial asset purchases, retained management/ownership and contingent seller payments are valid deal structures where legally applicable.
- Double House / Two-Family House is a supported later small multi-household rental form; land acquisition does not automatically authorize construction.
- NPC entrepreneurs can fill real market gaps when they can assemble the same necessary capabilities and resources.
- Open Frontier uses curated plausible procedural settlement names with player override where setup permits.
- Feed & Grain and Livery/Freight classification remain explicit research holds; Revision XVI does not yet supersede Revision XV on those business-form boundaries.
- Private Banking House is a valid institution form; incorporated/state/national forms can carry different capital, reporting and governance rules.
- Related-party bank lending remains real lending subject to underwriting, liquidity, concentration and solvency.

- Bank proprietary investments are institution-specific; supported debt securities are not cash and generic corporate stock speculation is not an ordinary bank power.
- Customer securities intermediation is conditional rather than universal and requires an institution, dated authority and outside market relationship that actually support it.
- The ordinary 1880-1905 tax model is primarily property-based, with dated local layers, exemptions, specific licenses/road obligations, special assessments, staged delinquency and a 1905 inheritance-tax event; no general retail sales tax or general South Dakota income tax applies during the normal horizon.
- The Lifetime Ledger records career history from Day One and supplies scenario completion records and the 1905 Historical Record.
- Liquid Owner Cash, business cash, bank cash, Book/Ledger Equity and Net Worth remain distinct concepts.
- Forecasts remain forecasts and Unknown is preferable to fabricated precision.

APPENDIX XVI-B - HISTORICAL RESEARCH BASIS

The Revision XVI research pass used period statutes, appellate decisions, federal/state reports, census instructions/occupation classifications, local historical surveys and period commercial records to test the playtest assumptions. The Canon records gameplay-relevant conclusions rather than turning these materials into a citation-heavy legal treatise. High-value anchors include:

- Dakota and South Dakota banking law/history, including the 1891 state-bank regime, private-bank continuity, reserve/concentration requirements and bank reporting.
- United States Comptroller of the Currency annual reports from the 1880s-1890s for Dakota/South Dakota bank balance-sheet categories and institutional conditions.
- South Dakota and federal banking sources distinguishing ordinary banking powers, institution-specific securities eligibility, correspondent/fee services and the narrower circumstances in which customer investment intermediation is historically supportable.
- U.S. Supreme Court national-bank cases from the 1890s limiting general corporate-stock dealing.
- Dakota territorial and South Dakota cases involving promissory notes, mortgage acceleration, foreclosure, renewals and secured obligations.
- Dakota mechanic's-lien law and documented 1880s Dakota construction contracts showing contractors, subcontractors, material suppliers, bids, extras and progress-linked payment.
- 1880 and 1900 U.S. Census occupation instructions/classifications distinguishing actual trades, clerical roles, transport roles, livery work, banking work, foremen and helpers.
- Dakota territorial and South Dakota constitutional/statutory material on real and personal property taxation, annual assessment, exemptions, local levies, road obligations, special assessments, delinquency/tax sale, statehood continuity and the 1905 inheritance-tax introduction.
- South Dakota/Black Hills local historical sources documenting private banks, bank buildings and the later Double House / Two-Family House form. Feed & Grain and Livery/Freight classification are deliberately deferred to Research Item 6.

APPENDIX XVI-C - SUPERSESSION & INTEGRATION MAP

This integration deepens Revision XV / preserved Revision XIV authority wherever the subjects overlap. The most important routing is:

- CAN 1.5 / opening scenarios and Open Frontier configuration -> stronger Day-One introductory calibration; exact starting figures remain tuning.
- CAN 1.7 and 13 / Expertise, traits, experience and workers -> universal transferable human capability, continuous learning and assessment.
- CAN 5 / projects, construction, condition and conversion -> staged physical construction, project stockpiles, sourcing and stage funding.
- CAN 6 / shared business architecture, cash and delegation -> owner locks, funding relationship distinctions, reserve/capital-budget separation and routine payment/draw automation.
- CAN 7-10 / businesses, supply and logistics -> Construction & Contracting and utilization diagnostics; Feed & Grain and Livery/Freight classification remain unsuperseded research holds.
- CAN 11 / acquisition and continuity -> owner-labor normalization, seller misrepresentation, transition knowledge, partial acquisitions and earn-out governance.
- CAN 12 / housing and occupancy -> Double House / Two-Family House and staged tract development.
- CAN 14 / markets and rivalry -> feasible NPC entrepreneurial entry and rival-intent evidence.
- CAN 15 / information design -> Lifetime Ledger, Liquid Owner Cash label, Book Equity vs Net Worth and zero suppression.
- CAN 18 / finance and banking -> private lending, private banking house progression, related-party constraints, debt-securities portfolio, fee services and taxation interactions.
- CAN 19 / executive management -> Capital Requests, Division/Operating Group layer and explicit zero-personally-managed-business end state.
- Historical Record / scenario completion authority -> use the persistent Lifetime Ledger as the common historical data source.

SUPERSESSION NOTE

Revision XVI explicitly deepens Revision XV and the preserved Revision XIV body on human capability, learning, construction, service quality, managerial authority, treasury, private lending, acquisition, housing, banking, taxation and career reporting. Feed & Grain and Livery/Freight classification remain pending Research Item 6 and are not superseded here. Where Revision XVI narrows a previously open or parked concept - especially universal customer securities brokerage, bank stock speculation, generic early-repayment interest treatment and the modern-only use of the duplex label - Revision XVI controls. All unrelated Revision XV authority remains unchanged.

LAND & LEDGERS
CANONICAL PROJECT BIBLE

Game Design, Mechanics & Historical Reference

ACTIVE CANON

REVISION XV

Scenario Architecture, Human Labor & Portfolio Logistics Pass

Revision date: August 29, 2026
Source baseline: Canonical Project Bible Revision XIV - August 29, 2026
Status: Active Canon

PRESERVATION GUARANTEE Revision XV preserves Revision XIV in full except where this integration pass explicitly deepens or supersedes its wording. The preserved Revision XIV body follows this integration section as the authoritative baseline. Where a conflict exists, Revision XV integration text controls. The separate Technical Implementation Bible remains subordinate for implementation architecture.

REVISION XV PURPOSE

This pass formalizes the design work completed after Revision XIV: scenario structure as a flagship play pillar, passive historical-news delivery, outcome-driven contracts and town-growth challenges, portfolio-wide transport assignment with ledger ownership, labor-hours and overtime, sleep/nutrition/fatigue/commuting, hidden mistake severity, employee traits/interests/experience, compensation growth, retention and poaching, and expertise-based talent recognition. The common rule remains that numbers may exist underneath the simulation without becoming gamey floating modifiers.

CANON LOCK - ONE SIMULATION, TWO FLAGSHIP MODES

- Scenarios and Open Frontier remain co-equal flagship structures using the same economy, AI, labor, transport, historical chronology and physical execution rules.
- Scenario development should normally deepen systems that also improve Open Frontier. Bespoke scenario-only economic physics are prohibited.
- Open Frontier remains a major focus and must remain compelling after the scenario catalogue is completed.

1.5J Scenario architecture - authored problems, systemic solutions

Scenarios are not arbitrary task lists. Each scenario should begin from a believable commercial situation that explains why the opportunity, obligation, pressure or deadline exists. Typical drivers include contracts, tenders, financing terms, creditor workouts, seasonal buying windows, land options, construction schedules, railway decisions, government procurement, investor patience, seller retirement, settlement promotion and historically fixed events.

CANON LOCK - The scenario states the result that must be achieved. The shared simulation determines the viable methods.

- Prefer "deliver X qualifying freight by [date]" over "own five wagons."
- Prefer "produce and sell X recoverable ore by [date]" over "own a mine, sawmill and blacksmith."
- Prefer "grow and sustain the settlement at X residents by [date]" over direct city-builder construction quotas.
- Supporting systems should matter because they become real bottlenecks: labor, housing, timber, tools, freight, finance, maintenance, processing, supplier reliability and working capital.
- Vertical integration is an option, not a requirement. A player may own supporting businesses, contract with NPCs, finance them or rely on market supply if that is economically viable.

1.5K Scenario phases, counterparties and failure

Substantial scenarios should usually evolve through roughly three to five meaningful stages rather than exposing a giant static checklist. A fixed premise may generate new pressures as the economy changes. The world should reveal constraints through counterparties, reports, shortages, labor behavior and visible conditions rather than prescribing a build order.

- A counterparty can renegotiate, penalize, reduce, terminate or reassign a contract when the world situation supports it.
- Failure can be graded. Missing a contract can cause lost payment, reputation damage or loss of future business without necessarily ending the entire save.
- Primary scenario failure remains distinct from Commercial Ruin.
- Completion records should acknowledge the actual economic history even when the primary target is missed.
- Optional Distinctions can reward speed, reliability, solvency, resilience, limited ownership, local development or other mastery without becoming mandatory clutter.

1.5L Deadline language

Player-facing deadlines should use clear inclusive language: **"By December 31, 1887"** means the player may complete the requirement through that date. When the next day begins without qualification, the deadline has been missed. Avoid confusing constructions such as "before January 1" when the intended final valid day can be stated directly. Exact dates remain scenario-specific and must be historically/economically justified.

1.5M Scenario families and cross-system focus

Scenarios may be grouped by a primary commercial family while carrying secondary system tags. The family identifies the central fantasy, not a restriction on methods. Current useful families include Commerce, Transportation, Mining, Agriculture & Ranching, Industry, Finance, Land & Property, Settlement Development, Infrastructure and Crisis & Recovery.

CANON LOCK - A focused scenario should still require economic interplay.

A mining scenario can remain unmistakably about mining while making success depend on timber, tools, labor, housing, processing, freight and finance. A transportation scenario can remain about moving goods while forcing decisions about vehicles, teams, employees, contracts, backhauls, maintenance and route capacity. The player should feel that they are assembling a functioning commercial pipeline, not collecting required buildings.

1.5N Settlement-development scenarios

Settlement-development scenarios are a signature Land & Ledgers inversion of the city-builder population challenge. The player does not control the town by decree. The player begins with limited private authority - often one business - and must create enough economic gravity that households and businesses choose to arrive, remain, invest and expand.

- Population targets are provisional tuning values, never historical claims merely because they appear in a scenario draft.
- Growth can require jobs, accommodation, affordable goods, construction supply, freight access, credit, services, infrastructure and local confidence.
- The player can buy businesses that have second-order town effects even when those businesses are not the highest standalone-profit investments.
- A weak sawmill may be strategically valuable because lumber scarcity is constraining housing. A freight company may unlock several NPC businesses. A bank may enable outside owners to expand without the player owning them.
- Population alone should usually be sustained for a meaningful period or paired with economic viability so a temporary boom does not count as durable success.
- Possible variants include Boomtown, Gateway Settlement, Second City, Save the Town, Railway Bypass and New Depot.

The scenario question is not "what should the mayor build?" but "why is this town failing to attract or retain people, and what can an owner/investor legitimately do about it?"

15.X Historical news, legal notices and technological milestones

Major public developments should be discoverable without repeatedly interrupting play. Forced full-screen newspaper popups are prohibited as the default presentation. The normal surface is a small, unobtrusive notification that the player may open for detail.

- Qualifying news includes major federal, territorial or state legal changes; proclamations and decrees; major financial/public events; and important technological or transportation milestones such as notable automobile, aviation, communication or infrastructure developments when historically relevant.
- News is not instantaneous omniscience. Information must travel through historically plausible channels and delay.
- Telegraph access, newspapers, post, rail connections, civic institutions and information networks can change how quickly and reliably news reaches the player or settlement.
- A town hall, courthouse, newspaper, post office, telegraph office or similar institution can become an information-distribution node where appropriate.
- Routine local business events should usually remain world evidence, correspondence, ledgers, notices or reports rather than "breaking news."
- The detailed news item may use a newspaper or official-notice presentation after the player chooses to open it; the game should not seize the screen merely because a story exists.

CANON LOCK - Historical news follows Information Distance. A fact may be true in Washington before the player legitimately knows it in Dakota.

9.X Portfolio logistics - ledger ownership and operational assignment

Transport assets should be legally and financially owned by a specific person or business ledger while remaining operationally assignable across the player-controlled portfolio when authority permits. Ownership determines whose books carry the asset, depreciation/condition burden, feed, maintenance and financing. Operational assignment determines what the asset is doing now.

CANON LOCK - Vehicles are not permanently welded to the business that owns them.

- A dairy farm can own a cart, horse team and employ the worker while the cart performs a route serving the General Store, the dairy and outside counterparties.
- Controlled wagons/carts/teams should be visible as a portfolio-wide available transport pool while retaining ledger ownership.
- The player can formally transfer an asset between Ledgers when long-term use changes. The transfer should be recorded rather than silently rewriting ownership.
- NPC owners with multiple businesses use the same underlying authority and asset rules, but their scheduling quality can vary through actual decisions rather than hidden efficiency multipliers.

9.XA Multi-stop routes and through-cargo

Routes may contain multiple stops and cargo may remain aboard through intermediate stops when its destination lies later in the route. This permits triangular and circuit logistics, backhauls and consolidation. Route orders should be policy-like rather than brittle scripts.

- Examples: load up to a quantity, reserve capacity for a later pickup, keep destination-A cargo aboard while visiting B, depart if cargo is unavailable, wait up to a defined time, or fill remaining capacity with eligible cargo.
- The route planner should expose expected capacity utilization, empty mileage, timing and likely bottlenecks without guaranteeing uncertain future inventory.
- Cargo remains physical. Capacity, cargo compatibility, loading time, vehicle condition, weather protection and team availability constrain what can be moved.
- External cargo requires a real agreement or freight contract. The player does not obtain free rights to haul NPC property merely because spare capacity exists.

9.XB Fleet composition and interruptions

Practical transport capacity is the combination of vehicle, suitable team/animal power, driver labor, route time and condition. Owning eight wagons does not imply eight usable departures if only six teams or qualified drivers are available. The player may interrupt a scheduled route for an urgent job, but missed pickups, contract breaches, shortages and schedule disruption remain real consequences.

6.1D Cross-business labor assignment and internal allocation

Employees remain employed and paid through a specific business/person ledger, but compatible labor may be temporarily assigned to another controlled business when authority, skill, travel and schedule permit. The game should not require formal rehiring for every cross-business task.

- A store clerk who has four hours of required store work may perform compatible hauling or warehouse work during remaining usable labor time.
- The cheaper worker is not automatically the best assignment: task skill, fatigue, travel, business coverage and opportunity cost matter.
- Internal labor and transport use can be allocated across Ledgers so the player can see which operation is consuming shared capacity.
- Owner-facing controls can support Automatic, Preferred and Manual assignment. Automatic scheduling should seek a sensible plan, not guarantee the globally optimal plan.
- The player is allowed to choose a more expensive or less efficient arrangement and bear the consequences.

6.1E Working time, overtime and cumulative strain

Work duration is a real causal input. Ordinary workday length, occupation, season and local practice require historical calibration rather than assuming a universal modern eight-hour shift. When compensation rules or agreements make overtime appropriate, additional work should increase payroll accordingly. Regardless of overtime pay, excessively long or repeated shifts create human costs.

- Longer shifts progressively increase fatigue, mistake risk, accident risk, irritability, absence pressure and turnover risk.
- Repeated 12-16 hour days should be materially worse than an occasional long day because recovery debt accumulates.
- Managers and automated schedules should be capable of producing imperfect staffing choices; the player can override them.
- No unexplained productivity penalty is required. Reduced performance should emerge from tired people, missed coverage, slower work, errors, injuries or departures.

13.X Sleep, nutrition, fatigue and daily time budget

People should have a lightweight daily human-condition model connecting work, housing, food, travel and health. The purpose is not life-simulation micromanagement. It is to make labor capacity and household circumstances causally meaningful.

- Sleep: people have restorative sleep needs/ranges. Long workdays, long commutes, poor housing, household obligations, noise, illness and traits such as insomnia can reduce actual recovery.
- Nutrition: quantity, quality and regularity of food can influence energy, recovery, illness resilience and physically demanding work. This links wages, household budgets, boarding/meals and local food supply to labor outcomes.
- Fatigue: accumulates from work duration, physical intensity, inadequate sleep, travel, illness and poor nutrition and recovers over time rather than resetting magically at midnight.
- Commute: travel time is part of the person's day. A worker who starts at 8:00 and needs one hour to walk should normally depart around 7:00. Walking, horse access, wagon transport, weather, road condition and distance alter the time budget.
- Punctuality: arrival is not perfectly deterministic. Travel disruption, fatigue, household events and behavioral traits can produce lateness.

CANON LOCK - The simulation may maintain exact numerical state underneath, but the ordinary player-facing experience should emphasize readable causes and consequences rather than floating "-17% fatigue" modifiers.

13.XA Mistake severity

Work errors should be able to resolve at multiple hidden severity bands such as minor, moderate and major. Severity and probability depend on the actual task and circumstances. Minor outcomes can include miscounts, wasted material or small delays; moderate outcomes can include damaged goods, incorrect records, missed deliveries or broken tools; major outcomes can include serious injury, machinery damage, lost cargo or fire-causing mistakes where physically plausible.

Relevant causal inputs can include competence, task familiarity, attention, fatigue, sleep, nutrition, shift length, supervision, equipment condition and environmental difficulty. The player does not need to see the hidden probability roll.

13.XB Employee traits, aptitudes, interests and experience

Workers should feel distinct without becoming full RPG characters. The preferred lightweight model uses a small number of visible or discoverable traits, work aptitudes/interests and experience earned through actual work.

- Traits can include Dependable, Quick Learner, Methodical, Distractible, Witty, Patient, Habitually Late, Insomniac and similar grounded characteristics.
- Avoid one deterministic "Low IQ" label as a catch-all. Where cognitive differences matter, use narrower underlying qualities such as reasoning, numeracy, memory, attention, learning speed or practical aptitude, and surface human-readable traits only when known.
- Traits may be correlated where plausible but should not be hard-paired into stereotypes. Intelligence does not inherently imply punctuality; conscientiousness or disorganization is a better causal route.
- Interests can include animals, machinery, numbers, customer-facing work, outdoor work, solitary work, mining, butchery or a family trade/background.
- Interest should mainly influence job attraction, retention, satisfaction and learning, with modest performance effects emerging through fit and experience rather than giant buffs.
- Experience grows from real task exposure. A retained clerk, teamster, butcher, bookkeeper or foreman can become materially more valuable over years.

13.XC Compensation growth, retention and poaching

Employees are participants in a labor market. As their experience, reputation and alternatives improve, they can request better compensation or accept competing offers. Valuable staff should create genuine retention decisions.

- Compensation can include wage, meals, board/lodging, bonus, schedule, stability, status/promotion, different duties and historically appropriate commission/profit-sharing arrangements where supported.
- A worker may approach the employer with a raise request or report a competing offer. The owner can match, exceed, bonus, promote, alter duties, refuse or allow departure.
- NPC businesses can poach player workers and the player can approach workers employed by NPC businesses.
- Job acceptance and retention consider compensation, work conditions, hours, commute, housing, household circumstances, interests, employer reputation, relationships and outside alternatives.
- Money should not solve every retention problem. A worker can prefer another occupation, safer work, shorter travel or a different environment.

1.7E Expertise and talent recognition

Owner Expertise may improve the player's ability to recognize useful employee information, but it must not magically improve the employee. A provisional expertise concept such as Talent Judgment / Eye for People can reveal progressively better evidence about observed workers. Final naming remains TBD.

- Low recognition may show only public role, tenure and obvious behavior.
- Greater expertise may reveal likely traits, work reputation, aptitude, experience, dissatisfaction or receptiveness to an offer where the player has a legitimate basis to know.
- Expertise improves observation and forecast confidence; it does not create +10% worker productivity or guarantee recruitment.
- Known facts remain exact when legitimately known. Uncertain retention or acceptance remains a forecast.

13.XD Employer reputation and organizational memory

Workers remember how employers behave. Reliable pay, reasonable treatment, promotion, kept promises and stable work can improve recruiting and retention. Missed payroll, chronic overwork, broken promises and poor conditions can damage the employer's labor-market position. The state may be numerical underneath but should normally surface through worker behavior, references, requests and recruiting outcomes rather than one giant reputation meter.

REVISION XV CROSS-SYSTEM CONSEQUENCE

These mechanics deliberately connect previously separate systems. Distant cheap housing can create long commutes; long commutes reduce sleep; poor sleep and long shifts increase fatigue; fatigue increases errors and turnover; labor instability reduces business throughput; weak throughput can create freight or supply failures; those failures can affect contracts, scenario objectives and settlement growth. Conversely, better scheduling, nearby accommodation, reliable food, retained experienced staff and intelligently shared transport capacity create real efficiency because the underlying operation improved.

APPENDIX XV-A - Scenario design reference

The Long Haul - Transportation

A major merchant or producer offers a preferred-carrier agreement. Move a large qualifying freight volume by an inclusive contract date. Success depends on route design, backhauls, teams, drivers, maintenance and customer commitments.

Gateway to the Hills - Transportation / Settlement

A gateway settlement wants dependable links to regional production and markets. Establish reliable freight/passenger service and enough commercial activity to strengthen the node before a competing route captures the traffic.

Black Hills Gold - Mining

An investor syndicate advances capital or grants an operating interest in exchange for a defined quantity/value of recoverable output by a contract date. Timber, tools, labor, housing, processing and freight become natural constraints.

The Deep Mine - Mining / Industry

A successful mine reaches a depth/cost threshold where water, hoisting, machinery, timber and treatment threaten viability. Maintain commercially sustainable production through the development agreement.

Timber Country - Lumber / Industry

A construction/mining boom creates a large supply agreement for lumber. Logging, hauling, mill throughput, storage, drying and labor must support the contract.

A Town Worth Staying In - Settlement Development

A land company, founder group or major owners need a weak settlement to become durable before financing or promotion collapses. Grow and sustain population by creating jobs, housing, services and access through private economic action.

The Panic - Finance / Crisis

Begin in 1893 with leverage and tightening credit. A creditor workout or maturing obligations create explicit milestones. Survive without magical protection, using asset sales, refinancing, collections, restructuring or distressed acquisition.

Main Street Merchant - Commerce

A wholesaler or merchant agreement rewards reliable regional demand coverage. Scale sales while solving supplier, credit, freight, staffing and storage constraints.

Iron, Wood & Steam - Infrastructure / Industry

A railway/mining construction tender needs timber, repair work, food, lodging and freight. The player can own, contract or finance the pipeline but must fulfill the real project schedule.

Second City - Settlement / Regional Power

A remote node has strong commercial potential. Build enough economic gravity for it to become a durable second regional center without receiving mayoral powers or direct population placement.

All names, dates, volumes, population thresholds and monetary values in this appendix are design examples and remain subject to historical research and balance calibration unless separately locked elsewhere.

APPENDIX XV-B - Revision XV canonical locks

- Scenarios are a flagship play pillar, not a side mode; Open Frontier remains equally important.
- Scenario goals are world-grounded commercial outcomes with believable reasons, counterparties and inclusive "By [date]" deadlines.
- Settlement-growth scenarios rely on private investment and economic gravity rather than city-builder authority.
- Major legal/government and technological news uses passive notifications; detail is opt-in and arrival obeys Information Distance.
- Controlled transport assets remain owned on a specific Ledger but may be assigned across the controlled portfolio when authority permits.
- Multi-stop routes, through-cargo, backhauls and conditional capacity planning are valid logistics mechanics.
- Employees belong to a paying/employing Ledger but compatible labor can be allocated across controlled businesses.
- Work duration, overtime where appropriate, fatigue, sleep, nutrition, commute and punctuality are causal labor inputs.
- Mistakes can have hidden severity bands and arise from real conditions rather than arbitrary random punishment.
- Workers have lightweight traits, aptitudes/interests and experience; they improve through work and may demand better compensation.
- Retention and poaching are two-way labor-market mechanics. Compensation is broader than wage alone.
- Expertise may reveal talent and improve judgment; it never creates unexplained worker bonuses.
- NPC inefficiency must come from poorer decisions/scheduling under the same rules, not hidden production penalties.
- Numerical state may exist underneath, but player-facing presentation should emphasize human-readable causes and consequences.

SUPERSESSION NOTE

This integration explicitly deepens Revision XIV Sections 1.5, 1.7, 6.1, 9-10, 13 and 15 and Appendix D wherever the subjects overlap. All unrelated Revision XIV authority remains unchanged.

- Mixed passages are split by meaning: the Canon states the required behavior; the Technical Bible states how that behavior is represented and executed.
- Stable section references remain useful. Technical sections should cite Canon authority such as CAN-12.6H rather than duplicating pages of research.
- The Canon may remain large. Revision XIV applies a depth-before-width rule: an important playable system should have enough causal decisions, failure states and cross-system consequences to justify its place before adjacent new systems are promoted into active scope.

EXECUTIVE SUMMARY

The Game in One Page

Land & Ledgers is a realism-first frontier business empire simulation set around a commercially ambitious Dakota-style region beginning in spring 1880. The player starts with a useful foothold and a starter business in a genuinely wagon-dependent regional economy, then lives through roughly twenty-five years of overlapping modernization as railheads approach, communication networks deepen, electric light appears, civic infrastructure accumulates, mining and agriculture mechanize unevenly, institutions mature and ownership grows from owner-operator to owner-manager and ultimately to capital command.

Ownership, not city painting: Growth comes through land, deals, projects, retrofits, staffing, supplier choice, finance, acquisition and capital allocation. The player represents one actual owner/investor in the world. Direct commands require real authority; settlements are shaped through ownership and economic gravity rather than god-mode municipal power.

A living regional economy: Households, workers, suppliers, rivals, lenders, settlements and businesses share real pressures. Nodes can rise, stall, decline or surpass the opening town.

Causal modernization: History defines the capability envelope. Rail, telegraph, telephone, electricity, water, sewer, mining processes and farm machinery appear when date, geography, capital, demand, legal authority and network economics support them.

World first, UI second: Prosperity, strain, neglect, crowding, labor stress, freight problems, infrastructure gaps and district change should be visible in the world before or alongside ledger explanation.

Two flagship play structures: Scenarios provide authored, difficulty-ranked commercial challenges with objectives and deadlines grounded in a specific starting situation. Open Frontier provides the large randomized 1880-1905 sandbox, generating a different regional economy from seed while preserving the historical capability envelope. Commercial Ruin remains a simulation state rather than a substitute for scenario failure or sandbox choice.

1905 frontier-industrial horizon: The authored historical envelope remains spring 1880 through 1905. Open Frontier normally begins in 1880 and can run through that full span; Scenarios may begin later or use shorter slices inside it. The strongest nodes can have rail, telephones, electric light, waterworks, banks and sophisticated mines while secondary towns and rural properties remain wagon-, horse-, well- and privy-dependent. Rare automobiles do not replace the core transport economy.

CONTENTS

I. Product Vision & Design Doctrine - core fantasy, historical frame, victory, loss and progression II. World, Land & Settlement Simulation - opening world, parcels, terrain, nodes and infrastructure III. Ownership & Property - property states, suitability, deal flow and projects IV. Shared Business Simulation - business authority, cash, continuity, maintenance and records V. Businesses & Industry - frontier businesses, agriculture, lumber, repair economy and mining VI. Supply, Trade & Logistics - suppliers, agreements, hauling, shocks and inventory VII. People, Households & Labor - households, kinship, accommodation, migration, labor and health VIII. Markets, Competition & Adversity - demand, pricing, rivals, persistent pressure and fire IX. Finance & Capital - lenders, investors, banks, legal chronology and distress X. Rail, Communications, Utilities & Capital Development - rail, telegraph, telephone, electricity and civic networks XI. Executive & Portfolio Management - auditors, delegated approvals and executive roles XII. Interface & Player Information Design - management surfaces, workflows, known facts and forecasts Reference Appendices - design locks, chronology, balancing anchors, research holds and system matrices

FEATURE STATUS BANDS

Core Identity - foundational to the ownership fantasy and unlikely to be cut. VP Core - needed to prove the ownership loop at commercial quality. Active VP Expansion - directly supports or extends the deep slice. Post-VP - important near-term expansion after the deep slice stabilizes. Rail Era - regional rail penetration begins mid-1880s; local service remains a corporate/corridor outcome. Late Capital Era - banks, mines, rail/utility ownership and executive layers become viable after wealth/delegation. Long-Term / Parked - historically valid destination layers whose production

priority is later or uncertain.

REVISION XIV DEPTH-BALANCE DOCTRINE

A system is not deep merely because it contains many variables, and the game is not broad merely because the Canon can name many businesses. Depth means the player can face meaningfully different situations inside the same system because demand, inputs, labor, property, information, capital, condition, timing and counterparties interact. Width means a genuinely different economic role has entered the simulation. Revision XIV therefore favors deepening existing high-value roles over adding another shallow storefront. New width should usually be promoted only when it creates a new economic relationship that existing systems cannot express.

REVISION XIV SYSTEMIC SPINE

Scarcity creates choices. Information makes those choices intelligible but imperfect. Counterparty agency makes external choices negotiated. Commitment gives decisions memory. Physical Truth determines whether promises can actually be fulfilled. Delegation determines how far the player's authority can reach. Alternatives determine bargaining power and resilience. Records and retained people allow organizations to persist. Failure removes options and recombines assets. Historical time makes delay irreversible. The design lock that follows from this is simple: progress should give the player control over more of the economy without ever giving them freedom from the economy.

PART I

Product Vision & Design Doctrine What the player is becoming, what the world must feel like, and how a campaign can end.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

CORE IDENTITY

1.1 Core fantasy

Land & Ledgers is a realism-first frontier business empire sim. The player rises from owner-operator toward owner-manager and then toward broader local and regional influence through business operation, land deals, repair, staffing, financing, supplier choice, strategic acquisition, and capital allocation. The town is not decorative; it is the main stage on which economic opportunity, labor strain, family burden, and regional usefulness become readable and playable. The player is not manually painting a town into existence. The fantasy is becoming the person who understands how this place works and profits by shaping it better than anyone else. Growth comes through ownership, deals, projects, retrofits, staffing, and capital allocation rather than click-to-place construction. The long arc moves from personal handling to delegation without turning businesses into passive-income abstractions. NPCs should increasingly feel like people living under the same economic gravity as the player rather than decorative extras.

1.1A Starting foothold, regional freedom, and settlement plurality

The starting frontier town remains the player's first proving ground, but it is no longer treated as a permanent dominant-town lock. The stronger canon direction is that the player begins with one useful foothold and one starter business, then gradually earns the freedom to buy, shape, and test land anywhere on the regional map. Some runs may remain centered on the original town for decades. Other runs may see a mine camp, freight stop, farm-service cluster, or rail-side settlement surpass the original town in gravity, capital, and regional importance. This does not turn Land & Ledgers into a freeform city-builder from minute one. The player is still not painting towns into existence by decree. The player creates or redirects settlement by buying land, adding services, supporting housing and boarding, creating jobs, improving freight access, and solving real local pressures. If the underlying conditions are weak, the experiment stalls or fails. If the conditions are strong and the player commits enough capital and operational support, a new node can become a hamlet, a town, or in exceptional cases the portfolio center of the whole run.

1.1B Player identity, authority truth, and economic power

The player is not an abstract government and does not receive command authority merely for being the player. The fiction treats the player as one actual business owner, investor and contracting person, even though the interface is camera-led rather than dependent on a third- person avatar. Every direct action should pass an Authority Test: what ownership, employment, contract, tenancy, lending, investment, management, legal right or other relationship permits this person to make the decision? If no legitimate authority exists, the player must influence the outcome indirectly through purchase, negotiation, finance, market behavior, petition, franchise, right-of-way assembly, hiring, contracting or another historically plausible channel. Late-game power should emerge because the player genuinely controls capital, jobs, land, credit, freight, utilities or communication - not because a hidden player-authority level grants city-builder powers. Other owners, workers, households, governments and corporations retain agency and can refuse, compete or choose differently.

1.2 Prestige-frontier presentation layer

The game should present itself as practical, weathered, ambitious, and commercially young. It should feel premium and serious, but not theatrical. World first, UI second: a thriving business should feel busy, cared for, and confident before the panel tells the player why. Big actions should feel substantial without becoming tedious. Weight comes from presentation, staging, visible state change, restrained pacing, and clarity of consequence. Details should communicate use, wear, prosperity, neglect, seasonality, and pressure rather than acting as random clutter. A store with healthy trade should feel distinct from a weak store before any ledger tooltip is read. A prosperous boarding house, crowded freight yard, or overworked sawmill should communicate strain through the world itself. The UI should feel like a premium frontier business ledger: clean, serious, practical, and readable.

1.2A World evidence doctrine

- Important economic conditions should preferably have a readable world expression before or alongside pure UI explanation.
- Prosperity, strain, neglect, crowding, labor stress, settlement instability, and district rise should appear through physical state, repeated behavior, and local perception.
- UI should clarify and compress; it should not carry the whole burden of truth.
- Physical Truth is part of world evidence: a local customer purchase, work shift, delivery or service call should correspond to the relevant person or transport asset actually being present and able to perform it before the transaction resolves.
- Do not turn dedicated merchandise capacity into manual shelf-by-shelf furniture placement; normalized allocation should capture the economic decision without store-layout busywork.
- Do not turn workload-based staffing into minute-by-minute employee micromanagement. The long-term model should calculate task pressure and usable labor while letting owners/managers set policy and schedules.
- Do not mistake detailed planning for automatic implementation scope. Preserve causal design depth, then classify it as owner-facing, delegated execution, background abstraction or parked detail.

1.3 Realism framework

Every system should be checked against five realism filters. Historical plausibility asks whether the thing could happen in this place and time. Economic plausibility asks whether the incentives, costs, and outcomes make sense. Social plausibility asks whether people in this setting would behave this way. Lifecycle plausibility asks whether the rule fits how a household or person changes across life. Gameplay tolerance asks what the smallest bend is when pure realism becomes too slow, too opaque, or too punishing for strong play. That final filter matters. Land & Ledgers should be realism first, not realism without judgment. The goal is not museum simulation. The goal is a grounded commercial historical sim that remains readable, playable, and worth mastering. Growth is not automatically prosperity. More people, more businesses, or more building frontage can also create overextension, speculation, freight strain, labor distortion, and fragility. The town should be allowed to boom badly, stall awkwardly, and recover unevenly.

1.3A Three-filter design standard and freedom doctrine

Every mechanic should also pass three standing production filters. First, historical accuracy: model what could plausibly happen in this place and period, and research uncertain details before inventing confidence. Second, player freedom: if an action is physically, legally and economically possible, prefer allowing it and letting consequences determine whether it was wise. The game should not forbid a store from opening at 1:00 a.m. merely because the choice is usually irrational; labor, lighting, customer schedules, safety and demand should decide whether it works. Third, premium systemic quality: aim for the depth and interconnection of a serious systems-driven strategy game with the presentation, world feedback and interface clarity expected from a high-end production. The player should feel free, informed and grounded rather than boxed into designer-approved answers. Depth-before-simplification rule: planning should understand the real process at high causal resolution before deciding what to abstract. A feature can later be cut or simplified for scope, but its surviving design should preserve the important physical and economic cause-and-effect rather than replacing reality with arbitrary bonuses.

1.4 Anti-drift rules

Do not reduce households to decorative customers or population filler. Do not reduce women to static domestic tags; treat wage work, boarding, teaching, widow ownership, and household labor as real mechanics. Do not treat professions as permanent immutable classes; age, injury, literacy, marriage, opportunity, and migration should matter. Do not make property automatically market-available just because the player wants it. Do not let acquisition bypass seriousness, seller motive, financing credibility, inherited problems, or rival pressure. Do not impose modern zoning-style restrictions when suitability, nuisance, frontage, adjacency, and conversion cost are the better answer. Do not let seasons become cosmetic reskins of the same loop. Do not let notices, Opportunity & Pressure, or recurring local orders feel detached from the underlying town simulation. Do not make ordinary actions slow for the sake of slowness. Favor grounded presentation over friction tax. Do not let rivals act without believable economic cause. Do not let off-map actors feel magical; outside finance, supply, and competition should still leave readable local effects. Do not let takeovers erase inherited weakness. Do not let notices surface raw abstract scores without human-readable causes. Do not assume all growth is healthy simply because counts went up. Do not use artificial prohibitions to keep the player inside an intended strategy when the action is historically and physically possible; let bad hours, bad allocation, bad credit policy, overstaffing, underpricing and other poor choices fail through their consequences. Do not grant city-builder god powers that the owner character has no legal or economic authority to exercise.

1.4A Design derivation rule

- Do not begin new mechanics from abstract feature brainstorming.

- Begin with a frontier reality, household situation, business strain, district change, ownership problem, or town condition.
- Ask what tension lives inside that situation.
- Ask what physical, behavioral, and social signals reveal it.
- Ask what threshold turns it from background condition into player-facing consequence.
- Ask what ownership response it wants: observe, prepare, intervene, or exploit.
- Only then define the mechanic, workflow, or UI surface.

1.4B Persistent Economic Consequence

CANON LOCK The player's economic world should accumulate history around important decisions rather than repeatedly resetting to a clean state.

A meaningful decision should change what is possible later. Buying property changes later site control. Training and retaining a clerk can create a future manager. Burning a supplier relationship can narrow future sourcing. Extending credit can create a long-lived account relationship or bad-debt history. Building a road can redirect practical market reach. Overleveraging can become decisive years later when refinancing disappears. A failed business can release workers, property, equipment, customers and knowledge into the economy for other actors to recombine. Recovery begins from the changed world produced by the failure; it does not restore the pre-crisis state.

1.4C Ownership converts dependency into responsibility

Scarcity must persist after ownership. Buying an asset or business should usually replace one outside dependency with a set of controlled internal responsibilities rather than erase the underlying problem. Owning a wagon gives control over a finite wagon, team and driver's time. Owning a sawmill gives control over processing but creates log supply, labor, maintenance, storage, utilization and market burdens. Owning a bank gives control over a financial institution but introduces liquidity, deposit, borrower, governance and concentration risk. Owning a railway does not solve transportation; it creates track, rolling stock, terminals, service, maintenance, traffic and capital requirements. This is a core progression doctrine: ownership expands legitimate authority while exposing new opportunity costs.

1.4D Shared causal grammar

Many Land & Ledgers systems should be understandable through a common sequence without being forced into one universal interface: WORLD TRUTH -> PLAYER/ORGANIZATIONAL KNOWLEDGE -> INTEREST -> AGREEMENT -> COMMITMENT -> PHYSICAL EXECUTION -> DISRUPTION -> RESPONSE -> CONSEQUENCE -> MEMORY. A supplier has stock; the player learns about it; terms are agreed; money, inventory or future capacity becomes spoken for; goods physically travel; weather or failure intervenes; the parties respond; the outcome changes future choices. The same grammar can underlie freight, property, lending, construction, employment, custom processing, partnerships, rail and utilities while each retains its own appropriate workflow.

1.4E Five design-review tests

Every major new mechanic should answer five questions before promotion into active scope.

- New-decision test: what decision does this create that cannot already emerge cleanly from the existing economy?
- Scarcity test: what becomes scarce, committed, risky or less discretionary because this system exists? A mechanic that only grants upside is likely shallow.
- Player/NPC parity test: can autonomous economic actors use the same meaningful lever under the same economic rules? If not, why is the exception legitimate?
- Physical-explanation test: if cost, output, speed, information or customer behavior changes, can the world explain what actually changed?
- Scale-transformation test: does scale change the nature of the decision rather than merely multiplying the same clicks? One store can use personal credit judgment; five stores require policy. One employee is personal hiring; a large organization requires managers, records and authority coverage.

2.1 Anchor setting

The recommended anchor is a Dakota-style frontier town in spring 1880. The settlement is in or near its early formal organization phase. It may have had a short pre-plat life as a trail stop, claims cluster, freight stop, creek crossing, or rough service node, but it is only beginning to harden into a true town. It should feel commercially ambitious, geographically well-sited, underbuilt in important ways, and not yet secure in its future. The wider world already contains railroads, telegraphy and extremely early telephone enterprise, but direct access remains

uneven. The opening town should therefore feel economically isolated by distance and infrastructure rather than technologically frozen in an earlier century.

2.1A Historical structural model

The strongest real-world structural analog remains a Rapid City-style gateway settlement: deliberately founded, commercially useful before local rail, shaped early by founders and merchant recruitment, sited near the edge of a resource region, and later amplified by transport and institutions. The new spring 1880 date makes that structure chronologically natural. The Black Hills still depend heavily on distant railheads and overland freight while the first local industrial railways and outside corporate approaches are only beginning to emerge. The player can therefore experience the transportation network moving closer over time instead of starting in a deliberately lagging 1892 town.

2.1B Indigenous-land and federal-policy context

The Dakota-style setting exists within land guaranteed to the Sioux Nation by the 1868 Fort Laramie Treaty and reduced after gold-driven trespass, military coercion and federal legislation. The 1877 Black Hills taking is only about three years old at the spring 1880 opening and must be understood as coerced dispossession rather than an ordinary voluntary land sale. The normal opening settlement nevertheless sits on land already inside the contemporary transferable United States public/private property system so the ownership simulation can function without making restricted land an ordinary commodity. Current title and historical provenance are separate authorities. A legally transferable town parcel can have a chain of title that ultimately derives from the 1877 taking. Reservation, federal, agency and trust- restricted land remain distinct from ordinary fee property and never enter the normal Make Offer workflow merely because they are visible on the map. Substantial Sioux reservation territory remains in 1880. Pine Ridge, Rosebud, Standing Rock, Lower Brule and other agencies operate as administrative and supply nodes inside larger inhabited landscapes. Indigenous people remain living economic actors: households may own livestock, farm or garden, haul freight, work for wages or federal agencies, travel, trade, buy town goods and receive federal supplies according to real household circumstances. Federal rations are physical support, not a generic income statistic. The Dawes Act of February 8, 1887 introduces federal allotment policy, but allotment is a survey-and-administration process rather than an instant conversion to private property. Trust allotments remain federally restricted and are not ordinary saleable collateral. The specific Sioux Act was enacted March 2, 1889 but became operative only after President Benjamin Harrison's proclamation of February 10, 1890. The resulting reservation reorganization and public-domain openings are external federal processes; the player cannot trigger, accelerate or profit through a special dispossession mechanic. The Ghost Dance crisis and Wounded Knee occur during the campaign. They must never become a generic Native hostility meter, raid system, combat mission or profit event. Wounded Knee on December 29, 1890 is treated as a massacre and as historical world context. Indigenous households continue to exist, work, trade and live in the regional economy after 1890. By 1900-05 the land system remains a mosaic of ordinary private property, federal public land, reservation land, individual trust allotments, agency or institutional reserves and corridor rights. The 1904 Gregory County portion of the Rosebud opening can alter migration, freight and settlement pressure externally, but newly opened federal land still requires entry and patent before it becomes ordinary private property.

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1.5 Play-mode architecture

CANON LOCK Land & Ledgers has two flagship ways to play the same simulation: Scenarios and Open Frontier. Scenarios are the authored campaign

layer. Open Frontier is the large procedural sandbox and primary replayability mode. They share the same people, businesses, markets, Physical Truth, Authority Truth, historical chronology, AI rules, maintenance, finance and economic causality. A mode changes starting state, configuration, objectives and presentation; it does not create a different economy.

Net Worth remains the primary capital-success metric wherever a scenario or optional sandbox victory uses a capital threshold. Liquid Cash and Net Worth remain visually distinct. Borrowing cash does not increase Net Worth because the corresponding liability offsets the asset. Capital qualification requires no arbitrary holding period unless a specific scenario has a world-grounded continuity requirement.

1.5A Scenario Campaign - authored commercial challenges

Scenarios are the formal campaign progression. The player chooses from a catalogue of authored commercial situations ranked by difficulty. Each scenario can specify a starting year, location, owned assets/businesses, debts, counterparties, historical pressures, time window, primary objectives and optional distinctions. Scenarios do not need to follow one continuous protagonist or save. Completing the scenario catalogue is the campaign-level achievement: the player demonstrates mastery across different frontier-commercial problems rather than following one scripted character through all of 1880-1905.

Scenario selection should be mostly nonlinear. Grouping by era or commercial theme is preferred over a rigid 1 -> 2 -> 3 chain, although selected introductory scenarios may be recommended first and some advanced challenges may reasonably expect familiarity with earlier systems. Difficulty is a communication aid, not permission to give NPCs magical bonuses.

1.5B Fixed premise, variable economy

A scenario should usually have a fixed dramatic premise but preserve procedural variation where it does not destroy the challenge identity. A Panic of 1893 scenario can always begin with credit contraction and a financially stressed portfolio while varying the exact debt structure, manager quality, customer failures, supplier relationships, property condition, labor market, surrounding owners, weather and available lenders. This prevents scenario replay from collapsing into memorized opening moves.

The simulation rules remain unchanged. The scenario establishes circumstances. It does not secretly change production, prices, AI competence or physical execution to manufacture difficulty.

1.5C Scenario objectives and difficulty doctrine

Primary objectives define completion. Optional Distinctions reward additional mastery without becoming mandatory checklist clutter. Objectives should prefer economic states and transitions over object counters.

- Weak objective: Own three freight wagons.
- Strong objective: Establish reliable scheduled freight between the agricultural district and the new depot while remaining commercially solvent.
- Weak objective: Build fifteen houses.
- Strong objective: Relieve the settlement's persistent accommodation constraint enough that employers can retain and recruit the required workforce.
- Weak objective: Own a mine.
- Strong objective: Establish a mining interest capable of operating on commercially sustainable terms without continual owner subsidy.

Scenario deadlines should have understandable causes: a historical window, seller retirement, creditor date, subscription period, winter, construction season, contract term, rail decision or the authored scope of the scenario. Avoid arbitrary timers whose only explanation is that the designer wanted one. Difficulty should arise through starting conditions and interacting pressures: liquidity, leverage, records, deferred maintenance, supplier concentration, management weakness, labor scarcity, geographic difficulty, commitments and time. Do not buff rival revenue or secretly improve NPC physics on Hard.

1.5D Scenario completion record

Scenario completion should preserve what actually happened rather than reduce mastery to one opaque score. A completion record can show the date, primary outcomes, major optional distinctions, ending financial condition, important businesses or assets and notable consequences. Difficulty medals, stars or equivalent presentation can summarize completion, but they should sit on top of the actual economic record rather than replace it.

1.5E Open Frontier - flagship procedural sandbox

Open Frontier is the major replayability mode. The normal form begins in spring 1880 and generates a new regional economy from a seed while preserving the historical capability and legal envelope. It should not feel like 'Scenario Mode with objectives removed.' World generation itself is a headline feature. Each seed can vary opening settlement geography, parcels and owners, household composition, business presence, property condition, seller motives, supplier relationships, workforce composition, rural production, local resources, mining prospects and information quality, road usefulness, rival holdings, debt, migration patterns, settlement-node potential, infrastructure proposals and the sequence in which opportunities become visible.

The historical skeleton remains stable enough to preserve the game's identity: spring 1880 remains the recommended opening, statehood and validated legal chronology occur when history says they do, the Panic of 1893 remains part of the wider financial era, modernization follows the supported 1880-1905 capability envelope and 1905 remains the normal authored Historical Horizon. What changes from seed to seed is the commercial history inside that frame.

1.5F Open Frontier freedom and configuration

Open Frontier has no mandatory victory condition by default. The player may simply build, survive, expand, contract, sell, reorganize or pursue a self- defined commercial life. Optional success rules can be enabled for players who want a destination: Capital Victory, a selected Legacy Victory, Capital + Legacy or another supported custom configuration.

Any Victory remains a custom option rather than the recommended identity of Open Frontier. Commercial Ruin can remain enabled or disabled as a separate sandbox setting if the final setup design supports it.

Useful Open Frontier setup controls can include region seed, opening-settlement scale, starting foothold/wealth, rival density, volatility/variance, property availability, optional victory rules and continuation preference. These settings should alter initial conditions or variance, not turn historical physics into arcade modifiers.

1.5G Open Frontier opening report

A generated world should begin with a concise commercially plausible briefing rather than spoilers. It can tell the player what a merchant or investor could reasonably know: settlement scale, broad economic character, important transport limitation, visible service gaps, known institutions, obvious property facts and rumors whose reliability is explicitly uncertain. It should not reveal hidden geology, rival books or the seed's optimal strategy. The opening report exists to make a new world legible while preserving discovery.

1.5H Legacy Victory doctrine and current status

Legacy Victories remain useful optional definitions of what kind of commercial position the player built and can also serve as scenario goals. They should measure durable economic outcomes and system control rather than trivial ownership counters. Exact thresholds remain TBD and should be tuned against the live economy. Qualification may use meaningful dimensions already established in the Canon - market reach, business activity, employees, settlement importance, sustained traffic, portfolio resilience, productive control and solved pressures - only where those measures fit the specific Legacy.

LEGACY VICTORY | CANONICAL MEANING

Merchant Prince | Build a major retail/service commercial network that meaningfully serves household and business demand rather than merely owning a fixed count of storefronts.

Industrialist | Control substantial production and industrial chains such as mining, lumber, processing, heavy hauling and internal supply flows.

Financier | Become a major banking, lending, deposit and regional-capital power through a genuinely significant financial institution or finance network.

Railroad Baron | Establish meaningful railway ownership/control, completed corridor value and sustained freight or destination importance. A nominal or half-built railway should not qualify.

Land Baron | Control a major valuable land/property portfolio. Strategic value and productive usefulness should matter so the condition does not reward buying enormous amounts of nearly worthless land.

Regional Power | Maintain meaningful economic presence across multiple settlement nodes and shape the regional network rather than remaining purely concentrated in one location.

Settlement Builder | Develop a weak camp, hamlet or emerging node into an important, durable regional settlement through real jobs, services, housing, freight and economic gravity.

BALANCING STATUS Exact qualification thresholds remain intentionally unresolved. Tune them against the live economy rather than inventing fixed counters merely to make the design look complete. Grand Empire is explicitly REWORK REQUIRED. The previous 'Net Worth plus multiple Legacy badges' formulation risks completionism and shallow last- minute qualification. If Grand Empire survives, it should represent genuine integrated multi-sector regional economic power rather than badge collection.

1.5I Historical Horizon, scenario windows and continuation

The authored historical envelope remains spring 1880 through 1905. Open Frontier normally experiences the whole period. Scenarios may start in later years, end earlier or focus on a particular historical/commercial slice. A scenario deadline belongs to that scenario; 1905 is not a universal win deadline for Open Frontier.

Reaching 1905 should trigger a Historical Record/retrospective for long-form Open Frontier and any scenario whose structure reaches the horizon. The retrospective records what happened - businesses founded, acquired, transformed, sold or failed; major people and managers; properties; crises; settlement changes; institutions; regional reach; peak and ending finances - rather than awarding one Legacy Score. If play continues beyond the authored historical envelope, the preferred presentation is Continuation Year 1, Continuation Year 2 and so on rather than pretending the unchanged asset and technology set accurately models arbitrary future calendar years. Existing 1905-supported capabilities may continue diffusing within the world, but no new post-1905 capability family is implied.

1.6 Campaign loss condition - Commercial Ruin

COMMERCIAL RUIN Defeat is based on actual loss of viable economic foothold and recovery capacity, not a universal debt or negative Net Worth threshold. Land & Ledgers must not end a campaign because a fixed cash, debt, or Net Worth number becomes sufficiently negative. Zero cash, negative cash, negative Net Worth, delinquency, default, judgment, or foreclosure on one asset are not automatically defeat. Commercial Ruin is the true campaign-defeat state and occurs only when the player has effectively lost the remaining economic foothold required to continue as an owner

and has no realistic recovery path. The proposed 1880-1905 campaign begins inside the 1878-1898 gap in federal bankruptcy legislation, so distress does not use a modern federal-bankruptcy reset. Exact Dakota Territorial remedies at the 1880 opening and changes after South Dakota statehood remain subject to the dedicated legal chronology pass. The authoritative loss check should consider whether the player retains an operating business, meaningful income-producing property, economically useful assets, legally usable collateral, sufficient liquid capital to restart, protected or unencumbered property, or a viable financing/restructuring path. Debt enforcement is creditor- and asset-specific: secured creditors pursue pledged property, unsecured creditors pursue debts against legally reachable property, and exemptions can leave an insolvent owner with a genuine path back. A player may therefore fall from a large portfolio to one viable foothold and continue. The authoritative loss check should consider whether the player retains an operating business, meaningful income-producing property, economically useful assets, usable collateral, sufficient liquid capital to restart, or a viable financing/restructuring path. A player may therefore fall from a large portfolio to one remaining business and continue. If that final business is pledged as collateral, the player defaults, the lender takes it, and the foreclosure leaves no meaningful business, productive property, collateral, capital, or realistic recovery route, Commercial Ruin is appropriate.

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1.6A Distress must precede defeat

Financial failure should be playable rather than instantaneous. Preserve and connect the existing business-cash, continuity, finance, workout, collateral, and foreclosure systems into a broad progression: strain -> payment due -> delinquency -> workout or restructuring -> default -> legal or negotiated enforcement -> sale / liquidation / foreclosure -> possible deficiency or recovery -> Commercial Ruin only if recovery capacity is exhausted.

- Recovery may include owner cash injection, inter-business cash movement, refinancing, revised schedules, additional collateral, investor capital, inventory liquidation, project delay, cost reductions, business turnaround, voluntary sale, or closure of weaker holdings.
- Foreclosure is a strategic loss and portfolio contraction event, not automatically a campaign-ending event.
- Commercial Ruin should be evaluated from the player's remaining economic capacity, not from an arbitrary universal debt ceiling that becomes meaningless at larger portfolio scale.

1.6B Debt enforcement and protected recovery - 1880-1905 legal lock

Dakota Territory already has a developed codified commercial-law framework at the spring 1880 opening. Real-estate mortgages operate as liens rather than automatic transfers of ownership, and default does not instantly give the mortgagee possession. Real-estate and chattel enforcement remain staged processes involving the applicable notice, judicial or authorized sale, seizure where relevant and redemption or finalization. The territorial real-estate framework includes a one-year redemption anchor after foreclosure sale; exact procedural detail can be compressed for gameplay without erasing the sequence. Historical homestead and personal-property exemptions already exist before statehood, so insolvency does not imply complete household dispossession. Separately owned spousal property is not silently reachable as the business debtor's property, and legally relevant homestead rights can require spousal participation in transfer or encumbrance. These protections remain part of the Commercial Ruin test because a distressed owner may retain a genuine recovery foothold. The Black Hills has a meaningful interest-law chronology. At the spring 1880 opening Dakota's general framework uses a seven-percent legal/default rate and permits written contractual interest up to twelve percent. From 1881 through June 30, 1887, covered Black Hills counties permit written contracts at any mutually agreed interest rate; the general ceiling returns July 1, 1887. These are legal boundaries, not automatic market rates: actual pricing still comes from capital scarcity, borrower strength, collateral, purpose, term, lender appetite and competition. Statehood on November 2, 1889 does not reset private law. Existing contracts, mortgages, corporations, lawsuits, bonds and rights continue. South Dakota institutions and statutes then refine the regime during 1890-92; the previously validated 1892 South Dakota creditor rules remain later-campaign anchors. Municipal finance also changes during the campaign: the federal act of July 30, 1886 places a four-percent general debt ceiling on territorial political subdivisions and sharply restricts public credit or subscription aid to private corporations; the South Dakota Constitution later uses a five-percent general local debt ceiling subject to later infrastructure provisions. The campaign begins inside the 1878-1898 federal bankruptcy gap. On July 1, 1898 the Bankruptcy Act of 1898 creates a new federal distress route for qualifying debtors. It is not a modern universal reset and should be represented only if it strengthens late-game distress gameplay. Commercial Ruin remains the campaign loss test throughout: legal bankruptcy, foreclosure or liquidation is not defeat unless viable ownership and recovery capacity are actually exhausted.

1.6C Legal regime authority

The simulation should query a small set of dated legal authorities rather than expose nineteenth-century filing work to the player: Dakota Territory versus South Dakota; interest/usury regime; mortgage and chattel enforcement state;

homestead and protected-property treatment; municipal debt capacity and public-aid eligibility; and federal bankruptcy availability. Law matters through what ownership, finance and civic structures are legally possible, not through a paperwork minigame.

1.7 Experience, Expertise, Commercial Standing & Information Progression

Land & Ledgers may use a progression layer, but the preferred direction is professional Experience / Expertise rather than a conventional RPG skill tree or player-researched technology tree. The player character is one actual commercial person whose judgment can improve over a twenty-five-year career. The system should feel like accumulated professional competence: operating stores, buying property, hiring people, extending credit, surviving shortages, handling freight, reviewing books, managing distressed businesses and entering new industries create relevant experience. The world technology chronology remains separate: electricity, pneumatic drills, cash registers, telephone systems, machinery and other technologies enter the market because history, geography and infrastructure make them available, not because the player spends expertise points.

1.7A Three progression authorities: Expertise, Standing, Modernization

- Expertise answers: how well can this person interpret, forecast, diagnose and administer this kind of problem? It grows from relevant experience, quality information and, where appropriate, education or professional exposure.
- Commercial Standing answers: how seriously do lenders, sellers, investors, corporations, workers and other counterparties take the player? It arises from net worth, liquid resources, income, payroll, completed deals, repayment record, business scale, reputation and regional importance.
- Modernization answers: what technology, service, institution or network capability actually exists in the world and can be purchased, contracted or connected? It follows historical availability and local economics, not character experience. These authorities may overlap in consequence but must not be collapsed. An inexperienced heir can possess enormous capital and standing without commercial expertise. A veteran merchant can be highly skilled while still lacking the net worth to finance a railroad. A brilliant 1882 businessman cannot research a 1902 technology into existence.

1.7B Experience Ledger and anti-grind rule

The simulation may maintain an internal Experience Ledger that records meaningful exposure: years personally operating a business, full seasonal cycles completed, acquisitions closed, employees hired and lost, supplier relationships managed, credit accounts administered, property improvements completed, freight routes run, crises survived, distressed businesses turned around and other domain-relevant events. Experience gain should be weighted by relevance, novelty, complexity, scale and direct involvement rather than by raw repetition.

- The first property purchase should teach more than the twentieth nearly identical low-value lot purchase.
- Managing the first employee resignation, first manager vacancy or first serious labor shortage can create more management experience than repeatedly hiring and firing trivial workers.
- Failure counts as experience. A bad credit decision, disastrous overstock, failed acquisition, foreclosure threat, equipment breakdown or winter feed shortage can improve later recognition of the same class of problem without reimbursing the economic loss.
- Delegated ownership provides commercial exposure but should not automatically create specialist mastery. Owning a mine for ten years does not turn the player into a metallurgist if every technical decision was handled by specialists.
- Grinding must be structurally unrewarding. Repeating low-complexity actions solely to raise an expertise meter should produce sharply diminishing learning value.

1.7C Known-fact versus uncertain-future doctrine

Known transactional facts should be shown exactly when the player or business legitimately possesses them. If the store bought an item for $0.12, the recorded known purchase cost is $0.12. If the freight invoice is $1.00, expertise does not change it to a range. If the books contain a confirmed $18.40 customer balance, the player sees $18.40. Artificially hiding known numbers behind low skill is prohibited. Expertise belongs primarily where reality is genuinely uncertain: future demand, future cash flow, likely travel time, property market value, maintenance burden, repayment risk, employee retention, production throughput, mine geology, construction cost before quotation and other forecasts. These may be shown as central estimates, ranges and confidence levels. Better expertise, better records, more relevant history and stronger information sources can narrow the range, but they do not guarantee the outcome.

- Example: last winter the town used 65 cords of known firewood through tracked local channels. Current population and household composition are known. Next winter consumption remains a forecast; an inexperienced owner may see 55-105 cords while an experienced merchant with good records may see 72-88 cords.

- Forecast explanations should expose drivers where useful: prior consumption, population change, household count, boarding capacity, known reserves, employment shifts, competitor activity and unresolved weather uncertainty.
- Expertise should sometimes widen caution rather than narrow confidence. A skilled owner can recognize that a new mine camp with no sales history is intrinsically difficult to forecast.

1.7D Expertise must not replace causality

No expertise node should directly create unexplained savings, output, demand, durability or negotiation success. A logistics skill does not reduce a $1.00 trip to $0.90 merely because the player leveled up. Expertise may reveal that two half-load shipments can be consolidated, that a scheduled recurring route could justify a lower carrier quote, or that a different route is cheaper. The player or manager then changes the actual arrangement, and the real arrangement changes the cost.

- No Merchandising +10% sales. Better merchandising analysis can reveal persistent workwear stockouts; the owner reallocates space and captures demand that was previously lost.
- No Property Expertise -10% purchase price. Better diligence can identify a roof defect or seller urgency that creates legitimate negotiating leverage.
- No Management +8% worker productivity. Better management expertise can identify uncovered workload, bad scheduling, weak retention or inappropriate staffing; changing wages, hours, coverage or delegation then changes results.
- No Maintenance Expertise +15% durability. Better maintenance expertise improves diagnosis, scheduling and repair-versus-replace judgment; the actual servicing preserves the asset.

1.7E Expertise and specialists

The player does not need to personally become expert in every domain. Information quality can come from the owner, a business manager, a superintendent, a bookkeeper, an engineer, an assayer, an attorney, an inspector or another historically appropriate specialist. A late-game portfolio should become smarter partly because the owner has built an organization containing competent people. Personal expertise helps the owner judge reports and ask better questions, but it does not replace professions whose work requires specialist training. This creates two legitimate growth routes. The owner can develop direct expertise by spending years operating in a domain, or can compensate for limited personal expertise by hiring people who possess it. Losing a long-serving specialist can therefore reduce organizational insight even when the underlying physical assets remain.

1.7F Potential expertise domains - oversized design pool

- Bookkeeping & Commercial Analysis: reports, reconciliation, cost attribution, trend recognition and financial interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Merchandising: assortment, turnover, stockout, seasonal and display-space interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Purchasing & Supply: supplier comparison, landed-cost interpretation, procurement timing and concentration risk. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Freight & Logistics: route, load, travel-time, utilization and transport-bottleneck analysis. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Property & Valuation: land/building value, redevelopment potential, repair burden and comparable-market interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Negotiation & Dealcraft: counterparty motive, urgency, deal structure and likely room for movement. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Employment & Management: staffing, retention, supervision, delegation and organizational diagnosis. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Credit & Collections: customer repayment capacity, receivables, delinquency and collection posture. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.

- Finance & Capital: debt capacity, financing structure, investment comparison and capital allocation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Risk & Resilience: concentration, reserve sufficiency, single points of failure and contingency recognition. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Retail Trade: industry familiarity with customer traffic, merchandise cycles and store operations. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Lodging & Hospitality: occupancy, room/board economics, meal demand and transient-customer patterns. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Agriculture: field work, seasonal windows, machinery use and crop-production constraints. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Livestock & Ranching: forage, water, herd composition, winter preparation and sale timing. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Industrial Operations: machinery, power, throughput, process bottlenecks and heavy-operation reporting. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Lumber & Sawmilling: log supply, mill throughput, drying/storage and lumber-chain constraints. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Mining: commercial mining reports, development economics, uncertainty and mine-operating structure. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Banking: loan portfolios, deposits, liquidity, credit quality and concentration. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Railway Operations & Promotion: corridor economics, freight projections, route strategy and railway negotiation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Utilities & Infrastructure: network load, capacity, coverage, maintenance and expansion economics. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Regional Markets: multi-node price, shortage, migration and market-reach interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Forecasting & Scenario Planning: confidence ranges, scenario comparison and recognition of forecast uncertainty. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Inventory Control: days-of-supply, target stock, excess inventory, stockout and reorder analysis. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Pricing & Margin Analysis: landed cost, gross margin, turnover and price-response interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Supplier Relations: supplier reliability, recurring relationships, volume commitments and priority arrangements. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Commercial Contracts: commitments, fulfillment exposure, renewal risk and unfavorable-term recognition. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Customer Accounts & Relationship Trade: repeat purchasing, account history, recurring demand and customer stability. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.

- Treasury & Cash-Flow Management: payroll, bills, reserves, liquidity collision and near-term cash forecasting. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Debt & Refinancing: repayment structures, collateral exposure, restructuring and refinance comparison. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Investment & Partnership Structuring: outside capital, profit sharing, control dilution and risk allocation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Acquisition Diligence: books, condition, supplier, staffing and hidden post-close obligation analysis. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Turnaround & Distress Management: recoverability, restructuring, closure, sale and weak-business diagnosis. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Maintenance & Reliability: service backlog, failure exposure, downtime and repair planning. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Asset Lifecycle & Replacement Planning: repair, overhaul, replacement and remaining-service economics. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Construction & Capital Projects: cost ranges, sequencing, materials, delay and project bottlenecks. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Land Assembly & Site Selection: frontage, access, adjacency, corridor and expansion-potential interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Survey, Title & Property Records: survey descriptions, title concerns and parcel-history interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Commercial Law & Legal Awareness: contractual/legal exposure and recognition of when professional advice is needed. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Insurance & Risk Transfer: coverage, retained risk, claim exposure and protection economics. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Labor-Market Intelligence: prevailing compensation, scarce roles, recruitment pressure and competitor demand. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Delegation & Manager Oversight: authority bands, escalation design, manager diagnosis and intervention timing. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Hiring & Talent Assessment: candidate suitability, reference interpretation and hiring-risk estimation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Workload & Scheduling: labor-hours, simultaneous task pressure, shift coverage and seasonal staffing. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Community & Institutional Relations: authority mapping, civic counterparties, institutional interests and cooperation needs. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Market Intelligence & Information Networks: newspapers, correspondence, telegraph, telephone and commercial-contact interpretation. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Seasonal Planning: distinguishing seasonal cycles from structural trends and preparing recurring annual needs. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.

- Business-to-Business Trade: anchor customers, recurring commercial demand, account exposure and supply commitments. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Vertical Integration & Portfolio Coordination: internal supply opportunities, shared bottlenecks and transferred rather than eliminated costs. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Crisis Response & Business Continuity: emergency alternatives, continuity plans and rapid recognition of cascading failure. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful.
- Opportunity Recognition & New Ventures: unmet demand, weak coverage, emerging nodes and potentially viable new enterprise identification. This is a candidate domain, not a guaranteed separate progress bar; related areas should be merged where playtesting shows the distinction is not meaningful. Current preferred consolidation is roughly eight to fifteen broad owner expertise tracks plus industry familiarity and specialist employee expertise underneath them. The fifty-domain list is intentionally oversized so future design can merge downward rather than discover missing professional areas too late.

1.7G Current high-priority expertise candidates

- Bookkeeping & Commercial Analysis: the strongest thematic fit for Land & Ledgers and the foundation for increasingly sophisticated owner reports.
- Forecasting & Scenario Planning: converts accumulated records and experience into progressively better ranges for uncertain future outcomes.
- Property & Valuation: makes repeated ownership/deal experience materially improve interpretation of land, buildings and acquisitions without price magic.
- Merchandising: turns years behind the counter into better use of assortment, display space, turnover and seasonal evidence.
- Employment & Management: improves the owner's ability to see staffing, retention and organizational problems rather than buffing workers.
- Purchasing & Supply: improves supplier, order, lead-time and landed-cost judgment while actual contracts and logistics create the savings.
- Credit & Collections: improves assessment of repayment risk while leaving household finances and future events genuinely uncertain.
- Freight & Logistics: improves route/time/load judgment while actual route, team, schedule and contract changes drive cost.
- Acquisition Diligence: improves the ability to identify weak books, deferred maintenance, hidden liabilities and post-close capital needs.
- Maintenance & Reliability: improves inspection interpretation, downtime planning and repair/replacement judgment without changing the physics of equipment.

1.7H Commercial Standing and natural unlocks

Net worth, income, deal history, payroll, repayment performance and portfolio scale should primarily create Commercial Standing rather than expertise. Standing can unlock counterparties and transaction sizes naturally: a tiny merchant may not be credible for a large property loan, while a wealthy regional employer with strong collateral and a clean repayment record can attract bankers, investors, railroad promoters and major sellers. These are natural unlocks because the outside actor has a reason to engage. Completed deals can also improve seller confidence independently of negotiation expertise. A buyer who has successfully closed fourteen transactions may be considered more credible than a first-time buyer making the same nominal offer. The game should prefer these evidence-based relationship effects to abstract character levels.

1.7I Expertise consolidation authority

The fifty-domain list in 1.7F is a decomposition/reference pool, not a proposed player-facing collection of fifty progress bars. The current preferred owner structure consolidates toward roughly eight broad Expertise families, with Industry Familiarity and specialist employee competence underneath them.

- Commercial Analysis - bookkeeping interpretation, cash flow, forecasting, scenario analysis and risk recognition.
- Trade & Merchandising - assortment, pricing, turnover, customer accounts, demand patterns and commercial presentation.

- Supply & Logistics - suppliers, landed cost, contracts, route economics, transport utilization and supply-chain resilience.
- Property & Dealcraft - valuation, site selection, title awareness, acquisition diligence, negotiation and redevelopment judgment.
- People & Management - hiring, compensation, workload, retention, delegation, manager oversight and labor-market interpretation.
- Credit & Capital - borrower quality, debt structure, refinancing, investment structure, liquidity and capital allocation.
- Assets & Operations - maintenance, reliability, production bottlenecks, replacement planning and capital-project interpretation.
- Regional Strategy - settlement change, regional markets, institutional relationships, infrastructure, concentration risk and new-venture recognition.

Industry Familiarity remains separate: retail, lodging, agriculture, livestock, timber, mining, banking, railway, utilities and other supported operating domains. A person can be excellent at commercial analysis without understanding mine metallurgy. The owner can also compensate for weak personal familiarity by retaining specialists.

1.7J Expertise outputs

Expertise should mainly improve five things: recognition, interpretation, estimation, exception detection and comparison of plausible responses. It helps the player notice a pattern sooner, understand likely causes, produce better-calibrated forecast ranges when evidence exists, detect unusual terms or missing information and compare trade-offs. Expertise can also recognize that uncertainty is irreducible and widen caution. It must never lower real prices, raise production, improve worker output, reveal hidden geology, prevent failure or force a counterparty to agree without a causal mechanism in the world.

PART II

World, Land & Settlement Simulation The regional substrate: settlement growth, terrain, parcels, node emergence, infrastructure and readable economic pressure.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

2.2 Starting scale and identity

The normal spring 1880 opening remains roughly 14 to 22 households and 55 to 95 residents, with a likely sweet spot near 70 to 85. This is an intentionally compressed younger gateway settlement, not a claim that real Rapid City itself had only that population in 1880. Historical Rapid City exceeded one hundred residents within months of its 1876 founding and then grew rapidly after spring 1880; the fictional opening uses the same merchant-gateway structure at a smaller scale. The town normally begins with no local common-carrier rail, no broad piped water or sewer network, no electric service and no normal public telephone exchange. Roads are dirt and weather-sensitive; limited grading, drainage ditches and short wooden walks or boardwalks can exist on the busiest frontage. Water normally comes from wells, hand or mechanical pumps, springs, creeks and cisterns; sanitation relies on privies and simple drainage; fire response is buckets, local water and ad hoc or very early volunteer organization. The General Store is the strongest normal starter business. A blacksmith, livery/freight service, boarding house or small hotel, butcher, saloon, bakery, drug counter or other service may exist according to the seed, but no fixed business roster is required. Boarding and transient lodging should be common enough to matter because teamsters, workers, travelers and new arrivals exceed the stable household count. A resident doctor is historically possible but not guaranteed; periodic or nearby medical service remains valid. A formal local bank is normally absent at this tiny scale even though sophisticated banking already exists in larger Black Hills nodes. Regular mail and an overland freight connection should be normal for a viable gateway. Stage or express service is appropriate when route economics justify it. The settlement uses a distant practical rail interface rather than local track. Urgent information can be carried to a telegraph node; a local telegraph office or private telephone line is exceptional rather than default. Industrial rail, electric light, piped water and other advanced systems can exist elsewhere in the region without belonging to the opening town. The surrounding economy should be ranch-heavy with crop farming concentrated in suitable watered or irrigable valleys. Hay is strategically important. Horse-powered farm machinery, barbed wire and pumping windmills are period-available but ownership is uneven. Regional gold hard-rock mining is already industrializing, yet the player's opening town is not required to be a mine camp. Federal forts, agencies, land offices and survey systems matter regionally without requiring a local federal institution in every seed. Indigenous travelers, workers, freighters, households and traders can appear where geography and law make that plausible while restricted land remains outside ordinary acquisition.

2.2A Starting foothold and later regional plurality

The opening town should still feel like the player's first serious commercial arena: the place where labor, demand, credit, and acquisition become readable. However, it should be understood as a starting anchor rather than a permanent destiny. The region can later contain multiple meaningful settlement nodes, and the original town can remain stable, stall, or even decline if another node becomes more useful, better connected, or better capitalized. This means the game now supports three broad forms of player expansion. First, the player can simply deepen the original town through more businesses, housing, and institutional weight. Second, the player can create satellite service nodes that remain dependent on the original town. Third, the player can build enough gravity in a remote area that a new town emerges and eventually rivals or surpasses the first one. All three outcomes should be valid, and none should be hard-forced by the rules.

2.3 Growth phases

Growth is overlapping modernization rather than a single infrastructure ladder. The opening years are organizing settlement and commercial foothold: thin margins, incomplete services, fragile boarding capacity and long wagon legs to distant rail interfaces. As nodes become more useful, transportation, communications, civic infrastructure, industrial technology and power can advance independently. Telegraph or telephone access can precede rail; waterworks can precede electric light; a railroad can bypass one town while another grows; a mine can possess electric light or private industrial rail while nearby households still use wells, privies and wagons. Regional railway penetration accelerates from the mid-1880s, but local service remains a corporate and corridor decision. Mining becomes deeper and more capitalized; treatment technology expands; farm and ranch operations gain machinery, fencing and water control; municipal and utility networks accumulate spatially; statehood and later federal bankruptcy alter the legal environment without resetting existing contracts. Growth can reverse. A town may boom, overbuild, lose population, suffer the Panic

of 1893 or drought and later recover while keeping some of the infrastructure created during its earlier peak. The normal authored historical horizon is spring 1880 through 1905, roughly twenty-five years. Open Frontier normally spans that period, while Scenarios can use shorter windows inside it. Optional sandbox victory can occur earlier and continuation can extend beyond the horizon, but 1905 is the preferred end-state boundary for authored historical capability. It provides enough runway for rail networks, telephones, electricity, waterworks, cyanide-era mining, professionalized finance and late rural communications to mature while horse and wagon transport still dominate ordinary rural life. Major Black Hills regional rail integration is still being built in 1905, and motor vehicles are only beginning to appear as rare registered machines rather than a replacement transport economy.

2.3B End-state envelope, not a modernization checklist

The 1905 horizon defines what can plausibly exist, not what every successful town must have. The strongest player-controlled node may possess rail, telegraph, telephone, electric lighting, waterworks, hydrants, partial sewerage, brick or stone commercial blocks, multiple financial and professional services and sophisticated industry. A secondary settlement may still have dirt streets, wells, privies and wagon- only freight. A mine camp may use compressed air, electric light and industrial rail while its housing and sanitation remain rough. A ranch may have barbed wire, windmills and a rural mutual telephone while still lacking electricity. Uneven adoption is the intended frontier- industrial texture. Automobiles and gasoline tractors are not core progression before the normal end. An occasional motorcar by 1905 is historically valid as rare/cutting-edge background in a strong node, but trucks do not replace wagon freight, tractors do not replace horse- powered agriculture and automobile road infrastructure does not become a core economic requirement.

2.3C Asset-Honesty Test for the historical horizon

When considering whether to extend the authored historical endpoint, ask whether a prosperous settlement at that date would visibly or economically require a major asset family the game does not honestly support. If yes, the horizon has probably moved too far. Open Frontier or scenario pacing problems should be solved through calendar pacing, delegation and late-game decision design rather than casually changing the historical boundary.

2.3A Regional settlement network and node succession

The original town's growth phases still matter, but they no longer exhaust the map's long-term story. As the portfolio broadens, other parts of the region can generate enough labor pull, freight burden, resource value, frontage demand, and household settlement pressure to justify additional nodes. Those nodes begin as local concentrations of life and work, then gradually differentiate into camps, hamlets, service settlements, or full towns depending on population, durability, and usefulness. The key rule is that the first town starts ahead; it does not stay ahead by fiat. A remote mine district, freight crossing, farm-service corridor, or mixed agricultural-industrial pocket can attract households and capital strongly enough that the regional hierarchy changes. This gives the player room to create a second or third town without erasing the special role of the opening settlement.

3.1 World generation and land structure

Generate the town from a deterministic seed so starts are repeatable and comparable. Core generated layers remain roads, plots, building anchors, ownership, households, businesses, and edge development parcels. Not every business type needs to exist in every seed. Missing services should create Opportunity & Pressure rather than generation failure. Growth should remain positive-feedback in believable ways: jobs attract people, people create demand, demand supports more businesses, and more businesses create more jobs. The player can accelerate this loop but should not be its only cause. Town siting should favor natural gateway logic: creek corridor, valley mouth, pass, road junction, or plains-to-hills interface. The early commercial strip may reflect founder or speculator pre-commitment. Prime frontage may exist before the rest of the town fully deserves it. The town should feel like it is in the right place to matter. Town core parcels should skew toward private households, active businesses, widow or estate holdings, and founder or speculator parcels. The inner expansion ring should mix private holdings and partially developed opportunity lots. The outer edge should support rougher new-development opportunities, remote production parcels, and simpler site conditions.

3.1A Survey frame, parcels, legal geography, and provenance

The regional map carries a survey-and-parcel skeleton beneath the visible world. Every parcel should conceptually know its survey identity, parcel type, acreage, current jurisdiction, ownership form, transfer status and historical provenance even if the player never sees all fields simultaneously. The town plat is carved inside the broader frame rather than replacing it. Current ownership and transferability are separate from provenance. Ordinary private fee property can enter acquisition when the owner is willing and the deal is credible. Federal public- domain or unpatented entry land is not an ordinary private sale. Tribal or reservation land, individual federal-trust allotments, federal agency reserves and other restricted parcels do not enter Make Offer. Railway or utility rights-of-way can confer a corridor right

without transferring surrounding fee ownership. Newly opened federal land becomes ordinary market property only after the applicable entry and patent process creates transferable private title. This structure supports town lots, rural holdings, assembled ranch land, claim parcels, creek fragments, depot strips, mining claims, federal land, reservation land, restricted allotments and specialized corridor rights under one authority without pretending every visible acre is for sale.

3.1B Tiled terrain, hydrology, and parcel-aware shaping

The visual world should be generated through a tiled terrain system backed by a deterministic seed. Terrain acts as the rendered ground layer while the hidden two-meter grid, road graph, parcels, anchors, and settlement nodes remain the real source of truth. Macro landform generation, hydrology, roads, parcel boundaries, building pads, and vegetation masks should happen in ordered passes so the region feels naturally sited instead of noise-based. Parcel-aware terrain shaping is now a canonical rule. Buildings, roads, yards, loading spaces, and civic lots should request practical grading from the terrain layer instead of forcing the player to accept awkward slopes. At the same time, the world should preserve natural relief, creek banks, hills, wet ground, and transition zones so the map feels premium and grounded rather than flattened into obvious game rectangles.

3.1C Settlement nodes and contextual growth

Every meaningful inhabited place should be modeled as a settlement node: opening town, mine camp, service hamlet, freight stop, timber outpost, ranch-service cluster or later town. Nodes track population, households, boarders, housing pressure, freight access, business mix,

service coverage, local confidence, infrastructure capacity and decline risk. Population is evidence and demand, not a magic unlock authority. Broad labels such as camp, hamlet, village, town or large town may summarize what the player sees, but rail, waterworks, banking, electricity, fire service and other systems arise from their own viability, legal and network conditions. Population bands can help summarize scale or tune AI expectations, but they must not directly trigger infrastructure merely because a threshold was crossed.

3.1D Regional emergence beyond the original town

The player may attempt to grow new settlement nodes anywhere on the map where they can control land and sustain real activity. The simulation should recognize those efforts when enough work, housing, service coverage, and freight logic gather in one place. Success should never be guaranteed. Poorly chosen sites can stall as thin outposts or collapse after a promising start. Well-supported sites can become self-reinforcing and eventually surpass the opening town. The region is therefore a living hierarchy of settlements rather than a single forever-dominant center. 3.1D-1 Economic geography doctrine and active research priority The player does not conquer territory; they change the map of economically practical relationships. Geographic distance must be evaluated through the purpose of the trip or relationship rather than one universal Distance Penalty. A mile means something different for groceries, medical care, ore, livestock, lumber, machinery, employment and banking. Practical market reach can depend on travel time, road condition, season, terrain, transport mode, value-to-weight ratio, payload, perishability, urgency, information and available substitutes. A rural household may buy staples locally, machinery farther away and sell livestock through a different node. Rail, warehouses, branch businesses, freight service, banking and communications can redraw these practical relationships without changing map ownership color.

Regional commercial geography remains one of the largest outstanding dedicated research/design passes. Exact market-area behavior, branch/wholesale organization, drayage/transshipment and whether routes require persistent economic identities should remain holds until that pass resolves them.

3.1E Town morphology, alleys, and civic improvements

Settlement form should derive from founding history, terrain, water, roads, and later transportation rather than one universal frontier grid. Founder-service towns may use surveyed blocks and reserved commercial frontage; mining or gulch settlements can grow linearly and irregularly; railroad townsites can orient around depots and freight land; road hamlets and mine camps can remain looser until permanence justifies formal platting. Front and rear geography both matter. Main Street frontage carries customer visibility and prestige, while alleys and rear yards can carry deliveries, waste removal, privies, sheds, fuel, stable access, utilities, and secondary entrances. Business receiving should therefore connect trip-based freight to an actual unloading interface rather than treating inventory as if it appears inside a building. Public improvements should accumulate causally. Young nodes can begin with dirt streets, uneven pedestrian edges, wells or pumps, privies, simple ditches and partial wooden walks. Traffic, mud, drainage, sanitation, fire protection, density and property value can justify boardwalks, grading, surfacing, curbs, gutters, water mains, hydrants, sewer or drainage works and later utility corridors. Improvements are spatial, can be bundled into corridor projects, and can impose special assessments or other real financing burdens rather than arriving as a free town-tier bonus. Nuisance-heavy uses such as liveries, corrals, large yards, workshops, warehouses, slaughtering, and

waste-intensive businesses should sort spatially through land cost, odor, manure, fire exposure, freight traffic, customer response, and yard need rather than through modern zoning hard locks. As the core becomes denser and more valuable, those uses may move outward or toward depot/freight land for economic reasons.

3.1F Civic infrastructure networks, finance, and project lifecycle

Formal waterworks, hydrants, organized fire protection and sewer construction are technologically possible in 1880 but are not normal requirements for a 70-85-person settlement. Young nodes normally use wells, pumps, springs or creeks, cisterns, privies, dirt streets and ad hoc fire response. Infrastructure appears when density, commercial value, fire pressure, sanitation problems, geography, capital and civic capacity justify it. Major civic projects use a lifecycle: visible need -> promoter or civic proposal -> legal authority/franchise and route -> financing -> source/engineering plan -> construction -> initial service district -> later extension, rebuilding or capacity expansion. Incorporating a company or approving a franchise does not mean a physical network exists. Water distinguishes source, storage, mains, hydrants and property connections. Sewerage distinguishes private disposal and drainage from formal sewer districts and property connections. Street, drainage, water and sewer work can be bundled. Coverage, condition and capacity remain spatial, allowing a Main Street service district while outer homes still use wells and privies. Financing can use taxes, special assessments, municipal debt, private franchise capital, merchant subscription or mixed arrangements where the applicable legal regime permits it. Legal capacity matters. After July 30, 1886 territorial local governments face a four-percent general debt ceiling and major restrictions on lending public credit to private corporations; after statehood South Dakota uses a five-percent general local debt ceiling subject to later infrastructure provisions. The player normally influences projects as property owner, lender, supplier, landholder, utility investor or civic advocate rather than as a mayor placing pipes.

3.2 Opportunity & Pressure

Opportunity & Pressure is one of the largest crossover systems in the project. It is the town's sensing-and-response layer for unmet need, blocked growth, surplus, bottlenecks, weak coverage, settlement strain, and contested openings. Opportunity means the town contains money-making or growth-solving potential that has not yet become urgent or contested. This can come from shortages, weak service coverage, recurring stockouts, unfilled orders, idle land, idle labor, underused buildings, blocked immigration, overcrowding, homelessness, boarded-up or weak businesses, supplier unreliability, freight delay, or even local surplus that could support a new trade or export path. Pressure begins when an opportunity worsens, becomes time-sensitive, or attracts visible action from rivals, lenders, suppliers, migrants, employers, or other town actors. A housing shortage is an opportunity until families begin failing to settle, rents spike, boarding saturates, or a rival starts buying parcels to answer it. A lumber shortage is an opportunity until stalled projects, backlog, price strain, or rival mill action makes it active pressure. The system should read town state across inventory, demand, labor, housing, service access, land, freight, migration, and continuity. It should care about both shortage and surplus. A town with too little lumber may justify a sawmill; a town with strong lumber output but weak town-facing retail may justify a lumber yard; a town with too many hides but weak processing may justify downstream trade.

3.2A Signal-and-threshold ladder

- Early signal: the town is leaning. The condition is visible, but not yet urgent.
- Mounting signal: the condition is repeating. The town is compensating badly or beginning to bend around the problem.
- Action signal: a threshold has been crossed and now demands response through ownership, migration, supplier change, expansion, or rival behavior. Major opportunities and pressures should ideally have physical signals, behavioral signals, and social signals. A boarding shortage can show through crowded porches, repeated full-house behavior, and gossip or notice-board language. Freight unreliability can show through idle wagons, delayed supplier arrivals, and price or stock consequences.

3.3 Outputs and design job

Opportunity & Pressure should never be a hidden number that merely alters the world in silence. It should surface as notices, immigration effects, startup leads, acquisition prompts, expansion prompts, stalled-growth explanations, market openings and rival-intent warnings. The player should be able to read not only that a signal exists, but why it exists. Repeated transactions, stockouts, failed purchases, missed orders, failed hires, vacancy changes, unstable accommodation, boarding saturation, route delay and business closure should accumulate into readable trends rather than isolated notifications. The town survives long-term by becoming regionally useful: not just bigger or richer, but useful enough that the surrounding region increasingly depends on it for supply distribution, repair capacity, lodging, livestock services, administration, education or transport coordination. Opportunity & Pressure is one of the main tools that should reveal this process.

PART III

Ownership & Property How land, buildings, deals, diligence and projects create the ownership loop.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

4.1 Ownership buckets and property states

Property and business acquisition remain signature differentiators. The player can always attempt an offer, but not every seller will treat the player as serious immediately. Ownership buckets include founder or speculator townsite holders, private households, business owners, widows or heirs or estates, and edge development parcels. Founder or speculator holdings may include reserved prime frontage, half-filled early commercial strips, or parcels held specifically for later merchant recruitment. Property states include owner-occupied and not interested, quietly open to a serious buyer, debt-pressured, widow-controlled, estate unsettled, inheritance disputed, publicly listed, and bank-held distressed.

4.2 Suitability and mixed use

Parcels are not permanently typed as residential or business lots. Current use matters, but future use is driven by frontage, frontage quality, parcel depth, rear or alley access, freight/loading access, water and drainage, yard or storage capacity, nuisance, neighboring uses, traffic, public right-of-way exposure, assembly potential, and conversion cost. A narrow Main Street lot can be excellent retail property and poor freight property; a deeper secondary lot can be the reverse. Mixed domestic-commercial use remains valid when the building, lot, household, and surrounding town make it workable.

4.3 Deal opportunity channels, seller motive, seriousness and deal structure

Deal opportunity channels include public listings, private approaches through clicked properties or introductions, rumor and whisper-market leads, triggered sales from death, debt, retirement, relocation, inheritance pressure, package deals and distressed or bank-held resale paths. Sellers should not all simply want the highest price. Motives can include maximum price, quick closing, trusted buyer, family preservation, clean handoff, local continuity, debt escape or hidden-problem transfer. Seriousness gates can include proof of funds, lender support, earnest money, references or prior clean deal history. Early access to a better deal may depend on documented deal history, relationship memory and whether the seller believes closing can actually happen.

4.3A Deal structure is part of negotiation

A serious offer should be more than one purchase price. Where appropriate, terms can include cash at closing, financed portion, seller-carried balance, earnest money, closing date, exclusivity, financing or diligence contingencies, included inventory/equipment, excluded personal property, treatment of existing tenants or accounts, transition assistance and package-versus-separate asset treatment. Not every transaction exposes every term. Two offers with the same headline price can carry different certainty, speed, risk and burden for the seller.

4.3B Negotiation is exchange, not persuasion dice

There should be no abstract negotiation attack that lowers prices. When one party asks the other to concede, the counterparty should care what they receive in return. A lower price can be paired with faster closing or more certain cash. Longer diligence can require stronger earnest money. Seller financing can justify a different total price or security. Included equipment, transition assistance or exclusivity can have their own value. Better Dealcraft Expertise improves interpretation of motive, terms and likely room for movement; it does not compel agreement.

4.3C Negotiation exists in time

A deal can improve or deteriorate while the player investigates and negotiates. Another buyer can appear, urgency can change, inventory can be sold, workers can leave, financing can tighten, a defect can worsen or a creditor can intervene. The player can lose a good deal by over-negotiating it. Patience can also help when the seller's alternatives weaken. Time is therefore part of acquisition strategy rather than a decorative progress bar.

4.4 Diligence, closing risk, and valuation

Due diligence remains layered: title, books review, condition survey, supplier review, staffing and management review and demand or market-fit review. Closing remains risky. Financing can fail, title problems can surface, suppliers can flee and labor can leave before transfer completes. Valuation should consider more than static land score: frontage, access, current improvements, traffic, neighbor strength, pressure from town growth, seller urgency, distress state and whether there is hidden operating baggage attached to the business or site.

4.4A Diligence answers specific uncertainties

Diligence should not be a generic completion meter. Title review asks what is actually being conveyed and what claims can interfere. Books review tests whether reported revenue, receivables, inventory and expenses are credible. Physical inspection estimates repair and deferred-capital burden. Employee review asks which key people are likely to remain. Supplier review asks which relationships survive transfer. Market review asks whether current earnings are likely to persist. The player can choose the questions worth buying time and information to answer.

4.4B Closing with incomplete information is allowed

The player may deliberately close before every uncertainty is resolved when speed, rival pressure or seller terms justify the gamble. Skipping an inspection or shortening books review should not create an abstract Risk +20% penalty. It means the buyer has chosen to own the same real asset while knowing less about it. Post-close consequences come from whatever condition, obligation, employee decision or market fact actually existed.

4.5 Workflow presentation lock

Tabs remain browsing, summary, and management hubs. Small actions stay inside tabs. Serious multi-step processes break into dedicated workflow screens. Land acquisition and business acquisition should share one workflow shell family. Loan application should use a separate finance-specific shell family. These workflow screens should feel like formal frontier business processes rather than enlarged inline button chains. Important decisions should not only be clear; they should carry the right ownership mood. Loan review, acquisition, emergency restock, closure, turnaround, and expansion should not all feel emotionally flat. A decision's tone is part of its design job, not just its function.

5.1 Projects as build mode

Because the player does not place prefabs by hand, projects are the equivalent of build mode. Projects should be concrete capital decisions with time, labor, material cost, and downtime implications. Core project types remain new build, expansion, repair, refurbishment, and retrofit or conversion. A project should feel like a real commitment with sequencing, cost buckets, and consequences, not a floating instant upgrade card.

5.2 Cost buckets and condition

Core cost buckets remain lumber, nails and hardware, labor, specialty fixtures and tools by business type, site preparation when required, and downtime cost when an operating business is interrupted. Condition measures wear, neglect, structural reliability, and maintenance

burden. Presentation measures visible care such as cleanliness, frontage quality, display appeal, and public confidence. Condition and presentation should be linked but not merged. Poor condition can also raise heating or fuel burden later for some properties.

CANON LOCK

5.3 Design rule and authored-building authority

Anything sensible into anything sensible. The game should reject conversions that make no practical sense, but otherwise allow broad reuse of frontier properties. Claimed ground area and usable site footprint remain separate from the authored visual scale of a building. Buildings should not visibly stretch or shrink merely to fit a different lot. If a building is a poor fit, solve the problem through site choice, footprint, yard arrangement or placement rather than distorting the structure.

PART IV

Shared Business Simulation One operating architecture for businesses, cash, continuity, turnaround and portfolio control.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

CORE IDENTITY

6.1 Shared shell and operating role coverage

Every business should inherit from one broad shared business model so the economy stays coherent. The shell distinguishes ownership, current managing authority, operating control, qualified-role coverage, task workload, labor terms, reputation and productive capacity rather than collapsing everything into one employee count. Identity includes business name, type, owner, parcel, status and founding story. Operations include workers, manager or foreman, operating hours, shift state, qualified-role coverage, task backlog and current throughput. Economy includes revenue, payroll, upkeep, input cost, output value, cash, debt exposure and policy settings. Asset state covers condition, presentation, display/storage/bulk capacity where relevant, overflow state, expansions, adjacent land, transport capacity and heating or fuel burden. Recurring commitments track buyer and seller obligations, due dates, priority and breach history. Operating roles remain a first-class hidden logic layer, but ordinary businesses should not require one full-time employee per narrow task. Compatible work can be covered by the same person when workload permits; specialization emerges when scale, skill or simultaneous demand makes it necessary.

6.1A Labor compensation and occupational markets

Employment should be evaluated as a compensation package rather than cash wage alone. Historically plausible arrangements can include wages, meals, board, lodging, seasonal terms, work schedule, and job stability. Employer-provided board or bunk space can materially improve recruitment but creates its own food, housing, maintenance, and capacity burden. Exact wages remain occupation-, place-, and season-dependent rather than one universal town wage. Labor is not one interchangeable pool. NPCs carry skills and acceptable work paths that determine which roles they can cover competently. General labor can move across many low-skill tasks, while teamsters, clerks, smiths, sawyers, miners, foremen, bookkeepers, and other skilled roles create narrower labor markets. Seasonal demand can temporarily pull compatible workers into harvest, construction, freight, mining, or other surges, creating wage pressure in the overlapping occupations rather than an automatic townwide wage multiplier.

6.1B Workload-based staffing, labor-hours, and task compatibility

Long-term staffing should be driven by real work rather than rigid job slots. Businesses generate task demand from customer traffic, production, deliveries, receiving, stocking, bookkeeping, cleaning, maintenance, hauling, opening hours and other business-specific work. Employees contribute available labor-hours and can cover multiple compatible tasks. A quiet General Store may be run by one owner or clerk who serves customers, takes payment, restocks and handles ordinary upkeep; a booming store can require multiple people because several tasks must happen at once. A logger, mill hand, teamster, miner and clerk should not all be treated as interchangeable eight-hour work units. Job type, skill, physical conditions, travel, season, daylight, machinery and workload determine practical labor capacity. MVP hiring may remain simpler, but the design should avoid hard-coding permanent employee counts that prevent later workload-based staffing.

6.1C Business Decision Authority and administrative work

Every meaningful owner-operated business should expose administrative decisions capable of changing performance even when two businesses own similar physical assets. The current managing authority is whoever actually administers the business: the player when personally managing a player-owned business, an NPC owner for an NPC-run business, or a hired manager when the owner delegates. Ordinary employees execute assigned work - selling, stocking, cleaning, hauling, cooking, sawing, shoeing, operating machinery and similar tasks - but do not silently decide merchandise allocation, supplier strategy, staffing policy, pricing posture, credit posture or capital use. Owners and managers make those administrative decisions. NPC authorities use the same decision space as the player and can overstock slow goods, understock fast sellers, become too generous with credit, refuse profitable credit, hire badly, hold too little cash or fail to react to a changing market. Player advantage should come from judgment, attention and portfolio coordination, not exclusive mechanics.

6.1D Operating schedules and temporal freedom

Operating hours are a policy choice wherever the business can physically function. The managing authority should be free to set unusual or inefficient schedules, including overnight retail hours, rather than being forced into designer-approved opening times. The simulation then supplies consequences through customer schedules, lighting/fuel needs, employee availability, wages, fatigue or shift structure where implemented, safety and actual demand. Production and outdoor businesses remain constrained by their real processes: a forest crew

cannot simply ignore visibility and weather, while an underground mine or adequately lit fixed industrial plant can be less dependent on daylight. The continuous 24-hour clock is authoritative; usable work time is job-specific.

6.2 Owner cash versus business cash

Owner cash and business operating cash remain separate authorities. Businesses should hold their own cash positions. The player can top up or withdraw from businesses, and automation rules can move cash at thresholds when a business is too full or too starved. A business should survive first and pay the owner later. This distinction becomes more important as the portfolio grows because it prevents every business from behaving like a personal pocket and allows the player to read which enterprises are truly healthy.

6.3 Net worth and portfolio maturity

Net worth remains the master success metric and should be treated as a real portfolio calculation rather than flavor text. Liquid Cash is immediately spendable owner cash. Net Worth is the grounded total economic position, including owner cash, business cash, owned asset value where supported, liabilities where supported, and later investment or debt exposure. This matters because the late game is not just richer versions of the early game. The player increasingly behaves like a portfolio owner directing capital across multiple operating entities, not a singlestore clerk who happens to own extra deeds.

6.4 Delegation and executive maturity

The business shell should support the project's long-arc delegation fantasy. Early in a business, the owner normally acts as the administrative authority. As the portfolio grows, a hired manager can take over routine administrative decisions such as ordinary hiring and replacement, merchandise-space allocation, reorder and supplier policy, routine pricing, customer-credit decisions within policy, scheduling and other day-to-day choices. The owner should be able to define authority bands and approval limits rather than surrendering the business completely. Capital projects, major debt, sale, acquisition, exceptional contracts or decisions outside delegated limits can still escalate. Managers themselves are real employees who can quit, be poached, relocate, become ill, retire or die; a management vacancy therefore creates a real administrative problem. Late ownership should mean bigger decisions at a higher level of abstraction, not passive autoplay or permanent micromanagement.

6.4A Organizational coverage, not Organizational Capacity points

Do not expose one universal Organization or Capacity score. A growing portfolio can fail organizationally in different ways: execution coverage, decision coverage, information coverage, administrative coverage and specialist coverage. A business may have enough hands to perform the physical work but no manager authorized to resolve an unusual supplier decision. A remote mine may operate well locally while owner reports are stale. A profitable group may have strong managers but books that no longer reconcile. These conditions should create concrete symptoms - waiting approvals, stale records, unresolved exceptions, missed maintenance, weak hiring, repeated travel needs - and causal responses such as hiring, promotion, delegation, better records, communication, central staff, sale or slower expansion.

6.4B Owner attention without action points

The owner should not receive an arbitrary daily action allowance. Attention matters because the player character and their organization cannot be physically and informationally everywhere. If the owner leaves the store to inspect property, ordinary store work needs coverage. If a manager has authority, routine decisions can continue. If a remote exception can be resolved through correspondence, telegraph or telephone, physical travel may be unnecessary. The useful distinction is physical attention, decision attention and information attention. Delegation changes the player's relationship with time by making longer periods safe to advance when competent people and reporting systems can handle routine work.

11.1 Diagnosis and transfer rule

Buying a weak business should be fast access to an opportunity and its problems, not a magic profit button. Diagnosis buckets remain condition, pricing, supply, labor or management, demand fit, and reputation. Old Inherited relationship memory and commercial history remain a drag. Owner Reputation modifies recovery speed and seriousness but does

not erase structural problems. New management should generate only a small curiosity bump, not a miracle cure.

11.2 Integration stance

Possible integration stances remain stabilize first, re-stock aggressively, restore trust before growth, replace management quickly, refurbish first, keep the old name or rebrand, keep staff or clean house, and honor old credit later or tighten policy immediately. Integration stance should influence early cash burn, relationship-repair speed, employee retention, and how visible the ownership change feels to the town.

11.3 Continuity logic

Business continuity should resolve through three checks: legal control, financial continuity, and operating role coverage. The outcome is softer than a hard game-state ladder. A business may continue normally, continue in reduced form, or become dormant depending on business type and staffing structure. Retail and manager-led operations can often limp along under clerks, spouses, adult children, foremen, or acting managers. Skill-critical trades without qualified operators cannot. This same logic should govern NPC businesses and player businesses alike so the town feels internally consistent. A purchased, inherited, or defeated business keeps its baggage. Weak supplier ties, backlog, staffing problems, poor stock discipline, worn condition, bad books, and damaged trust should survive transfer until actively repaired. A change in owner can create curiosity and maybe some renewed trust, but it should not behave like a magical cleanse.

11.4 Commercial succession and authority redistribution

Death, retirement, incapacity, relocation, partner exit or career change should be treated first as a redistribution-of-authority problem, not as automatic business deletion or an automatic sale listing. Resolve three distinct questions: who legally owns or controls the relevant interest, who can actually operate or manage the enterprise and whether the enterprise can financially survive the transition. Possible outcomes include intact family continuation, spouse or heir ownership with hired management, existing-manager continuation, partner settlement, manager/employee buyout, outside acquisition, asset breakup, liquidation or dormancy. Exact probate and partnership procedure remains a legal research hold where not already validated. The core design lock is that legal identity, commercial identity and operating capability are related but not identical.

11.4A Business identity across ownership changes

An acquired business should not become a historyless 'Player Store #4.' Preserve enough economically meaningful identity to recognize the firm across ownership changes: name, founder or founding story, founding date, premises, ownership history, major expansions/conversions, important long-lived relationships and serious failures where retained by the Campaign Record. The new owner can keep the name, modify it or rebrand, but acquisition does not erase the business's past.

6.5 Asset Condition, Maintenance & Reliability

Asset condition is a shared business system for buildings, wagons/carts, harness and transport equipment, farm machinery, sawmills, mine plant, utility equipment, railway assets and other maintainable property. Condition is a physical economic state, not merely a universal durability bar. Assets deteriorate according to their construction, use, load, environment, age, operating practice, storage/exposure and unresolved defects. Maintenance consumes appropriate labor, parts, tools/facilities, time and money. The owner or delegated manager chooses how aggressively to maintain; the simulation supplies the consequences.

6.5A Condition domains and readable state

Condition should stop at economically meaningful component groups rather than every individual screw. A wagon might distinguish wheels/running gear, axle/bearing condition, frame/body and harness. A steam sawmill can distinguish boiler/pressure system, engine/mechanical state, drive system, saw/cutting system and building/support condition. A ranch windmill or water pump has different domains again. Player-facing presentation should normally use human-readable states such as Excellent, Good, Worn, Poor, Very Poor, Failed or specific known defects rather than pretending a nineteenth-century owner knows a component is exactly 67.4% healthy. Different asset classes must interpret condition differently; a worn roof and a worn boiler do not share one universal failure rule.

6.5B Wear must come from work and exposure

- Use wear derives from operating time, workload, load, route/terrain, heat, vibration, lubrication, material characteristics and the asset- specific stresses actually applied.
- Environmental deterioration derives from storage, rain/snow, moisture, corrosion, sunlight/weathering and other supported exposures. Sheltered equipment lasts better because it is sheltered, not because a shed grants an abstract

lifespan bonus.

- Idle equipment can deteriorate more slowly through use while still suffering exposure, corrosion, decay or preservation problems where appropriate.
- Boom conditions can create maintenance fragility because assets are run harder and maintenance windows disappear; recession can create time for maintenance while simultaneously reducing the cash available to pay for it.

6.5C Service, inspection, repair, overhaul and replacement

Maintenance work is divided conceptually into routine service, inspection, minor repair, major repair, overhaul/rebuild and replacement. Routine lubrication, cleaning, tightening and ordinary care should normally happen under standing policy without owner clicks. Inspection converts uncertainty into better information. Minor/major repairs address specific defects. Overhaul restores a heavily used asset through substantial work. Replacement is a capital decision and may also change capability if newer equipment differs materially from the old asset.

- Maintenance keeps an asset doing what it already does. Capital improvement changes capability. Repairing a wagon wheel is not an upgrade; purchasing a larger or different freight asset is.
- Repairs require actual compatible skill, material and facilities. A local operator can perform ordinary service; a blacksmith, wheelwright, machinist, engineer or specialist may be required for more demanding work.
- Repair quality can distinguish proper, temporary and improvised work where economically meaningful. A field repair may restore a disabled wagon enough to return to town without making it equivalent to shop work.

6.5D Known defects and inspection confidence

Known defects should matter more than a generic condition penalty. A loose wagon wheel, scored axle, leaking boiler tube, worn drive belt, damaged roof, failed fence section or pump problem creates consequences specific to the affected system. Defects can progress from developing to noticeable, serious, critical or failed according to the asset; sudden accident damage can bypass that progression. The simulation can know actual condition while the business knows only what employees and inspections have revealed. Ordinary workers notice obvious problems. Skilled operators can diagnose more. Specialist inspection increases confidence for expensive or safety-critical assets. Inspection knowledge becomes stale as the asset accumulates new use. This creates uncertainty without hiding facts that have already been properly measured or recorded.

6.5E Maintenance policy and cash-reserve interaction

Routine maintenance should be administered through standing policy in the same spirit as inventory replenishment. The managing authority can set automatic repair/maintenance spending limits, cash-reserve protection, approval thresholds, safety-critical shutdown rules, inspection cadence and spare-parts targets. If a routine job would drive business cash below the protected reserve, the manager can escalate it instead of silently spending the money. The player may deliberately defer maintenance. The game should usually allow this if the asset can still physically operate. Deferred work does not trigger a generic punishment; it allows wear, inefficiency, defects, hazard exposure and future repair cost to accumulate through the actual asset. Some neglect can remain lucky for a long time, which is why short-term operators and distressed businesses may rationally gamble on it.

6.5F Efficiency, reliability and failure

Avoid Maintenance Low = -15% Production. Gradual performance loss is valid only where physical deterioration really causes it: fouled boiler surfaces can increase fuel/steam burden, dull cutting equipment can reduce effective throughput, worn running gear can reduce practical travel reliability. Failure exposure is then calculated from asset/design characteristics, condition, unresolved defects, current load, environmental stress and operating practice. Probability should generally decide when a plausible stressed component fails, not whether a healthy new machine is randomly destroyed for drama. Catastrophic failure should normally require a credible causal chain such as serious defect plus continued operation, dangerous load, inadequate support, improper handling or extreme circumstances. Rare uncontrollable failures can exist but must remain rare enough that maintenance feels like risk management rather than tribute to RNG.

6.5G Spare parts, redundancy and downtime

- Businesses may carry spare parts as real inventory. This ties up cash and storage but can turn a multi-week outside-order delay into an immediate repair.
- Redundant installed capacity is different from parts inventory. A spare wagon, reserve pump, additional boiler or other backup asset may let production continue while the primary asset is down, at the cost of capital, maintenance and idle capacity.

- Maintenance consumes time as well as money. A good manager schedules work during low demand, overnight downtime, seasonal lull or weather shutdown where the real process permits it; poor scheduling can remove critical capacity during a peak period.
- Seasonal maintenance is especially important for farm machinery and other assets whose failure during a narrow work window is much more costly than the same repair in the off-season.

6.5H Buildings, fire, safety and service businesses

Buildings use the same system at a slower cadence: structure, roof/weatherproofing, chimney/heating, loading platforms and specialized systems can generate actual maintenance needs. Maintenance and presentation remain separate. A freshly painted building can have a neglected boiler; a weathered shop can be mechanically excellent. Maintenance feeds the existing causal fire and accident systems. Cracked stovepipes, damaged chimneys, overheating bearings, poorly maintained boilers, failed mine support, defective hoists or later electrical faults can create the relevant hazards. There is no universal Poor Maintenance fire/injury modifier. The defect produces the exposure.

6.5I Asset history, used equipment, salvage and finance

Maintainable assets should preserve meaningful history: purchase/install date, major repairs, overhauls, accident damage and replacement of major components. History affects diligence, resale, confidence and attachment. Used-equipment markets can therefore offer cheaper assets with uncertain condition; the player can inspect, refurbish, operate, resell or salvage them. Failed equipment can retain parts or scrap value instead of vanishing. Condition and maintenance backlog affect acquisition valuation and collateral value. Two sawmills with identical current annual profit should not be valued identically if one carries a major deferred overhaul. Banks, buyers, auditors and insurers where implemented can care about the real physical state of pledged property.

6.6 Bookkeeping, Records, Reconciliation & Internal Controls

Land & Ledgers should distinguish the business's physical/economic reality from the quality of its records without creating artificial uncertainty around known facts. The simulation knows actual stock, cash movement, receivables, shipments, wages and physical events. The business ledger records what was properly entered. Missing entries, counting errors, unrecorded damage, bad receiving, cash-handling mistakes or theft can create discrepancies, but a confirmed invoice or correctly recorded twelve-cent purchase remains exactly twelve cents.

6.6A Bookkeeping workload rather than mandatory job slots

Bookkeeping generates administrative workload from transactions, customer accounts, supplier bills, payroll, inventory movements, leases, freight, debt and other business activity. A tiny owner-operated store may generate only modest evening record work that the owner or clerk can handle alongside other duties. A large merchant, mine, bank or portfolio can justify dedicated bookkeepers and later portfolio staff. Do not require one full-time bookkeeper merely because a business exists.

6.6B Recorded state, physical counts and reconciliation

- Known recorded transactions stay exact; the challenge is whether all relevant transactions and losses were recorded correctly.
- Periodic cash reconciliation compares expected cash against actual counted cash. Stocktaking compares recorded inventory against physical count. Receivables review checks customer balances and aging.
- A discrepancy is a problem to explain, not an automatic theft accusation. Causes can include damage/spoilage not written off, counting error, receiving mistakes, credit-entry mistakes, employee error or deliberate theft/fraud.
- Correcting a discrepancy updates the books; the game should not keep a permanent fuzzy value once the business has actually counted and reconciled the asset.

6.6C Record quality and owner information

Record quality can emerge from bookkeeping workload versus available time/skill, procedures, management attention and later office equipment. If administrative work consistently exceeds capacity, books can fall behind or discrepancies can remain unresolved. Better books do not increase real sales by themselves; they allow the managing authority to understand margin, turnover, receivables, stockouts, cash conversion, maintenance burden and other operating truths sooner and more accurately.

6.6D Internal-control decisions

- Who keeps the books: owner, clerk, manager, dedicated bookkeeper or later portfolio staff.

- How often cash is reconciled and inventory is counted, including partial/cycle checks versus full stocktake where appropriate.
- Who is allowed to extend customer credit, receive supplier deliveries, handle the till or approve write-offs.
- How aggressively unusual differences are reviewed and when an outside auditor/inspector is hired.
- Later cash registers, typewriters and office processes improve accountability/administrative throughput through real recordkeeping functions rather than creating demand bonuses.

6.6E Acquisition and delegated-management consequences

Books review becomes meaningful diligence. A seller can report strong profit while carrying uncertain receivables, stale inventory counts or unrecorded maintenance obligations. Poor records do not necessarily mean fraud; they increase uncertainty and may justify deeper diligence. After acquisition, inherited record weakness survives until corrected. Delegated businesses also create principal-agent risk. An owner may trust a manager to run routine operations but still choose review cadence, reconciliation thresholds and audit triggers. Late-game auditors become valuable because they compare reported performance against stock, cash, contracts, staffing and physical condition rather than functioning as magical truth buttons.

PART V

Businesses & Industry Frontier businesses, rural production, lumber, remote industry and mining.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

7.1 General Store

The General Store remains the first fully playable deep slice because it touches the widest set of shared systems at once. It is a broad merchant business rather than a forced specialization class. A sufficiently large, well-capitalized store may carry food staples, clothing, household goods, tools, hardware, ranch or farm supplies, work goods, selected locally produced goods and other period-valid merchandise simultaneously. What it actually carries should emerge from physical capacity, working capital, supplier access, freight economics, local demand and owner judgment. Store identity therefore emerges from assortment and depth rather than from a Ranch Store or Mining Store upgrade button. Special-order capability lets a smaller merchant offer access to goods it does not permanently stock. The current managing authority controls the policy; ordinary clerks execute sales, stocking and receiving inside that policy.

7.1A Display, Storage, Bulk/Yard, and dedicated merchandise allocation

General Store inventory should use normalized physical capacity at a scale large enough to represent a true broad-assortment frontier merchant. The three primary pools are Display Space, indoor Storage Space and Bulk/Yard Space where the property supports it. Every individual physical unit consumes space according to size; the smallest practical packaged goods can use one normalized unit while larger sacks, boots, tools, firearms, stoves, fencing, feed or agricultural equipment consume progressively more. Exact size values remain balancing data, not historical claims. A proper store may be tuned around a large display pool - approximately the order of 250 normalized display units rather than a tiny 20-slot inventory - with proportionally larger storage and bulk capacity, but exact capacities remain open until the merchandise catalogue and building footprints are tested. Capacity is tracked as Total, Allocated/Reserved and Occupied. Merchandise space is dedicated by product line or fixture rather than treated as perfectly fungible every moment. If ten firearms each require ten display units, a 100-unit firearm allocation remains 100 units even after one firearm sells; the newly empty ten units are reserved for the firearm line until the managing authority deliberately reduces or repurposes the allocation. Employees can automatically refill an allocated display from backstock, but they do not replace an empty firearm position with unrelated one-unit goods. The same rule applies to storage and bulk areas. Reallocation is an administrative merchandising decision and may consume ordinary worker time for rearrangement without requiring a floor-layout minigame.

7.1B Assortment breadth, sourcing difficulty, landed cost, and pricing

The store should be allowed to carry as broad an assortment as its property, capital and supply network realistically permit. Off-map sourcing is not a magical catalogue. Each product must have a viable supplier or market source, and supplier choice should expose price, lead time, reliability, volume, quality, route and freight burden. Common regional goods may be easy to replenish while specialized manufactured goods or bulky machinery can be expensive, slow or practical only by special order until transport improves. Local goods can be purchased from nearby households or producers at negotiated/current local value and can have very different economics from standardized manufacturer or wholesaler goods. For MVP retail pricing, the clean rule is acquisition/landed cost plus the managing authority's chosen markup. If an item costs the business $1.00 and the authority applies a 50% markup on cost, the retail price is $1.50. Landed cost should include the supplier purchase price plus freight, drayage/handling and other supported acquisition costs rather than pretending invoice price is the whole cost. Gross merchandise margin and final operating profit remain separate; wages, rent/property burden, heat/light, losses, bad debt and other expenses come afterward. Historical evidence supports wide variation by product and location, so exact category defaults and blended period margins remain a research/calibration hold rather than a universal canon percentage.

7.1C Replenishment policy, cash reserves, and seasonal profiles

Routine replenishment should normally run automatically under standing policy rather than require a purchase click for every sale. The managing authority can set target stock, reorder point or equivalent days-of-supply logic, preferred and backup suppliers, order quantity, emergency/rush tolerance and manual overrides. Each business can protect a cash-reserve threshold. While business cash remains above the reserve, approved routine reorders may execute automatically; if a purchase would push cash below the reserve or the business is already below it, the system can require manual approval and notify the current managing authority. This creates real judgment: a rigid manager can protect cash while allowing flour to stock out, while an aggressive manager can preserve availability and leave the

business unable to meet payroll. Inventory policy should support seasonal profiles. A simple Same All Year option keeps one policy; turning it off allows spring, summer, fall and winter overrides for stock targets, reorder behavior, allocation and whether a line is stocked at all. Firewood or heavy blankets may become important in winter and unnecessary in summer; seed or other seasonal goods can move in the opposite direction. Seasonal reallocation remains a deliberate administrative decision rather than an automatic demand bonus.

7.1D Store administration, operating hours, and manager delegation

The General Store is a principal test case for Business Decision Authority. The managing authority can choose assortment, display/storage allocation, markup and pricing posture, suppliers, reorder rules, cash reserve, customer credit, special-order policy, delivery policy, operating hours and ordinary staffing. NPC owners use these same controls and can make poor decisions. Later managers can receive broad routine authority within owner-set limits, including hiring replacement clerks, changing allocation, responding to fast- or slow-selling merchandise, adjusting reorders and handling ordinary credit. A stock clerk can refill the rifle rack; only the owner or delegated manager decides that the rifle rack should shrink so workwear can expand.

7.2 Blacksmith

The Blacksmith covers repair triage, shoes, plow points, wagon parts, hinges, chains, and practical fix work. It has emergency repairs, interrupt-driven workflow, medium-high injury risk, and strong spring and fall pressure. Continuity remains highly skill-dependent. Without a real journeyman or equivalent smith, normal operation should collapse quickly even if the parcel and tools remain.

7.3 Butcher

The Butcher handles fresh and preserved meat conversion with spoilage risk and sourcing pressure. It lives on perishables, summer spoilage, winter storage advantage, knife injury, and livestock sourcing. It can weaken into a narrower provision-counter form when processing skill or supply continuity breaks rather than always failing in one step.

7.3A Livestock, carcass and dressed-meat sourcing

A butcher should be able to operate through more than one sourcing posture. Where local conditions justify it, the business can buy live animals and carry the slaughter/processing burden itself, buy carcasses or quarters from another slaughtering source, or supplement local supply with preserved or dressed meat brought through wider trade. Those choices move costs and risks rather than erasing them. Live-animal purchasing requires holding space, handling labor, slaughter capacity and waste management but gives more control over timing and byproducts. Dressed-meat purchasing reduces those burdens while increasing dependence on supplier reliability, freight and preservation. A frontier butcher tied to an owned ranch can have unusually strong supply security, but the ranch's herd, winter losses and sale timing remain real constraints.

7.3B Whole-carcass economics and product balance

The butcher cannot turn one animal into only the most profitable cuts. A carcass produces a mixed set of saleable meat, fat, bones, hide and other material, with exact detail abstracted to economically meaningful product groups. Demand for those groups need not match the physical yield. A strong household market may absorb common fresh cuts while a boarding house, mine camp or restaurant can absorb stew meat and larger repeat orders. Slow-moving product creates markdown, preservation, wholesale, byproduct or spoilage pressure. The core owner decision is therefore not just 'how many cattle to buy' but how much carcass throughput the local customer mix can actually absorb before value is lost.

7.3C Fresh, preserved and seasonal product strategy

Fresh meat is the highest-turnover local line but is exposed to temperature, delivery timing and uncertain daily demand. Salted, cured, smoked or otherwise preserved products can extend sale life where the shop has the required skill, time, equipment and inputs. Ice and cold-storage access can materially widen the useful selling window without creating modern refrigeration. Winter can reduce spoilage pressure while increasing other supply and travel risks; summer can make ice, rapid turnover and disciplined buying much more important. The player can therefore run a high-turnover fresh shop, a more preservation-heavy provision business or a mixed operation according to local demand and capability.

7.3D Capacity, labor and operating flow

Butcher capacity should come from distinct but connected work: animal receiving or carcass receiving, slaughter where performed, cutting, preservation, cleaning, display/sales and delivery. A skilled butcher is not interchangeable with

every helper. Assistants can move stock, clean, wrap, deliver and support preparation, but loss of the skilled processing role can sharply narrow what the business can do. The business can become backlogged after a large livestock purchase, a Saturday demand spike, a mine or hotel contract, or worker absence. Excess throughput can be as bad as too little because unfinished or unsold perishable product deteriorates.

7.3E Wholesale accounts, delivery and cross-business demand

Household counter sales should not be the only market. Boarding houses, hotels, restaurants, mine camps, logging camps, stores and other repeat buyers can create account or supply-agreement demand where the local economy supports it. Large recurring buyers improve throughput visibility but can concentrate customer risk and reserve stock that might otherwise serve walk-in households. Delivery can be offered when labor and transport exist, creating a service advantage at a real operating cost. The player can choose whether scarce supply protects contract customers, household trade, internal portfolio buyers or the highest-margin outlet.

7.3F Byproducts, nuisance, sanitation and continuity

Hides and tallow remain the principal byproduct categories worth carrying at normal simulation depth. Bones, blood and other residues should enter only where a real buyer or disposal problem makes them economically meaningful. Slaughtering and meat handling also create odor, waste, water, cleaning and pest pressure. If slaughter occurs on-site, those burdens affect site suitability and neighbors; buying dressed carcasses shifts much of that burden upstream. Good cleaning and waste removal protect continuity through actual labor and service, not a hygiene buff. Fire, equipment failure, ice shortage, livestock shortage, skilled-worker loss or a bad spoilage run can each weaken the business through a different causal path. Acquisition diligence should therefore ask not only whether the shop has customers, but what part of its supply, skill, cold-storage and account base the buyer is actually inheriting.

7.4 Crop Farm and Ranch

Crop Farm and Ranch remain distinct because western South Dakota is not one uniform agricultural surface. Ranching and grazing form the broader West River baseline, while crop farming concentrates in suitable valleys, watered ground and irrigable sites. Hay and forage are strategic commodities because winter survival, work animals and livestock carrying capacity depend on them. Spring 1880 is already a horse-machine agricultural economy. Manufactured prairie plows, mowers, reapers, threshing machinery, emerging grain binders, barbed wire and mechanical pumping windmills are period-valid. The player does not research these inventions. Adoption depends on acreage, capital, draft animals, labor, repair access, freight and whether a machine can be used enough to justify ownership. Agricultural machinery changes seasonal work capacity rather than biological yield. Soil, water, seed, weather and crop condition determine potential production; plowing, planting, haying, harvest and threshing capacities determine how much can be completed before seasonal windows close. Expensive seasonal equipment can be owned, shared, borrowed or hired through custom work, creating a real ownership-versus-utilization decision. Ranch technology is primarily about control. Fencing is spatial infrastructure that can protect crops, divide pasture, control stock and reduce some herding burden but requires wire, posts, labor, gates and repairs. Windmills and wells expand watered range rather than applying a generic livestock bonus. Hay storage, sheltering terrain, herd condition and grazing pressure determine winter resilience. The winter of 1886-87 is an important historical stress test, not a scripted loss percentage. Overstocking, degraded range, poor autumn condition, insufficient hay and severe weather interact to determine mortality. Better-prepared ranches can fare substantially better than

highly exposed open-range operations. Rail changes agriculture through both market access and capital-goods access: livestock can reach distant packers, grain can aggregate at elevators where production supports them, and machinery, wire and replacement parts become cheaper to obtain. Wagons remain the local feeder system. Horse power remains the normal motive force through 1905; steam is specialized and gasoline tractors remain rare cutting-edge technology outside the core progression.

7.5 Business agency standard across industries

The General Store depth model is a template for agency, not a requirement that every business use shelves. Each serious business should expose owner/manager decisions rooted in its real operation so that two otherwise similar businesses can perform differently because they are administered differently. A blacksmith can choose service mix, backlog priority, contract customers, materials reserve and apprenticeship; a boarding business can choose occupancy density, room/board package, meal quality, rates, customer mix and food sourcing; a butcher can choose livestock/carcass sourcing, product mix, ice/cold-storage use and household versus business customers; a freight company can choose routes, cargo, rates, scheduled versus charter work, team reserves and contracts; a sawmill can choose timber source, mill utilization, product mix, storage/drying and customer commitments; a ranch can choose stocking, herd composition, water, fencing, hay reserve, winter feeding and sale timing. NPC authorities receive the same choices. The design target is not more clicks but more consequential policy.

7.5A Playable Business Depth Gate

CANON LOCK A business name is not enough to earn playable scope. Before a candidate business graduates from the future pool into an active

player-operable system, the design should be able to answer most of the following in a grounded way: who buys from it and why; what physical inputs or information it needs; what labor and specialist capability constrain it; what property, capacity or location matters; what recurring owner policies change its behavior; how cash, credit and working capital constrain it; what assets can wear or fail; what bad operation looks like before total closure; how rivals or substitutes compete with it; how its economics change across season, settlement growth or modernization; and which existing systems it feeds or depends upon. Not every business needs equal page count or equal UI, but a player business should have more than a unique revenue formula. If the answers are weak, keep the activity background-only, merge it into a broader service family or leave it parked until research justifies depth.

8.1 Boarding and settlement infrastructure

Boarding House or Hotel is a major settlement and labor-market bridge and remains an active VP expansion target. Boarding lets the town absorb single workers, temporary labor, and recent migrants before or instead of household ownership, while rental houses and rooms support longer-term family settlement. Housing capacity therefore affects the effective labor supply: a worker available somewhere in the region cannot become stable local labor if there is no practical place to live. Employer bunks, meals, and lodging can also form part of compensation. Boarding is consequently both a daily operating business and productive settlement infrastructure. The 1880 census itself distinguished boarders, hotel inmates and hotel cooks/waiters inside real shared-roof/shared-table arrangements, while Black Hills evidence includes boarding operations serving mine labor and proprietors who moved between cook, boarding-house and hotel roles.

8.1A Accommodation products and physical capacity

A boarding operation sells use of real accommodation rather than a generic occupancy percentage. Depending on the building, it can offer beds in shared rooms, private rooms, room-and-board packages, longer-stay boarding, short transient lodging and, where physically suitable, family rooms. The proprietor's own household can occupy part of the same building without becoming identical to the boarders. Capacity is bounded by usable sleeping space, beds, sanitation, water, heating, fire exposure and meal service. Crowding above a comfortable level may be physically possible, especially during a boom, but it increases workload, complaint, fire, illness and turnover pressure instead of being blocked by an arbitrary cap.

8.1B Board, meals and the second capacity bottleneck

'Board' should matter. A house that includes meals operates both accommodation and food service, creating demand for meat, flour, vegetables, coffee, fuel, water, kitchen labor and storage. Bed capacity and meal capacity are separate. A proprietor may have empty beds but no practical kitchen capacity to add full-board customers, or a strong kitchen may feed nonresident diners when local custom supports it. Menu detail should remain abstracted to meal quality, food cost, supply reliability and labor burden rather than becoming a cooking game. Supply agreements with butchers, stores, farms or owned businesses can stabilize service through real procurement.

8.1C Customer mix, stay length and demand pulses

The customer mix should follow the settlement's reason for existing. A freight stop can receive teamsters and short-stay travelers; a mine or railway construction surge can create concentrated single-worker demand; a growing town can support clerks, tradesmen and new arrivals who board for weeks or months before renting; a hotel in a mature node can draw commercial travelers and higher-paying transient guests. Different customers value rate, meal quality, privacy, proximity to work, stable terms and service differently. Major construction, mine expansion, railroad surveys or seasonal work can produce arrival pulses that temporarily make every bed valuable, followed by sharp vacancy if the project ends.

8.1D Rate, package and occupancy policy

The owner can set room-only versus room-and-board pricing where the establishment supports both, weekly or longer-stay terms, transient rates, meal quality, target customer mix, credit/account tolerance and reservation of rooms for expected contract or commercial customers. A low-price crowded house can earn strong short-run revenue while creating more wear and service pressure. A higher-price quieter house can preserve capacity but may sit partly empty in a weak market. The game should not assign one universally optimal occupancy rate; the right posture depends on local demand, labor, food cost, property quality and the owner's strategy.

8.1E Labor, household work and proprietor dependence

Boarding houses are labor businesses. Cooking, cleaning, laundry, bed turnover, heating, water hauling, receiving supplies, serving meals, bookkeeping and guest handling all consume work. Small houses can rely heavily on proprietor and family labor; larger hotels may need cooks, waiters, cleaners, clerks or other helpers. This creates a meaningful scale transition: adding rooms without adding service capacity can reduce quality and retention. Loss of a strong cook, housekeeper or proprietor can damage the establishment even when the building is undamaged. Women should be fully eligible as proprietors and managers; Black Hills evidence for Lucretia Marshbanks provides a direct regional example of a cook, mine boarding-house manager and hotel owner.

8.1F Boarding as labor-market and project infrastructure

A boarding house is economically productive because it converts regional or arriving workers into locally usable labor. Employers, mines or major projects can reserve or subsidize beds, operate their own lodging or rely on independent proprietors. A player who expands boarding ahead of a mine, rail or construction project can profit from the resulting labor surge if the project materializes; if it fails, the same rooms can become debt-heavy vacancy. The reverse also matters: a profitable mine may be unable to hire to plan because every acceptable bed is occupied. Housing pressure should therefore be able to make boarding expansion, employer lodging or permanent rental construction competing solutions to the same bottleneck.

8.1G Fire, sanitation, reputation and continuity

Dense sleeping, stoves, kitchens, lamps, stairs and transient occupancy can make boarding establishments important fire and safety exposures. Water, privies, waste removal, cleaning and crowding affect operating quality through real property and labor conditions. Reputation should emerge from actual experiences such as clean rooms, reliable meals, honored terms, theft/security problems, repeated overcrowding or bad service rather than from a flat hotel-quality upgrade. A major fire, illness wave, food-supply failure or proprietor death can displace residents and simultaneously damage local labor supply, making boarding continuity a settlement issue rather than only a business-income issue.

8.2 Lumber chain

Small Sawmill and Lumber Yard are active near-term chain priorities because they connect construction supply, hauling, fuel byproducts, and remote production. Small mill is the entry production site with modest intake, one main processing line, limited storage, and stronger vulnerability to labor or hauling bottlenecks. Large mill is the upgraded form with higher intake, more workers, more yard pressure, higher maintenance and injury exposure, and a larger role in construction supply. Lumber Yard is the town-facing storage and retail arm, either independent or paired with an owned mill.

8.3 Future pool and institutional anchors

The broader future pool remains Doctor's Office or Town Doctor, Livery or Freight Office, Grain Mill, Saloon, Tailor or Dry Goods, Bakery, Bank, Barber, Wheelwright or Repair Shop, Telegraph Office or communication-linked service, Telephone Company or private/exchange communications service, Water Company where private utility conditions justify it, Electric Light or Light & Power Company, and later luxury or investment-linked businesses that make sense once town prosperity becomes visible. Historical availability does not equal implementation priority: telephone and water enterprise can be period-valid early while still remaining post-VP systems. Towns can also gain seriousness through institutional anchors such as schools, courthouses, depot offices, hospitals, military or federal presence. These are not VP-core requirements, but they are valid later-phase growth multipliers. Doctor service remains active early as a grounded town service. The player does not begin with a doctor business. The doctor's office can be acquired later as a service business. It remains a small service business, not a hospital simulation, and matters because treatment access reduces labor loss, treatment delay, and medical burden across the town and later across the portfolio.

8.3B Future-pool width discipline

The future pool is a research and opportunity map, not a promise that every named enterprise receives its own shallow management screen. A bakery, saloon, barber, tailor, grain mill, telephone company, water company or prestige business should become directly playable only when it passes the Playable Business Depth Gate in 7.5A or when its ownership creates a genuinely new strategic relationship. Otherwise it can remain a simulated NPC service, a component of another business family or a parked candidate. Revision XIV explicitly prefers six deep businesses over twelve businesses that differ mainly in icon, recipe and revenue rate.

8.3A Secondary settlement services and remote civic emergence

The broader service pool should no longer be read only through the lens of the starting town. Remote nodes that cross enough population, boarding, freight, and permanence thresholds should be able to support their own local store,

butcher, blacksmith support, doctor outpost, schoolhouse, freight office, or other practical services. The exact combination should depend on why the node exists. A mine settlement, a farm-service hamlet, and a freight-side crossroads should not all evolve the same way even if they briefly sit in similar population bands. Institutional anchors matter here as well. Schooling, formal platting, civic office, a church meeting hall, a freight yard office, or a small town hall should emerge from local pressure and usefulness rather than as arbitrary unlocks. These responses should be softened by randomized trigger windows so that settlement life feels discovered and negotiated rather than gamified into exact breakpoints.

8.4 Byproduct markets

Keep byproducts practical rather than overbuilt. The important VP forms remain butcher hides and tallow plus sawmill slabs and offcuts as fuel wood or low-grade sale output. Later deepening of bark, sawdust, blood, bones, or other residues should happen only where it materially improves the economy rather than bloating the tree.

8.5 Logging, timber, sawmills, and lumber-chain additions

Logging and sawmills form a practical supply-chain layer rather than a decorative business type. Construction, repairs, mines, rail, storage, housing and industrial expansion depend on lumber, beams, boards and related byproducts. Timber extraction uses stand/worksite logic where supported, with terrain, vegetation, road access, hauling distance, season, daylight, weather and local timber quality affecting viability. Logging labor is outdoor work whose practical day can be limited by usable light, travel, physical conditions and available timber rather than a universal shift. Sawmills remain shared-shell businesses with log supply, power source, saw capacity, lumber output, yard or pond storage, green-lumber handling/drying, machinery condition, labor, maintenance and hauling strain. Mining progression increases the strategic value of timber, and large expansion can create real logging, hauling and sawmill booms rather than a flat lumber-demand modifier.

8.5A Sawmill power, water sites, log handling, and bottlenecks

Sawmill configuration should follow period technology and site economics rather than a linear Water -> Steam -> Electric upgrade ladder. Water-powered mills are historically valid where a site provides adequate flow and head and may require a dam or impoundment, mill pond, headrace/millrace, wheel or turbine arrangement and tailrace. Mill or log ponds can also hold raw logs before sawing. Steam-powered mills are already plausible at the 1880 opening in the Black Hills and can be portable or more fixed; they trade water-power geography for boiler, engine, fuel, water, maintenance, skilled operation and machinery-logistics requirements. Black Hills-specific prevalence and exact mill forms remain subject to targeted research before generation weights are locked. The causal chain should distinguish standing timber -> felling -> extraction/skidding/hauling -> log landing or pond -> mill intake -> power availability -> sawing -> green lumber handling -> stacking/drying -> finished inventory -> outbound freight. Felling, extraction, mill throughput, power, handling, drying/storage and outbound transport can each become the active bottleneck. Improving the saw does not create logs, and hiring more loggers does not help if the yard or mill cannot absorb their output. This is contained systemic depth: it reuses labor, transport, storage, property, time and machinery rather than creating a detached mill minigame.

8.6 Commercial, cold-storage, and office technology progression

Commercial modernization is gradual and should reinforce ownership problems rather than become a gadget tree. The 1880 General Store remains strongly ledger-, barrel-, sack- and crate-oriented while manufactured packaged goods already exist. Branded boxes, tins and standardized packaging become increasingly visible during the 1890s and early 1900s as rail distribution and national brands deepen. Natural ice harvesting, icehouses and iceboxes remain the normal local cold-storage system. Refrigerated railcars exist in the wider world before local Black Hills rail and later expand the practical market reach of perishables where the railway and icing network support them. Mechanical refrigeration remains capital-heavy and exceptional before 1905. Mail-order competition becomes a serious late-campaign pressure, especially in the 1890s. Catalog merchants compete strongly for standardized manufactured goods when mail, express and rail can deliver them, but the local store retains immediate availability, small quantities, emergency supply, local exchange and household credit. Cash registers exist at the opening edge of commercial technology but are not normal for a tiny frontier store. They become increasingly useful in larger late-1880s and 1890s businesses because they improve employee cash accountability and reconciliation rather than demand. Typewriters similarly exist before the campaign but become increasingly common in banks, railway offices, mines, utilities and major commercial offices as administrative scale grows.

21.1 Mine placement, districts, and opening chronology

Mines are remote industrial businesses tied to mineral districts, claims, processing and freight rather than magic resource nodes. The spring 1880 region is already industrializing: placer mining remains possible and small prospects exist, but substantial hard-rock mining and large stamp mills are already operating in the northern Black Hills. Mining is

therefore historically present immediately even though ownership of a serious mine remains a capital-heavy later player path. District generation should care about deposit family, geological confidence, access difficulty, extraction style, metallurgical character, treatment requirements, freight burden and boom potential. Claim logic comes first; settlement consequences follow. Mine camps can grow into nodes when labor, boarding, freight and services concentrate, and can later decline or convert if the deposit or economics weaken.

21.2 Mine identities and metallurgy

Gold remains the dominant Black Hills mining identity. Lead-silver and other polymetallic deposits are secondary and localized, with distinct smelter and treatment economics. Tin is speculative and should emphasize the difference between promotional samples, representative bulk grade and actual mill recovery. Coal is strategically important as fuel but often regional/imported or geology-conditional. Iron is niche and does not create a local integrated rail-steel economy. Ore grade alone is insufficient. A meaningful deposit also has metallurgical character and treatment compatibility. Free-milling gold may support stamp crushing and amalgamation, while refractory or sulphide-bearing material can require custom treatment, smelting, chlorination, cyanidation or other appropriate routes. Ore mined is not identical to recovered metal. The system should distinguish geological mineralization, physically mineable material, treatable material and economic ore. Technology and freight can lower the economic cutoff without changing the geology, allowing old stockpiles, dumps, tailings or marginal sections to acquire value later.

21.3 Functional bottlenecks and development

Mining technology removes actual constraints rather than granting mine levels. Important capacities include geological confidence, development rate, drilling/blasting, ground support, internal haulage, hoisting, water control, crushing, treatment, power, freight and technical labor. Improving one can expose another: pneumatic drilling can create more broken rock than the hoist or mill can handle; a larger

hoist can expose treatment capacity; a larger mill can expose fuel, water, timber or freight limitations. Stamp milling, amalgamation for suitable ore, steam machinery, blasting, timbering, mine cars and serious powered hoisting are already available at major operations near the opening. Heavy machinery can be purchased before local rail by sending it to a distant railhead and hauling it overland at high cost and delay. Rail improves delivered cost, replacement parts, fuel and bulk shipping rather than unlocking industrial machinery. Mine water is site- specific. Hydrology, season and workings determine inflow; pumping becomes necessary where drainage or existing capacity is insufficient. Timber demand likewise derives from geology, opening geometry and support needs rather than a universal per-ton rule.

21.4 Technology chronology, 1880-1905

Hand hammer-and-drill work remains credible at important operations well into the early 1890s. Compressed-air drilling and compressor plants become increasingly significant during the 1890s and mature toward 1900-05, raising development capacity while creating new power, maintenance and skill dependencies. Electricity enters mining first on a limited scale, especially lighting. Homestake's 1888 dynamo is a useful regional anchor. Steam, water and mechanical drives remain important for years; broader electric motive power becomes increasingly plausible near the turn of the century but does not replace steam universally by 1905. Gold treatment overlaps rather than advancing in one switch. Stamp/amalgamation is opening-era technology for suitable ore. Black Hills chlorination becomes significant around 1889; specialized matte smelting expands around 1890; an early Deadwood cyanide plant exists in 1892 but is intermittent; cyanidation becomes increasingly successful and widespread in the late 1890s and early 1900s. Availability, ore character, expertise, plant economics and reliability determine adoption. The Homestake interests also demonstrate that a private industrial narrow-gauge railway can become rational around 1881-82 when a company needs to connect mine openings, mills and timber supply before common-carrier rail reaches the region. Internal mine track, industrial railway and regional common-carrier rail remain three separate transport scales.

21.5 Labor, safety, and professionalization

Mines are labor-hungry and increasingly specialized. Early operations can rely on miners, laborers, teamsters, blacksmith/repair support, bosses and assayers; larger operations add hoist operators, millmen, boiler operators, machinists, timber crews and superintendents; later sophisticated plants may require engineers, metallurgists, compressor personnel and electricians. Regional mining education and technical professionalization deepen during the 1880s and 1890s without creating a player research tree. Safety is causal. Ground failure, blasting, hoisting, machinery, fire and water hazards arise from geology, equipment condition, maintenance, explosives handling, support, supervision, fatigue and site conditions rather than one universal annual accident percentage. The game should represent the ownership consequence without becoming a mine-engineering simulator.

21.6 Mining company and portfolio operation

Each mine remains an owned operating business inside the shared business model, with operating information for deposit family, claim/development state, accessible workings, metallurgical character, output, stockpiles, workers, foreman or superintendent, support/timber, power, hauling, road/rail access, treatment route, condition and shutdown/hold/expand/sell decisions. A portfolio mining overview can summarize output, shortages, hauling, claims, debt, investors, technical coverage and risk without merging every site into one hive mind. Starting a mine uses a staged path: district information or rumor -> prospecting/claim control -> assay and representative sampling -> feasibility -> financing/investor structure -> initial works -> production -> expansion. A mine may own its mill, use a custom mill, ship selected ore to a smelter, stockpile difficult material or later build specialized treatment. Vertical integration reduces real procurement and coordination burdens only where the player actually controls those inputs; it also transfers those operating risks into the portfolio.

7.2A Repair, Fabrication & Technical Services - shared economic system

The Blacksmith remains a core opening-era craft business, but Revision XI broadens the design from one shop type into a shared Repair, Fabrication & Technical Services economy. Actual maintainable assets generate work orders; shops satisfy that demand according to their people, tools, facilities, materials and capabilities. The design should not force historically overlapping crafts into artificial hard classes. Blacksmithing, horseshoeing, wagon repair, wheelwright work, harness work and later machine-shop repair can be combined or specialized according to settlement scale and market demand. Direct Black Hills evidence supports this width before the campaign even begins: an 1878 Deadwood business directory lists multiple blacksmith/repair shops, blacksmith-and-wagon-repair businesses, a wagon maker and saddle/harness makers. A tiny 55-95-person opening node may still have only one flexible craftsperson or no resident specialist; a larger mining/commercial node can support several competing specialists.

7.2B Capabilities rather than rigid shop classes

- Smithing capability requires a forge, appropriate tools/material and a qualified smith; it supports iron hardware, practical fabrication, tool/implement repair and other compatible work.
- Horseshoeing/farrier capability requires the relevant skill, tools and animal-handling setup; the work is recurring because the horse- powered economy physically uses shoes and hooves.
- Wagon/wheel capability requires woodworking/wheel skills, workspace and appropriate components; iron-tire work may overlap with the smith.
- Harness/tack capability requires leatherworking skill, workbench/tools, leather and hardware; harness is a real transport asset rather than invisible equipment.
- Machine-work capability becomes increasingly important as mines, mills, utilities and rail create demands beyond the traditional smithy; powered tools, machinists and larger facilities can emerge without a Blacksmith Level upgrade ladder.

7.2C Repair work orders and Physical Truth

Each meaningful repair is represented as a work order with customer, asset/component, reported problem, actual/diagnosed problem, urgency, required capability, materials, location, queue position, estimated labor and agreed price/terms. The job resolves only when the asset or repairperson reaches the correct location, the materials/tools are available and the labor is actually performed.

- A horse being shod physically visits the smith/farrier and is temporarily unavailable for work.
- A wagon with ordinary damage travels to a shop; a wagon immobilized miles away may require temporary field repair or a repairperson/tools to travel to it.
- Large fixed machinery can be repaired on site, or a component can be removed and hauled to a capable shop. The game should not teleport a broken pump or axle to town.

7.2D Diagnosis, queue, urgency and pricing freedom

Customers can report symptoms without knowing the defect. Diagnosis can reveal additional work and convert a rough estimate into a proper quote. Shop capacity is therefore a queue of real jobs competing for skilled labor, workstations, material and time. Routine shoeing, plow work and fabrication can be interrupted by a freight wagon or mine component whose downtime is economically urgent. The managing authority may set any price the market can plausibly accept. If an ordinary job commonly sells for $2, the player may ask $8; a desperate monopoly customer may pay, ordinary customers may wait or travel elsewhere and sustained exploitation can create a profitable opening for a competitor. Price limits should come from contracts, law where relevant and customer alternatives, not designer caps.

7.2E Accounts, recurring contracts and reserved capacity

Repair trades can use the shared commercial-credit system. Farms, ranches, freight companies and mines may hold running accounts where historically appropriate rather than paying cash for every shoeing or small repair. Larger customers can negotiate recurring service: priority repair, reserved capacity, agreed pricing or regular fleet/animal work. Guaranteed demand can stabilize shop income but reserving capacity for one major client can increase waiting time for everyone else.

7.2F Materials, parts, fabrication, refurbishment and salvage

Repair shops hold real materials and parts: forge fuel, metal stock, common hardware, wagon components, shoeing supplies, leather/harness inputs where supported and increasingly manufacturer replacement parts as machinery standardizes. A shop can choose to carry expensive slow-moving parts because emergency local availability is valuable; obsolete parts can become dead inventory.

- Repair restores an existing asset; fabrication creates a needed component; replacement installs a purchased new component; temporary repair restores limited function; refurbishment combines multiple repairs to restore a used asset for continued use or resale.
- A shop may buy broken or worn wagons/equipment, repair them and resell them, or dismantle uneconomic assets for parts/scrap. Used-equipment commerce therefore grows naturally out of the maintenance and repair systems.

7.2G Skill formation, apprenticeship and competitive labor

Craft continuity depends heavily on real people. Apprentices/helpers can initially perform support and simple compatible work, then gain capability through actual experience where historically calibrated. Exact apprenticeship terms, years and titles remain a research hold. A highly skilled employee can demand higher compensation, leave, be poached or eventually open a competing shop if capital/property become available. Buying a craft shop does not automatically buy the craftsperson. Diligence should ask whether the smith, wheelwright or machinist will remain after sale. Customer relationships may partly follow the individual whose skill created them, making key-worker retention strategically important.

7.2H Repair-market evolution, 1880-1905

The opening economy can rely heavily on flexible hand craftspeople. As standardized farm machinery, steam equipment, mines, mills, rail and utilities expand, more work shifts toward diagnosis, manufacturer parts, heavier repair and machine-shop capability. The traditional smith does not disappear by 1905 because horses, wagons and practical fabrication remain essential; the mix of work changes. A large mine or mill can internalize blacksmith/machinist capacity as a maintenance department while smaller operators continue to buy service externally.

7.4A Rural Production Economy - square-off doctrine

Crop Farm and Ranch remain the two core rural production businesses, but Revision XI broadens them into a Rural Production Economy. Agriculture should have comparable width and depth to mining without turning the game into a manual farming simulator. The player decides land use, herd/field strategy, seasonal priorities, capital equipment, hiring, storage, contracts, sales and risk posture; workers and animals physically perform the work. Farm and ranch identity emerges from land, water, biological production, market access and owner decisions rather than from one generic rural-business template.

7.4B Ten rural participation lanes

- Crop farming: produce marketable crops through land suitability, biological growth and successful completion of seasonal work windows.
- Hay/forage production: convert summer land and seasonal labor into feed that supports the horse/livestock economy and winter reserves.
- Cattle ranching: manage herd composition, forage, water, winter preparation, sales and biological growth.
- Sheep ranching: retain only if research and mechanics give it genuinely distinct output, labor, market and management characteristics rather than reskinned cattle.
- Local/market produce: supply nearby households, stores, lodging and service businesses through proximity and perishability.
- Custom agricultural work: own seasonal machinery/capacity and sell work to other farms when the player is not using it.
- Livestock commerce: buy, assemble, hold and move livestock for resale without necessarily owning the breeding operation.

- Agricultural supply: sell seed, wire, machinery, parts, tools and other historically supported inputs through existing merchant businesses or specialized operations where demand supports them.
- Processing/storage: mill, aggregate, store or otherwise handle agricultural output where the local production volume and historical business form justify it.
- Rural property/infrastructure: own productive land, water improvements, pasture, storage or supporting assets and profit from their real usefulness; leasing/use-right variants remain a targeted research hold.

7.4C Field authority and crop operating cycle

Cultivated land should be represented by meaningful field/parcel production state rather than a farm-wide output meter. A field or field group knows area, suitability/soil, moisture/water state, current crop/use, current seasonal operation, work progress, crop condition, access and ownership. The engine can use detailed values internally while the owner receives practical agricultural reports. Crop production separates biological potential from work completion. Soil, water, seed, weather and crop condition determine what the land could produce; plowing/ground preparation, planting, cultivation where supported, harvest, threshing/processing and hauling capacity determine how much of that potential is actually realized before the relevant windows close. A machine does not create yield; it lets the farm complete more work in time.

7.4D Seasonal capacity and priority

Seasonal jobs compete for workers, horses and machines. If 80 acres are ready but the current crew/equipment can harvest only 12 acres per day, rain, illness or a breakdown can make field priority consequential. The managing authority can choose which acreage is worked first according to value, vulnerability, access or owner judgment. Missed work reduces realized production through the crop/field state rather than a Farm Efficiency percentage.

7.4E Machinery ownership, hire and custom work

Expensive seasonal machinery creates a utilization problem. Ownership provides schedule control but ties up capital, storage, maintenance, horses and repair capacity. Hiring custom work avoids fixed ownership cost but exposes the farm to contractor queues and weather-driven

schedule shifts. An owner with excess machine capacity can sell custom service to neighbors, converting an underused capital asset into a seasonal business. The schedule itself becomes valuable. A contractor can book several farms; rain or breakdown at the first job propagates delay to later clients. Urgency rises as seasonal windows close. Exact period Black Hills custom-work practices and pricing remain a research/calibration question, but the ownership-versus-hire and utilization logic is canonical.

7.4F Hay and forage as first-class strategic commodities

Hay/forage is not merely another crop. It links summer land productivity to winter survival of horses and livestock. A ranch can produce its own hay, buy from nearby producers, maintain recurring agreements or enter the spot market during shortage. Self-sufficiency uses more land, machinery and seasonal labor; purchased feed frees land/capital but increases market and supplier exposure. Hay production should include enough physical stages to make timing meaningful - standing forage, cutting, drying/curing, collection/stacking or period-appropriate storage and movement - without requiring manual farming actions. Weather during the process can matter differently from weather during growth. Exact equipment and storage practices remain crop/period research items.

7.4G Herd structure and biological capital

Ranch livestock should not collapse into one Cattle count. Use cohorts or economic groups appropriate to the supported species: breeding animals, young stock, market animals and other meaningful categories. Exact cohort definitions and reproductive timing require historical livestock research, but the principle is that animals can be both productive biological capital and saleable assets. Breeding is therefore a retention/allocation decision rather than a reproduction buff. A cash-starved ranch may sell breeding-quality animals to raise money today and weaken future herd growth. A better-capitalized operation may retain them. Biological growth, deaths, purchases and sales change the herd over time.

7.4H Range, water and carrying capacity

Rural land value and usable livestock capacity depend on forage, water, terrain, access, season, weather and prior grazing pressure rather than acreage alone. A large tract with poor practical water access can be less useful than a smaller watered property. Wells, pumps and windmills expand usable range because they create actual water access; they do not grant a generic livestock bonus. Overstocking should be allowed. Excess grazing pressure reduces actual forage condition, which can weaken animal condition, winter resilience and future pasture availability. A distressed

rancher may knowingly maintain too many animals because an immediate sale is financially painful. The game allows the gamble and follows the land/animal consequences.

7.4I Fencing and spatial ranch control

Fencing remains physical infrastructure with gates, boundaries and maintenance. It can protect cultivated land, partition pasture, control stock and reduce certain labor burdens. The player chooses which boundaries justify capital. Broken or neglected sections create actual control problems rather than a flat ranch penalty. Barbed wire is opening-era available technology, but adoption remains property- and capital-dependent.

7.4J Winter preparation and the 1886-87 stress test

Winter preparation should become one of the most consequential recurring rural decisions. Before winter, the ranch can evaluate herd size, animal condition, stored hay, remaining pasture/forage, water reliability, sheltering terrain and forecast uncertainty. Expertise and records can improve the feed/resilience estimate, but future winter severity remains unknown. The owner can buy feed, sell animals, reduce exposure or gamble. Severe winter consumes reserves and changes animal condition over time; it does not apply a scripted mortality percentage. The 1886-87 winter remains a historical stress test in which poorly prepared, overstocked operations can suffer far more than well-prepared ones.

7.4K Livestock sales, holding and physical market movement

Livestock sale is a capital-allocation decision. Market-ready animals can generate cash; selling breeding stock may damage future production; holding animals keeps potential upside but continues feed/water/land burden. Animals sold remain physical animals requiring appropriate movement to buyers, aggregation points or later rail interfaces. Rail changes feasible market reach and transport economics rather than magically raising livestock value.

7.4L Agricultural storage and sale timing

Harvested crops, feed and other outputs require commodity-appropriate storage rather than entering a global inventory pool. Storage capacity and condition determine whether the producer can hold output, protect it and wait for a later market. A bumper harvest can create a storage/freight problem and weak harvest-time prices; selling immediately may be necessary for debt/cash even if the owner expects better prices later.

7.4M Rural credit and seasonal cash flow

Agriculture should be one of the strongest demonstrations of irregular cash timing. Seed, wire, feed, repairs, wages and machinery needs can occur before crop/livestock proceeds. Merchants and lenders can extend appropriate credit based on the real operation. A failed harvest or weak livestock sale can then propagate into overdue merchant accounts, rent/mortgage pressure, lender distress and reduced purchasing power.

7.4N Family labor, hired surges and workload

Family farms/ranches can combine owner/household labor with hired permanent and seasonal workers. Family members are not free generic labor slots; age, household duties, health, schooling where relevant and other employment still matter. Seasonal work can create temporary demand spikes far above the normal workforce, generating migration, boarding and wage pressure in the overlapping labor market.

7.4O Repair, weather and settlement feedback

Agriculture feeds the repair economy through wagons, harness, plows, mowers, pumps/windmills, fencing and other equipment. Repair pressure should become particularly visible before/during seasonal work windows, when a two-day delay can be far more expensive than the same repair in winter. Weather acts on the physical process it affects. Drought changes soil moisture, pasture growth, hay production and water pressure; rain can improve moisture while simultaneously delaying field work; hail or frost damages exposed crops; severe winter acts through forage, shelter and animal condition. Do not implement global Farm Output -30% events.

7.4P Agricultural settlement identities and downstream businesses

Rural production should shape settlement business mix without an Agricultural Town trait. Ranch-heavy hinterlands create demand for livestock service, hay, freight, repair, wire/hardware and lodging. Productive crop valleys can support storage, milling, seasonal labor, machinery/custom work and heavier freight. General Stores become agricultural suppliers through actual assortment. Grain-mill/storage and livestock-commerce business forms require targeted regional research before exact operational templates are locked.

7.4Q Rural land, leasing and water-right research holds

Rural property value should derive from usable acreage, soil, water, forage, terrain, access, improvements, fencing, storage/buildings and market distance rather than a single Land Quality score. Acquisition diligence inherits pasture condition, hay reserves, equipment condition, water-system reliability and other operating baggage. The project should research Dakota/Black Hills agricultural leases, grazing leases, crop-tenancy/share arrangements, ditch/irrigation organization and water rights before adding detailed mechanics. These have high width potential but must not be invented from generic western assumptions.

7.4R Rural square completeness test

Every serious rural enterprise should be able to answer ten depth questions: productive land/resource, biological state, seasonal timing, labor, draft/machinery, inputs, storage, maintenance, market/logistics and capital/authority. Across the width dimension, the current design pool is crop farming, hay/forage, cattle, sheep where distinct, local produce, custom work, livestock commerce, agricultural supply, processing/storage and rural property/infrastructure. This is a completeness matrix, not a requirement for one hundred exposed UI features.

21.7 Mining Economic Ecosystem - square-off doctrine

Mining remains one of the deepest physical systems in the canon, but Revision XI broadens it from primarily mine ownership into an economic ecosystem. Mineral discovery creates value and risk across claims, finance, technical information, treatment, trade, supplies, contracting, property and settlement. The player and NPCs may participate at any economically legitimate point without owning a producing mine. Vertical integration is an available strategy, not the assumed endpoint.

21.7A Ten mining participation lanes

- Prospecting & Claim Development: finance discovery, control legitimate claims, improve geological information and sell, partner, hold or develop the property.
- Mine Ownership & Operation: develop workings, extract material and manage the existing bottleneck-rich mine system.
- Mining Investment & Promotion: provide risk capital through historically appropriate ownership/investor structures, taking partial upside without operating every mine.
- Assay & Technical Services: sell credible analytical/technical information to prospectors, mines, investors, lenders and buyers.
- Custom Milling & Treatment: own treatment capacity and process other mines' material for fees or negotiated arrangements.
- Ore & Concentrate Commerce: buy physical mineral lots, assemble/route compatible material and bear working-capital, assay and market risk.
- Mining Supply & Equipment Trade: serve the district through historically supported tools, consumables, parts, clothing, hardware and industrial supply, usually through flexible merchant identity rather than a Mining Store class.
- Mining Contract Services: sell freight, timber, repair, construction and technical capability to mines through the same shared businesses used elsewhere.
- Mine-Camp Property & Services: profit from housing, boarding, stores, freight yards and other settlement demand created by mining without owning the deposit.
- Reprocessing, Salvage & Failed-Mine Redevelopment: acquire remaining value in claims, plant, dumps/tailings, machinery, roads, buildings, housing and other distressed industrial assets.

21.7B Exploration, development and production as separate capital uses

A mineral property should distinguish exploration (learning what may exist), development (physically exposing/accessing and understanding the deposit while preparing production) and production (extracting currently economic material). A cash-starved owner may exploit known high-value material and postpone learning; a well-capitalized company may deliberately fund development ahead of immediate output. These decisions change information and physical access rather than adding a Research percentage.

21.7C Claim information history and valuation uncertainty

Claims preserve evidence histories: discovery, surface work, trenching/development, assay reports, previous production, abandoned workings and prior-owner records where available. Mineral-property value can remain extremely uncertain because the future deposit is not fully known. Asking price can be exact while fair-value estimates remain wide. Better representative sampling and development narrow geological uncertainty; Expertise cannot reveal unobserved rock.

21.7D Work fronts, geological confidence and operating allocation

Operating mines can contain economically meaningful work areas or fronts rather than one mine-wide grade/confidence score. North Workings can be well understood and expensive to support while a deep extension remains uncertain and water-prone. Owner/superintendent allocation among fronts becomes a strategic decision without requiring the player to draw tunnels.

21.7E Ore classes, stockpiles and economic cutoff

Mine output should preserve enough classification to make treatment and future economics matter: high-value/special lots, normal mill feed, lower-grade stockpile, material requiring different treatment and waste where appropriate. The owner can stockpile material that is currently uneconomic rather than automatically discarding or processing it. Later freight, treatment or price changes can make old stockpiles, dumps and tailings valuable. Cutoff/selection decisions should affect what material is mined or processed under current economics, but exact geological implementation and period terminology require specialist research. Avoid a simplistic High Grade Mode bonus; the physical ore and mine plan should carry the consequence.

21.7F Assay and technical-information business

An assay office should be a real professional-service business where demand supports it. Samples are received, registered, queued, prepared/analyzed and reported. The key distinction is sampling quality versus analytical quality: a competent assayer can accurately test an unrepresentative hand-picked sample. Investors and mine owners therefore care both about who assayed the material and how the sample was obtained.

- Assay capacity creates queues during booms; rush turnaround can be sold if the office has capacity.
- Professional reputation affects how lenders, investors and buyers treat reports because credibility is the service being sold.
- Large mines can internalize assay capability; small operators can purchase it externally. The same information system supports both.

21.7G Custom milling and treatment as independent industry

The existing canon already allows a mine to use a custom mill. Revision XI makes the provider side fully playable. A custom mill has treatment compatibility, physical capacity, power, water, labor, maintenance, receiving/storage, customer queue and tailings handling. Its central financial problem is utilization: an oversized mill with insufficient ore feed can be a disastrous investment. Customer agreements can reserve capacity, establish price/fee structure and define priority subject to historical research on exact forms. When total customer demand exceeds mill capacity, the managing authority chooses how commitments, owned mines and spot work are prioritized. Treatment specialization means not every ore lot is economically suitable for every plant.

21.7H Mineral commerce and physical lots

Ore/concentrate commerce remains a research-sensitive lane but is preserved because the economic role is powerful. Mineral lots are physical: source, weight, location, sampled/assayed characteristics, treatment compatibility, ownership and intended destination. A trader who buys material ties up working capital until it is hauled, treated or resold. Value depends on expected recovery minus treatment/freight and on confidence in the sample/assay. Aggregation can create value by combining economically compatible small lots into commercially useful shipments where historical practice supports it. Exact Black Hills ore-buying, settlement and sampling arrangements remain a targeted research hold rather than being invented from a corporation name alone.

21.7I Mining supply and contract-service marketplace

Mining supply should mostly broaden existing merchants instead of creating a rigid Mining Store class. A district can create unusual value for locally stocked slow-moving parts or consumables because a small component can prevent days of industrial downtime. The merchant chooses whether to stock, special-order or require a deposit; exact catalogue items must be historically researched. Mines also issue real demand for freight, timber/lumber, repair, construction and technical work. The mine can self-perform, contract externally or acquire the supplier. Service companies compete on actual capacity, reliability, price and timing. Internal ownership moves the cost/risk into the portfolio rather than making the service free.

21.7J Mine-camp property and independent commercial ecosystems

A remote mine can generate enough workers and freight to create a settlement opportunity for landlords, boarding operators, merchants, repair shops and other services. Independent investors can speculate on transferable nearby property and build ahead of expected employment. If the mine disappoints, vacancy and business failure follow. A mine

company can instead provide some housing/services internally, trading greater capital/control for less independent local enterprise; detailed company-town arrangements require targeted regional research.

21.7K Failed mines, reprocessing and breakup value

Mine failure leaves physical and informational history rather than deleting the site. Claims/mineral rights, workings, roads, buildings, mill plant, machinery, stockpiles, dumps/tailings, housing and utilities can retain separate values. A distressed owner, lender or estate may sell them together or separately. Restart cost depends on actual condition: flooded workings, deteriorated support and exposed machinery can make an apparently cheap mine extremely expensive to reactivate. Later treatment economics can make surface material valuable even if underground production never restarts. Rights to tailings/dumps/reprocessing should be treated as distinct economic interests where historical/legal research supports them.

21.7L Mine management hierarchy and operating posture

A small prospect and a large industrial mine should not use the same management granularity. Small operations can be owner/boss led. Larger mines need superintendents, foremen and specialized operators. The superintendent can allocate labor among work fronts, manage routine purchasing/maintenance, schedule shifts and respond to ordinary operating conditions within authority; the owner controls major expansion, financing, treatment strategy, acquisition and shutdown. Mine operating posture can include development-heavy, normal production, reduced operation, temporary idle or closure rather than a simple on/off switch. Idling may still require pumping, inspection, security, maintenance or preservation. Shutting a wet mine without adequate pumping can allow water to make later restart progressively harder.

21.7M Shifts, safety and maintenance integration

Mining can use longer or multiple shifts where historically/operationally justified, but added scheduled hours do not automatically multiply production. Labor, supervision, hoisting, mill capacity, light, power and maintenance windows all need to support the schedule. Exact Black Hills shift patterns remain a research/calibration item. Safety should remain the outcome of actual ground support, equipment condition, explosives handling, water control, supervision, fatigue/workload and skill. The player can defer support/maintenance when physically possible. The game should not hide the risk behind a Safety Budget slider or universal accident modifier.

21.7N Mining labor, information and acquisition diligence

Mine expansion creates differentiated labor demand: general labor, miners, teamsters, timber/support workers, hoist/boiler operators, smiths/mechanics, mill workers, supervisors and later technical specialists. A district can have abundant labor but lack one critical skill. Better wages, board/housing, recruitment and training/promotion where appropriate are causal responses. Mining information should preserve provenance: rumor, visual showing, promotional sample, independent assay, development exposure, bulk test/treatment and actual production history carry different evidentiary weight. A major mine acquisition therefore becomes a premier diligence challenge involving claim/title review, records, independent assays, machinery/condition inspection and technical reports while rival bidders or time pressure can still exist.

21.7O Roads, water, power and industrial rail as capital forks

Remote access is a mine capital decision. The owner can expand wagon capacity, improve road/bridge/drainage, later pursue industrial rail or seek common-carrier connection. Water can be both resource and hazard: pump capacity and actual inflow determine accessibility over time. Power can come from steam, water where site permits, private generation or later purchased electric service. These choices alter physical operating costs and dependencies rather than applying mine modernization bonuses.

21.7P Mining boom, closure and regional contagion

A credible successful discovery should attract NPC prospectors, investors, landlords, freighters, merchants and service providers through visible evidence, raising claim prices, wages, housing pressure and service demand. Claim rushes therefore emerge from information rather than a scripted Gold Rush event. Rumor alone can produce speculation, but stronger evidence produces stronger capital response. Closure propagates in reverse. A large mine shutdown can reduce worker income, empty housing/boarding, weaken stores, free freight capacity, cut timber/repair demand and damage bank loans. The district keeps roads, buildings and distressed assets, allowing later redevelopment rather than resetting the map.

21.7Q Mining accounting and portfolio concentration

Mining reports should separate major economic causes such as exploration/development, extraction labor, support/timber, hoisting/haulage, pumping, power/fuel, treatment, freight, maintenance and administration. Bookkeeping/Expertise can improve owner interpretation, but the costs themselves come from the real operation. A vertically integrated mining empire can own mines, mills, timber, freight, repair, housing, industrial rail and power. This improves control and may reduce external margin/coordination costs, but it can create severe correlated exposure: a mineral downturn can hit nearly every supposedly diversified business at once. Risk analysis should recognize vertical concentration rather than treating business count as diversification.

21.7R Mining square completeness test

Every mining lane should answer ten depth dimensions: information, physical capacity, labor/skill, inputs/logistics, customers/contracts, cash/finance, condition/reliability, time/lifecycle, authority/delegation and regional consequence. Across the width dimension, the ten current lanes are prospecting/claim development, mine operation, investment, assay/technical services, custom treatment, mineral commerce, supply, contract services, mine-camp property and failed-mine/reprocessing. The square is a design-completeness test rather than a requirement for one hundred separate controls.

PART VI

Supply, Trade & Logistics How goods move, how supply chains are controlled and how shortages emerge from real causes.

READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

9.1 Supplier classes and sourcing logic

Stores and production businesses should not receive goods from nowhere. Supplier classes remain internal player-owned suppliers, other local businesses, hired local hauling or freight services, and off-map regional suppliers. Core sourcing logic remains unit price, hauling and transfer cost, lead time, reliability, quality, volume limit, and market reach. Owned hauling can lower marginal transport cost and provide stronger control when shipment volume keeps teams, wagons, and labor productively used, but it creates continuing feed, labor, maintenance, stabling, and storage costs even when idle. Hired hauling avoids those fixed burdens but charges a commercial rate and offers less direct control. The better mode therefore depends on route, volume, frequency, and utilization rather than ownership always being cheaper in the long run.

9.1A Landed cost and source geography

The economy should distinguish supplier quote from landed cost. Landed cost is the actual acquisition burden after supported freight, transfer, drayage/handling, route and other direct procurement costs are included. This distinction matters especially before local rail: an inexpensive distant item can be less profitable than a higher-priced regional or local source once overland transport is counted. Standardized manufactured goods may have relatively legible wholesaler/manufacturer pricing, while local food, timber, livestock, repair inputs and household-produced goods can be negotiated or driven by current local availability. Product profitability therefore emerges from source, route, turnover, credit and selling price rather than from one hard-coded margin property.

9.2 Recurring local orders

Recurring local orders belong in VP because they make the town economy feel connected without needing full formal freight contracts. Each agreement should store buyer, seller, goods category or item, target quantity, cadence, next due date, price mode, hauling responsibility, and breach history. Useful examples include boarding houses buying meat and staples, butchers sourcing livestock from a ranch, blacksmiths keeping repeat repair work with farms and ranches, and general stores supplying nearby service nodes on a repeat basis. Agreements can succeed, partially fill, arrive late, fail, or be cancelled; those outcomes should affect trust, Business Reputation, and future renewal odds.

9.2A Supply agreements, internal chains, and visible commitments

Recurring local orders should now expand into a full Supply Agreements system. Agreements are the standing policy layer between businesses, while actual shipments are the execution layer. A ranch can commit slaughter animals to a butcher, a butcher can commit preserved output to a general store or boarding house, a sawmill can commit lumber to a yard or project, and a store can maintain recurring inbound agreements with local or off-map suppliers. The player should be able to see the agreement, the route, the cadence, the target quantity or percentage, the reserve rule at the source, the priority rank, the hauling mode, and the recent fulfillment history. This system must work for owned-to-owned chains, owned-to-non-owned sales, non-owned-to-owned procurement, and background non-owned trade that still affects the player indirectly. The point is not just to move goods; it is to let the player deliberately shape supply chains, protect critical buyers, prioritize internal vertical integration, and decide where overflow goes when a preferred destination is full or absent.

9.2B Commercial Commitment Authority

Supply agreements are one instance of a broader Canon concept: meaningful commercial commitments consume future freedom. A commitment should answer who is obligated, what is promised, when or over what period, how much, where performance occurs, whose assets perform the work, who pays which costs, what is reserved or prioritized, which risks each side bears and what happens when circumstances change. Employment, freight service, leases, custom processing, construction, lending, partnerships, priority repair and other high-value relationships can use the same conceptual grammar without sharing one generic contract screen.

Contract state is not execution state. Signing a promise can reserve stock, money, labor time, processing slots, transport capacity or property use, but the promised outcome still requires the real people, assets, materials and travel

to perform it. This makes future capacity strategically saleable where the resource is real: a mill can reserve treatment time, a freight business can reserve haul capacity, a repair shop can reserve priority slots and a warehouse can reserve space. There are no generic Future Capacity points.

9.2C Discretionary-control reporting

For important resources, the player should be able to distinguish what exists, what is physically usable now, what is already promised or legally burdened, what ordinary operations require, what policy deliberately reserves and what remains genuinely redirectable. A wagon scheduled for tomorrow's contract is owned but not freely available. Cash pledged to a project is not equivalent to uncommitted cash. A leased room is owned but not vacant. A breeding herd may be saleable property while still being strategically necessary biological capital. These states should remain resource-specific rather than collapsing into one universal Capacity or Discretionary Control meter.

9.2D Performance, breach and recovery

Commitment performance should allow more than Success/Failure. Depending on the relationship, outcomes can include on-time performance, late performance, partial performance, substitution, renegotiation, cancellation, excused nonperformance where historically/contractually appropriate and unexcused breach. Communication, records, urgency, prior history and available alternatives can influence whether the parties renegotiate or end the relationship. Missed commitments should damage the specific relationship or commercial standing they actually affect rather than subtract from one global reputation bar.

9.3 Asset markets and rush response

Asset markets should cover wagons and transport assets, livestock classes, seed sources, and later more specialized equipment. A wagon is never enough by itself; it also needs animals, harness, upkeep, storage space, and assigned labor. Rush orders remain a meaningful emergency lever that can protect against stockouts, off-map leakage, and trust erosion at a real cost. In Land & Ledgers, rush response is one of the clearest ways supply, cash flow, and reputation collide in a readable management decision.

CANON LOCK

9.3A Trip-based logistics and non-teleporting freight

Important goods cannot teleport. A delivery order has a source, destination, cargo class, route, loading window, travel time and unloading step, and destination inventory updates only when the goods physically arrive or complete a justified off-map leg. A delivery can contain multiple transport legs. Distant goods may move by outside rail or river service to a regional transfer point, then complete the final journey by hired or owned wagon. Each local wagon leg derives practical capacity and travel time from vehicle type, draft team, cargo, load, route quality, grade, surface/weather, animal condition and the availability of water, feed, bridges, crossings and stopping infrastructure. Severe

conditions can stop a route rather than merely applying a speed percentage. For locally represented movement, the actual person, animal and vehicle state remains authoritative enough to validate departure and arrival. Agents should travel beyond the player's normal field of view toward a hidden regional boundary before converting to coarse/off-map simulation. Returning agents should re-enter from the corresponding boundary and continue physically. Abstraction changes simulation resolution, not identity, cargo, route or reality.

9.3B Transport assets, draft power, roads, and the Livery / Freight backbone

Freight wagons are already mature commercial technology at the opening. The 1880-1905 transport story is primarily changing route length, road quality, team economics, rail interface and commercial organization rather than Wagon I -> Wagon II -> Wagon III. Heavy freight, farm wagons, local drayage and passenger/stage equipment can differ by job without becoming a technology tree. Draft animals retain economic identities. Oxen are slow but strong and useful for heavy haul where speed matters less. Mules are hardy professional freight animals. Horses support faster local delivery, farm work and time-sensitive movement. Practical load depends on vehicle, team, grade, road condition, weather, cargo and animal condition rather than a universal capacity. Road improvement is a physical project. Horse- drawn scrapers and graders are already period-valid, but rural roads remain predominantly dirt and weather-sensitive through 1905. Grading, drainage, culverts, bridges and selective surfacing improve reliability, feasible load and travel time through real route conditions rather than an abstract freight bonus. As rail approaches, long wagon segments shorten; after a depot arrives, wagons shift toward drayage, farm/ranch collection, mine and timber feeder service and secondary settlements. Blacksmiths, farriers, wheelwrights, harness makers and local repair remain important throughout the campaign. Factory-standardized parts increasingly change repair from fabrication toward diagnosis, ordering and installation without eliminating local skilled work.

9.4 Business-to-business logistics and internal routing

Business-to-business routing should become a general ownership system, not a mining-only exception. The player should be able to direct outputs from one business to another, to off-map buyers, to projects, or to storage where the chain makes sense.

- Owned businesses may route output by percentage, fixed quantity, cadence, priority, reserve rule, destination, transfer price, and hauling responsibility. Cadence can be daily, weekly, monthly, or another supported interval.
- Same-owner transfers can use a flexible price modifier: free/0%, at cost, below market, market value, or above market where the current economy supports it. There should not be an artificial minimum price if both businesses are player-owned, but ledgers must remain honest so profit is not magically created.
- Split routing should be supported where feasible. A ranch can send part of livestock output to an owned butcher and the rest off-map. A sawmill can reserve some lumber for projects, send some to mines, and sell excess locally or off-map. A butcher can supply a general store, a boarding house, households, or outside buyers.
- Local inventory remains real. Global summaries can reveal portfolio-wide stock and shortages, but goods should not become a magic shared warehouse. Transfers require actual source stock, reserve checks, destination capacity, route access, loading/unloading ability, transport assets or hired freight, labor, cost, and lead time.
- Business-to-business logistics should support owned-to-owned, owned-to-non-owned, non-owned-to-owned, and background non-owned trade that affects player-facing supply, prices, relationship memory and opportunity pressure.

9.4A Internal allocation, opportunity cost and related-party governance

Wholly owned businesses do not need to fake-negotiate with one another. Their managers can advocate, warn and expose local consequences, but the owner with legitimate authority allocates. Internal ownership does not make the resource free. Physical cost, accounting transfer value and opportunity value are separate questions. Lumber sent internally still consumed logs, labor, mill capacity and freight, and it may displace an outside sale. Internal transfer pricing must never create consolidated profit from nothing.

Where outside or minority ownership exists, the rule changes. A player cannot use control of one partly owned company to strip value for the benefit of a wholly owned company without the other owners caring. Material related-party transfers, distributions, new obligations or asset sales should respect the control rights and protections established by the ownership agreement. This connects partial ownership, bookkeeping, internal transfers and governance.

9.5 Livery / Freight business and professional hauling

Livery / Freight should function as the professional hauling layer of the town and region. It is both infrastructure and a business opportunity. Businesses can self-haul if they have assets and labor; otherwise they can hire freight, rely on buyer pickup, or use a player-owned or NPC- owned hauling service.

- The hauling business should be able to serve multiple clients: player businesses, NPC businesses, households or institutions where relevant, construction projects, mine camps, sawmills, ranches, farms, stores, and later rail or depot interfaces.
- A hauling trip should account for cargo class, distance, route difficulty, grade, road condition, wagon/cart fit, animal and team requirements, labor, loading/unloading time, feed and upkeep, water or stopping access, storage or stabling capacity, and lead time. The shortest geometric route is not always the cheapest or most reliable route.
- Self-hauling can reduce out-of-pocket freight cost, but it still consumes paid labor, animal use, wagon time, feed, upkeep, and opportunity. It is cheaper only when the player has the assets and can spare the capacity.
- A player-owned Livery / Freight business can profit from the town economy's movement needs. As mines, sawmills, farms, butchers, stores, and construction projects grow, demand for professional hauling should rise.
- The system should support source-delivers, destination-pickup, hired freight, and route assignment modes where current implementation permits. It should avoid teleporting important cargo unless an early underlying simulation placeholder is clearly marked as temporary.

REVISION V INTEGRATED

9.6 Causal supply pressure and price formation

CAUSAL PRICING Supply pressure changes real output, availability, demand, transport capacity, reliability, or delivered cost first. Price reacts afterward. Supply pressure is a consequence, not an arbitrary event modifier. The game must not simply decide that livestock, grain, lumber, or another commodity costs more. A grounded condition must first change actual output, available supply, demand, transport capacity, reliability, or underlying delivered cost. Market and supplier prices then respond to those changed conditions. Do not implement logic equivalent to "cattle disease -> beef price +25%." Prefer a chain in which disease reduces usable livestock supply; procurement becomes tighter; buyers

compete for fewer animals; butcher throughput may fall; and price pressure emerges according to the actual market state.

REVISION V INTEGRATED

9.6A Livestock disease

- Livestock health problems may reduce animal condition, sale eligibility, reproduction, herd output, usable slaughter stock, or survivorship depending on the supported simulation depth. Named diseases should only be used where their period, geography, and transmission are historically justified; do not rely on one generic cattle-epidemic event.
- A disease affecting one ranch should normally damage that ranch and dependent buyers without automatically moving the entire regional market.
- Several affected local ranches can create a settlement/local shortage.
- Only an outbreak large enough to affect the broader regional supply pool should materially constrain off-map livestock availability or pricing.

REVISION V INTEGRATED

9.6B Crop failure and infestation

- Grounded causes can include infestation, crop disease, drought, excessive rain, frost, poor harvest, or other supported environmental conditions.
- The first-order effect is reduced yield or usable output. Supplier availability and price pressure follow from the resulting supply situation.
- Commodity specificity matters. A wheat problem does not automatically raise every agricultural price. Secondary effects may occur later through substitution, feed demand, transport competition, or connected supply chains.

REVISION V INTEGRATED

9.6C Geographic shock scale

- Supplier/business scale: one producer or supplier is impaired. Dependents may suffer even if the wider market barely moves.
- Local/settlement scale: enough nearby production or transport is impaired to move local availability. Off-map sourcing can remain an escape valve, usually with longer lead time and higher freight or acquisition cost.
- Regional/off-map scale: the problem is large enough to affect the broader supply pool. Outside capacity, price, and lead time can now deteriorate as well. These events should be much rarer because they remove a major player fallback.

REVISION V INTEGRATED

9.6D Diversification and vertical integration

Supplier concentration should create a real risk/reward tradeoff. One cheap primary supplier can maximize margin but increase exposure. Multiple suppliers can cost more while increasing resilience. Vertical integration is another response, but owning the ranch, farm, mill, or freight business transfers the underlying production and labor risk into the player portfolio rather than deleting it.

9.6E Market reach, information, cold chain, and price propagation

Each good or service has a practical market reach determined by value-to-weight ratio, perishability, transport cost, route reliability, information speed, customer mobility and available substitutes. A store stockout, settlement shortage, regional shortage and wider-market shock remain distinct. Rail can greatly expand market reach by reducing bulk freight cost; telegraph and telephone can reduce information delay without moving goods. Natural ice, icehouses and later refrigerated rail extend the practical reach of perishables where the required cold chain exists. Refrigeration should therefore change spoilage horizon and reachable buyers rather than act as one global technology toggle. Existing inventory and replacement cost remain separate. A merchant may hold stock bought cheaply while new supplier quotes rise, or may choose not to reprice until replacement cost becomes relevant. Outside-market influence grows as transport and communication

networks deepen, making late-campaign local prices more exposed to distant conditions without erasing local route, quality and shortage effects.

9.6F Shock severity, buffers, and recovery

Economic adversity should not carry a predetermined percentage effect. Severity emerges from the condition's geographic footprint, which assets or people are actually exposed, timing within a production cycle, available inventory or forage reserves, spare labor and route capacity, substitute suppliers or goods, the affected share of the relevant market, and the time required for recovery. Two businesses exposed to the same weather, illness, financial condition, or transport failure can therefore experience very different outcomes.

- Acute shocks include hail, injury, bridge loss, sudden supplier failure, and other fast-onset capacity losses.
- Developing shocks include drought, pasture deterioration, supplier unreliability, labor scarcity, and inventory depletion and should usually move through Early -> Mounting -> Action signals.
- Seasonal culmination occurs when months of exposure become decisive at harvest, wintering, roundup, or another timing- sensitive point.
- Structural shocks include oversupply, prolonged credit tightening, route decline, or persistent population loss and can remain for months or years rather than expiring on a short event timer.
- Surplus is a real pressure family. Bumper harvests, overbuilt services, or excessive livestock supply can weaken sale prices and storage economics even when physical output is strong.

10.1 Storage rules and spoilage

Interior storage remains safest and limited. Exterior or yard storage remains valid for appropriate bulky goods but can carry greater weather, loss, spoilage or security exposure. Every physical inventory unit should consume normalized capacity appropriate to its size and storage type rather than existing as weightless database stock. Spoilage should begin with perishables such as meat and vulnerable prepared goods. Spoiled goods should be removed automatically but reported clearly in the ledger so the player can understand why a business is bleeding value.

10.1A Capacity authority: total, allocated, occupied, and unallocated

Inventory-capable businesses should distinguish total physical capacity from allocated/reserved capacity and actual occupied capacity. Unallocated capacity is genuinely free for a new line. Reserved-but-empty capacity belongs to an existing line and stays unavailable to unrelated goods until the managing authority deliberately changes the allocation. This prevents unrealistic perfect packing after every sale and gives layout/assortment decisions economic meaning without requiring manual shelf placement. The same authority can be reused for display, backroom storage, warehouse space, yard/bulk storage, cold storage or other business-specific pools.

10.1B Stocking execution versus administrative allocation

Routine shelf stocking is execution work. Employees should refill displays from available backstock according to the owner/manager's target and allocation when workload allows. They do not decide to shrink a slow product line, add a new category, convert a firearm fixture to cigarettes or consume reserved capacity with unrelated stock. Those are administrative decisions. Temporary special orders may use receiving/holding capacity without establishing a permanent merchandise allocation, provided the business has somewhere to stage the item until collection or delivery.

10.2 Stock discipline and stockout memory

Households and business buyers should not simply fail a purchase and forget it. Repeated stockouts should contribute to weaker household preference, softer renewal odds on recurring orders, and broader erosion of local trust. Bad stock discipline should create a soft but persistent drag rather than only a one-time lost sale. This gives the player a reason to care about consistency, not just occasional high- margin spikes.

10.3 Saturday readiness

Saturday should remain a recurring readiness opportunity, not a guaranteed demand multiplier. Historical practice supports Saturday-night marketing and late store hours in many communities, but routine household shopping remains reserve-driven and does not occur on a rigid weekly script. Rural travel patterns, household routines, income timing, local custom, weather, seasonal work, and available cash determine whether a particular Saturday becomes unusually busy. Readiness still matters: stock, staffing, urgent requests, local supply stability, visible condition, and operating hours can turn a concentrated shopping day into opportunity or lost trust.

PART VII

People, Households & Labor People as households, workers and economic actors rather than decorative population.

READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

12.1 People simulation

Each visible NPC should be a simulated person with identity, age, relationships, origin and migration path, skill formation, employment, schedule, health, location, household burden, housing state, economic relationships, risk markers and succession flags. The individual person is the fundamental social and labor actor. The player should not know everything about a person immediately; some truths should emerge through references, employment history, observed performance, account records, reports and time rather than magical full-info UI. A person who works at the sawmill is the same person whose wage supports a household, who can shop, travel, become ill, quit, be poached, relocate, age or die.

12.1A Household economy versus individual action

The household is the fundamental ordinary consumption and domestic economic unit, while individuals perform the actual actions that make that economy work. Food, fuel, clothing, rent, household credit and many purchases are planned at household level so the simulation does not create one independent grocery need per person. A specific member can then be chosen to shop, work, deliver, travel or seek service according to schedule, location, ability and household circumstances. Household members can have separate jobs and life histories while contributing to shared resilience and dependency. Exact personal-versus-pooled cash detail can remain abstract unless a later mechanic needs a separate purse; the default design priority is truthful household cash flow and individually truthful work/movement.

12.2 Households as economic units

Households are labor systems, consumption units, care networks, and sometimes mixed domestic-commercial enterprises, but family, household, and dwelling should remain distinct authorities. A family represents kinship and dependency relationships; a household represents people sharing domestic and economic life; a dwelling represents the physical residential space. A boarding proprietor family and six unrelated boarders can therefore share one building without becoming one biological family. Household strength should depend on earning capacity relative to dependency load, reserves, housing burden, debt, and household production rather than on household size alone.

12.2A Household budgets, cash timing, substitution, and hardship

Households do not generate money because they have a need. They spend actual cash, use available account credit, sell goods or labor, draw reserves, substitute, delay purchases, reduce quantity or go without. Income timing matters as much as annual income: clerks and wage workers can receive relatively regular pay while ranch, farm, freight or business households can experience irregular large settlements. This makes merchant credit, rent, fuel and seasonal purchasing meaningful. Financial stress should propagate through real choices: unemployment can reduce discretionary buying, increase use of store credit, threaten rent, change work search and eventually contribute to relocation rather than applying one generic poverty modifier.

12.3 Settlement ladder and household improvements

Boarding is a major bridge between immigration and permanent settlement, but not the only bridge and not a mandatory first stage. New arrivals can board, rent, join kin, move directly into stronger accommodation, use employer housing, or arrive with enough means to buy. Rental housing is a normal durable tenure state rather than a temporary failure state. Vacant rental-capable properties, rooms above businesses, boarding capacity, and employer bunks therefore matter as distinct settlement infrastructure. Household improvements such as chicken coop, kitchen garden, root cellar, or boarder rooms remain useful where the game supports them.

12.4 Family labor and winter pressure

Helper-age children should begin contributing through chores and light work as soon as it is historically plausible. Family businesses should benefit from this, especially general stores, farms, ranches, and households with coops or gardens. Each household should carry heat retention, stove tier, fuel reserve, preferred fuel, and winter strain. Heat retention is driven by house type, size, and condition. Small tight houses can outperform larger drafty ones. When fuel

drops below critical, urgency rises and the household may overpay, shift to lower- quality fuel, or cut spending elsewhere. Winter strain should feed illness pressure, work efficiency, and financial instability rather than acting as a fail state.

12.5 Demographic generation and migration networks

Population composition should emerge from settlement function, migration history, housing, and employment rather than fixed statewide quotas. The 1890 South Dakota population was strongly migration-driven and somewhat male-skewed, but individual nodes should vary substantially. Mining camps, freight nodes, and construction booms can attract disproportionate numbers of single working adults; agricultural and mature service communities can support more families, children, renters, owners, and older residents. The opening target of roughly 14-22 households and 55-95 residents, with a likely sweet spot around 70-85, remains valid. The improvement is in composition: varied family sizes, single workers, couples, children and youth, widows or widowers, adult relatives, boarders, renters, owner households, and a small older population should combine into the total rather than cloning one average household. Historical regional averages are guardrails, not generator quotas. Migration should have a reason. Job vacancies, wages, kin relationships, land or business opportunity, institutional employment, displacement from another node, and existing community/language networks can all pull specific people. Origin and nationality shape biography and relationships, not predetermined skill bonuses. Growth changes composition as well as headcount: a mine or railroad project may first produce boarder-heavy labor growth, while stable work and housing later encourage kin migration and durable household formation. Decline should work in reverse as transient workers leave before deeply rooted households.

12.5A Household lifecycle, formation, aging, and succession

A twenty-five-year campaign requires households to change. People can marry, have children, age into new work or schooling stages, leave home, form new households, take boarders, combine with kin, become widowed, inherit, relocate or die. The town in 1905 should not be the 1880 roster with older portraits. Children at the opening can become adult workers, owners or managers by the late campaign. Exact schooling, child-work and age thresholds require historical calibration, but the lifecycle authority should be preserved from the start so demographic continuity can emerge without turning the game into a genealogy simulator.

13.1 Health matters economically first

Health should matter economically before it becomes medically elaborate. Minor illness reduces efficiency for roughly one to two weeks. Moderate illness causes missed work, household income dip, and a higher doctor-visit chance. Severe illness can create long absence, rare death chance, and possible lasting weakness.

13.2 Injury pools and job risk

Low-risk jobs include clerk, store work, and office work. Medium-risk jobs include butcher, delivery or cart work, and basic farm labor. High- risk jobs include blacksmith, ranch work, construction, and heavy hauling. Profession-based injury pools should remain grounded. Ranch and horse work risks kicks, trampling, falls, rope-hand injuries, crush injuries, and head trauma. Farm work risks tool cuts, wagon accidents, livestock injuries, strains, and crushes. Butchery risks deep cuts, infection, and finger loss. Blacksmithing risks burns, eye injuries, crushes, and repetitive hand damage.

13.3 Doctor service and employer relief

Medical service remains active early as a grounded town service. The player does not begin with a doctor business. The doctor's office can be acquired later as a service business. Office visits handle most cases. House calls are reserved for severe illness, severe injury, immobility, and later special cases. Employer relief should be a policy or choice mechanic rather than a universal legal guarantee. Some employers may cover treatment or partial wages after serious workplace injuries to retain labor, improve trust, or maintain employment relationship history.

13.3A Time, travel and practice capacity

A doctor primarily sells scarce professional time and judgment. Office care concentrates patients around the doctor; a house call consumes travel time as well as treatment time. In a small wagon-and-foot town the difference is economically important. A doctor who spends half a day riding to a remote mine cannot simultaneously serve ordinary office patients. The player should therefore see a queue and coverage problem rather than a magical healing radius. Roads, weather, horse availability, distance and later telephone access can change how efficiently calls are arranged without removing physical travel. Historical North American practice strongly supports the home-visiting model and the gradual efficiency advantage of office practice.

13.3B Urgency, triage and incomplete treatment

The medical layer should distinguish enough severity to create scheduling decisions without becoming a diagnosis simulator. Minor and routine cases can wait or self-manage; moderate cases are more likely to seek care; severe injury, immobility, childbirth complications where modeled and serious illness can justify urgent house calls or immediate attention. When demand exceeds capacity, some patients wait, travel elsewhere or go untreated. Treatment can reduce absence, complications or death risk where historically plausible, but it never guarantees recovery and should not erase the underlying severity.

13.3C Fees, accounts and who pays

Medical service can be paid at the visit, carried on an account, paid partly later, covered by an employer in selected workplace cases or supported through a specific institutional arrangement where historically justified. Exact fee levels and collection customs remain calibration items, but the economic consequence is clear: a busy doctor can hold substantial receivables while lacking cash. A generous credit posture can improve access and reputation but create bad-debt exposure. A mine or large employer can create dependable business through a service arrangement while also consuming priority during a serious accident or illness cluster.

13.3D Practice inputs, facilities and professional continuity

The doctor's office needs modest but real supplies, equipment, heat, clean working space and transport for house calls. The design should not track every medicine bottle, but shortages of basic dressings, commonly used medicines or transport can reduce what the practice can do. More important is the doctor themselves: professional skill and local trust are not contained in the building deed. If a doctor leaves, dies or is incapacitated, an acquired office may retain premises, records and some goodwill while losing the core professional capacity. Hiring or attracting another qualified practitioner is therefore a real continuity problem.

13.3E Reputation, information and referral

Households should learn about the doctor through direct treatment, neighbors, employers and local reputation rather than omniscient quality scores. Better medical expertise can improve treatment odds and recognition of serious cases, but uncertainty remains. Some cases can be referred or sent to a larger off-map center when local capability is inadequate and travel is feasible. A reputable practice can draw patients from nearby nodes, while repeated absence, poor outcomes or inability to respond can weaken trust even if the doctor remains technically competent.

13.3F Public-health load without a public-health minigame

Illness waves should matter primarily because they create simultaneous household and labor disruption. They can also overload the doctor, consume supplies, raise demand for house calls and expose the difference between one small practice and a settlement with deeper medical capacity. The player should not run a modern health department through the doctor screen. The useful ownership choices are staffing/support, transport, credit, employer arrangements, practice location, service priorities and whether capital is justified to expand or attract another practitioner.

13.4 Labor continuity

Businesses should feel labor strain through absences, reduced efficiency, coverage weakness, and continuity risk. A worker shortlist or watchlist is a valid owner-planning tool and should remain consistent with partial knowledge and staged discovery. Health therefore becomes part of labor realism, not a detached medical minigame.

13.4A Employee turnover, poaching, and management vacancies

Employees are not permanent building attachments. They can quit, accept a competitor's offer, relocate, retire, become unavailable through illness or injury, or die. Competitors can poach capable workers by making real employment offers, and the player can do the same. Compensation, working hours, role/status, conditions, stability, distance, housing and household circumstances can affect whether a worker changes jobs. Ordinary vacancies can be handled by a delegated manager within authority; losing the manager or another unique administrative/skill-critical person should escalate because the authority or function itself has disappeared.

REVISION V INTEGRATED

13.5 Player mortality gameplay-tolerance lock

GAMEPLAY TOLERANCE NPC mortality remains simulated. Random player mortality does not end a campaign.

NPC mortality remains simulated and can generate estates, widow ownership, inheritance, labor loss, property sales, succession problems and business-continuity consequences. Mortality should not be a uniform random employee-death roll. Where implemented, risk should arise from age, period-appropriate disease, childbirth, living conditions, exposure

and occupation-specific accidents, with exact cause weights calibrated from historical evidence. Western Plains mortality evidence supports meaningful infectious-disease and pneumonia risk

as well as exposure and accidental/violent death; dangerous work such as mining, heavy hauling, forestry, ranching and industrial machinery should carry different accident profiles from clerical or retail work. Random player mortality must not end the campaign. The player can remain a normal human within the fiction without subjecting a long economic save to random death Game Over.

REVISION V INTEGRATED

13.6 Human illness as an economic shock

Existing health rules remain authoritative, but illness should connect more directly to production, logistics, service coverage, and supply. The preferred causal chain is: illness -> worker absence or reduced capacity -> operating-role weakness -> lower output / lower hauling capacity / reduced service -> downstream delays, shortages, or price pressure where the market is tight enough.

- Illness among teamsters or freight workers can reduce transport capacity and reliability.
- Illness among sawmill workers can reduce lumber output.
- Illness among farm labor during harvest can reduce crop output or leave part of the harvest unrealized.
- Illness among miners can reduce extraction and increase remote-site strain.
- Illness among retail/service staff can reduce hours, service quality, and operating-role coverage even without changing commodity production.

REVISION V INTEGRATED

13.6A Timing and seasonal consequence

The same disruption should not have identical consequences at every point in the calendar. Illness, injury, transport failure, or labor loss can matter much more during harvest, roundup, Saturday trade, a construction surge, a mine expansion, or another capacity-sensitive period. Timing should modify the economic consequence through the work that fails to occur, not through an arbitrary "bad season" penalty.

PART VII - REVISION XII INTEGRATED ADDITION

12.6 Household, Accommodation, Kinship & Demographic

Lifecycle REVISION XII AUTHORITY This section is the governing interpretation layer for earlier Sections 12.1-12.5A and for the Residential Property system in 12.7 wherever those passages use household, dwelling, housing state or family more narrowly. It does not delete the earlier systems. It clarifies the entities and relationships they operate on. A person survives a change of household; a household survives loss of a dwelling; a family relationship can survive separation across households; and an accommodation can exist without conventional real-estate ownership. Core doctrine: A household is a social-economic domestic unit, not a building and not a parcel. Accommodation is the current relationship that answers where a person or household is sheltered. A building can contain several economically meaningful accommodation spaces and several households. Systems should modify relationships and states rather than treating a household as an object owned by the Housing system.

12.6A Household identity, minimum size and lifecycle output

A household may consist of one adult maintaining a distinct domestic economy or any larger group of people who actually pool important domestic resources, obligations and consumption. A one-person merchant above a store, widower in a cabin, laborer renting a room or adult boarder who keeps a distinct budget can therefore be a one-person household. Dependence on wages, remittances, kin support or credit does not by itself merge that adult into another household. Household size has no arbitrary fixed maximum. A household can legitimately contain parents, many children, a widowed parent, an orphaned niece or nephew and another dependent relative if those people actually operate as one domestic economy. Large households should remain uncommon because marriage timing, fertility variation, mortality, migration, children leaving home, widowhood, economic pressure and household splitting naturally limit them - not because the simulation imposes a maximum-member rule.

- Household size is generated by life events, not selected directly. Birth, marriage, death, widowhood, adoption/care placement, adult children staying or leaving, migration, household split, household merger and kin hosting change membership.
- Family and household remain separate. Related people can live in different households. Unrelated people can share a building or even accommodation without becoming one household if their domestic economies remain substantially separate.
- Household formation and residential formation are independent. Two people can marry and initially remain with kin or in boarding before establishing a new domestic unit or acquiring separate accommodation.
- Household split and merger can occur without migration. Adult children can establish a new household next door; a widowed parent can merge into an adult child's household; two households can share a dwelling temporarily without merging finances. The existing spring-1880 population envelope of roughly 55-95 residents, with 70-85 preferred, remains a stronger opening constraint than the older 14-22 household count. The household range becomes an indicative calibration target rather than a hard generator gate. A seed with many one-person workers or boarders may legitimately exceed it; another with several large family households may fall nearer the old band. The generator should preserve believable people and relationships rather than force an obsolete household count.

12.6B Kinship graph and family authority

Kinship should be explicit enough to support inheritance, dependency, migration, household formation, family labor and succession. Each local person can carry direct parent/child and spouse relationships where known. Siblings, grandparents, grandchildren, aunts/uncles, nieces/nephews and other useful relationships can then be derived rather than stored redundantly.

- Surname is evidence for generation, not proof during simulation. World generation may use matching surnames as one clue when intentionally creating related local families, but once the world exists the relationship graph is authoritative. A maternal uncle with another surname can be closer kin than an unrelated person sharing the child's surname.
- Off-map kin are valid lightweight records. A person can have a parent, sibling, aunt, uncle or adult child elsewhere in Dakota or beyond the simulated region without requiring the whole United States to be simulated. Such kin can matter for chain migration, emergency care, inheritance correspondence and later return migration.
- Kinship does not command household membership. The game should never automatically merge everyone related to one another. Family is a network; household is a current domestic/economic arrangement. This is a high-value shared authority: the same small relationship graph can support orphan placement, succession, estate claims, family migration, caregiving, remarriage, household creation and business continuity.

12.6C Dependent minors, loss of responsible adults and care resolution

When a child or other dependent minor loses the last locally responsible adult, the death system should not assign a random household directly. It should create a Dependency Resolution Case. The case preserves the minor, their current accommodation, sibling group, estate interests, known kin and immediate shelter while the simulation seeks a workable care arrangement. The preferred simulation search order is a social-plausibility priority, not a claimed statutory kin hierarchy: surviving parent or existing lawful guardian where one exists; suitable close adult kin; established non-kin relationships such as a household already closely connected to the child or parents; another suitable local household; then an off-map relative, public, charitable or institutional path where historically appropriate. Suitability considers accommodation, resources, existing dependents, health, willingness, relationship strength, travel practicality

and the ability to provide care. Sibling groups receive a strong preference for placement together, but the simulation may separate them when no plausible arrangement can take all of them. The receiving household does not automatically acquire the minor's inherited assets. Care, guardianship, estate ownership and property management are different relationships. A child can live with an aunt while an estate remains legally distinct and a guardian or other lawful authority administers the minor's property. When relevant status changes later, direct control can pass according to the legal regime. Historical anchor: the 1887 Compiled Laws of Dakota Territory contains a dedicated Chapter 13, Guardian and Ward, including an article on Guardians of Minors. This establishes formal guardianship as a real territorial legal institution by 1887. Exact spring-1880 procedures, appointment priority, county practice and property-management rules remain a LEGAL RESEARCH HOLD rather than being inferred backward from the 1887 compilation.

12.6D Child contribution, dependency and adult authority are separate

Do not use one Age >= X switch for work, independence and legal authority. The simulation should distinguish at least three concepts: economic contribution, residential/domestic independence and legal/contracting authority. A child can help a household or earn wages before being capable of an autonomous adult household or full legal control of property. The 1880 U.S. Census instructions required an occupation entry for each person age ten and upward and further instructed enumerators to record a younger boy or girl's occupation when that child was regularly earning money, contributing to family support or appreciably assisting in mechanical or agricultural industry. The same instructions also warned that ordinary chores or errands should not automatically be treated as a gainful occupation. This is an unusually useful design boundary: age ten is historically old enough that meaningful work can exist, but it is not a canon declaration that every ten-year-old works or that a ten-year-old becomes an independent household head.

- Very young dependent: primarily consumes care and cannot meaningfully cover ordinary productive workload.
- Child/helper stage: can perform appropriate chores and, where actual circumstances support it, meaningful family-business or agricultural work.
- Youth stage: broader wage work, apprenticeship or regular productive contribution becomes increasingly plausible, subject to occupation, schooling, health, household need and legal context.
- Independent adult stage: can normally form a one-person household and exercise the ordinary adult authority permitted by the current legal regime. Exact age boundaries, territorial/state legal majority, independent contracting capacity, schooling requirements and occupation-specific child-labor restrictions remain a targeted research hold. The important canon lock is that contribution begins before full independence and that those states cannot be collapsed into one threshold.

12.6E Fertility is person/couple based, not household based

Births arise from actual people. The household does not roll for a baby merely because two adults are present. The demographic system identifies a biologically compatible reproductive couple and evaluates conception over time using age, pregnancy state, fertility variation, time since a prior birth, health and whether the partners are actually co-resident or otherwise in a plausible relationship for conception. A multigenerational household can contain more than one reproductive couple; an elderly couple, widow household or one-person household does not generate births simply because the Household entity exists. Do not assign every couple a visible Desired Children = 6 quota and do not impose a universal child cap. Individual variation should emerge from marriage timing, age-specific fertility, spacing, underlying fertility, widowhood, separation, mortality, migration and chance within historically calibrated rates. Some couples will have no children, some one or two, many several and a minority very large numbers of births. The distribution should be calibrated at the population level rather than scripted at household creation. The simulation does not need a reproductive physiology minigame. A monthly or similarly low-frequency demographic review is enough. Conception creates a pregnancy state; birth creates a persistent new Person; subsequent conception becomes possible only when the previous pregnancy and a plausible recovery/spacing interval allow it. Exact spacing distributions are a research/calibration question rather than a fixed two-year rule.

- Birth has real crossover consequences. Dependency and domestic workload rise; consumption changes; accommodation suitability/crowding can change; the pregnant or postpartum person's available work can change for a period; later the child becomes a helper, worker, heir, migrant or business owner.
- Maternal and child risk are causal demographic risks. Pregnancy, birth, infancy and childhood must be materially riskier than in a modern setting, but outcomes should be historically calibrated and not staged as constant dramatic roulette.
- Birth count is not resident-child count. Children may die, leave home or live elsewhere. Family history and current household membership are therefore different datasets.

12.6F Historical demographic calibration anchors, not quotas

Revision XII adopts several broad historical anchors for automated calibration while explicitly refusing to treat national values as exact Black Hills rules.

- Fertility: the Social Security Administration's historical U.S. series gives total fertility of about 4.24 children per woman in 1880, 3.87 in 1890 and 3.56 in 1900, with a modest rebound to about 3.50 in 1905. The same series adjusted to count children surviving to age ten is about 2.96 in 1880 and 2.78 in 1900. Use these as aggregate order-of-magnitude targets and evidence that births and surviving children must not be conflated.
- Marriage timing: Census Bureau historical estimates place median age at first marriage around 26.0 for men and 21.9 for women in 1890, with very similar 1900 values. Use the late-nineteenth-century pattern - women generally early twenties, men generally mid-twenties - as a national calibration anchor, not a frontier hard age.
- Family-size context: the 1890 census reported 5.04 persons per census family nationally in 1880 and 4.93 in 1890. The 1900 census reported South Dakota rising from about 4.7 in 1890 to 4.8 in 1900 and specifically observed that newly settled communities could begin with smaller average families and grow as settlement matured. This supports a young frontier seed with many single or small domestic units without making it ahistorically childless.
- Census terminology warning: the 1880 census definition of family is not identical to the game's Household authority. Census instructions counted a person living alone as a family, but could also count all hotel inmates as one census family under a common roof/table. Land & Ledgers deliberately separates distinct domestic economies inside boarding/hotel buildings. Historical family-size values therefore require translation before balance use. These values are calibration envelopes. Opening age/sex structure, migration selectivity, settlement maturity, local mortality and household formation can move a particular seed away from national averages. The correct test is whether repeated 25-year simulations produce plausible distributions and trends, not whether every household resembles the national mean.

12.6G Opening population generator: people first, relationships second

World generation should not begin by selecting Family of Five, Single Miner or Large Ranch Family as opaque household templates and then filling them with decorative people. The preferred order is:

- 1. Generate the resident population envelope appropriate to the spring-1880 settlement, including age, sex, origin/migration status, occupation potential and the frontier-biased presence of single/transient adults.
- 2. Establish explicit kinship and marriage relationships consistent with ages and migration history.
- 3. Generate prior children/birth histories where parental ages and demographic calibration support them, including children who are deceased or off-map when that history matters.
- 4. Form households from actual domestic/economic relationships, allowing one-person, nuclear, multigenerational and kin-extended households.
- 5. Assign accommodation based on property, rent/board offers, employer lodging, kin hosting, temporary shelter, household size, affordability and suitability.
- 6. Validate aggregate outputs - residents, household-size distribution, single adults, child share, boarders, renters/owners, age structure, labor supply and housing pressure - then reroll or rebalance only when the seed is structurally implausible. This order gives every household an explanation. A 51-year-old couple with adult and teenage children looks different from a 25-year-old couple with two small children because their actual ages and history differ, not because a household archetype selected a sprite set.

12.6H Accommodation relationship ontology

A household does not require ownership or occupancy of a conventional dwelling to exist. Household identity persists through boarding, renting rooms, employer accommodation, kin hosting, doubled-up housing, temporary shelter, displacement and periods without stable shelter. Housing systems must therefore model the relationship between household/person and accommodation rather than embedding the household inside a house. The canonical relationship chain is conceptually: Person -> Household membership -> Residential Occupancy / Accommodation Arrangement -> Accommodation Space -> Building -> Parcel The layers answer different questions. A Person is one human being. A

Household is a domestic/economic grouping. Family/Kinship is the relationship network and does not require co-residence. Residential Occupancy records who is sheltered where and on what terms. An Accommodation Space is an economically meaningful place capable of sheltering people. A Building is the physical structure that may contain commercial, storage and accommodation spaces. A Parcel is the property geography and legal land interest beneath it. A change at one layer should not silently rewrite the others. A fire can destroy an accommodation space while the household, kinship, employment, debts and ownership rights remain. The resulting loss of accommodation then creates migration, boarding, construction, credit and labor consequences through their own systems.

12.6I Accommodation states and housing security

Residential arrangement is not a class. A household can move among forms over its history according to money, work, family size, availability and expected permanence. Supported states include owner-occupying, independent rental, partial-building or room rental, boarding/lodging, employer lodging, kin-hosted accommodation, doubled-up/shared occupancy, transient lodging, temporary shelter/camp, precarious accommodation and no stable shelter.

- A tent or rough temporary shelter can be real accommodation. Do not implement HasHouse = false -> Homeless = true. A tent can have a location, weather protection, heating limitations, storage limits, fire exposure and season-specific suitability even though it is not conventional real estate.
- Housing security is separate from tenure. Owner-occupation can be precarious under foreclosure pressure; employer lodging can be stable while the job lasts; kin hosting can be stable or temporary; boarding can be durable for a single worker. Use actual termination,

affordability, crowding and dependency conditions.

- No stable shelter is a pressure condition, not a permanent population state. Displacement should trigger attempts to board, rent, stay with kin, use temporary shelter, seek employer accommodation or leave the node according to actual options. A permanence ladder must never become a linear upgrade ladder. A transient worker may rationally prefer a bunk or tent despite being able to afford more; a merchant family may value a secure private dwelling strongly. Suitability is actor-specific.

12.6J Shared buildings, separate households and multi-location members

Two households can occupy the same physical dwelling or accommodation space without becoming one household. If the Carter household temporarily stays with the Thompson household after a fire but keeps separate earnings, debts, shopping and future plans, they are two households sharing accommodation. They merge only when their domestic economy actually merges. Conversely, not every member of one household must sleep at the same coordinates every night. A ranch family member can lodge near seasonal work or a worker can maintain weekday lodging near a mine while remaining economically attached to a primary family household. Architecture should therefore allow a household-level Primary Domestic Anchor plus person-level current residence assignments when needed. Most people will simply inherit the household's primary accommodation; exceptions should be supported without inventing a second household. Boarding houses demonstrate the distinction. One building may contain the proprietor household, several one-person worker households, a married couple household and transient people whose permanent household is elsewhere. Shared meals or a common roof do not automatically merge their finances. This deliberately differs from the 1880 census enumeration convention that could count a hotel and all inmates as one census family. Mixed-use buildings are equally valid. A General Store can contain a commercial operating space, owner quarters and one or more rentable rooms where the authored building plausibly supports them. Parcel owner, building owner, business operator and residential occupant can all be different parties. The simulation tracks only economically meaningful functional spaces; it does not need to model every bedroom, hallway or closet as a separate management object.

12.6K Accommodation Space capabilities and emergent suitability

Accommodation should expose real capabilities rather than a universal Housing Quality score. Depending on the space, relevant characteristics include shelter/sleeping capacity, weather protection, heat source/capability, water access, sanitation, cooking capability, storage, privacy/shared status, condition, fire exposure, yard/outbuilding access, proximity to work/services and seasonal accessibility. Suitability emerges from household/person x accommodation x season. A one-person worker can tolerate a summer tent that is unacceptable for a large family in winter. A room over a store may provide excellent town access but poor yard access. A remote employer cottage can be secure yet tie the household closely to one job. The property does not contain a magic Suitability = 54 value; the same physical facts are evaluated differently by different households. Crowding remains a consequence rather than an arbitrary lock. Accommodation capacity should represent practical sleeping/shelter ability and increasing strain, not a gamey hard maximum that prevents emergency doubling-up. Severe crowding can increase privacy loss, domestic workload, sanitation/fire pressure and the incentive to move or build without erasing the emergency option.

12.6L Accommodation pressure, migration and practical labor supply

Settlement housing pressure should be measured by accommodation match, not Population - Beds. A node can have two vacant cabins, three rented rooms, four boarding beds, employer bunk space and temporary camp capacity yet still face a severe shortage for households with several children. Single workers and family migrants therefore see different opportunity sets. Migration becomes a real search problem: can this person or household identify accommodation they can afford, access and accept? A single miner may accept a bunk. A merchant family may need stable private space. A wealthy entrant may purchase or build. A desperate household may double up with kin. The same town can attract transient labor while failing to retain families. This directly modifies effective labor supply. An employer may know that twenty-five workers are willing to take the wage, but only nineteen can relocate immediately because practical lodging is scarce. Others can commute if geography permits, wait, accept camp conditions or refuse. Employer-provided lodging therefore has real recruitment value because it changes the accommodation constraint, not because it applies a labor bonus. Temporary camps allow settlement to grow ahead of permanent housing. A mine or rail/construction surge can produce tents, rough shelters, saturated boarding and doubled-up dwellings rather than a hard No Housing -> No Migration gate. The resulting crowding, weather exposure, sanitation strain, fire exposure, lumber demand, food demand and construction opportunity should be visible and should create the economic pressure that causes the node to harden into a town - or fail if the employment pulse disappears.

12.6M Lifecycle continuity and relationship persistence

Demographic and household change should preserve the continuity of actual people and relationships over the full campaign. Birth, death, marriage, migration, household formation, household separation, displacement, guardianship and changes of accommodation should alter the relevant relationship without erasing unrelated history. A person remains the same person when they leave a household, move into boarding, change jobs or inherit property. A household remains the same domestic unit when it moves from one accommodation to another unless the domestic economy itself splits or merges.

The player should experience these changes through people, notices, leases, payroll, housing pressure, succession, estate events and visible town life rather than through administrative screens. The design must preserve enough historical continuity that later systems can answer who lived where, who worked for whom, who is related, who owes what, what property is owned and why a current arrangement exists.

12.6N Calibration loop and research-hold register

Demographic balance should be tested by running many full 1880-1905 simulations and evaluating distributions rather than hand-authoring a few plausible households. Calibration reports should include births per woman, surviving-child distribution, first-marriage timing, age structure, household-size distribution, one-person households, boarders/lodgers, renters/owners, widow/widower share, household splits/mergers, orphan/dependency cases, migration by life stage and total population growth. Targeted research holds before exact constants are locked: Dakota/Black Hills marriage and remarriage patterns; childlessness and parity distribution; birth spacing; maternal, infant and childhood mortality by place and subgroup; territorial/state legal majority and contracting capacity; school-attendance and child-work law chronology; spring-1880 guardianship procedure and county/charitable placement practice; detailed boarding/lodging household enumeration; and how census-family statistics should be translated into the game's narrower domestic-economic Household definition. The canon is already strong enough to build the architecture without inventing these constants. Where evidence is absent, preserve the causal relationship and label the calibration open. This follows the project-wide rule: understand real process depth first, then simplify only where the surviving abstraction keeps the important cause-and-effect.

REVISION XII RESEARCH BASIS - PRIMARY/HIGH-AUTHORITY ANCHORS

U.S. Census Bureau, 1880 Census Instructions to Enumerators: defines persons living alone in cabins, huts, tents or rooms above stores; defines census-family conventions; requires occupation reporting for persons age ten and upward and permits recording younger children doing regular productive work. U.S. Census Bureau, Compendium of the Eleventh Census: 1890 - Progress of the Nation: national persons per census family 5.04 in 1880 and 4.93 in 1890. U.S. Census Bureau, 1900 Census, Volume 2: Population, Part II: South Dakota average census-family size 4.7 in 1890 and 4.8 in 1900 and discussion of smaller average families in newly settled communities. U.S. Census Bureau historical first-marriage series: 1890 medians approximately 26.0 men and 21.9 women. Social Security Administration historical fertility series: total fertility approximately 4.24 in 1880 and 3.56 in 1900, with a survival-to-age-ten adjusted series around 2.96 and 2.78 respectively. Compiled Laws of the Territory of Dakota, A.D. 1887: Chapter 13 Guardian and Ward, including Guardians of Minors. These anchors support architecture and broad calibration; they do not erase Black Hills-specific research holds.

12.7 Residential Property, Tenancy & Housing Markets

Housing is now a full household, property and investment system rather than only a settlement-support variable. Every physical dwelling or residential unit should know its owner, occupant household/person, tenure arrangement, agreed rent/board terms where applicable, condition, maintenance, debt/encumbrance state and physical housing

characteristics. A household does not simply have HousingState = Rented; it occupies an actual unit owned by an actual person, estate, company or other entity.

12.7A Historical tenure calibration

The federal census did not systematically ask ordinary households whether homes were owned or rented until 1890, so there is no defensible exact 1880 Dakota renter percentage. The 1890 South Dakota nonfarm benchmark is nevertheless strong: approximately 45.12% of all reported nonfarm homes were rented; outside cities of 8,000 or more, about 44.13% were rented and 55.87% owner- occupied; in the larger-city category, renting was higher. The 1900 known-tenure nonfarm data still indicate roughly 40% renting. For the fictional tiny 1880 gateway settlement, the provisional design calibration is approximately 35-45% of independent nonfarm households renting and 55-65% owner-occupying, with roughly 40% renters as a useful initial target. This is an inference anchored to the nearest direct census evidence, not a claim that an 1880 census measured exactly 40%. Boarders, lodgers and transient workers are additional housing arrangements and should not be counted as independent renter households in that percentage.

12.7B Who owns rented housing

The census tenure tables identify occupancy tenure more reliably than landlord type, so Revision XI does not invent landlord percentages. Historically supported ownership forms include private individuals/investors, ordinary households owning an additional property, widows/heirs/estates, boarding proprietors, employers or mining companies in contextual settings, and lenders/receivers that acquire housing through distress. Black Hills records include purpose-used rental houses and boarding businesses, demonstrating that owner and occupant can be different real economic actors.

12.7C Residential units and tenure products

- Owner-occupied dwelling: the household owns the accommodation it occupies.
- Independent rental dwelling: a household rents a house or self-contained residential unit and runs its own domestic economy.
- Rented room/lodging: an individual or small household rents part of a larger property without necessarily receiving meals.
- Boarding: accommodation plus services, especially meals where included, with its own capacity and operating-business economics.
- Employer lodging: bunkhouse, cottage, room or board supplied as part of or alongside an employment relationship.
- Kin/doubled-up housing: people stay with relatives/another household without becoming a normal commercial tenancy; this can relieve housing pressure while increasing crowding. One building can support more than one residential relationship only where the authored physical structure supports it: owner residence plus rooms, commercial ground floor plus rooms above, boarding arrangement or multiple suitable units. The simulation structure should permit this without generating anachronistic apartment patterns by default.

12.7D Rent freedom and market formation

Landlords set actual asking rents rather than receiving a townwide rent modifier. The player may ask whatever amount they choose, subject to applicable period law or contract. Households compare available options and may refuse, move, crowd elsewhere or leave the node. Estimated market rent can exist as analysis, but it is not an enforced maximum. Average town rent is an output of real agreements. Housing shortage can cause landlords to ask more and households to accept more because alternatives are scarce; new construction responds with delay because it requires land, capital, lumber/hardware, labor and time. If many investors respond at once, a later housing surplus can create vacancies and rent competition.

12.7E Household housing choice

Household choice should consider affordability, size/suitability, crowding, location relative to work and services, condition, heating burden, water/sanitation arrangement, yard/outbuilding usefulness where relevant, relationship/kin factors and available alternatives. Cheap nominal rent is not always the cheapest household option if a drafty distant house consumes far more fuel and travel time. Suitability is household-specific. A small room may be reasonable for a single worker and poor for a family with children. The game should avoid a universal House Quality score as the sole decision input.

12.7F Crowding as consequence, not occupancy lock

Reasonable capacity can be estimated from usable residential space and sleeping arrangements, but it should not be a hard maximum. Families can double up, take relatives or accept crowded housing when scarcity demands it. Crowding

can increase wear, domestic burden and, where historically/medically justified, sanitation/health risk; exact health effects require targeted research. The important rule is freedom with consequence rather than Cannot Move In because capacity is exceeded.

12.7G Rental-property owner decisions

- Set rent and change it when the current agreement/law permits.
- Choose applicant-review policy and whether an agent/manager can accept ordinary tenants automatically under standing criteria.
- Set tolerance for arrears, partial payment or case-by-case exceptions; exact eviction/removal process remains a Dakota/South Dakota legal research hold.
- Maintain, repair, refurbish, expand, connect utilities, convert or sell the property.
- Decide whether rooms/properties are furnished or services/fuel are included where historically appropriate; exact common arrangements require research.
- At scale, delegate rent collection, vacancy letting and minor repairs while retaining approval for rent policy, major renovation, sale, debt and exceptional tenant situations.

12.7H Rent arrears and household cash allocation

Rent is an actual household obligation paid from actual cash or negotiated arrears. A household with $11.40 that simultaneously owes $8 rent, needs $5 of winter fuel and owes a merchant account cannot satisfy every claim magically. Household priority, urgency, credit access and landlord policy determine what is paid or postponed. Missed wages can therefore propagate from employer to household to landlord to lender and merchant. Rental ledgers should track current rent due, past-due balance, partial payments, last payment and arrears age where the system supports it. Arrears are structurally similar to commercial receivables but should follow the historically correct tenancy/legal process rather than automatically using merchant-credit rules.

12.7I Vacancy, turnover and physical moves

Vacancy is a real cost. When a tenant leaves, rent stops until a new household agrees and physically occupies the property. A landlord can re-rent immediately, hold the unit vacant for repairs/refurbishment or deliberately wait for a higher-paying market. High asking rent may increase income per occupied month but also increase vacancy/turnover risk. Moves follow Physical Truth. The household leaves one actual dwelling and enters another. Close moves can be represented at an appropriate abstraction with people and household goods/cart movement; off-map migration follows the established hidden-boundary transition. The simulation should not change address while the people visibly remain elsewhere.

12.7J Landlord maintenance, exploitative operation and market consequence

Residential property plugs directly into the Asset Condition system. Roofs, stoves/chimneys, structure, weatherproofing, privies/water arrangements and later utility connections can require real work. Vacancy can be a valuable repair window. A landlord may deliberately defer work and extract high rent during scarcity. The game does not need a Slumlord class or morality perk; exploitative landlord behavior emerges from high rents, crowding, low maintenance and short-term cash extraction. Such a strategy can genuinely succeed while housing is scarce. Longer-run consequences emerge through deterioration, fuel burden, fire/sanitation exposure, tenant willingness to remain, competition from better housing, future capital repairs and any historically grounded civic/legal pressure. The game should not apply an automatic moral penalty merely because the player chose an unpleasant strategy.

12.7K Employer housing and labor-market integration

Employers may build or control housing where geography and business scale justify it, especially remote mines or industrial sites. Housing can be offered as part of compensation or rented separately according to historically appropriate arrangements. It can make recruitment feasible where jobs exist but accommodation does not. It also creates dependence: shutdown can remove both income and employment- linked housing. Company housing is not a universal requirement. A sawmill beside a functioning town can rely on independent landlords; a remote mine twenty miles away may need bunkhouses or cottages. The choice depends on land, capital, labor need and existing settlement supply.

12.7L Residential construction, speculation and boom/bust

NPC and player housing construction should be an investment decision driven by land, expected occupancy/rent, construction cost, finance, maintenance and market outlook. The simulation should not spawn the exact housing

quantity demanded. Supply responds with delay and can overshoot. This makes speculative housing possible. A player can build near a proposed depot or mine before demand exists. If the project arrives, occupancy and land value can surge. If the railroad bypasses the town or mine fails, the player owns empty buildings with debt and maintenance. Housing surplus can then lower rents, improve household accommodation and push weak landlords toward sale/default.

12.7M Utilities, neighborhoods and nuisance without zoning

Water, sewer, electricity and telephone connections can become property-level landlord investments when the network reaches the site. Connection may support higher rent/lower vacancy or reduce household burden, but it costs capital and does not automatically pay back. Partial adoption allows wealthy and poor streets to modernize unevenly.

Residential preference can respond to nuisance, manure, smoke, industrial noise, rail yards, slaughtering, fire exposure and traffic without a modern zoning lock. Lower rents/land values can make undesirable areas economically acceptable to some households. Neighborhood form emerges from price, access, externalities and land use.

12.7N Distress, foreclosure, estates and tenant-to-owner movement

Rental property participates fully in the property/finance system. A distressed landlord can sell one house, a package or an entire portfolio, refinance, defer maintenance or default. Estates can inherit occupied rentals. Lenders can acquire property through the appropriate foreclosure process and then temporarily collect rent or resell where legally supported. Existing tenancy treatment through foreclosure remains a legal research item. Households are not permanently typed as renters. A renter can accumulate savings, obtain period-appropriate financing, buy an occupied house from the landlord or purchase elsewhere. An owner can lose property and become a renter. Tenure is a life/economic state, not a social class.

12.7O Residential holdings at portfolio scale

Early rental property can remain directly owned income-producing property rather than requiring a formal business entity. As holdings expand, the player can hire historically appropriate agents/property managers if research supports the title/structure. Portfolio UI should show property, occupant, rent, payment status, vacancy duration, condition, debt and exceptions so dozens of units remain manageable without losing the ability to inspect one household/property in detail.

PART VIII

Markets, Competition & Adversity Demand, rivals, pricing and conditions that can push back against accumulation.

READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

14.1 Household shopping logic

Households should buy from the business that makes the most sense, not automatically from the player. Demand resolves mainly at household level for clarity and performance, but a specific person performs the local trip. Household choice should consider price, stock, operating hours, distance or convenience, Business Reputation, remembered experience, credit availability, urgency and accessible substitutes. A local purchase should not complete before the shopper actually reaches the business and the business is still able to serve the purchase. If normal local options are closed, unavailable or unsuitable, the household can wait, substitute, seek another local merchant or, when urgency and travel economics justify it, obtain goods through an off-map trip or channel. Owner Reputation remains a smaller trust modifier or tie-breaker, not the main store-choice driver.

14.1A Customer urgency, information, and remembered relationships

Not every need has the same urgency or tolerance for delay. A household can postpone discretionary cookware, but food exhaustion, winter fuel shortage or a work-critical replacement can justify higher prices, unusual travel, emergency credit or inconvenient hours. Customers should act on information they plausibly possess rather than omniscience. Repeated successful purchases, stockouts, accepted credit and other direct experience can shape future choice without requiring one exposed loyalty score. The player likewise should not receive exact private household cash or rival books without an information source; ownership, account records, employment and later banking or professional reports determine what can be known precisely.

14.2 Reserve-based shopping and pricing

Target stock is what the household would ideally like to hold. Reserve stock is the point where the category joins the shopping list. Critical reserve is where urgency spikes and faster shopping attempts become likely. Routine shopping should remain threshold-driven rather than scripted to a rigid weekly day. Overpricing should push households toward alternatives when they exist; underselling can still sink the business through margin failure.

14.3 Commercial accounts, credit posture, and early-game action density

Customer credit is a shared commercial capability rather than a General Store-only mechanic, but it should appear only where historically appropriate for that business relationship. The current managing authority can choose cash-only versus credit, broad credit posture, customer/household limits, exceptions, collection tolerance and whether ordinary staff may approve small transactions. General Stores use

household book accounts; doctors and service trades can carry billed or running accounts; blacksmith/livery-type services can maintain repeat-customer accounts where appropriate; merchants and wholesalers can use business-to-business trade credit; sawmills or lumber sellers can extend account terms to repeat buyers. Ordinary store tabs do not automatically accrue interest unless a specific agreement warrants it. Credit creates accounts receivable, liquidity risk, customer retention and competitive differentiation. A generous NPC can report strong sales while starving itself of cash; a stingy NPC can drive reliable customers to a rival. Repayment emerges from actual cash-flow events, not a universal weekly script. Early-game action density still matters, but account decisions should be mostly policy-driven rather than a popup for every purchase.

14.4 Rival response and market causality

Rival openings and expansions should emerge from the same town logic the player reads. A rival should not open a business merely because a hidden number crossed a threshold. The rival should appear to be acting on visible backlog, weak coverage, rising wages, housing strain, supplier slack, newly available land, financing access, or a distressed acquisition opening. Rival action is one of the cleanest ways an opportunity becomes pressure. This preserves the player fantasy that the town is alive and reactive rather than procedurally decorative. It also ensures that defeating or buying out a rival yields inherited real conditions rather than fake clean resets.

14.5 Category markets, customer segments and practical market share

There should be no single universal Market Share number for the player's empire. Competition is specific to a good or service, a reachable customer group and a geography. A general store can dominate workwear while losing food staples; a butcher can own household trade while a rival controls hotel accounts; a freight firm can dominate one mine corridor and have little presence in farm hauling. Market share should therefore be a derived analysis built from actual transactions, unmet demand and reachable alternatives. Customer segments matter only where they change behavior: households, repeat business accounts, remote camps, institutional buyers, travelers and project demand can value different combinations of price, reliability, credit, location and volume.

14.6 Substitution, leakage and the outside option

A weak local market should not trap customers into buying from the nearest business. Households and businesses can substitute products, postpone demand, repair instead of replace, use household production, buy from another node or leak demand off-map when price, urgency and transport make that rational. Likewise, strong local merchants can capture demand from surrounding nodes. Rail, road improvement and communication can therefore increase competition as well as opportunity: the same connection that brings cheaper inventory can make outside sellers more accessible. Market reach is a two-way door.

14.6A Credible alternatives as bargaining power and resilience

Market power should be understood primarily through the counterparty's best credible alternatives, not through an abstract Influence or Monopoly meter. A worker with several employers has bargaining leverage. A mine with one practical treatment provider is vulnerable. A borrower with several willing lenders can reject poor terms. A rancher with one reachable merchant has fewer options. A town served by two viable freight routes is more resilient than one dependent on a single carrier. The same principle explains distress: as liquidity, suppliers, workers, buyers, lenders or sale options disappear, the actor's remaining choices become worse. Power and distress should therefore emerge from the structure of alternatives.

14.6B Bounded awareness and practical choice sets

People do not evaluate the entire simulation every day. A practical decision process is Trigger -> Awareness -> Feasibility -> Shortlist -> Choice. A person reconsiders a supplier, employer, store, lender or service when something gives them a reason. They know alternatives because of prior use, travel, direct observation, family/acquaintance networks, coworkers, advertising, directories, correspondence, solicitation or other supported information channels. Distance, hours, stock, transport, housing, qualifications, credit eligibility and urgency then determine which known options are actually feasible. Choice occurs among a bounded shortlist rather than through omniscient optimization.

14.7 Rival strategy and capital constraints

Rivals compete through the same kinds of levers available to the player: price, assortment, stock reliability, hours, location, customer credit, delivery, service specialization, recurring contracts, labor recruitment, supplier relationships, property acquisition and capital investment. They should not receive hidden demand, production or cost bonuses. A rival with deep cash reserves can survive a price war or carry more inventory; a weak rival may be unable to respond even when the opportunity is obvious. Rival strategy should reflect its assets, information, owner preferences, relationships and financing access. This makes an aggressive competitor dangerous because it is actually well positioned, not because difficulty mode grants it a multiplier.

14.8 Price competition, service competition and destructive rivalry

Price is one competitive dimension, not the whole market. Two stores at similar prices can differ through stockouts, credit, hours, delivery, location and trust. A price war is valid when businesses deliberately cut margin to defend or capture trade, but it burns cash and can make both firms weaker. Deep pockets therefore matter, as do purchase cost and supplier terms. Conversely, a high-price business can remain viable if it offers scarce goods, better reliability, convenient credit, superior location or a service package customers actually value. The game should allow predatory-feeling strategies to fail economically without requiring an invisible fairness rule.

14.9 Advertising and commercial information

Advertising should change who knows an offer exists, not manufacture demand from nothing. The Black Hills press already carried business advertising in the opening era, including hotels, banks, blacksmiths, groceries, restaurants and professional services. Newspaper notices, signs, handbills and later faster communications can therefore announce openings, new stock, special terms, routes, services or sales to customers who might otherwise remain unaware. Advertising works best when the underlying offer is useful. Repeated promotion of empty shelves or bad service can waste money or even spread disappointment faster. The value of advertising rises when the reachable market is large enough to justify the cost and when information is otherwise fragmented.

14.10 Entry, overcapacity, exit and market succession

Profitable unmet demand should attract capital eventually, but entry requires a suitable property, startup capital, labor, supply and a willing owner or investor. New businesses therefore arrive with delay and can overshoot. Several merchants may all respond to the same boom, creating later overcapacity, margin pressure and vacancy. Weak firms can narrow their service, seek credit, merge or sell, change location, accept a buyer, close or fail through distress. Their customers, workers, supplier commitments and property then become opportunities for others. This creates market succession: a strong player can win not only by undercutting rivals, but by surviving a downturn, buying a distressed location, hiring displaced workers or absorbing abandoned accounts.

14.10A Entry requires a business thesis

NPC entry should not occur merely because a margin statistic crossed a threshold. A plausible entrant needs awareness of the opportunity, an operator or investor willing to act, capital/credit, property or equipment, labor, supplier access, customer reach and an expectation that the enterprise can survive likely incumbent response. If nobody can assemble a credible business thesis, a profitable-looking gap may remain open. Successful player dominance therefore does not require the simulation to spawn a challenger for balance alone.

14.11 Competitive memory and reputation repair

Customers and counterparties should remember economically meaningful experiences rather than accumulate one permanent loyalty score. Successful fills, fair handling of credit, reliable contracts, repeated stockouts, surprise closure, poor repair, bad lodging or missed deliveries can all shape future choice. Memory should fade when the event is old or when new evidence contradicts it, allowing businesses to recover from a bad period through sustained better operation. Ownership change can reset expectations only partially. A new proprietor may create curiosity, but inherited employees, location, stock, debts, service failures and local stories remain until real operation proves otherwise.

14.11A Commercial memory types

The economy should preserve memory selectively rather than invent one permanent Reputation score. Personal memory is what a person remembers or has learned. Relational memory is what two counterparties know about dealing with each other. Recorded memory is preserved in books, correspondence, contracts, account histories and reports. Procedural memory is embedded in experienced employees and routines. These forms survive different disruptions. Losing the founder may remove personal knowledge while records and staff preserve continuity. Losing both staff and records can genuinely reduce organizational capability.

14.11B Reputation-score interpretation lock

Where earlier Canon wording uses Business Reputation, Owner Reputation, employment relationship history or similar shorthand, Revision XIV interprets that term as a summary of specific commercial evidence and remembered history, not as one universal permanent scalar that mechanically modifies every interaction. A seller can trust the player's closing record; a worker can distrust one employer's treatment; a bank can value repayment history; households can remember stockouts. Those facts may be summarized for readability, but the causes remain relationship-specific.

REVISION V INTEGRATED

3.2B Strategic failure doctrine

The simulation should be willing to let the player permanently lose individual battles without ending the campaign. Strategic failure is an essential source of consequence and should arise through the same grounded economic causality as success.

- Property or acquisition opportunities can be lost to rivals or seller decisions.
- Businesses and collateral can be sold, seized, foreclosed, closed, or otherwise lost.
- Important workers can resign, die, become unavailable, or accept competing opportunities.
- Supplier relationships or scarce supplier capacity can be lost.
- Household preference, customers, market share, and recurring commitments can shift to competitors.
- Rail corridors, financing opportunities, investment opportunities, and settlement momentum can be lost.
- A settlement can stall, decline, ghost, or be surpassed without automatically creating player defeat.

REVISION V INTEGRATED

3.2B-1 Distress as deterioration of alternatives Business distress should be legible as a shrinking set of credible responses rather than a universal Business Health meter. Cash disappears, lenders tighten, suppliers shorten terms, managers leave, collateral is already pledged, customers defect, asset buyers vanish or a dependent market collapses.

The business becomes dangerous because fewer good moves remain. One entity can lose its options, be sold, closed, liquidated or taken by creditors while the owner continues elsewhere.

3.2B-2 Failure creates recombination Failure should often redistribute useful economic pieces instead of deleting them. A closed enterprise can release property, machinery, wagons, inventory, workers, managers, customer relationships, supplier capacity and local demand. Creditors, rivals, employees, managers or the player can recombine those pieces into new ownership structures. Distress can therefore be both destruction and opportunity, giving the economy continuity across booms and downturns.

3.2C Adversity as part of the ownership loop

The economy is not considered proven if the dominant loop is only profit -> expansion -> more profit. MVP must include enough grounded adversity to demonstrate that external and internally generated forces can weaken the player, create difficult ownership decisions, produce strategic losses, and occasionally push a portfolio toward distress. Adversity should primarily emerge from simulation state and readable economic cause. It should not become a random disaster deck or a constant popup stream. MVP adversity families

- Rival competition: rivals can open or expand businesses, acquire distressed competitors, compete for parcels, recruit workers, compete for supplier capacity, alter prices, and act on visible profitable openings. Rival causality rules from Revision IV remain authoritative.
- Supply pressure: real production, supplier, transport, or availability problems can make procurement harder and can eventually affect prices. Revision V causal-supply rules below are mandatory.
- Labor pressure: illness, injury, resignation, competing wage offers, and skilled-worker scarcity can reduce operating-role coverage. Severity depends on the business and the role lost.
- Financial pressure: payroll, debt service, inventory purchases, expansion, project costs, and disruption can create genuine liquidity stress that makes owner cash, business cash, refinancing, transfers, and asset sales meaningful.

REVISION V INTEGRATED

3.2D Persistent conditions, not notification spam

Economic problems should normally exist as persistent conditions that evolve through the existing signal ladder. The player should be interrupted when a meaningful threshold is crossed, not every time the same underlying problem updates.

- Early signal: condition is visible but not urgent.
- Mounting signal: repeated evidence shows the town or business is compensating badly.
- Action signal: the condition has crossed a threshold where ownership response is warranted. One continuing freight or supply problem should mature from weakening to mounting to critical rather than generating a fresh alert every day. UI should explain cause and trend while the world provides physical or behavioral evidence where practical.

3.2E Fire, conflagration, and physical recovery

Fire must not operate as a universal annual disaster probability. Ignition comes from credible building, heating, lighting, cooking, industrial, machinery, or environmental conditions; spread then depends separately on combustible contents, construction, spacing, neighboring uses, wind, detection, water access, and firefighting response. Randomness may decide whether a credible hazard actually ignites, but it must not bypass the physical conditions that make the fire plausible. Wood-frame frontier construction is inexpensive and historically appropriate but creates greater conflagration exposure when buildings are dense. Brick and stone can materially improve containment without making a building fireproof. Water mains, hydrants, organized volunteer fire companies, hose or hook-and-ladder equipment, and response distance are real settlement infrastructure and should protect only the properties they can practically reach. Fire damage should resolve against physical authorities separately: inventory can burn while a structure survives; a building can be lost while the business continues in temporary premises; workers or residents can be injured or displaced; neighboring properties can suffer partial losses; and utility or freight assets can be damaged independently. Insurance can exist as a later risk-transfer product, but exact period premiums remain a tuning/research question. Major fire can also create opportunity and restructure a node. Rebuilding can drive lumber, masonry, hauling, labor, finance, temporary-premises, and land demand; strong towns may rebuild quickly and in more permanent form while weak nodes can lose businesses or population. Wildland and prairie fire should remain a separate environmental hazard family whose behavior depends on vegetation, dryness, wind, terrain, and ignition rather than the urban structure-fire model.

3.2F Fire service progression and municipal fire rules

Fire protection can mature from ad hoc bucket response to an organized bucket brigade, volunteer hook-and-ladder or hose company, hydrant-supported response and, only in sufficiently large late-campaign towns, paid professional service. Volunteer organization should remain normal through most of the campaign. Fire-company capability, water coverage, equipment and actual response distance affect suppression; the existence of a town fire organization is not a universal percentage reduction. Growing territorial towns can impose specific fire-prevention rules without adopting modern zoning. Historically grounded powers include regulation of dangerous stoves or stovepipes, ashes, explosives and combustible storage, requirements for practical fire equipment, inspection and, where local pressure justifies it, more fire-resistant construction in vulnerable commercial blocks. These rules should emerge from actual fire pressure and civic action rather than a settlement-rank unlock.

PART IX

Finance & Capital Credit, banks, investors, collateral, distress and the 1880-1905 legal environment.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

18.1 Finance before player banking

Off-map finance should exist before on-map banking. Early in the game the player is borrowing from an established outside finance system that already functions as part of the world. The first on-map bank will usually arrive because the town has matured enough to support it, not because the player immediately has the capital to found one. This keeps banking from arriving as a fantasy power too early and better supports the town's own growth arc.

18.2 Multiple banks and lender personalities

The finance market should support more than one lender over time, but not necessarily multiple equal full-service banks from the first moment the town becomes bankable. The grounded version is one local bank or banker first, followed later by a second bank, private banker, stronger outside lender presence, or a player-founded bank once population, rail access, and regional seriousness justify it. Different lenders can have different reputations, collateral appetite, payment schedules, willingness to finance land versus business versus infrastructure, and tolerance for distress or speculation. This makes finance a real strategy layer rather than a flat interest-rate slider.

18.2A Deposits, confidence, and regional credit ecology

The lending market is a regional credit ecology, not merely a source of outbound loans. Banks compete for deposits, hold different reserve postures, carry different loan quality and sector exposure, and depend to varying degrees on correspondent or outside capital access. Solvency and liquidity are separate: a bank can own sound loans yet still be unable to satisfy a sudden concentration of withdrawals. Bad debt, overexposure to one boom sector, rumors, nearby suspensions, wider money-market stress, or repeated failures can therefore tighten lending through real balance-sheet and confidence channels. The UI may summarize these interacting conditions through a readable Regional Credit Climate such as Easy, Normal, Cautious, Tight, or Crisis, but the label must be derived from actual lender states rather than acting as a global modifier. Individual banks can still react differently. A strong conservative bank may continue excellent collateral lending during a tight climate; an overexposed rival may stop new credit or suspend payment. Acute bank-run liquidity stress and a longer credit depression are different phases and should not share one short event timer.

18.3 Founding versus acquiring a bank

Both bank paths are valid. The player can eventually acquire an existing bank if conditions, seller motives, or strategic pressure make that possible. The player can also found a bank once wealth, collateral, labor, and reputation justify it. Founding one should be capital-heavy and prestige-heavy, not a quick side-business. Acquisition may be cleaner in some runs; founding may be more attractive where there is no strong incumbent or where the player wants tighter policy control.

18.3A Silent partners, external investors, and subscription capital

Not all growth should be financed through debt alone. Before the full late-capital stock-and-bond layer arrives, the game should support outside investors, silent partners, subscription drives, and profit-share funding for suitable projects. These structures are especially useful for riskier ventures such as mine development, mill expansion, remote service clusters, boarding expansion, or speculative district plays where a conservative lender may be unwilling to advance enough capital. The important distinction is that lenders protect downside while investors buy into upside. Debt preserves control but burdens repayment. Equity-like outside capital reduces ownership purity but shares risk and can accelerate growth. This should become a meaningful strategic fork in the player's financial life rather than a flavor variation on the same loan mechanic.

18.4 Loan review, automation, and delegated thresholds

Bank ownership should create two overlapping play styles. First, the player can personally review important loans, read the application, inspect the borrower's creditworthiness, collateral, trade prospects, and existing obligations, and approve or deny the request. Second, the player can automate ordinary lending through staff thresholds. Small loans under a chosen amount can be processed or denied automatically according to policy. Larger loans break the threshold

and trigger a player review notice. This mirrors the project's business-cash automation logic: ordinary repeatable work can be delegated, while high-consequence exceptions still surface to the owner.

18.5 Collateral, bad debt, and instruments

Collateral should be real, varied, and instrument-specific. A borrower may pledge land, a building, a business, inventory, tools, livestock, wagons, machinery, or later paper instruments where law and local practice support them. Each debt should know whether it is secured or unsecured, what property supports it, priority where relevant, and whether the borrower carries personal liability. A pledged asset can remain in productive use before enforcement; missed payment does not instantly make the lender its owner. Bad debt should happen, and creditor recovery should proceed through workouts, judgments, seizure, chattel or mortgage foreclosure, negotiated surrender, and sale as appropriate.

18.6 Banking additions

Banking remains a late-capital and institutional power layer. Off-map finance exists before player banking; local banking arrives when a settlement becomes economically serious enough; player banking becomes a way to shape regional credit rather than a simple money button.

- Add a settlement or node bankability ladder: too small for formal banking, regional lenders watching, private banker likely, ready for first local bank, strong enough for competing banks, and regional finance hub. The ladder should consider population, business count, payroll volume, deposit demand, land value, debt demand, rail or industrial seriousness, and local confidence.
- Use explicit loan categories: working capital, property purchase, business acquisition, construction or project, rescue liquidity, infrastructure, speculation, mine or rail development, and refinance or consolidation. Each category should have distinct lender appetite, collateral preference, repayment logic, and risk.
- Expose lender posture in player-readable terms. Lenders may prefer land collateral, known borrowers, commercial expansion, distress lending, relationship lending, rail or mine development, or conservative low-risk loans.
- Player-owned banks should support policy controls and automation thresholds. Routine loans can be handled by staff policy; significant, risky, strange, or strategic loans should trigger owner review.
- Loan review should include borrower identity, loan purpose, repayment source, collateral quality, existing debt, reputation, market outlook, strategic value, and risk to the player portfolio.
- Add competitive-conflict review. If an applicant seeks funding to open, expand, or strengthen a business that competes with a player-owned or player-controlled industry, a player-owned bank can deny, reduce, delay, counter-condition, or approve the loan. This is a real advantage of owning a bank, but repeated self-serving denial can damage bank reputation, civic trust, deposit confidence, local growth, or invite rival lenders.
- Banking should create opportunities and problems across the wider sim: funding rivals, rescuing weak businesses, creating distressed assets, supporting mines or rail, accelerating secondary settlements, tightening credit after shocks, and exposing the bank to sector or node risk.
- Before enforcement, banks may attempt workouts such as revised schedules, partial payment, added collateral, refinance, or negotiated voluntary asset transfer. Manager replacement, tighter operating control, or debt-to-ownership structures should appear only where agreement, acquisition, receivership, or the business structure makes them legitimate rather than as automatic creditor powers. Recovered assets should become gameplay through resale, temporary operation, strategic hold, split liquidation, or acquisition paths.

18.7 Credit enforcement and institutions across the 1880-1905 campaign

The spring 1880 economy begins under a developed Dakota Territorial legal framework. Mortgages, liens, exemptions, separate property, corporations, banks and municipal institutions already exist. Real-estate mortgages are liens rather than automatic title transfers, and foreclosure or chattel enforcement is asset-specific and staged. A one-year real-estate redemption period is a strong territorial anchor; exact notice and court procedure can be compressed where gameplay does not need the detail. The Black Hills interest regime changes during the campaign: the general opening framework uses seven percent as the legal/default rate and permits written contracts up to twelve percent; from 1881 through June 30, 1887, covered Black Hills written contracts may set any mutually agreed rate; the general ceiling returns July 1, 1887. Lenders still price loans according to actual risk and liquidity rather than automatically charging the legal maximum. Banking and corporate organization predate statehood. A First National Bank operated in Deadwood before the campaign, demonstrating that the absence of a bank in the player's tiny opening town is an economic bankability issue rather than a legal or technological one. Congress's July 30, 1886 territorial restrictions matter to civic finance and railroad promotion: territorial local governments face a four-percent general debt ceiling and cannot treat municipal credit or corporate subscription as an unlimited subsidy instrument. South Dakota statehood on November 2, 1889 preserves existing contracts, corporations, rights, litigation and obligations rather than resetting them. The state constitutional framework uses a five-percent general local debt ceiling and later amendments expand

flexibility for specified infrastructure. The campaign begins in the 1878-1898 federal bankruptcy gap, so most early distress resolves through workouts and territorial/state creditor remedies. The Bankruptcy Act of 1898 takes effect July 1, 1898 and adds a federal bankruptcy route for qualifying late-campaign debtors. It is not modern Chapter 7/11 and should not become a universal reset. Commercial Ruin remains based on the actual exhaustion of ownership and recovery capacity.

18.8 Bank balance sheet and liquidity doctrine

A player-owned bank must operate as a business with both assets and liabilities. Deposits are money the bank owes to depositors, not free owner cash. Loans and other earning assets can generate income while tying up funds. Cash and liquid correspondent balances protect the ability to meet withdrawals and settle obligations but usually earn less. Recovered property can have value without being liquid. The central banking decision is therefore how aggressively to turn available funds into earning assets while preserving enough liquidity to survive ordinary and stressed withdrawals. Do not reduce this to one Reserve slider that directly grants profit or safety; policy should matter through the actual composition and timing of the bank's commitments.

18.9 Loan-book construction, concentration and maturity mismatch

Loan quality is portfolio-specific. A bank can be profitable yet dangerously concentrated in one mine district, cattle market, railroad project, speculative property boom or connected borrower group. The player should be able to review exposure by borrower, purpose, collateral family, settlement node and sector without receiving perfect future knowledge. Short working-capital lending, longer property debt and major infrastructure financing create different timing. A bank that funds long, illiquid assets with deposits that can be withdrawn quickly carries liquidity risk even when most borrowers remain solvent. Seasonal agriculture, payroll cycles, construction and tax or trade timing can create predictable cash-demand periods that a prudent banker learns to prepare for.

18.10 Deposit relationships and confidence

Deposits should come from actual households, businesses, institutions and counterparties that choose where to keep money when local banking exists. Their choice can reflect convenience, safety perception, relationship, service, confidence, the bank's owner reputation and available terms where historically supported. Large depositors create useful funding but concentration risk: losing one mine payroll account or major merchant balance can matter more than many small withdrawals. A bank run is therefore an extreme acceleration of a real liability, not a random event card. Rumor can matter only because depositors already have funds they can attempt to withdraw. Confidence can recover through credible liquidity, continued payment, outside support, improved balance sheet condition and time; a cosmetic reputation purchase should not stop a real run.

18.11 Correspondent access, outside capital and panic transmission

Local banking remains connected to wider finance. Correspondent or outside balances can support settlement, remittance and liquidity, while dependence on those relationships can transmit distant stress inward. Historical banking panics show interior banks withdrawing balances from money-center banks and money-center strain then restricting currency back to the interior. Land & Ledgers should preserve that two-way dependency without simulating the national financial system bank by bank. During an outside panic, correspondent availability, withdrawal demand, asset saleability and new-loan appetite can all worsen through separate channels. A conservative local bank may remain open but tighten credit; an overextended bank may suspend or fail.

18.12 Bank income, operating cost and the cost of being conservative

A bank earns through interest, fees and other supported financial services while paying staff, premises, communication, legal/collection and loss costs. Holding more liquidity can reduce earnings because fewer funds are deployed; pushing nearly every available dollar into loans can increase short-run income while making withdrawal stress dangerous. Lending to familiar insiders may reduce information cost and also create concentration or conflict. The player should face real tradeoffs between growth, liquidity, credit quality, customer service and strategic influence rather than maximizing a single interest-rate control.

18.13 Workouts, foreclosure assets and bad-debt resolution

A defaulted loan does not instantly turn collateral into cash. Workouts can preserve value when a viable borrower needs time, but they can also delay an inevitable loss. Enforcement can leave the bank holding a house, farm, store, inventory, machinery or business that requires management, maintenance, rent collection, sale effort or temporary operation. Recovered assets can depress reported loan losses while creating new liquidity and management problems. The player can liquidate quickly at a discount, hold for a better market, appoint management, restructure the borrower or sell to another investor when law and the asset permit it. This makes bad debt part of ownership gameplay rather

than a red number that disappears after foreclosure.

18.14 Bank governance, conflicts and institutional reputation

Owning a bank creates power and scrutiny. The player can finance projects, rivals, suppliers, customers and settlements, but repeated self-serving credit denial or unsafe insider concentration can weaken depositor confidence, relationships and institutional standing even when technically legal. The game should not apply a moral score to aggressive banking. Consequences should arise because counterparties notice patterns, rival lenders recruit customers, depositors protect themselves, borrowers seek alternatives and the bank's own risk becomes concentrated. Delegated officers can handle ordinary underwriting inside policy while large connected loans, strategic conflicts, workouts and unusual collateral rise to the owner.

18.15 Banking completeness test

Banking is deep enough when the same nominal level of deposits can support very different outcomes depending on liquidity posture, loan quality, sector concentration, maturity, confidence, correspondent access and management. A bank should be capable of failing while owning valuable assets, surviving a panic while reducing new credit, earning less because it deliberately holds more liquidity, or becoming regionally influential because its underwriting and deposit relationships are strong. It must never function as an infinite-money business merely because the player owns the institution.

PART X

Rail, Communications, Utilities & Capital Development Networks that reshape geography, information, power and capital command.

READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

CANON LOCK

2.4 Railway arrival design lock

Rail is not present locally at the spring 1880 start and should never be treated as a cosmetic flip. The wider Black Hills economy still relies heavily on distant external railheads and long overland wagon legs, while industrial rail inside the Hills and outside common-carrier penetration are only beginning to develop. Regional railway expansion therefore affects the player before local track arrives: the nearest practical rail interface can move closer, shortening wagon legs, changing delivered cost, supplier choice and market reach. Local service remains an independent corporate decision shaped by network strategy, engineering, land access, rival towns, expected traffic, townsite opportunity and railroad finance.

2.5 Telegraph follow-through and electricity timing

Telegraph, telephone, electricity and civic infrastructure should no longer be written as sequential steps behind rail. Telegraph already exists unevenly in the Black Hills by 1880; telephone lines and exchange enterprise also appear exceptionally early; electric lighting becomes commercially available during the 1880s; waterworks already exist in exceptional dense towns. Rail commonly accelerates all of these by improving transport, capital and corridor economics, but each network has its own adoption logic and can move ahead or lag behind the others.

20.1 Private but public-facing rail

Railways in Land & Ledgers should feel private, profit-seeking, and strategically independent while remaining deeply entangled with land politics, approvals, town pressure, freight demand, and capital markets. The player can eventually become a railway promoter, land assembler, financier, founder, owner, or controlling industrial customer once wealth and opportunity justify it. Outside railway companies should evaluate the region for their own network reasons; a commercially strong town can gain negotiating leverage and still be bypassed.

20.2 Corridor projects rather than freehand city-builder track

The player should not become a city-builder track painter. The stronger design is corridor projects. Inside town, the player secures rights-of- way, acquires strips, chooses depot and station sites, and approves corridor segments and sidings. Outside town, expansion becomes an abstracted rail project to named off-map destinations. Distance, grading difficulty, bridges, river crossings, tunnels, and route-choice modifiers affect cost and viability. Some routes will be straightforward, some will be expensive, some will demand a bridge or tunnel, and some may simply not be worth the capital at the current stage.

20.3 Required facilities and commercial geography

A rail company should need more than track alone. Core facilities can include depot, freight handling area, maintenance shed or light service building, fuel and water support, sidings and later more specialized service infrastructure. Rail arrival creates new commercial geography, but depot location is only potential gravity: road access, available land, distance from the old core and established business strength determine whether a second commercial center actually forms.

20.4 Materials integration and owned supply chains

Rail construction and operation should benefit from owned supply chains, but only where the portfolio actually controls a historically plausible input. Player-owned timber, lumber, grading/earthwork capacity, hauling, land, blacksmith repair, local hardware, and in some runs regional fuel can make a project cheaper or more reliable. Rails, heavy machinery, and much finished iron or steel normally remain wider- market manufactured inputs; owning a local iron-bearing claim does not create an integrated rail-steel industry. Internal ownership should reduce real middleman, transport, and coordination costs rather than granting a generic construction discount.

20.5 Off-map expansion logic

The player should be able to expand from the current named town to off-map destinations without the project needing a full simulated region of hand-built towns. Off-map destination projects should calculate distance, base materials and labor needs, route modifiers, and

expected commercial value. The line therefore becomes a strategic frontier infrastructure project rather than a requirement to fully simulate every settlement along the way.

20.6 Railroad corporate strategy, negotiation, and bypass

Outside railroad interest should become visible before commitment. A plausible progression is regional observation -> preliminary interest -> survey activity -> competing-route consideration -> local negotiation -> conditional commitment -> construction. Surveyors, newspaper discussion, land speculation, rival-town lobbying, depot negotiation, and right-of-way requests should make the decision readable before track appears. The player can materially improve Commercial Attractiveness and reduce Construction/Acquisition Burden by generating recurring freight, assembling or donating right-of-way, providing depot land, supporting grading or local materials, contributing capital, or organizing public aid. The player cannot directly command an outside company's Strategic Interest. A railroad may choose cheaper terrain, a rival town, a railroad-controlled townsite, or another network direction even when the player's town is commercially useful. A bypass can create a new depot node whose warehouses, hotels, yards, businesses, and housing pull activity away from the original settlement. Valuable regions can also attract competing railroads. Separately, a large mine, logging operation, smelter, or industrial portfolio may justify a captive or industrial railway before a full public common-carrier route becomes attractive.

20.7 Moving railheads and the 1880-1905 regional chronology

Every settlement should know its current practical rail interface rather than only whether local track exists. Useful access states include remote from rail, distant railhead access, nearby regional railhead, approaching/surveyed corridor, local depot service and junction or competing service. Freight can change railheads as new lines open, so a store may receive cheaper or faster goods years before its own town gains a station. The regional historical guide is deliberately non-scripted: distant railheads dominate around 1880; Homestake-style industrial narrow-gauge operation becomes plausible around 1881; outside rail reaches the southern Black Hills in 1885, Rapid City in 1886 and Whitewood in 1887; local/interurban lines appear by the late 1880s; major Deadwood, Belle Fourche, Hot Springs and Hill City connections develop around 1890-91; and by 1900 the Hills possess a substantial multi-company network. Major east-west South Dakota integration is still being built in 1905: work on the Chicago & North Western temporary Missouri River crossing at Pierre begins in November 1905 and the first official train crosses shortly before Christmas, with the permanent bridge following later. Industrial rail, local/interurban rail and regional common-carrier rail are distinct business scales. Wagons remain important after rail by shifting toward depot drayage, feeder routes and secondary-node distribution. These dates govern plausibility and outside-world pressure, not automatic calendar unlocks for the fictional town.

20.8 Railway operation after construction

> CANON LOCK Completing track begins a railway operating business; it does not finish one. A working line must turn traffic into enough revenue to cover crews,

fuel, water, stations, maintenance, rolling stock, administration, debt and periodic capital repair. A lightly used but expensive corridor can therefore become a financial burden even if it was strategically satisfying to build. The railway's core management loop should be traffic, rate/service policy, capacity, reliability, maintenance and capital allocation - not full train dispatch.

20.9 Freight classes, tariffs and rate policy

Rail freight historically used explicit rates and commodity/classification systems. The game should compress this into a manageable set of economically distinct traffic classes rather than price every historical tariff line. Bulk low-value freight, livestock, ordinary merchandise, higher-value goods and unusually difficult or perishable traffic can differ in rate, handling and capacity burden. Rates should normally be quoted by route and class with enough visibility that a mine, ranch, merchant or competing town can compare rail with wagon or another carrier. Cheap rates can stimulate traffic but consume capacity at low margin; high rates can protect margin while driving shippers toward alternatives or political pressure.

20.10 Shipper commitments, traffic base and line utilization

A new line should be bankable partly because someone is expected to ship on it. Mines, mills, ranches, merchants, grain shippers, lumber businesses and later connecting railroads can provide anchor traffic through expected volume or

formal commitments where appropriate. The player can build speculative capacity ahead of demand, but the financial risk remains. Line utilization should emerge from actual traffic rather than a generic railway-level stat. A railway may be busy in one season and weak in another, profitable on one segment and underused on a branch, or dependent on one mine whose closure suddenly destroys the traffic base.

20.11 Service frequency, reliability and rolling capacity

Customers care not only that track exists, but how often and how reliably service actually moves. More frequent service can reduce inventory waiting and improve shipper value, but it requires locomotives or motive power, cars, crews, fuel/water and maintenance capacity. Too little equipment creates delays and lost traffic; too much creates idle capital. Breakdowns, washouts, bridge trouble, fuel shortage, crew shortage or congestion can reduce service through real operating constraints. The player should decide broad service posture and capital additions rather than route individual trains minute by minute.

20.12 Depot handling, sidings and wagon interface

The depot is a transfer machine between rail and the rest of the regional economy. Freight still has to be received, stored where necessary, loaded, unloaded, documented and moved by wagon to its final local destination. Siding, platform, yard and storage capacity can become bottlenecks even when the main line has spare capacity. Good road access and drayage services increase the practical value of a depot. This preserves the wagon economy after rail arrival and gives freight firms a new role rather than making them obsolete.

20.12A Rail transforms old businesses rather than merely obsoleting them

Rail arrival should recontextualize the wagon economy. Long-haul wagon routes can decline while depot drayage, farm/ranch collection, mine feeder service and secondary-node hauling grow. Warehouses and commercial frontage may relocate. The original store can gain access to distant manufactured goods while facing mail-order or outside competition. Old freight businesses can pivot into local distribution. This is the preferred modernization pattern throughout Land & Ledgers: history asks new questions of existing assets before it adds a replacement business category.

20.13 Passenger, mail and express as secondary service layers

Passenger, mail and express traffic can exist where corridor scale and historical circumstances justify it, but they should not displace freight as the main commercial reason most player rail projects matter. Passenger service can support settlement access, commercial travel and town prestige while carrying its own equipment, schedule and station burden. Mail or express contracts can provide useful recurring revenue where researched. These layers should enter only when they create a meaningful operating decision rather than because every railway menu is expected to contain every historical department.

20.14 Competition, interchange and route capture

A second railroad can change the economics of the first even without duplicating every track. Competing service can force better rates, improve shipper options, redirect through traffic or weaken a monopoly depot district. Junctions and interchange can add traffic but also make the line dependent on another carrier's connection. Large shippers can gain bargaining leverage when carriers compete for their traffic. A captive industrial line has different incentives from a public common carrier: its main purpose may be lowering an owned mine or mill's logistics cost rather than maximizing third-party freight revenue.

20.15 Interstate Commerce Act boundary after 1887

Where a player common carrier participates in interstate commerce, the Interstate Commerce Act of February 4, 1887 changes the legal envelope. The Act requires just and reasonable charges, public rate schedules, prohibits special rebates and undue preferences, restricts long-haul/short-haul discrimination and creates the Interstate Commerce Commission. Historical enforcement was imperfect and contested, so this should constrain available rate practices rather than create perfect regulation. A purely local industrial or intrastate operation may sit under a different legal regime. The player should not be asked to litigate rate cases routinely, but a railroad strategy that relies on secret discriminatory rates should become legally and reputationally riskier after 1887.

20.16 Maintenance, renewal and capital discipline

Track, bridges, culverts, water/fuel facilities, rolling stock and depots are durable assets that wear and can become unreliable. Deferred maintenance can preserve cash during a bad year while increasing delay, failure and later repair burden. Heavy traffic can accelerate wear and justify stronger maintenance capacity. Expansion, replacement and debt service compete for the same capital. This ties railway ownership back into the shared condition, bookkeeping, finance

and executive systems instead of making the railroad a separate empire game.

20.17 Railway completeness test

Railway ownership is deep enough when a player can make a good corridor fail through bad rates, weak service, overbuilding, poor maintenance or dangerous debt, and can make a modest corridor valuable through strong traffic commitments, disciplined capacity, reliable interchange and useful depot geography. The intended fantasy is owning and directing a transportation business, not painting track or dispatching every train.

22.1 Telegraph timing and corridor logic

Telegraph is already part of the wider Black Hills communications environment in spring 1880. Deadwood and other important corridor nodes received telegraphic service in the 1870s, so the opening town is information-poor because direct access is uneven, not because the technology does not exist. Telegraph tends to follow useful stage, military, mining, federal, commercial or railway corridors. A line passing a region, practical access to an office and a functioning local commercial office are separate states.

22.2 Gameplay effects

Telegraph changes the speed and reach of information, not the physical movement of goods. Market quotations, supplier availability, distressed-sale notices, bank or lender messages, railroad negotiations, and other short commercial information can arrive much sooner, while actual inventory still follows the freight system. Information should carry an age and source: a market quote received by telegraph this morning is different from a newspaper or letter several days old. Telegraph access does not create omniscience; messages can be brief, costly, incomplete, mistaken, promotional, or strategically misleading.

22.3 Business implications

A telegraph office or communication-linked service can exist as its own business or service node where demand supports it. The normal 70- 85-person opening settlement should not have a direct office unless a specific stage, military, mining, federal or through-corridor reason justifies it, but it can still use indirect service by physically carrying urgent messages to the nearest wired node. Faster communication improves supplier negotiation, emergency sourcing, credit decisions and rail operations while also accelerating rumor and financial contagion.

22.4 Telegraph access states and information age

Telegraph access should progress from mail/traveler only -> courier-to-wire access -> regular access through a nearby telegraph node -> local telegraph office -> rail-integrated/mature communications node. The first and last leg of a message can therefore remain physical even when the long-distance portion travels by wire. Important information should retain source, sent time, received time and information age. Telegraph reduces delay rather than granting a market-awareness bonus. It remains best for short urgent written messages; letters, invoices, contracts, catalogues and other detailed documents continue to use physical mail. National telegraph coverage expands rapidly through the 1880s and 1890s, so the outside information network becomes denser even when the player takes no direct action.

22.5 Telephone as a parallel communications network

Telephone technology is already exceptionally early in the Black Hills at the spring 1880 opening. Deadwood sources record telephone lines by 1878, the 1880 federal census records a Deadwood telephone exchange with subscribers, and a Rockerville/Rapid City telephone and telegraph company was incorporated in April 1880. The exact organizational date of the earliest Deadwood exchange is disputed, so canon locks only the broader fact that regional telephone service exists by the opening years of the 1880s. Telephone and telegraph are complementary. Telegraph is the stronger mature long-distance written network; early telephone is primarily immediate spoken communication among local or shorter-distance connected endpoints. A telephone call can eliminate messenger delay between a store, bank, freight office, hotel, manager or mine, but goods, workers, doctors and wagons still travel physically. Spoken calls also do not automatically create the documentary record of a telegram or signed letter. Telephone service should be subscriber-based and physical. Adoption can begin with a private point-to-point line between high-value sites, then grow into a local exchange with switchboard and operators, expanded subscriber coverage, toll/inter-town connections and later regional or rural/mutual service. Network value rises as more useful subscribers join, while sparse extensions still carry real pole, wire, maintenance and staffing costs.

22.6 Telephone ownership, competition and late-campaign growth

A Telephone Company is a legitimate infrastructure ownership path. Early networks can face stronger proprietary or licensing barriers; independent-company formation becomes much more plausible in the mid-1890s as key patent restrictions weaken. Competing Bell- associated, independent or mutual systems may have different coverage and

need not automatically interconnect, making subscriber base, compatibility, toll access and acquisition meaningful business assets. Rural/mutual telephone expansion becomes increasingly plausible around 1900-05, allowing remote farms or ranches to gain voice service before they gain electric utility service. Telephone should also support the executive-management transition. As the portfolio grows, connected managers can report and coordinate remotely, giving the later UI a physical historical reason to move from constant owner travel toward delegated oversight. Technology reduces communication delay; it does not create free automation or perfect knowledge.

CANON LOCK

23.1 Electricity eligibility and narrow rollout

Electricity is absent from the normal spring 1880 opening settlement for both chronological and economic reasons. Commercial electric lighting is only beginning nationally, so period technology availability and local viability must be treated separately. The player does not research electricity as an invention; equipment and engineering become purchasable through the wider economy over time, while concentrated paying demand, capital, generation opportunity and distribution economics determine whether a particular node adopts it.

23.2 Generation paths and existing industrial power

Generation depends on local resources and existing industrial infrastructure rather than one universal coal-steam template. Black Hills precedent strongly supports waterpower, steam power, and combined water/steam systems, including dynamos added to mills or mining facilities. A dedicated steam central station remains valid where fuel supply and demand justify it, but coal should not be the mandatory first fuel. Large mines, mills, hotels, or other concentrated users can justify isolated generation before a broad town utility exists and may later become the foundation of wider service.

23.3 Ownership and business model

The first broad town service will most often be a privately financed Electric Light or Light & Power Company operating under a local franchise or permission to use streets and rights-of-way, although municipal ownership remains possible. Founding or acquiring the utility requires a viable plant site, generating equipment, capital, anchor demand, operator coverage and physical distribution. Early effects begin with illumination rather than universal motor power. Arc lighting is especially appropriate for streets and large spaces; incandescent lighting expands the useful interior customer base as equipment becomes available. Electricity does not create a universal productivity modifier or instant rural modernity.

23.4 Distribution geography, load, and reliability

Electricity is delivered through generation, poles/wires and specific connected customers rather than a townwide boolean. Early local systems are geographically compact, making plant location, customer density, water/fuel access and distribution buildout strategically important. Connected load matters more than customer count: a mine, mill, hotel or street-lighting contract can justify more capacity than many small houses. Improved alternating-current equipment during the late 1880s and 1890s gradually makes more distant generation and larger service territories economical without requiring the player to configure voltage, frequency or transformer engineering. Early service need not operate continuously. Some plants can produce primarily during evening lighting hours or industrial shifts. Generator, boiler, water, machinery, operator, pole, or line failures should affect the customers actually dependent on that system. Kerosene, candles, and other lighting fuels should decline gradually in connected districts rather than disappearing from the regional economy.

23.5 Electrical technology progression, 1880-1905

Spring 1880 is near the beginning of commercial electric-light adoption. The normal opening settlement has no electric service. Deadwood electric lighting in 1883, Rapid City and Homestake-era late-1880s installations, Hot Springs wiring around 1891, Spearfish hydroelectric development around 1893, Black Hills AC-era projects in the 1890s, the Deadwood-Lead electric local railway around 1901 and

utility/steam-turbine consolidation around 1904-05 define a widening capability envelope rather than scripted unlock dates. Arc lighting can precede broad incandescent interiors; electricity begins primarily as lighting rather than instant motorization. Steam engines, water wheels, belts and line shafts remain useful for years. By 1905 a strong town or mine can plausibly possess a meaningful electrical network, but rural electric service remains unusual and a nearby primitive settlement can still have none. The intended end state is a mixed-power frontier- industrial economy rather than a universal electrified modern world.

23.6 Utility consolidation, labor and changing markets

Electrification creates new skilled work for operators, electricians and line-maintenance staff. Small early utilities and isolated generators can later be purchased, merged, enlarged, relocated or retired as networks mature. A player can therefore evolve from an isolated mill or mine dynamo into a local light company and eventually a multi-node utility, while old generating equipment may retain resale or relocation value. Electricity should alter existing markets gradually. Connected businesses can operate effectively later into the evening; lighting-fuel demand weakens first in serviced commercial districts while rural and unconnected demand persists; and obsolete inventory risk can emerge for merchants that overstock products whose demand is being displaced. These effects arise from actual service coverage and customer behavior rather than generic modernization bonuses.

24.1 Capital-command phase

Late game in Land & Ledgers is the capital-command phase. The player has delegated routine operations to managers and trusted staff, relies heavily on reports, audits, and portfolio oversight, and begins shaping the town through institutions and infrastructure rather than only through individual storefront optimization. Major late-game ventures include banks, railway companies, mines, utility plants, remote industrial chains, and later luxury or investment-linked commerce.

24.2 Luxury, investing, and civic-commercial prestige

As wealth grows, new late-game opportunities become plausible: jewelry or precious-goods trade linked to gold and silver output, investment office functions linked to stocks or bonds, finance-market intermediation, or other prestige commercial institutions that make sense once the town and the player both command substantial capital. These should not arrive as random rich-person toys. They should grow out of the same ownership-led logic as everything else: control of supply, financing power, information advantage, and regional seriousness.

24.3 The town still matters

Even in the late game, the player should not mentally leave the town behind. The town remains the proving ground. A railway changes its geography. A bank changes its credit environment. A mine changes its labor and service pressure. A utility changes its night economy. The fantasy is not escaping the simulation; it is becoming powerful enough to redirect it.

24.4 Final late-game lock

The late game should not simply be bigger menus. It should be the phase where earlier businesses, land, supply chains, and reputation finally pay off in institutions that reshape the town and region. That is the full commercial arc this edition is intended to preserve. Appendices The appendices below are intentionally practical. They are not decorative filler. They are quick-reference surfaces meant to preserve current locks, future late-game identity, and scope-aware caution.

PART XI

Executive & Portfolio Management Delegation, diagnosis and owner attention at portfolio scale.

> READING RULE This part is the authoritative home for the systems listed here. Related appendices compress the same rules for

routing and summary; they do not override the body.

19.1 Why the executive layer exists

Late game in Land & Ledgers should not mean the player manually revisits every store to check margins one by one forever. Once the portfolio is large enough, the player should be able to rely on trusted staff and contracted specialists. This is the difference between owning several businesses and running an actual frontier business empire.

19.2 Auditor and inspector roles

An auditor, inspector, or efficiency specialist is a strong late-game role. The player can hire one as payroll staff or contract one for a one-off review. That specialist can inspect a business, mine, livery, hotel, or other operation and return a grounded report on margins, staffing, stock discipline, supplier drag, local demand fit, condition, management weakness, and whether the business should be stabilized, expanded, sold, or closed. The auditor should not be a magical truth button. They are an informed professional lens that shortens diagnosis and helps the player act at portfolio scale.

19.3 Delegated approvals and policy bands

Delegated authority begins below the executive-office scale. A business manager can receive owner-defined permission over routine staffing, operating hours, merchandise allocation, reorder policy, supplier changes, pricing bands, customer-credit decisions and minor repairs. The owner can set thresholds or require approval for new product lines, large customer exposure, major supplier changes, capital projects, debt, asset sales or other strategic commitments. Executive office roles later extend the same principle across the portfolio: small cash transfers, low-risk loans, routine supply contracts or modest repairs can be handled under policy while high-value acquisitions, major distress actions, railway corridors or infrastructure commitments rise to the owner. Delegation should preserve owner agency by making exceptions and policy boundaries meaningful rather than turning management into a flat automation bonus.

19.4 Closure, sale, and turnaround recommendations

A core job of late-game staff is not just optimizing healthy businesses but identifying weak ones before they become permanent sinkholes. An auditor or portfolio agent should be able to recommend soft turnaround, management replacement, aggressive re-stock, controlled wind- down, sale to another owner, or full closure. These recommendations should be grounded in observed conditions and local evidence rather than raw abstract ratings. A weak business in a rising district might deserve patience; an overbuilt service in a stagnant pocket might be better sold before more capital is wasted.

19.5 Office roles beyond auditor

Other executive roles can emerge over time: portfolio bookkeeper, acquisitions agent, rail superintendent, bank manager, mine superintendent, legal or title runner, or utility manager. The design goal is not to create a bureaucratic layer for its own sake. The goal is to make late-game delegation legible, grounded, and deeply tied to ownership fantasy.

19.6 Exception management and owner attention

Late-game scale should create an attention problem without inventing a visible 'attention points' currency. The portfolio generates exceptions: a manager requests a large repair, a business misses its cash reserve, a supplier fails, a bank exposure breaches policy, a mine superintendent asks for capital, or an acquisition reaches a decision deadline. Delegation absorbs routine work when it stays inside policy. The owner becomes involved when something crosses a threshold, conflicts with another policy or requires authority that was never delegated. A well-run organization therefore feels quieter because it produces fewer unresolved exceptions, not because the game simply hides more information.

19.7 Manager competence, style and organizational memory

Managers should differ through experience, domain familiarity, judgment and policy preference rather than through universal percentage buffs. One manager may be conservative with inventory and cash, another aggressive on sales and credit, another unusually good at labor retention. Their choices still operate through the same causal systems. Long-serving staff can accumulate valuable knowledge of suppliers, workers, recurring customers, property and local seasons. Losing them can therefore create an information and continuity shock even when the replacement has similar nominal skill. The player can standardize policy across the portfolio but should still feel that people are administering it.

19.8 Shared portfolio services and earned synergies

A large portfolio can centralize work such as bookkeeping, purchasing, legal/title review, acquisitions, treasury, maintenance coordination or logistics planning only when the player actually hires the staff and creates the relationships needed to do it. Central purchasing can combine recurring demand into stronger supplier terms if orders, transport and storage really can be consolidated. A portfolio bookkeeper can improve reporting across several businesses because one office is processing the records. A repair department can serve multiple properties if it has workers, tools, materials and travel capacity. No 'corporate office' building should grant a generic empire efficiency bonus. Synergy is earned through shared capability and scale.

19.9 Treasury and capital allocation

The executive layer should make capital allocation one of the central late-game decisions. The owner can retain cash, transfer capital between owned businesses where legally and structurally permitted, pay down debt, support a distressed but strategic operation, expand a strong business, fund a new corridor, hold liquidity for a downturn, buy a rival or distribute/withdraw owner cash. Managers can recommend uses of capital but should not silently move large sums across business boundaries without authority. Portfolio performance should therefore be more than the sum of individual profit margins: timing and allocation of scarce capital determine which parts of the empire survive and compound.

19.9A Talent allocation and portfolio triage

Capital is not the only scarce late-game resource. A player can have enough money to acquire another business but no trustworthy person capable of running it. Senior clerks, foremen, managers, bookkeepers, superintendents and specialists can therefore compete for assignment across the portfolio. The principal may promote internally, recruit, relocate a manager, supervise personally, delay the acquisition or sell another operation.

Mature play must also make divestment legitimate. The owner should increasingly ask what not to own. Sale, closure, liquidation, lease, restructuring, manager buyout, minority sale, capital withdrawal and strategic abandonment can all be rational responses. A portfolio that only grows by accumulating permanent holdings will eventually become the same panel multiplied twenty times; capital command requires pruning as well as acquisition.

19.10 Portfolio policy, local autonomy and escalation

The player can set portfolio-wide standards - minimum cash reserve, maximum customer-credit exposure, maintenance escalation, acquisition return expectations, manager approval limits or preferred internal suppliers - while allowing local managers to deviate when authorized. Too much centralization can produce slow or poorly fitted decisions; too much autonomy can create hidden risk and inconsistent strategy. Exceptions should explain why local conditions make the standard policy awkward. The player can then approve the deviation, change the policy or replace management. This creates governance rather than automation.

19.11 Partners, minority capital and governance conflict

Where a business or project includes outside investors, silent partners or other meaningful co-owners, the player should not behave as if 51 percent economic interest necessarily means unlimited freedom from all agreed obligations. Partnership terms can constrain distributions, new borrowing, sale, major expansion or related-party deals where the financing structure makes that appropriate. The design does not need a corporate-law simulator, but outside capital should preserve the central bargain from 18.3A: sharing risk can also mean sharing control or upside. A partner dispute should emerge from a real strategic or financial conflict rather than a random relationship event.

19.11A Economic interest and control rights are separate

Shared ownership should distinguish economic interest, capital responsibility, ordinary management authority, reserved decisions, distributions and exit/buyout rights where the arrangement makes those terms meaningful. A forty-percent interest does not automatically grant forty percent of every decision, and a majority economic interest does not erase contractual protections given to other investors. This is not an excuse for a corporate-law minigame; it is the minimum needed for partnerships, syndicates and minority capital to create real trade-offs rather than functioning as free money.

19.12 Communication infrastructure and portfolio reach

The executive transition should have a physical information basis. Telegraph can accelerate written reports and urgent instructions; telephone connections can allow rapid conversation between connected businesses, managers, banks, depots and industrial sites; improved rail and roads can shorten personal inspection travel. These networks reduce delay but do not make remote management omniscient. An unconnected mine or distant property can still produce stale reports and require travel or a trusted superintendent. This keeps modernization relevant to how the player manages, not only to what the town looks like.

19.12A Information distance and organizational truth

Remote ownership should distinguish at least four conceptual layers when the distinction matters: World Truth, Local Recorded/Observed Truth, Last Reported Owner Knowledge and Management Interpretation/Forecast. The simulation can know a remote business's exact current state while the owner possesses yesterday's telegraph summary or last week's written report. This should not become an annoyance that hides every owned number. Use information delay where it creates a real economic choice, exception or risk. Communication technology changes how quickly authority can be exercised; it does not move goods or make local records automatically correct.

19.12B Central offices solve real coordination problems

A central office, headquarters or portfolio office is justified when it consolidates records, correspondence, treasury, specialist review, purchasing support or exception reporting for a geographically distributed organization. It should not grant a generic Headquarters +20% Efficiency bonus. Its value comes from reduced duplicated administrative work, faster comparison, more reliable reporting and concentrated specialist capability, while also creating salaries, overhead and possible single points of failure.

19.13 Executive failure and key-person risk

Delegation creates new failure modes. A dishonest or weak manager can conceal problems if records and oversight are poor; an overworked portfolio bookkeeper can fall behind; a superintendent can leave for a rival; one acquisitions agent can become a bottleneck; a centralized purchasing office can create a single point of failure. The answer should be better controls, redundancy, staffing and inspection where justified, not a permanent executive-risk debuff. Small portfolios should not need an expensive bureaucracy; large portfolios should feel the value and cost of building one.

19.14 Executive completeness test

The executive layer is deep enough when growth changes what the owner does rather than merely reducing clicks. Early play is dominated by direct operating choices. Mid-game play adds manager policy and exceptions. Late play increasingly asks where capital, authority, specialist attention and information should flow across a portfolio. The player should still be able to inspect one store or household in detail, but the primary late-game skill becomes diagnosing and governing interconnected assets without receiving magical automation.

PART XII

Interface & Player Information Design How deep simulation is compressed into readable ownership decisions without exposing implementation detail.

> READING RULE This part is the authoritative home for player-facing information hierarchy, management surfaces and workflow presentation.

Technical architecture, persistence, system cadence and Unity implementation live in the separate Technical Implementation Bible and do not belong to the design canon except where a technical constraint is explicitly elevated into a design decision.

15.1 Core UI structure

The UI should read like a clean frontier business ledger, not parchment fantasy clutter. Deep simulation should live beneath compact, readable management surfaces. The always-visible HUD should show Liquid Cash, Net Worth, date, day of week, season, time speed, and alert strip. Businesses should share one common UI shell with business-specific panels layered on top. Household panels should stay lighter and focused on composition, reserves, pressure, and settlement state.

15.2 Tabs and workflows

Tabs remain overview, browsing, and launch-point surfaces. Loans and acquisitions should break into dedicated workflow screens once the player enters a serious multi-step process. Acquisition workflow should support land and business purchase under one family with staged process flow and a framed live property preview or photograph-like render area. Loan workflow should read like a formal application under review rather than instant cash from a tab button.

15.3 Information priority and readability

Property and acquisition surfaces should show owner name, owner type, sale status or property state, current use and suitability, conversion notes, land scorecard, lead source, confidence, urgency, exclusivity, rival interest, and deadline where known. Management readability rules matter. Prefer grouped cards, scrollable content regions where needed, and confident player-facing copy over dense internal-sounding dumps. Tooltips should behave intelligently near edges and corners. General Store information should remain specific to the store; renting and boarding should have their own surface. Government or civic information should remain separate from player holdings where practical. The game should proactively surface what the player can act on soon: today's pressure, tomorrow's risk, this week's opportunity, and this season's drift. Information surfaces should ideally summarize conditions the world has already begun to imply, rather than replacing world readability outright. The notice layer should separate opportunities from active pressures. Opportunities can include merchant recruitment, empty prime frontage, boarding shortage, blocked settlement, freight bottleneck, underused supply, post- boom vacancy, and town not yet serious enough for outside capital. Pressures can include worsening shortages, rival movement, pending closures, time-sensitive deals, transport deterioration, civic petitions becoming urgent, or corridor-value shifts tied to outside interest.

15.3A New management surfaces: logistics, supply, and settlement nodes

The management UI should now surface three additional families of information. First, each business should expose a Logistics / Transport section showing owned hauling assets, assigned labor, storage/stabling capacity, active routes, and hauling mode. Second, the portfolio should expose a Supply & Trade surface where the player can read and edit standing supply agreements, route priorities, reserve rules, and overflow policies. Third, the map and settlement UI should expose node-level summaries for population, boarding, business mix, freight burden, service coverage, and current growth or decline pressure. These surfaces should follow the same design philosophy as the rest of the project: compact, ledger-like, grouped, and player-facing. The goal is not to bury the player in freight microdata. The goal is to make the consequences of logistics, regional growth, and inter-business chaining legible enough that the player can make deliberate ownership decisions instead of guessing at hidden simulation.

15.3B Business policy, capacity, and delegated-authority surfaces

Business management UI should distinguish operations from administration. The General Store should be able to show Total / Allocated / Occupied capacity by Display, Storage and Bulk/Yard; merchandise-line allocation; stock target and reorder state; supplier and landed cost; selling price/markup; turnover and stockout history; cash reserve; seasonal policy; operating hours; customer-account exposure; and who currently has authority to change each policy. The interface should make fast-selling, dead-stock, overallocated, understocked and cash-constrained conditions legible

without prescribing one correct answer. Manager delegation should expose permissions and approval thresholds clearly enough that the player understands what the manager may decide and what will escalate.

15.3C Maintenance, records, expertise and housing management surfaces

Revision XI adds four management-surface families that should remain consistent with the clean ledger presentation. Asset/maintenance views should list maintainable assets, readable condition/known defects, maintenance backlog, next action, required skill/parts and expected downtime rather than health bars. Bookkeeping views should distinguish historical actuals, physical counts/reconciliations, receivable aging and forecast values. Expertise views should show professional experience and newly available analysis without presenting unexplained RPG bonuses. Residential holdings should summarize tenant, rent, arrears, vacancy, condition, debt and maintenance while allowing property-level inspection.

15.3D Forecast presentation standard

Reports should visually distinguish Actual / Known from Forecast / Estimated. Historical sales, purchase costs, invoices, cash counts and confirmed balances are exact. Future sales, demand, market value, repair cost before quotation, cash at month end, employee retention and other uncertain outcomes can use central estimate, likely range and confidence. Where useful, the player can inspect the drivers behind the forecast so expertise assists judgment rather than replacing it.

REFERENCE APPENDICES

The main body remains authoritative; appendices compress design locks, research and routing.

APPENDIX A

Canon Lock Register

- Spring 1880 is the authoritative opening; do not revert to 1892.
- The authored historical horizon is 1905. Open Frontier normally begins in spring 1880 and has no mandatory victory by default; Scenarios may use shorter historical windows.
- No dialogue checks for deals; use grounded business logic.
- No modern zoning-style parcel hard locks; use suitability, nuisance, frontage, adjacency, access, legal status and conversion cost.
- Businesses share one shared business model; specialized industries extend it rather than becoming disconnected minigames.
- Owner cash and business cash remain separate authorities. Net Worth includes liabilities; borrowing cash does not create Net Worth.
- Commercial Ruin requires actual exhaustion of viable ownership and recovery capacity; one default, foreclosure or bankruptcy is not automatic defeat.
- Supply, price, fire, disease, weather and other shocks must arise through real footprint, exposure, reserves, substitutes, networks and recovery rather than universal percentage penalties.
- Spring 1880 begins without local common-carrier rail; practical railheads can move closer before local service. Industrial, local/interurban and regional rail remain distinct.
- Telegraph and telephone are parallel networks already regionally plausible at the opening. Information has source and age; communication does not move goods.
- The normal opening town has no electric service, broad piped water/sewer network or normal public telephone exchange. Utilities are spatial networks with coverage, capacity, condition and project lifecycles.
- Mining hierarchy remains gold dominant; lead-silver/polymetallic secondary; tin speculative; coal strategic but often regional/conditional; iron niche. Mining progression is functional bottlenecks and metallurgy, not mine levels.
- 1880 agriculture is already partly mechanized by horses. Machinery changes seasonal work capacity, not biological yield. Barbed wire and pumping windmills are opening-era technologies; tractors are not normal core progression by 1905.
- Dakota Territory already has codified property, mortgage, exemption, banking, corporate and municipal law. Statehood in 1889 preserves existing obligations; the 1898 Bankruptcy Act adds a late federal distress route.
- Restricted Indigenous/reservation/trust land never enters ordinary Make Offer or normal collateral merely because it is visible. Current title, transfer status and historical provenance are separate authorities.
- Allotment is a process, not an 1887/1890 global conversion. The specific Sioux reorganization becomes operative February 10, 1890. The player never drives federal dispossession.
- No generic hostile-Native raids, Indigenous hostility meter, Civilization score or Wounded Knee profit/combat event.
- Automobiles may appear rarely by 1905 as background/cutting-edge machines; motor freight, tractor farming and automobile-road modernization remain outside the normal campaign.
- Historical availability never overrides production priority. Late systems protect the destination and do not become immediate VP scope.
- If the implemented game already has a stronger, clearer, more grounded or more production-ready mechanic, update canon upward rather than downgrading the game.
- Every mechanic is judged through historical accuracy, maximum plausible player freedom and premium systemic readability. Prefer consequences over arbitrary prohibitions.
- The player is one actual owner/investor, not an omnipotent municipal authority. Direct commands require a legitimate ownership, employment, contractual, financial, management or legal basis.
- Physical Truth is canonical: locally significant purchases, work, deliveries and services resolve only when required people/assets actually reach and satisfy the physical interaction; off-map abstraction changes resolution, not reality.
- The current managing authority - player owner, NPC owner or hired manager - controls administrative business policy. Ordinary workers execute tasks; they do not silently redesign the business.
- Player and NPC businesses use the same meaningful operating levers. NPCs may succeed or fail through their own allocation, pricing, credit, staffing, sourcing and cash-management decisions rather than hidden handicaps.
- General Store inventory uses dedicated physical capacity. Every item consumes space; reserved-but-empty merchandise allocation is not automatically available to unrelated goods until administratively reassigned.

- Routine replenishment can be automatic under standing targets and supplier policy, but cash-reserve protection can force manual approval. Seasonal merchandise and target profiles are valid owner/manager controls.
- Labor is ultimately workload- and occupation-dependent rather than one rigid employee-per-task or universal workday model. Continuous 24-hour time and seasonally changing daylight are canonical inputs.
- Customer credit is a shared commercial capability where historically appropriate, with real receivables and liquidity risk; ordinary store tabs do not automatically bear interest.
- Households are the primary domestic economic unit; individual people remain the actual workers, shoppers, travelers, health actors and life-cycle participants.
- Canonical revisions should not decrease in total page count or word count from the immediately previous numbered revision. Replace obsolete material when necessary, but preserve useful detail and add meaningful depth rather than shrinking the master reference.
- Known facts remain exact when legitimately possessed. Expertise may improve interpretation and forecasting but must not blur confirmed purchase costs, invoices, counted cash, recorded balances or other known values into artificial uncertainty.
- Experience/Expertise, Commercial Standing and world Modernization are separate progression authorities. Expertise cannot research future technology into existence; Standing cannot substitute for skill; skill cannot create unexplained discounts or physical bonuses.
- Asset maintenance is causal. Use/load/exposure/age/defects create deterioration; service and repair require real labor, parts, facilities and downtime. Deferred maintenance is allowed and produces asset-specific consequences rather than a universal penalty.
- Repair work is a real service market generated by actual assets. Blacksmithing, horseshoeing, wagon/wheel work, harness work and later machine repair are capabilities that may combine or specialize according to people, facilities and demand.
- Residential renting is a major historically supported tenure state, not an edge case. Opening independent nonfarm renter share should provisionally calibrate around 35-45% rather than a token percentage, with boarders/lodgers tracked separately.
- The player can set aggressive or irrational rents, maintenance policies, business hours, prices and other lawful/physically possible policies. The simulation should answer through vacancy, demand, deterioration, labor, competition and other consequences rather than arbitrary designer prohibition.
- Mining is an economic ecosystem. The player may profit from prospecting/claims, finance, assay, treatment, mineral trade, supply, contracting, camp property and redevelopment without owning a producing mine.
- Agriculture is a seasonal land-and-capital economy. Machinery changes work capacity rather than biological yield; livestock remain biological capital; weather acts through the physical crop, forage, water, animal or work process actually affected.

APPENDIX B

Feature Status & Production Priority Core Identity - Ownership-led growth, real households, flexible parcel use, shared business shell, realism-first simulation and visible world evidence. VP Core - General Store deep slice, deal-flow acquisition, household shopping, notices, Net Worth, staffing continuity and doctor node. Active VP Expansion - Boarding/settlement growth, recurring orders, survey/terrain groundwork, workflow UI, node logic, early logistics visibility and adversity proof. Post-VP - Broader business pool, richer property markets, stronger manager policies, rural production chains and remote-node maturation. Rail Era - Moving railheads, depot/bypass competition, feeder-freight transition, corridor projects and rail-linked communications. Late Capital Era - Banks, mines, railway companies, utility acquisitions, executive staff, investment workflow and mature regional communications. Long-Term - Deep utility operations, municipal infrastructure ownership, mature electric power, toll/rural telephone, richer weather/street systems and formal freight contracts. Parked - Specialized reservation trading, deeper commercial-credit/legal edge cases, richer gossip, tenant operators, scheduled wagon rounds and other valuable layers not in active production. Visual / Background at 1905 edge - Occasional automobile, extremely rare gasoline tractor, cutting-edge mechanical refrigeration or other period-valid technology that should not replace the horse/wagon economy. Outside normal 1880-1905 campaign - Widespread motor freight, ordinary tractor farming, automobile-oriented road systems, universal rural electricity, modern zoning and universal professional municipal services. Replaced - Dialogue checks for deals, parcel hard-typing, owner-presence- heavy business logic and teleport-heavy supply assumptions.

APPENDIX C

Canon Navigation & Design Routing

TASK / QUESTION | PRIMARY HOME | REFERENCE
Core fantasy, tone, realism | Part I / Sections 1.1-1.4A | Appendix A / N
Scenarios, Open Frontier, optional victory, loss and play-mode setup | Part I / Sections 1.5-1.6 | Appendix D / E / O / X
Settlement nodes, population, regional plurality | Part II / Sections 2.2-3.3 | Appendix H / I
Land, parcels, acquisition, valuation | Part III / Sections 4-5 | Appendix F
Business operations, cash, net worth, continuity | Part IV / Sections 6 and 11 | Appendix A / M
General Store, blacksmith, butcher, farm, ranch, boarding | Part V / Sections 7-8 | Appendix J
Mining, logging, sawmills, remote camps | Part V / Sections 8.5 and 21 | Appendix J / U
Supply agreements, hauling, routing, cargo, Livery/Freight | Part VI / Sections 9-10 | Appendix G
People, households, kinship, accommodation, labor, health | Part VII / Sections 12-13 | Appendix H / S / V
Demand, pricing, rivals, adversity, fire | Part VIII / Sections 14.1-14.11 and 3.2 | Appendix A / O
Loans, banks, investors, collateral, workouts | Part IX / Section 18 | Appendix K
Rail, depot, telegraph, telephone, electricity, utilities | Part X / Sections 20, 22-24 and 2.4-2.5 | Appendix L
Auditors, executive staff, delegated approvals | Part XI / Section 19 | Appendix M
Player-facing UI, workflows, forecasts and information hierarchy | Part XII / Section 15 | Appendix Q / R / S

ROUTING RULE

For design or research work, start with the narrowest authoritative body section and then use its appendix for compressed locks, open questions and calibration. Technical architecture, repository state, save/load and implementation sequencing belong in the separate Technical Implementation Bible.

APPENDIX D

Play Modes, Scenario Campaign & Optional Victory Reference

MODE | RULE

Scenarios | Authored, difficulty-ranked commercial challenges. Fixed premise, variable economic detail where useful. Same underlying simulation rules as Open Frontier.

Scenario Campaign | The formal campaign is completion across the scenario catalogue rather than one continuous protagonist/save. Scenarios can be grouped by era or commercial theme and need not be strictly linear.

Open Frontier | Flagship procedural sandbox. Recommended opening spring 1880, seed-generated regional economy, no mandatory victory by default and full historical simulation through the 1905 horizon.

Open Frontier + Optional Victory | Sandbox may enable Capital, Selected Legacy, Capital + Legacy or supported custom conditions without changing the underlying economy.

Continuation | After the authored horizon, preferred labels are Continuation Year 1, Continuation Year 2, etc.; existing supported capabilities can diffuse but no new post-1905 technology family is implied.

Commercial Ruin | Separate economic defeat state. It is not the same as failing a scenario objective and need not be enabled in every sandbox configuration.

SCENARIO DESIGN LOCKS

- Objectives should describe economic outcomes or transitions, not arbitrary object counts.
- Optional Distinctions reward mastery without becoming required checklist clutter.
- Difficulty comes from starting conditions, commitments, scarcity, information, time and intertwined pressures - never hidden NPC production/revenue buffs.
- Scenario deadlines should have a world, historical or scenario-scope reason.
- Scenario completion records the actual economic result, not only one score.
- Scenario randomization should preserve the premise while varying enough detail to prevent memorized opening scripts.

OPEN FRONTIER REPLAYABILITY LOCKS

- World generation is a headline feature, not merely objective-free play.
- The historical skeleton is stable; commercial history is variable.
- Seed variation can alter geography, ownership, households, businesses, suppliers, labor, condition, resources, rivals, debts, migration, settlement potential and infrastructure proposals.
- Opening information should be commercially plausible but should not reveal hidden truth or an optimal strategy.
- The sandbox must remain worth playing even after every scenario is completed.

OPTIONAL LEGACY VICTORIES

Merchant Prince | Major retail/service network with meaningful household and business service.

Industrialist | Substantial production/industrial control with real operating chains.

Financier | Major lending/deposit/regional-capital position.

Railroad Baron | Meaningful railway control plus sustained traffic/destination importance.

Land Baron | Major valuable and strategically useful land/property portfolio.

Regional Power | Meaningful economic presence across multiple settlement nodes.

Settlement Builder | Durable development of a weak/emerging node through real jobs, services, housing and freight.

Grand Empire | REWORK REQUIRED; do not implement as simple multi-badge collection.

APPENDIX E

Commercial Ruin & Distress Reference LOSS RULE: zero cash, negative Net Worth, missed payment, default, foreclosure on one asset or a late-campaign bankruptcy filing are not automatic defeat. Commercial Ruin occurs only after viable ownership and recovery capacity are effectively exhausted. Financial failure remains playable: strain -> due payment -> delinquency -> workout/restructuring -> default -> applicable enforcement -> voluntary or forced sale / foreclosure -> possible redemption, deficiency or recovery -> Commercial Ruin only if viable economic capacity is exhausted. Opening law is Dakota Territorial, not informal frontier practice. Mortgages and chattel security are asset-specific; exemptions and separate property can preserve a recovery foothold. Statehood November 2, 1889 preserves existing obligations. The federal Bankruptcy Act becomes available to qualifying debtors July 1, 1898 but is not a modern universal reset. Recovery tools can include owner cash injection, inter-business cash movement, refinancing/revised schedules, additional collateral, investor capital, inventory liquidation, project delay, cost reduction, turnaround, voluntary sale and closure of weaker holdings.

APPENDIX F

Property & Acquisition Reference

AREA | CANONICAL HANDLING | DESIGN NOTE

Ownership buckets | Founder/speculator, private household, | Seller motive and seriousness determine access; business owner, widow/heir/estate, edge | restricted land cannot enter ordinary Make development parcel, federal/public-domain, and | Offer flow.
legally restricted/provenance states where applicable.

Property states | Owner-occupied/not interested, quietly open, | Do not make property magically available debt-pressured, widow-controlled, estate | because the player wants it.
unsettled, inheritance disputed, listed, bank-held distressed.

Suitability | Frontage, depth, rear/alley access, freight | Use physical suitability and externalities instead access, yard/storage, water/drainage, nuisance, | of modern zoning locks.
adjacency, traffic, assembly, ROW exposure, conversion cost.

Deal channels | Public listings, private approaches, rumor leads, | Rivals can contest deals where readable
triggered sales, package deals, distressed resale. economic cause exists.

Diligence | Title, books, condition, suppliers, staffing, | Closing risk and inherited baggage survive market fit. | transfer.

Workflows | Tabs are hubs; serious acquisitions use formal | Land and business purchase share a workflow workflow screens. | shell family.

APPENDIX G

Supply, Cargo, Internal Routing & Freight Reference

SYSTEM | CANONICAL RULE | DESIGN PURPOSE

Supply agreements | Standing policy: goods, quantity/percent, | Lets the player deliberately shape supply chains.
cadence, reserve, priority, hauling mode, fulfillment history.

Delivery orders | Source, destination, cargo class, multi-leg route, | Inventory updates on arrival; distant goods can transfer point, loading window, travel time, | use rail/river trunk plus local wagon leg.
unloading; practical wagon capacity derives from vehicle/team/route/condition.

Internal routing | Percentage/fixed quantity, cadence, priority, | Supports ranch-to-butcher, sawmill-to-mine, reserve, destination, transfer price, hauling | project supply, overflow.
responsibility.

Transfer price | Free/0%, at cost, below market, market, or | Ledgers remain honest; no magic profit above market where supported. | creation.

Cargo classes | Packaged goods, perishables, live cargo, bulk | Different carts/wagons and handling rules materials. | matter.

Self-haul | Requires animals, wagons/carts, harness, labor, | Lower cash freight cost but real feed, upkeep, storage/stabling. | capacity/opportunity cost.

Livery / Freight | Professional hauling for businesses, projects, | Infrastructure plus business opportunity.
mines, mills, farms, stores, households/institutions where relevant, rail/depot interfaces.

Supply causality | Shock changes output, availability, demand, | Market price responds afterward.
transport, reliability, or delivered cost first.

Shock scale | Supplier/business -> local/settlement -> | Magnitude matches actual affected supply.
regional/off-map.

Diversification | One cheap supplier can increase concentration | Vertical integration transfers risk into the risk; multiple suppliers can improve resilience. | portfolio rather than deleting it.

APPENDIX H

Settlement Nodes & Population Bands

BAND / MODEL | TYPICAL FEEL | DESIGN READING
0-9 people | Fragile occupation, camp services, improvised | Almost no guaranteed local service.
shelter.

10-24 people | Thin but meaningful local demand. | Boarding pressure, simple trade, more reliable
supply.

25-49 people | Stronger service case. | Small store, blacksmith support, durable
housing patterns.

50-99 people | Rising civic pressure. | Schooling pressure, clearer commercial
frontage, formal platting potential.

100-199 people | Greater differentiation. | Service specialization and coherent local center.
200+ people | Begins to act like a true town. | Institutions and freight needs can influence the
wider region.

Gateway-settlement model | Deliberately founded, commercially useful, | Town matters through supply, repair, lodging,
geographically well-sited, boom/bust capable. | livestock support, administration, education,
freight.

BAND / MODEL | TYPICAL FEEL | DESIGN READING
Demographic composition | Actual NPC households generate | Node summaries are outputs of people
sex/age structure, boarders, renters, | and settlement function, not target
owners, dependents, skills, and migrant | quotas.
mix.

APPENDIX I

Survey, Terrain, Hydrology & Vegetation Reference

LAYER | CANONICAL RULE | DESIGN REASON

Survey / parcels | Every region area knows survey identity, parcel | Supports valuation, acquisition, type, acreage, ownership/provenance, transfer | Indigenous/federal land distinction, land status, and restricted/federal status where | history, and conversion logic. applicable.

Terrain tiles | Render landform, hydrology, textures, scenic | The visible world should reflect the underlying economic and geographic truth. detail.

Hydrology | Generated early enough to influence siting, | Water is not a decorative afterthought. value, vegetation, prospecting, freight.

Parcel-aware shaping | Roads, pads, yards, loading spaces request | Avoid awkward slopes without flattening the practical grading. | whole world.

Vegetation chunks | Efficient distant cover, scenic props, harvestable tree stands, | Supports premium look and later forestry landmark groves. | without simulating every tree at all times.

World-evidence transport | Visible wagons/carts correspond to real shipment jobs | Transport shown in the world should be economic evidence, rather than unrelated decoration. | not unrelated decoration.

APPENDIX J

Remote Industry, Mining, Logging & Sawmill Reference Mine staging - district information/prospect -> claim control -> representative assay/feasibility -> financing/investor structure -> initial works -> production -> expansion. Deposit identities - gold dominant; lead-silver/polymetallic secondary; tin speculative; coal strategic but often external/conditional; iron niche. Ore authority - grade plus metallurgical character, treatment compatibility and confidence. Geological mineralization is not automatically economic ore. Functional mine capacities - development, drilling/blasting, ground support, internal haulage, hoisting, water control, crushing, treatment, power, freight and skilled labor. Improvement moves bottlenecks rather than granting a mine level. Treatment chronology - stamp/amalgamation opening-era for suitable gold; chlorination important by 1889; specialized smelting around 1890; early intermittent cyanide 1892; successful cyanidation expands late 1890s/early 1900s. Mechanical chronology - serious steam/hoisting opening-era; hand drilling remains credible into early 1890s; compressed-air drilling expands in the 1890s; electric lighting appears before broad electric motive power. Industrial rail - private mine/mill/timber narrow-gauge can precede common-carrier rail when freight concentration justifies it. Mine as business - preserve common cash/income/ownership shell plus local claim, workings, ore, treatment, labor, timber, power, haulage and condition state. Mine camps - can become settlement nodes, stall, decline, ghost or convert as underlying economics change. Timber extraction - stand/worksite logic with terrain, road, haul distance and quality. Sawmills feed building, mine support, rail, farms and repairs; mining can materially raise timber demand.

APPENDIX K

Finance, Banking & Legal Chronology Reference Bankability - banking exists regionally before 1880; the opening town may still be too small or weak for a local bank. Bankability depends on deposits, payroll, business volume, land value, debt demand, rail/industrial seriousness and confidence rather than a population unlock. Loan review - purpose, repayment source, collateral quality, existing debt, reputation, market outlook, strategic value and portfolio risk. Workouts - revised schedules, partial payment, extra collateral, refinance and negotiated asset transfer can precede enforcement. Collateral - secured and unsecured creditors differ; mortgages/chattel security remain asset-specific; protected property and separate ownership can remain outside creditor reach. Interest chronology - opening Dakota general framework: 7% legal/default, up to 12% written contract; covered Black Hills written-contract freedom 1881 through June 30, 1887; general ceiling returns July 1, 1887. Municipal finance - federal territorial act July 30, 1886: 4% general local debt cap and strong limits on public credit/subscription aid to private corporations. South Dakota statehood: 5% general constitutional local debt cap subject to later infrastructure provisions. Statehood - November 2, 1889 changes institutions but preserves existing contracts, rights, corporations, litigation, bonds and debts. Bankruptcy - no federal bankruptcy system from 1878 until the Bankruptcy Act of 1898 takes effect July 1, 1898. Late bankruptcy is a qualifying distress route, not a universal reset. Credit climate - deposits, liquidity, loan quality, sector exposure, outside capital and confidence remain underlying states; a climate label summarizes rather than imposes a global modifier.

APPENDIX L

Rail, Communications, Utilities & 1905 Boundary Reference Rail - no local common-carrier rail at the normal 1880 start. Practical railheads move closer over time; industrial rail can precede public regional service; local service depends on company strategy, route economics, land and traffic rather than a tier. Regional rail chronology - outside Black Hills penetration 1885, Rapid City 1886, Whitewood 1887, substantial late-1880s/1890-91 expansion, multi-company network by 1900, major east-west South Dakota connection work still underway in 1905. Telegraph - already unevenly regional in 1880; access progresses from mail/traveler and courier-to-wire toward nearby node, local office and mature rail-integrated service. Information has source and age. Telephone - regionally present exceptionally early; private line -> exchange -> expanded subscriber network -> toll/inter-town -> regional/rural mutual. Property connection matters; networks can compete and fail to interconnect. Electricity - normal opening town has none. Lighting-first adoption expands during 1880s-1900s according to load, capital, generation and distribution economics; steam/water/mechanical power persists. By 1905 strong nodes can have meaningful electric systems without rural universality. Water/sewer - young nodes use wells/pumps/privies/drainage; formal source/storage/mains/hydrants/sewer districts/property connections expand spatially. Partial coverage can persist for decades. Fire service - ad hoc/buckets -> volunteer hook-ladder/hose -> hydrant-supported volunteer service -> mature volunteer department; paid/professional service only where unusually large/complex late nodes justify it. 1905 boundary - rail, telephone, electric light, waterworks, cyanide-era industry and sophisticated finance can coexist with horse/wagon rural life. Automobiles are rare background, not a new core transport regime.

APPENDIX M

Executive Oversight & Portfolio Cadence

ROLE / CADENCE | PRIMARY OUTPUT | ESCALATION BOUNDARY
Auditor / inspector | Diagnose weak businesses and recommend | Owner approves sale, closure, heavy action. | turnaround.
Portfolio bookkeeper | Track cash health and threshold alerts. | Cross-business liquidity stress escalates.
Acquisitions agent | Surface leads and early diligence. | Earnest money or serious commitment escalates.
Bank manager / officer | Routine underwriting, monitoring, operations. | Large loans, distressed workouts, strategic conflict escalates.

ROLE / CADENCE | PRIMARY OUTPUT | ESCALATION BOUNDARY
Rail superintendent | Corridor operations and service readiness. | Expansion, site change, major service failure escalates.
Mine superintendent | Remote-site throughput and labor stability. | Capital expansion or severe accident/disruption escalates.
Daily review | Urgent failures, threshold breaks, exceptional | Immediate notices. opportunities.
Weekly review | Saturday readiness, stock discipline, staffing | Manager notes and owner choices. gaps, small corrective actions.
Monthly review | Net worth drift, debt, integration outcomes, | Major escalation, sale, closure, capital underperformers, investments. | allocation.

APPENDIX N

Anti-Bloat & Non-Goals

- Do not turn the game into a full railroad-dispatch simulator.
- Do not require a fully simulated region of mapped towns before off-map rail expansion is allowed.
- Do not make banking a constant paperwork chore once the portfolio is large enough for thresholds and staff.
- Do not treat electricity as a universal modernity switch that erases frontier texture.
- Do not let stocks, bonds, or luxury goods arrive so early that they undermine the ownership-first climb.

DESIGN DERIVATION QUESTIONS

- What frontier reality is this modeling?
- What tension makes it interesting?
- What physical, behavioral, and social signals reveal it?
- What threshold changes it from background to consequence?
- What ownership response does it ask for?
- What structure does it strengthen?
- When in the game timeline does it matter most?
- What place or district logic changes its meaning?
- Does it create readable town consequence?
- Does it fit the prestige-frontier tone?
- Who has legitimate authority to make this decision?
- What person, asset, location or physical movement must actually exist before the action can resolve?
- Can the player be allowed to make a bad or unusual choice and learn through consequence instead of prohibition?
- What does the ordinary employee execute, what does a manager administer, and what remains an owner-level capital decision?
- Have we understood the real process deeply enough before choosing the abstraction?

APPENDIX O

Open Balancing & Calibration Questions

- Capital Victory Net Worth target or target presets.
- Exact numeric qualification thresholds for optional Legacy Victories and the replacement definition for Grand Empire if retained.
- Precise engine test for when all realistic Commercial Ruin recovery paths are exhausted.
- Rival aggressiveness, response time and market-entry frequency.
- Occurrence, propagation and warning calibration for disease, infestation, drought, hail, severe winter, illness and wildland fire without universal annual percentages.
- Quantitative boundaries between inventory, settlement, regional and wider-market shocks by commodity reach.
- Market elasticity, outside-reference influence, replacement-cost behavior and supplier-quote response speed.
- Year-appropriate 1880-1905 retail prices, markups, cash-discount practice and merchant-account default calibration where exact values matter.
- Wagon rated/practical loads by vehicle, team, grade, weather and cargo; freight tariffs.
- Farm machinery work rates, custom-service pricing, fencing costs, windmill pumping and hay/storage tuning.
- Mine development, hoisting, treatment, timber, recovery and machinery capacities where exact balance numbers are required.
- Boarding/rental prices, wage bands and board/lodging valuation.
- Exact legal procedural timing, exemption values and lien-priority edge cases only where gameplay requires greater specificity than the canonical regime.
- Fire-insurance premiums and exact local fire/sanitation distances where authority is needed.
- Electric, telegraph, telephone and water utility tariffs, schedules, capacities, reach and franchise/rate details. CALIBRATION INSTRUCTION Do not invent these numbers merely to complete a feature. Tune them against the live economy after the causal systems exist and can be observed.
- Maintenance wear/service parameters by asset class; defect progression, inspection confidence, downtime and catastrophic-failure calibration.
- Repair parts/material catalogue, blacksmith/wheelwright/harness/machine-shop labor rates and business-capacity defaults by settlement scale.
- Bookkeeping workload rates, discrepancy frequency and internal-control effectiveness without turning ordinary known facts into artificial uncertainty.
- Expertise consolidation: which of the fifty candidate domains become broad tracks, industry familiarity or specialist-only competence; exact experience accumulation and diminishing-return rules.
- Forecast confidence/range calibration so better expertise improves usefulness without becoming clairvoyance or deterministic prediction.
- 1880 opening renter-share calibration by settlement type around the 35-45% provisional range; landlord portfolio distribution and boarding/rental overlap.
- Dakota/South Dakota tenancy terms, rent cadence, arrears, notice/removal law, foreclosure-tenancy treatment, furnishing and landlord/tenant repair obligations.
- Rental construction cost, vacancy, turnover, household housing-choice weights and relationship between condition, heating burden, rent and location.
- Mining assay-office capacity/pricing, custom-mill terms, ore-buying/aggregation practice, investor structures, reprocessing rights and claim/lease legal details.
- Mining work-front, stockpile and operating-posture balance; mine-shift patterns; restart/dewatering and failed-mine breakup economics.
- Rural field work rates, crop/forage list, hay curing/storage, herd cohort/reproduction model, livestock market movement and custom- work contracting.
- Black Hills/West River agricultural leasing, grazing arrangements, irrigation/ditch organization and water-right mechanics before canonizing rural tenancy/use-right systems.

APPENDIX P

- Normalized merchandise size values and representative General Store Display / Storage / Bulk capacity by building footprint; approximately 250 display units is a current tuning-scale example, not a historical lock.
- General Store category markups, landed-cost defaults and blended gross/operating margin calibration. Current source work supports wide variation and a plausible mid-20s blended retail gross-margin reference, but no universal store percentage is canonized.
- Reorder targets, cash-reserve defaults, seasonal stocking profiles and manager decision quality/response speed.
- Workload-to-labor-hour rates by business and task; occupation-specific hours, seasonal daylight usage, shift structure and fatigue only where economically meaningful.
- Employee turnover, poaching responsiveness, retirement, mortality and period-appropriate disease/accident weights; avoid one generic annual worker-death chance.
- Household budget pooling, shopping-trip frequency, urgency thresholds, information memory and off-map purchasing/travel tolerance.
- Black Hills sawmill prevalence by steam/water configuration, mill crew sizes, throughput, fuel/water use, drying/yard needs and seasonal operating practice. 1880 Opening & 1905 End-State Audit NORMAL SPRING 1880 OPENING Population/households - 55-95 residents, 14-22 households, likely sweet spot 70-85; intentionally smaller than real Rapid City by 1880. Built form - predominantly log/frame/wood with rough outbuildings, yards, stables and mixed use; major brick commercial blocks are not normal. Streets - dirt/ruts, limited grading, drainage ditches, partial boardwalk/wood walks where traffic justifies them. Commerce - General Store strongly expected as starter; blacksmith, lodging/boarding, livery/freight and other services seed contextually rather than from a fixed roster. Institutions - mail/postal identity and schooling can exist early; a dedicated schoolhouse, church, library, newspaper, doctor or bank is not guaranteed at this tiny scale. Transport - long overland wagon/stage logistics to distant practical railhead; no local common-carrier rail. Communications - indirect telegraph access likely; local office conditional; telephone network exceptional; mail remains important. Utilities - no normal electricity, piped water/sewer or hydrant network; wells/pumps/creeks/cisterns, privies and ad hoc fire response. Rural economy - ranch-heavy; crop farms where soil/water justify them; horse machinery, barbed wire and windmills available but unevenly owned. Industry - major gold mining exists regionally, but the opening town is not required to be a mine camp. Law/land - Dakota Territorial codified law; transferable town property separated from federal/reservation/trust/restricted land and from historical provenance. STRONGEST PLAUSIBLE 1905 PLAYER TOWN Scale - potentially hundreds to several thousand residents; no fixed target and no requirement for monotonic growth. Built form - denser frame construction plus brick/stone commercial blocks, multi-story business buildings, warehouses, yards and more differentiated districts. Transport - local rail depot likely if the town won service; sidings, freight yards or competing connections possible; wagons remain essential for drayage and feeders. Communications - telegraph office and subscriber telephone exchange/toll links plausible; rural mutual networks may connect surrounding farms/ranches. Utilities - compact electric-light/power network, formal waterworks/hydrants and partial sewer or improved street drainage plausible; coverage can remain incomplete. Fire/civic - mature volunteer fire service likely in a strong node; paid/professional service only exceptional. Schools, civic halls, newspapers, libraries and medical institutions can deepen with actual local support. Commerce/finance - one or more banks, specialized retail/professions, mail-order competition, cash registers/typewriters in larger businesses, ice/cold-storage nodes and regional wholesalers where viable. Industry - deep capitalized mines can use compressed air, sophisticated hoisting, rail, cyanidation and electric lighting; timber, freight and power chains deepen. Rural hinterland - still horse/wagon centered, fenced and increasingly windmill-watered; electricity can remain absent even where telephones exist. Land - ordinary private, federal public, reservation, trust allotment and specialized corridor statuses still coexist.

SECONDARY NODES IN 1905 Secondary towns, mine camps, ranch hamlets and farm-service clusters can remain far less modern than the strongest center. Some may have rail but no electricity; telephone but no sewer; a mine generator but privies; a bank but dirt streets; or neither rail nor piped utilities. Network inequality is canonical. TECHNOLOGY BOUNDARY Core by 1905 where economics justify it: rail, telegraph, subscriber telephone, compact electricity, water/hydrants/partial sewer, advanced hard-rock mining, horse-powered agricultural machinery, fencing/windmills, ice/cold-chain access, office equipment and mail-order competition. Rare/cutting-edge: automobiles, gasoline tractors, broad mechanical refrigeration, extensive electric motive power outside major industry. Outside normal campaign: motor trucks replacing wagon freight, ordinary tractor agriculture, widespread automobiles, automobile-oriented roads, universal rural electricity, universal sewer/paving/professional fire service and modern zoning.

HISTORICAL SOURCE BACKBONE

- A. T. Andreas, Historical Atlas of Dakota (1884), especially contemporary Rapid City development and schools.
- Rapid City Public Library local-history collections for founding, gateway function, banking, civic institutions and rail arrival.

- South Dakota State Historical Society archives and historic-context studies for rail, ranching, roads, municipal development, 1905 vehicle registration and regional institutions.
- U.S. Census historical population tables for Black Hills town scale and non-uniform growth.
- U.S. Geological Survey and period mining journals for mining/milling technology.
- Territorial/state statutes, South Dakota constitutional sources and federal statutes/cases for legal chronology.
- National Archives, National Park Service and federal statutes/proclamations for treaty, reservation, allotment and federal land chronology.
- U.S. Census, Report on the Mortality and Vital Statistics of the United States (1880 Census), especially Western Plains cause-of-death patterns used to ground non-uniform disease, exposure and accident mortality.
- National Archives historical logging materials for dawn-to-dusk, six-day winter logging context; use as work-pattern evidence rather than a universal labor-hours rule.
- National Park Service mill-race and lumber-heritage studies for dam, millrace/headrace, tailrace, mill/log pond and water-powered mill site structure.
- Harvard Business School Historical Collections, Buy Now, Pay Later: Pre-Industrial Credit in Practice, for widespread book credit in agricultural/frontier retail and irregular settlement timing.
- U.S. Congressional Serial Set, Adjustment of Osage Traders' Claims (1902), for contemporary inland General Store testimony on freight/drayage, selling expense, variable merchandise markups and the distinction between invoice cost and delivered retail economics.
- U.S. Census, 1890 real-estate mortgage/home proprietorship tables for South Dakota nonfarm owner-versus-renter tenure; use as the nearest direct tenure anchor while recognizing that 1880 census schedules do not provide the same systematic home-tenure measure.
- U.S. Census 1900 home-tenure tables for South Dakota as a second-period check showing renting remained a major nonfarm tenure state.
- Deadwood/Lawrence County historical-property documentation and local preservation records for purpose-used rental houses, boarding operations, mortgages, ownership changes and landlord/occupant separation.
- 1878 Wolfe's Black Hills/Deadwood business directory for multiple blacksmith/repair shops, blacksmith-and-wagon-repair businesses, a wagon maker and saddle/harness makers before the 1880 opening.
- Historic American Engineering Record / Library of Congress shop studies for combined blacksmith-wheelwright facilities and the late- nineteenth-century transition toward powered machine-shop repair as machinery became more complex.
- Period steam-engineering and boiler-operation manuals plus surviving inspection reports for continual cleaning, lubrication, inspection, repair, boiler condition and the economic/safety importance of competent maintenance.
- Period agricultural machinery-care advice for repair, shelter and seasonal preservation of farm implements; use as evidence for condition/exposure mechanics rather than a universal annual wear rate.
- South Dakota historical incorporation records, congressional assay-office material and USGS mining reports for prospecting/mining companies, mining-and-milling enterprise, assay demand, treatment and mineral-commerce research direction. CHRONOLOGY STATUS All ten targeted 1880-1905 research passes remain complete. Revision X does not reopen chronology; it integrates the subsequent business-agency, inventory, labor, household, physical-world and management-design decisions. Remaining questions are balancing, implementation prioritization and targeted historical calibration.

APPENDIX Q

Maintenance, Records & Information Authority Reference

Q1 Maintenance chain

Use / environment / age / operating practice -> condition and specific defects -> observation/inspection -> maintenance or deferred work -> parts/labor/facility/downtime -> repaired condition, continued deterioration or failure. Player-facing reports should emphasize readable state and known defects.

Q2 Record chain

Actual simulation event -> properly recorded transaction or missing/incorrect record -> reconciliation/stocktake/audit -> discrepancy explanation/correction -> improved historical information. A known transaction stays exact; bookkeeping quality affects completeness and interpretation rather than inventing fake price uncertainty.

Q3 Forecast chain

Available historical evidence + current state + relevant owner/manager/specialist expertise + information quality -> forecast central estimate / range / confidence. Future shock and genuine unknowns remain capable of producing outcomes outside the expected range.

Q4 Common maintenance owner decisions

- Automatic routine-service spending threshold and cash-reserve protection.
- Inspection cadence and which defects trigger shutdown/owner review.
- Repair now, temporary repair, defer, overhaul or replace.
- Stock spare parts or rely on outside sourcing.
- Own reserve capacity or accept interruption risk.
- Schedule maintenance during low-demand windows versus continued production.

APPENDIX R

Experience, Expertise & Commercial Standing Reference

R1 Progression doctrine

Expertise grows from relevant lived commercial experience; Commercial Standing grows from actual wealth, scale and track record; modernization grows from the historical world. None should substitute for the others. Expertise primarily reveals, estimates, compares, diagnoses, warns and recognizes patterns. Any real cost/output improvement must be explainable through a changed contract, route, allocation, staffing decision, equipment choice, maintenance action or other causal process.

R2 Oversized candidate pool

- 1. Bookkeeping & Commercial Analysis - reports, reconciliation, cost attribution, trend recognition and financial interpretation.
- 2. Merchandising - assortment, turnover, stockout, seasonal and display-space interpretation.
- 3. Purchasing & Supply - supplier comparison, landed-cost interpretation, procurement timing and concentration risk.
- 4. Freight & Logistics - route, load, travel-time, utilization and transport-bottleneck analysis.
- 5. Property & Valuation - land/building value, redevelopment potential, repair burden and comparable-market interpretation.
- 6. Negotiation & Dealcraft - counterparty motive, urgency, deal structure and likely room for movement.
- 7. Employment & Management - staffing, retention, supervision, delegation and organizational diagnosis.
- 8. Credit & Collections - customer repayment capacity, receivables, delinquency and collection posture.
- 9. Finance & Capital - debt capacity, financing structure, investment comparison and capital allocation.
- 10. Risk & Resilience - concentration, reserve sufficiency, single points of failure and contingency recognition.
- 11. Retail Trade - industry familiarity with customer traffic, merchandise cycles and store operations.
- 12. Lodging & Hospitality - occupancy, room/board economics, meal demand and transient-customer patterns.
- 13. Agriculture - field work, seasonal windows, machinery use and crop-production constraints.
- 14. Livestock & Ranching - forage, water, herd composition, winter preparation and sale timing.
- 15. Industrial Operations - machinery, power, throughput, process bottlenecks and heavy-operation reporting.
- 16. Lumber & Sawmilling - log supply, mill throughput, drying/storage and lumber-chain constraints.
- 17. Mining - commercial mining reports, development economics, uncertainty and mine-operating structure.
- 18. Banking - loan portfolios, deposits, liquidity, credit quality and concentration.
- 19. Railway Operations & Promotion - corridor economics, freight projections, route strategy and railway negotiation.
- 20. Utilities & Infrastructure - network load, capacity, coverage, maintenance and expansion economics.
- 21. Regional Markets - multi-node price, shortage, migration and market-reach interpretation.
- 22. Forecasting & Scenario Planning - confidence ranges, scenario comparison and recognition of forecast uncertainty.
- 23. Inventory Control - days-of-supply, target stock, excess inventory, stockout and reorder analysis.
- 24. Pricing & Margin Analysis - landed cost, gross margin, turnover and price-response interpretation.
- 25. Supplier Relations - supplier reliability, recurring relationships, volume commitments and priority arrangements.
- 26. Commercial Contracts - commitments, fulfillment exposure, renewal risk and unfavorable-term recognition.
- 27. Customer Accounts & Relationship Trade - repeat purchasing, account history, recurring demand and customer stability.
- 28. Treasury & Cash-Flow Management - payroll, bills, reserves, liquidity collision and near-term cash forecasting.
- 29. Debt & Refinancing - repayment structures, collateral exposure, restructuring and refinance comparison.
- 30. Investment & Partnership Structuring - outside capital, profit sharing, control dilution and risk allocation.
- 31. Acquisition Diligence - books, condition, supplier, staffing and hidden post-close obligation analysis.
- 32. Turnaround & Distress Management - recoverability, restructuring, closure, sale and weak-business diagnosis.
- 33. Maintenance & Reliability - service backlog, failure exposure, downtime and repair planning.
- 34. Asset Lifecycle & Replacement Planning - repair, overhaul, replacement and remaining-service economics.
- 35. Construction & Capital Projects - cost ranges, sequencing, materials, delay and project bottlenecks.

- 36. Land Assembly & Site Selection - frontage, access, adjacency, corridor and expansion-potential interpretation.
- 37. Survey, Title & Property Records - survey descriptions, title concerns and parcel-history interpretation.
- 38. Commercial Law & Legal Awareness - contractual/legal exposure and recognition of when professional advice is needed.
- 39. Insurance & Risk Transfer - coverage, retained risk, claim exposure and protection economics.
- 40. Labor-Market Intelligence - prevailing compensation, scarce roles, recruitment pressure and competitor demand.
- 41. Delegation & Manager Oversight - authority bands, escalation design, manager diagnosis and intervention timing.
- 42. Hiring & Talent Assessment - candidate suitability, reference interpretation and hiring-risk estimation.
- 43. Workload & Scheduling - labor-hours, simultaneous task pressure, shift coverage and seasonal staffing.
- 44. Community & Institutional Relations - authority mapping, civic counterparties, institutional interests and cooperation needs.
- 45. Market Intelligence & Information Networks - newspapers, correspondence, telegraph, telephone and commercial-contact interpretation.
- 46. Seasonal Planning - distinguishing seasonal cycles from structural trends and preparing recurring annual needs.
- 47. Business-to-Business Trade - anchor customers, recurring commercial demand, account exposure and supply commitments.
- 48. Vertical Integration & Portfolio Coordination - internal supply opportunities, shared bottlenecks and transferred rather than eliminated costs.
- 49. Crisis Response & Business Continuity - emergency alternatives, continuity plans and rapid recognition of cascading failure.
- 50. Opportunity Recognition & New Ventures - unmet demand, weak coverage, emerging nodes and potentially viable new enterprise identification.

R3 Current likely consolidation

The final game will probably use fewer than fifty tracks. Current preference is roughly eight to fifteen broad owner expertise areas, separate industry familiarity and specialist employee competence. Bookkeeping/Analysis, Forecasting, Property/Valuation, Merchandising, Management, Purchasing/Supply, Credit, Logistics, Acquisition Diligence and Maintenance/Reliability are the strongest current candidates.

APPENDIX S

Residential Property & Tenancy Reference

S1 Tenure calibration

1890 South Dakota nonfarm housing provides the nearest direct census anchor: about 45% rented overall and about 44% rented outside cities of 8,000 or more. 1900 known-tenure nonfarm homes remain around 40% rented. Because the opening game is 1880 and the census did not yet collect the same tenure table, use approximately 35-45% renter households as a provisional opening calibration for independent nonfarm households; boarders/lodgers remain additional.

S2 Core entities

- Property/building owner.
- Residential unit.
- Occupant household/person.
- Tenure arrangement.
- Rent/arrears/payment history.
- Condition/maintenance.
- Debt/encumbrance.
- Housing suitability and service connections.

S3 Landlord strategy without classes

Stable landlord, high-rent operator, cheap/crowded housing investor, premium landlord, speculative builder, employer landlord, mixed-use owner, distress buyer and land assembler are emergent strategies. The game should not require the player to select a landlord archetype or Slumlord perk.

S4 Open historical questions

- 1880s Dakota/Black Hills rent cadence, written versus oral tenancy and deposit practice.
- Notice/removal/eviction procedure and tenant treatment through foreclosure.
- Typical division of repair, fuel and furnishing responsibilities.
- Distribution of landlord types and portfolio sizes in Black Hills settlements.

APPENDIX T

Repair, Fabrication & Technical Services Reference

T1 Capability families

Opening-era: smithing, horseshoeing, wagon/wheel work, harness/tack and general repair. Growth-era: larger implement/industrial work, manufacturer parts and internal mine/mill maintenance. Late campaign: machine-shop and railway/utility specialized work where capital and industrial demand justify it.

T2 Work-order chain

Asset/customer need -> reported problem -> physical arrival or on-site service -> diagnosis -> quote/terms -> queue/prioritization -> materials/parts -> skilled labor -> completion -> payment/account -> asset returns to service -> repair history updated.

T3 Key business tensions

- Routine work versus rush interruption.
- Walk-in customers versus recurring contracts/reserved capacity.
- Fabricate locally versus order a manufactured replacement.
- Stock slow-moving parts versus preserve cash/storage.
- Train/retain skilled workers versus risk losing them to competitors.
- Repair customer assets versus buy/refurbish/salvage used equipment.

APPENDIX U

Mining & Rural Production Square Matrices

U1 Mining width x depth

Width lanes: Prospecting/Claims; Mine Operation; Investment; Assay/Technical Services; Custom Treatment; Mineral Commerce; Mining Supply; Contract Services; Mine-Camp Property; Reprocessing/Failed-Mine Redevelopment. Depth dimensions: Information; Physical Capacity; Labor/Skill; Inputs/Logistics; Customers/Contracts; Cash/Finance; Condition/Reliability; Time/Lifecycle; Authority/Delegation; Regional Consequence.

U2 Rural width x depth

Width lanes: Crop Farming; Hay/Forage; Cattle; Sheep where distinct; Local Produce; Custom Agricultural Work; Livestock Commerce; Agricultural Supply; Processing/Storage; Rural Property/Infrastructure. Depth dimensions: Land/Resource; Biological State; Seasonal Timing; Labor; Draft/Machinery; Inputs; Storage; Maintenance; Market/Logistics; Capital/Authority.

U3 Square rule

The square matrices are completeness tests. If one lane has only a name and no answers to the depth dimensions, it is decorative width and needs design work or removal. If one central activity accumulates many more detailed mechanics than the surrounding economic ecosystem, it is excessive depth and the design should widen before adding more microscopic detail. The goal is balanced causal richness, not equal UI complexity for every lane.

APPENDIX V

Household, Accommodation, Kinship & Demographic Lifecycle Reference Status: Revision XII canonical reference appendix. This appendix compresses Section 12.6 for routing, balance and future balancing. It does not override the body.

V1. Relationship authority map

Entity / relationship Answers Key rule Person Who is the individual? Persists through household, work and accommodation changes. Family / Kinship Who is related to whom? Explicit parent/child/spouse graph; does not require co-residence. Household Who shares a domestic economy? One adult or larger group; size is lifecycle output, not building capacity. Residential Occupancy Who is staying where and on what terms? Relationship object; can represent rent, boarding, employer lodging, kin hosting or temporary shelter. Accommodation Space What physical shelter is usable? Capabilities and condition; can be room, dwelling, bunk space, rough shelter or tent/camp element. Building What physical structure contains functions? Can contain commercial, storage and multiple accommodation spaces. Parcel What land/legal geography underlies it? Ownership/transfer authority remains separate from occupancy.

V2. Accommodation arrangement matrix

Arrangement Typical authority/terms Simulation implication Owner occupied Household/owner controls occupancy subject to debt/legal rights. Security depends on finances, condition and legal exposure, not ownership label alone. Independent rental Lease/tenancy relationship. Longer-term household settlement; rent, maintenance and removal rules apply. Room / partial building Household rents only part of mixed-use or multi-unit property. Supports dense Main Street and family/single-person options without one-building-one-household assumption. Boarding / lodging Bed/room and often meals/services. Crucial for single workers, transients and early migration; distinct households may share one house. Employer lodging Employment-linked bunk, cottage or board arrangement. Recruitment value and job-dependence emerge causally. Kin hosted / doubled-up Host household or shared occupancy. Emergency/low-cost capacity; households may remain financially separate. Temporary shelter / camp Tent or rough shelter with real location/capability. Allows booms ahead of permanent stock; exposes weather, fire, sanitation and crowding pressure. No stable shelter No reliable current arrangement. Triggers search/migration pressure; not a passive permanent status.

V3. Dependent-minor resolution matrix

Trigger: a dependent minor loses the last locally responsible adult. Resolve care through a dedicated dependency process rather than directly assigning the child to a household. Priority Search lane Rule 1 Existing lawful responsibility Surviving parent or existing guardian/caretaker relationship where valid. 2 Suitable close kin Adult sibling, grandparent, aunt/uncle or other meaningful kin, evaluated for actual suitability. 3 Established non-kin Known household/associate/neighbor with strong social connection and ability to care. 4 Suitable local placement Other willing household when no stronger relationship resolves the case. 5 Off-map/public/charitable path Distant kin, county/public/charitable or institutional route where historically appropriate. Non-negotiable: placement priority is a game decision architecture, not a claim that 1880 Dakota law imposed this exact order. Siblings are preferentially kept together. Care/guardianship does not transfer inherited property. Exact legal procedure remains a research hold.

V4. Demographic calibration anchors

Measure Anchor Use Confidence / caution Opening residents 55-95; preferred 70-85 Canonical fictional settlement scale. High design confidence; intentionally compressed vs real Rapid City.

Measure Anchor Use Confidence / caution Opening households Legacy 14-22 becomes indicative Check domestic-unit plausibility, not hard validation. Revision XII supersedes hard-band interpretation. U.S. TFR 1880 ~4.24; 1890 ~3.87; 1900 ~3.56 Aggregate conception/birth calibration. Historical reconstruction; not Dakota quota. Survival-adjusted TFR 1880 ~2.96; 1900 ~2.78 to age 10 Prevents births = surviving children. Broad national series. First marriage 1890 medians: men ~26.0, women ~21.9 Marriage formation calibration. National late-19th-century anchor. Census family size U.S. 1880 5.04; 1890 4.93; SD 1890

4.7 -> 1900 4.8

Distribution sanity check. Census family != game Household; translate carefully. Child work evidence 1880 occupation reporting age 10+; productive work could be recorded younger Separates contribution from adulthood. Does not imply universal employment or legal independence.

V5. Demographic calibration expectations

- Across repeated 25-year campaign simulations, compare births per woman and surviving-child distributions against broad historical envelopes.
- Track age at first marriage, widowhood/remarriage, household splitting/merging and adult children remaining/leaving.
- Track one-person households, boarders, renters, kin-hosted households, employer lodging, temporary shelter and accommodation insecurity separately.
- Check whether family-sized households are disproportionately blocked when a node has beds but the wrong type of accommodation.
- Measure orphan/dependency cases, sibling separation frequency, local versus off-map resolution and estate/guardianship continuity.
- Check whether boom nodes can grow temporarily through boarding/camps without permanent housing, then either harden into durable settlement or contract when jobs disappear.
- Avoid calibration that requires hidden household-size caps, fertility buffs/debuffs or teleporting placement. Fix the causal parameters instead.

V6. Research holds carried forward

Demography: Black Hills/Dakota parity and childlessness distribution; birth spacing; age-specific fertility and remarriage by settlement type; maternal, infant and childhood mortality by place/subgroup; prevalence of multi-generation and fostered/kin-care households. Children and legal status: territorial/state majority and contracting authority; school-attendance chronology; occupation-specific child-work restrictions; apprenticeship practice and transition to independent wage work. Guardianship and dependency: exact spring-1880 guardian appointment law and procedure; county poor-relief/charitable practice; property administration for minors; local versus off-map care patterns; terminology appropriate to Dakota courts and counties. Accommodation: Black Hills boarding/lodging composition, tenancy cadence, oral versus written arrangements, furnished/unfurnished room practice, eviction/removal chronology, fuel/repair responsibilities and how period census definitions should map into the game's relationship ontology. These holds do not block the relationship architecture. They block only false precision. Future balancing should preserve configurable calibration data so validated historical values can replace placeholders without changing the entity model.

V7. Source anchors

U.S. Census Bureau: 1880 Census Instructions to Enumerators; Compendium of the Eleventh Census: 1890 - Progress of the Nation; 1900 Census, Volume 2: Population, Part II; historical Median Age at First Marriage: 1890 to Present series. Social Security Administration: historical U.S. fertility series with survival-to-age-ten adjustment. Dakota Territory: The Compiled Laws of the Territory of Dakota, A.D. 1887, Chapter 13, Guardian and Ward. These sources are used as architecture/calibration evidence, not as a claim that every national statistic directly describes the fictional opening node.

APPENDIX W

Revision XIV Depth-Balance Audit & Research Anchors

W1 Systems deliberately deepened in Revision XIV

- Butcher - expanded from a short business identity into sourcing, carcass balance, preservation, capacity, accounts, byproducts and continuity.
- Boarding House / Hotel - expanded into accommodation products, board/meal capacity, customer mix, labor, project demand and safety/continuity.
- Doctor - expanded into time/travel capacity, triage, fees/accounts, practice continuity, reputation/referral and illness-wave load.
- Markets / Competition - expanded into category markets, substitution, rival strategy, price/service competition, advertising, entry/exit and memory.
- Banking - expanded into balance-sheet liquidity, loan concentration, deposits/confidence, correspondent stress, acquired assets and governance.
- Railway ownership - expanded from corridor development into tariffs, traffic, service, depot handling, competition, regulation and maintenance.
- Executive / Portfolio - expanded into exception management, manager differences, shared services, treasury, governance, talent allocation and communication reach.
- Acquisition / Dealcraft - expanded into deal structure, exchanged concessions, time pressure and question-based diligence.
- Expertise - consolidated from an oversized reference pool toward broad owner families plus industry familiarity and specialist knowledge.

W2 Systems reviewed but not expanded materially

Shared maintenance and records, household/kinship/housing, rural production and mining already carry strong causal depth relative to their width. Telegraph, telephone, electricity and civic networks also already contain sufficient adoption, geography, ownership and reliability logic for the current Canon stage. Further additions to those systems should answer a specific design problem rather than increase detail for its own sake.

W3 Verified research anchors used for the Revision XIV pass

- U.S. Census Bureau, 1880 Instructions to Enumerators: hotels and boarding-style shared-roof/shared-table arrangements, boarders, hotel cooks/waiters and nonstandard dwellings are explicitly recognized in contemporary enumeration practice.
- U.S. National Park Service, Lucretia Marshbanks: direct Black Hills evidence for a woman serving as cook and manager at the DeSmet mine establishment and later purchasing a boarding house in Sawpit Gulch in 1883, renamed Rustic Hotel.
- Library of Congress, Custer Chronicle overview (1880-1890): opening-era Black Hills newspaper advertising included hotels, blacksmiths, banks, a surgeon, groceries, restaurants and other local businesses, supporting advertising as a real information channel rather than a modern marketing abstraction.
- 'A Doctor in the House: The Architecture of Home-offices for Physicians in Toronto, 1885-1930' (PMC): North American physician practice retained a strong home-visiting model; travel burden and later transport/telephone access helped make office practice more efficient. The Canon uses this as structural evidence, not as a Black Hills fee or staffing calibration.
- FDIC historical timeline, 1850-1899, and Federal Reserve History, Banking Panics of the Gilded Age: nineteenth-century banking depended on deposits, reserves, loan quality and interbank relationships; panics could produce withdrawals, bank suspensions and transmission between interior and money-center banks. Statutory national-bank reserve rules are not applied automatically to every possible player bank form.
- Smithsonian Libraries / National Museum of American History, 1877 freight tariff and classification and Sinclair's 1869 Freight Computer: freight charges were explicitly calculated from weight/rates and railroads used freight classifications and tariffs well before the game's opening.
- U.S. National Archives, Interstate Commerce Act (1887): interstate common carriers faced requirements for just/reasonable and published rates plus restrictions on rebates, undue preferences and long/short-haul discrimination under the new Interstate Commerce Commission regime.

W4 Ongoing depth discipline

When future design work identifies a broad new business or institution, first compare it against the depth dimensions in 7.5A and the square-matrix doctrine in Appendix U. If it adds a new name but no new causal relationship, keep it parked. If it adds a relationship the existing economy cannot express, research the real process deeply enough to understand its inputs, decisions, failure, lifecycle and cross-system consequences before promoting it to active scope.

APPENDIX X

Revision XIV Commercial-Consequence & Play-Mode Synthesis

X1 Canon-ready systemic doctrines added in this revision

- Persistent Economic Consequence: important decisions alter later possibilities and the world retains economically meaningful history.
- Ownership-as-Responsibility: ownership converts outside dependency into controlled internal scarcity, obligations and maintenance rather than solving the problem.
- Commercial Commitment Authority: agreements reserve real future resources/capacity and remain distinct from physical execution.
- Credible Alternatives: bargaining power and resilience derive from feasible alternatives rather than Influence, Monopoly or Capacity meters.
- Bounded Awareness: actors reconsider through Trigger -> Awareness -> Feasibility -> Shortlist -> Choice rather than omniscient optimization.
- Advertising-as-Information: advertising changes awareness of a real offer; it does not create demand by modifier.
- Organizational Coverage: execution, decision, information, administrative and specialist coverage remain distinct causal problems.
- Owner Attention Without Action Points: travel, authority and communication determine what the principal can personally resolve.
- Information Distance: remote organization can know local reality before the principal receives or interprets it.
- Capital and Talent Allocation: mature play increasingly allocates scarce money, managers and specialist attention between competing uses.
- Partial Ownership and Control Rights: economic interest, governance, capital responsibility and exit rights need not be identical.
- Internal Opportunity Cost: wholly owned businesses do not fake-negotiate, but internal use preserves real physical and forgone-market cost.
- Distress as Option Loss: failure becomes dangerous as credible responses disappear; entity failure need not equal player-career failure.
- Recombination: failed firms release workers, assets, property, customers and knowledge into the wider economy.
- Commercial Succession: legal ownership, operating capability and financial continuity are resolved separately when key people leave or die.
- Selective Economic Memory: persist identity-changing events; summarize routine execution.
- Economic Geography: the player changes practical relationships across the map rather than conquering territory.

X2 Play-mode authority added in this revision

Scenarios are the formal authored campaign layer. They are ranked by difficulty, built around specific commercial situations and judged through primary objectives plus optional distinctions. The same simulation rules apply as everywhere else. Scenario randomness is encouraged when it preserves the premise.

Open Frontier is the flagship procedural sandbox. It should be aggressively replayable, generating a different 1880 regional economy from seed while preserving the same historical capability envelope. It has no mandatory victory condition by default. Optional Capital/Legacy goals can be enabled, but they do not define the mode.

The two modes solve different player desires. Scenarios ask: can you solve this particular commercial problem? Open Frontier asks: what kind of economic history will emerge from this world? Neither is a lesser version of the other.

X3 Scenario-family concept pool - design direction, not locked production order

Possible families include Foundations (merchant, property, freight, settlement growth), Industrial Frontier (timber, ranch/winter, mine boom, custom processing), Network Era (railhead, bypassed town, telegraph/telephone, electric light), Capital & Crisis (Panic of 1893, banking, distressed empire, railroad promotion) and Capital Command (regional power, succession, late-portfolio restructuring). Exact scenario names, count, objectives, prerequisites and difficulty ratings remain future design work.

X4 Active research/design holds

- Regional commercial geography, overlapping market areas and whether routes need persistent economic identity.
- Wholesale/branch-business structure, central purchasing, local books/managers and internal freight.
- Rail-depot drayage/transshipment and the transformation of long-haul wagon businesses after rail arrival.
- Exact ordinary B2B agreement conventions where stronger Black Hills evidence is still needed.
- Probate, partnership death, inheritance and married-property legal details not yet validated.
- Employee-to-partner/owner pathways and period-specific compensation/ownership arrangements.
- Exact Legacy qualifications and Grand Empire replacement.
- Final Open Frontier setup parameters, procedural ranges and scenario campaign catalogue/progression.

X5 Production filter

Future additions should usually deepen the network of consequences connecting systems the player already manages. A new business or institution earns active scope when it introduces a genuinely new economic verb, asset, relationship or risk that cannot be expressed cleanly by existing systems. The game does not need to simulate everything about frontier life. It should simulate the economic consequences of the things it chooses to represent unusually well.

END OF ACTIVE CANON - REVISION XIV